

frontmatter

The complete record · 238k words · 2026-08-31

121	237.7k	56	518	38	7
sections	words	diagrams	tables	research agents	rounds

v2.1 adds the four-page digest that opens this document, and the compact typeset. No finding was added, changed or removed from v2.0.

Supersedes FRONTMATTER-PRD-2026-08-29.md (v1.1), FRONTMATTER-BUILD-PLAN-2026-08-29.md , and FRONTMATTER-DECISIONS-2026-08-29.md . Those three are now historical; this is the single governing document.

Status: pre-development. This is the input to the build, not a report on it.

Read this. Open the rest only when you need the working. Every claim below is expanded, sourced and evidence-tagged in the section named beside it.

The product. A markdown editor with a deliberately simple surface and a deep engine. You prompt any LLM anywhere; the outputs that matter land as structured, rendered, agent-legible markdown in files you own.

- The law (§5).** The file is the only source of truth. Every app-like thing — board, calendar, decision card, dashboard, published site, and every AI edit — is a deterministic, reversible projection of that file, owning no state of its own.

The category sentence. Notion made the app the source of truth and trapped your data inside it. frontmatter makes the *file* the source of truth and lets every app be a disposable lens over it — provably, byte for byte, reversibly.

The problem, in one line (§2). Six of the nine problems share one shape: *the tool reports success while quietly losing something*. Sync said green and lied. Import said 47 succeeded and lost 4. The editor said it preserved comments and deleted them. **So: be the tool that does not lie about what it did to your file — and can prove it.**

What is actually proven

CLAIM	EVIDENCE
Our engine does not corrupt strangers' data	8,513 files, 7 vaults, 0 corruption, 0 throws MEASURED , byte-pinned and re-verifiable via <code>npm run corpus</code>
Competitors do	Three write paths executed : Front Matter CMS deletes YAML comments while its own source says it preserves them; Hubble deletes reference links with their text; OpenKnowledge's frontmatter permanently drifts MEASURED
Every rich-text framework is lossy by design	<code>blocksToMarkdownLossy()</code> is a real shipping API name FETCHED
The internal substrate works at scale	124 SKILL.md automations, 24,539 gate decisions, 5,014 trace rows MEASURED

What is not proven

Nobody has paid us anything. And developers are the hardest freemium audience there is — median dev-focused free-to-paid is 5%, **half the non-developer rate** FETCHED . Every conversion, churn, ticket-rate and cost figure in §23–26 is an assumption, stated so it can be replaced by measurement.

The biggest risk is not technical: a fidelity guarantee may be something engineers admire and nobody buys.

The decisions, all of them

#	DECISION	WHY	§
1	No new markdown format	Profiles over valid CommonMark that degrade in a dumb renderer	0.2
2	Carrier = blockquote callout for prose, fenced code for opaque data	A callout has no closing marker to lose ; an unclosed fence swallows the rest of the document (CommonMark §4.5)	8.2
3	Sync = git-merge + splice journal + compare-and-swap. Never a CRDT	A CRDT cannot own the bytes, cannot refuse, and interleaves — convergence buys byte-identical garbage (arXiv 2305.00583)	31.1
4	Never a tree-of-record	The market's own confession, above	0.2
5	The eval lane is refused	It ends the corruption guarantee <i>and</i> turns every prompt injection into RCE on infrastructure holding all customer documents and keys	9.2
6	Control plane in Postgres, zero document bytes; <code>workspace_id</code> from day one	Retrofitting tenancy post-launch is the highest-cost change on the board	7
7	MDMAX wires in as four ordered seams	Currently 1 symbol from 1 of 13 files is used	7.3
8	Publishing v1 = narrow, paid, <code>noindex</code> , no uploads	Paid + noindex + random slug removes the entire commercial abuse motive	44
9	Not a Notion-style PM tool	Founder boundary	0.2

The numbers that decide things

Foreign-vault refusal rate today	83% — one cause, zero-indent YAML sequences. Every fidelity number we would market is false until NF-1 and NF-3 land (§28.1, §58)
FX	₹95.4/\$, not ₹83. So ₹699 = \$7.33, above every India AI anchor → ₹599 (§24.1)
RBI ceiling	₹15,000/transaction is an architectural constant. Indian cards get exactly one payment attempt (§45)
Contribution margin	57–73% on every INR tier. Margin was never the problem (§23.2)
₹20L/mo requires	113,507 free signups , ~1.26M visitors . A distribution problem, not a pricing problem (§25.3)
Support wall	46.4 founder-hours/month at 10,000 users — before engineering (§25.1)
Cloudflare R2	No object versioning . The standard DR recipe does not port (§32.1)

The plan

R0 engine truth → T0 trust surface → T1 tenancy → T3 AI protocol → T4 capture → v1. R0 alone is ~9 weeks and is the part most likely to be wished away. Renders (T2) are out of v1 except two one-line cleanups.

First week (§28.8): rotate the two PATs · stand up CI and **make it fail once** before trusting a green · NF-3 red proof (set-only, on a synthetic fixture — the corpus has zero bare-CR fences and cannot prove it) · NF-1 fix · re-derive the recovery rate to replace an inference with a measurement · write the NFC/NFD key-equality decision.

What only you can decide (§54)

The name · whether documents leave the device · BYO vs platform key · does the agent get write authority · CJK in or out of v1 · does `café` in NFC equal `café` in NFD for key addressing — **the one engine unit an agent cannot start**. And the two unrotated PATs, today.

What we may not say in public (§58)

Any fidelity percentage · "NF-1 recovers 99.98%" (an inference, not a measurement) · anything about Obsidian's onboarding funnel (the widely-repeated claim is **unsourced**) · "first AI attribution" (Cursor and Grammarly exist) · "serve markdown to agents and get cited" (refuted) · any `ss`-tagged number.

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97.1 Threat model · 97.2 Secrets management · 97.3 HTTP headers and the exact CSP · 97.4 XSS and the sanitisation boundary · 97.5 SSRF · 97.6 Rate limiting · 97.7 Bot and scraper defence on published pages · 97.8 GitHub App token blast radius · 97.9 Prompt injection and the lethal-trifecta gate · 97.10 Dependency and supply chain · 97.11 Vulnerability disclosure · 97.12 Deliberately not doing yet

98 Caching and content delivery

98.1 The five layers and what belongs in each · 98.2 Content-addressed caching: the SHA is the cache key · 98.3 The GitHub rate-limit budget · 98.4 The snapshot problem · 98.5 Publish, unpublish, and immediate revocation · 98.6 Stale-while-revalidate, and where it is forbidden · 98.7 The cache-decision table · 98.8 CDN choice and prices

99 Scaling, load balancing and capacity

99.1 The capacity model, with the arithmetic · 99.2 What we build vs. what we rent · 99.3 Database connections: the real serverless bottleneck · 99.4 Queue design and the backpressure rule · 99.5 The GitHub API: a limit we do not own · 99.6 The degradation ladder — what we shed, in order · 99.7 Where this architecture stops working

100 Observability, availability and disaster recovery

100.1 The constraint that shapes the whole section · 100.2 Stack selection · 100.3 Telemetry topology · 100.4 The correlation envelope, from day one · 100.5 The three scrubbing mechanisms, plus the gate · 100.6 Retention · 100.7 SLOs a solo founder can honestly commit to · 100.8 Alerting · 100.9 Status page and incident comms · 100.10 Incident runbook skeleton

00 Is this real? # the 30-second check that rules out a false alert**01 Stop the bleeding # the ONE reversible action: feature flag, rollback, disable route****02 Confirm blast radius # query with workspace_id + time window, paste the numbers****03 Communicate # status page template text, pre-written, copy-paste****04 Diagnose # links: Sentry saved search, Axiom APL query, Vercel deploy list****05 Fix or escalate # vendor support links + account IDs****06 Verify recovery # the assertion that must go green, run it 3 times (LR#63)**

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§0 How to read this document

What it is. The complete product record: where the idea came from, what it solves, what the market cannot do, what we will build, how it is written on disk, how it is priced and sold, what it costs to run, what can kill it, and what must be decided before anyone writes code. It consolidates **twelve research rounds, 105 reports, 374,866 words**, plus the engineering record of this repository.

Audience. The build team. §1–5 for why · §6–10 for the substrate · §11–20 for the product · §21–27 for the business · §28–49 for the build · §50–60 for operating and deciding.

0.1 Evidence tags — the difference between a fact and a lead

Every factual claim carries one. This is not decoration; §57 exists because we once did not.

TAG	MEANS	PUBLISHABLE AS FACT
MEASURED	Executed on this machine against live code or a real corpus	Yes
FETCHED	Primary source opened and read	Yes
DERIVED	Computed here; the arithmetic is shown	Yes, with the arithmetic
SS	Search summary — nobody opened the page	No. Verify or omit
INFERENCE	Reasoning from the above	Labelled as judgement

0.2 Settled verdicts — do not re-litigate

Each cost weeks and survived every round since.

#	VERDICT	COST OF REOPENING
1	No new markdown format. Every extension is a profile over valid CommonMark that degrades to readable text in a dumb renderer	Settled 2026-07-29. A formally-specified markdown successor launched Aug 2026 sits at 2 stars FETCHED
2	MDMAX is a library subordinate to the editor, not a compiler product with its own destiny	—
3	Never a tree-of-record. No feature may introduce a parse tree as the source of truth	<code>blocksToMarkdownLossy()</code> is a real API name in a shipping framework — the market's own confession SS
4	Sync is git-merge + a splice journal + compare-and-swap. Never a CRDT	§31.1 gives three independent disqualifications
5	The render carrier is a blockquote callout for prose and a fenced code block for opaque data	§8.2. This corrects PRD v1.1 §9
6	The eval lane is refused — no arbitrary client-side code execution	§9.2
7	Not a Notion-style project-management tool	Founder boundary

0.3 Three rules that govern the build

- 1. **Red proof before green.** A test on a rare fault proves nothing until it fails against the unfixed code. If you cannot make it fail, the test does not cover the bug — say so rather than reporting a pass.
- 2. **Agents propose, the founder merges.** Capture `git rev-parse HEAD` before and after every agent run and reconcile. The instruction not to commit is advisory; the reconciliation is the gate.
- 3. **Re-derive every number at write time.** §57 is a list of twenty numbers that went stale because we did not.

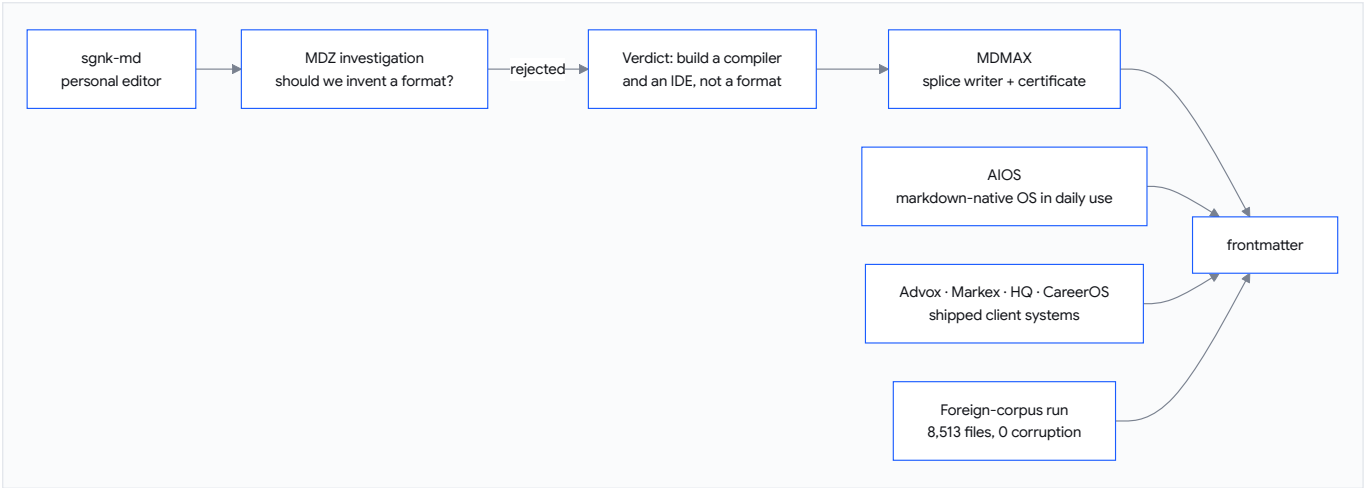
0.4 How this document is executed

Through `specs/`, not through prose. A lane is done when its spec reaches `state: verified`, which the harness writes only after every `verify:` command exits 0 and a red proof exists.

```
npm run spec      # the contract gate
npm run corpus    # the fidelity corpus, 8,513 files, byte-pinned
npm run verify    # typecheck → lint → test → build → arch → spec
```

§1 Where the idea came from

In one line: we tried to invent a better markdown, proved that was the wrong move, and were left holding the two pieces that turned out to be the product.



STAGE	WHAT HAPPENED	WHAT IT LEFT BEHIND
sgnk-md	Personal Obsidian alternative, live at md.sgnk.ai. 416 commits, frozen since 2026-07-17 MEASURED	The editor chassis: four view modes, tree, tabs, slash commands, vim keys, wikilinks, KaTeX, Mermaid, PWA, Tauri
MDZ investigation	Asked whether a markdown successor was the play	Rejected. "Build a compiler and an IDE, not a format"
MDMAX	The compiler half, built instead	13 files, 3,614 lines: splice writer, OffsetMap, degradation certificate, construct detectors MEASURED
AIOS	An orchestrator running this studio entirely on markdown	124 SKILL.md automations, an 892-line self-amending constitution, 5,014 trace rows, 24,539 gate decisions MEASURED
Client systems	Advox, Markex, HQ, CareerOS	Reusable spines: entitlements, pooled-RLS tenancy, fail-closed verifiers, publish queues
Foreign-corpus run	Ran the writer against seven strangers' vaults	8,513 files, 0 corruption, 0 throws MEASURED

The insight that closed it. The two things MDMAX shipped — a writer that provably cannot corrupt, and a certificate that proves how a file degrades — are exactly the enforcement mechanism for a much bigger idea than a format: **the file can be the only source of truth, and every app-like view can be disposable**. We built the enforcement before we named the law.

§2 The problem

#	PROBLEM	EVIDENCE	ANSWERED IN
1	AI work evaporates. An hour of thinking ends as a chat log. ChatGPT's only exit is an account ZIP emailed with a 24h link; NotebookLM did not persist chats until Jan 2026; an exporter-extension economy exists (one at 400,000+ installs) and all of it ends in a stale one-shot dump	SS/FETCHED	§12
2	Sync silently destroys data. The single most-cited pain: 561 of 3,220 HN comments. Fresh, not historical — the Obsidian forum in the last six weeks carries a file showing "fully synced" while missing its final Korean characters	SS + FETCHED	§31
3	Editors rewrite bytes you never touched. Architectural, not a library choice — three teardowns, three codebases, one identical failure	MEASURED	§71, §19
4	Nobody can tell you what the AI changed. Word, Docs, Cursor, Grammarly and Lex all show AI edits before accept and lose the distinction forever after	SS	§11
5	Everything you own is trapped in a view. Coda export returns disconnected strings with formulas stripped; Notion formulas cannot aggregate across rows; under 4% of GitHub notebooks reproduce	SS	§5
6	The review loop does not exist on files you own. Google's public API cannot create suggestions at all; Lex's track-changes is "in development"; OpenKnowledge comments are machine-local and never committed	FETCHED	§13
7	Import is a lie. The #1 failure class is not broken formatting — it is reports claiming success while losing files . Obsidian has posted a \$500 bounty for a detailed import log and \$5,000 for Notion database conversion	FETCHED	§20
8	Price betrayal. Notion cut free AI to 20 responses <i>for life</i> and raised Business ~20%; Microsoft's +43% Copilot bundling drew a CMA probe; Cursor apologised for an opaque reprice; Windsurf churned twice	SS	§24.8
9	Complexity fatigue. Markdown tools overwhelm the people they are meant to serve	see note	§18

Note on problem 9. The widely-repeated claim that "*most of Obsidian's million-plus downloaders never get past their first note*" is **unsourced** — it traces to an unattributed pull-quote in a 2025 author blog post with no Obsidian source, no methodology and no denominator FETCHED. **Do not cite it.** The complexity problem is real; that particular statistic is not evidence for it.

The shape shared by problems 1, 2, 3, 4, 6 and 7: the tool reports success while quietly losing something. *Sync said green and lied. Import said 47 succeeded and lost 4. The editor said it preserved comments and deleted them.*

Product thesis: be the tool that does not lie about what it did to your file — and can prove it.

§3 The market

3.1 Category map

CATEGORY	PLAYERS	WHAT THEY OWN	WHY THEY CANNOT TAKE OUR POSITION
Office suites	Word, Google Docs, LibreOffice	Track changes, 40 years of review UX	Not file-native; Google's API cannot create suggestions FETCHED
Workspace / blocks	Notion, Coda, Craft, Airtable	Databases-as-views	The view IS the data; export lossy by architecture SS
Markdown PKM	Obsidian, Logseq, Bear, Ulysses, iA Writer, Typora	Local files, plugins	Single-player by design; no review loop, no provenance, no fidelity proof
Markdown collab	HackMD, Hubble, OpenKnowledge, GitBook, Mintlify	Team markdown, docs sites	All regenerate the file; OpenKnowledge needs a running daemon MEASURED
AI editors	Cursor, Windsurf, Lex, Grammarly	Inline AI, diff review	Provenance evaporates at accept; not markdown-native SS
Chat-to-artifact	Claude Artifacts, Canvas, v0, Bolt, Perplexity Pages	Generation, versioning	The artifact lives in their store. OpenAI removed Canvas in May 2026 SS
Agent memory	mem0, claude-mem	Facts for agents	Atomises the journey; the finished document is not first-class
Free self-hosted	AppFlowy 76,040★ · AFFINE 71,976★ · SiYuan 46,023★ · Logseq 44,669★ · Outline 40,362★ · Trilium 37,624★ · Docmost 21,501★ — all pushed within 24h FETCHED	Zero cost, file-native, active	This is the price floor. \$5–8/seat must be justified against \$0

3.2 The gap map

GAP	EVIDENCE OF ABSENCE	WINDOW
Byte-exact structured editing on both halves of the file	All three teardowns regenerate; the two strongest each solved fidelity only on the half their editor does not model MEASURED	Frontmatter-splice claim durable; general claim erodes
Fidelity without a running daemon	OpenKnowledge's byte contract exists only inside a live CRDT server — agent edits error out without it MEASURED	Durable — architectural
O (edit) not O(document)	Their own source comment: full serialize+parse per edit, " <i>unbounded by doc size</i> ", and fixing it " <i>needs a real incremental parser</i> " they lack FETCHED	Durable
Render-fidelity certification	Nobody certifies cross-engine degradation. Litmus at \$500/mo proves the <i>testing</i> layer monetises while the <i>data</i> layer stays free SS	6–12 months. Publish the dataset before the design
The review loop on files you own	Google's API cannot create suggestions FETCHED ; Lex "in development" SS ; OK comments never committed FETCHED	6–12 months
Byte-anchored, portable, reader-visible AI provenance	Absence check precise: admin telemetry (Cursor), cloud report (Grammarly), repo sidecar (Agent Trace). Writing editors have nothing SS	Open
The rendered, evolving AI-output document	Memory tools store facts; consumer apps store sources; exporters store dumps	Open — the category to name
Session interchange between AI tools	No standard. OpenAI and Claude exports are mutually unreadable; every parser is reverse-engineered FETCHED	Open — whoever publishes the open shape becomes default
A file-native home screen	Obsidian's Homepage plugin: 1,294,057 downloads , ~20× every bespoke dashboard plugin FETCHED . No product ships it natively	Open
Claim-level method + confidence	Across OKF, SKILL.md, llms.txt v2, MyST, .prompty — no spec claims it; OKF argues <i>against</i> stored scores FETCHED	Open; a renderer is the right shape
"An agent can read and write my markdown"	Commodity. Bear, Craft, Notion, MDflow, GitBook ship it; obsidian agent-skills 47,418★; the skills CLI did 9.3M downloads in one week FETCHED	0 months — do not position here

3.3 The window

- **Now:** generic agent-markdown is commodity.
- **3–6 months:** "markdown editor with an agent harness" crowds. OpenKnowledge ships ~100 releases/week; 3,239 → 3,673 stars in 27 days FETCHED .
- **6–12 months:** fidelity, the review loop and the Obsidian-power-user lane stay open, because every incumbent's incentives point elsewhere — Notion toward blocks, OpenAI toward artifacts-as-output, Google toward Docs, OpenKnowledge toward the team wiki.
- **Structural hedge:** OpenAI removed Canvas. If "delegate, don't edit" wins, the durable asset is the verification layer, not editor chrome. **The roadmap keeps weight on the engine.**

Sequencing law: the differentiation tracks (R0, T2, T3 — §28) must land inside the 6–12 month window.

§4 Is the idea good? The honest verdict

Yes, with two conditions: ship fidelity and the review loop inside the window, and do not position on the commodity sentence.

QUESTION	ANSWER	CONFIDENCE
Is the problem real?	Yes. Nine problems with public complaint evidence; three verified by executing competitors' code	High MEASURED
Is anyone else solving it?	Partially, each capped by an architectural choice they cannot cheaply reverse	High MEASURED
Do we have an unfair advantage?	Yes — a byte-preserving writer and a certificate that already pass on strangers' data	High MEASURED
Will people pay?	Unproven. Zero paying users. And developers are the hardest freemium audience: median dev-focused free-to-paid is 5%, half the non-dev rate	Low FETCHED
Can one person build it?	The v1 scope yes. Teams, SSO and compliance no — §53	Medium
Is the timing right?	Yes, and tight	High FETCHED

Proven **MEASURED** : the engine does not corrupt foreign data (8,513 files, 7 authors); competitors do (three executed teardowns); the internal markdown-OS substrate works at scale (24,539 gate decisions, 124 automations); the campaign pipeline produces multi-surface output from one file (10 episodes in ~1 week).

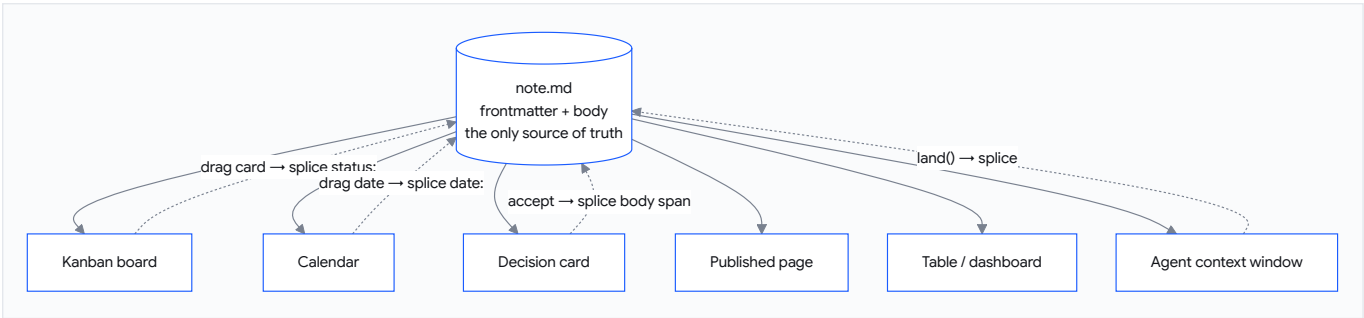
Assumed — must be tested early: that anyone pays; that the review loop is a purchase trigger rather than a nice-to-have; that India is viable top-of-funnel; that funnel 2 exists at all. **Every conversion, churn, ticket-rate, storage and CPU figure in §23–26 is an assumption stated explicitly so it can be replaced by measurement.**

Refuted — do not repeat: "serve markdown to agents and get cited"; "first AI attribution" (Cursor and Grammarly exist — the defensible claim is narrower, §11); the Obsidian first-note statistic (§2).

The single biggest risk is not technical. It is that a fidelity guarantee is a thing engineers admire and nobody buys. GitBook's lost-work complaints prove the pain exists; they do not prove anyone pays a premium to avoid it. §21 and §27 are the mitigation.

§5 The core concept

The file is the only source of truth. Every app-like thing — the board, the calendar, the decision card, the dashboard, the published site, and every AI edit — is a deterministic, reversible projection of that file, owning no state of its own.



Why this is a law, not a slogan. It separates the tools that survived twenty years from the tools that became traps.

SURVIVORS – THE VIEW IS COMPUTED, STORED NOWHERE	TRAPS – THE VIEW ATE THE DATA
org-mode's agenda, running since 2003, computed on the fly from date tags in plain text SS	Coda's export loses formulas, buttons, canvas properties SS
Obsidian Bases (2025): the <code>.base</code> file saves only how you want to look SS	Notion formulas cannot aggregate across rows SS
Potluck (Ink & Switch): "a clear separation between text and annotations... the original text freely editable" FETCHED	Jupyter: meaning lives in invisible kernel state; under 4% of GitHub notebooks reproduce SS

What is new is making it the explicit product primitive. Everyone else rediscovered it as a side effect. And its two hardest requirements are already built here: write-back must be provably reversible (the splice writer) and any view must strip back to portable markdown (the certificate).

The category sentence:

Notion made the app the source of truth and trapped your data inside it. frontmatter makes the **file** the source of truth and lets every app be a disposable lens over it — provably, byte for byte, reversibly.

It composes with the AI story rather than sitting beside it. If every view is a projection, an AI edit is a projection running backwards: a splice with its byte range recorded, reviewable and reversible. The AI gets the same path as a human dragging a card, and the same guarantee.

Prior art to respect: Potluck is the near-exact precedent and is unmaintained; Obsidian Bases is market validation that the demand is real. Neither claims the law as a product primitive; neither can prove reversibility.

§6 Principles and non-goals

6.1 Principles

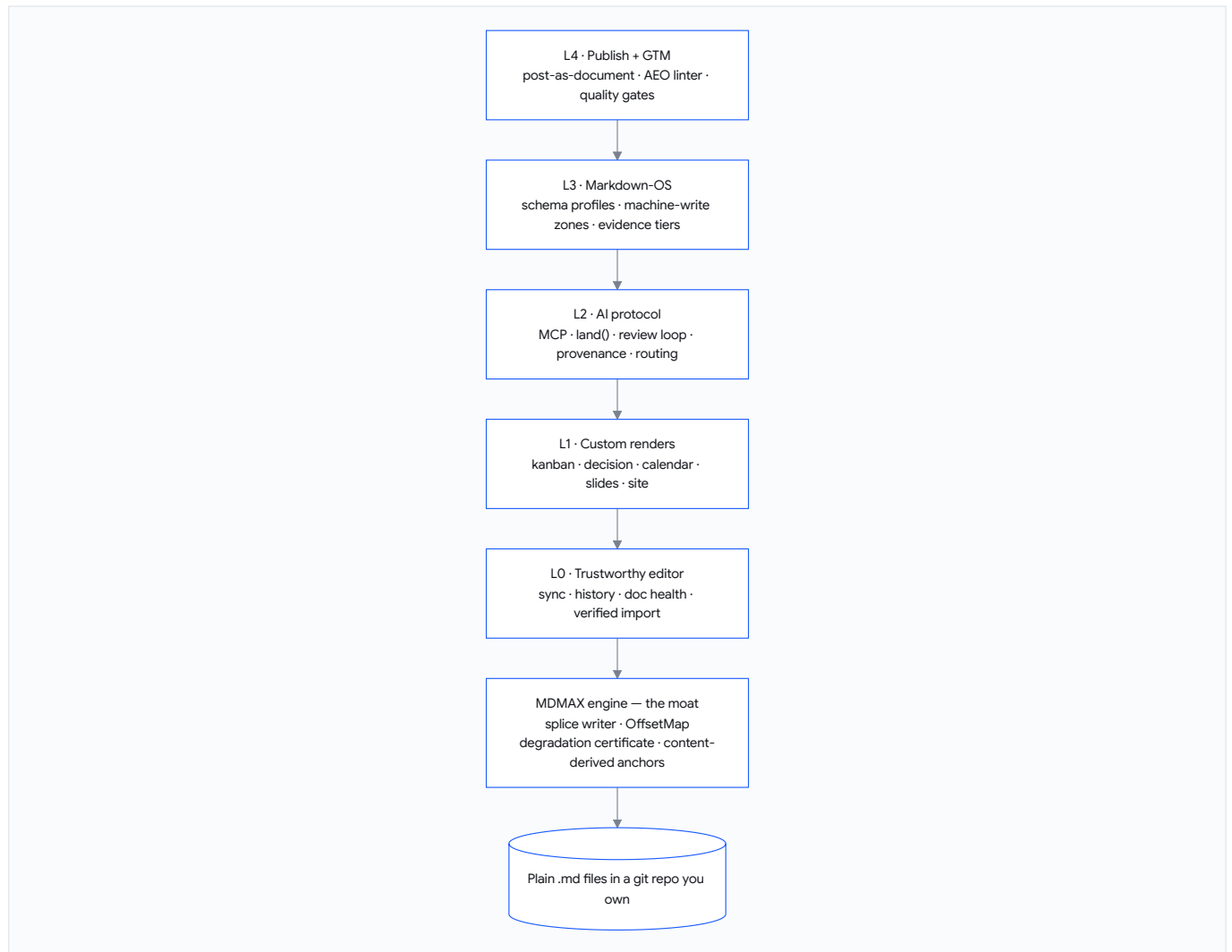
1. The file is the truth; views are disposable.

- 2. **Refusal is a first-class outcome.** Cannot do it safely → return the input unchanged and say why. Never guess.
- 3. **Never lie about what happened.** Every count is re-derived from disk, never from the loop that did the work.
- 4. **Degrade, never break.** Every extension is valid CommonMark.
- 5. **The model gets a data slot, never a canvas.**
- 6. **One grammar for every change** — human suggestion, AI edit, sync conflict.
- 7. **Measure the document, not the user.** Implicit feedback only.
- 8. **Gates, not scores.** Named binary checks.

6.2 Non-goals

The expensive mistakes are all additions.

NEVER	WHY
A new format, sigil, or dialect	Settled 2026-07-29
A tree-of-record, even once	<code>bLocksToMarkdownLossy()</code> is a real API name SS
A plugin marketplace	VS Code's own wiki names extensions the #1 performance suspect; Typora's most-requested feature (251 votes) sits against a product loved <i>because</i> it has none SS .
The eval lane — arbitrary client-side code	It ends the corruption guarantee and converts every prompt injection into remote code execution on infrastructure holding every customer's documents and API keys (§9.2, §51)
Block IDs in files, columns-in-markdown, synced-block bytes	Pollutes the text
Likert writing scores	Schema-rejected internally; Grammarly's opaque score is the anti-pattern
Ambient AI buttons, un-disableable AI	Notion users write ad-blocker rules against AI buttons SS
Silent auto-landing into a curated vault	Auto-capture only into an inbox lane
Opaque credit repricing	Cursor apologised; Windsurf churned twice; Notion got roasted
Streaks, badges, soundscapes, format-on-save default	Noise
A bespoke canvas as centre of gravity	Spatial state is not linear text
Notion-style project management	Founder boundary. A markdown editor with deep AI, not a PM tool
AIOS internal machinery as consumer UI	§15.4
Graph-view investment, the mother-markdown container	Zero of 89 feature requests mentioned graph view; 45 transclusions in 25 MB of our own vault were all documentation of the feature, never use MEASURED
Any "first" or "only" claim without one more verification round	We already caught one



7.1 The engine

The splice contract. Locate the byte range, replace exactly those bytes, leave every other byte bit-identical. Never regenerate from a parse tree. If the range cannot be located unambiguously, **refuse** and return the input unchanged. Grounded in lens theory (Foster et al., TOPLAS 2007).

Regenerate — every competitor measured

Parse to in-memory model

Mutate model

Serialise whole file

LOST: comments, anchors,
spacing, reference links,
leading zeros

Splice — frontmatter

Locate byte range

cannot locate safely

Replace those bytes only

REFUSE
return input unchanged

Every untouched byte
bit-identical

MDMAX: 13 files, 3,614 lines, 10 tests [MEASURED](#).

FILE	LINES	CAPABILITY, WITH ITS MEASURED NUMBERS
<code>domain/constructs.ts</code>	618	19 constructs detected; returns UTF-8 byte ranges; conservative detectors behind a skip mask
<code>infrastructure/bench.ts</code>	538	7 engines : remark-app, react-markdown, marked, markdown-it, markdown-it-html-true, commonmark, kramdown-jekyll. <code>benchId</code> = sha256 over the full option set. A missing engine is a hard refusal
<code>domain/fold.ts</code>	430	<code>mdmax/fold@1</code> versioned equivalence fold. Prototype 705/1613 strict vs 69/1613 folded (43.71% → 4.28%) over 40 files × 6 engines — explicitly not attachable to <code>fold@1</code> until re-measured
<code>domain/verdict.ts</code>	411	<code>![[Some Note]]</code> → VOID on marked 16.4.2 and commonmark 0.31.2; 21 of 24 <code>{{#id}}</code> leak ; front matter LEAKs in 23 of 24 bench configurations
<code>application/certify.ts</code>	375	Layer-2 inventory + layer-3 differential; never throws
<code>domain/offsets.ts</code>	315	Branded U16Offset / ByteOffset / GraphemeIndex. Only 67 of 1,080 corpus files have bytes == UTF-16 units (93.8% diverge) ; 103 of 2,314 contain non-BMP
<code>domain/shape-gate.ts</code>	174	<code>MAX_BYTES</code> 4MB, <code>MAX_LINES</code> 200,000. Measured quadratics: <code>WIKILINK_RENDER</code> k=1.98, 36,865 ms on 320 KB ; <code>mdast-util-from-markdown</code> 12,429 ms vs micromark 1,207 ms (10.3x) . Strict UTF-8 decode, refuse never repair
<code>domain/cert-contract.ts</code>	168	4 verdicts PASS / STRIP / CORRUPT / VOID; 3 classes LEAK / DESTROY / MUTATE. The artifact is a JSON sidecar, never written into the <code>.md</code>
<code>domain/placement.ts</code>	154	Setext defect: <code>"Heading\n---"</code> → h2; blank-line isolation does not fix it → refuse, never repair
<code>domain/targets.ts</code>	133	15 targets ; 7 of ~12 surfaces unprobeable ; Slack and Discord modelled as lossy sinks
<code>domain/frontmatter-prepass.ts</code>	121	170 of 907 home frontmatter blocks are invalid YAML = 18.74% . Never writes, never throws
<code>domain/slug.ts</code>	115	Over 393 anchors: github-sluggier 21.88% , dash-collapsing 95.42% , either 98.73% — caveat, 85.2% come from one file
<code>domain/normalize.ts</code>	62	99.627% re-anchoring across 384 configurations

7.2 Current state — verified

Metric	Value	Measured
Modules under	src/modules/	14 — 175 files
TypeScript files in	src/	226 · 25,407 lines
Test files		98 · 1,575 tests passing
App routes		30
Commits on branch		40+ (engine/plan-and-diagnostics)
npm run verify		Green — typecheck, lint, 98/98 test files, build compiled, arch "violations": [], specs 4 scanned 0 errors
CI		None. .github/ does not exist

Stack: Next 16.2.6 · React 19.2.6 · CodeMirror 6 · unified/remark/rehype · gray-matter · yaml · mermaid 11 · minisearch · zustand · zod · next-auth v5 · AI SDK v6 with five providers · Tauri v2 · puppeteer-core.

7.3 The MDMAX wiring plan — four ordered seams

Currently one symbol from one of thirteen files is used in product code: decodeStrict, imported by get-snapshot.ts and search-index.ts. The other twelve files have zero product importers.

#	Seam	Call Site	Contract	Blocked By
1	Ingress gate	live	decodeStrict refuses, never repairs	done
2	Write gate	commit-changes.ts, share-writer.ts	every write passes shape-gate + placement before a blob is created; a refusal is a 4xx with a reason, never a silent pass	NF-1, NF-3
3	Splice engine	replaces splice-frontmatter.ts's single-key scanner	one implementation for kanban drag, calendar drag, land() and share	NF-4 design
4	Certificate	background job → JSON sidecar in R2, surfaced in Doc Health	certify.ts never throws; the artifact never enters the .md	\$38 jobs

The boundary rule: MDMAX exports pure functions over bytes and imports nothing from src/modules/*. eslint-plugin-boundaries plus the arch report (with a minimum-files-scanned floor so the gate cannot pass by going blind) enforce the direction.

Do not wire seam 2 before NF-1 and NF-3 land. At the measured refusal rate the write gate would reject 83% of foreign vaults' publishes — an availability incident wearing a correctness costume.

7.4 Three known truths that contradict our own older docs

- docs/mdmax/PLAN.md says MDMAX "shipped". It is implemented and unit-tested; twelve of thirteen files are unreachable by product code.
- There is no CI, in a product that intends to sell document CI.
- The shipped design system is not our design system. src/app/globals.css is byte-identical to sgkn-md's, self-labels "Linear-style modern SaaS", and ships --accent: #18181b with 6/8/12px radii — neither the canonical #1a5cff accent nor the square-corner rule. A body-faint token at 2.14:1 fails WCAG AA while being in the shipped set.

§8 The markdown substrate

Position: frontmatter is a CommonMark editor with a profile layer. Everything below defines what the floor is, how a profile is written on disk, which extensions are in scope, which frontmatter keys we read and write, and how typed data lives in plain text without becoming a database.

8.1 The spec landscape and what the floor actually is

Spec / Dialect	Status and Version	Governance	Extension Mechanism	In Our Floor?
CommonMark	0.31.2, released 2024-01-28; still the current release as of 2026-08 [SS]	commonmark.org spec, John MacFarlane et al.	None. Deliberately no extension syntax.	Yes — the floor
GitHub Flavored Markdown (GFM)	Living spec, versioned against CommonMark 0.29	GitHub, cmark-gfm reference impl	Fixed extension set, not user-extensible	Yes — the practical floor
MyST	Markdown for scientific/technical docs, myst-parser	Executable Books / Jupyter Book	Directives + roles (colon-fence :::)	Profile target, not floor
Pandoc Markdown	Pandoc ≥ 3.x	John MacFarlane	Extensions toggled by name (+pipe_tables) and fenced_divs	Degradation target
Obsidian Flavored Markdown	Rolling, tied to app version	Obsidian (proprietary)	Callouts, wikilinks, %%comments%%, block refs	Degradation target
Markdoc	0.4.x, Stripe	Stripe (MIT)	{% tag %} curly-brace tags	Studied, rejected as carrier
MDX	3.x	Vercel/unified	JSX embedded in markdown	Rejected — violates the eval-lane ban
remark / micromark	unified collective	Extension via micromark syntax extensions + mdat utils	The implementation substrate we will use	Implementation

The floor is not “CommonMark” in the abstract — it is CommonMark 0.31.2 plus the four GFM extensions that every one of the seven certified engines implements: tables, task list items, strikethrough, and autolink literals. [inference, from the engine matrix in §5]

FLOOR RULE	STATEMENT	ANTI-RECOMMENDATION
F1	Every byte frontmatter writes MUST parse as valid CommonMark 0.31.2 and MUST render as readable prose in a parser that knows nothing about us.	Do NOT emit a construct whose fallback rendering is garbage (raw <code>{% %}</code> tags, orphan <code>:::</code> colons, bare JSX).
F2	Extensions live inside constructs CommonMark already defines — block quotes (§5.1), fenced code blocks (§4.5), link reference definitions, HTML blocks (§4.6).	Do NOT introduce a new block-level token. That is a new format, and a new format is settled-against.
F3	GFM tables, task lists, strikethrough, autolinks are assumed present. Anything beyond those four is a profile feature and must be degradation-certified.	Do NOT assume footnotes, definition lists, or heading anchors. GFM's footnote support is not in the four-extension core.
F4	We target <i>round-trip byte preservation</i> , not <i>round-trip AST equality</i> . The splice engine never reserializes an untouched region.	Do NOT adopt a normalize-on-save model. Normalizing is a whole-file rewrite and destroys the one property that distinguishes us.

What would falsify the floor choice: if the degradation certificate shows that a majority of the 7 engines fail one of the four GFM core extensions on a realistic corpus, the floor drops to bare CommonMark 0.31.2 and tables become a profile feature. [inference]

Where sources disagree: the research reports treat "CommonMark compliance" as a binary, while the engine-certification work in this repo treats it as a per-construct matrix with partial credit; the matrix view is the one the product implements, because a certificate that says "97% compliant" is not actionable and a certificate that says "this engine drops your callout title" is. [inference]

8.2 The carrier decision — how an extension is written on disk

This section corrects the earlier PRD claim that the fenced-code info string is the single dispatch mechanism for frontmatter extensions. It is not. It is one of two carriers, and it is the wrong one for anything a human is meant to read. The correction is forced by a structural asymmetry in CommonMark itself, described in the failure-mode row below.

Carrier scorecard. Six candidates were evaluated against seven criteria. Scoring is per-criterion pass / partial / fail.

CARRIER	LITERAL FORM	VALID COMMONMARK?	DEGRADES TO READABLE?	UNCLOSED-STATE FAILURE	HUMAN-EDITABLE	NESTING	ECOSYSTEM PRECEDENT	VERDICT
Blockquote callout	<code>> [!NOTE]</code> + <code>> body</code>	Yes — §5.1 block quote with a text first line	Yes — renders as an indented quote with a visible <code>[!NOTE]</code> label	Structurally impossible — a block quote has no closing marker; it ends when the <code>></code> prefix stops [measured]	Yes — every line is prose	Partial — nesting requires <code>> ></code> , which is legal but ugly	Very high — GitHub, Obsidian, Bear, Zed, Docusaurus	ADOPT — prose carrier
Fenced code block	<code>```fm-data</code> + payload + <code>```</code>	Yes — §4.5 fenced code block with an info string	Yes — renders as a code block; content is visible, inert, and copy-pasteable	Catastrophic — an unclosed fence "runs until the end of the containing block" per CommonMark §4.5, swallowing the remainder of the document [fetched, spec §4.5]	No — payload is machine-shaped	No — fences do not nest (only outer-fence tricks)	Very high — Mermaid, PlantUML, math, mermaid / dot / vega all dispatch on info string	ADOPT — opaque-data carrier only
Colon fence / directives (<code>:::</code>)	<code>:::note\nbody\n:::</code>	No — bare CommonMark renders <code>:::note</code> as a literal paragraph	Partial — orphan colons visible as text	Same swallow risk as fences in implementations that treat it as a fence	Yes	Yes — the best nesting story of any candidate	Medium — MyST, Docusaurus, remark-directive, the unmerged CommonMark generic-directives proposal	REJECT as primary; accept on input
Curly tags (Markdoc <code>{% %}</code>)	<code>{% callout type="note" %}</code>	No	No — raw <code>{% %}</code> visible as noise	Unclosed tag corrupts the block	Partial	Yes	Low outside Stripe	REJECT
HTML block	<code><div class="fm-callout"></code>	Yes — §4.6	No — stripped or escaped by most sanitizers; renders as nothing or as raw tags	Unclosed tag consumes until blank line (§4.6 condition 6)	No	Yes	High but declining	REJECT
Link reference definition abuse	<code>[fm:note]: #</code>	Yes	Yes — invisible (definitions are not rendered)	None	No	No	None	REJECT — invisible is worse than ugly

The asymmetry that decides it. A block quote is delimited by a per-line prefix, so there is no state to leave open — the construct terminates the moment the prefix stops, and a truncated or half-typed callout costs exactly the lines that carry `>`. A fenced code block is delimited by a matching closing fence, and CommonMark §4.5 specifies that if none is found the block runs to the end of the containing block; a single dropped backtick line therefore turns the rest of a 4,000-line vault note into code. [fetched, CommonMark §4.5]

Decision.

- Prose that a human reads → blockquote callout.** `> [!NOTE]`, `> [!DECISION]`, `> [!TASK]`. Recognized by GitHub, Obsidian, Zed, Docusaurus; degrades to a labelled indented quote everywhere else. **Anti-recommendation: do NOT put machine payloads in a callout** — one stray unprefixed line silently ends the block and orphans the rest of the payload as body prose.
- Opaque machine data → fenced code block with an info string.** ````fm-board`, ````fm-query`, ````fm-schema`. **Anti-recommendation: do NOT put reader-facing prose in a fence** — it renders monospaced, is excluded from search and word count in most tools, and is not editable in a WYSIWYG surface.
- Accept `:::` directives on input, never emit them.** They arrive from MyST and Docusaurus vaults and must not be corrupted; converting them to our carriers on write would be a whole-file rewrite and violates the splice law.

FAILURE MODE	BLOCKQUOTE CARRIER	FENCE CARRIER	MITIGATION WE SHIP
Unclosed construct	Impossible [measured]	Swallows rest of document [fetched §4.5]	Fence writer emits open + close in one splice, never two; validator refuses a file with an odd fence count
Lazy continuation	A non- <code>></code> line ends the quote; body text below is orphaned but visible	N/A	Writer prefixes every emitted line; reader treats first unprefixed line as terminator
Nested inside a list	Legal (<code>- > [!NOTE] </code>); indentation-sensitive	Legal; fence must be indented to the list content column	Splice computes the container indent from the byte range, never assumes column 0
Engine strips the marker	Renders as an ordinary quote — content survives	Renders as a code block — content survives	Both are certified as <i>lossy-but-readable</i> , never <i>destructive</i>
Tab/space indent in container	Callout body may fall out of the quote	Fence close may not match	REFUSE and surface the byte offset rather than guess

What would falsify this decision: if a measured sweep of real-world vaults shows callout-marker recognition below roughly half of the certified engines *and* those engines mangle rather than merely ignore the `[!NOTE]` line, the prose carrier moves to plain bold-labelled blockquotes with no marker. The fence decision is falsified only if a future CommonMark release changes §4.5 unclosed-fence behaviour, which would be a breaking spec change and is not expected. [inference]

8.3 The extension catalogue — what we support, what we refuse

Ranked by (ubiquity across the 7 certified engines) × (value to the projection law). Tier 1 ships in v1; Tier 2 is profile-gated; Tier 3 is input-tolerated only; the refusal list is permanent.

Tier 1 — ship in v1, assumed present in the floor

#	EXTENSION	LITERAL SYNTAX	WHY IT EARNS V1
1	GFM tables	<code>'''\\n Col Col \\n --- --- \\n a b \\n'''</code>	The board and the decision-card projections both read tabular rows; tables are the only tabular construct with universal support
2	Task list items	<code>'''\\n- [] open\\n- [x] done\\n'''</code>	The task projection's entire state lives in one byte (<code> </code> vs <code>x</code>) — the cleanest possible splice target
3	YAML frontmatter	<code>'''\\n---\\nkey: value\\n---\\n'''</code>	Document-level typed data; see §8.4
4	Blockquote callouts	<code>'''\\n> [!NOTE] Optional title\\n> Body line.\\n'''</code>	The prose carrier from §8.2
5	Fenced code with info string	<code>''''''fm-board\\n{"group":"status"}\\n'''</code>	The opaque-data carrier from §8.2
6	Strikethrough	<code>'''\\n~~struck~~\\n'''</code>	GFM core; free
7	Autolink literals	<code>'''\\nhttps://example.com\\n'''</code>	GFM core; free

Tier 2 — profile-gated, degradation-certified before enable

#	EXTENSION	LITERAL SYNTAX	GATE
8	Footnotes	<code>'''\\nText[^1]\\n[^1]: Note.\\n'''</code>	GFM + Pandoc + Obsidian yes; not in CommonMark. Certify per-engine before a projection depends on it
9	Wikilinks	<code>'''\\n[[Note Title]]\\n[[Note alias]]\\n'''</code>	Obsidian/Logseq/Foam native; renders as literal brackets elsewhere. Enable only in a vault profile
10	Definition lists	<code>'''\\nTerm\\n: Definition\\n'''</code>	Pandoc/kramdown yes, GFM no. Degrades to two paragraphs — acceptable
11	Math	<code>'''\\n\$inline\$ and \$\$display\$\$\\n'''</code>	GitHub and Obsidian yes; degrades to visible TeX source, which is readable
12	Mermaid / diagram fences	<code>''''''mermaid\\ngraph TD; A-->B;\\n''' '''</code>	Already an info-string fence; costs us nothing and validates the carrier
13	Heading IDs	<code>'''\\n## Title {#custom-id}\\n'''</code>	kramdown/Pandoc syntax; GFM auto-generates instead. Conflict — see below
14	Highlight	<code>'''\\n==marked==\\n'''</code>	Obsidian/Pandoc; renders as literal <code>==</code> in GFM

Tier 3 — parse on input, never emit

#	EXTENSION	LITERAL SYNTAX	RULE
15	Colon-fence directives	<code>'''\\n:::note\\nbody\\n:::\\n'''</code>	Preserve byte-for-byte; project as a callout in our UI; write back unchanged
16	Obsidian block references	<code>'''\\nSome text ^block-id\\n'''</code>	Preserve; never rewrite the id
17	Obsidian comments	<code>'''\\n%hidden%\\n'''</code>	Preserve; treat as content, not metadata
18	Embeds/transclusion	<code>'''\\n![[Note#Heading]]\\n'''</code>	Preserve; do not resolve on write
19	Markdoc tags	<code>'''\\n{% callout %}...{% /callout %}\\n'''</code>	Preserve; never generate
20	Front-of-line HTML blocks	<code>'''\\n<details><summary>x</summary>...</details>\\n'''</code>	Preserve; never generate

Permanent refusals, with the reason each is refused

REFUSED	REASON	SETTLED BY
MDX / JSX in markdown	Requires arbitrary client-side evaluation	The eval-lane ban
Any new block token of our own	Would constitute a new markdown format	"No new markdown format"
A tree-of-record / AST-as-truth model	Contradicts the projection law	"No tree-of-record"
Third-party syntax plugins at runtime	Contradicts the no-plugin-marketplace decision, and makes degradation certification impossible (an uncertified syntax cannot be certified)	"No plugin marketplace"
Normalizing rewriters (prettier-on-save style)	Whole-file rewrite defeats byte-preserving splice	Splice law
Setext headings on <i>output</i>	<code>== / ---</code> underlines collide with frontmatter delimiters and thematic breaks; ATX (<code>#</code>) is unambiguous. Parse on input, never emit	[inference]

Anti-recommendation for the whole catalogue: do NOT enable a Tier 2 extension because one engine supports it well. The gate is the degradation certificate across all 7, and a projection may only depend on a construct that has a certified fallback. **The falsifier for any Tier 2 promotion is a single certified-destructive result — lossy is acceptable, destructive is not.**

8.4 Frontmatter key conventions — being a good citizen

We read the ecosystem's keys and we write a namespaced subset. The rule is: **read broadly, write narrowly, never reformat a key we did not write.**

Master key table

KEY	TYPE	ORIGIN ECOSYSTEM	WE READ	WE WRITE	NOTES
title	string	Universal (Jekyll, Hugo, Obsidian, Astro, Zola, Eleventy)	Yes	Only if absent and user asks	The single most portable key
date	date	Jekyll, Hugo, Eleventy	Yes	No	Format conflict — see below
tags	list<string>	Universal	Yes	Yes	Accept both flow ([a , b]) and block (- a) styles; preserve the style found
aliases	list<string>	Obsidian, Hugo	Yes	No	Obsidian-critical; never reorder
draft	bool	Hugo, Astro, Zola	Yes	No	Hugo's canonical publish gate
published	bool	Jekyll (published: false)	Yes	No	Jekyll's inverse of draft — conflict below
description	string	Astro, Hugo, Docusaurus	Yes	No	
slug	string	Hugo, Astro, Docusaurus	Yes	No	
author / authors	string list	Hugo (authors), Jekyll (author)	Yes	No	Singular/plural conflict — below
categories	list	Jekyll, Hugo	Yes	No	
layout	string	Jekyll, Eleventy	Yes	No	
permalink	string	Jekyll, Eleventy	Yes	No	
weight	number	Hugo, Docusaurus (sidebar_position)	Yes	No	
cssclasses	list	Obsidian	Yes	No	
publish	bool	Obsidian Publish	Yes	No	Third spelling of the same concept
id	string	Docusaurus, Logseq	Yes	No	
sidebar_position	number	Docusaurus	Yes	No	
status	string	Convention, not spec	Yes	Yes	Our primary board-grouping key
due	date	Convention (Tasks plugin, Dataview)	Yes	Yes	ISO 8601 date only
fm.profile	string	Ours	Yes	Yes	Which frontmatter profile governs this file
fm.version	string	Ours	Yes	Yes	Profile schema version
fm.projections	list<string>	Ours	Yes	Yes	Which views this file opts into
fm.cert	string	Ours	Yes	Yes	Degradation-certificate id for this file's constructs

Named conflicts — where ecosystems disagree and we must not silently pick

CONFLICT	ECOSYSTEM A	ECOSYSTEM B	OUR RULE
Publish gate	Hugo/Astro: draft: true means unpublished	Jekyll: published: false means unpublished; Obsidian Publish: publish: true means published	Read all three. Never write any of them. If two disagree in one file, REFUSE and surface both.
Author cardinality	Jekyll: author: Name (scalar)	Hugo: authors: [A, B] (list)	Read both; normalize in-memory to a list; write back the shape found.
Date format	Jekyll accepts 2026-08-28 10:00:00 +0530 ; Hugo prefers RFC 3339 2026-08-28T10:00:00+05:30 ; YAML 1.1 auto-typing turns unquoted dates into timestamps	Obsidian users routinely write 2026-08-28 plain	Preserve the literal bytes. Parse permissively for projection; never rewrite the serialization.
Tag delimiter	YAML list	Obsidian also allows inline #tag in body, and space-separated strings in some vaults	Read the frontmatter list as canonical; index body #tags as secondary; write only to the list.
Heading anchors	GFM auto-slugs headings	kramdown/Pandoc use explicit {#id}	Never generate anchors into the file. Projections compute them.
Delimiter	--- ... --- (YAML) is universal	TOML +++ (Hugo, Zola); JSON { } (some Eleventy)	Support --- fully; detect +++ /JSON and REFUSE to write rather than convert.
Key casing	snake_case dominant	camelCase appears in Astro content collections	Preserve the casing found. Our own keys are always lowercase, dot-namespaced.

Rules for the build team

- Namespace every key we invent under fm. — do NOT add bare top-level keys, because a bare status or type collides with three existing ecosystems and makes our writes indistinguishable from the user's.
- Preserve key order, comments, blank lines, quoting style, and indentation on every write. A frontmatter edit is a splice into the value bytes, not a YAML re-dump. **Anti-recommendation: do NOT use a load-then-dump YAML round trip, even a "round-trip-safe" one — it normalizes quoting and folds long lines, which is a whole-file rewrite by another name.**
- On a duplicate key, REFUSE. YAML 1.1 parsers disagree on last-wins vs error, so any choice we make is wrong somewhere.
- On an unparseable frontmatter block, treat the file as having no frontmatter and edit only the body. Never repair.

8.5 Typed and queryable data in plain text

Three layers, in strict precedence order. A projection reads from the highest layer present and never writes to a layer it did not create.

LAYER	WHERE IT LIVES	LITERAL SYNTAX	SCOPE	TYPED?	QUERY COST
L1 — Document metadata	YAML frontmatter	<code>'''n---\nstatus: doing\ndue: 2026-09-01\nfm.profile: board\n---\n'''</code>	Whole file	Yes, via profile schema	O(1) per file
L2 — Block metadata	Inline key-value on the line	<code>'''n- [] Ship the parser (due:: 2026-09-01) (owner:: sagnik)\n'''</code>	One block	Yes, via profile schema	O(lines)
L3 — Opaque payload	Fenced block with info string	<code>''' fm-query\nfrom: tasks\nwhere: status == "doing"\n'''</code>	Where the fence sits	Yes, schema-validated	O(1) per fence

Design rules

- **The file is the database. There is no index of record.** Any cache we build is a projection: derivable from the bytes, discardable, and rebuilt without asking the user. If a cache and a file disagree, the file wins and the cache is wrong. [projection law]
- L2 syntax follows Dataview's `key:: value` inline field convention because it is the largest existing installed base for typed inline data in markdown and it degrades to visible parenthesised text. **Anti-recommendation: do NOT invent a shorter inline syntax** — a novel sigil is a new format, and the settled position forbids one.
- Queries are **declarative and non-Turing-complete**. A query is a filter + sort + group over parsed fields, evaluated by our engine. Do NOT ship a query language with function calls, arbitrary expressions, or user-supplied code — that is the eval lane by another name, and it is settled against.
- Type coercion is explicit and lossless-or-refuse: a value that does not parse as its declared type is surfaced as a typed error on that byte range, not coerced to null and not silently skipped. A silently-skipped row is indistinguishable from a real negative. [Learned Rule #59 class]

Type set (deliberately small)

TYPE	LITERAL	REFUSAL CONDITION
<code>string</code>	any scalar	never
<code>number</code>	<code>42</code> , <code>3.14</code>	non-numeric text
<code>bool</code>	<code>true</code> / <code>false</code>	any other spelling, including <code>yes</code> / <code>no</code> — YAML 1.1 coerces these and YAML 1.2 does not, so we refuse rather than pick
<code>date</code>	ISO 8601 <code>2026-09-01</code>	any other format
<code>datetime</code>	RFC 3339 with offset	naive datetimes without an offset
<code>list<T></code>	YAML sequence or flow	mixed element types
<code>enum</code>	value from the profile's declared set	value outside the set
<code>link</code>	<code>[text](target)</code> or <code>[[wikilink]]</code>	unresolvable target is <i>not</i> a refusal — it is a warning; links to non-existent notes are legitimate in vaults

Anti-recommendation for the layer: do NOT promote L2 inline fields into L1 frontmatter automatically “for queryability”. That is a rewrite of lines the user did not edit, it changes the rendered output, and it is the exact behaviour that makes users distrust editors that touch their files.

What would falsify this design: if profiling shows that a full-corpus L2 scan over a realistic vault (order 10⁴ notes) cannot complete inside an interactive budget, L2 gains a persisted index — still a projection, still discardable, still with the file as the sole source of truth, but materialized. That is a performance change, not a change to the law. [inference]

§9 The rendering system

LAYER	COMPONENT	DECISION	GOVERNING MEASUREMENT
Parse	@lezer/markdown 1.7.2 (editor) + unified / remark (preview)	Keep the split, vendor Lezer, fix the preview's conformance	Lezer is the only JS parser with exact offsets on inline marks <i>and</i> incremental reparse — 3.58x on a 56 KB doc [measured]
Address	U16Offset / ByteOffset / GraphemeIndex branded types, one OffsetMap	Keep: UTF-16 code units internally, bytes only at git-blob and hash edges	mdast root end offset 36 vs UTF-8 length 41 on the same string [measured]
Compute	Four lanes (C0 client-pure, C1 client+library, C2 server, L LLM), one refused (arbitrary client eval)	Ship A/B, gate C, refuse D permanently	4.03% of 1,159,166 notebooks reproduce [fetched]
Project	render: switch + fm-view fenced lane + body data	Three-slot rule; frontmatter carries the switch only	One key render: board degrades to <hr> render: board [measured]
Emit	HTML → Chrome 151 PDF (+ paged.js) → Marp / DOCX / EPUB	Buy fidelity inside the browser we already ship; no second layout engine	paged.js closes both native Chrome gaps on the same binary [measured]

9.1 The parser layer and why byte-accurate positions decide everything

A splice editor can only edit what its parser can address; every other parser property is negotiable, and position fidelity is not.

TIER	ENGINE	POSITION API	UNIT	INLINE COVERAGE	SPLICE-SAFE?
A	pulldown-cmark 0.13.4	<code>into_offset_iter()</code> → <code>(Event, Range<usize>)</code>	UTF-8 bytes	every event	Yes — the position API is the splice API [fetched]
A	goldmark v2.0.0 / v1.8.5	<code>text.Segment{Start, Stop, Padding}</code>	UTF-8 bytes, <code>Stop</code> exclusive	<code>block</code> <code>Text</code> <code>Lines()</code> + inline	Yes, if <code>Padding</code> is re-added; otherwise splices drift inside blockquotes and lists [fetched]
B	comrak 0.54.0	<code>Sourcepos{start,end}</code> <code>LineColumn</code> , 1-based	UTF-8 bytes by default; <code>parse.sourcepos_chars</code> silently flips to chars	block + inline	Yes, after building a line index; the global unit switch invalidates every stored anchor [fetched]
B	cmark 0.31.2 / cmark-gfm 0.29.0.gfm.13	<code>cmark_node_get_start_column</code> etc.; <code>CMARK_OPT_SOURCEPOS</code>	bytes	block only	Block-level only [fetched, cmark.h L408-420]
C	@lezer/markdown 1.7.2	from / to on every node incl. <code>HeaderMark</code> , <code>EmphasisMark</code> , <code>LinkMark</code> , <code>URL</code>	UTF-16 code units	full	Yes — <code>src.slice(from,to)</code> round-trips exactly [measured]
C	unified/remark-parse 11.0.0	<code>position.{start,end}.offset</code>	UTF-16 code units	full	Yes [measured]
D	markdown-it 15.0.1 / markdown-it-py 4.2.0	<code>Token.map = [line_begin, line_end]</code> , 0-based half-open	lines	block-open and inline only; <code>_close</code> and all 9 inline children measured <code>map: null</code>	No [measured]
D	commonmark.js 0.31.2	<code>node.sourcepos</code>	characters (C cmark counts bytes — a direct disagreement between the reference implementation and its own JS sibling)	block only; every inline <code>NO-SOURCEPOS</code>	No [measured]
D	kramdown 2.5.2	<code>options[:location]</code>	start line only	—	No [fetched]
D	pandoc 3.10.2	<code>sourcepos</code> extension, <code>commonmark</code> reader only	line:col	wraps non-attribute elements in a synthetic <code>Div / Span</code> , changing the tree being measured	No [fetched]
E	marked 18.0.11, Python-Markdown 3.10.3, mistune 3.3.4, showdown 2.1.0	none (<code>raw</code> string, <code>ElementTree</code> , token dicts, regex substitution)	—	—	No [measured/fetched]
?	markdown-rs 1.0.0	<code>unist::Point.offset</code> documented as a character index while the parser operates on <code>&str</code>	disputed	—	Record as disputed, never assume Tier A [fetched]

Decisions:

- **Keep @lezer/markdown in the editor and unified in the preview.** Lezer is the only JS parser giving exact inline-mark offsets *and* incremental repare; 56 KB doc, one-character insert: 16.88 ms full repare vs 4.71 ms incremental = 3.58× [measured/derived]. **Anti-recommendation:** do not consolidate on one parser to reduce surface area — markdown-it returns `map: null` on every inline node, so a single-parser stack would delete inline splice capability outright.
- **Vendor @lezer/markdown into the repo this quarter.** Its README now says only that the repository has moved to `code.haverbeke.berlin`; 147 stars, single maintainer, npm still publishing (1.7.2, 2026-07-15) [fetched]. It is load-bearing and has no substitute with equivalent position fidelity. Budget for owning a fork.
- **Fix the preview before shipping any render profile.** Bare remark+rehype-raw scores 498/652 = 76.4% on CommonMark 0.31.2; the shipped app pipeline (`remark-gfm` + `remark-breaks`) scores 439/652 = 67.3%; `remark-gfm` costs 2 tests, `remark-breaks` costs 57 = 8.7 points [measured/derived: 498-441]. Make `remark-breaks` a render-time toggle the certificate records as a declared deviation, or drop it. Certifying seven engines against a preview less CommonMark-conformant than any engine it certifies is not a certificate.
- **Incremental parsing is a requirement, not an optimisation.** 1,417,778 B parsed in 173.3 ms by `marked` (7.80 MB/s) and 229.7 ms by `markdown-it` (5.89 MB/s) [measured/derived]; a 1 MB note therefore re-parses in ~130 ms, above the 100 ms interaction budget on every keystroke. `remark-parse` is 60.2 ms on 67,200 B vs `commonmark.js` at 4.0 ms — **15.05×** [measured/derived], extrapolating to ~3.6 s at the 4 MB shape-gate ceiling [derived, linear, unverified at that size].
- **Add three non-JS engines to the bench before adding a single construct:** `cmark-gfm` pinned at `0.29.0.gfm.13`, `goldmark` at both `v1.8.5` (Hugo master's actual pin) and `v2.0.0` (released 2026-08-27), `pulldown-cmark` `0.13.4` (142,120,888 total / 42,632,672 last-90d downloads) [fetched]. The bench today is 6/7 JavaScript and measures **4 distinct parsers**, since `remark-app` and `react-markdown` share micromark and `commonmark` is cmark's algorithm in JS. Beyond coverage: goldmark's and pulldown-cmark's byte offsets are an **independent oracle for the splice machinery** — a second implementation that disagrees about where a construct starts is a bug report the JS-only bench structurally cannot produce.
- `targets.ts` **currently maps** `github-blob` (fidelity: `requires-push`) and **every declared surface** — Obsidian, Notion, Typora, Bear, Slack, Discord — to `engineId: 'remark-app'` [fetched, L92-115]. A declared row backed by a different engine's output is a placeholder. Either back it or mark it uncertified.

Falsifier for this section: if a profile ships that needs inline positions and validates cleanly on the markdown-it targets, the Tier D classification is wrong and the bench is not exercising inline anchors.

9.2 The computation budget — four lanes, one refused

LANE	CONTENT	DETERMINISM	CAPABILITY SURFACE	VERDICT
A — client-pure (C0)	<code>fmm-query</code> : declarative, non-Turing-complete, read-only over the vault index; no network, no FS, no DOM; hard row and wall-clock caps	pure function of bytes	none	Ship
B — client + library (C1)	Fixed-vocabulary renderers bundled once: <code>chart.js</code> (12,893,359/wk), <code>leaflet</code> (6.9M/wk class), <code>mermaid</code> (15,313,390/wk, 89,973★), plus spreadsheet-grade expression evaluation over named Lane-A results	pure	none	Ship, bundle-cost gated
C — server (C2)	Cross-file index; out-of-process execution via an external engine (Quarto 1.10.18 / jupyterx 1.19.5 / papermill), never in the editor process, never given the document store; results written back as a fenced block	pure per input hash	server-scoped, per-document opt-in	Gate
D — arbitrary client code (the eval lane)	Arbitrary JS/WASM with host bindings; rendering stored <code>text/html</code> or <code>application/javascript</code> as live DOM	—	full	Refuse permanently
L — LLM	Generative	non-deterministic	—	Not a render lane at all: a projection must be a pure function of bytes, and an LLM cannot be [inference]

A frontmatter document may DECLARE computation; it may never CARRY a capability.

What the refusal genuinely costs:

LOSS	DEMAND EVIDENCE	RECOVERABLE IN LANE A/B/C?
Reactive dashboards (Observable Framework 3,606★ / 49,972 npm-mo, marimo 22,531★, Pluto.jl 5,368★)	[fetched]	Partially — fixed-vocabulary panels only
Live values inline in prose (MyST <code>{g1ue}</code> , <code>nb_execution_mode: inline</code> , Quarto inline expressions)	[fetched]	Yes, Lane B
Runbooks that execute the README's fences (Runme 2,155★)	[fetched]	No
Vault-scoped queries (Dataview 4,857,171 downloads)	[fetched]	Yes — and this is the only loss that actually bites a markdown editor, and it needs no arbitrary eval at all [inference]
<code>.ipynb</code> / <code>.qmd</code> / percent / MyST import and export	<code>jupyter</code> , <code>quarto convert</code> , <code>marimo export md</code> all round-trip with zero execution [fetched]	Not a loss

The strongest argument the refusal is wrong, stated at full strength:

- The stated justification is a non-sequitur. CVE-2026-42557 (high, 2026-05-06, GHSA-mqcg-5x36-vfcg) fires on "a notebook or a Markdown file... an arbitrary command upon a single click... No kernel needs to start", and CVE-2024-22420 (medium, 2024-01-19) triggers on "opening a malicious notebook with Markdown cells, or Markdown file using JupyterLab preview" [fetched]. CVE-2026-44727 (critical, 2026-06-18) is a non-sandboxed nbconvert render path ending in kernel RCE. Every one of these is an *output-channel* or *host-binding* bug, not a compute escape. Refusing the eval lane buys none of the claimed safety, and frontmatter is exposed to that class today, unchanged.
- Corruption-safety and execution are orthogonal. Livebook proves a file can be simultaneously valid Markdown ("every Live Markdown is valid Markdown"), byte-stable, splice-friendly, and executable [fetched]. CommonMark 0.31.2 §4.5 makes fence content literal text — the splice engine never has to understand a `{python}` fence to edit around it.
- Therefore the refusal pays the category's full cost for a safety property it does not deliver.

Verdict: **the refusal stands, but on a different justification than the one it was written with.** What breaks the argument above is blast radius, not sandboxing: an editor holding the whole vault, with an agent writing into it, makes eval a confused deputy over every file — and Livebook concedes the limit in its own docs, that stamping "only takes care of Livebook resources: when you execute the notebook, the code in the notebook will still have access to the current machine" [fetched]. WASM solves the wrong half; a WASM sandbox does not sanitize a `text/html` output [derived from the advisory mechanisms]. Dataview is the natural experiment: the declarative lane and `dataview.js` ship together and only the second carries vault-API capability [fetched]. We take the first and decline the second. Population-scale evidence that in-document arbitrary execution does not buy reproducible documents: Pimentel et al., MSR 2019, 1,159,166 notebooks from 264,023 repositories — **24.11% executed without errors, 4.03% produced the same results** [fetched]. Recorded discrepancy, not resolved: those headline percentages use a denominator of 863,878 while the attempted set reconstructs to 788,813 (570,476 exceptions + 9,982 timeouts + 208,323 finished + 32 corrupt), which would give 26.41% and 4.42% [derived].

Mechanism, six parts:

- Info-string profile, no new syntax.** Directives after the first word of a fence are spec-guaranteed inert (CommonMark 0.31.2 §4.5: "this spec does not mandate any particular treatment of the info string"; Example 143). Per-cell options take **Quarto's** `#| comment form`, never MyST's `:opt: form — measured: #| echo: false` degrades to a valid Python comment; `:tags: [hide-input]` degrades to a visible syntax error [measured].
- Capability stamp, Livebook-shaped.** Trailer `<!-- fm:{"stamp":"...", "offset":N} -->`, HMAC-SHA256 over bytes `[0, offset]`. It grants capabilities (which paths Lane A may read, whether Lane C is enabled), never execution. Unstamped or foreign ⇒ silently degrade to Lane A. The splice engine treats the trailer as one invalidated range recomputed on every write; the `offset` field is what makes that O(1) rather than a whole-file rehash.
- Output is data, never markup.** Persist results as fenced `text/plain` / CSV, or SVG only after sanitisation. Never store or render `text/html` or `application/javascript`. This single control kills the CVE-2026-42557 / CVE-2026-44727 class and is worth shipping *regardless* of the lane decision.
- Render computed output in a sandboxed frame with an explicit CSP, allowinglist no data-attributes.** JupyterLab's failure was an allowlisted `data-commandlinker-command` attribute reaching a global click handler, not a missing sanitiser [fetched].
- Determinism gate.** Every computed block carries `hash=<sha256 of declared inputs>`; a stale hash renders the stored text greyed and never auto-executes — Quarto `freeze: auto` / jupyter-cache semantics without a kernel.
- Agent rule.** The agent may emit Lane A/B blocks; it may not emit or alter the stamp. Stamping is a human action. That is the cut that stops prompt-injection → capability escalation without banning the feature.

Anti-recommendation: do not ship Lane C as a default-on background renderer to look competitive with Quarto. Falsifier: if, six months after Lane A ships, the top unmet request is still "run my Python here" rather than "query my vault here", the lane boundary is drawn in the wrong place.

9.3 The render possibility space — every surface a file could become

The three-slot rule, and the measurement that forces it: a **single** frontmatter key `render: board` degrades to `<hr><h2>render: board</h2>` in marked 16.4.2, markdown-it 15.0.0 (commonmark preset) and commonmark 0.31.2 — a setext H2 that outranks the document's own H1 in every non-frontmatter-aware renderer [measured]. Cost is linear in key count: 4 keys ≈ 60 chars of H2 junk, 12 keys ≈ 180 [derived]. **The intuitive design — a rich view spec in frontmatter — is the degrading one.**

- Slot 1, frontmatter: the switch only.** 1–3 scalar keys, `render:` plus at most a profile name.
- Slot 2, a fenced fm-view lane: the view spec.** Arbitrarily large, degrades to contained code text, zero sigil leakage. Obsidian shipped exactly this shape for Bases — a `.base` YAML with `views: [{type, name, limit, groupBy, order, filters, summaries}]`, "embedded in a code block" [fetched].
- Slot 3, the body: the data,** in constructs that already degrade — headings, lists, GFM tables, task items.

Write-back taxonomy: **BODY-MOVE** (splice a contiguous body span) · **KEY-SET** (one frontmatter scalar) · **CELL-SET** (one table cell) · **FENCE-SET** (inside the `fm-view` lane) · **SIDECAR** (no textual home) · **NONE**.

v1 shortlist, ranked:

#	SURFACE	DRIVER	LANE	DEGRADES TO	WRITE-BACK	DEMAND	EFFORT
1	Kanban board	<code>render: board + ## Lane + - []</code>	C0	outline of headings and checkboxes, fully readable	BODY-MOVE — the drag is a pure body splice, so byte-identical write-back is provable	obsidian-kanban 2,601,160 downloads on 2.0.9-beta , repo 4,483★ last pushed 2026-03-06 [fetched]	S
2	Enhanced table + charts over the same table	GFM table + <code>fm-view</code> for sort/width/type; <code>chart</code> reads the adjacent table, never its own fence	C0 / C1	the plain GFM table stays readable	CELL-SET, byte-exact	table-editor-obsidian 3,148,583; obsidian-charts 320,106; chart.js 12,893,359/wk [fetched]	S / M
3	Slides	<code>render: slides + --- breaks</code> (Marp-compatible)	C0	continuous document split by thematic breaks	NONE	advanced-slides 835,057; marp 12,420★; @marp-team/marp-core 91,989/wk [fetched]	M
4	The XS bundle: SOP/checklist, meeting notes, ADR, changelog, CV, data dictionary, bug/triage	one <code>render: value + a heading</code> convention each	C0	near-perfect — a changelog is already Keep-a-Changelog; a CV is the best degradation in the list	KEY-SET, CELL-SET, or the 1-byte <code>[] → [x]</code> splice	obsidian-tasks 4,114,650 [fetched]; the rest are conventions with no plugin incumbent	XS each — ship the profile mechanism once and these are configuration
5	<code>render: cert</code> — the document renders its own cross-engine degradation verdict	<code>render: cert over targets.ts</code> (15 targets, fidelity: 'local' 'declared', <code>uncertifiableShare()</code>)	C0	the verdict table, readable	NONE	0 of 7,020 registry plugins match provenance , degrad , evidence , or compare ; <code>cert</code> → 1, unrelated [measured over fetched data]	S
6	Calendar	<code>render: calendar</code> or <code>## YYYY-MM-DD</code> + <code>date: per file</code>	C2	dated headings in chronological order	KEY-SET on <code>date:</code> when a card is dragged	calendar 3,048,022; full-calendar 455,970 [fetched]	M — first surface that costs real architecture (vault index)

v2 shortlist: dashboard/status page (C1) · Gantt over a task table with `start / end` , not mermaid (C0/C1, CELL-SET on drag) · map via leaflet (map-view 165,018, leaflet-plugin 308,151) · pivot (C0, derived, NONE) · flashcards with scheduling state in a sidecar, never in the document (spaced-repetition 586,736) · roadmap · org chart · database-view-over-vault, Bases-shaped (C2, L effort, the largest v2 item) · `render: ink` (byte-anchored provenance heatmap) · `render: review` (suggestions as a projection over a sidecar of hunks — Google Docs' public API cannot create suggestions; CriticMarkup's toolkit last moved 2021-03-04) · `render: evidence` · `render: context` .

Carrier decision, and a recorded disagreement: m10 measures `<!--fm ... -->` as fully suppressed, 0 visible characters, in all three of its engines — but those runs used markdown-it's `commonmark` preset, i.e. `html:true` . At markdown-it's `default` `html:false` , the same comment escapes to visible `<!--fm: profile=x-->` ; [measured, m6/m8]. The two reports do not disagree about the bytes; they disagree about which configuration counts as "a dumb renderer". **We therefore keep the settled carrier — a blockquote callout for prose, a fenced code block for opaque data — and permit COMMENT-SET only where the certificate explicitly records the `html:false` visibility for that target.** Per-item state that must survive a drag lives in the body where it already degrades (the checkbox byte, the table cell, the heading section) or in the `fm-view` lane keyed by a stable anchor. **Anti-recommendation:** do not adopt the invisible-comment carrier globally on the strength of the cleaner measurement; it is clean in exactly two of the three configurations we certify.

Traps, with the number that makes each tempting:

TRAP	THE SEDUCTIVE NUMBER	WHY IT FAILS
<code>:::</code> container directives as the extension grammar	<code>micromark-extension-directive</code> 7,336,465/wk; the CommonMark "Generic directives" thread opened 2014-09-06, 173 posts, 79,614 views, last post 2025-04-03 — open 11.6 years [fetched/derived]	Not in CommonMark 0.31.2; leaks two sigil lines per block in all three dumb engines [measured]. Leaf directives (<code>:::name{...}</code>) are worse — full line leaked, zero fallback content. Ban leaf directives at registration.
Mermaid as the diagram strategy	89,973★, 15,313,390 npm/wk, 35 diagram families [fetched]	Degrades to <i>source code</i> , the weakest degradation of any lane. Adopt as a C1 lane; never let a first-party surface depend on it for degradation.
Canvas / whiteboard as a render	excalidraw 7,578,223 — the largest single number on the page [fetched]	Positions have no textual home; 100% sidecar, so the render carries none of the value.
Forms and surveys	write-back into frontmatter is elegant	modalforms 68,493 — weakest demand of any C1-effort surface; multi-respondent state is a database, not a file
Invoices, quizzes, itineraries, habit trackers	they demo well	budget-app 1,003; habit-tracker 10,237 [fetched] — zero or near-zero registry signal
Knowledge graph	—	juggl 134,747, 3d-graph 71,000, extended-graph 68,259 — every graph plugin is small [fetched]

Recorded disagreement, unresolved: obsidian-kanban shows 2,601,160 downloads against a repo last pushed 2026-03-06, and Bases now ships natively while Dataview still shows 4,857,171 downloads against a repo last moved 2025-11-17 [fetched]. Whether these slots are *open* or *just closed by the platform vendor* is not settled by these data points. Falsifier for the v1 ranking: if Obsidian ships a first-party board with splice-clean write-back before we do, item 1 loses its ranking rationale entirely.

9.4 Output targets and their fidelity limits

target	tier	mechanism	fidelity ceiling
HTML, styled standalone	v1	ours	Not a target — the intermediate . Every downstream loss is introduced after HTML [inference]
PDF via existing Chrome 151.0.7922.174	v1	@page margin boxes	Page numbers, page counts, @page: first , static running heads, repeating thead — all already available in the binary we ship, all currently unused [measured]
PDF via Chrome + paged.js 0.4.3	v1, flagged	921,161-byte MIT polyfill injected via addScriptTag	Closes both native gaps: string(chap) running headers and target-counter resolving to seeref [XREF 3] [measured]. Cost: full JS re-layout, margin:0 in page.pdf()
Marp slides	v1	HTML-comment directives + --- breaks	Measured to degrade to invisible in a plain CommonMark renderer — the only slide format satisfying the profile constraint without inventing syntax [measured]
Social cards	v1	reuse page.screenshot()	Avoids satori 0.33.4's Yoga limits: no CSS Grid, display limited to flex block contents none -webkit-box , WOFF2 unsupported [fetched]
DOCX via pandoc 3.10.2	v2	--reference-doc , custom-style on Div/Span/Table, math → OMML	Pandoc's own statement: its AST is "less expressive than many of the formats it converts between... complex tables may not fit" [fetched]. ~150 MB Haskell binary, off the serverless path
EPUB via pandoc	v2	--split-level=N , --epub-cover-image , Dublin Core metadata; EPUB 3.3 is a W3C Recommendation	Reflowable HTML+CSS in a zip. Page numbers do not exist ; anything authored against @page is meaningless here [inference]
Confluence storage XML	v2	writer, not export	"XHTML-based... technically XML": every rich element is an ac:structured-macro . Source editor ships from version 10.2.3, disabled by default [fetched]
SSG bundles (Hugo / Docusaurus / GFM)	v2	convention adapter, no compiler	Inverted fidelity risk: the SSG re-parses our markdown with its engine — exactly the cross-engine problem we already certify [inference]
Notion	no work needed	imports .md and .markdown natively [fetched]	—
Slidev	never	—	Per-slide front-matter measurably corrupts: ---\nlayout: center\n--- renders as <hr> then layout: center background: /bg.png [measured]
ODT · Deckset · man pages · MJML 5.4.0	never	—	DOCX covers ODT's demand; Deckset is a proprietary Mac app with no headless driver; roff drops images, tables and links; markdown→MJML is a lossy re-authoring into mj-section / mj-column , not a compile [fetched/inference]
Prince 16.2 · LaTeX · Typst 0.15.1 · WeasyPrint 69	never in-product; optional external hook	—	Each adds a second layout engine with its own styling language. Prince: USD \$3,800 /server one-time, \$2,660/unit at 5+, Desktop \$495, site licence "around USD \$2000 per year" [fetched]. LaTeX: GB-scale, disqualified for a serverless function on size. WeasyPrint: Python+Pango runtime, and cannot do right-to-left or bidirectional text at all [fetched]
PDF/X commercial print	never in v1-v2	—	Chrome emits RGB only, no /OutputIntent , no bleed, no crop marks, no ICC [inference from measured output]

Paged-media capability, measured and fetched:

capability	chrome 151 native	paged.js 0.4.3	weasyprint 69	prince 16.2
@page margin boxes	yes [measured]	yes [measured]	yes [fetched]	yes [fetched]
counter(page) / counter(pages)	yes [measured]	yes [measured]	yes, "known limitations #93" [fetched]	yes [fetched]
Running header from content (string-set)	no — CSS.supports("string-set", "x content()") → false [measured]	yes [measured]	yes [fetched]	yes [fetched]
Cross-reference with page number (target-counter)	no — XREF count 0, computed ::after content is none [measured]	yes [measured]	yes [fetched]	yes [fetched]
Footnotes (float: footnote)	no [inference]	partial [SS]	yes [fetched]	yes [fetched]
orphans / widows	unverified — parsed, but paragraph→page assignment was byte-identical at 1 vs 20 ; the test never produced a straddling paragraph, so it did not exercise the feature (LR#68: unverified, not unsupported) [measured]	via re-layout [SS]	yes [fetched]	yes [fetched]
Repeating table header	yes — header printed 3x across a 4-page table [measured]	yes [SS]	yes [fetched]	yes [fetched]
Tagged PDF + outlines	page.pdf({tagged:true, outline:true}) emits /StructTreeRoot and /Outlines — one flag away from a11y-usable output, currently unset [measured]	—	PDF/A and PDF/UA generation, validity "not guaranteed" [fetched]	yes [fetched]

PDF with real page numbers, content-derived running heads and page-numbered cross-references, generated from plain CommonMark with no LaTeX distribution, no \$3,800 licence and no Python runtime, is measured achievable on the binary we already ship — and no competitor in this survey offers it from a markdown editor.

Two recorded disagreements. MDN's @page page states that only margin , page-orientation and size are "implemented by at least one browser" and lists counter-reset / counter-increment as "not supported by any user agent yet" [fetched]; the measurement contradicts this for Chrome 151, corroborated by chromestatus feature page-margin-safety , desktop_first milestone 150, whose summary references authors wanting @page margin boxes for custom headers and footers [fetched]. Trust the measurement; MDN is stale. Second, inside one engine: CSS.supports("content", "target-counter(attr(href url), page)") returns true while the declaration is dropped and produces nothing [measured] — CSS.supports is not a usable capability probe for generated-content functions.

Shipped fragilities in the existing path, all fixable, all currently live [measured]:

- src/app/api/export/pdf/[...path]/route.ts calls exactly page.pdf({format:"A4", printBackground:true, margin:{...}}) — no displayHeaderFooter, no tagged, no outline .
- src/modules/export/presentation/print-css.ts is 156 lines whose only paged rule is @page { margin: ... } ; zero occurrences of break-inside , orphans , widows , counter , running , table-header-group anywhere in src/ , docs/ , test/ .
- waitUntil: "load" then a fixed setTimeout(2000) for the mermaid CDN — a wall-clock race, not a readiness signal.

- `document.fonts.ready` depends on Google Sans downloading from the network inside the lambda; blocked egress silently changes the typeface.
- `@sparticuz/chromium` ships **Open Sans only — Latin, Greek, Cyrillic**; fonts must arrive via `/var/task/.fonts`, `/opt/fonts`, or `/tmp/fonts` [fetched]. Locally, CJK fell back to STSongti-SC-Regular, a macOS system font [measured]; on Lambda that font does not exist and the same document renders as tofu [derived]. Installed `puppeteer-core` 25.0.4 (latest 25.9.0), `@sparticuz/chromium` ^148.0.0 (latest 149.0.0) [fetched].
- `process.platform === "linux"` gates the production binary, so the production renderer is never the one developers see — different font set, different failure mode.

Recommendation: ship a font layer with the PDF lane before shipping the PDF lane. **Anti-recommendation:** do not add WeasyPrint or Prince to close the footnote gap; each buys fidelity by adding a second styling language that profile authors would have to reason about alongside CSS. The genuine differentiator is not PDF quality — it is the **per-target degradation certificate extended from engines to output targets**: "this footnote survives HTML and PDF, degrades to a bracketed link in DOCX, is dropped in EPUB." Falsifier: if paged.js re-layout exceeds the `maxDuration = 60` budget on a representative 100-page document, the v1 flag becomes a v2 server job.

9.5 The hard edges — what markdown genuinely cannot do

Tables. GFM's own normative text: "Block-level elements cannot be inserted in a table" [fetched]. There is no colspan or rowspan syntax in CommonMark or GFM; Pandoc `grid_tables` buys spans and costs readable degradation — a dumb renderer emits `<p>+-----+-----+ | Fruit| Note |...</p>`, a wall of pipes [measured]. HTML fallback splits the bench: `marked` passes `<table><tr><td colspan=2>` through, `markdown-it` at its default `html:false` escapes it into visible `<table>`; `text` [measured]. Cell splitting happens **before** inline parsing, so `'x|y'` becomes two cells in both engines and GFM requires `\\` even inside code spans — meaning source `x\\y` renders `x|y` and a naive splice round-trip is lossy [measured]. Ragged rows are silently padded or truncated to the header count, with no warning [measured]. A caption placed on the line immediately after a table is absorbed as a data row (`<td>Table: my caption</td>`); with a blank line it degrades readably to `<p>Table: My caption.</p>` [measured]. Source columns align by character count while displays align by East Asian Width — is 4 characters and 8 display columns [derived], so every naive prettifier misaligns CJK tables.

RTL. `marked` emits **no** `dir=` for Arabic or Hebrew input in paragraphs, headings, lists, links or tables [measured]. Markdown has no direction primitive at all. GFM table alignment survives as `align="right"`, but logical start/end in RTL is the mirror of physical left/right, so `:--` selects the wrong edge [measured]. A mixed run such as `مرحباً بالعالم MDMAX 42 מודים שווים` renders as one undelimited sequence [measured]. Obligations: `dir="auto"` on block containers, page-level `dir` from front matter, first-strong isolation for mixed runs. WeasyPrint 69 cannot do RTL or bidi at all, which removes it from consideration for any Arabic or Hebrew customer [fetched].

CJK. CSS Text 3 (CR Draft, 2026-08-14) §4.1.3: for collapsible breaks a segment break "is either transformed into a space (U+0020) or removed depending on the context... **The rules for this operation are UA-defined in this level**" [fetched]. renders as `<p> </p>` in both engines [measured]; whether the reader sees a spurious space is browser-dependent and not specifiable by us. Word counting: `countWords` at `src/modules/editor/presentation/EditorPane.tsx:31` splits on `/\s+/` and undercounts against `Intl.Segmenter` by **20.00× on Chinese prose, 17.00× on Japanese, 12.00× on Thai, and 1.19× on a markup-heavy mixed document**; `ko/hi/en` are exact [measured]. Recorded disagreement: the brief said 1.7–2×; the ratio is a function of the Latin-and-markup fraction, not a constant — publish it as a range with the corpus named, never as one number. Search: MiniSearch's default tokenizer is `/[\n\r\p{Z}\p{P}]+/u` (`node_modules/minisearch/dist/es/index.js:2002`), and Chinese has neither `\p{Z}` nor `\p{P}` between words, so a whole sentence becomes one token — measured recall **3/7 = 42.9% overall and 1/5 = 20.0% on CJK-only queries**, rising to **7/7 = 100%** with an `Intl.Segmenter` tokenizer applied to *both* index and query [measured]. UAX #29 §4.1 says outright that reliable word boundaries in Thai, Lao, Chinese or Japanese "requires the use of dictionary lookup" [fetched]; `Intl.Segmenter` is that dictionary and is already in Node 24 and every current browser — zero new dependency. Grapheme integrity: `𐄂` is 4 UTF-16 units, 4 code points, **1 grapheme**, 12 UTF-8 bytes; `𐄂` is 11 / 7 / 1 / 25 [measured]. Any splice, truncation or column measurement that indexes by UTF-16 unit or code point will split a Devanagari cluster — LR#68 one level up.

Accessibility. WCAG 2.2 §5.2 makes AA all-or-nothing: "For Level AA conformance, the web page satisfies all the Level A and Level AA success criteria" [fetched] — one unlabelled image fails the page. Measured against this repo: **11 of 14 images have empty alt = 78.6%** (1.1.1); GFM tables emit **no** `<caption>`, **no** `scope=`, **no** `<colgroup>` (1.3.1); wide tables scroll in two dimensions, violating 1.4.10 Reflow at 320 CSS px; **35 heading-level skips and 5 of 186 files with more than one** `<h1>` (2.4.6); `marked` emits **no** `lang=` and **no** `dir=` for any input (3.1.1/3.1.2); GFM task lists emit `<input disabled type="checkbox">` with no accessible name (4.1.2) [measured/fetched]. Math is the one bright spot: `katex@0.17.0` `renderToString` emits MathML with `<semantics>` and `<annotation encoding="application/x-tex">`, so the TeX source survives into the a11y tree [measured] — **never ship image-rendered math; that is 1.1.1 with a text alternative you cannot generate**. Crossref and the arXiv API returned no study of markdown-rendered accessibility specifically [measured]; treat "markdown a11y is well-studied" as false and ship our own numbers, marking any first-to-measure claim unverified (LR#72).

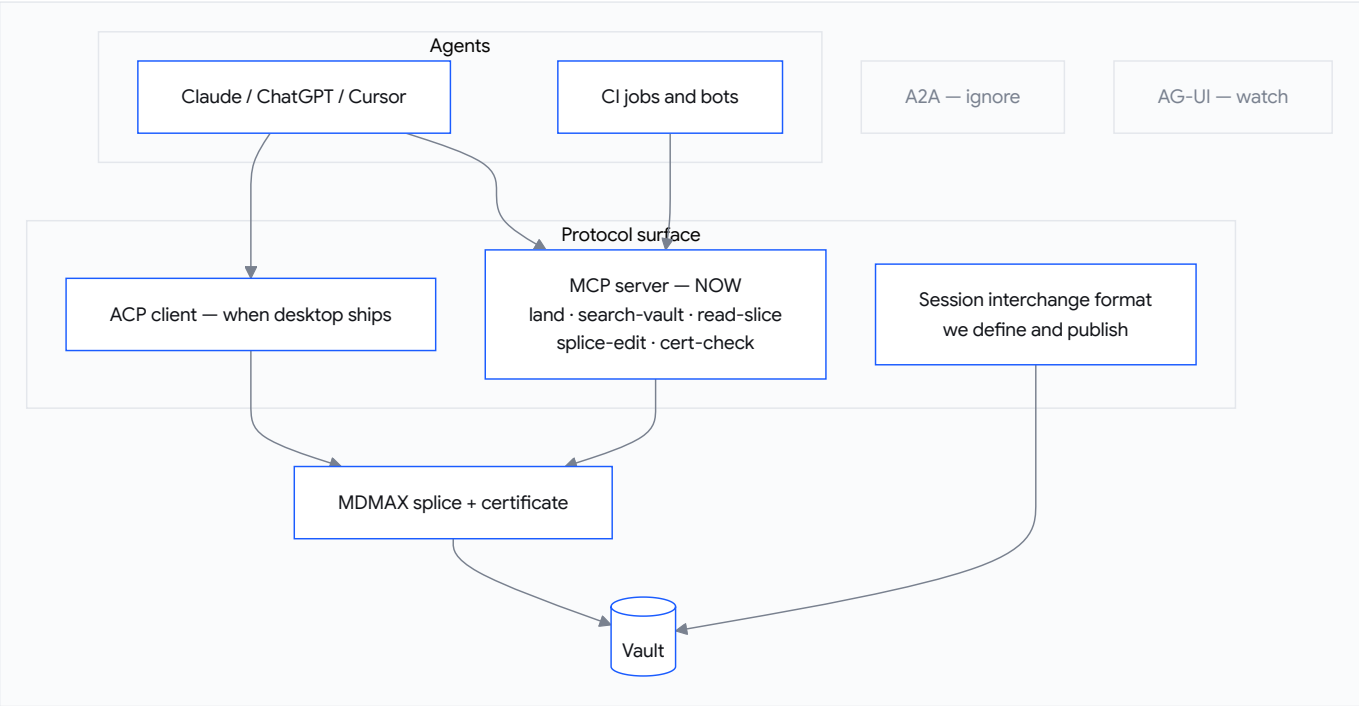
Tell users these are impossible rather than faking them:

CLAIM WE REFUSE TO MAKE	REASON
Merged cells	No colspan/rowspan in any CommonMark-compatible profile
Block content inside a table cell	GFM forbids it in normative text; a list in a cell is a lie
Multi-column pages, floats, sidebars, wrapped pull quotes	No block-container syntax exists; anything invented renders as literal <code>:::</code> noise
Numbered auto-resolving cross-references ("Figure 3")	Requires a numbering pass no dumb renderer will run: <code>{#fig-x}</code> leaks as visible text [measured]
Portable footnotes	Not in the GFM spec — <code>grep -ci footnote</code> on <code>cmark-gfm's spec.txt</code> = 1, an intro mention; the syntax lives in <code>test/extensions.txt:702</code> [measured]. Neither <code>marked</code> nor <code>markdown-it</code> renders <code>[^a]</code> by default. Promise degradation ("the note body stays readable"), never portability
A stable table caption position	After the table with a blank line, or it is absorbed as a data row
Guaranteed CJK line-joining	UA-defined per CSS Text 3; we can normalise our renderer, not GitHub's
Correct word counts without a segmenter	Say "approximate for Chinese, Japanese, Thai" until <code>Intl.Segmenter</code> ships
Sanitised HTML by default	<code>marked</code> ships no sanitiser and no URL-scheme filter — it renders <code><script>alert(1)</script></code> verbatim and <code>[click](javascript:alert(1))</code> as a live <code>javascript: href</code> ; <code>markdown-it</code> refuses dangerous schemes even at <code>html:true</code> [measured]. Two of our seven bench engines disagree by construction: that is a certification axis, not a bug. GFM's <code>tagfilter</code> filters exactly nine tags and states "All other HTML tags are left untouched"; GitHub compensates with private post-processing [fetched] — the spec is not a security boundary; the platform is . Any "renders like GitHub" claim must exclude that layer
Round-trip-exact pipes and tabs	<code> </code> in a cell and tab-vs-4-space indentation are semantically equal and byte-different; the splice engine must state which it preserves. CommonMark: "tabs behave as if they were replaced by spaces with a tab stop of 4" [fetched]
WCAG AA out of the box	No <code>scope</code> , no <code>caption</code> , no <code>lang</code> , no <code>dir</code> , no accessible name on task-list checkboxes. Conformance is something the publisher adds, never something the format supplies

Two structural hazards for the splice model specifically. Reference links are **non-local**: CommonMark's own author writes that `[foo][bar]` has four possible meanings "depending on whether the references... are defined elsewhere (perhaps later)" and that this makes "accurate syntax highlighting nearly impossible" [fetched]; confirmed locally — the same source renders as text, or `/x`, or `/y` [measured]. And the whitespace cliff: nesting depends on the content column, where indent 1 is a sibling, 2–5 nests, and **6 stops the bullet being a bullet** and turns it into prose [measured]. One space changes document structure invisibly, which is a lint obligation at save time, not a publish-time check. Recorded, unresolved:

marked and markdown-it agreed structurally on 6/6 setext-vs-thematic-break cases and 5/5 list/indent cases, differing only in <hr> vs <hr /> and align= vs style="text-align:" [measured] — the ambiguity class is specified and implemented consistently, so the risk here is authors, not engines. **Anti-recommendation:** do not build an auto-fixer that silently normalises ambiguous indentation or reference-link placement; refuse and surface, per the engine's own contract.

§10 Formats, protocols and interoperability



10.1 Format posture

POSTURE	FORMATS	RATIONALE
READ	OKF bundles (v0.2, own Google repo, 356 adopter repos, +24%/27d FETCHED) · SKILL.md folders · MyST keys · textbundle	Zero parser cost; recognise sources / generated / verified / status / stale_after and derive trust tiers
EMIT	llms.txt v2 + markdown twins + rel="alternate" type="text/markdown" (Chrome Lighthouse audits for it FETCHED) · OKF export · Skill export — a curated collection becomes a SKILL.md folder consumable by 41+ agents	"The journey becomes a reusable capability" is the strongest interop story found
BUILD	Session interchange format — markdown-native, open, documented	No standard exists; the absence is the opening FETCHED
WATCH	.prompty · AG-UI	Not load-bearing yet
IGNORE	Promptfile (dead) · langchain-hub (archived) · A2A and server-cards as document formats	Dead or wrong layer

10.2 Protocol posture

PROTOCOL	POSTURE	REASON
MCP	Ship now, tools-first	The July 2026 revision went stateless and deprecated Roots, Sampling and Logging — tools are the lowest common denominator FETCHED . Hard-cap under 20 tools; verdict-first responses; refusals that name the rule and give a corrected example
ACP	Client, when a desktop surface exists	40 registered agents including an Anthropic-co-authored Claude adapter; session/load resume list is the only shipped multi-vendor session-continuity semantics anywhere FETCHED . ACP agents are local subprocesses over stdio
A2A	Ignore	Wrong layer for a document tool
Session interchange	Build and publish ours	§10.1

The plumbing is table stakes. The pitch is what rides on it: every agent edit goes through the splice writer, so the agent cannot corrupt what it edits.

10.3 The open territory

Claim-level method and confidence is claimed by no spec, and OKF explicitly argues *against* stored scores. The compatible design is signals, not scores, derived at render time — which is a renderer's game, and therefore ours.

§11 The AI layer — what AI-native actually means

11.1 The retention evidence, before any feature list

Active-install proxy = peak single-version downloads, across 7,020 plugins with stats and 142,701,824 cumulative downloads. MEASURED, OBSIDIANMD/OBSIDIAN-RELEASES, 2026-08-29

CAPABILITY CLASS	PEAK-VERSION SUM	CUMULATIVE	MEMBERS
Deterministic projection (Dataview, Kanban, Calendar, Tasks, TaskNotes)	7,289,307	15,840,093	5
Deterministic structure (Linter, Outliner, Tag Wrangler, Table Editor, Templater)	2,019,220	12,057,166	5
Agentic in-vault (Copilot, realclaudian, agent-client, smart-composer)	463,062	4,133,966	4
Ambient related-notes (Smart Connections, second-brain, lookup, ExcaliBrain)	293,333	1,631,201	4
Generation / chat (textgenerator, bmo, chatgpt-md, local-gpt, ollama, gpt3-notes)	202,778	924,287	6
Voice capture (whisper, transcription)	46,478	87,502	2

- **Deterministic projections out-install every AI capability combined by 7.25x** **DERIVED: $7,289,307 \div 1,005,651$** — which is the strongest available validation of the projection law, arriving from outside our own thesis.
- **697 of 7,058 plugins (9.9%) describe an AI capability but take only 6.09% of cumulative downloads and 3.44% of the peak-version sum** — AI is over-supplied and under-installed by ~2.88x on the active-install measure. **MEASURED** **DERIVED**
- Source disagreement, recorded not resolved: the popular "AI plugins get abandoned" story is **false on the supply side**. AI-described plugins are *less* stale — **11.3% (78/688) unreleased >12 months vs 20.9% (1,464/7,020)**, median 50 days since last release vs 71. The cohort is younger, which partly confounds it. The abandonment is demand-side: actively maintained products that few people install. **MEASURED** **INFERENCE**
- The largest file-native editor in the category ships no AI at all: **zero occurrences of the token "AI" across 9,502 chars** of its full Active/Planned/Launched roadmap back to July 2023. What it does ship is Bases (data in local markdown properties, views described in valid YAML), Kanban and Calendar views for Bases, an Airtable→Markdown import, a CLI, and a Keychain for plugin API keys. **MEASURED** **FETCHED**

11.2 Ranked capabilities — build in this order

RANK	CAPABILITY	RETENTION EVIDENCE	SPLICE SHAPE	COST / OP
1	Transformation verbs on a selection — summarise, expand, restructure, translate, change register, extract entities, to-table	Strongest of the set: Bing Copilot telemetry over 200k conversations finds "the most common and successful AI-assisted work activities involve information work — the creation, processing, and communication of information" (arXiv 2507.07935v6, upd 2025-12-22) FETCHED . Notion's shipped surface is "Edit with AI" on a highlighted range FETCHED	Perfect — output <i>is</i> a byte range	\$0.004–\$0.03
2	Frontmatter-key fill and repair — tags, status, dates, typed links	Structure bucket 2,019,220 active installs, won today by deterministic tools MEASURED	Perfect — a YAML key is the addressable unit; profiles already read them	<\$0.002
3	Agent edits arriving as suggestions (MCP <code>land()</code> into the review loop)	Fastest-growing AI bucket at 463,062; <code>realclaudian</code> is #13 by cumulative downloads across all 7,058 plugins (1,940,612). Users are filing "Improve Agent Mode review and consent controls" (2026-08-19) against the market leader — they are asking a competitor for our differentiator MEASURED FETCHED	Perfect — a splice against <code>baseSha</code>	per-op
4	Q&A over the open file with byte-anchored citations	Real but shallow: ChatPDF claims "10M+ users" and "1,000,000+ Q's answered every day" = 0.1 queries per registered user per day DERIVED — signup scale, not habit scale	Perfect — a citation <i>is</i> a byte range	~\$0.005
5	Heading/structure repair as a proposed diff	Currently won by determinism: Obsidian Linter 112,395 active installs, no AI MEASURED	Good — do the deterministic 80% first, AI only for what regexes cannot decide	~\$0.006
6	Scoped multi-document synthesis — explicit N files, never "the vault"	Real and throttled by its own vendor: Gemini Notebook caps at 50 chat queries/day , 100 notebooks × 50 sources × 500,000 words FETCHED	Acceptable — bounded input set, ideally a projection's filter result	\$0.01–\$0.10
7	Voice capture into the file	Real category, tiny in-vault (46,478). Two companies pivoted <i>out of</i> the document to chase it — Tana's meeting platform is now "a separate product from Tana Outliner"; Mem is now "Your AI chief of staff" FETCHED	Acceptable, as capture-to-inbox only	external ASR
8	Ambient related-notes / duplicate detection	Highest maintenance burden per install in the category : Smart Connections 5,407★ with 489 open issues = 9.05% issues-per-star vs Copilot's 1.30% (99/7,640) ; the most-commented open issues are all silent index breakage — "Pane is always loading" (39), "Embeddings no longer function (linux)" (24), "Doesn't seem to look at my notes" (22) MEASURED FETCHED	Poor — requires a second index beside the file	\$38.79/user/month
9	Ghost text / continuous completion	Declining cohort (202,778, below both agentic and ambient); <code>textgenerator</code> last released 2026-04-27, <code>gpt3-notes</code> 2023-07-07, <code>vault-chat</code> 2023-06-03 MEASURED	Poor — writes without being asked; contradicts propose-first at the mechanism level	continuous

11.3 Cost arithmetic

Corpus: `knowledge` n=590 md files, p50 **479 words**, p90 **3,584**, total 1,980,166 words; `frontmatter` repo n=186, p50 2,439, p90 10,328. **MEASURED** Prices: Haiku 4.5 **\$1/MTok in, \$5/MTok out**; Sonnet 5 **\$2/\$10**; Opus 5 **\$5/\$25**; batch Haiku **\$0.50/\$2.50**. The same page warns Claude 4.7+ tokenizers emit "approximately 30% more tokens for the same text." **FETCHED**, **DOCS.CLAUDE.COM/PRICING** Assume 1 word ≈ 1.33 tokens **INFERENCE**.

OPERATION	TOKENS	HAIKU 4.5	NOTE
Summarise a p90 note → 200 w	4,767 in / 200 out	\$0.0058 DERIVED	Cheaper than one Notion credit (\$0.01) FETCHED
Restructure a p90 note in place	4,767 / 4,767	\$0.0286 DERIVED	Output is 83% of cost
Same on Sonnet 5	4,767 / 4,767	\$0.0572 DERIVED	2× for a formatting-shaped task
One full pass over a 590-note vault	2,633,621 in	\$2.63 DERIVED	Input only
Ambient re-rank on every save, 20 neighbours @ p50, 100 saves/day	1.274 MTok/day	\$38.79/user/month DERIVED: $1.274 \times \\$1 \times 30.44$	The line that kills rank 8

11.4 Explicit refusals

REFUSE	BECAUSE
No persistent semantic index of the vault	A second source of truth beside the file is what the projection law forbids, it costs \$38.79/user/month at 100 saves/day DERIVED , and the market leader's most-discussed open issues are all silent index failure MEASURED
No "rewrite in my voice" as a headline feature	Across 3 studies, 7 datasets, >880,000 texts, LLM writing assistance is linked to a statistically significant 21–50% reduction in writing-complexity variance ($p \leq .05$), "emphasizing conformity over individuality" (arXiv 2502.11266, upd 2026-08-24) FETCHED . Ship register-shift as a one-shot verb the user asks for, never as a default or ambient suggestion
No unsourced summarisation	Entity hallucination in abstractive summarisation is documented to at least 2021 (arXiv 2102.09130), and Gemini Notebook's own help page lists "information not in sources" as a first-class failure FETCHED . Every summary carries the byte ranges it came from or it does not render
No ambient AI buttons, no ghost text by default	66% of ~33,000 developers name "AI solutions that are almost right, but not quite" as their top frustration; only 3.1% highly trust AI output (29.6% somewhat trust, 26.1% somewhat distrust, 19.6% highly distrust), and favourability fell to 59.7% from "70%+" in 2023 and 2024" FETCHED , STACK OVERFLOW 2025 . An unrequested suggestion spends trust the product cannot refill
No AI that decides or publishes	"Don't plan to use AI for this task": deployment/monitoring 75.8%, project planning 69.2%, committing and reviewing 58.7% FETCHED . AI may propose a status change; the human commits it. This is also the boundary that keeps us out of Notion-style project management
No meeting bot, no chief-of-staff agent	Tana and Mem both left the document to chase it. That is a different company FETCHED
No standalone AI SKU	On 2024-06-01 Notion sold AI at \$8/member/month annual, \$10 monthly, "Now with Q&A". On 2026-08-29 that add-on does not exist: Free and Plus get only a trial, Agent and Enterprise Search sit inside Business at \$20/member/month, and Custom Agents meter at \$10 per 1,000 monthly Notion credits. Standalone document-AI became bundled table stakes plus a metered agent line in ~26 months FETCHED , TWO DATED SNAPSHOTS DERIVED . Price AI as a metered line inside a free-forever editor

11.5 The instrumentation contract

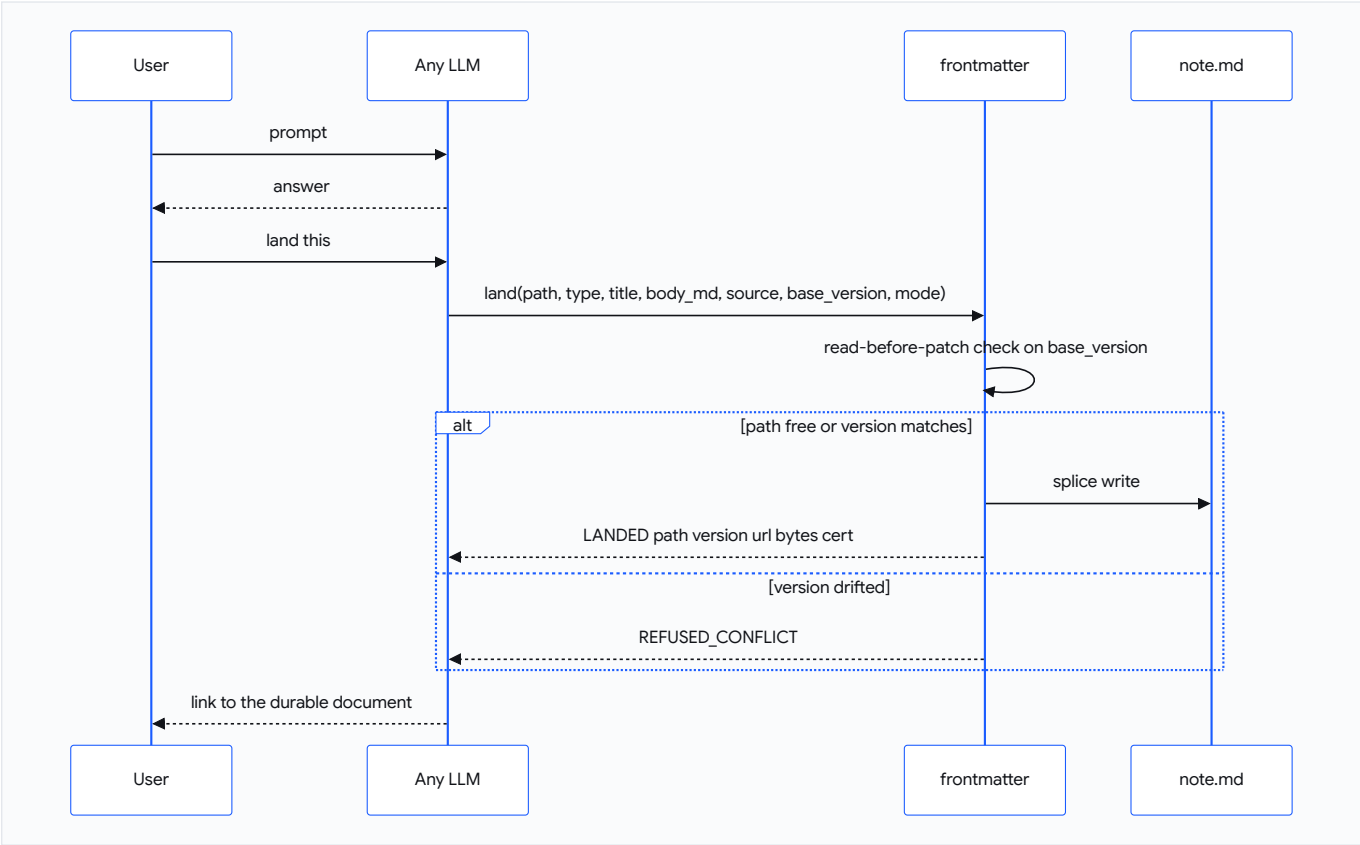
- Report **Strong Acceptance** — accepted only if <50% of the proposal was edited *and* the edits do not change critical parts — not raw acceptance. The only published figures for a propose-first assistant are Ansible Lightspeed: **49.08% strong acceptance on multi-line suggestions, Day-30 retention 13.66%** (10,696 users, 3,910 returning; arXiv 2402.17442, pub 2024-02-27, upd 2024-10-22, self-described as the first code-completion tool to publish N-day retention). FETCHED
- That is a small model on one narrow YAML dialect, so 49.08% is a ceiling for a constrained verb, not a general rate. INFERENCE
- This supersedes the **"~30% acceptance rates ss" line currently in** `docs/FRONTMATTER-MASTER-PLAN-2026-08-28.md:82` , which is a search summary and must not be published as fact. MEASURED, GREP OF THE PLAN FILE
- Record accept / reject / partial / edit-distance-after-accept **per verb, locally, publish nothing**. The entire ranking in 11.2 was reconstructed from install counts, version distributions, issue ratios, and roadmap absences — never from a survey — which is a working demonstration that behaviour is readable without asking. Telling a user who somewhat distrusts AI (26.1%) that you are measuring their rejections is the same conversation twice. MEASURED INFERENCE
- Kill rule per verb: below 20% strong acceptance or below 10% D30 after 90 days of data, the verb is removed, not tuned.

11.6 What "AI-native" means, in the words a user gets

AI-native does not mean the app writes for you. It means your markdown file is the only thing that exists, and the AI is a set of precise verbs you point at a piece of it — summarise this section, turn these paragraphs into a table, fill in the tags, translate this, restructure these headings — where every verb comes back as a proposed change you can see, byte for byte, before it touches the file, and lands through the same splice writer that guarantees nothing you did not select gets rewritten. **Nothing runs in the background, nothing is indexed behind your back, nothing is published or decided on your behalf, and if you delete every model key tomorrow the file, the board, the calendar and the site all still work exactly as they did — because those were never AI features, they were always just your file, read a different way.**

Decision falsified by: transformation verbs failing to clear 20% strong acceptance in the first 90 days, or a measured demand signal for ambient related-notes that survives the \$38.79/user/month line.

§12 The capture loop



```
land({ path?, type, title, body_md,
      source: { tool, model, conversation_url, session_id },
      base_version?, mode: create | patch | rewrite })
→ { LANDED | VERSIONED | REFUSED_CONFLICT | NEEDS_TARGET,
    path, version, url, bytes_written, cert }
```

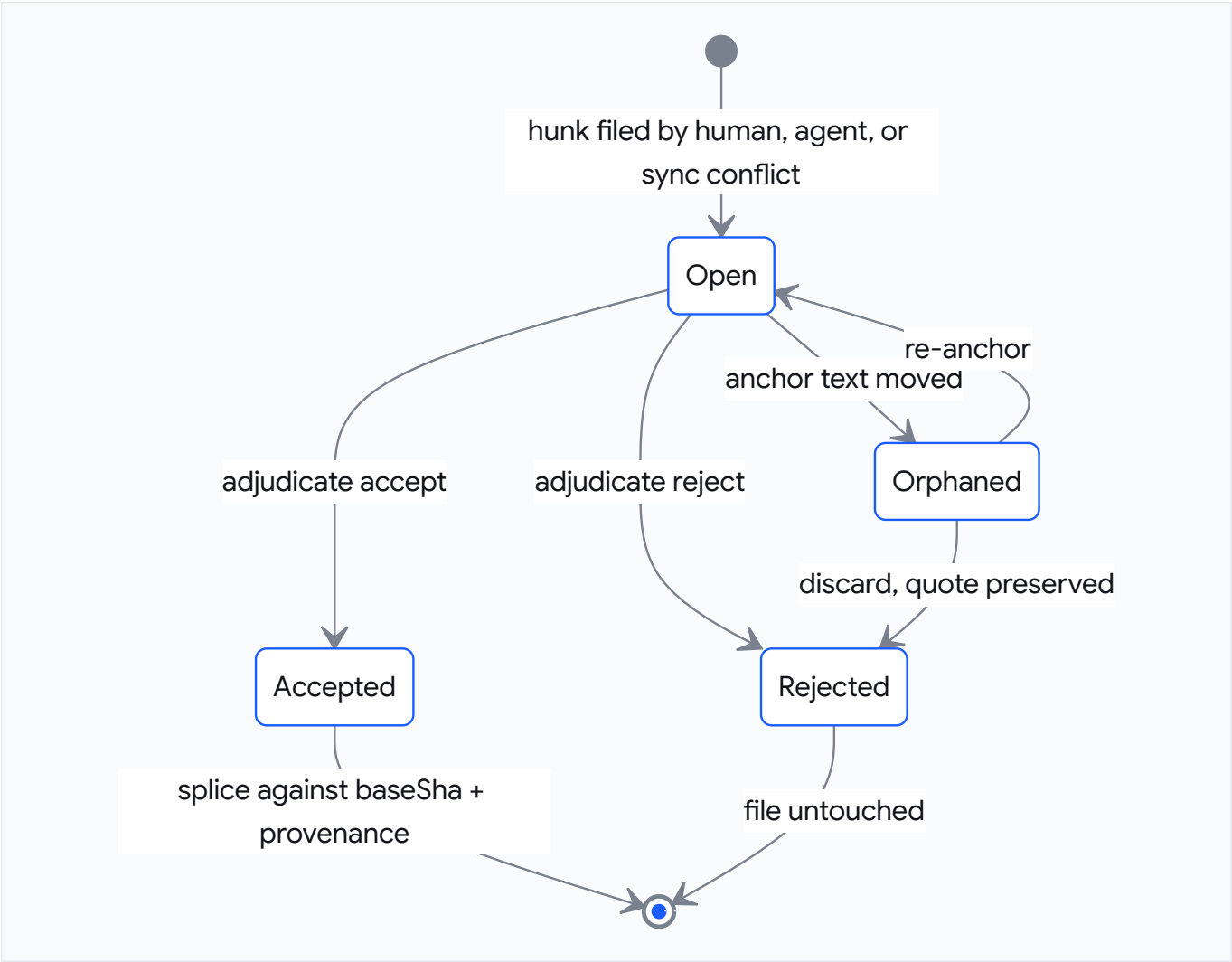
RULE	WHY - EACH FROM A DOCUMENTED FAILURE
Identity is the path, explicit in every call	On claude.ai, where identity is inferred from phrasing, "it made a new artifact instead of updating mine" is the #1 documented failure
Two edit sizes are protocol modes, not prompt etiquette	patch vs rewrite — otherwise the model guesses
base_version enforces read-before-patch	Structurally kills the drift bug where the user hand-edits and the model keeps talking about the version it remembers. Bolt's own system prompt: always edit the latest content FETCHED . The file is the memory; the model's memory of the file is a cache to invalidate
Counts re-derived from disk	The anti-lying-report rule
Typed at capture	type: picks schema, render and home grouping. Granola: \$1.5B valuation on template-typed capture with per-line provenance SS
Users see the nouns, never "artifact"	Bolt's system prompt literally forbids the word FETCHED
Auto-land only into an inbox lane	Never silently into the curated vault

Distribution: an MCP server for MCP-capable tools · a chat-side skill for everything else ("land this" → typed fenced block → one-paste inbox) · ZIP importers for ChatGPT and Claude where the verification report is the demo · retro-capture parsing one conversation into several typed documents.

The promotion loop is what makes it a home rather than a filing cabinet: landed documents become retrievable, citation-gated context — draft → active → source-of-truth → superseded .

§13 The review surface

One grammar for every change: human suggestions, AI edits and sync conflicts all arrive as hunks. The deepest single extraction of the research — 40 years of office software, sourced to the Google Docs API discovery document and pandoc's docx reader **FETCHED** .



RULE	DETAIL
Suggesting is a mode	Edit / Suggest / View dial. The suggester role has no byte-writing code path
Hunks coalesce at word grain	Git's line grain is the wrong resolution for prose
Markup / Final / Original are pure render projections	The OnlyOffice save-in-preview-deleted-changes bug is the negative spec
The adjudication ladder	per hunk → per suggestion → all-shown-under-filter (filter by author, <i>including AI agents</i>), with accept-and-advance and preview-before-bulk
Resolution is an authored thread event	<code>replies.action ∈ {resolve, reopen}</code> — never a silent boolean
Anchors are quote-preserving with visible orphaning	Word silently deletes orphaned comments; Docs orphans opaquely. We badge, preserve the quote, offer re-anchor
Accept = splice against <code>baseSha</code>	Plus a <code>Co-authored-by</code> trailer
Governance via branch-protection semantics	No document freezes
Interchange	CriticMarkup + pandoc <code>--track-changes</code> spans, both directions

The two moves that beat the incumbents: **programmatic and AI suggestion authorship** — Google's public API cannot create suggestions at all `FETCHED`, while ours is a plain sidecar schema any CI job or agent can file into — and **durable provenance through accept**, where theirs evaporates.

Anchoring implementation: Hypothesis's `match-quote.ts` + `approx-string-match` (BSD-2 / MIT) is the mechanism that lets an AI target survive human edits.

Treat the review surface as a breaking-API contract. Cursor and Windsurf both regressed per-hunk control and both got publicly burned. Per-hunk accept is the single most-demanded feature in AI editors `FETCHED`.

§14 Documents that build things — the spec-driven lane

The founder's question for this lane: can frontmatter be the place you write a spec, an AI reads it, and output is generated? The answer this section defends is *yes for the document half, no for the execution half* — and the entry is narrow, cheap, and gated on one measurement we can take ourselves.

HALF OF THE LANE	WHO OWNS IT TODAY	FRONTMATTER'S POSITION
Authoring, review, rendering, drift display, requirement identity	Nobody. Not one tool surveyed ships a way to read or edit a spec other than a plain text editor [fetched, 14 tools]	Enter here
Execution, test generation, approval gates, agent dispatch	AWS Kiro (IDE-only), spec-kit /implement , the coding agent's own loop [fetched]	Do not enter

14.1 The market as it stands

Census — GitHub, 2026-08-28 UTC [measured, api.github.com]

REPO	STARS	FORKS	LICENCE	CREATED	LAST PUSH	STATE
anthropics/claude-code	143,296	22,918	none declared	2025-02-22	2026-08-28	live
github/spec-kit	132,035	11,865	MIT	2025-08-21	2026-08-28	live
cline/cline	67,077	7,241	Apache-2.0	2024-07-06	2026-08-28	live
Fission-AI/OpenSpec	66,571	4,583	MIT	2025-08-05	2026-08-28	live
gsd-build/get-shit-done	64,629	—	MIT	2025-12-14	2026-05-31	dead redirect
bmad-code-org/BMAD-METHOD	52,419	5,964	NOASSERTION (npm says MIT)	2025-04-13	2026-08-28	live
continuedev/continue	35,666	5,295	Apache-2.0	2023-05-24	2026-08-28	live
cursor/cursor	33,190	2,287	none	2023-03-12	2026-05-12	issue tracker
Kilo-Org/kilocode	27,054	3,096	MIT	2025-03-10	2026-08-28	live
RooCodeInc/Roo-Code	24,322	3,415	Apache-2.0	2024-10-31	2026-05-15	ARCHIVED
agentsmd/agents.md	23,968	1,813	MIT	2025-08-19	2026-08-25	live
open-gsd/gsd-core	8,849	—	MIT	—	—	live successor
buildermethods/agent-os	5,343	831	MIT	2025-07-16	2026-05-05	quiet 3.8 mo
kirodotdev/Kiro	4,231	310	none (proprietary AWS)	2025-06-17	2026-08-27	issue tracker
gemini-cli-extensions/conductor	3,713	—	Apache-2.0	2025-12-17	2026-08-11	live
gotalab/cc-sdd	3,644	—	MIT	2025-07-17	2026-05-20	live
Privacy-ai/spec-kitty	1,572	—	MIT	2025-10-09	2026-08-28	live
tesslio/cli	71	—	—	—	—	live

Disagreement recorded: the repos endpoint returns OpenSpec at 66,571; the search endpoint, same minute, returns 66,570 [measured]. Both stand.

Distribution [measured, api.npmjs.org + pypi.org, week 2026-08-21→27]

PACKAGE	LAST WEEK	LAST MONTH (2026-07-29→08-27)	LATEST	LICENCE FIELD
@fission-ai/openspec	464,621	1,645,308	1.11.0	MIT
bmad-method	16,941	81,735	6.11.0	MIT
@tessl/cli	6,034	19,794	0.102.0	SEE LICENSE.md
specify-cli (PyPI)	n/a	n/a	1.0.1, 56 releases	null

- [derived] OpenSpec ÷ BMAD weekly = 464,621 ÷ 16,941 = **27.43×**; OpenSpec ÷ Tessl weekly = 464,621 ÷ 6,034 = **77.00×**; OpenSpec stars ÷ tesslio/cli stars = 66,571 ÷ 71 = **937.6×**.
- VS Code marketplace installs [measured, gallery API]: GitHub.copilot 74,459,915 · cline 5,131,205 · roo-cline 1,961,704 (lastUpdated 2026-05-15, identical to its GitHub archive date) · Kilo-Code 1,473,000.

Convention adoption [measured, GitHub code search 2026-08-28 — approximate bucketed totals over a subset of public repos, every value factoring as odd×2ⁿ]

CONVENTION	INDEXED FILES
CLAUDE.md	774,144
AGENTS.md	460,800 (repo root) / ≈858,112 (unrestricted)
.claude/skills/**/SKILL.md	≈380,928
.cursor/rules/**/*.mdc	186,112
.github/copilot-instructions.md	146,944
openspec/specs/**/spec.md	117,504
GEMINI.md	65,152
.kiro/specs/**/requirements.md	33,728
.kiro/specs/**/tasks.md	≈30,592
.specify/memory/constitution.md	11,264
memory-bank/activeContext.md	2,728

Disagreement recorded: two research passes returned 460,800 and ≈858,112 for AGENTS.md on the same day — the first restricted to repo root, the second not; CLAUDE.md matched exactly at 774,144 across both. Quote the root-restricted figure when comparing like for like. A third, independent number exists: agents.md the site claims "over 60k open-source projects" [fetched] — different unit, different index, also stands.

Independent census — SPECMINE, arXiv 2608.25202, 2026-08-25 [fetched]: 470,795 `spec.md` / `specs.md` files across **73,030 repositories**, attributed to **17 named tools**; a separate Kiro census of 98,574 files across 12,910 repositories; 2,421,323 typed references indexed. [derived] $470,795 \div 73,030 = 6.45$ **spec files/repo**. Its PR sweep found spec-and-implementation co-changing in one PR in **581 repositories** — [derived] $581 \div 73,030 = 0.80\%$, a **lower bound only** because the sweep covered 11 tools' ≥ 10 -star repos, not the census. Do not quote it as a co-change rate.

The shape of the category — what all of them do the same way

- Markdown is unanimous.** Every artifact in every tool surveyed is a `.md` file. Zero use a formal or binary spec language [measured across 14 tools].
- Three-artifact skeleton**, reached independently by Kiro, spec-kit, OpenSpec, agent-os and BMAD: `requirements/spec` → `design/plan` → `tasks` [fetched].
- A hidden dot-directory keyed to one vendor** — `.specify/`, `.kiro/`, `openspec/`, `agent-os/`, `.cursor/`, `.claude/`, `memory-bank/`. The convention is the lock-in; the response is an install matrix (spec-kit 37 agents, OpenSpec 36 tools [measured]), not a shared location.
- Given/When/Then plus RFC-2119 is the shared dialect**: OpenSpec `### Requirement: / ### Scenario:` with SHALL/MUST, Kiro EARS, spec-kit `**Given**/**When**/**Then**` [fetched].
- Human approval is the gate, not a machine** — Kiro three-phase approvals, Factory `ExitSpecMode`, Claude Code plan-mode approval [fetched].
- Nothing is machine-verified against code except one product.** Kiro extracts properties from EARS requirements, generates property-based tests, runs them and links failures back to the requirement — and its own docs limit the claim to "evidence of correctness, not a proof... not formal verification", IDE-only per its own capability matrix [fetched]. `openspec validate --archived` exits non-zero on an unticked `tasks.md` checkbox — a genuine CI gate, on the markdown [fetched]. Everything else is an LLM pass that does not block.

What none of them do — no durable identity for a requirement (spec-kit's `/analyze` maps tasks to requirements "by inference by keyword / explicit reference patterns", i.e. by grep, because nothing better exists [fetched]); no machine binding from requirement to code range (SPECMINE had to *reconstruct* it from 2.42M typed references); no deterministic drift detection; no review surface; no cross-tool portability across 17 named tools.

The cost of the identity gap is measured. arXiv 2606.30689 [fetched]: only the per-line-citation condition (`traceSDD`, hierarchical `REQ-XXX.Y.Z` IDs) enables automated hallucination detection — **TDR 86.4% (Claude Sonnet 4.6) / 88.0% (GLM-5-turbo)**, against 0% for Spec Kit, OpenSpec and the uncited baseline, FPR 0% in both. The same paper prices it: citation *reduces* output determinism (Claude $d=-0.76$, $p=0.003$; GLM $d=-0.72$, $p<0.001$).

Verdict on whether a markdown editor is a credible home

FOR THE EDITOR	FOR THE IDE
The artifact is 100% markdown in 100% of tools; whoever owns markdown editing owns the file type [measured]	Verification lives where the code and the test runner live — Kiro's PBT loop must <i>run tests</i> , and that is the only differentiated capability in the category [fetched]
The IDE has <i>conceded</i> the artifact: Claude Code makes plan a session permission mode with no file; Factory Spec Mode is ephemeral; Cursor ships rules, not specs [fetched]	Approval gates are conversational and belong in the agent loop [fetched]
Four conventions are legible to any editor with zero integration — [derived] $460,800 + 117,504 + 33,728 + 11,264 = 623,296$ indexed files [measured]	Distribution flows through the coding agent: spec-kit installs into 37 agents in one command [measured]

Decision: enter as a viewer, reviewer, renderer and anchor-provider for specs other tools already write; do not become tool #18. Do not define a new spec format or a `.frontmatter/specs/` directory — that is convention #18 in a field with 17 and no interchange standard. Do not build an approval workflow, do not run or generate tests, do not build spec↔code semantic verification. Sell to the OpenSpec and Kiro user: 464,621 weekly npm downloads is a distribution channel, not a competitor.

The strongest argument against entering at all. The category has no retention evidence and three of its biggest names have already left. Tessl — the best-funded pure-play, the company that coined the spec-as-source-of-truth pitch — pivoted its CLI to a skills registry; its current CLI reference contains **no spec command at all**, and its CLI repo sits at **71 stars against OpenSpec's 66,571, a 937.6× gap** [measured]. Roo Code is **archived** on 1,961,704 installs [measured]. `get-shit-done` accumulated 64,629 stars and became a redirect stub [measured]. SPECMINE observes spec-and-code co-change in **581 repositories** against a 73,030-repo census [fetched] — consistent with specs being generated once and abandoned. Every artifact we would render is written by an MIT-licensed CLI that costs nothing; willingness-to-pay is unproven precisely because the incumbent price is zero. And our actual moat is *fidelity* — byte-exact splice, the degradation certificate, the machine-view differ — and nothing in this category has been observed to churn over fidelity. The honest counter-case is that spec-driven development is a 12-month prompt-scaffolding fashion whose artifacts are write-once, and that building a renderer for write-once files is building a museum. [inference on measured/fetched inputs]

What would falsify the decision to enter: on a corpus of real `openspec/` and `.kiro/specs/` repositories, measure the fraction of spec files edited more than once after creation. If that fraction is low, the museum argument wins and the correct move is to ship the renderer as a feature and never as a product. Take this measurement before any further investment; it is one we can take ourselves.

14.2 The document canon we render, not invent

TYPE	CANONICAL SOURCE	SECTIONS	FRONTMATTER KEYS	MACHINE-READABLE SCHEMA	RENDER PROFILE
ADR — MADR	<code>adr/madr</code> template, 1422 B [fetched]	6 H2 (Context and Problem Statement; Decision Drivers*; Considered Options; Decision Outcome; Pros and Cons*; More Information*) + 2 fixed H3 (Consequences*, Confirmation*); minimal variant = 3 H2 + 1 H3; 4 template variants ship	5, all optional in-file: <code>status</code> , <code>date</code> , <code>decision-makers</code> , <code>consulted</code> , <code>informed</code>	NO — <code>schema.json</code> → 404 at both paths [measured]	<code>decision</code>
ADR — Nygard	2011 article [fetched]	5: Title, Context, Decision, Status, Consequences	none — plain headings	NO	<code>decision</code>
RFC — IETF	RFC 7322 §4 [fetched]	23 listed elements , 10 marked [Required] + 1 conditional (IANA) + 1 on-demand	first-page header, not YAML	YES, two — RFC 7991 XML vocabulary; <code>rfc-index.xml</code> (2,324,273 bytes) [measured]	<code>spec</code> + <code>proposal</code>
RFC — Rust	<code>rust-lang/rfcs/0000-template.md</code> [fetched]	9 H2: Summary; Motivation; Guide-level explanation; Reference-level explanation; Drawbacks; Rationale and alternatives; Prior art; Unresolved questions; Future possibilities	4-item bullet list , not YAML: Feature Name, Start Date, RFC PR, Rust Issue	NO (process state tracked externally by rfcbot)	<code>proposal</code>
RFD — Oxide	RFD 1, 47,089 B [fetched]	none prescribed — content prompts only	4 AsciiDoc attributes : <code>:authors:</code> , <code>:state:</code> , <code>:discussion:</code> , <code>:labels:</code>	NO heading schema: header parseable	<code>decision</code>
RFC — company-internal	Squarespace + Uber posts [fetched]	Uber backend 11, Uber mobile/web 11, literal-string overlap 5 [derived]; Squarespace adds 3	Squarespace 7 header fields; approval enum is 2 values by design : <code>yes</code> <code>not yet</code>	NO	<code>proposal</code>
Design doc — Google	industrialempathy.com [fetched]	5 top-level; explicitly refuses to prescribe — "Rule #1 is: Write them in whatever form makes the most sense"; 10-20ish pages, or a 1-3 page mini	none	NO	<code>proposal</code>
Gherkin	cucumber.io reference + <code>gherkin-languages.json</code> [fetched/measured]	Primary keywords: <code>Feature</code> , <code>Rule</code> , <code>Example / Scenario</code> , <code>Given / When / Then / And / But / *</code> , <code>Background</code> , <code>Scenario Outline</code> , <code>Examples</code> ; 4 secondary (<code>""</code> , <code> </code> , <code>@</code> , <code>#</code>)	none — tags are the metadata channel	YES — <code>gherkin-languages.json</code> , 80 languages, English entry 11 keyword keys + <code>name</code> + <code>native</code> = 13 [measured]	<code>behavior</code>
OpenAPI	OAI repo, <code>versions/</code> [measured]	Root fixed fields: 3.2.0 = 11, 3.1.1 = 10, delta = 1 (<code>\$self</code>); 9 spec files in <code>versions/</code>	the document <i>is</i> the frontmatter (<code>info</code>)	YES, strongest — <code>spec.openapis.org/oas/3.1/schema/2022-10-07</code> → 200, 3535 B; <code>oas/3.1/dialect/base</code> → 200, 345 B. <code>.../schema/latest</code> → 404 for both 3.1 and 3.2 — pin the dated URI [measured]	<code>api</code>
Changelog	Keep a Changelog 1.1.0, 8394 B [fetched]	# Changelog → ## [Unreleased] → ## [version] - YYYY-MM-DD → ### <type> ; exactly 6 types (Added, Changed, Deprecated, Removed, Fixed, Security); exactly 7 principles	none	NO — principle #1 is literally "Changelogs are for humans, not machines"	<code>release</code>
Postmortem — Google SRE	SRE Book App. D (2016) + SRE Workbook Ch. 10 (2018) [fetched]	The two disagree . Book: 12 top-level + 3 sub, incl. Timeline. Workbook: 9 body sections, adds Recovery Efforts + Glossary, action items gain <code>Priority</code>	Workbook metadata: Owner, Shared with, Status, Incident date, Published	NO — Google's own parser (Requiem) is internal	<code>incident</code>
Runbook	no standards body [inference]	Skelton Thatcher template: 65 headings, 10 H2 [measured, <code>grep -cE '^#{1,4} '</code>]	our invention	NO	<code>ops</code>
Test plan	IEEE 829-2008 superseded by ISO/IEC/IEEE 29119-3:2013	IEEE 829 defines 10 document types , not one	none standardized	NO	<code>test</code>

Types with no canonical form — we would be inventing

TYPE	CANONICAL SOURCE?	WHAT ACTUALLY EXISTS
PRD	NONE	Vendor blog templates only; every "PRD standard" claim is [SS] at best
FRD	NONE	Defence/consulting folklore; overlaps SRS "Specific requirements"
TRD	NONE	Closest standardized neighbour is SRS
SRS	Partial, paywalled — ISO/IEC/IEEE 29148:2018 exists, <code>iso.org</code> returned 403, clause list never opened; IEEE 830-1998 withdrawn	The circulating section list is [SS] from a secondary encyclopedia page citing SWEBOK
Runbook	NONE	Skelton Thatcher template; Google says "playbook", not runbook — the terminology is not shared
User-story wrapper (As a / I want / So that)	NONE — Connextra-format folklore	Gherkin covers acceptance criteria only
Design doc	NONE, by design	Google's article explicitly refuses to prescribe
Company-internal RFC	NONE	Squarespace and Uber are two data points sharing 5 literal strings

Four of the fourteen requested document types have no canonical form at all, and a fifth (SRS) has one nobody could open — so every template we ship for those must carry the label "house convention", never "canonical".

Ship order, because it is the order in which a template can be validated rather than merely rendered: (1) OpenAPI and (2) Gherkin first — both have a fetched, versioned, machine-readable normative artifact. (3) MADR and (4) Keep a Changelog next — no schema, but a fully specified section list and enum from a single primary source. (5) IETF RFC — fully specified but heavy. (6) Google SRE postmortem — ship the **Workbook** variant and label the version, because the two Google templates disagree.

Anti-recommendations, all load-bearing:

- Do **not** label anything "IEEE 829" — superseded, and it defines 10 documents.

- Do **not** publish a 29119-3 or 29148 clause list until someone opens the paywalled text; both are [SS] and both hosts returned 403.
- Do **not** attach YAML frontmatter to `CHANGELOG.md` or to `.feature` files. A `---` fence at line 1 makes a `.feature` file unparseable by every Gherkin implementation; if metadata is needed it must be tags or a sidecar.
- Do **not** silently pick an ordering where sources conflict: Nygard's article orders Title → Context → Decision → Status → Consequences, the widely-copied `joelparkerhenderson` rendering puts Status second [fetched, both]. Render what the file says.
- Do **not** default the runbook template to 65 headings — that is a system-operation manual, not an incident procedure. Default to the 10 H2 skeleton, offer the full set as an expand variant.
- Do **not** bullet-ize a Nygard ADR: the source states bullets are "acceptable only for visual style, not as an excuse for writing sentence fragments" [fetched].
- Do **not** ship a "blameless" lint that flags usernames. Google's own bad-postmortem example is blameful by containing judgement, not by containing names — names appear throughout the good one.
- Do **not** inline a full OpenAPI document inside a design doc; the Google guidance warns such definitions "quickly get out of date". Link it; render it separately.

14.3 Prompts, specs and skills — three artifacts or one?

Decision: one substrate, three lifecycles. Same file skeleton (`---` YAML + CommonMark, closed key set, splice-safe), different required keys, different verifier. Do not merge them into one file type; do not give them three unrelated formats.

Evidence the substrate is already shared [fetched 2026-08-28]:

- VS Code ships **four** sibling markdown-plus-YAML types — `*.instructions.md`, `*.prompt.md`, `*.agent.md`, `SKILL.md` — discovered by the same walker with the same parent-repo rule.
- Microsoft is collapsing prompt files into skills in production: "Agents running on the Agent Host don't use prompt files. To use an existing prompt with the Copilot agent, convert it to an agent skill", behind `chat.customizations.promptMigration.enabled`.
- PromptLayer, a *prompt registry*, now versions **SKILL.md** folders alongside prompt templates (Free 1 collection/30 files, Pro 5/50, Team unlimited/100, hard 5 MiB per file).
- promptfoo, a *prompt test runner*, ships a `skill-used` assertion and `trajectory:tool-used`.

	PROMPT DOC	SPEC DOC	SKILL DOC
Loaded	on invoke, fully	by human/agent on reference	progressively — L1 metadata always, L2 body on trigger, L3 files on demand
Addressed by	name + version	path	<code>description</code> match — a retrieval key, not documentation
Versioned by	semver + eval gate	git history	<code>metadata.version</code> , informal
Tested by	assertions over outputs	review/approval	with-vs-without A/B
Fails by	wrong output	ambiguity	not triggering at all
Cache role	is the cached prefix	is cached <i>content</i>	changes the tool set → invalidates the whole prefix

Design consequences the build team implements directly:

- A skill's `description` is linted for **trigger coverage**; a prompt's `inputs` for **fixture completeness**; a spec's headings for **stable anchors**. Three linters, one parser.
- Closed key set with one `metadata` escape hatch.** [measured, local corpus of 697 `SKILL.md`, 695 parseable] $293/695 = 42.2\%$ carry at least one key outside the 6-field `agentskills.io` spec, and every one **hard-errors** on upload rather than degrading: `Unexpected key(s) in SKILL.md frontmatter: argument-hint. Allowed properties are: allowed-tools, compatibility, description, license, metadata, name`. An open key set silently manufactures unportable documents.
- Cap `description` at **1024** chars, not 1536: the spec caps it at 1024 while the listing truncates `description + when_to_use` at 1,536 — the smaller cap is the portable one. [measured] $17/695 = 2.45\%$ of local files already exceed 1024; resolved length median 379, p90 719, max 1,885.
- Adopt `.prompty` **\$2.7 rich input kinds** (`thread`, `image`, `file`, `audio`) — the four things that break naive `{{var}}` substitution — and make `example` mandatory when `required: true`, because that is what makes a prompt file self-testing.
- Put **version** in the file. Every registry surveyed has a version model and no file format does; Langfuse (integer + labels), Braintrust (hex + environments), LangSmith (commit hash + tags) and PromptLayer (version + release labels) each own it server-side with a different primitive. Adopting one is adopting its lock-in.
- Declare cacheability in the document (`breakpoints`, `ttl`, `min_tokens_assert`), because an under-minimum Anthropic prefix is **silently uncached with no error** — the only signal is `cache_creation_input_tokens == 0 && cache_read_input_tokens == 0`. Minimum cacheable prefixes differ per model: 512 (Opus 5 / Fable 5 / Mythos 5), 1,024 (Opus 4.8, Sonnet 5/4.6/4.5, Opus 4.1/4, Sonnet 4), 2,048 (Mythos Preview, Opus 4.7, Haiku 3.5), 4,096 (Opus 4.6, Opus 4.5, Haiku 4.5); 4 breakpoint slots, 20-block lookback, order `tools → system → messages`. OpenAI: min 1,024 tok (GPT-5.6+) / 2,048 (older), write $1.25\times$ / read $0.1\times$, max 4 cache writes per request, `ttl` sole legal value `30m`. [all fetched]

Anti-recommendations:

- Do not invent a new prompt file extension.** `@prompty/core` gets 755 npm downloads/month against promptfoo's 2,536,364 — [derived] **3,359×** — and `.prompty` self-labels as "v2 Alpha... may change". Two unrelated projects already both claim the name "Promptfile" (132★ / 12 downloads a month). The extension is not the moat.
- Do not add frontmatter to `AGENTS.md`.** It has no schema by design, 23,968★, median local size **145 bytes** across $n=49$ [measured]. The absence of keys is the property that produced the adoption.
- Do not put YAML frontmatter in `llms.txt`.** The spec's only required element is a single H1 and it explicitly permits a leading BOM; a `---` block is a violation, not an extension.
- Do not use full Jinja2.** Prompty published a normative 10-feature subset with `trim_blocks` / `lstrip_blocks` forced off precisely because identical templates rendered differently across runtimes; promptfoo uses **Nunjucks**, not Jinja2, so templates are not portable between the two.
- Do not render user input without pre-render role-marker neutralisation** (Prompty §6.3: a nonce injected at every role boundary plus post-parse validation). Without it, any `{{var}}` containing `user:` on its own line silently forges a message boundary.
- Do not let a prompt document read arbitrary files.** `$$file:` must reject absolute paths, `..` and symlink escapes, and frontmatter must never grant itself additional roots. A prompt document is untrusted content the moment it is shared.
- Do not build a prompt *library* surface before a prompt *test* surface.** Every format surveyed that lacks a testing model (`llms.txt`, `AGENTS.md`, both `Promptfiles`, `.prompty`) is adoption-flat or ships no verifier; every artifact with real volume (promptfoo 2,536,364/mo, LangSmith 26,474,529/mo, Langfuse 27,044,017/mo PyPi) is a testing or versioning surface whose file format is an afterthought.

14.4 The AI build loop and where files beat chat

The six-phase model — named by everyone, persisted by almost no one [derived across 11 systems]

#	PHASE	NAMED BY	PERSISTED AS A FILE
0	Constitution — standing rules that outlive the task	Kiro steering, spec-kit constitution, Cursor/Codex/Devin/Windsurf rules, CLAUDE.md	✓ everywhere
1	Prompt — the raw ask	nobody names it	✗ nowhere
2	Spec — what + why, testable	Kiro requirements.md (EARS), spec-kit spec.md	2 of 11
3	Plan — how, reviewable before edits	Cursor, Cline, Copilot, Jules, Antigravity, Claude, Kiro design.md , spec-kit plan.md	named by 8 of 11, filed by 2
4	Tasks — decomposition with state	Kiro, spec-kit, Roo boomerang	2 of 11
5	Verify — evidence the thing does what \$2 said	Kiro PBT, spec-kit analyze , Antigravity Walkthrough	1 of 11; zero systems produce a signed pass/fail record
6	Memory — what to carry forward	Claude auto-memory, Devin Knowledge, Windsurf Memories, Cline memory-bank	file in 2 of 4; vendor DB in 2

The industry converged on phase names and diverged on persistence — and the phases that evaporate are the prompt, the plan, and the verification, which are exactly the three a team needs six months later.

What evaporates, with the vendor’s own words [fetched 2026-08-28]

ARTIFACT	WHERE IT GOES
Cursor plans	"Plans are saved by default in your home directory. Click 'Save to workspace'..." — not the repo. Checkpoints "stored locally and are separate from git"
Jules plans	reviewed in-session and auto-approved on a timer if you navigate away
Roo subtask context	"only this summary returns to the parent"
Antigravity Implementation Plan / Walkthrough	sidebar review pane; the Walkthrough is explicitly "a concise summary... to remind the user of what has happened" — a reminder, not a record; no documented on-disk path
Windsurf Memories	auto-generated, ~/.codeium/windsurf/memories/ , machine-local, per-workspace, "not committed to your repository"; the Devin Local agent "does not persist memories" at all
Devin Knowledge / Copilot Memory	vendor DB and managed service respectively — not in the repo
Claude Code plan	a permission mode in the Shift+Tab cycle, session-scoped, no file. Checkpoints deleted at 30d

Measured on this machine, one repo (~/Desktop/GitHub/frontmatter) [measured/derived]

QUANTITY	VALUE
Session transcripts	267 .jsonl
Transcript bytes	427,336 KB total – 20 KB memory dir = 427,316 KB
Durable auto-memory	20 KB
Ratio ephemeral : durable	427,316 ÷ 20 = 21,365.8 : 1
ExitPlanMode invocations across all 267 transcripts	0 (542 of 562 raw string hits are subagent tool-exclusion lists)
Human-authored .md tracked in the repo	128

[derived] Instruction-file adoption ÷ spec-file adoption ≈ 858,112 : 30,592 = 28.05 : 1. The industry has agreed on how to instruct an agent and has not agreed on how to record what it did.

The literal opening. Even spec-kit, the most file-native system in the set, encodes lifecycle state as prose bold-keys, not frontmatter: spec-template.md , plan-template.md , constitution-template.md and checklist-template.md all begin with # ... and carry **Status**: Draft as body text; only tasks-template.md opens with --- [measured, 5 templates fetched and first-line-tested]. The state machine exists and is unaddressable.

What we build. A loop document family under .frontmatter/loop/<id>/ — prompt.md , spec.md , plan.md , tasks.md , verify.md , plus constitution.md at vault root — sharing one frontmatter key set: id , kind , state ∈ {draft, proposed, approved, superseded, abandoned} , loop , derives_from: [<doc-id>@<sha>] , base_sha , supersedes / superseded_by , actor {human|agent, tool, model, session_ref, prompt_digest} , approved {by, at, mode: explicit|timer|policy} , verified {by, on, until, method, result: pass|fail|partial} , stale_after . Transitions are splice-enforced: any edit to an upstream document flips every downstream document to stale and appends the diff to its derives_from ledger — Kiro’s manual "Sync Files" button made automatic and auditable. Terminal states are superseded (new file plus LATEST pointer) and abandoned ; nothing is deleted, which supplies the back-edge Kiro states it does not have ("Can I switch workflows after starting a spec? No ... create a new Feature Spec" [fetched]).

Build order, and why:

- verify.md first, not spec.md . Spec and plan have two credible incumbents; verification has zero durable implementations across 11 systems [derived]. Its body is a table of requirement IDs × evidence, each row {claim, command, exit_code, output_digest, at} — information Kiro’s PBT and spec-kit’s /analyze already produce and discard into a pane.
- prompt.md second. The prompt is the only artifact literally no system persists, and it is the cheapest to capture.
- Make derives_from: <id>@<sha> mandatory and refuse the write without it. This is the structural fix for the staleness class Kiro papers over with a Sync button and Cline papers over with "update memory bank".
- Render, don’t relocate. Keep .kiro/specs/ and specs/NNN-*/ where they are and project frontmatter over them. Migration is the reason spec tools die.
- Record approved.mode including timer and policy , so a rubber-stamped plan is visibly rubber-stamped.
- Emit AGENTS.md and SKILL.md so the loop is legible to Claude Code, Codex, Devin, Jules, Cursor and Windsurf with no adapter.

Anti-recommendations:

- Do not build plan mode. Eight of 11 systems have it; it is table stakes and it lives in the IDE’s input loop, which we do not own.
- Do not build a rules/instructions file. AGENTS.md ≈858K and CLAUDE.md 774,144 — that fight is over.
- Do not invent an extension gate. Cursor’s own docs concede plain .md is silently ignored inside .cursor/rules ; that friction is a documented resentment, not a moat.
- Do not store lifecycle state as prose bold-keys. That is precisely the defect being exploited.
- Do not replicate memory-bank’s six-file taxonomy — 2,728 adoption against ≈858,112 for AGENTS.md. The shape did not win.

- **Do not require a running daemon.** Statelessness is the differentiator against the CRDT competitor.

Absorption risk, stated honestly. The loop *documents* will be absorbed — Kiro ships three and spec-kit six, and spec-kit was pushed 2026-08-28. Cursor is one config flag from durable plans; assume it lands within 12 months and do not build the business on plan capture. What no single IDE can absorb: (a) the **cross-vendor projection**, because no vendor will make its memory readable by a competitor — Windsurf memories are machine-local, Devin Knowledge is a vendor DB, Copilot Memory is a managed service, and a team on all three today has three incompatible stores and zero shared verification record; (b) the **byte-exact write guarantee** on files the vendor also edits; (c) the **verification record**, because a vendor certifying its own agent's output is the second-opinion-from-the-first-opinion problem in commercial form. Build on those three; treat `spec.md` and `plan.md` as commodity surfaces we render, not products we sell.

14.5 What conformance we can honestly claim

MECHANISM	PROVES	CANNOT PROVE
Schema validation (JSON Schema / ajv)	Structural + per-field constraint conformance of an instance	Cross-field arithmetic, temporal or business invariants. [measured] ajv 6.15.0, schema <code>{total≥0, discount≥0}</code> , 3 payloads → 3 schema-valid, 1 of 3 valid while violating the English sentence "a discount must never exceed the order total"
Property-based testing (Hypothesis 8,918★, fast-check 5,121★)	Falsification — a minimised counterexample to a stated invariant	Absence of counterexamples ≠ proof; the property itself is unverified against the English spec
OpenAPI conformance fuzzing (Schemathesis 3,565★)	Responses deviate from the schema; 5xx on generated inputs. 8 fuzzers × 16 services: "the only one to handle more than two-thirds of our target services without a fatal internal error", 1.4×–4.5× more unique defects than second-best [fetched arXiv 2112.10328v1]	That the schema <i>is</i> the spec
Consumer-driven contracts (Pact)	Provider responses satisfy each identified consumer's recorded expectation	Verbatim scope limits: "Pact is about checking the contents and format of requests and responses"; "Pact does not test the side effects of a request"; explicit not-good-for list includes public APIs and load testing [fetched]
Mutation testing (Stryker 3,060★, PIT 1,856★)	Your test suite detects seeded faults	Spec conformance at all — it grades tests, not code-vs-spec; equivalent mutants: "the only solution is by finding these by hand" [fetched]
Model checking (TLA+ 3,019★, Alloy 863★)	A <i>design</i> satisfies stated invariants in a bounded state space	Newcombe et al., AWS, verbatim: "How do we know that the executable code correctly implements the verified design?" — "The answer is that we don't." [fetched PDF]
Tag-based traceability (OpenFastTrace 163★)	Every normative item has a marker of the required artifact type; stale links detected (Orphaned / Outdated / Predated / Covered-*)	That the marked code <i>implements</i> the item
LLM-as-judge	A cheap, correlated, non-deterministic opinion	A verdict stable under reordering, prompt length, or rerun

Nothing measured in this survey takes a natural-language requirement plus a codebase and returns a sound verdict; every deployed system reduces the problem to a human having written a test, a formal property, or a tag — the human is always the oracle.

Supporting numbers, and one recorded disagreement:

- LLM judges, sources disagree: MT-Bench reports strong judges reach "over 80% agreement, the same level of agreement between humans" [fetched arXiv 2306.05685v4]; Judging the Judges (13 judges × 9 exam-takers) finds them "still quite far behind inter-human agreement", off by "up to 5 points", with leniency bias [fetched arXiv 2406.12624v6]. Both stand. Code-specific evidence is worse: 8 LLMs over **1,405 Java methods** and **1,281 Python functions** — "even the best-performing LLM frequently misjudges the correctness of the code" [fetched arXiv 2507.16587v1]; CodeJudgeBench, **26 judge models** — "all models still exhibit significant randomness", and "simply changing the order in which responses are presented can substantially impact accuracy" [fetched arXiv 2507.10535v2].
- The test oracle itself is weak: SWE-Bench+ found **32.67%** of successful patches involved solution leakage and **31.08%** passed on weak tests; filtering both dropped SWE-Agent+GPT-4 from **12.47% → 3.97%** [fetched arXiv 2410.06992v2], [derived] **8.50 absolute points**, a **3.14×** reduction. A second study found **7.8%** of patches counted correct while failing the developer-written suite and net **6.2 absolute points** of inflation [fetched arXiv 2503.15223v2].
- Formal methods lite is a *design* verifier: [derived] AWS's six published TLA+/PlusCal specifications total **3,031 lines** (804 + 645 + 939 + 102 + 223 + 318) and found roughly ten design bugs, with the code-to-design link stated as unproven by the authors.
- Automated link recovery is not gate-grade: SpecMap reaches "up to **73.3%** file mapping accuracy" [fetched arXiv 2601.11688v1]; NL-PL traceability gains over SOTA are **+3.68%** (HGT) and **+8.84%** (Gemini 2.5 Pro) F1 across 12 projects [fetched arXiv 2509.05585v1].
- The market has let this category die twice [fetched]: **Optic** (spec-vs-runtime drift) `archived=true`, last push 2026-01-08, "Optic Labs is now part of Atlassian" — [derived] 233 days stale; **Dredd** (OpenAPI-to-implementation conformance) `archived=true`, last push 2024-05-11 — [derived] 840 days stale. [derived] 2 of 18 repos queried are archived = **11.11%**, and both are in the spec-conformance category specifically, while every property-testing, schema-validation, mutation-testing and traceability repo queried was pushed within 79 days. [derived] npm last-week: ajv 378,773,799 ÷ fast-check 37,512,910 = **10.10×**; fast-check ÷ @stryker-mutator/core 2,320,443 = **16.17×**; ajv ÷ @pact-foundation/pact 600,095 = **631.19×**. The world buys shape validation, sparingly buys falsification, and has twice refused to buy conformance.

What frontmatter ships — all document-side, all provable from the file alone:

CAPABILITY	MECHANISM	HONEST LABEL
Revision-pinned requirement identity	Adopt OpenFastTrace <code>id=revision</code> semantics; editing an item's normative text bumps the revision and voids every outstanding coverage claim . OFT's guide calls this "one of the most useful safeguards in OFT" [fetched]	requirement identity
Content-derived anchors as the durable ID	Our existing anchors resolve at 99.627% with 0.050% false over 41,642 block-versions [measured, internal] — the durable <code>REQ-XXX.Y.Z</code>	stable anchor
Normative-vs-informative segmentation	We own the AST; classify and count normative items deterministically instead of an LLM inferring them by keyword the way spec-kit <code>analyze</code> does	normative item count
Coverage matrix	Emit "N of M normative items have ≥1 declared covering artifact; K links are Outdated; J are Orphaned"	link coverage — never conformance
Drift display	Anchor resolution failure is a deterministic staleness signal. No LLM, no false confidence	stale link
EARS-shaped criteria linting	<code>While <pre-condition>, when <trigger>, the <system name> shall <system response></code> — first published 2009, users named include Airbus, Bosch, Dyson, Honeywell, Intel, NASA, Rolls-Royce, Siemens [fetched]	syntax lint
Change-impact broadcast	On a revision bump, emit the closure of dependent items, tests and previously-linked artifacts	impact set
Machine-checkable extraction	Where a requirement contains a schema, table, enum or numeric bound, extract it into a JSON Schema or test fixture the code side runs	generated gate

The division is: **documents generate gates; gates decide**. The local precedent is already in this repo — `specs/harness/` holds 5 `.mjs` gates that exit non-zero, e.g. `import-boundary-report.mjs` "Fails (exit 1) if any source file still references `@/server` , `@/Lib` , or `@/components *` [measured].

Anti-recommendations — what would be over-claiming:

- **Do not claim "verifies that code matches the spec."** AWS, with 3,031 lines of formal spec and a model checker, states outright it does not know that. A markdown tool claiming it is claiming more than TLA+.
- **Do not ship an LLM conformance verdict as a gate.** If a judge ships at all, it ships its own kappa and precision against a human gold set, and it is labelled triage.
- **Do not present link coverage as conformance.** "100% of requirements are linked" is compatible with 0% of them being implemented. Say so in the UI copy, not the footnotes.
- **Do not build automated spec→code link inference as a truth source.** 73.3% file-level accuracy means roughly one file in four is mis-assigned; as a suggestion with human confirmation it is useful, as a matrix cell it is a fabricated audit trail.
- **Do not market to DO-178C / IEC 62304 / ISO 26262 compliance.** Those clause texts are paywalled and were not opened [SS]; the cited FDA baseline — "General Principles of Software Validation", January 2002, Docket FDA-1997-D-0029, status Final [fetched] — is [derived] **24 years** old; and tool qualification in a regulated toolchain is a formal evidenced process. This is the highest-liability over-claim available to us.
- **Do not build a test runner, mutation engine, or coverage tool.** Those categories are saturated and healthy. The two archived repos are exactly the conformance category — that is a warning, not a vacancy.
- **Do not equate a green LLM convergence report with verification.** spec-kit's `converge` loops an LLM to a fixpoint and reports  "Converged"; its stopping condition is the judge's own opinion [fetched].
- **Do not claim novelty for the traceability matrix.** It is 2002-era regulated practice with live OSS implementations (OpenFastTrace 163★ GPL-3.0, StrictDoc 370★, Sphinx-Needs 299★, Doorstop — all pushed 2026-08-28). The credible claim is *ergonomics and freshness in Markdown*, not invention.

What would falsify the conformance position: if a published, reproducible method takes an English requirement plus a codebase and returns a verdict whose agreement with human review is measured above inter-human agreement and is stable under response reordering, then the document-side-only boundary is too conservative and we should revisit. Until then, we ship identity, coverage, freshness and extraction — and we let a gate that exits non-zero make every decision.

§15 The internal system, productised

QUESTION	VERDICT	FALSIFIED BY
Sell the orchestrator (AIOS) as its own product?	No. The substrate has zero tenancy fields in the trace ledger and 2 of 151 files containing any HTTP-listener code [measured]	≥1 tenant field in the schema, ≥1 non-self paying customer, and ≥10,000 chained ledger rows — all three, not any one
Embed part of it in frontmatter?	Yes — the artifact-facing half. Ship what is about <i>the document</i> ; keep what is about <i>the agent</i>	A user-visible feature whose value survives without a claim about how the AI learns
Keep the rest internal?	Yes. It is real QA value aimed at frontmatter's own code, and a trust liability aimed at users	An internal loop that has actually fired, been attributed, and been calibrated against human labels

The whole productisation reduces to one filter: an internal asset ships if its output is a fact about the user's file, and stays internal if its output is a claim about the software's own intelligence.

15.1 What already runs, measured

All counts read from live files on this machine on 2026-08-29, read-only.

STORE / SURFACE	LIVE COUNT	TAG
~/ .sgnk/bin/ entries	149 → 147 executables (79 .sh + 66 .py + 2 extensionless; AGENTS.md and __pycache__ are not tools)	[measured] / [derived: 79+66+2=147]
~/ .sgnk/bin/ total bash+python lines	21,481	[measured]
— of which self-declare READ-ONLY / OFFLINE / PROPOSES-ONLY / RETRIEVAL-ONLY	32 of 147	[measured, grep]
~/ .sgnk/gates/ scripts	69 = 33 assert-* + 34 break-* + 2 harness meta	[measured]
~/ .claude/skills-src/ SKILL.md	124 across 27 category dirs	[measured]
~/ .claude/skills/ SKILL.md (incl. plugin-vendored)	309	[measured]
settings.json hooks	28 across 9 events (SessionStart 9, UserPromptSubmit 5, PreToolUse 6, PostToolUse 3, Stop 2, SubagentStop 1, SessionEnd 1, PreCompact 1)	[derived: 9+5+6+3+2+1+1+1=28]
traces/ daily ledger	63 .jsonl files, 5,018 rows, 30 keys/row	[measured]
— accepted non-null	181 / 5,018 = 3.607%	[derived]
— skill: "unknown"	3,946 / 5,018 = 78.637%	[derived]
state/complexity-gate-log.jsonl	24,669 rows, 2026-07-17T21:01:09Z → 2026-08-28T20:20:22Z = 42 days = 587.357 rows/day	[measured] / [derived]
— floor / strong	95.1194% / 4.8806% (23,465 / 1,204)	[derived]
— single / workflow	99.1163% / 0.8837% (24,451 / 218)	[derived]
— rule2_gated: true	10,191 = 41.311%	[derived]
— floor and guarded	38.068% (9,391 / 24,669)	[derived]
state/ total entries	43,703 (21,679 .gate-tier.json + 15,758 .turn-meta.json + 6,144 .count = 43,581 of them)	[measured] / [derived]
Machine-fed stores	propensity 44,279 · subagent-reconcile 29,081 · routing-journal 26,037 · trifecta-decisions 8,088 · session-evals-rc 6,476 · injection-hits 1,707	[measured]
Human-fed stores	assertions 725 · trigger-log 388 · spawn-log 255 · PREFERENCE-LOG.jsonl 232 · self-heal 139 · routing-shadow 83 · ledger-chain 59 · regression-gates 40 · casebank-ops 5 · prereg 2 · shadow-log 1	[measured]
PREFERENCE-LOG.jsonl composition	188 accepted / 44 rejected = 81.034% / 18.966%; 223/232 = 96.121% carry no skill attribution	[derived]
regression-gates.jsonl	40 rows, 40 proven_nonvacuous: true, last registered 2026-08-13T01:41:14Z — 16 days stale	[measured] / [derived]
assertions.jsonl	725 rows: 386 true / 339 false = 53.2414% pass; test:py 393 · build:js 201 · test:js 75 · typecheck:ts 26 · lint:js 16 · typecheck:py 8 · lint:py 6	[derived] / [measured]
injection-hits.jsonl	1,707 — Bash 1,667 · WebFetch 34 · WebSearch 6	[measured]
calibration.json (2026-08-28T20:10:09Z)	low n=14 rate 0.0 · mid n=0 · high n=26 rate 0.269 vs a 0.20 bar → CALIBRATION_DRIFT condition met	[measured] / [derived]
skill-health.json (2026-08-28T18:27:54Z, 7-day window)	131 tracked: active 0, dormant 5, dead 116, infrastructure 10 — contradicted by 446 trace rows for sgnk-drift-watch in the same window	[measured]
mdmax engine	13 files / 3,614 lines; 15 targets, 19 constructs; uncertifiableShare() = 8 of 15 (53.3%) not locally probeable; benchId 51947c2e88127bfcd80125c705404f0b9f4c54165b95c8cbd4f19fdfe071d360 ; fold mdmax/fold@1	[measured]
mdmax live run on docs/mdmap/MAP.md	24 blocks × 7 local targets = 168 cells: PASS 104 · STRIP 2 · CORRUPT 13 · VOID 49	[measured]
mdmax histogram over docs/mdmap	21 files, 517 blocks, 259 with BROKEN = 50.10%, 21/21 files affected	[measured]
Other markdown assets	knowledge/ 5 scripts / 709 lines, 203 notes, 0 errors, 94 warnings; sgnk-campaign/ 16 py files / 5,807 lines (7 generators, 6 gates, 3 infra); docs/mdmap 21 .md / 0 lines of code; fm preview/ 21 files / 2,306 lines; fm export/ 5 files / 646 lines; graphify 0.7.9, 3,848 nodes / 3,737 links	[measured]
npm names	mdmax 404 (free) · frontmatter 200 (taken) · aios 200 (taken) · sgnk 404 · frontmatter-cert 404 · gray-matter 8,989,723 weekly downloads (2026-08-21→27)	[fetched]

Disagreements, recorded not reconciled:

- bin/ count: 149 entries under one exclusion rule vs 151 non-__pycache__ files / 133 executable under another [both measured]. Neither is wrong. Do not publish a single number without stating the rule.
- Floor rate: CLAUDE.md LR#35 says 90% / 10% at n=2,064 (2026-07-28); the master plan §4 L2 says 90% floor at 23,778; live is 95.1194% at 24,669 [derived]. Do not quote "90% floor" anywhere externally — re-derive at write time.
- Trace-ledger schema: a 31-field row in one report vs a 32-key union across rows ranging 16–30 keys in another [measured]. The rows are sparse JSONL; "the schema" is not a fixed row shape.
- baselines/ ; 2,382 vs 2,381 [both measured] — one file's drift between two reads on the same day.
- "110 daily trace files" appears in three prior grounding docs. It counts 46 .lock files and a steps/ subdir. The daily ledger is 63 files. Any deck quoting 110 is quoting lockfiles.
- Engine count in mdmax: 6 engine implementations pinned (unified 11.0.5, react-markdown 10.1.0, marked 16.4.2, markdown-it 15.0.0 ×2 configs, commonmark 0.31.2, kramdown 2.5.2 + kramdown-parser-gfm 1.1.0) across 7 local targets of 15 declared [measured]. Where product copy says "7 engines" it means 7 local targets.

The single structural fact this section establishes: machine-fed stores all exceed 1,000 rows; every human-fed store is under 250 — roughly 176,000 machine rows against 232 preference rows [derived]. Build nothing whose value depends on the user rating something.

15.2 Internal asset to product feature

Lift: S ≤1 week · M 2–6 weeks · L ≥1 quarter.

INTERNAL ASSET	WHAT IT DOES TODAY	PRODUCT FEATURE	LIFT	D2C OR B2B OR INTERNAL
Trace ledger, 11 tools, 5,018 rows, <code>files_touched</code> + <code>files_sha256</code> + <code>prompt-to-line.py</code> [measured]	One <code>gen_ai.*</code> -mapped row per task, machine-local	AI link : per-document <code>.frontmatter/trace.jsonl</code> sidebar; hover any paragraph for <code>{contributor, model, promptDigest, sessionRef, kept/reverted}</code>	M	BOTH
<code>ledger-chain.jsonl</code> , 59 rows [measured]	Hash chain over ledger entries	Tamper-evident AI-edit export : the review artifact a compliance reader accepts	M	B2B
33 <code>assert-*</code> + 34 <code>break-*</code> + <code>regression-gates.jsonl</code> 40/40 <code>proven_nonvacuous</code> [measured]	Every registered check has a recorded command that makes it fail	Proven-non-vacuous badge : a check with no falsification proof renders as <i>unproven</i> , not green	S	BOTH
F4 assertion producers (8 tools), <code>assertions.jsonl</code> 725 rows [measured]	Wrap any command, log pass/fail, composite-veto verdict	<code>npx mdmax cert</code> + GitHub Action + <code>--fail-on=BROKEN</code> : docs fail the build when they render broken for a named consumer	S	B2B
mdmax, 13 files / 3,614 lines, 15 targets × 19 constructs, <code>mdmax/fold@1</code> [measured]	4-verdict differential across engines; missing engine = hard refusal	The degradation certificate itself — the one capability with no competitor in the surveyed field	S (shipped)	B2B
Complexity gate, 24,669 rows, 95.1194% floor [derived]	5-D verdict routes each call to the cheapest sufficient model	Visible AI meter in currency + per-action line ("small model · 0.4¢ · [redo on the large model]")	S	D2C surface, B2B policy
<code>escalation-ladder.sh</code> , 7 deterministic rungs [measured]	<code>sonnet/medium</code> → <code>sonnet/high</code> → <code>opus/medium</code> → <code>opus/high</code> → <code>opus/xhigh</code> → <code>opus/max</code> → <code>fable/high</code>	"Try harder" button: one rung up, cost delta shown before it runs, both outputs diffed	S	D2C
F7 reward mining, <code>accepted_asis</code> / <code>edited_kept</code> / <code>abandoned</code> survival verdicts [measured]	Mines git + edit distance for whether an AI edit survived	Edit-survival telemetry — never a rating prompt; local-first, visible, deletable, exportable	M	D2C
<code>PREFERENCE-LOG.jsonl</code> , 232 rows [measured]	Accept/reject pairs, 96.121% unattributed	"What I've learned about your edits" panel: per-row delete, one-click export, global off switch	S log / M panel	D2C
F11 drift-watch + 2,382 baselines [measured]	Baseline → compare → alert with dedup and severity	Doc Health , deterministic checks only: broken link, broken anchor, duplicate heading, expired <code>verified.until</code> , unreviewed AI edit, cited file newer than the doc	M	BOTH
F10 session lifecycle (9 tools) + snapshot cards <code>@0-KEY.md...@6-conversation.md</code> + <code>GLOBAL-REGISTRY.md</code> 58 repo rows [measured]	Snapshot-on-exit, KEY on start, PreCompact preservation	Sessions as documents + ChatGPT/Claude ZIP importer where the <i>verification report</i> is the demo	L	D2C
11 weekly <code>rules-hygiene-*.md</code> reports; <code>MEMORY.md</code> index shape [measured]	Report-only; the rules file is never auto-edited	Memory as an ordinary markdown file the user opens, edits, deletes — plus a hygiene <i>review prompt</i> , never an auto-prune	S file / M hygiene	BOTH
<code>aio-s-library</code> , 14 docs-as-skills [measured]	A doc declares the conditions under which it should be read	Docs that route themselves : a frontmatter key registering trigger conditions	M	BOTH
<code>freeze</code> / <code>unfreeze</code> / <code>guard</code> skills [measured]	Breadth-limit an agent's write authority	Document freeze : lock a section; AI and collaborators both bounce off it	S	B2B
<code>knowledge/scripts/validate.py</code> : 6 required keys, 5 enums, 14 <code>item_type</code> values, 203 notes / 0 errors / 94 warnings [measured]	Hard vs soft findings, exit-coded	<code>fm lint --schema</code> : user-declarable frontmatter contract, validated in-editor and in CI	M	B2B
<code>build_catalog.py</code> :146 <code>replace_block</code> , <code><!-- CATEGORIES:START -->...END</code> [measured]	Regenerates blocks inside a hand-edited file	<code><!-- fm:generated:x --></code> regions : byte-stable, survive concurrent human edits, refuse on drift	M	BOTH
<code>apply-patches.py</code> : NOT FOUND / AMBIGUOUS / NO-OP refused, APPLIED re-read from disk [measured]	Four-verdict find/replace applier	The refusal contract for every AI edit and toolbar transform — this is the splice writer's missing verdict vocabulary	S	INTERNAL → BOTH
6 campaign gates: <code>check-render.py</code> (COLLISION/EDGE/CLIPPED), <code>check-overflow.py</code> , <code>check-clarity.py</code> [measured]	Measure <i>rendered</i> pixels, not source opinions	Export gates on frontmatter's own PDF/HTML path, which today has none	M	B2B
<code>check-provenance.py</code> hard rule [measured]	Every numeric token in the derived artifact must appear in a source	Numeric-provenance linter over any transform: excerpt, summary, share snapshot, AI rewrite	S	BOTH
Red-proof evidence discipline (<code>MEASURED</code> / <code>FETCHED</code> / <code>[SIMULATED]</code>) [measured]	Convention in prose, enforced by nobody	Provenance chips as a rendered, lintable inline tag with <code>{value, source, tier, re_verify_cmd}</code>	M chips / L execution	BOTH
<code>scan_paste.py</code> 44-extension triage + <code>type-handlers.md</code> 14-format extraction matrix with an honest <code>fidelity</code> field [measured]	Classifies an arbitrary folder tree	Vault import triage : drag a folder in, get a typed plan before anything is written	M	D2C
<code>incident-to-eval.py</code> + <code>regression-synth.py</code> + <code>sgnk-regression-gate.sh</code> [measured]	Turns a defect into a permanent, registered check	Incident → permanent check block with an executable <code>broke:</code> field — a document primitive no competing editor has	M block / L CI	B2B
<code>sync_to_md.py</code> link rewriter + <code>slug-decision-audit.mjs</code> [measured]	The only working slug↔title link rewriter here; the audit measures anchor swing but applies nothing	<code>fm rewrite --dialect obsidian github commonmark</code> with splice-level byte-identity proof on untouched bytes	M	BOTH
<code>injection-hits.jsonl</code> , 1,707 rows [measured]	Counts injection-shaped patterns per tool	Counted paste/import risk log with a per-hit reason — never a green "you are safe" shield	S	B2B
F1 bandit (13 tools), F3 test-time compute (9), <code>debate/consensus/reflexion</code> , <code>skill-health.json</code> , 5 <code>UserPromptSubmit</code> hooks, raw <code>traces/</code> store, <code>explore-budget.sh</code>	See 15.4	None	—	INTERNAL

Ship order, by evidence quality: memory-as-a-file (S) → cost meter (S) → "Try harder" (S) → proven-non-vacuous badge (S) → `mdmax cert` CI (S) → trace provenance (M) → provenance chips (M) → Doc Health checks (M) → regression blocks, B2B (M) → schema lint (M) → importer (L).

Anti-recommendation: do not ship any composite score — vault health, document grade, routing confidence, quality number. Every scalar in this stack is contradicted by at least one other ledger in the same stack: `skill-health.json` says **active 0 of 131** while the trace ledger records **446 rows** of that same window's activity [measured]. A user-facing scalar hides that disagreement instead of resolving it.

15.3 User-authorable automations as markdown documents

CONSTRAINT	MEASURED / FETCHED VALUE	DESIGN CONSEQUENCE
Official SKILL.md spec keys	6: name , description , license , compatibility , metadata , allowed-tools [fetched]	Editor writes only these
Keys in use locally	20 distinct across 124 files [measured]	14 of them are decoration
Files that would fail packaging today	54 / 124 = 43.5% [derived: 124 - 70 spec-clean]	Non-spec keys are a hard error , not a warning: Unexpected key(s) in SKILL.md frontmatter: argument-hint. Allowed properties are: allowed-tools, compatibility, description, license, metadata, name [fetched]
Inert keys with zero readers anywhere	preamble-tier 28 files, gbrain 6, benefits-from 4, triggers 36, version 36 [measured]	No free-form frontmatter — comments belong in the body
The one home-grown key with a real consumer	capabilities → sgkn-skill-track.sh:48 [measured]	Ship it as a logged product feature , not as a key users maintain
Hard caps	name ≤64, description ≤1024, compatibility ≤500 [fetched]	Enforce at the keystroke with a live counter, not at export
Over-cap files locally	17 / 124; sgkn-mobbin at 1,885 chars is visibly truncated with ... in this session's own listing [measured]	The failure is live, not theoretical
Listing budget	description + when_to_use truncated at 1,536 combined chars; total listing budget = 1% of the model context window; least-invoked skills lose their descriptions first [fetched]	Put the <i>when</i> before the <i>what</i> ; surface a listing-budget meter; cap library size per user
Body size	n=124, min 36, median 266, max 3,058 lines [measured]	Recommend 266 lines as the split-to- references/ threshold
Negative scoping	70 / 124 descriptions carry NOT for: naming the sibling to use instead [measured]	The editor generates this clause; it is the only collision-prevention mechanism in a large listing
Inline triggers	21 / 124 open with a literal Trigger: list, while the separate triggers: key is inert [measured]	Triggers go in prose, in description
Shell injection	0 / 124 use the !`cmd` syntax [measured]	Refusing it in imported documents costs this corpus nothing and closes arbitrary execution on open
Invocation locks	11 / 124 set disable-model-invocation: true , and every one is destructive or outward(sgkn-approve , sgkn-md-update , ship , land-and-deploy , deploy-to-vercel , vercel-cli-with-tokens) [measured]	That ratio is the design pattern: outward or destructive ⇒ never auto-fires
Name/dir integrity	3 name ≠ parent-directory violations locally, which the spec forbids [measured]	Rename key and folder atomically
Tool-grant lifetime	allowed-tools is experimental, support varies between implementations, and Claude Code clears the grant on the user's next message [fetched]	The UI must say "for this turn"; the host permission system is the boundary, never the key

The authoring model, three tiers:

- **Typed by the user (2 keys + 1 selector):** name , description , and a *Run mode* selector. Run mode compiles to disable-model-invocation: true for the local runtime and is **dropped on export**, because that key is a Claude Code extension, not spec [fetched].
- **Written by the editor, never hand-typed:** allowed-tools (space-separated, generated from checkboxes — the spec says space-separated; 37 local files use **YAML-list form** and 4 use **commas** [measured], and only the space form survives export), metadata , compatibility .
- **Refused by the editor:** every other key. This removes all 54 export-failures, all 17 cap violations, and all 3 name/dir violations at once.

Three gates a non-programmer can run:

1. **Lint** — 6 keys, two length caps, the ^[a-z0-9]+(-[a-z0-9]+)*\$ name regex, name == dirname. Deterministic, zero model calls.
2. **Trigger test** — the author writes 3 phrases that should fire it and 3 that should not; the harness checks selection against the *whole* existing listing. This is the only failure mode an author cannot self-diagnose, because it depends on everyone else's descriptions rather than on theirs.
3. **Dry run** — execute under a read-only tool grant and show the transcript. Never a first run with write tools.

Sharing is file-out, file-in, peer-to-peer. Export a directory (SKILL.md + optional scripts/ references/ assets/) as a zip; on import, disclose the allowed-tools grant and every path the body references, and default imported automations to manual-only for the first N runs. Imported document text is data, never instruction — a shared SKILL.md is untrusted third-party content, and body text asserting authority or pre-authorization is the primary attack surface the moment sharing exists.

Anti-recommendations — each names the mechanism that converts a document format into the banned marketplace:

- No central index, gallery, or search over other users' automations. Discovery-of-strangers'-code *is* the marketplace. Local symlink promotion (~/.claude/skills/<name> → skills-src/<category>/<name> , no build step [measured]) is the correct ceiling.
- No install-by-identifier. The moment a user obtains an automation without reading it, a package manager has replaced the document.
- No versioning, upgrade, or dependency resolution. 36/124 files carry version: and nothing reads it [measured] — making it functional means resolvers, lockfiles, and transitive trust.
- No requires / parent / composes-with typed edges. A dependency graph needs a registry to resolve it.
- No ratings, download counts, verification badges, or "featured" surface. Reputation primitives only make sense when the population is strangers.
- No executable payload by default. The corpus already ships 119 committed .pyc files [measured] — unreadable by the person consenting. If scripts/ is supported at all, it is opt-in per import, per file, source shown.
- No paid tiers or entitlement fields in metadata , which the spec explicitly offers for "entitlement or catalog fields" [fetched] — an invitation to decline.

The line to hold: a user may write an automation, run it, and hand the file to a specific person; a user may not browse, install, rate, depend on, or update someone else's. [inference]

15.4 What must stay internal, and why

ASSET	LIVE STATE [MEASURED]	WHY IT IS A TRUST BURDEN IF SHIPPED
Learned-rules ledger (RULE 1–8 + numbered rules)	74 rules / 892 lines in the live CLAUDE.md; the latest of 11 weekly hygiene reports parsed 69, so the report is 5 rules stale [derived]; 69/69 carry zero citations; a second count puts it at 76 — recorded, not reconciled	The user-facing sentence is “the AI keeps a private file of rules about you, and its own auditor is out of date about what is in it.” A ledger whose every entry is uncited cannot be defended when a user disputes one, and no demand signal for it exists anywhere in the record. Ship memory as a file the user owns and can delete (15.2) — never an agent-authored ruleset about the user.
The bandit's learning claim (F1, 13 tools)	10 arms; 3 of 10 carry "seeded": "offline-0.5x" — synthetic, not live evidence; largest arm is literally named __unattributed__ (opus a=24.1 b=5.9 → posterior mean 0.80333 [derived]); shadow-log.jsonl 1 row; prereg.jsonl 2 rows; routing-shadow.jsonl 83 rows; 96.121% of preference rows carry no skill attribution	"It learns which model suits you" is a claim with no attributable evidence behind it, and the record contains three documented incidents of prematurely calling this loop live (LR#60/#62/#63). The moment a user asks "learned from what?", the honest answer is an arm named __unattributed__. The negative class is worse: the system's own epoch entry measures precision[rejected] = 0/8 = 0.000 — every rejection it has ever emitted was wrong — so acting on it is worse than acting on nothing. Ship the deterministic 7-rung ladder as a button; ship zero learning claims.
Debate panels and multi-agent adjudication (sgnk-consensus,sgnk-debate-panel,sgnk-reflexion-step,bon-*x5,poll-aggregate,codex-judge,red-team,sgnk-fault-inject)	Excellent against frontmatter's own code; ~/sgnk/insights/ does not exist after 14 months — infrastructure built and then never fed is this system's documented failure mode	Every one of these multiplies model calls per user action, which is COGS a bundled-inference consumer tier cannot absorb, and each adds a latency and an explanation the user did not ask for. Worse, LR#16 already holds that debate convergence is not correctness — so a converged panel cannot authorise anything, which means the user is paying for deliberation that carries no authority.
Rawtraces/ as a data store	30 keys per row over a 32-key union, rows ranging 16–30 keys, including cwd — absolute filesystem paths into client repositories; zero tenancy fields (no tenant / org / user / account / workspace / team key exists across 45 sampled rows)	Publishing or syncing this store exfiltrates client identity through directory names before it delivers a single feature, and there is no column to scope a redaction to. Ship the schema and the per-document sidecar; the machine-local store never leaves the machine. Retrofitting tenancy later means re-keying 24,669 gate rows, 44,279 propensity rows, 26,037 routing-journal rows and 2,382 baseline files that are currently addressed by filesystem path and machine-local session UUID.
skill-health.json as a dashboard	131 tracked: active 0, dormant 5, dead 116, infrastructure 10 — while the trace ledger shows 446 rows for one of the "dead" skills in the same 7-day window	Correct internally (dormant ≠ dead, LR#51) and catastrophic as a screen a customer sees about their own automations: it tells a paying user that 116 of the 131 things they built are dead, on the strength of a metric another ledger in the same system contradicts.
The 5 UserPromptSubmit interruption hooks (sgnk-nudge,sgnk-reward-gold-nudge,sgnk-skill-suggest,sgnk-digression-guard,sgnk-pref-capture)	5 of 5 prompt-submit hooks are interruption hooks;sgnk-digression-guard has 0 *.contract.json files and 0 trace rows across all 63 trace files — it has never fired here	This is the ambient-writing-coach pattern the master plan §10 already bans, and 28 hooks in an editor is 28 chances to interrupt a sentence. A loop that has never fired cannot be productised honestly, and a background model call judging a user's prose against a goal it inferred is both a cost line and a paternalism problem. Ship a brief: key the user writes themselves, checked on request.
break-*.sh as a user-runnable action	34 scripts that deliberately break the thing so the paired assert must go red	Ship the property — a check is displayed as unproven until a paired failing fixture is recorded — never the button. A user-triggered "break my document" action is a data-loss vector.
Exploration routing on customer work (explore-budget.sh)	Auto-disarms internally; gated OFF for PO and client tasks	Randomly routing a paying user's document to a non-default model to gather counterfactual signal is indefensible, whatever it is called in the UI.
Trifecta / injection detector as a badge	trifecta-decisions.jsonl 8,088 rows, E=true P=true U=true on 8,088/8,088 = 100% [derived]	A detector that has never once said "no" cannot back a security badge. Ship the counted hit log (1,707 rows, Bash 1,667 · WebFetch 34 · WebSearch 6) with a per-hit reason; never a green shield.
Any judged quality score	sgnk-complexity-gate eval: n=20, kappa 1.0, gold_status: "synthetic-unverified", meets_kappa_bar: false, self-labelled "IN-DISTRIBUTION / TAUTOLOGICAL"; adversarial re-run n=16, kappa_AB 0.863; calibration's top bucket corrects at 0.269 against a 0.20 bar [derived]	A number the user cannot dispute, backed by gold the file itself calls tautological. Ship binary gates with the failing lines linked (LR#4), and if a judge is ever used, its precision / recall / kappa ship with every verdict (LR#5). Never a Likert writing score — schema-rejected internally at every layer.

15.5 Is the orchestration layer itself sellable?

Verdict: no. Do not build, package, or sell AIOS as an observability, routing, eval, or agent-orchestration product. Embed the five document-facing hooks named in 15.2 and interoperate with the incumbents by emitting OTEL to their endpoints.

VENDOR	ENTRY PAID TIER	FREE TIER	TAG
Langfuse	Core \$29/mo (100k units, +\$8/100k); Pro \$199/mo; Teams add-on \$300/mo; Enterprise \$2,499/mo	Hobby, 50,000 units/mo	[fetched]
Laminar	Starter \$30/mo (3GB, +\$2/GB); Pro \$150/mo (10GB, 6-mo retention)	1GB, 7-day, 1 seat	[fetched]
LangSmith	Plus \$39/seat/mo (10k base traces); LCU \$1.50, LSU \$1.00	Developer \$0, 1 seat, 5,000 traces	[fetched]
Portkey	Production \$49/mo (100k logs, +\$9/100k)	10k logs/mo	[fetched]
Arize AX	Pro \$50/mo (50k spans, 10GB, 30-day)	25k spans, 1GB, 15-day	[fetched]
W&B Weave	Pro from \$60/mo	\$0/mo	[fetched]
Helicone	Pro \$79/mo; Team \$799/mo	10k requests, 1GB, 1 seat	[fetched]
Galileo	Pro \$100/mo (billed yearly)	\$0/mo, unlimited users	[fetched]
Braintrust	Pro \$249/mo (\$249 credits, 5GB, 50k scores, 30-day)	Starter \$0 (\$10 credits, 1GB, 10k scores, 14-day)	[fetched]
HoneyHive · Maxim · AgentOps	no price extractable — JS-rendered pages; unmeasured ≠ absent	—	[fetched]

- Median entry paid tier = \$50/mo. Values [29, 30, 39, 49, 50, 60, 79, 100, 249], n=9, median = 5th value [derived].
- 9 of 9 priced vendors ship a free tier [derived from the table]. Price floor is \$29/mo against a \$0 self-hostable option at 33,864 stars.
- Scale check 1. The entire lifetime complexity-gate log is 24,669 rows. Langfuse's free tier includes 50,000 units/month. 24,669 ÷ 50,000 = 0.49338 — 49.338% of one month of one vendor's free allowance [derived].
- Scale check 2. The entire trace ledger is 5,018 rows. LangSmith Developer (free) includes 5,000 traces/month. 5,018 ÷ 5,000 = 1.0036 — the whole ledger is one month of a free tier [derived].

- **Distribution gap.** LiteLLM 57,495 stars; Langfuse 33,864; Portkey gateway 12,845; Phoenix 11,227. npm last-month: `langsmith` **26,474,529**, `@langfuse/core` **8,735,327**, `langfuse` **8,003,800**, `braintrust` **5,951,758** [fetched].
- **The model vendor has absorbed four of the six candidate capabilities** [fetched, docs.claude.com]: `org / group / member spend limits` with daily CSV export; `claude.ai/analytics/claude-code` plus a Claude Code Analytics API and an Enterprise Analytics API; `CLAUDE_CODE_ENABLE_TELEMETRY=1` exporting OTEL metrics, logs and traces (beta) to any OTLP endpoint; `GET /v1/skills` with `source=custom|anthropic` and **Versions**; and a multi-tenant nav listing Managed Agents, Environments, Sessions, Deployments, Vaults, Memory Stores, Workspaces, Service Accounts, Federation, Rate Limits and Tunnels. Their published benchmark is **~\$13 per developer per active day, \$150–250 per developer per month, below \$30 per active day for 90% of users**.

The strongest argument against this verdict, stated at full strength. The compliance wedge is separable, has a regulatory forcing function, and embedding it caps its addressable market. Anthropic's analytics give *aggregates* — spend per user, per model, CSV, daily [fetched]. They do not give **byte-level attribution of which bytes in a repository a model wrote**, nor a tamper-evident chain over that record. Both primitives already exist here: `files_sha256` and `files_touched` in the trace row, and `ledger-chain.jsonl` as a hash chain [measured]. **No vendor in the eleven-row table above sells "prove which bytes an agent wrote, and prove the log was not edited"** — that is a real, named, unoccupied gap in a well-funded field, and an enterprise forced to answer *"which of these commits are model-authored, under which policy, and can you prove the log is intact"* will pay more than \$249/mo and will not care that the answer arrives inside a markdown editor. Embedding it therefore shrinks the buyer set from *every regulated engineering org* to *frontmatter users*. That is the honest case for selling it separately, and it is not weak.

Why the verdict survives it. `ledger-chain.jsonl` holds **59 rows** — a prototype, against a free tier that ingests 50,000 units a month [measured/derived]. There are **zero tenancy fields** in the schema, and a compliance buyer's first question is "scoped to which org" [measured]. **2 of 151 files contain any HTTP-listener code**; this is a filesystem-coupled single-operator rig, not a service [measured]. The compliance checklist — SOC-2 Type II, HIPAA, SAML SSO, RBAC, audit logs, data residency, DPA, BAA, uptime SLA — appears in the Langfuse, Helicone, Braintrust, Arize and Portkey feature tables and in none of ours [fetched]. And the precedent is inside our own record: `shadow-log.jsonl` = 1 row, `~/sgnk/insights/` absent after 14 months [measured]. Building infrastructure and then not feeding it is this system's documented failure mode, and a compliance SaaS is the most expensive possible instance of it.

Falsification bar, in Rule #32 shape. Reopen this decision only when all three hold: ≥1 tenant/workspace field in the trace schema, ≥1 non-self paying customer, ≥10,000 chained ledger rows. Until then the compliance product is a hypothesis, not a roadmap item.

Recommendations, with their anti-recommendations:

- **Ship the artifact-conformance slice as the product surface** — `npx mdmax cert`, `--fail-on=BROKEN`, GitHub Action, priced as document CI. Position against nobody, because every vendor above instruments *the call* — prompts, tokens, latency, spans, scores — and **none certifies the artifact against downstream renderers** [inference from the eleven pricing and feature pages opened]. **Anti-recommendation:** do not price or position it as observability; that puts it beside a \$29 floor with a free self-hostable alternative.
- **Register `mdmax` on npm now** — `registry.npmjs.org/mdmax` returns **404** [fetched]. **Anti-recommendation:** do not claim `frontmatter` as an npm name; it returns **200** and is taken [fetched].
- **Add a tenant/workspace field to the trace schema now if the compliance wedge is even a maybe.** It is cheap today — the rows are sparse JSONL already varying 16–30 keys [measured] — and expensive after 100,000 rows.
- **Emit OTEL from frontmatter's agent layer to the customer's existing endpoint.** Interoperate with Langfuse, Phoenix, Braintrust; do not compete with them. **Anti-recommendation:** do not build a storage or trace-viewing UI of our own.
- **Keep the complexity gate as an unmarketed internal COGS mechanism.** Its value is margin on frontmatter's own AI calls, and it is what makes a bundled-inference consumer tier survivable — not a SKU. **Anti-recommendation:** do not sell model routing or a gateway; LiteLLM is free at 57,495 stars and Portkey bundles it at \$49/mo [fetched/measured].
- **AIOS's real contribution to the shippable slice is discipline, not code** — fail-first gate validation, floors rather than equality pins, preflight-the-environment, producer/consumer field-name contracts. That discipline is *why* a green certificate can be trusted, and that is the product claim [inference].

§16 Feature inventory and the honest v1

16.1 The completeness matrix

Legend: IN PRD = named in `docs/FRONTMATTER-PRD-2026-08-29.md`; CODE = present in `src/` today; parity keys OB Obsidian · TY Typora · iA iA Writer · BE Bear · HM HackMD · DM Docmost · SY SiYuan · AF AppFlowy · OU Outline. * = parity asserted without opening the source `INFERENCE`.

Search and navigation

FEATURE	IN PRD	CODE	NEAREST PARITY	DEMAND EVIDENCE	VERDICT
Full-text vault search	Yes (§14 Home band 1)	Yes, MiniSearch	Universal	—	shipped
Search operators (<code>path:</code> , <code>file:</code> , <code>OR</code> , quoted phrase, exclusion)	No	No	OB ships them as a core-plugin doc <code>FETCHED</code>	OB documents a whole operator language <code>FETCHED</code>	v1, build
Saved searches	No (0 term hits)	No (<code>savedSearch</code> 0 hits) <code>MEASURED</code>	OB: a search is one of 7 bookmarkable types <code>FETCHED</code>	—	v1, build
Bookmarks / pins	1 mention	Yes (<code>editor/bookmarks</code>)	OB groups + reorder <code>FETCHED</code> ; BE pin-notes <code>FETCHED</code>	—	v2
Command palette, <code># / @ / :</code> goto	Yes §13 L0, §14.1	Yes	TY "Open Quickly" <code>Cmd+Shift+O</code> <code>FETCHED</code>	—	shipped
Backlinks + unlinked mentions	1 passing mention	Yes (<code>preview</code> , 21 files)	OB two collapsible sections, sort + filter <code>FETCHED</code>	OB frames unlinked mentions as "discover links you aren't aware of" <code>FETCHED</code>	v1, document
Sort + group (name, modified, created, frontmatter field)	No (0 hits)	No (only <code>tree-order.ts</code>) <code>MEASURED</code>	OB Bases sort/group/filter <code>FETCHED</code>	—	v1, build
Tag browse / rename / merge	17 mentions, no tag <i>operations</i>	Partial	BE devotes 5 FAQ pages to tags alone <code>FETCHED</code>	—	v1 browse, v2 rename
Graph view	Yes (§16, 3 files)	Yes	OB, SY, Logseq *	3,848 nodes / 3,737 edges internally <code>MEASURED</code>	v2 polish

Editing mechanics

FEATURE	IN PRD	CODE	NEAREST PARITY	DEMAND EVIDENCE	VERDICT
Undo/redo across mode switch	Only as a <i>risk</i> (likelihood 3 / impact 5) MEASURED	Yes (<code>historyKeymap</code>)	Universal	Named in the risk register, not the feature list MEASURED	v1, harden + name
Find/replace in file	No (0 hits)	Yes (<code>searchKeymap</code> , <code>openSearchPanel</code>)	TY <code>Ctrl+H</code> class FETCHED	SY 3, AF 4 issues+PRs titled "find and replace"; Logseq 0 MEASURED	v1, document
Find/replace across vault	No	No (0 hits)	SY has it as an <i>open ask</i> MEASURED	An open request inside a mature competitor is the cleanest demand signal available MEASURED	v2 (L — must route through splice + one review surface)
Complete + remappable keyboard map	5 keys in §14.1; <code>shortcut / keyboard</code> = 0 elsewhere MEASURED	Partial	TY publishes ~40 keys plus "Change Shortcut Keys" FETCHED	OU: 25 issues+PRs titled "keyboard shortcut" MEASURED	v1 map, v2 remap
Table keybindings (row/cell select, insert, delete, align)	<code>table</code> ×30	Partial	TY <code>Ctrl+L</code> / <code>Ctrl+E</code> / <code>Ctrl+Shift+Backspace</code> ; iA Smart MD Tables; BE tables FAQ FETCHED	Three independent products ship dedicated table keys FETCHED	v1
Multi-cursor, folding, zen	Yes §13 L0	Yes	iA ships Folding and Dynamic Outline Windows only FETCHED	iA shipping folding on one platform is evidence it is hard, not unwanted INFERENCE	v1
Autosave + crash recovery, as a named promise	No (0 hits)	Yes (IndexedDB + localStorage dirty index) MEASURED	Universal *	—	v1, name it
Paste as markdown / plain	Yes	Partial	TY <code>Cmd+Shift+V</code> , <code>Cmd+Shift+C</code> FETCHED	—	v1
Snippets, reopen-closed-file	No	No	TY <code>Ctrl+Shift+T</code> FETCHED	Weak	v2

Files, safety, export, media, craft, platform

FEATURE	IN PRD	CODE	NEAREST PARITY	DEMAND EVIDENCE	VERDICT
Trash + restore	No (0 hits)	Yes (<code>TrashModal</code> , <code>api/vault/delete</code> , <code>api/vault/restore</code>)	OB, BE ✱	Delete-with-no-undo is a top churn cause INFERENCE	v1, document
Multi-select bulk move/delete/set-field	"bulk" ×2, unspecified	No — <code>multiSelect</code> / <code>selectedFiles</code> / <code>bulk</code> = 0 hits in <code>src</code> MEASURED	BE documents ⌘-click and two-finger swipe select FETCHED	BE documents selection gestures purely to enable bulk export FETCHED	v1, build
Rename with link rewriting	<code>rename</code> ×3	Partial (<code>repository/create-rename-merge</code>)	OB ✱	—	v1
Duplicate file (<code>Cmd+Shift+S</code> class)	No	No	TY "Save As / Duplicate" FETCHED	—	v1
Whole-vault backup	No	git, for GitHub-backed vaults only	BE dedicated backup-restore FAQ FETCHED	—	v1 — explain, do not build
Encryption / per-note lock	1 mention (BYO-key); <code>E2EE</code> 0	No	BE per-note password + Face/Touch ID, "we cannot see or reset it", gated behind Pro FETCHED	BE monetises it FETCHED	v2
Conflict handling	Yes, strong (conflict inbox, <code>merge3</code> , TO gate)	Yes	BE conflicted-notes FAQ; HM avoids it via realtime FETCHED	—	v1
Supported file types, published	Not enumerated	Partial	OB enumerates <code>.md</code> <code>.base</code> <code>.canvas</code> , 8 image, 6 audio, 5 video, pdf FETCHED	OB treats the list as a support contract FETCHED	v1, publish the list
PDF fidelity (paper size, page breaks, header/footer, metadata)	<code>print-CSS</code> + <code>pdf-doc.ts</code> ; options unspecified	Partial	TY exposes every one plus a LaTeX/Pandoc engine FETCHED	OU 14 issues+PRs titled "export pdf", top hit <i>"Images are missing in pdf export"</i> ; DM 64 titled "export" MEASURED	v1
Print (<code>Cmd+P</code>)	1 mention	No	TY prints via the PDF pipeline FETCHED	—	v1
DOCX / EPUB / RTF / ODT / LaTeX	No	No	TY ships 16 targets; BE gates <code>html/docx/pdf/jpg/epub</code> behind Pro FETCHED	Two products make the export matrix the paid tier FETCHED	v2 (shell to Pandoc)
Image paste / drag-drop upload	No	Yes (<code>api/vault/upload</code>)	TY, HM FETCHED	Universal FETCHED	v1, document
Attachment path policy (relative, <code>./</code> / prefix, escaping)	No	Implicit	TY needed 8 sub-settings FETCHED	Exactly where portability breaks INFERENCE	v1 — one rule, no dial
Word count / read time	No	Yes (<code>N words</code> · <code>N min</code> , <code>ceil(words/200)</code>)	OB core plugin, CJK-aware FETCHED	OB shipped CJK counting specifically FETCHED	v1, + CJK
Spellcheck	No	Yes (browser-native toggle)	iA: system on Mac/iOS, Hunspell on Windows FETCHED	—	v1 native only
Focus mode / typewriter	<code>typewriter</code> ×1, <code>focus mode</code> 0	Yes, shipped toggle MEASURED	iA Focus Mode on all platforms — its oldest differentiator FETCHED	—	v1, document
Authorship / AI-vs-human provenance	Yes §10.3, §13 L3	Partial	iA Writer alone : AI text dimmed, other-human underlined FETCHED	The only competitor analogue FETCHED	v1
Tabs / split panes	<code>tabs</code> 3 / <code>split</code> 5	Partial	TY New Tab is macOS-only FETCHED	—	v1
Mobile editing	1 mention, inside one T1 roadmap cell MEASURED	No	OB marks help pages <code>mobile: true</code> per feature; iA has a full iPhone/iPad column; BE ships iOS shortcuts, widgets, Siri; AF iOS+Android; HM mobile modes FETCHED	Every product opened has a first-class mobile story FETCHED	v1: read + light edit PWA
Offline	7 mentions; T6	Partial	AF 8 issues+PRs titled "offline"; AFFiNE 16, top hit <i>"chore(electron): remove offline mode"</i> MEASURED	Sources disagree: AFFiNE <i>removing</i> an offline mode is a counter-signal, recorded not smoothed MEASURED	v1 local vault only
Real-time multiplayer	Deliberately deferred to T6	No	HM and DM both lead their docs with it FETCHED	—	never in v1
Accessibility (screen reader, focus order, contrast)	0 mentions MEASURED	Unknown	Unverified for all competitors	Legal exposure selling to EU/US orgs INFERENCE	v1 baseline
Localisation	0 mentions MEASURED	No	iA UI in 10 languages; SiYuan README in 4 FETCHED	Selling globally from India with an English-only UI INFERENCE	v2

Two ground-truth caveats the build team must resolve before trusting line-level references above: the code audit recorded `git rev-parse HEAD = 5e0d5a585cf0090451c809c1e2781a5f9a5a587b` while a same-window audit recorded HEAD unchanged at `d714fb5f0305ea1e5f8612b43e271d3b4e95fe54` MEASURED, DISAGREEMENT PRESERVED ; and GitHub `search/issues` totals include pull requests and closed items, so every count above is a ceiling, not an issue count MEASURED .

16.2 Missing entirely — what users expect and we had not listed

Ordered by churn risk INFERENCE , each anchored to a fact in 16.1.

#	GAP	WHY IT IS A GAP
1	Mobile	One sub-clause in one roadmap cell against a universal competitor platform matrix FETCHED
2	Vault-wide find and replace	Zero in PRD, zero in code; an open request even inside SiYuan MEASURED
3	Keyboard shortcut map	5 keys specified; Typora publishes ~40 with a remap path FETCHED
4	Search operators	Obsidian ships <code>path: / file: / OR /quoted/escaped as core</code> FETCHED
5	Saved searches	Obsidian makes a search a bookmarkable object FETCHED
6	Sort and grouping controls	No spec, no code MEASURED
7	Bulk / multi-select	Zero code hits MEASURED
8	Shipped-but-undocumented set: trash, restore, word count, read time, spellcheck, in-file replace, image paste, focus mode, autosave	Unowned features rot; a feature nobody wrote down is a feature nobody regression-tests INFERENCE
9	Attachment path policy	The most common cause of a vault that stops rendering elsewhere; Typora needed 8 settings for it FETCHED
10	PDF fidelity contract	Outline's top PDF issue is missing images; Docmost's is attachment paths MEASURED
11	Print	1 mention, no implementation MEASURED
12	Accessibility	0 mentions, and the only gap here carrying legal exposure INFERENCE
13	Localisation	0 mentions while selling globally MEASURED
14	Tag management (rename, merge, nested)	tags x17, tag <i>operations</i> 0 MEASURED
15	Encryption / note lock	Bear monetises exactly this FETCHED
16	DOCX export	Bear and Typora both gate it as paid FETCHED
17	Published supported-file-type list	Obsidian treats it as a support contract FETCHED
18	Duplicate file, reopen closed file	Trivial, universally expected FETCHED

16.3 What no competitor has — the real differentiators

DIFFERENTIATOR	NEAREST THING ANYONE ELSE HAS	DISTANCE
Byte-preserving splice edits — every mutation is a span replacement, never a re-serialisation	Hubble.md regenerates the whole body, deletes reference links along with their visible text, and is not even a fixed point MEASURED , \$15	Categorical. No opened competitor claims byte preservation
Cross-engine degradation certification (<code>mdmax cert</code> , <code>--fail-on=BROKEN</code>)	None found in any opened source	Categorical
Typed evidence tiers <code>{value, source, tier, re_verify_cmd}</code> with a working linter	iA Authorship dims AI text and underlines other-human text FETCHED — presentation, not typed data	iA proves the need; nobody types it
Machine-write zones — fenced regions an agent may rewrite and outside which it may not, splice-enforced	Obsidian Bases writes derived <i>views</i> , never guarded regions FETCHED	Categorical
Document CI — 34 assert/break gate pairs, 725 assertions, 40 regression gates MEASURED	GitBook has publishing CI * ; nobody gates document <i>content</i>	Near-categorical
Every view is a deterministic reversible projection of one file	Obsidian Bases stores views in <code>.base</code> files or embedded code blocks FETCHED — still a second artifact	Bases is the competitive answer. Ship before it becomes the default expectation
Token-budget meter per agent-facing file	None found	Categorical
Staleness / drift detection over 2,382 baseline files, >15% alert	None found	Categorical
Section-level restore with no silent expiry	JetBrains' 5-day wipe is the named anti-pattern	Strong
Automations authored as documents (124 <code>SKILL.md</code> files)	Plugin marketplaces everywhere; nobody makes the automation format <i>be</i> the document format	Categorical — and it survives the settled no-marketplace constraint

16.4 The honest v1 minimum

Tier 0 — already built, must be named, documented and regression-tested (near-zero cost): undo/redo · in-file find/replace · word count and read time · spellcheck · trash and restore · image paste and drop upload · templates and daily notes · backlinks and unlinked mentions · command palette · four view modes · focus mode · HTML and PDF export.

Tier 1 — must build, in this order:

#	ITEM	SIZE
0	Fix the three self-contradictions first (§7.2): <code>mdmax/</code> is imported by zero product files while <code>docs/mdmax/PLAN.md</code> says "shipped"; there is no CI in a product that sells document CI; the shipped design system is not our design system MEASURED	—
1	Search operators (<code>path:</code> , <code>file:</code> , OR , quoted, exclude)	S
2	Saved searches, including bookmarking a search	S
3	Sort and group controls on tree and Home	S
4	Multi-select → move / delete / set frontmatter field, routed through splice and the single review surface	M
5	Complete keyboard map, published as a <code>.md</code> file inside the vault	S
6	Table keybindings (row/cell select, insert, delete, align)	M
7	Attachment path policy: one rule, relative, <code>./</code> -prefixed, documented	M
8	PDF and print fidelity, plus an images-present regression test	M
9	Mobile read + light edit on the existing PWA	L
10	Rename with link rewriting	M
11	Autosave and crash-recovery promise, stated	S
12	Accessibility baseline: keyboard-only reachability, focus order, contrast	M
13	Supported-file-type list, published as a contract	S

Any one of the three self-contradictions in row 0, found by a first customer, costs more than every feature below it.

Explicitly out of v1, and say so publicly: real-time multiplayer · comments · canvas · encryption · DOCX/EPUB · grammar check · writing goals · localisation · graph-view polish.

Anti-recommendations, each a founder-visible refusal:

- **No canvas or whiteboard.** Obsidian needed a new file format (`.canvas` , JSON Canvas) to do it FETCHED . A canvas is a tree-of-record by another name and breaks the settled "the file is the only source of truth" constraint.
- **No fuzzy duplicate detection.** Exact-hash duplicate *listing* is fine; similarity scoring is a probabilistic verdict on a user's own writing, inside a product whose pitch is determinism.
- **No own spelling dictionary.** iA bundles Hunspell only because Windows lacks a system service FETCHED .
- **No real-time multiplayer in v1.** It costs a CRDT layer that fights byte-preserving splices head-on, and CRDT sync is settled out.
- **No settings dial for anything with a defensible default** — reveal policy, attachment paths, export margins.
- **Do not publish a fidelity number in marketing before R0 lands** (§24 R0) MEASURED , and **do not claim offline-first before the T0 gate passes** — two-device offline-edit convergence, zero loss, watched by a user MEASURED .

Falsification: if a paying user in the first 20 cancels citing an item marked v2 or "never" above, that row is mis-tiered and the tiering must be re-run against churn reasons rather than parity counts.

16.5 Reconciling a full feature set with a simple surface

The constraint governs **surface area, not capability**. Four disposal routes; every v1 item lands in exactly one, and an item with no route does not ship.

ROUTE	RULE	WHAT SHIPS HERE, HIDDEN
Visible — ≤7 permanent affordances	Earns pixels only if a first-time user needs it in session one	Four modes + <code>Cmd+E</code> · search field · file tree · right rail (Properties first) · sync chip · the three AI verbs (Accept / Discard / Try again)
Invoked — palette-only, zero chrome	Discoverable by typing, invisible otherwise. <code>Cmd+K</code> is the settings menu we never draw	Saved searches · sort and group · bulk operations · rename-with-rewrite · duplicate · reopen-closed · export formats · print · vault-wide replace (v2) · every keyboard command
Ambient — status bar or hover only	One line, no panel, no dot until it means something	Word count and read time ($N_{\text{words}} \cdot N_{\text{min}}$) · doc-health count, which opens a panel only when non-green · token-budget meter · staleness · autosave state
Contextual — appears on the object, dies with it	Never persistent	Table controls on caret-in-table · image resize and align on selection · attachment actions on the embed · AI prompt at the cursor (<code>Space</code>) · per-hunk accept

Specific hidings, each a decision with its anti-recommendation attached:

- **Search operators are typed, never a filter-builder.** No advanced-search modal, ever. Obsidian teaches them in one doc page FETCHED .
- **Bulk operations get no toolbar.** Multi-select in the tree plus `Cmd+K` . A bulk-action bar is the fastest route to looking like the project-management tool we settled against being.
- **Attachment path policy is a decision, not a setting.** Typora needed 8 sub-settings FETCHED ; we ship one rule, with the escape hatch in the versioned settings *file*, not the UI.
- **The existing toggles list is the ceiling for the entire preferences surface** — spellcheck, focus mode, line numbers, vim, ghost text already share one list MEASURED . Anything that wants an eighth toggle must displace one of the seven.
- **Accessibility is invisible by construction** — zero surface cost, and the only v1 item carrying legal exposure.
- **Mobile is a different surface, not a shrunken one.** Read, capture, light edit. Do not port the rail.
- **The keyboard map ships as a `.md` file in the vault** — the product documenting itself in its own format, with zero UI.

§17 First run, activation and retention

Activation = the first projection-mediated write to a Markdown file that frontmatter did not create. One event, not two-part, measured against a 7-day window because 7 is the mode of time-bounded milestones in the only benchmark opened DERIVED FROM FETCHED MODE=7 .

Why this act and not another:

PROPERTY	BASIS
It is the product thesis in a single act — the user edited through a projection and watched their own file change correctly	INFERENCE
It structurally entails the retention substrate: a file we did not create means the user pointed at their own corpus, and corpus presence is what makes session 2 happen	INFERENCE
It is movable by a solo founder with first-run copy, picker ordering and empty-state wording	INFERENCE
It can happen inside the first minute — milestones that take three sessions cannot be moved by design changes	FETCHED: "TOO LATE" FAILURE MODE
It has a falsifiable 2× test	FETCHED

Rejected candidates and the named failure mode each hits: app launched or vault opened — "too early", the canonical mistake [FETCHED](#) ; first note created in frontmatter — measures the sample, not the corpus [FETCHED+INFERENCE](#) ; ran a degradation certification — proves engine quality, not personal value [INFERENCE](#) ; published a site — "too late" [FETCHED](#) ; N files across M sessions — "too complicated" and unmovable [FETCHED](#) .

Benchmarks, and why they do not transfer unmodified. Single opened source: Lenny Rachitsky and Yuri Timen, "What is a good activation rate", 2022-10-25, n=500+ self-reported products [FETCHED](#) .

METRIC	VALUE
Definition	<code>activated / users who completed signup flow</code>
All products	mean 34%, median 25%
SaaS only	mean 36%, median 30%
"GOOD" / "GREAT"	60th / 80th percentile
Time-bounded milestones	~6% of products; median window 10 days, mode 7 days
Two-part milestones	~10%
Valid-milestone threshold	activated users must retain ≥2× non-activated

frontmatter has no signup, so the benchmark's denominator does not exist; substituting first launch makes the denominator strictly larger
[DERIVED: SIGNUP-COMPLETERS ≤ FIRST-LAUNCHERS, SO ACTIVATED/FIRST-LAUNCHERS ≤ ACTIVATED/SIGNUP-COMPLETERS](#) . **Anti-recommendation: never publish or target "30% activation" for frontmatter — report** first-run completion rate **and** day-7 return rate **under those names**. Two further guardrails: the per-type "B2B prosumer SaaS" figure exists in that source only as an image and was never read [MEASURED](#) ; and no published quantitative abandonment study for note-taking apps was found at all, so any churn percentage quoted in marketing is quoting nothing [MEASURED](#) . The circulating claim about what some share of Obsidian downloaders never do is refuted at source — [obsidian.md](#) and [obsidian.md/about](#) publish no download, user or install count, and the help corpus has no usage-statistics page [FETCHED](#) ; it must not appear in any deck, page or PRD revision.

First-run flow, designed against the evidence. No email field, no OAuth wall, no network call required to reach step 5.

#	STEP	BASIS
1	Launch lands on a folder picker with "Open an existing folder" first and visually primary , "Start a sandbox" second. Obsidian orders these the other way because its vault model forces it; we have no such constraint	FETCHED: OB OFFERS EXACTLY TWO CHOICES, "CREATE NEW VAULT" / "OPEN FOLDER AS VAULT"
2	No walkthrough, no tour, no "What's New" modal	FETCHED: NN/G - TUTORIALS "INTERRUPT USERS, DON'T NECESSARILY IMPROVE TASK PERFORMANCE, AND ARE QUICKLY FORGOTTEN"; PUSH REVELATIONS "DRAMATICALLY OVERUSED"
3	On folder open, show what was found as a status line — file count, which projections are available, which are not and why. Never show a definitive empty state while the scan is still running	FETCHED: NN/G NAMES THIS THE MOST DAMAGING EMPTY-STATE PATTERN
4	Every projection's empty state carries its own pull revelation: the board on a corpus with no status keys names the key it looks for, shows the exact line it would add, and offers one button that adds it to one file	FETCHED: NN/G GUIDELINES 2 AND 3
5	The activating act is a one-line in-place edit through a projection on a file the user already owns — drag one card, write one key. The write is a splice, the diff is shown, undo restores byte-identity	FETCHED: THE PRAISED ARCGIS COUNTER-EXAMPLE REQUIRED THE USER TO *PERFORM* THE TASK
6	Show the byte-level diff of the very first write, once, inline — then never unprompted again	INFERENCE
7	Sandbox is a real folder of real <code>.md</code> files on disk, disposable, opened on demand from a persistent Help affordance, never auto-launched	FETCHED: OBSIDIAN SANDBOX VAULT + LINEAR'S RESET-ON-REFRESH DEMO BOTH SATISFY THE NN/G CRITERIA - THREE UNRELATED SOURCES CONVERGING
8	Segment implicitly by which door they opened, as Linear does with its admin/member and team-size splits. Corpus shape is the signal: dated files → calendar first; status-keyed files → board first; neither → reader	FETCHED + INFERENCE
9	Ship a copy-pasteable agent prompt so an AI coding agent can perform setup	FETCHED: VERCEL'S 3-STEP GETTING-STARTED DOES EXACTLY THIS
10	One first-run friction to design out explicitly: in Obsidian, clicking a folder then "new note" creates the note in the vault root , not the selected folder	FETCHED, OBSERVED: ARXIV:2509.20187V1, N=7, 2025-09-24

Retention mechanics, and the banned list:

- **Correctness is the retention mechanic.** One silent corruption ends the relationship permanently.
- **Templates are organization starters, not decoration** — the strongest supported recommendation here, because two unrelated sources land on it: the survey verbatim *"Metric spiked after introducing templates to choose from in the onboarding flow"* and the paper's *"starting from a 'blank page' might be overwhelming"* FETCHED x2.
- **Serve the retrieval strategy** — the paper's central finding is that intended retrieval determines creation and organization behaviour, so search must be instant and exact-match-first FETCHED.
- **Let the content carry the cadence.** A calendar over the user's dated files pulls on their deadlines; a board pulls on their work. This is the opposite of a streak, which manufactures a cadence the content does not have.
- Near-zero re-entry cost: reopen exactly where the file was, no login, no sync spinner.
- Reversibility receipts on demand — surface the degradation certificate before a risky act (first write to a foreign vault, first cross-engine round-trip), never on a schedule.

Banned outright: streaks · badges · XP or levels · "you haven't opened X in N days" · red-dot unread counts · weekly digest (there is no email field, and adding one to enable a digest would be the single worst trade in this document) · note-count or graph-node vanity displays · first-launch tour · "What's New" modal FETCHED: NN/G NAMES THIS PATTERN AS HARMFUL · artificial trial countdown · referral loop.

Instrumentation under implicit-telemetry-only — no identifiers, no event stream, no phone-home per action:

SIGNAL	SOURCE	ANSWERS	CANNOT ANSWER
Version-check requests per day per version	The update check the app already makes	Install-base trend; day-N return as checks on day N+1 . . N+7 / checks on day N	Anything per-user
.frontmatter/state.json , plain text, never transmitted	Local, user-deletable, documented	Everything for that user, and only when they paste it into a bug report	Any aggregate
Degradation certificates attached to bug reports	User-initiated	Which foreign vaults break, which projections were reached	Frequency — reporters self-select
Generator marker in published site output	The projection already writes the file	Count of installs reaching the furthest downstream act	Who, or how many tried
Docs page-hit ratios between consecutive pages	Docs server logs	Where the documented flow leaks	Whether they are users
Refunds and non-renewals	Billing	The only hard churn number available	Cause

Test the 2× requirement as a **release** experiment, not a user experiment: change one first-run element, hold everything else, compare day-7 return between version cohorts FETCHED + INFERENCE. Anti-recommendation: do not add an anonymous opt-in analytics prompt at first run — it is a consent modal in the exact slot the empty-state research says destroys trust, it violates implicit-only, and the version-check denominator already yields the one ratio that matters.

§18 Simplicity as an engineering discipline

Simplicity here is a set of numbers with owners and CI gates, not a taste claim. The governing constraint is the projection law: the file is the only source of truth, every view is a deterministic reversible projection owning no state — which means simplicity is enforceable, because anything that would need its own configuration surface is already forbidden by the architecture.

RULE	STATEMENT	ENFORCED BY
S-1	Feature count is not the variable; at-rest visible controls (V0) and concepts-to-learn (C) are	§18.2 counters, CI
S-2	Disclosure depth never exceeds 2 levels [fetched: NN/g, <i>Progressive Disclosure</i> , pub. 2006-12-03 — "designs that go beyond 2 disclosure levels typically have low usability because users often get lost"]	§18.3 depth walk
S-3	Exactly one descent path per command; exactly one for destructive commands [fetched: NN/g — "it's rarely a good idea to offer multiple ways to progress to secondary options"]	P script, CI
S-4	A projection is offered iff the file's own bytes entail it. Never predicted, never configured	§18.5
S-5	Zero-config correctness is a simplicity metric, not an engine metric	Z , foreign-corpus gate
S-6	Latency is a correctness property with a p99 budget, not a polish item	§18.4 budget table

A **nine-control window over a file the engine refuses to open is not simple, it is broken** — today the engine refuses 6,613 of 6,614 foreign files on zero-indent block sequences, 83% aggregate [project-measured, `docs/FRONTMATTER-MASTER-PLAN-2026-08-28.md`:135; not re-verified in this pass]. Fix Z Z before any surface work.

18.1 The mechanisms for hiding depth, and what each costs

#	MECHANISM	WHAT IT BUYS	WHAT IT COSTS	WHEN IT BREAKS	FRONTMATTER
M1	Sensible defaults — ship the answer, not the question	Zero decisions before first value; the config surface stops being a surface	Every default is a bet that is wrong for some cohort; changing it later breaks muscle memory	When a default rewrites bytes. A formatting default that mutates untouched bytes is not a preference, it is data loss [inference, grounded in the byte-preserving splice constraint]	Use
M2	Progressive disclosure — core visible, advanced one level down	Improves 3 of 5 usability components: learnability, efficiency, error rate [fetched, NN/g]	Two hard requirements: the right split, and one obvious strongly-scented path down [fetched]	Past 2 levels [fetched]	Use, capped at D=2
M3	Staged disclosure — wizards, linear sequence	Each step simple, purpose clear; good for one-time setup	Everyone pays the sequence every time; NN/g distinguishes it from M2 explicitly (hierarchical vs linear) [fetched]	On re-entry. Fine twice a year, a tax daily	Reject
M4	Command palette — the depth valve	Unbounded command count at zero at-rest pixel cost; searchable, so nothing memorised	Pure hidden navigation. NN/g measured discoverability "cut almost in half" by hiding main navigation, later first use, higher perceived difficulty, and worse on desktop than mobile (n=179, 6 live sites, with WhatUsersDo) [fetched]; transfer to app command surfaces is [inference] — their study was web nav	Use, never as the only path	
M5	Expert mode / modal keymaps (Vim mode)	A second interaction language for ~0 surface	Splits the product in two for support, docs, tests	When expert mode changes semantics rather than input. Fine for cursor motion; fatal for what is written to disk	Bury at D=2
M6	Opinionated single-path design	One correct way ⇒ nothing to choose ⇒ nothing to explain. iA Writer: "has fewer features, by design. But each one is intentional" [fetched, ia.net/writer]	Every user whose path is the other one leaves; you have chosen your non-customers	When the opinion contradicts the file. If the file says X and the single path renders Y, the opinion is a bug	Use
M7	Feature gating by usage — instrument, then bury or kill	NN/g prescribes it: "instrument the code to record how often people use various features" [fetched]	Analytics alone lie; NN/g requires supplementing them "with observational usability testing to discern whether a page gets many hits because users want it or because they simply enter the page by mistake" [fetched]	Local-first products with no telemetry have no signal at all — Arc's problem without Arc's data	Substitute (see §18.2, U)
M8	Hidden-until-relevant UI — contextual reveal	Control appears only where its object exists; at-rest count stays flat as features grow	Users cannot form a model of what the app can do; capability becomes folklore	When relevance is <i>predicted</i> rather than <i>entailed</i> . "A table is under the cursor" is entailment; "we think you want a table" is a slot machine	Use, entailed only
M9	Training wheels — block advanced states for novices	Carroll & Carrithers, CACM 27(8):800–806, 1984, DOI 10.1145/358198.358218 ; companion <i>Blocking Learner Error States in a Training-Wheels System</i> , Human Factors 26(4):377–389, 10.1177/001872088402600402 [fetched via Crossref metadata only — the papers were not opened, ACM DL is Cloudflare-gated; do not quote effect sizes]	Blocked states must be explicable or the block reads as a bug	Anything that refuses a legal file. Blocking a <i>learner error</i> is not refusing valid YAML	Reject
M10	Progressive summarisation of chrome — focus mode, fade-on-type	Zero learning cost, reversible, cheapest thing in the table	Almost none	Only when it hides <i>state</i> (unsaved, syncing, conflicted) rather than controls	Use

The two NN/g conditions are the whole discipline: get the split right, and make the descent obvious and singular [fetched]. Every mechanism above is an implementation of one of those two.

Anti-recommendations: do not use M2 to justify a third level — three levels means cut features, per NN/g's own instruction [fetched]; do not ship M8 on prediction; do not build M9 into the file-open path; do not adopt M3 anywhere, including first-run.

18.2 A measurable definition of simple

Each counter is a number produced by a script and re-derived at write time, never a hand estimate.

Metric	Definition	Measurement Procedure	Target
V0 — visible controls at rest	Interactive affordances rendered in the default window: vault open, one .md selected, no hover, no selection, 1440×900	Build, screenshot, hand-count once, then pin with a DOM assertion (document.querySelectorAll('button,[role=button],a,input,select') inside the shell subtree) in CI	≤ 9
K1 — keystrokes to first value	Discrete input events from cold launch to the first user word durably on disk	Instrumented run	≤ 3 (launch → type → autosave)
S0 — settings exposed by default	Toggles reachable without an "Advanced" affordance	Static count	≤ 6 (today: 5 toggles + 1 mode segment [measured])
C — concepts to learn	Distinct nouns required to use the product at all; anything not nameable on the first-run screen is a concept being smuggled	Enumerate from first-run copy	≤ 3: file, folder, view
D — disclosure depth	Max levels from at-rest to the deepest shipped capability	Walk the tree	≤ 2 [fetched, NN/g ceiling]
P — path multiplicity	Distinct routes to a given command	Cross-reference palette ⇝ menu ⇝ shortcut ⇝ toolbar	≤ 2; exactly 1 for destructive [fetched]
Z — zero-config correctness	% of foreign vaults that open, render and round-trip byte-identically with no setting changed	Foreign-corpus suite	≥ 99.9% (today 17% pass; 6,613/6,614 refused on zero-indent sequences [project-measured])
U — usage floor	% of active users touching each shipped feature	No telemetry will ship. Substitute: support requests + issue mentions per command per quarter, and say in the doc that it is a substitution so nobody later reports it as usage data	Bury below 5%; delete below 1%
R — reversibility	% of mutating actions with single-keystroke undo	Enumerate mutating commands	100% — non-negotiable when the file is the source of truth

Benchmark, competitors where the data exists:

Product	Countable Depth	V0 Evidence	Verdict
IA Writer	38 support-page headings incl. Settings, Content Blocks, Templates, Custom Templates, URL Commands, Apple Shortcuts [measured, ia.net/writer/support]	"No buttons, no popups, no title bar" — its own claim [fetched]	Succeeded, drifting. Watch D
Bear	73 modifier-bearing shortcut tokens across 14 sections [measured, bear.app/faq/mac-keyboard-shortcuts]	Native app, not measurable via curl	Succeeded. The reference implementation of enormous depth behind an invisible surface — depth lives entirely in the keymap
Notion	44 block types on the <i>basics</i> page alone [derived: 13 basic + 5 views + 6 media + 12 embeds + 8 advanced = 44]; 11 product surfaces in site nav [fetched]	n/a	Failed at simple, won at market. Both are true; recorded, not resolved
Obsidian	175 en/*.md help pages, 28 under en/Plugins [measured, GitHub API obsidianmd/obsidian-help@master]	Blank pane at first run	Mixed
Arc	5.52% of DAUs use more than one Space regularly; 4.17% use Live Folders; 0.4% use Calendar Preview on Hover [fetched, The Browser Company, <i>Letter to Arc members 2025</i> , 2025-05-26]	n/a	Failed. "for most people, Arc was simply too different, with too many new things to learn, for too little reward" — the novelty tax [fetched]
Evernote	Peek (2011) → Market → Work Chat (Oct 2014) [fetched, usefyi.com/evernote-history — secondary, vendor blog]	n/a	Failed. The canonical accretion death
frontmatter (today)	14 palette commands; 14 slash commands; 5 toggles; 14 src/modules ; 29 routes; 971 TS/TSX/JS/MJS files outside node_modules [measured, engine/plan-and-diagnostics]	Proxy only — 11 <button> in EditorPane.tsx (973 lines), 1 in Toolbar.tsx , 1 in VaultWorkspace.tsx [measured; a source-text proxy, not a rendered count]	On the right side, unverified. Notion ships 3.14× frontmatter's insert vocabulary (44/14) [derived] — that ratio is the position to hold

The claim that most Obsidian downloaders never get past their first note is refuted and must not be repeated. It traces to a Goodreads author-blog post by Red Tash dated 2025-05-21, where it appears as an unattributed pull-quote — "Over 1 million people have downloaded Obsidian—but most never get past creating their first note" [fetched, verbatim]. No Obsidian source, no methodology, no denominator. The nearest real artefact is Obsidian forum topic 90236, 2024-10-21, an anecdote [fetched via Discourse search.json]. Use Arc's published percentages instead; they are the only first-party feature-adoption figures in the corpus.

Source disagreement, recorded not resolved: IA Writer markets "fewer features, by design" [fetched] while its own support index carries 38 headings [measured]. Both are honest, and the gap is exactly the target state — which is why V0 and C must be counted separately from feature count.

Falsification: if a build passes V0 ≤ 9, C ≤ 3, D ≤ 2 and still produces support requests dominated by "where is X", the counters are measuring the wrong thing and M4 is being over-relied on; re-derive from U before adding surface.

Anti-recommendations: do not report V0 from grep -c '<button>' ; do not build a settings tab structure (tabs are level-2 disclosure spent on the lowest-value surface); do not treat Notion's 44 blocks as a target to approach; do not add a formatting toolbar, even collapsed, even on hover, even "just for tables" — markdown's proposition is that the syntax *is* the toolbar, and a bold button admits the syntax failed.

18.3 The surface at rest — what is visible, what is one keystroke away

At rest = a real vault open, one .md file selected, no hover, no selection, default settings, 1440×900.

Visible, V0 = 9. Anything else on screen is a violation.

- 1. The text. Full-bleed, one column, no border.
- 2. File tree, collapsed to a rail not a panel: folder names only, no icons, no counts, no sizes.
- 3. Breadcrumb path of the current file, which is also the rename affordance (click to edit in place).
- 4. **One** mode control: three-state segment Edit · Read · Split .
- 5. **One** view control: the projection selector Document · Board · Calendar · Card · Site , disabled-but-visible when the file's bytes do not entail that projection.
- 6. Save/sync state. Text, not a spinner. Three states only: Saved / Saving / Conflict .
- 7. Word count.
- 8. Search entry point (Material Symbol search , inline SVG per house standard).

9. Right-pane toggle — properties / outline / backlinks, which **cycles** rather than fanning out.

Not visible at rest, and this is the design: no formatting toolbar, no font picker, no theme switcher, no plugin surface, no AI button, no export button, no share button, no graph button, no settings gear.

One keystroke away (D = 1):

KEY	OPENS	RATIONALE
	Command palette — the single depth valve. Every non-at-rest command lives here and only here	M4; NN/g single-descent-path [fetched]
	File switcher, fuzzy, filenames only	Object selection, not command invocation
	Insert menu — 14 items today [measured]. Cap at 16	M8
	In-file find	Frequency
	Vault search	Frequency
	Settings — cap at 6 toggles, one screen, no tabs	S0
	Collapse everything back to at-rest	Reversibility of the surface itself

Buried at D = 2, reachable only from the palette or a properties field: Vim mode; line numbers; spellcheck; AI ghost text; import; export vault; trash/restore; version history; link doctor; share; publish; graph; degradation certificate; splice diagnostics; conflict merge.

Never surfaced (engine-internal): byte-preserving splice mechanics, offset maps, cross-engine degradation certification across 7 markdown engines, the foreign-corpus gate. **These are the depth, and they must be felt as "it never corrupted my file" and never seen.**

Build actions, in order: (1) convert V0 from proxy to measurement and pin the at-rest screenshot into the repo as a golden image — a diff on it is a design-review trigger; the repo has no `.github/` at all today [project-measured, master plan L164]; (2) cap the palette by policy at **40** commands with `MAX_RESULTS` held at 20 [measured: 14 today, `MAX_RESULTS=20`] — past 40 a palette is a menu you cannot see; (3) enforce $P \leq 2$ with a script that cross-references palette `u` shortcuts `u` menus `u` toolbar and fails CI on a third route.

Anti-recommendations: do not add a second route to any destructive command; do not add a menu bar item for something already in the palette; do not defend the blank first run with the refuted Obsidian statistic — defend it, or don't, on K1.

18.4 Latency as the quality signal

Perceptual bands, each with its real source:

BAND	CLAIM	SOURCE
~10 ms	"produce each frame in an animation in 10 ms or less" — the 16 ms frame minus browser paint	web.dev/articles/rail [fetched]
16.7 ms	One frame at 60 Hz; smoothness requires 60 new frames per second	web.dev [fetched]
20–30 ms	MIDI audio latency "enough to disrupt your performance"	Fatin, <i>Typing with pleasure</i> [fetched]
~40 ms	"Human visual system needs about 40 ms to process its input"; also Fatin's minimum inter-keystroke interval	Fatin [fetched]
50 ms	Process input events within 50 ms, because idle work can queue input for the other 50 ms	web.dev [fetched]
100 ms	"0 to 100 ms — users feel like the result is immediate"	web.dev [fetched]
150 ms	Fatin's own average inter-key interval "during my own relatively fast typing"	Fatin [fetched]
~200 ms	Full sense→consciousness→muscle round trip; SpeechJammer disrupts speech at ~200 ms	Fatin [fetched]
1000 ms	"users lose focus on the task they are performing"	web.dev [fetched]
10000 ms	"users are frustrated and are likely to abandon"	web.dev [fetched]

Two structural facts decide the budget. Latency is **not masked** by the 40 ms vision delay — "any 'external' delay is added on top of the vision delay, not 'masked' by it" — and typing is feedback-dependent below conscious perception: "One does not necessarily need to perceive latency consciously to be affected by it" [fetched, Fatin]. Jitter is a separate axis: the nervous system adapts to constant delay, but "any irregularities in delay durations (so called jitter) pose additional problem because of their inherent unpredictability", with GC/JIT named as a jitter source specific to JS engines [fetched].

Hardware floor is not free: typical keyboard 14 ms avg (matrix 0.5, debounce 8.5, USB poll 4, transmission 1) plus typical monitor 12 ms (refresh 8, pixel response 4) $\approx$ 26 ms; "ideal" is 3 ms [fetched]. **Software budget to stay inside RAIL's 100 ms response bound = 100 – 26 = 74 ms** [derived].

The latency budget. Targets are p99 on a 300 KB note, not the mean.

OPERATION	TARGET P99	PERCEPTUAL JUSTIFICATION
keypress → caret + glyph	≤ 8 ms	Must fit inside one 16.7 ms frame alongside paint; GVim achieves 0.9 ms avg [fetched]
keypress → highlight of edited line	≤ 8 ms	Same frame; lezer incremental measured 1.36 ms @ 294 KB [measured]
keypress → live-preview conceal update	≤ 10 ms	RAIL animation budget [fetched]
cursor move / selection change	≤ 8 ms	Same frame, and it fires more often than edits [inference]
keypress → split-preview text update	≤ 50 ms	RAIL input-processing bound, leaving 50 ms for queued idle work [fetched]
keypress → status bar / word count	≤ 2 ms	Decoration must not compete with the caret [inference]
scroll one viewport	≤ 10 ms/frame	Animation budget [fetched]
paste 100 KB	≤ 100 ms	Single discrete action, "feels immediate" band [fetched]
keypress → draft durably persisted	≤ 400 ms, unconditionally within one burst	IDB cost is per-transaction (~2 ms), not per-byte; >1 s loses task focus [fetched, rxdn.info + web.dev]
open a warm note (draft in IDB)	≤ 100 ms	Immediate band [fetched]
open a cold note (network)	≤ 1000 ms, rendered skeleton before 100 ms	Focus-loss threshold [fetched]
open a 1 MB document	≤ 1000 ms to first interactive frame	CM6 renders only the viewport [fetched]
refuse a hostile document (shape gate)	≤ 50 ms	Must be cheaper than the attack it prevents [inference]

The repo's own stated keystroke budget is wrong and must be re-derived, not defended: `src/modules/mdmax/domain/shape-gate.ts:27-35` declares `BUDGET_MS.keystroke = 250`, which is 2.5× RAIL's 100 ms response bound and 5× its 50 ms processing bound [derived, fetched].

Measured on this machine, 2026-08-29, mean of 3–50 iterations after warm-up, synthetic markdown at three sizes:

OPERATION	9 KB	58 KB	294 KB
<code>@lezer/markdown</code> full parse	1.19 ms	4.05 ms	16.78 ms
<code>@lezer/markdown</code> incremental, 1-char edit, fragments reused	0.25 ms	0.22 ms	1.36 ms
<code>micromark</code>	3.8 ms	22.3 ms	105.8 ms
<code>mdast-util-from-markdown</code>	4.0 ms	20.4 ms	110.9 ms
<code>remark-parse</code> + <code>remark-rehype</code> (repo's plugin set)	11.0 ms	41.6 ms	212.7 ms
<code>react-markdown</code> full pipeline + <code>renderToString</code> (repo's plugins)	20.8 ms	91.5 ms	534.1 ms
<code>countWords()</code> as written	—	0.24 ms	1.32 ms

Derived: preview render at 294 KB is 534.1 / 16.7 = **32 dropped frames**; at 58 KB one render is 91.5 / 100 = **91.5% of the entire RAIL response budget**; lezer incremental beats the remark pipeline by 212.7 / 1.36 = **156×** and the react-markdown pipeline by 534.1 / 1.36 = **393×**. `Text.toString()` on a 580 KB CodeMirror `Text` is **0.207 ms** (mean of 200) [measured] — materialising the whole doc string per keystroke is *not* the bottleneck, which corrects the obvious hypothesis.

Named defects to fix, in priority order [all measured/read in `src`]:

- `editor/presentation/EditorPane.tsx:373-405` — `MarkdownPreviewPane` **unmounts the entire preview on every keystroke** (`setRenderContent(null)` → `<PreviewPending>` → `scheduleAfterPaint` remount at :331-354): lost DOM identity, lost scroll anchoring, KaTeX/highlight re-init, visible placeholder flash. This is the single largest feel defect.
- `preview/presentation/Markdown.tsx:41-61` — `components` and `rehypePlugins` produce new identities every keystroke; every downstream memo is defeated by construction.
- Same file — the preview parses the whole document with a second, non-incremental engine while CodeMirror already holds an incrementally-maintained lezer tree of the same bytes. Two parsers over one file is two truths, which contradicts the projection law. **Project the preview from the lezer tree CM already maintains, and window it.**
- `EditorPane.tsx:596` + `live/LivePreview.tsx:35` + `live/block-split.ts:31-36` — Live mode passes the *deferred* content into a full `remark-parse`: 212.7 ms per keystroke at 294 KB [measured].
- `CodeMirrorEditor.tsx:243-262` writes the full doc into the Zustand store synchronously inside CM's dispatch, pulling React reconciliation into CM's documented write phase [fetched, CM6 guide]: :266 runs a second redundant `toString()`; :253's "throttle" is a trailing debounce that never fires during a burst at Fatin's 150 ms inter-key average [fetched]; :277-306 runs a char-by-char prefix/suffix scan on every store change this view itself causes.
- There are no web workers in `src` at all** — `grep -rn "new Worker("` returns zero hits [measured] — while `shape-gate.ts:27` documents budgets "enforced by the caller via `worker.terminate()`". The time budget is declarative fiction.

Correct as-is, do not touch: `editor/presentation/live-preview.ts:52-70` (iterates `view.visibleRanges` only, per CM6 guidance [fetched]) and `editor/presentation/split-scroll-sync.ts` (250 ms intent window, no per-frame work).

Benchmark suite: Playwright + CDP tracing, headed Chromium, fixed CPU throttle, reporting **p50, p99, max and SD** — **never a bare mean**, each spec run ≥ 3 times. Specs: `keystroke` (150 ms and 40 ms intervals into 1 KB/10 KB/100 KB/300 KB/1 MB in all four modes; assert p99 keypress→glyph ≤ 8 ms, p99 keypress→preview-commit ≤ 50 ms, max ≤ 100 ms, SD ≤ 5 ms, zero long tasks > 50 ms); `reparse-cliff` (type `` then <!-- then \$\$ then an unclosed [[at the top of a 300 KB note — lezer's documented worst case, where "even a tiny document change... can require a big part of the document to be re-parsed" [fetched]); `adversarial` (320 KB of [[, -, nested >, *, assert the shape gate refuses in ≤ 50 ms and regression-lock k ≤ 1.05 for every regex touching user content); `mount` (assert preview DOM node identity is stable across a keystroke); `persistence` (≤ 3 IDB transactions per 100 keystrokes, zero synchronous `localStorage` on the keystroke path); `open`; `scroll` (≥ 58 fps on 1 MB, memory within ~2× file size). Every assertion is a floor or ceiling, never an equality.

Quadratic regression, reproduced [measured]: pattern `/(!?)\[[^\]]+\]\]/g` against N KB of `[[` — 40 KB → 662 ms; 80 KB → 2,607 ms; 160 KB → 10,740 ms; 320 KB → **42,679 ms**, zero matches; `k = log₂(42679/10740) = 1.991`. Source disagreement, recorded not resolved: `shape-gate.ts:8` states 36,865 ms at 320 KB and `k = 1.98` — same shape, ~16% apart, different machine and day. Scope correction: `shape-gate.ts:9` cites a 10.3× `mdast-util-from-markdown` vs `micromark` gap; on ordinary prose the two are within noise (4.0 vs 3.8; 20.4 vs 22.3; 110.9 vs 105.8) [measured] — the gap is specific to the adversarial flat-list shape and the comment should say so.

Anti-recommendations: do not lower the 400 ms draft debounce toward 0 (IDB cost is per-transaction — 1,000 documents in one transaction ≈ 80 ms vs ≈ 2 s one-per-write [fetched, rxdn.info]); flush on burst-end, `visibilitychange` and `pagehide` instead; do not replace CodeMirror or write a custom contenteditable or a WASM/native buffer (VS Code found a C++ text buffer "could lead to significant memory savings, but we didn't see the performance enhancements we were hoping for. Converting strings between a custom native representation and V8's strings is costly" [fetched]); do not reach for `useDeferredValue` or `startTransition` as the fix — React 19.2 docs state the background re-render "is interruptible: if there's another update to the value, React will restart the background re-render from scratch" [fetched], so a 534 ms render restarted per keystroke commits zero

frames; do not add a second or third markdown engine or a "fast path" parser; do not virtualise the preview with a generic list virtualizer (block heights are unknown until rendered — try `content-visibility: auto + contain-intrinsic-size` first, measured); do not report a median anywhere.

Falsification: if the keystroke spec holds $p99 \leq 8$ ms with $SD \leq 5$ ms and users still report lag, the budget is wrong at the tail — re-derive from max, not p99, per Fatin's stated position that maxima are the important number [fetched].

18.5 The affordance problem — discovering that a file is also a board

The literature, and what it licenses:

FINDING	NUMBER	SOURCE
Raskin's test: an interface is modal w.r.t. a gesture when the state is not the user's locus of attention and the same gesture produces different responses depending on it	—	[fetched, Wikipedia <i>Mode (user interface)</i> , rev. 1327521622, last edited 2025-12-14]
Kinesthetic mode-switch method reduced mode errors	<code>F(1,11)=20.74, p<.001</code>	Sellen, Kurtenbach & Buxton 1992, <i>HCI</i> 7(2):141–164, DOI 10.1207/s15327051hci0702_1 [fetched, full PDF]
Visual feedback reduced mode errors	<code>F(1,11)=11.40, p<.003</code>	same [fetched]
Variance explained — mode-switch method	15.6% liberal / 11.0% conservative (ω^2)	same [fetched]
Variance explained — visual feedback	4.8% liberal / 4.1% conservative (ω^2)	same [fetched]
User-maintained state vs a visual indicator	$15.6 \div 4.8 = \mathbf{3.25\times}$; $11.0 \div 4.1 = \mathbf{2.68\times}$	[derived]
Variance explained by neither channel	$100 - (15.6 + 4.8) = \mathbf{79.6\%}$	[derived]
The visual condition was not subtle	It "involved changing the entire screen area pink ", and the authors warn this has "important implications for systems which rely on more subtle visual cues such as changing the shape of the cursor or the color of the menu bar"	[fetched]
Experiment 2 isolates the mechanism	A latching pedal was statistically indistinguishable from the keyboard (<code>p < .33</code>); it was not the pedal, it was that the state had to be actively sustained	[fetched]
Reversibility beats confirmation	Modal confirmation is "criticised as ineffective... due to habituation. Actually making the action reversible (providing an 'undo' option) is recommended instead"	[fetched, citing Norman 1983, <i>CACM</i> 26(4):254–258]

Citation-count disagreements, recorded not resolved: Sellen 1992 — Crossref 95 vs Semantic Scholar 179; Sarter & Woods 1995 (*Human Factors* 37:5–19) — 614 vs 987; Baldonado 2000 (AV/110–119) — 454 vs 899. Baldonado is the canonical multiple-views guideline paper; its **ACM PDF returned HTTP 403 and its rules are deliberately not quoted here**.

The invariant that dissolves most of the problem: typing always inserts text, in every projection. The board never captures the keyboard as a control surface; drag-and-drop and card-click are gestures that do not exist in text view. Zero gesture overlap means modeless under Raskin's test, so the 79.6%-unexplained-variance problem never arises because there is no mode to be wrong about [inference, grounded on Raskin's definition and Sellen's effect sizes].

Where the control lives:

- One control in the document title bar, left-aligned, present on every file, labelled with the **current** state — `Text`, `Board`, `Calendar`, `Decision`. Copy Google Docs' pattern where the button reads "Editing" and you "click **Editing** → select **Suggesting**" [fetched], not a toggle labelled with its destination.
- Clicking lists only the projections this file's bytes can satisfy, each with a computed reason line: "Board — 3 `status` values, 12 list items". Non-satisfiable entries show disabled with the missing precondition. **The menu is the schema documentation; there is no settings page.**
- Keyboard: one chord per profile (`#1` text, `#2` board, ...) plus a **held-key peek** (`#0` held) that renders the projection over the text and snaps back on release. The peek is the quasimode, and per Sellen it is worth roughly three visual indicators at a fraction of the surface cost.
- Nothing else. No tab strip, no `+ New view`, no view names.

How a user learns a file can be a board — five rungs, no tour: (1) the control exists on every file from first launch, so its location is learned once; (2) it enables itself when content qualifies — self-evidence, but weak alone at 4.8% of variance, so never the only rung; (3) **one inline ghost line at the point of authorship**, at most once per workspace, permanently dismissible — immediately after the third distinct `status:` value or the second `## Column`, a dimmed line appears at that block reading "3 columns · `#2` for board", which is recognition inside the locus of attention rather than a toast in a corner; (4) palette entries named `Board`, `Calendar`, `Decision card`, shown disabled-with-reason so the disabled state teaches the precondition; (5) templates where `new board` emits a file whose *text* is already board-shaped, so the user learns the markdown shape, not a feature.

What a board file must be in a dumb editor: `## Column` headings plus `- [ ] item`; `calendar` = items carrying `date: / due:`; `decision` = MADR headings — all correct in GitHub, Obsidian, Bear, `less`, `vim`. The projection adds **zero bytes**. Last-used projection, column widths, collapse state and scroll are app-local, keyed by path + content hash, never in the file. `render:` is a **hint, not an instruction**: the client owns the default, following VS Code's `workbench.editorAssociations` model shipped in v1.44 alongside `View: Reopen with` — where "You can still use **View: Reopen Editor With...** to switch an individual editor back to the text view" [fetched] — so a file arriving from someone else's repo opens as text with a one-line offer. A mode set by a stranger is Sarter & Woods' worst case.

The six rules: (1) one gesture, one meaning, across every projection; (2) the control names the current state; (3) **preview the byte before writing it** — dragging a card shows `status: doing → done` in the drag chip before drop, generalising Excel's Recommended-Charts preview and Docs' review-preview [fetched]; (4) undo, not confirmation — every projection write is one splice and one `#Z` on a stack shared with text view; (5) the escape is bidirectional and lives in the same control on every platform; (6) two panes on one file are both live and consistent, or the second is read-only with a stated reason.

Failure modes to test, each with its evidence anchor from `obsidian-kanban` (4,483 stars, pushed_at 2026-03-06, 600 open issues [fetched, GitHub API 2026-08-29]): F1 round-trip mutates bytes (`sha256` before/after text→board→text×20; issue #644, open 2022-09-27, 14 comments — board→note→board breaks until restart); F3 split panes diverge or wedge (#578, closed 2022-07-01, "Can't get out of error state if Kanban visible in more than one pane"; #666, 2022-10-20); F5 one-way switch (#732, open 2023-02-09, 7 comments: "When switching to markdown view, the menu entry to switch back is missing"); F6 stale per-file projection memory across rename/move/restart (PR #1221, open 2026-07-09, "clear stale `kanbanFileModes`"); F7 unknown `render:` blanks the editor (assert DEGRADE to text with a named reason); F8 body stops satisfying the profile mid-edit (assert REFUSE + zero writes); F9 line-ending/BOM/bare-CR corruption on projection write; F10 multi-byte truncation at a splice boundary (CJK/emoji column names, combining marks); F11 column rename half-writes (heading and every `status:` value rewrite atomically, or refuse); F14 a third-party `render:` auto-entering a projection on first open; F16 override-preservation surprise, anchored on Figma's own worked example where a fill change from `#1BC47D` → `#F531B3` is preserved on Step 3 and not Step 4 because the target variant started from `#FFFFFF` [fetched]; and **F15, the silent-keystroke event** — a user types while a projection has focus and the character goes nowhere. F15 is frontmatter's vi insert/command error and is the single number to instrument and watch.

Anti-recommendations: no onboarding tour, coach marks, spotlight overlays or "what's new" modals — they are confirmation dialogs wearing a hat and habituation kills them; **no `+ New view` tab strip**, because that single affordance is the on-ramp to Notion's view registry → per-view filters → sorts → grouping → sharing → **Lock views** → a permission tier that exists solely to stop colleagues breaking each other's filters, and Notion's own doc concedes "anyone with editing access can toggle this lock on or off at any time" [fetched]; no per-view saved filters stored in the file; **no app-owned settings block in the document** — `obsidian-kanban` writes `%% kanban:settings ```{json} ```%%` via

settingsToCodeblock [fetched], visible junk in every foreign renderer; no sidebar view file (.base -style splits the source of truth and imports the rename-desync class already reported in that ecosystem [fetched]); no badges, dots or status-bar chips as a discovery strategy (whole-screen pink bought 4.8% of variance; a 12px dot buys less); no content-sniffing auto-switch on open; no hover-only controls as the sole affordance (Notion's inline databases hide "controls and menus... until you hover" [fetched]); no user-defined view DSL, which is a code-execution surface with a friendly name; and do not call it "Mode", "View" or "Profile" in the UI — the user-facing word is the projection's own name, with "render profile" kept internal.

Falsification: if F15 instrumentation shows a non-trivial silent-keystroke rate after the modeless invariant ships, the invariant has been violated somewhere in the gesture map and the projection must lose keyboard focus entirely until it is found.

18.6 Mobile and cross-device

Current posture, measured in this repo 2026-08-29:

SURFACE	STATE
src/app/manifest.ts	display: "standalone", scope: "/", orientation: "any", 3 icon entries (192, 512, 512-maskable) all pointing at the same /favicon.png
Manifest gaps	no share_target, shortcuts, file_handlers, display_override, protocol_handlers — grep returned 0 hits
public/sw.js	98 lines, CACHE = "sgnk-md-v2", SHELL = ["/favicon.png", "/theme-init.js"]; HTML navigations network-only; /_next/static/* stale-while-revalidate; /api/ and /login never intercepted
Offline reading	impossible today — note content arrives via /api/, which the SW explicitly passes through [derived]
Mobile-specific code	visualViewport 0, inputMode 0, safe-area 0, touchstart 0, accessory 0; pointerdown 5; max-width 15
CSS media queries, whole app	3 total across 1,057 CSS lines: 2 × prefers-color-scheme, 1 × max-width: 1023px
src-tauri/tauri.conf.json	bundle.targets: ["dmg", "app"], macOS.minimumSystemVersion: "11.0", no iOS or Android block, frontendDist and window url = https://md.sgnk.ai

In one line: an installable desktop-web PWA with a responsive-ish layout, zero offline content, zero mobile input model, and a Tauri config that is a remote-URL macOS wrapper [derived].

What mobile users of this category actually do (App Store metadata, itunes.apple.com/search, US storefront, read 2026-08-29) [fetched]:

APP	VERSION	RATING	RATINGS N	PRICE
Google Docs	1.2026.34101	4.79	3,211,357	\$0
Microsoft OneNote	16.113.2	4.71	1,059,746	\$0
Apple Notes	2.0	4.84	632,431	\$0
Notion	1.7.331	4.78	89,887	\$0
Drafts	53.0	4.79	10,733	\$0
Bear	2.9.3	4.68	6,853	\$0
Craft	3.5.6	4.84	6,596	\$0
Working Copy	6.9.4	4.85	3,740	\$0
Obsidian	1.13.7	4.48	2,689	\$0
Ulysses	40.4	4.58	2,095	\$0
iA Writer	8.0.6	4.56	1,531	\$19.99

Derived: Google Docs ÷ Obsidian = 3,211,357 ÷ 2,689 = 1,194.3x. Apple Notes ÷ (Bear + Craft + Ulysses + iA Writer + Obsidian = 19,764) = 32.0x. Drafts ÷ iA Writer = 10,733 ÷ 1,531 = 7.01x. Working Copy, a git client, outranks Obsidian mobile on rating volume. Within markdown-native tools, capture-shaped apps outsell authoring-shaped apps roughly 7:1, and the whole markdown-native mobile category is about 1/32 the size of the default OS notes app [inference]. Craft's own front page states the split: "capture ideas instantly across all your devices, then refine them when you're ready" [fetched]. Obsidian forum thread volumes name the real complaints — "Make Obsidian Sync work in background (on Mobile)" 124 posts (opened 2021-10-20), "Obsidian Sync incorrectly duplicates sections of files" 105 posts (2025-01-12), "Mobile, startup: reduce time until the user can write" 64 posts (2023-04-03) [measured, Discourse search.json; views returned null and like_count 0 on every row, so those fields are unmeasured, not zero]. The two dominant families are sync trust and time-to-caret, not missing features.

Decision: ship PWA for v1; do not ship a native or wrapped mobile app.

FORCE TOWARD PWA	EVIDENCE
Every v1-scope capability is y on iOS Safari 26.6: Service Workers, IndexedDB, Wake Lock, env(); VisualViewport safari: 13; navigator.share safari: 12.1	[fetched, caniuse + MDN BCD]
A Home Screen Web App gets the full 60%/80% origin/overall quota, identical to the browser app; a wrapped WKWebView (Capacitor, Tauri iOS) gets 15%/20% — strictly worse, inverting the usual "native gives you more storage" intuition	[fetched, webkit.org/blog/14403/updates-to-storage-policy/]
App Store guideline 4.2 is a live rejection risk for exactly the artefact we would ship first: an app "should include features, content, and UI that elevate it beyond a repackaged website"; 4.2.2 names "web clippings, content aggregators, or a collection of links"	[fetched, developer.apple.com/app-store/review/guidelines/]
Tauri iOS costs Apple Developer Program enrolment, code signing and a macOS CI leg before one user benefits; @tauri-apps/cli 2.11.4 (registry modified 2026-06-28)	[fetched]
We are Next.js 16 with live /api/ and /login routes, so there is no static webDir to hand @capacitor/core 8.5.0 (2026-08-28) — a Capacitor build today ships a shell loading remote content, which is the 4.2 shape	[derived]

Survivable gaps, each with the mitigation: no showOpenFilePicker / showSaveFilePicker (safari: false ; caniuse "n" at every iOS version 3.2 → 26.6) — our source of truth is GitHub-backed, we never needed the picker; no Background Sync or Periodic Sync (safari: false , impl bug webkit.org/b/182565) — sync on foreground/visibilitychange only, and this is precisely Obsidian's 124-post complaint; no beforeinstallprompt (safari: false) and no Add-to-Home-Screen in iOS Chrome/Firefox — an instructional sheet on iOS Safari, and count the loss; eviction is LRU by last user interaction — call navigator.storage.persist() once and keep the server as the durable copy. Quota is a non-issue: 10,000 notes × 4 KB = 40 MB against the worst case (WKWebView, 15% of 128 GB = 19.2 GB) is 0.21% [derived]; eviction is the entire risk.

Source disagreements, recorded not resolved: MDN BCD reports FileSystemFileHandle / FileSystemDirectoryHandle as safari: 15.2 while caniuse reports the File System Access feature as "n" at every iOS Safari version through 26.6 — both correct at different granularity, so cite the picker, not the interface, when saying we cannot open the user's

folder. Push: BCD says `version_added: 16.4`; caniuse says `a` (partial, note 7). Chrome Android `showOpenFilePicker`: BCD says `132`, the FSA caniuse entry says `and_chr 151: "n"` — unreconciled; do not build on either number without re-testing.

Minimum credible v1 mobile scope: (1) **read offline, selectively** — cache note bodies in IndexedDB for the N most-recently-opened plus explicitly pinned, mirroring Google Docs' shipped opt-in model ("Make recent files available offline", a per-file toggle, and a separate Offline list) rather than Obsidian's whole-vault one [fetched]; (2) **capture** — one tap from the home-screen icon to a new note at one configured path, justified by Drafts 7.01× iA Writer [derived]; (3) **light edit, single-writer** — reuse the byte-preserving splice path, guard every write with a precondition check against the file's known SHA, and on mismatch **refuse and show the diff, never auto-merge**; (4) **projections read-only on phone** — render and tap through to the source line, no drag, no inline mutation; (5) the accessory row; (6) manifest repair — real 192 and 512 PNGs, `shortcuts` for "New note" and "Today", and keep `display: "standalone"` because a non-default `display` is the documented precondition for `Notification` ever working on iOS [fetched]; (7) **time-to-caret as the headline mobile metric**.

The accessory row. `position: fixed; bottom: 0` does not track the iOS keyboard — the visual viewport shrinks and the layout viewport does not, and `env(keyboard-inset-bottom)` is `safari: false` [fetched ×2]. Subscribe to `visualViewport.resize + scroll` and position with `transform: translate3d(0, Δ, 0)` where `Δ = layoutHeight - vv.height - vv.offsetTop`; transform, not `bottom`, to avoid per-frame reflow [inference]. Keyboard down, pad with `env(safe-area-inset-bottom)` (`safari: 11`); keyboard up, zero it. Editor font-size ≥ 16px suppresses focus auto-zoom.

SLOT	ACTION	BACKED BY
1	Heading cycle H1→H2→H3→none	<code>prefixLine</code> [measured, exists]
2–4	Bold, italic, inline code	<code>wrapSelection</code> [measured, exists]
5	Link	<code>insertLink</code> [measured, exists]
6	List toggle <code>- ↔ 1. ↔ none</code>	<code>prefixLine</code>
7	Task checkbox <code>[] ↔ [x]</code>	<code>prefixLine</code>
8–9	Outdent / indent	CodeMirror commands
10–11	Caret <code>← / →</code> , long-press = by word	<code>moveByChar</code> / <code>moveByGroup</code>
12–13	Undo / redo	<code>@codemirror/commands</code>

Behaviour: **collapsed cursor must work** — wrap actions with no selection insert the delimiter pair and place the caret between them, because requiring a selection first is the largest markup-entry friction on touch; caret-nudge buttons exist to avoid iOS selection handles entirely, which is why they earn slots ahead of tables, quotes or images; long-press is key repeat, never a submenu; one row that fits one screen width so it never scrolls, sitting outside the CodeMirror scroller. Textastic and Working Copy both ship exactly this pattern at 4.70 and 4.85 [fetched] — the pattern is settled, do not innovate on it.

Explicitly deferred: multi-note offline write queue; any conflict-merge UI; background sync; push; `share_target`; `file_handlers`; native/Capacitor/Tauri mobile builds; CRDT; attachments and image capture; multi-vault; iPad multi-window; graph view on phone.

Anti-recommendations: do not ship a Tauri or Capacitor shell that loads `https://md.sgnk.ai` remotely — that is the current `tauri.conf.json` shape [measured] and the literal 4.2 "repackaged website"; do not design any feature depending on Background or Periodic Sync; do not ship a custom install banner driven by `beforeinstallprompt`; do not make board/calendar/decision-card editable on phone, because drag-to-mutate on a 375 px viewport is where a reversible projection quietly becomes a Notion-style PM tool; do not introduce a CRDT (`yjs 13.6.32` and `@automerge/automerge 3.4.1` are current and healthy [fetched], and both impose an internal document model — a tree-of-record through the back door); do not add swipe gestures for editor commands; do not add `maximum-scale=1` or `user-scalable=no` (our viewport correctly omits both today [measured]); do not "fix" split view on phone (already disabled at ≤767 px, which is right); do not queue offline writes across multiple notes in v1, which is what produces 105-post data-duplication threads; do not make the accessory row customizable; do not treat IndexedDB or OPFS as the record; and do not benchmark against Obsidian's mobile feature list — its 4.48 rating is the lowest in the comparison table while its feature count is the highest [derived].

Falsification and revisit triggers, written down now: background sync becomes a top-2 support theme; measured install-to-home-screen conversion on iOS Safari falls below 15%; or a paying customer requires Files.app integration. Any one of those makes native a live question; none of them is true today.

\$19 Competitor teardowns — executed, not read

The most load-bearing evidence in this document. We ran their write paths.

PRODUCT	SCALE	WHAT HAPPENED	VERDICT
Front Matter CMS	80,527 installs <code>FETCHED</code> , VS Code, since 2019, owns our name	Edited one title field → deleted YAML comments, resolved anchors, stripped leading zeros. Its own source comments <i>"Do our own parsing to keep the comments"</i> — then builds the comment-preserving Document object and throws it away . Output differs warm vs cold	<code>MEASURED</code> The bolt-on was tried and shipped broken
Hubble.md	Markdown collab	Body fully regenerated on any edit; reference links deleted along with their visible text ; the round trip is not even a fixed point; the properties panel silently drops valid keys containing colons	<code>MEASURED</code>
OpenKnowledge	Most serious competitor; ~100 releases/week; 3,239 → 3,673★ in 27 days <code>FETCHED</code>	Body path is byte-perfect — respect it . But frontmatter permanently drifts (<code>tags: [alpha, beta]</code> → <code>tags: [alpha, beta]</code> , never restored); requires a running CRDT daemon; <code>O(document)</code> per edit <i>by their own docblock</i> ; comments machine-local, never committed; GPL + CLA dual-licensing	<code>MEASURED ON THEIR SHIPPED BUILD</code>

Joint verdict: the failure is architectural — lossy in-memory models — not a library choice.

Say this publicly. OpenKnowledge is not careless. Their body path is genuinely byte-perfect. They chose an architecture that caps them where we are not capped. Overstating this is the kind of claim that gets refuted in public.

Editor frameworks — every mainstream rich-text framework is lossy by design `FETCHED`:

FRAMEWORK	LOSS MECHANISM
ProseMirror	Fixed CommonMark schema; serialise is per-node functions writing fresh syntax. Reference links normalised away by construction
<code>@tiptap/markdown v3.30.5</code>	Explicitly beta . Documents that comments <i>"may be lost"</i> and that table cells allow <i>"only one child node per cell as the Markdown syntax can't represent multiple child nodes"</i>
Milkdown	Autolink backslashes double every round trip, exponentially (open); inline <code>
</code> silently deleted (open)
Lexical	Transformers over EditorState — helpers, not storage
BlockNote	Exposes <code>blocksToMarkdownLossy()</code>

This is why we use CodeMirror over a syntax tree and never a document model.

§20 Migration — what survives, what does not

Build order is 1 → 2 → 3 : Obsidian, Notion-zip, Evernote. Ship the verification report for Obsidian first — it is the only source where the report can honestly print IDENTICAL on nearly every row, which is how the report earns the standing to deliver bad news about Notion.

#	SOURCE	VOL	FEAS	SCORE	BASIS, PROXIES FETCHED
1	Obsidian	4	5	20	No conversion at all; the only 5 on the list
2	Notion (zip export)	5	3	15	58 notion -label issues, 55 closed (94.8%) DERIVED ; notion-to-md 1,358,239 npm downloads in the window 2026-07-29→2026-08-27 FETCHED ; \$1,000 + \$5,000 bounties FETCHED
3	Evernote (.enex)	3	5	15	DTD-defined, self-contained, base64 attachments inline — but yarle carries 85 open issues FETCHED
4	Apple Notes	5	2	10	Highest bounty after the API job (\$2,500, #15); no export format; macOS-only; iOS refused (#342, open since 2024-12-24)
5	Logseq	2	4	8	78,308 B reference implementation to copy semantics from
6	Notion (API)	5	1.5	7.5	253,994 B = 5.07× the zip path DERIVED: 253,994 ÷ 50,126 = 5.067 ; OAuth; 3 req/s ceiling
7–13	Bear · Roam · Confluence · OneNote · Google Docs · Word · Craft	—	—	8→1	Confluence's best OSS converter is archived since 2021-06-24, 143 stars; OneNote .one needs 150,662 B of parser including an LZX decompressor and a CAB reader; Craft #27 has been open 2 years 1 month with zero progress FETCHED

Per-source fidelity, construct by construct. Classes: SURVIVES · DEGRADES (recoverable, not identical) · LOST (absent from the export itself) · REFUSE (we decline before starting).

Obsidian → frontmatter

CONSTRUCT	VERDICT	EVIDENCE
Note bodies, YAML frontmatter, tags	SURVIVES byte-identical	INFERENCE
[[wikilinks]] , ![[embeds]] , [[note#heading]] , [[note#*blockid]]	SURVIVES as text; resolution semantics are Obsidian-specific and must be certified, not assumed	INFERENCE
Dataview / Templater / Bases queries	DEGRADES → inert code fences, listed explicitly	INFERENCE
.canvas files	REFUSE — JSON, not markdown	INFERENCE
Plugin-specific frontmatter keys	SURVIVES as data, LOST as behaviour	INFERENCE
.obsidian/ config, attachment folder settings	LOST — not content	INFERENCE
Bare-CR line endings; zero-indent YAML sequences	REFUSE with a named reason, never silent repair	[project-context, engine-known]

Notion (zip: PDF · HTML · Markdown & CSV) → frontmatter

CONSTRUCT	VERDICT	EVIDENCE
Page text, headings, lists, code	SURVIVES	FETCHED
Callouts	DEGRADES → HTML, rewritten to > [!info] ; icon and colour LOST	FETCHED Notion: callouts export as HTML, "as there is no Markdown equivalent"
Databases	DEGRADES → one CSV + one MD per row page	FETCHED
Database views	LOST — only current or default view exports, "Exporting all views at once isn't supported"	FETCHED
Form views	REFUSE — cannot be exported at all	FETCHED
Comments, page and block level	LOST — present only in the HTML export, which we do not import	FETCHED ; importer #311 open since 2024-10-09
Relations / rollups / formulas	DEGRADES → last rendered text, stops updating	INFERENCE ; the API path needs formula-converter.ts (19,397 B) and database-helpers.ts (43,237 B) FETCHED
Toggle blocks and toggle headings	DEGRADES, historically dropped children	FETCHED #469, #458
Synced blocks	DEGRADES → duplication	INFERENCE SyncedBlockRequest is special-cased FETCHED
Nested table inside a list	LOST silently — first column only	FETCHED #216, open since 2024-02-29
Numbered lists	DEGRADES → repeated 1.	FETCHED #566
Attachments and paths	DEGRADES — Windows MAX_PATH 260 breakage on nested subpage folders	FETCHED
Filenames	DEGRADES — Korean Unicode decomposition freeze (#348), trailing dot or space (#456), titles >255 chars lose data (#381)	FETCHED
Created / updated timestamps	LOST by default	FETCHED #478, #479
Pages the exporting account cannot see	REFUSE-gate — silently absent, and Notion does not report how many	FETCHED
Export availability	REFUSE-gate — admins can toggle Disable export; workspace export takes up to 30 hours; the link expires after 7 days	FETCHED

Evernote (.enex, ENEX 4.0 / ENML 2.0 DTD) → frontmatter

CONSTRUCT	VERDICT	EVIDENCE
Title, content, created, updated, tags	SURVIVES — the DTD is normative	FETCHED
Body markup	DEGRADES via HTML→MD; ENML is "a subset of XHTML... intentionally broadened"	FETCHED
Attachments	SURVIVES — <code><data encoding="base64"></code> , inline	FETCHED
Attachment MIME range	Constrained to <code>image/gif</code> , <code>image/jpeg</code> , <code>image/png</code> , <code>audio/wav</code> , <code>audio/mpeg</code> , <code>application/pdf</code> , <code>application/vnd.evernote.ink</code>	FETCHED
Ink notes	LOST — no markdown target	INFERENCE FROM THE FETCHED ALLOWLIST
<code>note-attributes</code> (geo, author, source-url, place-name, reminders, subject-date)	SURVIVES into frontmatter if we map them ; <code>source-url</code> is a known drop	FETCHED ; #48 open since 2023-08-07
Internal Evernote note links	LOST/DEGRADES — the single most-reported defect	FETCHED yarle #653, #684, #655, #478, #209, #357
Non-ASCII tags	CORRUPTED in the leading converter — Chinese characters skipped, accents skipped	FETCHED yarle #638, #637
Notebook / stack structure	DEGRADES	FETCHED yarle #678, #540, #353
Client-side encrypted text (RC2-64)	REFUSE	FETCHED yarle #632
Hard caps	REFUSE-gate — content ≤ 5,242,880 chars, resource binary ≤ 25MB, title 1–255 chars	FETCHED DTD
Tasks	SURVIVES — <code>task*</code> is in the DTD	FETCHED

Remaining sources, compressed: **Apple Notes** — no export format at all; SQLite `NoteStore` plus protobuf attribute runs; tables and internal links survive, highlight colours degrade (the reference implementation substitutes emoji), HEIC/HEIF degrades (#497), password-locked notes and drawings REFUSE, iOS REFUSE (#342) FETCHED. **OneNote** — free-form canvas positioning has no markdown target and is LOST; ink degrades to an SVG approximation; the binary path costs 150,662 B DERIVED. **Roam** — outline blocks survive, `((uid))` block refs degrade to links or embeds, `{{[[query]]}}` is LOST and was historically corrupted by the rewrite itself, and 17 Roam-only markup tokens are on an explicit scrub list FETCHED SOURCE. **Logseq** — the outline-to-prose de-indent alone is a dedicated 9,550 B module; block IDs, `key::value` properties and `NOW/LATER` all need explicit mapping; queries and flashcards LOST FETCHED. **Confluence** — Normal HTML space export excludes blogs, inline comments and attachments; Word export includes "only the first 50 attached images" and opens only in Microsoft Word; drafts are never exported; the export needs the *Export Space* permission FETCHED. **Google Docs, Microsoft Word, LibreOffice, Craft** — not supported, and the reason is measured: 47 of 58 open issues in the largest markdown-importer ecosystem (81% DERIVED) are requests for formats with no importer at all, the oldest being Craft #27 at 2 years 1 month FETCHED.

The pre-flight refusal contract — shown as a blocking screen per source, before any file is read, with counts where the export permits counting and the literal word "unknown" where it does not.

Universal, every source: comments and discussion threads do not survive · revision and version history does not survive · permissions and sharing state do not survive · anything the exporting account cannot see is silently absent and **no export reports how much was withheld**, so our count is a count of what we received, not of what you own · formulas, rollups, queries and live embeds arrive as their last rendered text and stop updating.

Obsidian, before starting: `.canvas` is not markdown and will not be converted; Dataview/Templater/Bases blocks become inert text; plugin behaviour does not transfer, plugin frontmatter keys do, as data. **Everything else should be byte-identical, and if any file is not, the report names it and the byte offset — that is the promise the report exists to keep.** **Notion, before starting:** one view exports, the rest are gone; Form views cannot be exported at all; callouts lose icon and colour; the export can take up to 30 hours and the link expires in 7 days; if your admin has toggled *Disable export* there is nothing we can do from our side; on Windows, export with *Create folders for subpages* off. **Evernote, before starting:** RC2-64 encrypted note text cannot be decrypted and those notes are listed and skipped; ink has no markdown representation; notes over 5,242,880 characters and resources over 25MB are outside the format's own limits; internal note links are the most fragile construct and we report resolved/unresolved counts rather than guessing; non-ASCII tags are where incumbent tools corrupt data, and we list every tag we alter. **Apple Notes:** macOS only, Full Disk Access required, locked notes unreadable, drawings lost. **Confluence:** blogs, inline comments and attachments excluded; 50-image cap on Word; drafts never exported. **OneNote, Google Docs, Word, Craft:** we do not support these — say it on the pricing page, not after the upload.

Anti-recommendations: do **not** build the Notion API importer first (5.07× the code, plus OAuth, a 3 req/s ceiling, cursor invalidation and documented per-run non-determinism where 1.8.1 loaded a different page list than 1.8.0) FETCHED ; do **not** build Craft, Google Docs, Word or LibreOffice importers; do **not** parse the `.one` binary; do **not** promise comment migration for any source; do **not** ship "Notion databases → boards" as a launch feature, because a projection from a lossy single-view CSV is a projection of a lie; do **not** let the report be a summary — per-construct counts or it is not a verification report; and do **not** auto-repair the source, because silently normalising input to make it importable destroys the byte-preservation claim outright.

§21 Market segments and ICPs

Two motions, one substrate. Acquisition is D2C-shaped; monetisation is B2B-shaped. By month 24 the median solo **B2B** founder's revenue is more than 4× the median solo B2C founder's SS — but through *self-serve* teams of 2–20 paying by card, not procurement.

#	ICP	HOOK DOCUMENTS	WHY THEM	PRICE ANCHOR
1	Dev-tool / API startups, 2–20 — the beachhead	Docs site, changelog, status page, runbooks, ADRs (no commercial tool exists)	They already live in this substrate: <code>gray-matter</code> does 35,782,970 npm downloads/month, <code>front-matter</code> 18.41M, <code>js-yaml</code> 1,228,031,655 FETCHED	\$20–40/editor seat, or \$150–300/team
2	Agencies and studios, 2–15	One file per client rendering as a client-facing status dashboard, agent-updated from work logs	Zephyrus's own shape — we can dogfood it honestly	Flat \$29–49/mo, or \$9–19/seat
3	Support-heavy SMBs	Knowledge base, help centre	Cleanest ROI: deflection 18% median (40–60% with AI), a SaaS ticket costs \$25–35 SS . Fierce incumbents — land after the substrate story is proven	\$99–249 per knowledge base
4	SMB internal ops	Wiki, SOPs, runbooks	Broad, shallow	\$8–15/internal seat

The consolidation wedge, at live prices DERIVED : Mintlify Pro **\$450** + GitBook Premium **\$65/site** + Statuspage Business **\$399** + LaunchNotes Growth **\$249** = **\$1,163/mo**, plus 5 GitBook users × \$12 = **\$1,223/mo**. But the honest *light* version of the same stack — Mintlify Starter \$0 + Statuspage Hobby \$29 + Beamer \$49 — is **\$78/mo**. **Quote the range, never only the top.**

Two framing corrections.

1. **Do not lead with deflection or "structured data."** Intercom, Zendesk, Document360 and GitBook all sell exactly that. It is our *proof* after a team is inside, not the opening line. **The wedge is one non-corrupting file that is editor, AI workspace, rendered surface and audit trail at once.**

2. The audit trail is what incumbents cannot copy quickly. Because the splice attributes changes at byte level, the file is its own audit log. One vendor's case study cut a sales cycle from four months to six weeks; Vanta and Drata charge \$7,000–\$30,000/yr for continuous audit trails ss. We get a slice as a byproduct of *how we write files*. "The AI edited your ops file, here is cross-engine proof it corrupted zero bytes" is a governance claim no Notion or Confluence AI can make.

Do not chase until there is a team or funding: SSO, SCIM, SOC 2, the enterprise knowledge-base market.

§22 The academic and research segment

22.1 Sizing, with the confidence stated

MEASURE	VALUE	TAG	CONFIDENCE
arXiv submissions, 2025	284,486	[measured] from arXiv <code>get_monthly_submissions</code> CSV, 422 rows, read 2026-08-29	High
arXiv CAGR 2019→2025	$(284,486 / 155,866)^{(1/6)} - 1 = 10.55\%/yr$	[derived]	High
arXiv 2026 Jan–Aug	226,607 → 28,326/mo → 339,910 annualised, +19.5% vs 2025	[derived]	Medium — an 8-month run-rate extrapolation
arXiv cumulative, all time	3,148,951	[measured]	High
CSL styles in the canonical repo	10,860 <code>.csl</code> files = 2,861 independent + 7,999 dependent	[measured], API <code>truncated: false</code>	High
Zotero Connector, Chrome Web Store	8,000,000 users	[fetched 2026-08-29]	Medium — store installs, rounded to 1M, not actives
Mendeley Web Importer, Chrome Web Store	3,000,000 users	[fetched 2026-08-29]	Medium, same caveat
Overleaf users	"over 20 million" (<code>/about</code>) vs "Over 25 million" (<code>/for/universities</code>)	[fetched 2026-08-29]	Low as a number. Same company, same day, two figures. Recorded, not resolved.
India higher-ed enrolment 2023-24	4.50 crore = 45,000,000; GER 30.0; STEM 10,200,000; faculty 1,732,000	[fetched] PIB PRID=2282525 / AISHE	High as a statistic, worthless as a denominator (§22.4)
Dollar market size for reference-management or academic-writing software	None obtained	[measured] — no analyst report opened; pypistats returned <code>429 RATE LIMIT EXCEEDED</code>	Do not quote a figure. There is not one.

Relative size, using Obsidian as the closest analogue surface [measured from `community-plugin-stats.json`, 2,096,248 B, read 2026-08-29]: 7,061 plugins, 142,701,824 cumulative downloads; the academic cluster (Zotero/citation 1,041,113 + Pandoc/export 995,130 + LaTeX/math 1,014,880) = 3,051,123 = 2.138% of all plugin downloads [derived]. That field is cumulative across every version, so it rewards release cadence, not installs. On a peak-single-version proxy against Dataview 0.5.68 = 2,114,034: Citations 0.4.5 = 165,583 → 7.8% [derived]; Zotero Integration 3.2.1 = 277,090 → 13.1% [derived]; Pandoc Plugin 0.4.1 = 493,054 → 23.3% [derived]. The source report states this bracket as 7.8%–20.9%; the 20.9% upper bound is not reproducible from its own three published peaks, so treat the honest range as 7.8%–23.3% and see §55.

22.2 Jobs to be done

JTBD	WHAT IT ACTUALLY MEANS	EVIDENCE IT IS THE JOB
Citation insert	Type <code>[@smith2020]</code> , get it right, never retype	Most-wanted RMS feature at 66.9%, n=121 [fetched, PMID 32395773]
Bibliography to a journal's style	2,861 independent styles exist because every journal differs	10,860 CSL files [measured]
Reference <i>correctness</i>	Fabricated references reaching manuscripts	<i>J Educ Eval Health Prof</i> 2026;23:2 names "inability to verify reference authenticity" as the critical RMS limitation [fetched, PMID 41549369]
Cross-references and numbering	<code>@fig:setup</code> → "Figure 3", stable under insertion	<code>obsidian-pandoc-reference-list</code> peak version 64,131 [measured]
Equations	<code>\$. \$ / \$\$. \$</code> rendering that survives the trip to the publisher	LaTeX/math cluster 1,014,880 cumulative; Latex Suite peak version 161,328 [measured]
Supervisor collaboration	Track changes, margin comments, accept/reject, asynchronous over months	Overleaf's university pitch leads on it; Manchester case study claims "310% jump in new projects" [fetched — vendor-reported, unaudited]
Journal submission format	Not "a PDF" — the publisher's <code>.cls</code> , or Word with named styles	Overleaf's moat; 20–25M users [fetched]
Reproducible computation	Executable cells → figures that regenerate	Quarto 5,967★ + Jupyter Book 4,273★ + rmarkdown 3,054★ [fetched 2026-08-28/29]

22.3 Architecture fit

JTBD	FIT	WHY
Citekey autocomplete from a declared <code>bibliography:</code>	CAN	Read a local <code>.bib</code> /CSL-JSON, splice-insert <code>[@key]</code> . No format invention, no plugin, no code execution
Rendered reference list	CAN	A pure function of (file, <code>.bib</code> , <code>.csl</code>) → the projection contract exactly. The view renders; the file keeps <code>[@key]</code>
Figure/table/equation numbering and cross-refs	CAN — best fit in the segment	Labels live in the file, numbers live in the projection. Insert a figure, every number in the view moves, zero bytes rewritten
KaTeX math rendering	CAN	A renderer, not user code
Degradation certificate for <code>\$...\$</code> , <code>[@key]</code> , <code>{#fig:x}</code> across 7 engines	CAN, and no competitor ships it	These are precisely the constructs that die silently between engines
Reference acquisition (browser capture, PDF library, dedup)	SHOULD NOT	Zotero Connector = 8,000,000 installs, free [fetched]
Track changes / suggestion mode / real-time co-editing	STRUCTURALLY CANNOT	Needs a CRDT/OT layer (a second source of truth) or a comment sidecar (a tree-of-record renamed). Both are settled against
Publisher <code>.cls</code> / journal Word template	CANNOT OWN	Unbounded compatibility surface plus a TeX toolchain. Delegable to Pandoc, never ownable
Executable code cells	STRUCTURALLY CANNOT	The eval lane is settled closed; Quarto and Jupyter Book own it at \$0
Shipping all 10,860 CSL styles	CAN but SHOULD NOT	Data, not code — architecturally legal, a maintenance tail with no revenue attached

Of the eight jobs, frontmatter can serve five well and refuses the three that decide the purchase: supervisor markup, journal format, and reproducible cells.

22.4 Verdict — serve incidentally, do not target

GROUND	EVIDENCE
The buying trigger is owned. A thesis is bought by "my supervisor comments and the journal accepts". Overleaf sells exactly that half	[fetched] 20–25M users
Everything we <i>can</i> do here is already free	Pandoc 46,053★, Zotero 15,094★, <code>jgm/citeproc</code> 182★, Quarto, Typst — all \$0 [fetched]
The India-student angle is the weakest monetisation surface on the board, not the strongest	Overleaf India-geolocated: Student ₹201.75/mo billed annually = ₹2,419/yr (₹241 month-to-month); Standard ₹421.75/mo = ₹5,059/yr; Professional ₹880/mo = ₹10,559/yr [fetched 2026-08-29; ×12 checks to within ₹2 rounding, [derived]]. That ceiling is set by a firm amortising over 20–25M users
Revealed preference is hostile to paid conversion	n=121 Tunisia: only 26.5% used any RMS, 81% preferred free/OSS, 50.4% did not know Zotero was free [fetched, PMID 32395773]. Fieldwork 2016 — low generalisability to India; do not extend without a second source
45,000,000 enrolled students is a vanity denominator	At an implausibly good 0.1% paid conversion = 45,000 users [derived], and nothing in the evidence supports 0.1%

The strongest argument against this verdict. Academia is the only market where "the file is the only source of truth" is a compliance requirement rather than a taste, and the only one where degradation certification has a named buyer: a university library or graduate school that must guarantee a deposited thesis still renders in fifteen years. The CSL repo having more forks (4,142) than stars (3,885) [fetched] is a field that already accepted "declarative data, deterministic renderer" as its architecture, and the AI-fabricated-reference problem is a *verification* problem — the exact shape of a certificate. It is answered on go-to-market, not on value: library and graduate-school procurement runs 9–18 months and gates on SSO, accessibility conformance, and security review, none of which a solo founder can service while also selling globally to individuals — so it justifies **keeping the door open**, not walking through it.

Falsifier. If one paying institutional customer names cross-engine degradation certification as a procurement line item without us marketing to universities, re-open this verdict in full.

22.5 The incidental feature set — build these five, in this order

- `bibliography:` / `csl:` / `title:` / `author:` **frontmatter awareness**. A file authored in frontmatter becomes valid Pandoc *and* Quarto input with zero edits. The export story is *hand it to pandoc, never we typeset*.
- Citekey autocomplete from the declared `.bib` /CSL-JSON. Read-only sidecar, splice-inserts `[@key]` . Serves the 66.9% job without becoming a reference manager.
- Cross-reference and numbering projection (pandoc-crossref syntax `{#fig:x}` / `@fig:x`), numbers computed in the view, never written to the file.
- KaTeX rendering for `$...$` and `$$...$$` .
- A degradation-certificate row set for academic constructs — `$...$` , `$$...$$` , `[@key]` , `{#fig:x}` , `@fig:x` , footnotes, definition lists — across GitHub, CommonMark, Pandoc, Obsidian, Quarto.

22.6 Anti-recommendations

- Do not build a reference manager. Read a `.bib` ; never own the library.
- Do not build track changes or co-editing to chase the supervisor relationship. It is the segment's most-demanded feature and the most direct contradiction of byte-preserving splice. Losing it is correct.
- Do not build LaTeX/PDF typesetting or `.cls` support. Publisher class files are an unbounded tail with the same shape as the 10,860-file CSL tail.
- Do not bundle more than ~20 CSL styles. Accept a user-supplied `csl:` path.
- Do not chase Typst (55,717★). It is a new *format* competing with LaTeX; "no new markdown format" is settled, so Typst is orthogonal — not a competitor, partner, or threat.
- Do not price against India-student economics, and do not put "45 million Indian students" in any deck.

§23 Business model and unit economics

23.1 Revenue models compared

MODEL	FIT	RISK	SUPPORT-LOAD EFFECT
Subscription (settled)	High — matches Obsidian Sync/Publish, HackMD, GitBook, Craft	Monthly churn; India monthly economics punished by flat fees; competes with subsidised India AI	Highest per rupee
Flat perpetual licence (Obsidian model)	High for the Work SKU. Obsidian: app free, Catalyst \$25 one-time , Commercial \$50/user/yr and explicitly NOT required FETCHED	No recurring floor	Lowest — buyer expectation is "supported, not serviced". 90.3% contribution DERIVED
Usage / credits	High for frontier AI; makes COGS ≤ revenue by construction. Mintlify charges \$0.01/credit overage FETCHED	Meter anxiety suppresses the differentiating feature	Medium — "why was I charged" is the worst ticket category
Seat-based B2B	Medium-high	Pulls toward the SSO/SCIM/SOC-2 gauntlet	High and lumpy
Marketplace / templates	Medium	Requires a community that does not exist; India's markdown community is greenfield SS	Low direct, high indirect
Services	Low-medium	Non-scaling	Extreme — services IS support

Anti-recommendations: do NOT bundle unlimited AI at any INR price · do NOT enter India's subsidised-AI price war · do NOT regional-price a future team tier · if you sell publishing, price the site, not the seat · do NOT build enterprise KB, SSO, SCIM or SOC 2 before there is a team or capital.

23.2 Unit economics per tier

INR prices are GST-inclusive; base = price ÷ 1.18, GST 18%. **Razorpay 2% + 18% GST on the fee = 2.36% effective** FETCHED . Infra \$0.153/paid user/month at 1,000-user scale DERIVED, \$25 .

TIER	GROSS	RAIL	FEE	GST	OUT	NET	INFERENCE	INFRA	CONTRIBUTION	% GROSS
India Pro ₹299	₹299	Razorpay	₹7.06	₹45.61		₹246.33	₹38.16	₹14.62	₹193.55	64.7%
India Pro ₹299, cheap-model default	₹299	Razorpay	₹7.06	₹45.61		₹246.33	₹14.31	₹14.62	₹217.40	72.7%
India Power ₹599	₹599	Razorpay	₹14.14	₹91.37		₹493.49	₹95.40	₹14.62	₹383.47	64.0%
World Pro \$5	—	Paddle 5%+50¢	\$0.75	MoR		\$4.25	\$0.40	\$0.153	\$3.697	73.9%
World Power \$10	—	Paddle	\$1.00	MoR		\$9.00	\$1.00	\$0.153	\$7.847	78.5%
Work \$50/yr flat	—	Paddle	\$3.00/yr	MoR		\$3.917/mo	\$0	\$0.153	\$3.764	90.3%
Free (BYO-key)	₹0	—	₹0	—		₹0	\$0 by construction	\$0.0112	-₹1.07	—

Every INR tier clears 57–73% contribution. Margin was never the problem; volume is (\$25.3).

23.3 Inference cost — the real ceiling

One assist = 3,000 in + 700 out. All prices FETCHED 2026-08-29 .

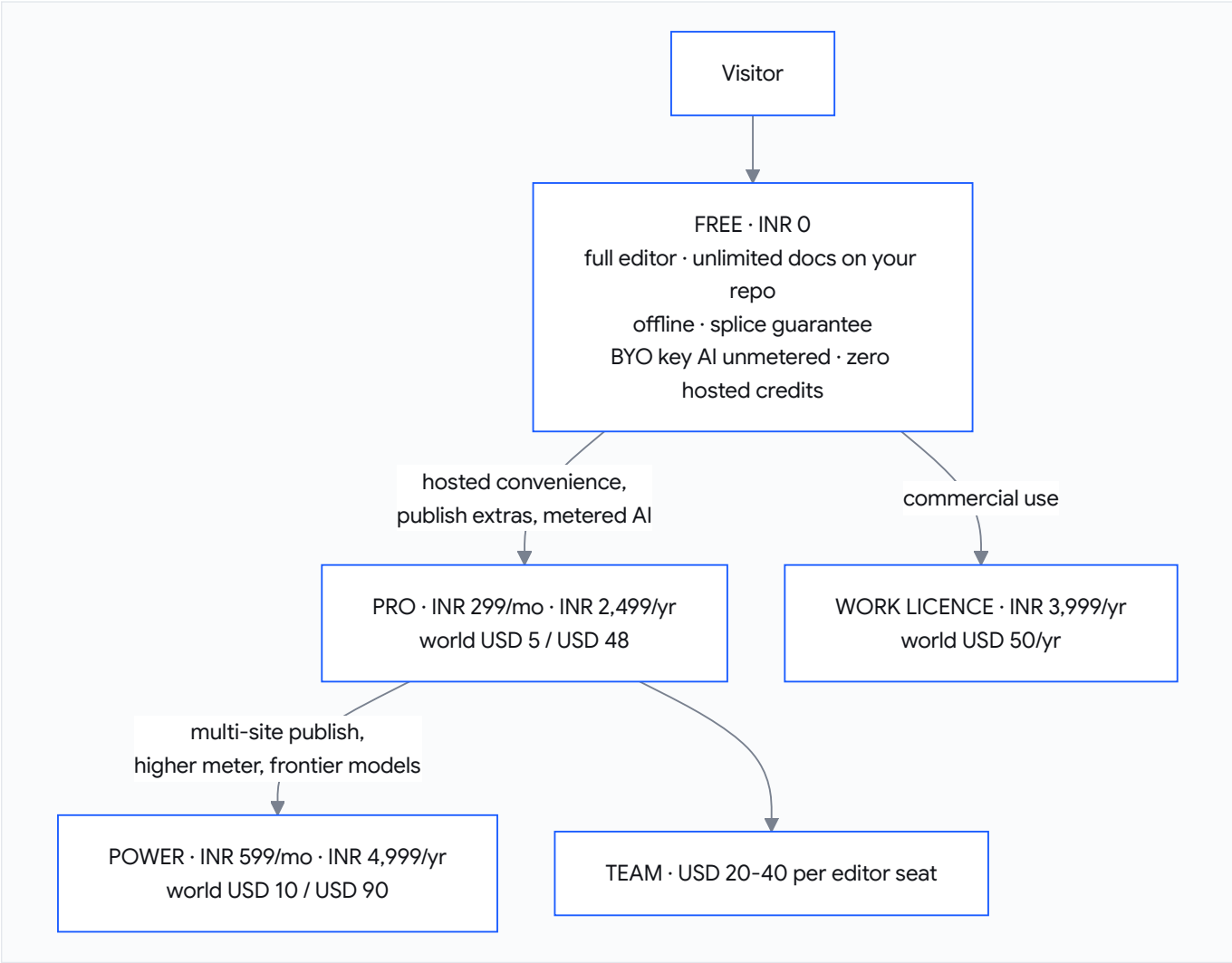
MODEL (IN/OUT PER MTOK)	\$/ASSIST	ASSISTS PER \$0.40
DeepSeek V4-Flash off-peak \$0.22/\$0.66	\$0.001122	356
Gemini 3.1 Flash-Lite \$0.25/\$1.50	\$0.001800	222
DeepSeek V4-Flash peak \$0.44/\$1.32	\$0.002244	178
Claude Haiku 4.5 \$1/\$5	\$0.006500	62
Claude Sonnet 5 \$2/\$10	\$0.013000	31
Claude Opus 5 \$5/\$25	\$0.032500	12

Four corrections to earlier assumptions, all FETCHED : Claude Sonnet 5 is **\$2/\$10, not \$3/\$15**, and the scheduled increase **will not occur** — our frontier assumption was 50% high. No "Gemini 2.5 Flash \$0.15/\$0.60" SKU exists. DeepSeek's \$0.22/\$0.66 is the **off-peak** rate; peak is 2×. **Gemini 3.7 Flash doubles on 2027-01-01** — meter in dollars, never in credits pegged to a model.

Prompt caching is the single largest COGS lever, because a document workspace re-sends the same file every turn. Cached heavy assist on Sonnet 5: **\$0.025 vs \$0.070 uncached** — 2.8× more assists per rupee FETCHED .

Tokenizer tax: Claude 4.7 and later produce approximately **30% more tokens** for the same text FETCHED . Any budget set on an earlier model understates cost by ~30%.

Free-tier inference is genuinely **\$0 only under BYO-key**. Gemini's own free tier is free but "content used to improve our products" — a fact the free tier must disclose, not hide.



24.1 The FX correction

The founder draft assumed ~₹83/\$. Live: ₹95.39 and ₹95.59 from two independent sources FETCHED. So ₹299 = \$3.13 and ₹699 = \$7.33, which sits above every India AI anchor (ChatGPT Go ₹399, Gemini AI Plus ₹199–399, Netflix Premium ₹649) and only 19–27% below a \$9–10 global tier. ₹699 becomes ₹599.

24.2 The tables

INDIA TIER	MONTHLY	ANNUAL (LEAD WITH THIS)	CONTENTS
Free	₹0	₹0	Full editor, unlimited docs on your own repo, offline, splice guarantee, BYO-key AI unmetered, zero hosted credits
Pro	₹299 (A/B ₹249 and ₹399)	₹2,499	Publish extras, live editing, hosted convenience, small metered AI on a cheap model with a visible meter
Power	₹599	₹4,999	Multi-site publish, higher meter incl. metered frontier, priority
Work	—	₹3,999/yr flat	Commercial-use licence — support and compliance, not more features

WORLD TIER	MONTHLY	ANNUAL	ANCHOR FETCHED
Free	\$0	\$0	Obsidian free-forever, no sign-up
Pro	\$5	\$48	Obsidian Sync \$4 annual/\$5 monthly; HackMD Prime \$5/seat annual (\$8 monthly)
Power	\$10	\$90	Obsidian Publish \$8/site annual — per SITE, not per user
Work	—	\$50/yr flat	Obsidian Commercial parity

24.3 Why ₹299 — and the counter-evidence

The anchor is what Indians already pay for AI productivity: ChatGPT Go ₹399; Gemini AI Plus ₹199→₹399; Netflix India ₹149/199/499/649; the Indian micro-SaaS starter band ₹299–499 SS.

Confirmed FETCHED: Notion has no India pricing — the page served to an India IP contains zero ₹ strings.

Counter-evidence worth taking seriously FETCHED: Craft already India-prices Plus at ₹526.7–₹658.3/mo — a direct competitor has decided the India individual price is 1.8–2.2× ₹299. That validates INR pricing and raises whether ₹299 leaves money on the table. Test ₹299 against ₹399 as well as ₹249.

24.4 The rail, and the ₹15,000 constant

RAIL	FEE ON ₹299/MO	NOTE
Razorpay (India domestic)	2.36%	2% + 18% GST on the fee. The flat-fee argument barely applies here
Dodo, India domestic	4% + 15¢ = 9.29%	The 15¢ flat is 4.79% of ₹299 — not the 12.76% an earlier note computed using the US 40¢ rate
Paddle / Lemon Squeezy	5% + 50¢ ≈ 15%	Paddle's own comparison prices a non-MoR stack at "≈7% and above"

Annual still wins — ₹2,499/yr drops the Dodo fee to **5.07%** **DERIVED** — but on Razorpay-domestic the annual argument is churn and convenience, not fees.

₹15,000 per transaction is an architectural constant, not a pricing input. RBI's 2026 Framework allows recurring authorisation without additional-factor authentication up to ₹15,000; software is not in the ₹1,00,000 carve-out. Full consequences in §45.

24.5 Which market first

Distribution India-first, revenue global-first. India has 21.9M GitHub contributors, +5.2M in a year, and +35% YoY consumer app spend **SS**. Against: **zero category-specific willingness-to-pay evidence for markdown tools in India** **SS**; the AI price here is collapsing to zero; ₹299 nets **\$2.58** against **\$4.25** for the identical feature at \$5; and **₹20L/mo needs 6,689 paying Indians versus 4,193 at \$5 — 59.5% more paying humans for identical revenue** **DERIVED**.

What matters most is not the discount — it is that UPI exists at checkout. International gateways without it lose 30–40% of Indian checkouts **SS**, **VENDOR-SOURCED**. Paddle and Lemon Squeezy do not support UPI.

24.6 Pricing policy — published, and treated as a promise

Free-forever editor · dollars not credits · BYO-key at every tier · one cheap surface unlimited · never auto-migrate plans · never reprice opaquely · commenters never bill.

§25 Cost structure and funnel math

25.1 Infrastructure

Assumptions stated so they can be replaced by measurement: 4% free→paid · 9,000 requests/user/mo · 8 CPU-ms/request · 20 MB/free user, 250 MB/paid · hosted inference for paid only, capped \$0.40 · support 0.02 tickets/free-user/mo, 0.10/paid, 12 min each. Rates **FETCHED**: Cloudflare Workers \$5/mo minimum, then +\$0.30/M requests and +\$0.02/M CPU-ms; **R2 storage \$0.015/GB-mo, Class A \$4.50/M, Class B \$0.36/M, egress FREE**.

LINE	100 USERS (4 PAID)	1,000 (40 PAID)	10,000 (400 PAID)
Workers	\$5.00	\$5.84	\$42.80
R2	free tier	\$0.29	\$69.03
Egress	\$0.00	\$0.00	\$0.00
Infra subtotal	\$5.00	\$6.13	\$111.83
Hosted inference	\$1.60	\$16.00	\$160.00
Total variable	\$6.60	\$22.13	\$271.83
Infra per paid user	\$1.250	\$0.153	\$0.280
Support load	2.3 tickets/mo	23.2	232 = 46.4 founder-hours

Three conclusions. Egress-free R2 is the load-bearing architectural choice — a document workspace's dominant byte flow is reads. The \$5/mo Workers minimum dominates below ~1,000 users, making infra per paid user **8.2× worse at 100 than at 1,000**. And **support, not compute, is the wall**.

25.2 Conversion benchmarks **FETCHED**

MOTION	GOOD	GREAT
Freemium self-serve	3–5%	6–8%
Freemium + sales-assist	5–7%	10–15%
Free trial	8–12%	15–25%

"The median conversion rate for developer-focused companies was 5% — half that of companies that do not sell to developers." **FETCHED** This governs us directly. We chose the hard half deliberately.

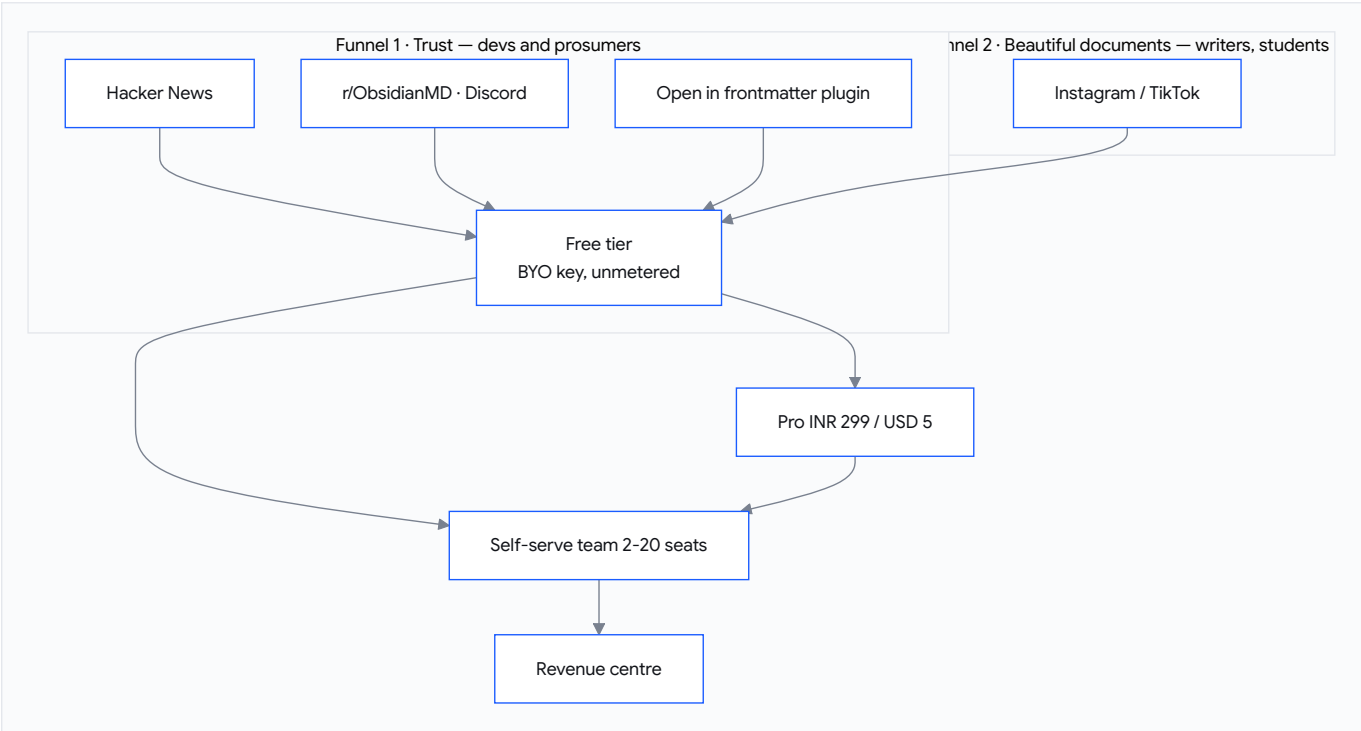
25.3 What each milestone requires **DERIVED**

Gross ARPU mix A (70% India ₹299 / 30% World \$5) = ₹352.40. Zero churn assumed.

MILESTONE	PAID USERS	FREE @5% (DEV MEDIAN)	CUMULATIVE VISITORS @9% SIGNUP
₹1L/mo	284	5,675	63,060
₹5L/mo	1,419	28,377	315,298
₹20L/mo	5,675	113,507	1,261,193

₹1L/mo is reachable at typical rates. ₹5L/mo needs ~28,000 free signups. ₹20L/mo needs 113,507 signups and ~1.26M cumulative visitors — a distribution problem, not a pricing problem, and the point where the plan stops being solo-founder-shaped.

§26 Go-to-market



	FUNNEL 1 – TRUST	FUNNEL 2 – BEAUTIFUL DOCUMENTS
Line	"The markdown source of truth AI can't corrupt"	"Beautiful documents from plain text"
Channels	HN, r/ObsidianMD (~344,000), Discord (~195,000), X, YouTube	Instagram, TikTok
Content	Evidence-first: the corpus result, a 10-second demo diff against a competitor's shipped product	Custom renders in 15 seconds of vertical video
Confidence	High. Our campaign practice already manufactures exactly this MEASURED	Low. No dev tool has grown Instagram-first, and our content pipeline mines the <i>system record</i> , so it would never surface aesthetic material on its own

The window is warm. The review-loop wedge was independently validated twice in four weeks — Markleft (Show HN) and OzBrain (92 points), whose founder pitches *literally* "the diffing, versioning and audit log of what was changed, by what agent and why" as the paid product FETCHED. And "Serve Markdown to AI Agents with Accept Headers" hit 175 points, 108 comments on 2026-08-26 FETCHED.

Channel discipline. The Obsidian community's code of conduct means the only admissible framing in the biggest watering hole is "integrates with your vault", never "Obsidian competitor" — break it and the launch is removed rather than debated. The "Open in frontmatter" plugin is the distribution play; Relay proved the path with 172,544 downloads of a commercial service's bridge plugin SS.

Cadence, grounded in measured supply. Three LinkedIn posts a week, a weekly canonical blog post, two Instagram carousels. The campaign system built 10 complete multi-surface episodes in about a week — supply is not the constraint MEASURED. What is *not* proven is sustained cadence: exactly two posts have ever shipped, one with a visible defect since 2026-08-10, three more on editorial hold MEASURED. So the survival mechanisms matter more than the plan: never miss twice; queue depth ≥ 2; read no metrics before post 20.

A publishing constraint to design around MEASURED: LinkedIn cannot post a PDF document through the current integration — the publisher implements images and videos only, and one item cannot carry a PDF for LinkedIn and PNGs for Instagram. The Documents-API entitlement probe exists, is read-only, takes ~5 minutes, and has still not been run.

Run the launch calendar inside frontmatter itself. The GTM engine becomes a continuous product demo, and the content pipeline's biggest gap — a sustained capture habit — becomes a product feature.

Never say "markdown editor." Every Show HN with that name becomes a thread of free alternatives. The category winners renamed the category: Obsidian sold ownership, Notion a workspace, Linear speed.

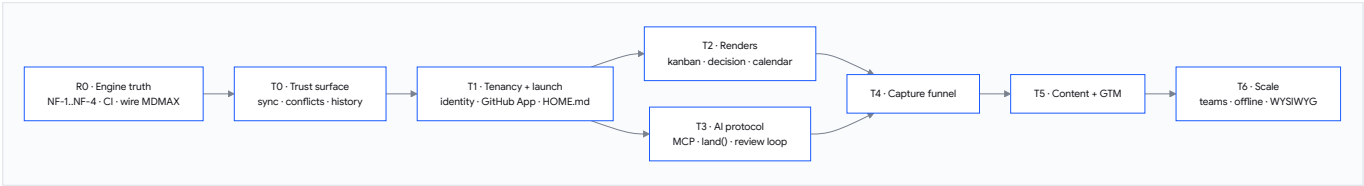
§27 Moat and defensibility

Ranked by hold-time, not by how good it feels.

RANK	MOAT	HOLDS	WHAT ERODES IT
1	Community / ecosystem	Years, compounding	Nothing external — but it does not exist yet, is the slowest to build, and India is greenfield SS
2	Byte-fidelity engine	18–36 months	An incumbent shipping a byte-exact writer. The work is hard but finite and increasingly LLM-assistable. GitBook's dominant complaint cluster is <i>reliability and lost work</i> — the gap is real today SS
3	Degradation certificate	12–24 months as an exclusive; longer as a standard	Commoditisation — which is also the win condition if we set the standard. Zero defensibility if the certificate is not independently checkable
4	Provenance / byte attribution	12–24 months	Platform content credentials shipping natively
5	Brand	Slow to build, durable once built	Naming risk (§52). "frontmatter" is the generic name of the substrate — <code>gray-matter</code> alone does 35.78M downloads/month
6	File-native data / no lock-in	Structural but non-exclusive	Every markdown tool claims it; seven free self-hosted alternatives sit at 21K–76K stars (§3.1)
7	Switching cost	Near zero, by design	This is the anti-moat. The portability that earns trust removes lock-in. Retention must be earned every month by the product, not by hostage-taking

Honest summary: the moat is engineering depth in a narrow place, plus timing. It is not distribution, brand, or lock-in — and it never will be. The go-to-market must convert engineering credibility into an outcome a buyer can name (§21).

§28 Roadmap



Points: $XS=1 \cdot S=2 \cdot M=4 \cdot L=8 \cdot XL=16$. One XL $\approx$ one substantial shipped surface $\approx$ 14–21 calendar days solo, so 1 pt $\approx$ 1.09 calendar days. Calendar days, not working days — the measured 25% active-day density on this branch is *inside* that rate, not a multiplier on it.

28.1 R0 — engine truth · 58 pts · the declared first lane

ID	DELIVERABLE	SIZE	SPEC
R0.1	NF-1 — a <code>-</code> item at column 0 is a continuation of the preceding key	M	<code>engine/nf-001-zero-indent-sequence</code>
R0.2	NF-2 — flow-seq closing <code>]</code> at column 0	S	—
R0.3	NF-3 — <code>FM_OPEN</code> misses a bare <code>---\r</code> ; a lone set prepends a second frontmatter block. Set-destructive and invisible to a round-trip oracle. Red proof first	M	<code>engine/nf-003-bare-cr-fence</code>
R0.4	NF-4 — quoted-key <code>SAFE_KEY</code> . Blocked on a decision , not on code	L	founder
R0.5	CI — port md's <code>ci.yml</code> ; all four harness scripts already exist	S	<code>platform/ci</code>
R0.6	Foreign-corpus standing gate	—	DONE — 8,513 files pinned, verify red-proofed
R0.7	Six construct-detector defects	M	—
R0.8	CJK — <code>countWords</code> segmenter, MiniSearch bigram tokenizer	M	gated on D17
R0.9	Wire MDMAX seams 1–3 (§7.3)	L	<code>engine/mdmax-seams</code>
R0.10	Reconcile <code>globals.css</code> with the design system, and fix the AA-failing token	M	—
R0.11	mdmax audit Tier 3/4 residue	M	—
R0.12	Replace the <code>npm run budget</code> stub with a real byte ceiling	S	—
R0.13	Close the open decisions. PAT rotation is today, not a milestone	M	founder

28.2 The remaining lanes

LANE	PTS	CONTENTS
T0 Trust surface	32	sync chip · conflict inbox · named-version history · since-you-last-opened banner · background auto-sync · local history + section restore · the two-device rig
T1 Tenancy + launch	42	identity + multi-tenancy <code>XL</code> · GitHub App <code>L</code> · multi-vault · mobile pass · HOME.md · quick capture · companion plugin
T3 AI protocol	44	MCP + <code>land()</code> <code>L</code> · review loop <code>L</code> · provenance + implicit telemetry · differ · cert distribution <code>L</code> · citation-gated answers · session continuity · publish the session format
T4 Capture	20	chat-side skill + paste inbox · ChatGPT/Claude importers <code>L</code> · promotion loop · retro-capture
T2 Renders	42	out of v1 except two <code>XS</code> cleanups: the <code>/Language-(\w+)/</code> hyphen fix and deleting dead <code>editable-table.tsx</code>
INFRA/LEGAL	25	MoR before the first <i>paid</i> signup · EU Art. 27 rep before the first EU <i>free</i> signup · CA + lawyer · ToS/privacy · auth, error tracking, R2 — all buy
GTM	34	land Markex's tree · the LinkedIn Documents-API probe · post-as-document · launch calendar in-product · 20 posts <code>XL</code>

28.3 Critical path

R0.3 → R0.1 → R0.2 → R0.4 → R0.9 → T0(auto-sync → local history → two-device rig)
→ T1(identity → GitHub App → HOME.md) → T3(MCP+land()) → review loop → cert distribution
→ T4(importers → promotion loop) → v1

R0.5 (CI) is not on the path but gates the credibility of everything after it — do it in the first 48 hours. T2 is off the path entirely. Legal items are date-gates, not effort-gates. R0.4 is the one engine unit an agent cannot start — it needs a written decision on Unicode key equality.

28.4 Milestones — every definition of done is executable

M	GATE	DONE MEANS
M0	CI live	A PR runs typecheck-lint-test-build-arch-spec and fails on a deliberately broken commit. <code>npm run budget</code> asserts a real byte ceiling. One red run is required before the first green is trusted
M1	Engine truth	Over the pinned corpus: <code>refused ≤ 2 of 7,969, changed = 0, threw = 0</code> . NF-3's set-only assertion fails on <code>HEAD~1</code> and passes on <code>HEAD</code> . <code>npm run corpus</code> exits 0 at gate time
M2	MDMAX wired	Seams 1–3 live; ≥8 of the 12 unreached files have a product importer; <code>npx mdmax cert --fail-on=BROKEN</code> exits 1 on a seeded fixture inside CI
M3	Trust surface	Two devices, both offline, both edit the same file, both reconnect: zero bytes lost and every divergence surfaced as a reviewable hunk, watched by a human, recorded. Section-level restore returns the file to a prior sha with every untouched byte identical
M4	Tenancy	A second GitHub account, never used in development, signs in via the GitHub App, connects a repo, edits, commits — no founder intervention, no shared secret
M5	Agent protocol	An external agent edits a real vault through <code>land()</code> ; every change appears in the review surface and is rejectable; a rejected change leaves the file byte-identical
M6	Capture	One ChatGPT export ZIP → durable typed documents in one click, and the verification report enumerates every dropped construct by count
M7	v1 public	M0–M6 green + MoR live + EU rep appointed + ToS published + one paying non-founder account

Reject these as a DoD: "kanban feels good" · "import works" · "fidelity is high" · anything whose evidence is a screenshot · anything a `grep` over source can satisfy.

28.5 Who does what

AN AI AGENT CAN OWN (DISJOINT ARTIFACTS)	ONLY THE FOUNDER
NF-1/NF-2 walk change + fixtures	The NF-4 key-equality decision
Corpus runner and gates	All open decisions
The six construct detectors, one agent each	PAT rotation
CJK segmenter + tokenizer	GitHub App registration, secrets, callback
CSS reconciliation	MoR, CA, lawyer, EU representative
Tier 3/4 residue	Watching the M3 two-device run
Test authoring	Accepting or rejecting every agent diff

Do not parallelise NF-1 and NF-4 against the same file, and do not fan out T0's sync work — it is one creative target carrying implicit architecture decisions.

28.6 Calendar

1 pt = 1.09 calendar days, from 2026-08-29. The assumption is stated so it can be argued with.

MILESTONE	CUM. PTS	DATE
R0 complete	58	2026-10-31
+ T0	90	2026-12-05
+ T1	132	2027-01-20
+ T3 (v1 subset)	176	2027-03-09
+ T4 = code-complete	196	2027-03-31
+ legal/infra = shippable	221	2027-04-28
+ full GTM	255	2027-06-04

R0 alone is 9 weeks. That is the part most likely to be wished away.

28.7 The ten things most likely to blow this

#	RISK	EARLY WARNING — CHECK WEEKLY
1	"NF-1 recovers 99.98%" is an inference from bucketing refusal causes, not a measured result of the patched writer	First patched run refuses >10 files. If the residual is >2, NF-1 was never one bug
2	Corpus loss or drift	Any run whose file count ≠ 8,513. Mitigated: pinned and red-proofed
3	NF-4 is a design task wearing a regex costume	Two weeks pass with no written NFC/NFD decision
4	"Wire MDMAX in" is an integration surface, not a wiring task	Week 2 of R0.9 still has 0 new product importers. Count them
5	CI's first green will be false. Four gates in this repo already reported green while blind	CI passes on a deliberately broken commit
6	Cadence is bursty — 8 active days in 32	Two consecutive weeks with zero commits
7	T1 identity is the only XL, scored from analogy	A week of schema churn instead of a signed-in second account
8	Open decisions gate scheduled work	Any lane started before its gating decision is written down
9	Legal lead times are vendor-clock	EU-reachable free tier with no Art. 27 representative
10	The name is unresolved	Any brand spend before \$52 closes

28.8 The first week, concretely

DAY	DO	PROVES
1	Rotate the two PATs. Then CI: port <code>ci.yml</code> , add <code>spec</code> and <code>corpus</code>	Only you can do the first
1	Make CI fail on purpose, once, before trusting any green	That the gate can see
2	NF-3 red proof — a set-only assertion on a synthetic bare-CR fixture that fails on today's code	Set-then-delete cancels out; that is how NF-3 stayed invisible. And the corpus cannot prove it — zero bare-CR fences in all 8,513 files
2–4	NF-3 fix, then NF-1 red proof and fix	The 83% refusal rate is the number the fidelity claim rests on
4	Re-derive the recovery rate from the patched writer	Replaces the 99.98% inference with a measurement
5	Write the NF-4 decision — one page on NFC/NFD key equality	Unblocks the only agent-blocked unit
5	The two <code>XS</code> cleanups	Unblocks every render profile

§29 Performance budgets and the testing strategy

29.1 The performance budget

All measurements below were executed on 2026-08-29 on Node v24.6.0 / darwin 25.6.0 / arm64, V8 `heap_size_limit` 4,288 MB [measured]. Repo state at the time: 79 `*.test.ts` files, 9,929 LOC of tests, vitest 4.1.7 installed (4.1.11 current), `fast-check` absent, `.github/workflows` absent [measured].

#	OPERATION	BUDGET	MEASUREMENT METHOD	ENFORCEMENT
P1	Byte splice, document ≤4 MB	≤5 ms p99	vitest bench, fixed corpus, 20 iterations	CI fails at p99 > 10 ms (2× headroom). Measured 4.73 ms end-to-end decode→splice→encode at 4 MB [measured]
P2	UTF-8 decode + validate on open	≤2 ms/MB	same	fail > 4 ms/MB. Measured 1.16 ms at 4 MB [measured]
P3	Any single parse of a user document	≤200 ms, else refuse	<code>shape-gate.ts</code> budget-ms	Refusal, not slowdown. Live constants: MAX_BYTES 4 MB, MAX_LINES 200,000, MAX_LIST_MARKER_LINES 20,000 [measured]
P4	Any regex run over user bytes	growth exponent $k \leq 1.05$ over 4 doublings	doubling harness; report the exponent, not the milliseconds	CI fails at $k > 1.2$
P5	Keystroke → paint	≤50 ms p95	Playwright trace + <code>PerformanceObserver</code> longtask	fail if any longtask > 50 ms across a 200-keystroke script
P6	Search, as-you-type prefix	≤50 ms p95 at 10,000 notes	headless bench against a frozen corpus	measured 12.64 ms at 10k; fails at 50k (130.90 ms) [measured]
P7	Search, exact term	≤5 ms p95	same	measured 1.64 ms at 50k [measured] — not the risk
P8	Cold index build, 10,000 notes	≤8 s, off main thread	Worker bench	measured 5,613 ms single-threaded [measured]
P9	Index heap	≤10× corpus bytes	<code>--expose-gc</code> heap delta	measured 9.25× at 50k (1,036 MB / 112.0 MB) [derived] — at the line, no headroom
P10	Cold start → first keystroke accepted	≤1,500 ms p75	Lighthouse + Playwright <code>page.type</code> timestamp	the editor must accept input <i>before</i> the index exists
P11	INP, field data	≤200 ms p75 [fetched, web.dev]	web-vitals RUM beacon	dashboard alert only — field data cannot gate a PR
P12	JS bundle, editor route	≤350 KB gzip	<code>next build</code> output	<code>npm run budget</code> , today a stub printing "No bundle budget configured yet" [measured]

Thresholds anchor to RAIL: 100 ms response, 50 ms task chunk, 1,000 ms focus loss, 10,000 ms abandonment; frame budget 16.7 ms [derived: 1000/60] [fetched, web.dev]. INP is scored at the 75th percentile of field page loads, mobile and desktop segmented, discarding the single highest interaction per 50 [fetched].

Gate on the exponent (P4) and on refusal correctness (P3), not on wall-clock. Two independent runs of the identical `WIKILINK_RE` benchmark over the same 320 KB input disagree by 2.63× — the repo archive records 36,865 ms at $k=1.98$, this machine measured 96,890.7 ms at end-to-end exponent 2.37 [measured][derived: 96890.7/36865]. Both stand; neither is corrected; the exponent is the invariant and the milliseconds are the machine. Anti-recommendation: a wall-clock gate on a shared runner is the classic flaky-red generator, so only P1/P2/P6/P7 carry ms thresholds, each with 2× headroom, each pinned to one runner class.

Published scale ceiling. MiniSearch 7.2.0, synthetic 400-word notes [measured; extrapolations derived from 0.554 ms/doc and 20.7 KB/doc]:

Vault	Build	Heap	Prefix Search	Published Status
1,000	0.55 s	21 MB	1.52 ms [measured]	Supported
10,000	5.5 s	207 MB	12.64 ms [measured]	Supported
50,000	27.7 s	1.04 GB	130.90 ms [measured]	Degraded — named, not hidden
100,000	~55 s	~2.07 GB	~357 ms [derived: 130.90 × 2 ^{1.45}]	Refused in the browser

- Publish this sentence verbatim: "Tested to 10,000 notes. Functional and honest about it to 50,000. Above 50,000 the web client refuses and tells you why; the desktop build is the supported path."
- The ceiling is set by heap, not CPU: 2.07 GB against a 4,288 MB V8 limit leaves nothing for the document, CodeMirror or React, and a browser renderer is stricter than Node [measured][inference].
- Anti-recommendation: do not publish a single "max notes" number. Obsidian 1.13.7 in restricted mode with zero community plugins and ~10,500 files reports a ~280 ms renderer stall every ~2 s while typing, attributed to `getAllPropertyInfos()` [fetched, forum.obsidian.md]. Users hit unusability long before they hit a cap; publish the stall behaviour.
- Anti-recommendation: do not buy headroom by shrinking the index (dropping body text, aggressive stemming). That converts a stated limit into a silently worse product, which is the exact failure the REFUSE doctrine exists to prevent.
- Route parsers by input *shape*, not by library name. On a 100 KB realistic note `mdast-util-from-markdown` 2.0.3 is 0.96× `micromark` 4.0.2 — identical; on 32,000 flat bullets (404.9 KB) it is 29.36× slower, exponent 4.00 in the 16k→32k band [measured][derived]. Recorded and not explained: micromark itself goes flat across that band (794.6 → 801.3 ms), which is not credible as linearity and is probably a JIT/GC artifact — do not cite micromark as linear.
- The splice thesis, quantified: 4.73 ms to decode, splice and re-encode 4 MB versus 23,527 ms to reparse a 405 KB pathological list is **4,974×** [derived]. The architecture argument is four orders of magnitude, not an aesthetic.

29.2 The test strategy and its pyramid

Proportions are stated as shares of CI wall-clock, not as test counts; 79 files and 9,929 LOC [measured] say nothing about coverage of the byte contract.

Layer	Wall-clock	Owns	Never Owns
L0 Property + differential	25%	splice algebra, offset arithmetic, refusal totality, UTF-8 boundaries, line endings, degradation certificate	anything requiring a DOM
L1 Unit (vitest)	15% (≈55% of test count)	pure functions, error enums, parser helpers	integration ordering, timing
L2 Corpus / golden	20%	real foreign vaults — the zero-indent-sequence and bare-CR classes; <code>npm run corpus</code> exists [measured]	synthetic inputs
L3 Integration (jsdom + fake-indexeddb, both installed [measured])	15%	CodeMirror↔engine sync, repository, R2 persistence	rendering fidelity
L4 E2E (Playwright), ≤20 specs	20%	five data-loss flows only: open, edit-save, offline→reconnect, export, share-revoke	assertions a byte test can make
L5 Visual	5%	KaTeX and Mermaid render regressions only	anything textual

Decision	Choice	Evidence	Anti-recommendation
E2E runner	Playwright	<code>@playwright/test</code> 1.62.1 at 57,911,488 weekly vs <code>cypress</code> 15.21.1 at 7,569,177 — 7.65× [fetched npm, derived]. Cypress Cloud: Free 500 test results, Team \$67/mo or \$799/yr, Business \$267/mo or \$3,199/yr [fetched cypress.io/pricing]; Playwright sharding is in the free OSS runner	The advantage is cost and CI shape, not authoring ergonomics. Cypress's time-travel debugger is better and we are giving it up
Visual regression	5% ceiling, KaTeX/Mermaid only	Chromatic Free \$0, paid tiers \$179 and \$399 [fetched chromatic.com/pricing; snapshot allowances did not parse — unverified]	Never visual-test prose. Font hinting and GPU rasterization differ per runner; a text-heavy diff suite is all noise and gets switched off inside a month
Mutation testing	<code>@stryker-mutator/core</code> 10.0.0, nightly, scoped to <code>src/modules/mdmax/domain/</code>	2,320,443 weekly [fetched npm]	Never on the PR path. Whole-repo mutation over 9,929 LOC is hours of runner time for a quarterly reading, and the score is not a gate
Contract testing	Schema-validated fixtures	Both contracts (engine↔CodeMirror, client↔Worker/R2) are in-process or single-team	Do not adopt Pact or a broker. Consumer-driven contracts pay across <i>team</i> boundaries; a solo founder has none

CI shape and cost, at published GitHub rates [fetched docs.github.com/en/billing/reference/actions-minute-multipliers, 2026-08-29]: Linux 1-core slim \$0.002/min, Linux 2-core x64 \$0.006, Linux 2-core arm64 \$0.005, Windows 2-core \$0.010, macOS 3/4-core \$0.062; included allowances GitHub Free 2,000 min + 500 MB, Pro 3,000 min + 1 GB.

Job	Trigger	Runner	Est. Min
typecheck + lint + unit + property (pinned seed) + <code>arch</code> + <code>spec</code>	every push	Linux 2-core	12
corpus (foreign vaults)	every push	Linux 2-core	3
Playwright, 3 shards	every push	Linux 2-core ×3	6
perf bench (P1/P2/P6/P7 + exponent P4)	every push	Linux 2-core, pinned class	4
property soak, random seed	nightly	Linux 2-core arm64	30
Stryker on <code>mdmax/domain</code>	nightly	Linux 2-core arm64	25
Tauri macOS build	weekly + tag	macOS	20

- PR path 12+3+6+4 = 25 min/push; 40 pushes/mo = 1,000 min. Nightlies (30+25) × 30 = 1,650 min. Linux total 2,650 min/mo; 650 billable over the 2,000 free allowance × \$0.006 = **\$3.90/mo**. macOS 20 min × \$0.062 × 4.33 runs/mo = **\$5.37/mo**. Total ≈ **\$9.27/mo** [derived]. macOS is 10.33× Linux per minute and is 58% of spend on 1% of the minutes [derived: 0.062/0.006].
- Under the free allowance with nightlies removed: 2,000/25 = 80 pushes/mo at zero cost [derived].

- Every CI duration above is an estimate, not a measurement — there is no workflow file to measure [measured]. The arithmetic is shown so each row can be replaced with a real number on the day the first workflow runs.
- Anti-recommendation to arm64 nightlies (17% cheaper [derived: 1-0.005/0.006]): arm64 is a different runner class, so any ms-threshold job must stay pinned to one architecture or P1/P6 stop meaning anything across runs.
- The correct first commit in this area is a workflow file, not a test. None of the existing 9,929 LOC is property-based, none is E2E, and none runs in CI, because there is no CI [measured].

29.3 Property-based testing for a byte-preserving writer

fast-check 4.9.0, 37,512,910 weekly downloads [fetched npm, 2026-08-29]. Model: "for any (x,y,...) such that precondition(x,y,...) holds, predicate(x,y,...) is true" [fetched fast-check.dev]. Two authoring rules from that documentation are binding here: the predicate must not mutate its inputs or shrinking degrades and the reported counterexample is wrong; and constrained arbitraries beat .filter , which generates-then-discards.

Generate at the byte layer — fc.uint8Array() , never fc.string() . A string arbitrary cannot produce invalid UTF-8, a lone continuation byte, or a bare CR, and those are precisely the classes that have already shipped as defects. Derive ranges as start = fc.nat(len) then end = start + fc.nat(len - start) ; never fc.tuple(nat, nat).filter(([a,b]) => a <= b) .

#	PROPERTY	STATEMENT	DEFECT IT WOULD HAVE CAUGHT
S1	Outside-range invariance	out.subarray(0,s) ≡ D.subarray(0,s) and out.subarray(s + p.length) ≡ D.subarray(e) , compared with Buffer.compare(...) === 0 , never string equality	The leading-BOM bug fixed in f47555f . String equality passes with a stripped BOM; byte comparison does not
S2	Length algebra	out.length === D.length - (e - s) + p.length	The OffsetMap off-by-3 fixed in cc1d451 . An off-by-N is a total arithmetic falsehood and needs no oracle
S3	Identity splice	splice(D, s, e, D.subarray(s,e)) ≡ D	The cheapest total oracle in the suite; the off-by-3 dies here around run 4
S4	Exact inverse (undo)	splice(splice(D,s,e,p), s, s + p.length, D.subarray(s,e)) ≡ D	Makes "deterministic, reversible projection" a theorem rather than a slogan
S5	Disjoint commutation	for non-overlapping ranges, applying in either order with offsets shifted yields identical bytes	Batch-edit offset drift; the multi-cursor class
S6	Refusal totality	every input yields {ok:true, bytes} xor {ok:false, code ∈ closed enum} ; ok:false ⇒ bytes === undefined . No third state	The "guessed instead of refusing" class the whole doctrine exists to prevent
S7	Refusal determinism	the same input twice yields the same code , asserted across two calls inside one predicate	A non-deterministic refusal cannot be reported or reproduced, which is worse than a wrong one
S8	UTF-8 boundary refusal	if s or e indexes a continuation byte (0b10xxxxx), the result must be ok:false , never valid-looking output	The multi-byte truncation class, which reproduces in ~0.34% of natural artifacts and therefore never surfaces in a sampled corpus
S9	Line-ending census	counts of CRLF, lone LF and lone CR outside [s,e] are unchanged	The queued bare-CR set-destruction class
S10	No-op idempotence	splice(D, s, s, empty) ≡ D for all s , including 0 and D.length	Boundary and empty-document handling
S11	Projection commutes	an edit applied through a view equals the equivalent splice applied to bytes	The Projection Law, as an executable assertion
S12	Fence isolation (model-based, fc.commands)	a sequence of body splices never moves the --- fence byte offsets, and the converse	Frontmatter boundary drift across an editing session
S13	Metamorphic certificate stability	for a splice certified non-degrading, the cross-engine render of before and after differs only inside the spliced region	Regressions in the degradation certificate itself

- Runner discipline: the PR path runs a pinned seed and a fixed numRuns ; a random-seeded property test on a PR is a coin-flip gate. Nightly runs random seeds at 100× numRuns . S1, S2, S3 and S6 — the four cheap total oracles — run on the PR path at low numRuns ; S4–S13 run nightly. The explicit cost: a property defect in S4–S13 can merge and sit for up to 24 hours.
- Every shrunk counterexample is promoted by hand into a deterministic unit fixture and never deleted. A shrunk failing input is the most expensive artifact the suite produces and the cheapest to lose.
- Anti-recommendation: properties will not find defects that live in interpretation. "Should a bare CR terminate a setext heading" is a spec question, and a property will happily confirm whichever answer was encoded. S1–S13 protect the byte contract; layer L2 protects the semantics.
- Anti-recommendation: do not property-test the parser's output shape. mdast trees are large, shrinking over them is slow, counterexamples are unreadable, and the assertions degenerate into a second implementation of the parser.

§30 Data model, schemas and migration

30.1 Everything that needs a version

Live state: exactly one file in the repo carries \$schema — specs/_schema/spec.schema.json , which requires spec: { const: 1 } and sets additionalProperties: false [measured, 2026-08-29]. Existing in-code carriers: Certificate.schema: 'mdmax/cert@1' (cert-contract.ts:132), Certificate.fold: { version } (:137), FOLD_VERSION = 'mdmax/fold@1' (fold.ts:45 , with the comment at fold.ts:40 stating that changing any rule requires bumping it because the certificate quotes it), and per-engine version / sha in bench.engines[] (cert-contract.ts:81,83) [measured].

#	ARTIFACT	BYTES HELD BY	VERSION CARRIER	PLACEMENT	MAY WE REWRITE IT?
1	Document frontmatter (user's own keys)	user file	none of ours	—	never unattended
2	User-defined frontmatter schema	user file (<code>.frontmatter/schema.yml</code>)	<code>fm_schema: 1 + \$schema</code> URI	first key, in-band	explicit action only
3	Render profile definition	user file or R2	<code>profile: 1 + \$schema</code> URI	first key	explicit action only
4	Degradation certificate sidecar	sidecar / R2	<code>schema: "mdmax/cert@1"</code> [measured — exists]	first key	regenerate, never edit
5	Fold ruleset	engine constant	<code>mdmax/foLd@1</code> [measured — exists]	code, quoted into the cert	N/A
6	Splice refusal vocabulary	engine; surfaced in cert + UI	<code>refusals@N</code> — missing today	code + cert	N/A
7	Session interchange format	exported file, published spec	<code>\$schema</code> URI + <code>session: N</code>	first key	never — it is an artifact
8	Saved searches / views	server row + optional export	column <code>v</code> int; export <code>saved_view: 1</code>	row + first key	yes, we hold it
9	Server-side object metadata	R2 custom metadata, DB	<code>x-fm-schema: fm/objmeta@1</code>	HTTP metadata	yes, we hold it
10	Workspace config	user file	<code>workspace: 1</code>	first key	explicit action only
11	Engine/target registry (bench)	repo + cert copy	per-engine <code>version / sha</code> [measured — exists]	embedded in cert	N/A
12	Export artifacts (HTML/PDF/print)	user's disk	generator stamp in comment/XMP	header	no — immutable output
13	Local index/cache (Tauri)	user's disk, derived	<code>cache: N</code> , rebuild on mismatch	sidecar DB pragma	yes — it is derived
14	AI tool-call schemas	wire	API version, not file version	request header	N/A

- Placement follows three tiers [inference, modelled on Avro]: (a) in-band, first key, self-describing for anything a user holds — Avro's rule is that "the original schema must be provided along with the data" [fetched, avro.apache.org 1.12.0, Schema Resolution], and we cannot ship a registry to an offline laptop; (b) sidecar header for derived artifacts we generate beside a document; (c) transport metadata for objects we hold.
- Absence rule, written down once and never re-interpreted: a missing version key means v1, permanently.** JSON Schema 2020-12 states that when `$schema` is "absent from the document root schema, the resulting behavior is implementation-defined" [fetched, json-schema.org draft/2020-12 core §8.1.1] — implementation-defined is exactly the hole, so we define it.
- Anti-recommendation: never infer a version by sniffing which keys are present. Key-shape sniffing makes every future additive change retroactively alter the parse of old files.
- Source disagreement, recorded: JSON Schema permits unversioned documents and leaves the behaviour open; Avro requires the writer's schema to accompany the data at all times [fetched both]. The specs disagree on whether unversioned data is admissible at all; we side with Avro for user-held files and with JSON Schema's leniency for reading foreign ones.

30.2 The compatibility contract

Anchors: semver — MAJOR for incompatible changes, MINOR for backward-compatible additions, PATCH for backward-compatible fixes, and "Major version zero (0.y.z) is for initial development. Anything MAY change" [fetched semver.org]. Confluent — BACKWARD (the default), FORWARD, FULL and their TRANSITIVE variants; BACKWARD checks only the previous version, BACKWARD_TRANSITIVE checks all [fetched docs.confluent.io].

RULE	STATEMENT	ANCHOR
C1	Every artifact declares BACKWARD_TRANSITIVE , not BACKWARD. A user opens a 2021 file, not last quarter's file	[fetched] Confluent: BACKWARD "ensures that consumers using the new schema X can process data written by producers using schema X or X-1, but not necessarily X-2"
C2	User-held artifacts additionally promise FORWARD for one MAJOR: an older build opens a newer file readably, degrading unknown keys, never erroring	[inference] The user's other machine is on the old build; we do not control rollout
C3	Unknown keys are preserved verbatim on write, never dropped	[fetched] Avro ignores unknown writer fields; we must go further and round-trip them, because our file is the source of truth, not a wire frame
C4	A new required field is a MAJOR change. New fields are optional or carry a reader-side default	[fetched] Avro signals an error for a reader field absent in the writer with no default; protobuf.dev: "Required fields are considered harmful by so many they were removed from proto3 completely"
C5	Never re-use a key name for a different meaning. Retire names into a reserved list	[fetched] protobuf.dev: "Never re-use a tag number... You can also reserve names to avoid recycling now-deleted field names"
C6	Never change a key's type in place. Add a new key, dual-read, retire the old	[fetched] protobuf.dev: "changing a field's type can be difficult to roll out safely even when the new schema can successfully parse old data"
C7	New enum members (verdict classes, refusal reasons, profile modes) require a reader-side default. An unknown member is never a crash	[fetched] Avro symbol-resolution rule; protobuf.dev: "Do Include an Unspecified Value in an Enum"
C8	Documentation and annotation fields never participate in compatibility checks	[fetched] Avro: "A schema's <code>doc</code> fields are ignored for the purposes of schema resolution"
C9	Two axes, never conflated: format version (monotonic integer, in-band) and product version (semver, in the changelog). A file says <code>session: 2</code> ; the app says <code>1.7.3</code>	[inference] semver's "0.y.z anything MAY change" is a statement about APIs, not about bytes on someone else's disk
C10	Any schema we publish pins its dialect: <code>\$schema</code> is a normalized URI with a scheme, at the document root	[fetched] json-schema.org 2020-12 core §8.1.1
C11	<code>additionalProperties: false</code> is banned in user-facing schemas	[measured] correct in <code>specs/_schema/spec.schema.json</code> because it is internal; fatal in a user file, because a closed schema makes an old build reject a new file instead of degrading it, breaking C2
C12	Anything the certificate quotes (<code>foLd.version</code> , engine <code>version / sha</code>) is part of the compatibility surface. Bumping it invalidates prior certificates rather than re-labelling them	[measured] <code>foLd.ts:40</code> ; [fetched] protobuf.dev: "Never Rely on Serialization Stability Across Builds"

- Anti-recommendation to C2: do not promise forward compatibility forever. One MAJOR, then a "this file was written by a newer version, please update" refusal. A refusal is honest; silently dropping keys the old build did not understand is the failure C3 exists to prevent.
- Anti-recommendation: do not use semver on file formats. MINOR permits additions an old reader cannot see, which is exactly the wrong promise for bytes on disk.
- Source disagreement, recorded: Confluent ships BACKWARD as its default while the same page recommends BACKWARD_TRANSITIVE for Protobuf "as adding new message types is not forward compatible" [fetched, same document]. We take the transitive line.

30.3 Migrating data we do not hold

Confluent's registry, GitLab's post-deployment migrations and Avro's writer-schema-with-data all assume the migrator can reach the bytes. We cannot: a Tauri user's vault on an offline laptop is unreachable, permanently [inference]. **No user-held format may ever require a migration to remain readable; migration is optional cleanup, never a precondition**

for correctness.

Read-old-write-new, stated as three phases [fetched martinowler.com/bliki/ParallelChange.html, Danilo Sato, 13 May 2014 — "expand, migrate, and contract"]:

PHASE	BEHAVIOUR
Expand	The reader accepts old and new. The writer still emits old . Ship, wait
Migrate	The writer emits new only for files the user's own edit already dirties. Never a sweep
Contract	The reader drops old support only after the deprecation clock expires — and "Ignoring and dropping columns should not occur simultaneously in the same release" [fetched docs.gitlab.com]

The touch budget. When a user edits a document we already rewrite the byte range they touched; a frontmatter migration may ride along **only if it sits inside a range that edit dirties anyway**, and may never widen the dirty range. The consequence is deliberate: migration converges asymptotically over months of ordinary editing, and never produces an mtime change the user did not cause.

When we may rewrite a user file — four gates, conjunctively. All four, or we do not write.

GATE	REQUIREMENT
G1 Consent	The user pressed a button whose label names the change. Precedent: Obsidian's Format converter is a core plugin, opt-in, whole-vault, and warns "Back up your Obsidian files before you perform the conversion" [fetched help.obsidian.md]
G2 Preview	A byte diff is shown, per file, before anything is written. Anti-recommendation: never a bare count ("412 files will be updated") without the diff for at least the first file
G3 Reversibility	An undo artifact — original bytes, hashed, recoverable — exists before the first write
G4 Refusal on ambiguity	If the old form cannot be mapped deterministically the file is skipped and reported , not guessed. This is the splice contract applied to migration

The certificate constrains migration mechanically, not just by policy: `Certificate.file: { path, sha256, bytes }` [measured `cert-contract.ts:138`] makes any byte change a cert-invalidating event, and `CertBlock.byteRange: [number, number]` [measured `:118`] means a frontmatter change that alters block length shifts every downstream offset — re-derive certificates, never arithmetic-shift a stored one. A migration that cannot be expressed as a splice is a migration we are not allowed to run.

- Anti-recommendation, and the most tempting shortcut available: do not add a "normalize on save" mode that reserializes frontmatter through a YAML round-trip. It would break every certificate in the corpus at once, reorder keys and destroy comments — and it makes every migration trivial to implement, which is why it will keep being proposed.
- Anti-recommendation: do not auto-migrate on open. It changes mtimes the user did not cause, triggers their sync client, and invalidates certificates invisibly.
- Anti-recommendation: do not rename an identity field. Astro's `slug` → `id` rename in the v5 Content Collections change needed a `legacy.collections` escape flag rather than a converter [fetched docs.astro.build/en/guides/upgrade-to/v5/]. Identity renames are not migratable; they are forks. The legacy tier's own wording is the model for the un-migratable: features "no longer recommended and... in maintenance mode... will eventually be deprecated, and then removed entirely" [fetched].

Deprecation ladder. Both gates are mandatory, version and date — GitLab encodes exactly this pair as `ignore_column :updated_at, remove_with: '12.7', remove_after: '2019-12-22'` [fetched docs.gitlab.com].

STAGE	GATE	MINIMUM DURATION
D0 Announce	Changelog + in-app notice naming the key and its replacement. The converter ships here	—
D1 Dual-read	Reader accepts both; writer emits old; old form documented as deprecated	≥1 MINOR
D2 Dual-write	Writer emits new on touched files only; reader still accepts both	≥2 MINOR
D3 Warn	Opening a file with the old form shows a one-line dismissible notice with a "convert this file" action	≥1 MINOR
D4 Remove	Reader stops accepting. Requires a MAJOR and an elapsed-date gate	MAJOR only

- Minimum wall-clock window for anything user-held: **18 months**. The observed comparable is 22.0 months — Obsidian's deprecated-properties help section was created 2023-08-04 and the 1.9.2-era docs commit is dated 2025-06-05, a span of 671 days [derived: $671 \div 30.44$]. That figure bounds announce→docs-updated, **not** announce→removal: the exact 1.9.0 release date could not be fetched because obsidian.md/changelog returned HTTP 404 on both RSS and sitemap on 2026-08-29 [measured]. Treat 22.0 months as unverified for the interval it is often quoted as.
- Ship the converter at deprecation, not at removal. Obsidian deprecated `tag`, `alias` and `cssclass` in 1.4, dropped support in 1.9, and shipped the automated converter "As of Obsidian 1.9.3" [fetched] — the same line that removed support, which makes the warning window unactionable.
- Anti-recommendation: do not run a fixed removal calendar. A calendar forces removals with no user benefit and manufactures MAJOR bumps with nothing in them. Removal is event-driven: the old form's measured share of live files must fall below a stated threshold **and** the clock must have expired.
- Source disagreement, recorded: GitLab mandates a fixed three-release ladder with both gates; Astro's legacy path is open-ended with no stated clock [fetched both].
- Migration needs the same closed refusal vocabulary the engine already has — `MIGRATE_REFUSED_AMBIGUOUS`, `MIGRATE_REFUSED_UNKNOWN_VERSION` — versioned as `refusals@N`, which does not exist today [measured].

§31 Sync architecture

Decision: git three-way merge over a stored true base, mediated by server-issued revisions and compare-and-swap, with an append-only splice journal as a checkable derivative. Never a CRDT. This is settled; the subsections below record why, so it is not re-litigated.

31.1 The decision and its three disqualifications

#	DISQUALIFICATION OF A CRDT	EVIDENCE
D1	A CRDT cannot own the file bytes. The projection law says the file is the only source of truth and every view is a reversible projection owning no state. In a CRDT architecture the CRDT document <i>is</i> the record and the <code>.md</code> file becomes the projection — the axiom inverted. You also inherit history GC, snapshot compaction, and a second persistence format to back up.	[inference] on the stated product law; decision matrix scores CRDT 2/5 on byte fidelity for exactly this reason
D2	Convergence buys byte-identical garbage. <i>The Art of the Fugue: Minimizing Interleaving in Collaborative Text Editing</i> , arXiv 2305.00583v3, published 2023-04-30: when two users concurrently insert text at the same position, the merged outcome may interleave the inserted passages, "resulting in corrupted and potentially unreadable text... The problem has gone unnoticed for decades, and it affects both CRDTs and Operational Transformation."	[fetched] Loro's README credits Fugue for its text layer [fetched]; Yjs's YATA and Automerge's list algorithm predate it. Convergence is not the same property as zero loss [inference]
D3	A CRDT cannot refuse. Refusal is not in the algebra — the entire design goal is that conflicts never surface. frontmatter's engine contract is locate-the-range, replace-those-bytes, REFUSE rather than guess. A substrate that structurally cannot refuse cannot implement that contract.	[inference]; CRDT scores 1/5 on both "conflict visibility" and "fit with refuse rather than guess" in the weighted matrix

Weighted decision matrix (1–5, 5 best; weights ×3 for convergence, offline, byte fidelity, refuse-fit; ×2 for conflict visibility and solo-founder cost) [derived from the scored table]:

OPTION	WEIGHTED TOTAL / 80
Git-merge + splice journal + CAS	74
Git-only as transport	69
LWW + conflict copy (baseline)	62
CRDT (Yjs / Loro / Automerge)	51
Server-authoritative OT	40
Fuzzy patch — what Obsidian ships today	32

Measured: what fuzzy patching does to markdown. `diff-match-patch@1.0.5` from unpkg, executed in node, instance defaults read live: `Match_Threshold 0.5, Match_Distance 1000, Patch_DeleteThreshold 0.5, Patch_Margin 4` [measured].

CASE	SETUP	RESULT	FLAG RETURNED
Replay	Patch <code>one two three → one two three four</code> , applied to an already-advanced doc, then applied a second time	<code>one two three four four</code>	<code>results = [true]</code> both times [measured]
Fuzzy misapply through changed context	Base <code>The quick brown fox jumps...</code> ; patch changes <code>jumps → leaps</code> ; server text meanwhile says <code>brown cat</code> plus an appended sentence	<code>The quick brown cat leaps...</code> — applied an edit whose anchoring context no longer existed	<code>[true]</code> [measured]
Repetitive markdown	Six identical <code>- []</code> task lines, tick the first; server has only three lines left	Ticks a line that is not the one that was ticked	<code>[true]</code> [measured]

The library states the intent itself: "Use best-effort to apply patch even when the underlying text doesn't match", and its own API wiki concedes the success flags are unreliable because large patches "may get broken up internally... with no way to figure out which patch succeeded or failed" [fetched, github.com/google/diff-match-patch README + wiki/API]. The replay result reproduces the exact shape of forum.obsidian.md topic **94732** (created 2025-01-12, 105 posts, 4,342 views, "Obsidian Sync incorrectly duplicates sections of files", reporter in Restricted Mode with no plugins, **no merge-conflict notification and nothing in File Recovery**) [fetched]. Recorded disagreement, not resolved: Obsidian's docs say auto-merge "saves all edits"; topic 94732 reports content *added that was never typed* with no conflict logged [both fetched] — both cannot be true of the same algorithm.

Measured: what git's three-way merge does to markdown. `git version 2.50.1` (Apple Git-155), run via process substitution, no files written [measured].

CASE	INPUT	RESULT	EXIT
M1	A edits line 1, B edits line 3	both applied cleanly	0
M2	Both append at EOF	conflict markers, nothing lost	1
M3	A edits para 1, B edits para 2	both applied cleanly	0
M4	No trailing newline, identical sides	bytes <code>41 0a 42</code> — trailing-newline absence preserved	0
M5	CRLF file, one side edits	bytes <code>41 0d 0a 42 32 0d 0a</code> — CRLF preserved verbatim	0
M6	<code>--union</code> on same-line conflict , <code>title: mine</code> vs <code>title: theirs</code>	emits both lines — silently invents a document with two <code>title:</code> keys	0
M7	Empty base, no common ancestor	conflict markers, no guessing	1
M8	Whitespace-only divergence, <code>item\t</code> vs <code>item_u</code>	conflict markers — invisible-character divergence surfaced, not swallowed	rc masked by pipe
M9	File already contains <code><<<<<< HEAD</code>	nested markers ; result unparseable as a merge	1

git's merge is byte-faithful and conservative: it preserves CRLF and missing-final-newline exactly (M4, M5) and refuses (M2, M7, M8) precisely where a splice writer would want to refuse [measured]. `git merge-file --union` **is BANNED in every code path — it is the one mode that guesses, and M6 shows it manufacturing a document with two `title:` keys and exit 0** [measured]. M9 is a live hazard for a markdown editor whose users write about git.

Honest counter-argument. The reason Obsidian uses diff-match-patch is that conservative merging produces conflicts users hate. M8 is the proof against our own position: a tab-versus-two-spaces divergence — invisible to the human — produces conflict markers and a scary artifact [measured]. If frontmatter's conflict rate is materially higher than Obsidian's, "refuses rather than guesses" reads to a paying user as "loses my flow", they churn, and the churn is never attributed to it. Mitigation that makes the counter-argument falsifiable: ship conflicts per 1,000 syncs, sliced by cause, as a first-class metric, publish it, and hold a budget. If the measured rate exceeds budget, the fix is **finer merge granularity, never a fuzzier apply**. What would falsify the whole decision: a measured conflict rate on real prose vaults that finer granularity cannot bring under budget. That number is currently **unmeasured** — `git merge-file` was not run against a large real vault.

Anti-recommendations, each with its evidence:

- Do not adopt Yjs, Automerge or Loro for the document body. Library facts read 2026-08-29: `yjs` **13.6.32** (published 2026-08-04T08:18:16Z), `loro-crdt` **1.15.0** (2026-08-27T01:45:21Z), `@automerge/automerge` **3.4.1** (2026-08-12T08:28:28Z) [fetched]. `yjs/dist/yjs.mjs` is 299,797 B raw / **62,586 B gzip**; `loro-crdt` bundler wasm is 3,181,087 B raw / **1,046,181 B gzip** — $1,046,181 \div 62,586 = 16.71\times$ [measured, derived]. Automerge 3.0's headline is cutting memory "by over 10x" versus Automerge 2, an admission of how heavy full-history CRDTs were [fetched]. *Narrow exception:* if live multi-cursor collaboration ships later, use Yjs **ephemerally** for the in-session channel and persist only splices to the file.
- Do not use diff-match-patch anywhere in the write path. It is acceptable as a read-only visual diff renderer only.
- Do not build server-authoritative OT: 1/5 on true-offline, 2/5 on refuse-fit, the transform-function matrix is the canonical solo-founder time sink, and per Fugue it does not escape interleaving anyway [fetched].

- Do not ship git-as-transport (`isomorphic-git` 1.41.9, published 2026-08-23T11:26:35Z [fetched]) on mobile. Obsidian Git's own README: mobile is "very unstable", no SSH auth, limited repo size from memory restrictions, and "Obsidian may crash on clone/pull, create buffer overflow errors, run indefinitely... I don't know how to fix this" [fetched]. Use git's *merge algorithm*, not git's *transport*.
- Do not resolve conflicts by timestamp. Syncthing's documented last resort is the "larger value of the first 63 bits of device ID" [fetched] — that is what timestamp ordering degenerates into once clocks tie.
- Do not market "conflict-free". Visible conflicts are the deliberate choice; the honest line is that frontmatter will sometimes ask a question and will never answer one for you.

31.2 The design

COMPONENT	RULE	RATIONALE
Server role	A byte store with compare-and-swap. It never merges.	R2's Workers API supports <code>put(..., { onlyIf: { etagMatches } })</code> , and "if the condition check for <code>put()</code> fails, <code>null</code> will be returned" [fetched, developers.cloudflare.com/r2/api/workers/workers-api-reference] — a real optimistic-concurrency primitive
Serialization point	Durable Object per document for the monotonic revision counter and per-document lock	DOs are "single-threaded and cooperatively multi-tasked" with "durable, transactional, and strongly consistent storage... accessible only within that object" [fetched]
Ordering	Server-issued revision, total. No wall clock on any correctness path.	Removes clock skew, VM resume and travelling users from the ordering decision entirely [inference]
Client state per document	<code>base_bytes</code> (verbatim bytes of the last synced revision), <code>working_bytes</code> , and an append-only splice journal of <code>{seq, base_digest, offset, deleted_len, inserted_bytes, result_digest}</code>	These are the records the splice engine already produces — no second document model [inference]
Journal authority	The journal is a checkable derivative, never authority . <code>fold(journal, base_bytes) == working_bytes</code> must hold at every boundary; if it does not, refuse to sync and fall back to whole-file conflict copy	Preserves "the file is the only source of truth" while still yielding an oracle [inference]
Merge granularity	line-based diff3 over a <i>sentence-normalised</i> token stream, then re-emit original bytes for unchanged regions	Markdown prose is often one paragraph per line, so raw-line diff3 conflicts on every co-edited paragraph [inference]
Clocks	HLC-style timestamps for display and tie-breaking only	Hybrid Logical Clocks give causality plus bounded closeness to physical time without TrueTime's GPS/atomic infrastructure and its δ ms ϵ [fetched, Kulkarni & Demirbas]. In practice unused, because the server revision is total

Reconnect protocol, ordered:

- Fetch server bytes and etag.
- If `etag == my base etag` → CAS-upload `working_bytes`. Done.
- Otherwise run a three-way merge **with the true base that was stored** — never a diff against the winner, which is what fuzzy patching does.
- Clean merge → CAS-upload merged bytes. CAS failure → return to step 1 (another device won the race).
- Conflict → **REFUSE**. Write `note (conflict <device> <server-rev>).md` holding *your* bytes untouched, leave the server bytes as the file, surface it in the UI.
- Never write conflict markers into the user's `.md`. M9 shows why: a document already containing `<<<<<< HEAD` produces nested markers and an unparseable result [measured].

Mobile and background sync:

- Background Sync API is unavailable on the two hardest targets.** `browser-compat-data` `api/SyncManager`: Chrome 49, Edge/Opera mirrored, **Firefox** `false`, **Safari** `false` (WebKit bug 182565), `safari_ios` mirrors Safari, `webview_android` `false` (crbug 40449796); MDN labels it "Limited availability... not Baseline" [fetched].
- Safari storage is evictable by default.** WebKit policy (Aug 10, 2023; Safari 17 / iOS 17): origin quota up to 60% of disk for browser apps and 15% for other apps, overall quota 80% / 20%, eviction LRU by origin and whole-origin, triggered by overall quota, storage pressure, or ITP-driven non-interaction; persistence "based on heuristics like whether the website is opened as a Home Screen Web App" [fetched].
- Native iOS gives no guaranteed window either: `BGAppRefreshTaskRequest` is "a short refresh task", `BGProcessingTaskRequest` "a processing task that can take minutes" — both are requests the system schedules [fetched]. The market leader does not attempt it: Obsidian's FAQ answers "Is my data being synced in the background?" with **"No"** [fetched].
- Do therefore: sync on `visibilitychange`, on `pagehide`, and on foreground focus; use Background Sync only as a Chrome-only optimisation behind a feature check; attempt `BGProcessingTask` in the Tauri v2 iOS app as best-effort, never as a correctness dependency; call `navigator.storage.persist()` and **surface the result** in plain words when it is denied.
- Never display "Fully synced" unless the server has ACKed a revision whose digest equals the local file's digest.** Forum topic 116380 (created 2026-07-23, 3 posts, 65 views; Obsidian 1.12.7 / installer 1.9.14 / Windows 11, Restricted Mode, default theme) reports the final 10–15 Korean characters missing while the log said only "Fully synced", healing never [fetched]. That is a UI that lied.
- Anti-recommendation:** do not build Web Push-triggered sync to work around the missing Background Sync API. It requires notification permission for a non-notification purpose, will be denied by most users, and yields a worse funnel than an honest "open the app to sync."

31.3 The convergence oracle — proving zero loss, not observing it

Two devices, offline, reconnect, human watching is a *demonstration*. The proof is five mechanical layers.

LAYER	ASSERTION	WHAT IT CATCHES — AND WHAT IT DOES NOT
1. Digest equality	SHA-256 of final bytes equal on both devices and on the server	Catches divergence. Catches no loss — both devices can converge on a truncated file
2. Splice-conservation invariant	For every journal record acknowledged to the user, <code>inserted_bytes</code> occurs as a contiguous byte subsequence of the converged document or appears byte-identically inside a named conflict artifact with a stable id. Formally $\cup \text{inserted}(A) \cup \cup \text{inserted}(B) \subseteq \text{bytes}(\text{converged}) \cup \text{bytes}(\text{conflict artifacts})$, each element attributable to exactly one destination	Loss = any inserted run present in neither. This is what "zero loss" means; digest equality is not it
3. Deletion-intent invariant	Bytes a device deleted are absent from the converged doc unless concurrently re-inserted by the peer; a byte reappearing with no re-insertion record is a resurrection bug	The M6 / <code>--union</code> failure class [measured]
4. Seeded deterministic network simulator	Single process, virtual clock, scriptable partition schedule. Random splice programs against a corpus of real markdown: frontmatter, CRLF files, no-final-newline files, files containing <code><<<<<<</code> , mixed indentation, emoji/CJK spanning multi-byte boundaries. $\geq 10,000$ seeded schedules per release; every failure checked in as a fixture with its seed	Rare faults that a live demo will never draw (LR#68)
5. Permutation-invariance and idempotence	Apply the same op set in k sampled orders, assert identical digests. Assert explicitly that replaying an already-applied op is a no-op	Non-commutativity is a bug even when nothing is lost. The measured <code>four four</code> result is an idempotence violation that DMP's own <code>true</code> flag hid [measured]

The rule for the harness itself: the oracle must first be shown to FAIL against a deliberately broken merger. Wire diff-match-patch in as a known-bad backend in CI; if the suite goes green on it, the suite is measuring nothing (LR#60, LR#68). Anti-recommendation: do not accept a passing suite as evidence of correctness until that red run is on record for the current release.

31.4 Failure modes to test

#	FAILURE	TRIGGER	ASSERTION
F1	Editor buffer not flushed before sync reads the file	forum 116380 [fetched]	Bytes handed to the sync engine == bytes in CodeMirror's doc, digest-compared, on every upload
F2	Patch replay duplication	Reconnect retries, at-least-once delivery	Idempotence: applying rev N twice yields an identical digest [measured failure in DMP]
F3	Fuzzy apply into changed context	Base drifted	Merge refuses unless base digest matches exactly
F4	Both append at EOF	Classic offline case	Conflict artifact, both texts recoverable (M2 shows git does this correctly)
F5	Torn or partial write	Crash mid-save, iOS jetsam	Write temp → fsync file → atomic rename → fsync directory; SQLite's rollback-journal discipline is the reference model [fetched, sqlite.org/atomiccommit.html]
F6	Conflict markers already in the user's prose	Markdown about git	Refuse to auto-merge any file matching <code>^(<<<<<< ===== >>>>>>) (M9)</code>
F7	Multi-byte character split at a splice offset	CJK / emoji, LR#68	Every offset lands on a UTF-8 boundary; the corpus must contain the Korean case from 116380
F8	Frontmatter duplicate keys	Any union-style merge	Post-merge parse: duplicate top-level YAML key ⇒ refuse (M6)
F9	Whitespace-only divergence	Tab vs spaces	Must be classified, not silently resolved (M8)
F10	Delete-vs-edit race	One device deletes, other edits	Never delete; materialise as a conflict artifact (Syncthing's documented behaviour) [fetched]
F11	Third-party sync layered on the vault	User has iCloud or Dropbox over the folder	Detect and warn — Obsidian documents that on-demand / online-only files are read as <i>deleted</i> and removed from the remote vault [fetched]
F12	Clock skew, device clock set backwards	Travelling user, VM resume	No correctness path may read <code>Date.now()</code>
F13	Storage evicted under the app	Safari / PWA, LRU whole-origin eviction [fetched]	Unsynced local edits must be detectable as missing, never silently treated as "nothing to sync"

Incumbent behaviour, for reference when specifying these tests [all fetched 2026-08-29]:

SYSTEM	MECHANISM ON CONCURRENT EDIT	TEXT MERGE	LOSS SURFACE
Obsidian Sync	Markdown merged with <code>diff-match-patch</code> ; all other types including canvas are last-modified-wins. Since 1.9.7, a per-device toggle: <i>Automatically merge</i> (default) or <i>Create conflict file</i> named <code>note (Conflicted copy <device> YYYYMMDDHHMM).md</code>	Yes, fuzzy	Docs warn auto-merge "may sometimes create duplicate text or formatting problems". The setting is device-local , so two devices can disagree about policy. Documented hole: a note created locally and re-downloaded within a couple of minutes keeps the remote version "without merging"
Syncthing	Never merges. Older mtime renamed <code><name>.sync-conflict-<date>-<time>-<modifiedBy>.<ext></code> ; ties broken by device-ID bits	No	Wall clocks decide the winner; conflict copies propagate as ordinary files and multiply across the cluster
Dropbox	Conflicted copy; "The last version saved will always appear as the conflicted copy"; manual merge recommended (page updated May 15, 2025)	No	Auto-save apps generate spurious conflicts
iCloud Drive	"one file is chosen as the current version and any other versions are tagged as being in conflict"; the app must resolve via <code>NSFileVersion.unresolvedConflictVersionsOfItem(at:)</code>	No	If the app never resolves, conflicts persist server-side, invisible in most editors
git	Three-way merge against a real common ancestor, <code>ort</code> default, conflicts marked, exit code = conflict count	Yes, line-based, refuses on overlap	Needs a stored base; line granularity is coarse for prose
Resilio	Unverified. <code>help.resilio.com</code> returned HTTP <code>000</code> on 3 URL variants and <code>resilio.com/blog/file-conflict-resolution</code> returned 404 [measured] — not paraphrased from memory	—	—

Open questions carried into build, none of which may be reported as settled: Obsidian's *server-side* algorithm beyond the docs' DMP claim (no post-mortem published for topic 94732); whether the 116380 loss is a CodeMirror flush bug, an IME-composition boundary bug, or a sync-read race (no maintainer diagnosis in the thread as of fetch); and the conflict rate of `git merge-file` on a large real prose vault — the single number that decides whether §31.1's counter-argument survives.

§32 Backup, restore and disaster recovery

32.1 The finding that reorders this — R2 has no object versioning

FACT	EVIDENCE
<code>GetBucketVersioning</code> ❌, <code>PutBucketVersioning</code> ❌, <code>GetObjectLockConfiguration</code> ❌, <code>PutObjectLockConfiguration</code> ❌ on the R2 S3 API compatibility matrix (page "Last updated Jul 31, 2026"). No versioning page exists in the R2 sidebar or in <code>r2/llms.txt</code>	[fetched 2026-08-29]
<code>DeleteObject</code> , <code>DeleteBucket</code> , <code>AbortMultipartUpload</code> are free operations — a runaway delete loop costs the bug nothing	[fetched]
"Emptying a bucket is irreversible. All objects in the bucket are permanently deleted." "Objects removed during this process cannot be recovered." Bucket lock rules must be removed first	[fetched]
Audit logs are control-plane only: "Logs for data access operations, such as <code>GetObject</code> and <code>PutObject</code> , are not included." There is no object-level forensic trail by default	[fetched]
R2 durability doc, verbatim: "Durability does not prevent intentional or accidental deletion of data." Eleven nines protects against disks, not against you	[fetched]

The standard cloud DR recipe does not port. AWS's own whitepaper recommends S3 Cross-Region Replication "while providing versioning for the stored objects so that you can choose your restoration point" [fetched] — on R2, versioning must be built in the key layout, not bought. Consequence for the build: L0 is content-addressed immutable keys

`u/{uid}/d/{docid}/v/{ts}-{sha256}` that are never overwritten, plus a tiny `HEAD` pointer object per document. Overwrite ceases to be a failure mode because overwrites do not exist; deletion becomes the only loss vector, and it is free and untraced. Every layer below exists to answer deletion. Anti-recommendation: do not design as if a versioning toggle will arrive — if Cloudflare ships `PutBucketVersioning`, treat it as defence in depth on top of the key layout, never as a replacement for it.

32.2 RPO and RTO by data class

Definitions verbatim from AWS Well-Architected REL13 [fetched]: RTO is “the maximum acceptable delay between the interruption of service and restoration of service”; RPO is “the maximum acceptable time after the last data recovery point”. The same page names “You select arbitrary recovery objectives” as an anti-pattern — hence the reasoning column.

CLASS	RPO	RTO SINGLE USER, SELF-SERVE	RTO WHOLE TENANT	RTO PLATFORM	REASONING
Document bytes	0 for any acknowledged write; ≤5 min for the off-provider copy	≤60 s	≤4 h	≤24 h	The engine's contract is byte-preserving splice-or-REFUSE; a backup that loses the last N minutes contradicts the product's own promise [inference]. R2 returns 200 “only when data has been persisted to disk” [fetched], so RPO 0 on the write path needs no extra work — the RPO actually under our control is the <i>second copy</i>
Version history	≤5 min	≤5 min	≤4 h	≤24 h	Undo across sessions is what customers think “backup” means; cheaper to keep as immutable content-addressed objects than to reconstruct [derived]
Metadata (doc index, titles, folders, share state)	≤5 min	≤5 min	≤1 h	≤4 h	Make metadata a <i>projection of the object keys</i> , rebuildable by <code>ListObjects</code> , exactly as every view is a projection of the file. Metadata loss then becomes a rebuild, not a restore [inference]
Published pages	= document RPO	≤15 min	≤4 h	≤24 h	Deterministic renders. Regenerate, never restore. The only irreducible state is the URL↔document binding, which belongs in metadata [inference]
Billing records	0	n/a	≤24 h	≤72 h	Keep the payment processor as system of record and hold a read-mirror only; total loss of our copy is a re-sync, not a reconstruction [inference]. Retention period is a statutory question not verified — check Companies Act 2013 §128 before setting it
Auth / identity	0 for identity rows; sessions expendable	n/a	≤1 h	≤4 h	Forced re-login is an annoyance; a lost identity row orphans a paying user's documents [inference]

- **Anti-recommendation: do not set RPO 0 on the backup tier.** Continuous replication “may not protect against disaster events such as data corruption or malicious attack (such as unauthorized data deletion) as well as point-in-time backups” [fetched, AWS DR whitepaper]. A zero-lag mirror faithfully replicates your DELETE.
- Does the user holding their own copy reduce the obligation? **Legally no:** DPDP §8(5) places the duty on the Data Fiduciary to protect personal data in its possession or under its control by taking reasonable security safeguards; Schedule item 1 penalty “May extend to two hundred and fifty crore rupees”; no clause conditions the duty on whether the Data Principal holds a copy [fetched, §8 and Schedule read in full]. **Operationally yes:** for a Tauri v2 desktop user the canonical file is on their disk and R2 is our backup of their primary, so the RTO that matters is re-sync; for a browser-only user R2 *is* the primary and no relief exists [inference]. **Commercially it is the strongest claim available and the easiest to void** — make the local copy a *tested* export, a scheduled verified plain-directory-of-`.md` mirror with a visible last-verified timestamp per device. Anti-recommendation: the export never substitutes for L3/L4. GitLab's users all held local clones and repositories survived because they were stored separately, and the company still lost roughly 5,000 projects, 5,000 comments and 700 users of database state [fetched].

32.3 The layered architecture and what each layer does NOT cover

LAYER	PROTECTS AGAINST	DOES NOT PROTECT AGAINST
L0 — R2 primary, content-addressed immutable keys <code>u/{uid}/d/{docid}/v/{ts}-{sha256}</code> , never overwritten, plus a <code>HEAD</code> pointer per doc	Disk and datacentre failure (eleven nines, erasure coding, synchronous writes) [fetched]; accidental overwrite, because there are none	Deletion of any kind. <code>DeleteObject</code> is free [fetched]. Account compromise. Bucket empty
L1 — Bucket lock, <code>Age</code> -bounded, on the <code>u/</code> prefix	Delete and overwrite of locked objects per prefix, “for a specified period — or indefinitely” [fetched]	Nothing once the lock is removed — and removal is <i>required</i> before emptying a bucket [fetched], so this is a speed bump with a documented removal procedure, not a vault. <code>Indefinite</code> also blocks DPDP §8(7)(a) erasure
L2 — Lifecycle rules, transition <code>v/</code> older than 30 d to Infrequent Access	Cost only	Data loss. Lifecycle is a <i>deletion engine</i> : “the expire (or delete) lifecycle transition takes precedence” on conflict, and <code>LifecycleDeletion</code> is one of only two delete triggers [fetched]. Misconfiguring L2 is itself a top-3 loss scenario
L3 — Nightly restic-format repo → second R2 bucket, separate account, separate API token, write-only	Bugs in our own code; single-token compromise; bucket-empty on the primary	Cloudflare-wide account action. The Feb 6 2025 incident took 100% of R2 APIs down 08:14→09:13 UTC because a <i>routine abuse remediation</i> disabled the Gateway [fetched] — one blast radius covers both buckets
L4 — Weekly full → Backblaze B2, different provider, different country, different credential store	Provider-level loss or account termination — the UniSuper failure mode: “one input parameter was left blank”, default 1-year term, automatic deletion, “No customer notification was sent” [fetched]	Ransomware resident longer than retention. Correlated encryption-key loss
L5 — Encrypted monthly archive on physical media, offline	Everything above, plus total credential compromise	Being forgotten. Requires a human hand [inference]
git — application code, schema, IaC, runbooks; NOT customer documents	Config and code loss; auditable change history	Customer documents. GitHub Docs, verbatim: “Git is not designed to serve as a backup tool” [fetched]

Why git is not the document substrate [all fetched unless marked]: files >100 MiB are **blocked**, >50 MiB warned, browser upload capped at 25 MiB, repos “ideally less than 1 GB, and less than 5 GB is strongly recommended”; pasted images and attachments defeat delta compression and grow packs monotonically [inference]; history rewrite destroys silently, since `gc.reflogExpire` defaults to **90 days**, `gc.reflogExpireUnreachable` to **30 days**, and `gc` calls `prune --expire 2.weeks.ago` [fetched; confirmed unset locally on git 2.50.1 [measured]]; the reflog is local and unpushed, so a fresh clone of a force-pushed repo has no undo record at all [inference]; and `git filter-repo` is simultaneously the correct DPDP erasure tool and the correct data-destruction accident — same command shape, opposite intent [inference]. **Anti-recommendation: do not put per-user document repos on GitHub for the free versioning** — it adds a data processor under DPDP, caps documents at 100 MiB, and buys a version store a single `gc` can prune.

Recorded disagreement, unresolved: CISA's 3-2-1 rule requires “2 different media types to protect against different types of hazards” [fetched]. An all-object-storage stack (L0–L4) satisfies three copies and one offsite but violates the two-media clause literally. L5 exists only to satisfy that clause; whether the clause is meaningful in 2026 is a judgement call, not a fact.

The statutory conflict, which must be named in the runbook: DPDP §8(7)(a) requires erasure on withdrawal of consent or when the purpose is no longer served, §8(8) deems the purpose no longer served after a prescribed inactivity period, and §12(3) adds erasure on request [fetched]. An `Indefinite` bucket lock makes compliance impossible. Resolution [inference]: bucket-lock only the *backup* prefixes with a bounded `Age` equal to the stated retention window, never `Indefinite`, and make deletion a scheduled tombstone that propagates to L3/L4 within that window — the maximum lag between an erasure request and true erasure then becomes a number that can be stated publicly. **The DPDP Rules 2025**

breach-notification deadline and the prescribed inactivity period under §8(8) were NOT opened — meity.gov.in returned empty bodies, PRS and egazette returned 404/000 [measured]. Do not write a 72-hour figure into the runbook from memory.

Cost, so no layer is skipped on a price argument. Assumptions stated so they can be attacked: 200 docs/user, 5 active docs/user/day, mean doc 25 KB [measured — 590 markdown files in the local corpus: mean 24,851 B, median 4,023 B, p90 25,916 B], 8 coalesced saves/active doc/day, 40 GETs/active doc/day, 30-day version retention, nightly snapshot to IA ≈ 1× live, one B2 copy of live+snapshot. Prices [fetched 2026-08-29]: Standard \$0.015/GB-mo, IA \$0.010/GB-mo, Class A \$4.50/M (IA \$9.00/M), Class B \$0.36/M (IA \$0.90/M), egress free, free tier 10 GB-mo + 1 M Class A + 10 M Class B (Standard only); B2 \$6.95/TB/30-day = \$0.006787/GB-mo.

USERS	LIVE	VERSIONS (30 D)	STD STORAGE	CLASS A	CLASS B	IA SNAPSHOTS	B2	TOTAL/MO	PER USER
100	0.48 GB	2.86 GB	\$0.00	\$0.00	\$0.00	\$0.18	\$0.01	\$0.19	\$0.0019
1,000	4.77 GB	28.61 GB	\$0.35	\$1.80	\$0.00	\$1.85	\$0.06	\$4.06	\$0.0041
10,000	47.68 GB	286.10 GB	\$4.86	\$58.50	\$18.00	\$18.48	\$0.65	\$100.48	\$0.0100

[derived; arithmetic executed on the inputs above]

At 10,000 users storage is \$23.34 and operations are \$76.50 [derived] — save granularity, not bytes, is the cost model: 48 saves/active doc/day = 72,000,000 writes/mo = \$319.50 Class A; 8 = 12,000,000 = \$49.50; 1 = 1,500,000 = \$2.25 [derived]. Per-object snapshot copies cost 2,000,000 Class A (IA) = \$18.00/mo against \$2.70/mo for one packed archive per user per night [derived]. Do coalesce versions on a debounce plus a semantic boundary and pack nightly snapshots — a ~6× swing at 10 k users. **Anti-recommendation: do not coalesce past the stated RPO** — every minute of coalescing sells a minute of the customer's RPO for roughly a hundredth of a cent [derived] — and do not put anything needing a hurried restore into Infrequent Access, which carries a **30-day minimum storage duration** and \$0.01/GB retrieval [fetched]; a full 48 GB restore at 10 k users is \$0.48, trivial, but a monthly verification pass that reads all data pays it every time, and that is the fee that quietly discourages the drill that must not be skipped. Total backup spend at 10,000 users is ~\$100/mo against a DPDP §8(5) ceiling of ₹250 crore — that ratio is not an argument to spend more, it is the reason no cost argument may ever appear in a decision to skip a drill.

32.4 The restore drill

Runbook, ordered; every step names its verification.

- Declare and freeze.** Disable the write path via a flag whose store is *not* R2. The Feb 2025 lesson, verbatim: "this tooling was unavailable because it relies on R2" [fetched] — the recovery lever must not sit behind the broken thing.
- Classify:** single-doc | single-tenant | metadata-only | whole-bucket | provider-level. Pick the shallowest layer that covers it; L4 restores are not for one deleted file.
- Restore to a NEW bucket**, never over the damaged one. Prevents compounding and preserves the corrupt artifact as evidence.
- Verify before cutover, byte-level, not object-count.** `sha256` every restored object against the manifest recorded at backup time; refuse cutover on any mismatch. Object count alone passes when every object is zero bytes — GitLab's S3 bucket was simply *empty* [fetched; inference].
- Rebuild metadata from keys**, do not restore it. If L0's key layout is the addressing scheme, this is a `ListObjects` sweep.
- Cut over and record actual RPO and RTO in wall-clock**, not "we met target". GitLab's honest numbers: 6 h 10 m of lost writes (17:20→23:30 UTC) and roughly 18 h of copying at ~60 Mbps [fetched].
- Notify.** DPDP §8(6) requires intimation to the Board *and each affected Data Principal* in the prescribed form and manner; failure carries a ceiling of **two hundred crore rupees** (Schedule item 2) [fetched]. The prescribed deadline lives in the Rules, which could not be opened — confirm before writing it in.

CADENCE	DRILL	PROVES	CONFIRMED BY
Every backup run	Restore one randomly chosen object, compare sha256 to source	The write landed, with these bytes	Machine gate; the run <i>refuses</i> and pages on mismatch
Weekly	Restore 1 synthetic tenant (~200 docs) into a scratch bucket, diff the whole tree, measure seconds	Single-tenant RTO is real	Machine; publishes a signed <code>last-good-restore</code> record with duration
Monthly	<code>restic check --read-data</code> on L3 and L4 — default <code>check</code> verifies structure only; <code>--read-data</code> is required to verify pack files on disk [fetched]	The bytes, not just the index	Machine
Quarterly	Cold restore from L4 into a brand-new provider account, from the runbook only, laptop wiped of ambient credentials, timed	The runbook is complete and the credentials are recoverable	Founder, timed, written up
Annually	Full DR assuming the Cloudflare account is gone; restore from L5 media	Independence from the primary provider	Founder plus one named external witness
Annually	Restore-refusal drill — corrupt one pack deliberately, confirm the gate goes red	The verifier can fail	Founder

A drill that has never failed is not evidence the drill works — break it on purpose once a year, or the green is uninterpretable (LR#68: a green suite is the expected result of running it, not evidence of a fix). For a solo founder, no drill's outcome may be a human judgement: every drill emits a machine-readable record and a *second, separately-hosted* watcher pages when that record goes stale, because absence of an alarm is not a pass. **Anti-recommendation: do not host that watcher on Cloudflare** — a Cloudflare-wide event takes storage and watcher together, which is exactly the shape of the Feb 2025 incident [fetched].

Silent-failure modes to instrument against. This studio has already lived the canonical one; reproduced read-only [measured, 2026-08-29]: under `set -e , a log()` whose last line is a trailing `[ "$QUIET" = "0" ] && echo "$1"` returns 1, kills the caller, and produces exit 1 with zero output and neither subsequent line run; rewritten as `if [ ... ]; then ... fi`, both lines run and exit is 0. Nightly backups were disabled for weeks by exactly this (LR#55).

MODE	HOW IT STAYS SILENT	INSTRUMENT
Trailing <code>[] && cmd</code> under <code>set -e</code>	Function returns 1, <code>set -e</code> kills the caller before the upload; exit code is 1 but nothing watches it [measured]	<code>bash -n</code> in CI; lint-ban trailing conditionals; assert bytes written , never exit code
Write silently denied (sandbox, permission, quota)	<code>>></code> fails and the tool prints its success line having stored nothing (LR#67)	Count objects and bytes before and after; refuse if the count did not move
Major-version tool mismatch	GitLab: <code>pg_dump</code> 9.2 against PostgreSQL 9.6; "S3 bucket was empty" [fetched]	Restore-and-diff every run. A dump you have not read back is a file, not a backup
The alarm channel is the broken thing	GitLab: cron failure emails "rejected by the receiver" because DMARC was not enabled — "we were never aware of the backups failing, until it was too late" [fetched]	Dead-man switch: the backup <i>pushes</i> a heartbeat, an independent watcher on a different provider pages on <i>absence</i> . Test the pager monthly by not sending
Field-name mismatch producer↔consumer	<code>bool(missing_key)</code> is False, indistinguishable from a real failure (LR#59)	Versioned manifest schema; the verifier reports which key carried the number ; missing ⇒ UNKNOWN ⇒ refuse, never a falsy default
Substring gate	<code>grep -qE '0 FAIL'</code> matches inside <code>10 FAIL</code> (LR#60, LR#73)	Word-boundary anchors; numeric comparison, never text match
Green because the check is a proxy	A check that greps source proves nothing about behaviour (LR#60)	Every gate must execute a restore; label any source-grep check as PROXY in its own output
Backing up nothing	Source path moved or renamed; the job succeeds over an empty tree	Assert object count within ±X% of live, and assert a floor , never an equality (LR#66)
Correlated deletion	A lifecycle rule plus a delete-propagating sync remove from every tier at once	L4 uses a write-only, delete-denied token and lags L3 by ≥7 days by design [inference]
No forensic trail	R2 audit logs exclude <code>GetObject</code> / <code>PutObject</code> [fetched]	Wire R2 event notifications (<code>object-create</code> , <code>object-delete</code>) to a Queue and archive them, up to 100 rules per bucket [fetched]; alarm on delete-rate anomaly

§33 Search and retrieval

33.1 Decisions

#	DECISION	ANTI-RECOMMENDATION	FALSIFIED BY
S1	Keep <code>minisearch@7.2.0</code> as the only index at 1k–10k notes	Do not migrate to Orama, FlexSearch, or Lunr; do not add a second engine "for CJK"	A profiled query p95 >50 ms at 10k notes on the target hardware
S2	Ship a ~15-line CJK bigram <code>tokenize</code> , passed identically to index-time and search-time	Do not ship <code>Intl.Segmenter</code> ; do not bundle a wasm morphological analyser	Bigram top-1 falling below segmenter top-1 on a named, human-labelled CJK query set
S3	Diacritic folding scoped to <code>\p{Script=Latin}</code> plus an explicit <code>ß→ss / æ→ae / ø→o</code> map	Do not ship <code>.normalize("NFD").replace(/[\p{Mn}]/gu, "")</code> globally	Any regression where a Latin-scoped fold changes a non-Latin byte
S4	Per-file incremental indexing keyed on blob SHA, with <code>MiniSearch.loadJSON</code> persistence	Do not keep the current whole-index rebuild on HEAD change	Blob-SHA keying missing an edit that HEAD-keying would have caught
S5	Ripgrep in <code>src-tauri</code> is the permanent exact/regex fallback path	Do not compile Tantivy to wasm; do not adopt DuckDB-wasm	Fallback latency exceeding index latency on a 74 MiB corpus
S6	No persistent vector store. Embeddings permitted only as a query-time re-rank of ≤30 BM25 candidates	Do not build local RAG as a retrieval path; do not add an ANN index	Re-rank moving top-1 by ≥10 pp on the wikilink relevance harness — below that, delete it

33.2 Engine comparison, read 2026-08-29

ENGINE	VERSION / PUBLISHED	SIZE OR DOWNLOADS	QUERY	CJK SUPPORT	VERDICT
MiniSearch	7.2.0, 2025-09-16, unpacked 826,513 B	2,659,659 npm/wk; 6,115★	0.88 ms @4,548 notes MEASURED	None built in; <code>tokenize</code> / <code>processTerm</code> are function arguments	Keep. The only engine whose fix is an argument, not a migration
Lunr	2.3.9, 2020-08-19	7,542,972 npm/wk; 9,200★	11,527 q/s FETCHED , VENDOR	<code>lunr-languages</code> + TinySegmenter fork	Reject. Index is immutable after <code>build</code> — fatal for per-file incremental
FlexSearch	0.8.212, 2025-09-06, unpacked 2,334,755 B	1,407,279 npm/wk; 13,784★	50,955,718 q/s FETCHED , VENDOR	<code>Charset.CJK</code> built in FETCHED	Fastest on paper; the vendor's own harness claims a 2,390× gap over MiniSearch, which is not credible as stated
Orama	3.1.18, 2025-12-19, unpacked 2,192,356 B	1,304,705 npm/wk; 10,537★	29,445 q/s FETCHED , VENDOR	32 locales, zero CJK ; <code>@orama/tokenizers/japanese</code> and <code>/mandarin</code> are separate packages FETCHED	Reject. You would migrate engines and still write a tokenizer
Fuse.js	7.5.0, 2026-07-13	13,851,708 npm/wk; 20,462★	422 q/s — slowest in the vendor table FETCHED	None; no BM25, no analyzer	Reject as primary. A fuzzy matcher, not an IR engine
Tantivy	0.27.0 repo / crates 0.26.1, 2026-04-21	3,783,112 dl/90d; 16,005★	"~2× faster than Lucene" FETCHED , VENDOR README LINE 132	<code>tantivy-jieba</code> 0.20.0, <code>cang-jie</code> , <code>lindera-tantivy</code> 5.0.1, Vaporetto	Right engine, wrong runtime. <code>src-tauri</code> only
SQLite FTS5	@sqlite.org/sqlite-wasm@3.53.0-build1 , 2026-04-21, 2,829,040 B	1,045★	—	<code>unicode61</code> does not segment CJK; <code>trigram</code> is the substring escape hatch FETCHED , SQLITE.ORG/FTS5.HTML	The 100k-note target. Caveat: <code>detail=none/column</code> forbids tokens >3 chars
DuckDB-wasm	1.33.1-dev57.0, 2026-06-22	unpacked 149,377,663 B (142 MiB)	—	via extension	Reject. 142 MiB of package to run BM25 over 74 MiB of prose
Ripgrep	14.1.1 (rev 939d4325be)	67,675★	linear scan; <code>grep -rF</code> over 74 MiB = 1.24 s MEASURED	UTF-8 native; never tokenises, so CJK works by construction	Keep as fallback. No index means no staleness bug
<code>sql.js</code> / <code>wa-sqlite</code> / <code>PGLite</code>	1.14.2 (24,151,707 B) / 1.0.0, 2024-01-05 , no npm licence field / 0.5.8 (25,437,263 B)	—	—	—	Superseded, legally unreviewed, and overkill respectively

Recorded disagreement: the FlexSearch table is the vendor's own harness and puts MiniSearch at 5,849 q/s against Orama's 4,454 — a 1.3× gap — while claiming 13,981,110 for itself. Treat the ordering as informative and the magnitudes as not. [FETCHED](#)

33.3 Measured baseline — what ships today

ITEM	VALUE
Index fields, ladder	<code>title, tags, body, code</code> ; 3-pass AND→OR→fuzzy(1) MEASURED , SRC/MODULES/VAULT/INFRASTRUCTURE/SEARCH-INDEX.TS
Default tokenizer	<code>/[\n\r\t\u{Z}\u{P}]+/u</code> — splits on space and punctuation only, not <code>\u{S}</code> , not script boundary MEASURED
の で を り た	tokenises to 1 token, 13 chars MEASURED
, , ,	0 hits each MEASURED
Why anything matches	pass 1 sets <code>prefix</code> on the last term, so only document-initial CJK substrings hit MEASURED
Diacritics	<code>processTerm</code> is <code>toLowerCase()</code> only; <code>cafe</code> → <code>café</code> , <code>resume</code> → <code>résumé</code> , <code>maximo</code> → <code>máximo</code> all MISS MEASURED
GFM tables	rows written <code> Name Type </code> index as ONE token; no cell is retrievable MEASURED , IN-REPO COMMENT LINES 80-85
Incremental	none — <code>ensureCache()</code> keys on <code>getHeadSha()</code> and rebuilds from a fresh zipball on any HEAD change: 7.0 s plus a zipball download per commit at 4,548 notes MEASURED

Reference corpus: **4,548** `.md` files, **77,549,893 bytes (74.0 MiB)**, mean **17,051 B/note**; 358 files (7.9%) contain CJK, 9,577 CJK characters total. [MEASURED](#)

33.4 The CJK fix

Harness: every CJK n-gram of length 2–4, keep the **2,389** occurring in exactly one file (of 2,854 distinct), deterministically sample 200, gold = that file, run the shipped 3-pass ladder. Node v24.6.0, macOS 25.6.0. [MEASURED](#)

TOKENIZER	RECALL@30	TOP-1	ANY-RESULT	MEAN QUERY MS	INDEX JSON
default (shipped)	85/200 = 42.5%	39.0%	54.0%	0.13	28,296,206 B
CJK bigram	200/200 = 100.0%	99.5%	100.0%	0.03	28,383,334 B
CJK bi+trigram	100.0%	99.5%	100.0%	0.03	—
<code>Intl.Segmenter</code> (ICU/UAX-29)	196/200 = 98.0%	81.0%	100.0%	0.04	28,304,189 B

- Delta on this harness: **+57.5 pp recall@30**, **+60.5 pp top-1**, **–77% query latency**, **+0.31% index size** (87,128 B on 28,296,206 B). [DERIVED](#)
- Recorded disagreement, unresolved:** our stated baseline is 18.1%; this harness measures 42.5% on the same engine. The auto-derived n-gram query set over-samples document-initial runs that the `prefix` pass rescues, which plausibly explains the gap. Against 18.1% the delta would be **+81.9 pp** [DERIVED: 100.0 – 18.1](#). **Do not average them, do not publish a range; re-run one harness over one named corpus before either number is quoted externally.**
- Ship the bigram, not the segmenter.** `Intl.Segmenter` scores 18.5 pp worse on top-1 (81.0% vs 99.5%) because dictionary boundaries do not align with the arbitrary substrings people type: `の で を り た` segments as `の` | `で` | `を` | `り` | `た`, and `の で を り た` as `の` | `の` | `で` | `を` | `り` | `た`. [MEASURED](#) It is also absent from Firefox before 125 (Chrome 87, Safari 14.1, Node 24), so a bigram fallback is required regardless. [FETCHED](#), [MDN BCD](#)
- Implementation, no new dependency, no wasm, no download: split on `/[\n\r\t\u{Z}\u{P}]+/u` as today; for any part matching `/[\u3040-\u30FF\u3040-\u4DBF\u4E00-\u9FFF\uAC00-\uD7AF]/`, emit every character **and** every adjacent bigram; else emit the part unchanged. Pass the same function as the

- search-time `tokenize` .
- Rejected on size: `kuromoji@0.1.2` (2018-03-19, 41,263,301 B), `jieba-wasm@2.4.0` (16,126,591 B), `budoux@0.9.1` (2,659,103 B, a phrase-breaker not an index). `Lindera-wasm@6.0.0` (1,842,304 B) is the only defensible wasm option and is still unnecessary. `FETCHED`
- ### 33.5 Diacritics and Indic — a separate defect
- Devanagari uses spaces, so it is **not** the CJK failure; `भारत की राजधानी नई दिल्ली है` tokenises correctly today. `MEASURED`
 - The real Indic failure is normalisation: `हिंदी` (anusvara) does not match `हिन्दी` (conjunct) — 0 hits. `MEASURED`
 - The obvious fix is a trap: `t.normalize("NFD").replace(/[\p{Mn}]/gu, "")` fixes `cafe` → `café` but **corrupts Devanagari** (`हिन्दी` → `हिंदी`) and still does not fix `strasse` → `Straße`. `MEASURED`
 - Fold only within `\p{Script=Latin}`, add the explicit `ß/æ/ø` map, and handle Indic with NFC plus a small anusvara↔conjunct equivalence table. Accent-insensitivity is the single largest user request in the category: forum.obsidian.md/t/1655, opened 2020-06-11, **760 likes, 151 posts, 11,376 views**. `FETCHED`

33.6 Architecture by scale, and where each tier breaks

TIER	STACK	MEASURED / DERIVED COST	BREAKS AT
1k notes (~3–17 MiB)	MiniSearch in-process, CJK bigram, Latin-scoped folding, per-file replace, serialized index in IndexedDB	946 ms build, 4.1 MiB index (0.34× corpus), 0.18 ms/query, 52.4 MiB heap <code>MEASURED</code>	Nothing. Do not add a second system
10k notes (~30–170 MiB)	Same engine, indexing moved to a Worker, rebuild strictly per-file	~15.5 s full build, ~60 MiB index, ~490 MiB heap <code>DERIVED AT 95 MS/ MiB, 0.36×, 3.0×</code>	The full-rebuild path , not query. A 15 s stall per commit is the failure; second break is heap on a 4 GB iPad
100k notes (~300 MiB–1.7 GB)	SQLite FTS5 via <code>@sqlite.org/sqlite-wasm</code> on OPFS with a bigram-expanded content column (or <code>trigram</code>), ripgrep in <code>src-tauri</code> as the exact path, Tantivy native only if FTS5 ranking proves insufficient	At 17,051 B/note: ~27 min build, ~614 MB index JSON, ~5.1 GB heap — a tab does not survive it. Even at 3 KB/note: ~29 s, ~108 MB index, ~900 MB heap <code>DERIVED</code>	OPFS write throughput on first index. FTS5's <code>unicode61</code> still does not segment CJK, so the bigram expansion is required at every tier <code>FETCHED</code>

Scaling is linear in **bytes**, not notes: 78.8 ms/MiB at 1k, 95.0 ms/MiB at 4,548 `DERIVED: 946/12.0, 7027/74.0`. Terms grow 21,978 → 114,711. `MEASURED`

Constant across all three tiers: the index is a projection of the files, never a source of truth, deletable and rebuildable at any moment — the same rule the board, calendar, and decision-card views already obey.

33.7 Local RAG over the vault — distraction as a retrieval path

- The published evidence does not support replacing lexical retrieval: BEIR (arXiv 2104.08663, 2021-04-17, rev 2021-10-21, 18 datasets, 10 systems) finds BM25 a robust baseline with dense retrievers often underperforming and re-rankers winning "at high computational costs"; *Lucene Is All You Need* (arXiv 2308.14963, 2023-08-29) finds a dedicated vector store unnecessary; BM25S (arXiv 2407.03618, 2024-07-04) reports up to **500×** over the popular Python BM25. `FETCHED`
- Seven Failure Points of RAG* (arXiv 2401.05856, 2024-01-11) states validation "is only feasible during operation." For an engine whose differentiator is cross-engine degradation certification, shipping a component validatable only in production is a category error. `FETCHED` `INFERENCE`
- RAG vs long context* (arXiv 2407.16833, 2024-07-23) finds RAG's only durable advantage is **cost** — precisely the regime that does not bind for one user's vault. `FETCHED`
- Cost if we ever do it: `Xenova/all-MiniLM-L6-v2` int8 is **22,972,370 B (21.9 MiB)**; this vault is ~37,900 chunks at 512 tokens = 13.9 MiB of int8 vectors `DERIVED`. Storage is not the constraint (Chromium allows 60% of disk per origin `FETCHED`, `MDN`); the first-run embed pass is.
- Verdict: permitted only as a hidden query-time re-rank of the ≤30 BM25 candidates — no persistent store, no first-run pass, no settings — and deleted if it does not move top-1 by ≥10 pp on the wikilink-derived relevance harness.** `INFERENCE`
- No study cited here measures a personal markdown vault; anyone claiming embeddings win for personal notes is extrapolating from web/QA benchmarks. `INFERENCE`

33.8 Anti-recommendations

- Do not expose search settings. Every top complaint in the category is a **default** being wrong; a "CJK mode" toggle converts a fixed bug into a permanent support surface.
- Do not build a plugin API for custom analyzers — already excluded by the no-marketplace and no-eval constraints, and it would let a third party silently break the degradation certificate.
- Do not add an ANN index: `voy-search@0.6.3` (2023-09-20) and `hnswnlib-wasm@0.8.2` (2023-07-08) are ~3 years stale, and a flat scan under 100k vectors is faster than the index build. `FETCHED`
- Do not build persistent RAG before the tokenizer ships. Between 57.5 and 81.9 pp of free recall is sitting unclaimed behind a 15-line function. `MEASURED`

33.9 Build order

- CJK bigram `tokenize`, index and search side, one function, zero dependencies.
- Latin-scoped diacritic folding plus the `ß/æ/ø` map.
- Per-file incremental indexing on blob SHA plus `MiniSearch.loadJSON` persistence — removes the 7.0 s rebuild per commit.
- Add `|` to the split class, fixing the documented GFM-table defect.
- Build the wikilink-derived relevance harness as a permanent gate; refuse any search change that does not move it.
- Only then evaluate the 30-candidate re-rank against its pre-committed ≥10 pp threshold.

§34 Collaboration model

The sync substrate is settled: git-merge plus splice journal plus compare-and-swap, never a CRDT. This section specifies the collaboration model that rides on it.

34.1 Model scorecard

(1–5, 5 best; "splice fit" = compatibility with `locate byte range → replace those bytes only → every untouched byte bit-identical → else REFUSE`)

	REAL-TIME CRDT (PEER)	SERVER-AUTHORITATIVE OT	GIT BRANCH-AND-REVIEW	PATCH EXCHANGE
Conflict handling	3 — never blocks, merges wrongly and silently	4 — one authority, degrades on long offline branches FETCHED	4 — 0 conflicts on disjoint paragraphs, 1 on same-line edits, 1 on reflow-vs-edit MEASURED , SYNTHETIC 3-WAY FIXTURE	4 — same engine, explicit sender-side rebase
Offline	5	2	5	5
Byte fidelity	1 — encoded overhead +52.5% (Yjs) to +146.3% (Loro) over the plain text; the .md degrades to a projection DERIVED	2	5 — bytes on disk are the document	5
Auditability	2 — per-keystroke ops, no human-legible unit	3 — server log, not user-facing	5 — commit, author, message, signature, Co-authored-by	5
Implementation cost	1 — new sync engine, relay, storage	2 — bespoke server; hardest thing on this list	4 — merge3 + baseSha already exists in the codebase	3 — engine free, UX expensive
Splice fit	1 — actively hostile . REFUSE has no meaning inside a CRDT	3	5 — native . A splice is a patch	5
Total /30	13	16	28	27

34.2 Why the category does not ask for live cursors

EVIDENCE	NUMBER	SOURCE
Async git vs commercial real-time multiplayer, in our exact beachhead	obsidian-git 3,067,376 vs system3-relay 190,728 = 16.08x; vs peerdraft 18,796; vs both combined FETCHED , OBSIDIAN-RELEASES STATS 2026-08-29 DERIVED	
Ranking	Git rank 6 of 7,020 ; Relay rank 118; obsidian-livesync (async replication, not co-typing) 894,228 at rank 29 FETCHED	
The "fishbowl effect"	"Writers don't want first drafts visible to the editor." Interviewees reported putting devices into airplane mode to stop edits being shared FETCHED , INK & SWITCH, UPWELLING, MARCH 2023	
Prior art on the same finding	Wang, Tan, Lu (2017), <i>Why Users Do Not Want to Write Together When They Are Writing Together</i> , DOI 10.1145/3134742, 53 citations FETCHED , CROSSREF	
Unit economics	Liveblocks \$0.002/realtime-minute → 2 h/day × 22 days = 2,640 min = \$5.28/user/month , above both the \$4/mo individual and \$5/seat team prices before any other COGS. Even 1 h/day × 20 d = \$2.40 = 60% of the \$4 plan. SOC 2 begins at the \$500/mo tier FETCHED DERIVED	
Line discipline in the real corpus	4,702 markdown files / 445,836 prose lines : median 52 chars, mean 74.8, 81.96% ≤ 100 chars, only 5.99% > 200 — already near semantic linefeed, so git's line-granular merge is a good fit, not a poor one MEASURED	

Three independent commercial products converged on proposal-and-adjudication rather than presence: Google's suggesting mode (overlay in a distinct colour, deletions struck through, owner emailed, accept/reject one-by-one or Accept all / Reject all FETCHED, SUPPORT.GOOGLE.COM/DOCS/ANSWER/6033474), GitBook's change requests ("based on the concept of branching, and feels familiar to anyone who uses pull requests" FETCHED), and Decap's publish_mode: editorial_workflow (save draft = commit to branch cms/collectionName/entrySlug plus open PR; approve = merge PR and delete branch FETCHED).

Suggesting mode is asynchronous review implemented on top of a real-time substrate — the substrate is not what users are buying, the proposal-to-adjudication loop is. INFERENCE

34.3 What we lose without live cursors, stated honestly

GENUINELY LOST	NOT LOST
Live-call co-editing of meeting notes	Multi-device (git sync)
The "I'm on line 40, you take line 80" coordination signal	No data loss (commits)
The trust cue that a document is not stale	Review (branches, per-hunk accept)
A demo moment that sells screenshots	Attribution (Co-authored-by , signatures)
	Offline (the repo is a primary copy)

Ship a **presence badge without a cursor** — "Priya has this file open, last edit 2m ago" — which recovers the coordination value at roughly 1% of the cost and does not create the fishbowl. Anti-recommendation: do not build live cursors as a trust signal.

34.4 Strongest counter-argument, at full strength

Ink & Switch — the most credible primary source here and the one this section otherwise leans on — found async-only insufficient: they "experimented with an asynchronous-only collaboration model but found that while drafts often have a single primary author, those authors often find it useful to be able to share an in-progress draft for initial feedback." FETCHED GitBook independently ships Live edits alongside Change requests. FETCHED The honest reading is that async-only fails at the limit, and the failure mode is *two people on one draft for twenty minutes* — served by an ephemeral live layer over a shared branch, not by peer CRDT, per-keystroke persistence, or a bespoke sync engine. INFERENCE

34.5 Staged migration

STAGE	SHIP	REVERSIBLE?
0 — now	Rename the model "branch and review" in all internal docs; remove CRDT rows from the pillar tables; move <code>Multi-user concurrent state, presence</code> to SERVER · not this cycle	n/a
1	Suggestions as splices. A suggestion is <code>{baseSha, byteRange, replacement, author}</code> in the sidecar; accept = a splice, reject = a delete. Materialise to JSON always; CriticMarkup optionally and only as a byte format we own — <code>criticmarkup.com</code> is a parked domain and CriticMarkup/CriticMarkup-toolkit (848★) last committed 2021-02-27 FETCHED	Fully — deleting the sidecar leaves a valid repo
2	Branch and review. Drafts as real git branches, <code>Co-authored-by</code> on session-batched commits, review UI = per-hunk accept over the splice list	Fully — it is git
3 — only if evidence demands	Ephemeral live layer. Presence plus a short-lived shared buffer over one branch; persist on session close as one splice batch, never per keystroke	Yes, by design
4 — exit hatch	If Stage 3 proves insufficient, adopt Loro (<code>loro-crdt</code> 1.15.0, published 2026-08-27, Fugue + Eg-walker) as an ephemeral in-session buffer only . The persisted artifact stays the <code>.md</code>	The one-way door is persisting CRDT state. Never cross it

34.6 Anti-recommendations

- **Never persist CRDT state as the document.** This is the single irreversible decision in this section and it deletes the splice guarantee: a CRDT always converges, so REFUSE has no meaning inside one. The measured failure in a competitor is exactly this — a byte contract that exists only inside a live CRDT server, where agent edits error out without it. MEASURED
- Do not adopt Automerge on the strength of its 3.0 blog post alone. The "memory usage cut by over 10x" and "700 MB → 1.3 MB" claims are the vendor's FETCHED; the only independent table available still pins automerge 2.1.10, yjs 13.6.11, ywasm 0.9.3, loro 0.10.1 against today's 3.4.1 / 13.6.32 / 1.15.0. The numbers are not reconciled and must not be quoted as if they were.
- Do not build a bespoke OT server. OT won in the field because Google, Microsoft, and ShareDB already paid for it (`sharedb` 6.0.2, 6,535★); a solo founder should not re-pay. FETCHED INFERENCE
- Do not model on Prose.io: dead since commit `9ef717ea94d1`, **2024-02-09**, README openly seeking maintainers. The git-CMS graveyard is real, and Decap and GitBook survived by adding **review**, not real-time. FETCHED
- Do not ship Pijul-style first-class conflicts as a user-facing concept — right theory (`~1.0.0-beta` after years), wrong surface for a deliberately simple product. FETCHED
- Patch exchange (git send-email, Radicle 1.10.2 released 2026-08-26, Sourcehut still "public alpha") is a source of design ideas, not a shippable UX: every project in that lineage is beta, alpha, or single-maintainer. FETCHED INFERENCE

Decision falsified by: a beachhead cohort where real-time plugin installs exceed git plugin installs, or a measured merge-conflict rate above 5% of multi-author sessions on the real corpus.

§35 Accessibility

35.1 Legal obligation by jurisdiction, with dates

JURISDICTION	INSTRUMENT	WHAT IT REQUIRES OF US	DATE	EVIDENCE
EU	Directive (EU) 2019/882 (EAA), Art. 2(2)	Applies to services provided to consumers after 28 June 2025 ; Art. 2(2)(f) puts e-commerce services in scope, defined at Art. 3(30)	2025-06-28, live	[fetched]
EU	EAA Art. 32(1)	Transitional period ends 28 June 2030 ; service contracts agreed before 28 June 2025 may run to expiry, max 5 years	2030-06-28	[fetched]
EU	EAA Art. 4(5)	Microenterprises providing services are exempt from the Art. 3 accessibility requirements and any obligation relating to compliance with them	live	[fetched]
EU	EAA Art. 3(23)	Microenterprise = fewer than 10 persons and turnover ≤ EUR 2 million or balance sheet ≤ EUR 2 million	live	[fetched]
EU	EAA Art. 2(4)(d)	Out of scope: third-party content neither funded, developed by, nor under the control of the operator	live	[fetched]
EU	EN 301 549 V3.2.1 (2021-03)	The harmonised standard "reflects the content of the W3C WCAG 2.1 Recommendation " — not 2.2	2021-03	[fetched]
EU	EN 301 549 clause 11.8	Authoring-tool clauses apply to us directly: 11.8.2 enable/guide conforming content; 11.8.3 preserve accessibility information in transformations ; 11.8.4 repair suggestions; 11.8.5 at least one conforming template	live	[fetched]
US	DOJ ADA Title II web rule, technical standard WCAG 2.1 Level AA	Binds state/local government, reaches us through procurement flow-down	fact sheet 2024-03-08; IFR published 2026-04-20 ; compliance 2027-04-26 (pop. ≥50,000) and 2028-04-26 (pop. <50,000)	[fetched]
US	ADA Title III	No technical standard promulgated ; exposure is judge-made	—	[SS]
US	<i>Robles v. Domino's</i> , 9th Cir. No. 17-55504	ADA applies to the services of a public accommodation, not services in a place of one; WCAG 2.0 compliance was a possible remedy , not the legal duty	filed 2019-01-15	[fetched]
India	RPwD Act 2016 s.46	"Service providers whether Government or private shall provide services in accordance with the rules on accessibility formulated by the Central Government under section 40 within a period of two years from the date of notification of such rules "	clock start unverified	[fetched]
India	RPwD s.89 / s.90	First contravention up to ₹10,000 ; subsequent not less than ₹50,000 up to ₹5,00,000 ; s.90 attaches liability to the person in charge — a solo founder-director	live	[fetched]

- **Net position today:** a one-person company under EUR 2M turnover meets Art. 3(23) and is exempt under Art. 4(5) [fetched + inference]. **The exemption is a headcount cliff, not a grace period — it disappears on the tenth hire or at EUR 2M with no transitional runway, so build to WCAG 2.2 AA now and treat the exemption as budget relief, not as permission to skip the work.**
- Do not claim Art. 2(4)(d) for our own chrome. The user's document body is third-party content; the page template, navigation, share dialog and theme are ours and are in scope [fetched + inference].

- **Anti-recommendation:** do not commission an ATAG 2.0 conformance claim. EN 301 549 clause 11.8.0 cites ATAG as **informative only** — “information that can be of interest to those who want to go beyond these requirements” [fetched]. Full Part A + Part B conformance buys nothing any jurisdiction above asks for.
- **Source disagreement, recorded not resolved:** EN 301 549 v4.1.1 returned **404** at ETSI's predictable deliver path [measured], so whether a version newer than V3.2.1 is the currently OJEU-referenced harmonised standard is unverified. The India s.40 ICT standards PDF returned an HTML error page [measured] — **do not state a compliance date for India**.
- **Falsifier:** if a state university or EU public body enters procurement, the exemption analysis is void and WCAG 2.1 AA becomes contractual on the date of that contract, not on a statutory date.

35.2 The criteria a markdown editor most commonly fails

SC (LEVEL)	FAILURE MODE	FIX	ANTI-RECOMMENDATION
1.4.3 Contrast (Min) AA	muted / secondary / error text under 4.5:1	Raise the tokens; gate contrast in CI as arithmetic	Do not add a "high contrast mode" and leave the default failing — a compliant alternative does not cure a non-compliant default
1.4.11 Non-text Contrast AA	focus rings and control borders under 3:1	Ring ≥3:1 against both adjacent colours; borders that <i>identify</i> a control ≥3:1	Do not raise decorative dividers; 1.4.11 covers boundaries needed to identify a component
2.1.1 / 2.1.2 Keyboard, No Trap A	Tab captured for indentation traps keyboard users	CM6 leaves Tab as focus-move unless <code>indentWithTab</code> is added; <code>indentWithTab</code> appears 0 times in <code>src/</code> [measured] — keep it that way	Do not "fix" a trap by documenting the escape key in a help modal; 2.1.2 requires discoverability at the point of trap
4.1.2 Name, Role, Value A	<code>role="dialog"</code> without <code>aria-modal</code> or an accessible name	Measured: 10 × <code>role="dialog"</code> , 9 × <code>aria-modal</code> ; <code>SgnkAiButton.tsx</code> has the role and not the attribute [measured]	Do not sprinkle <code>aria-label</code> on containers to silence a linter
1.3.1 Info and Relationships A	visual headings that are not real headings, or skipped levels	Preview already emits real <code>h1</code> – <code>h6</code> with <code>rehype-slug</code> ids [measured, components.tsx:111–114]; add a heading-order lint surfaced as an author <i>warning</i>	Do not auto-renumber the user's headings. Byte-preserving splice is the thesis; rewriting <code>###</code> to <code>##</code> violates it
1.1.1 Non-text Content A	un-alt'd image silently asserted decorative	Measured: components.tsx:180 renders <code>alt={alt ?? ""}</code> [measured]. Distinguish <i>absent</i> from <i>empty</i> : absent → warning + repair suggestion (EN 301 549 11.8.4); explicit <code></code> → honour as decorative	Do not generate alt text with the AI SDK and write it into the file. Auto-alt is the highest-volume false-accessibility generator there is
2.5.8 Target Size (Min) AA	20px icon buttons; SC text requires "at least 24 by 24 CSS pixels" [fetched]	Measured: <code>.sgnk-icon-btn</code> is 28×28px [measured] — passes	Do not add overlapping invisible hitboxes; the Spacing exception requires non-intersecting 24px circles
2.4.11 Focus Not Obscured (Min) AA (new in 2.2)	sticky toolbar covers the focused element	<code>scroll-margin-top</code> equal to sticky header height	Do not remove the sticky toolbar; the SC requires <i>not entirely hidden</i> , not <i>never overlapped</i>
3.3.8 Accessible Authentication (Min) AA (new in 2.2)	cognitive test in the auth flow	next-auth v5 OAuth with a paste-able field satisfies it [inference]	Do not add a CAPTCHA to the share page
3.1.1 Language of Page A	<code>src/app/layout.tsx:67</code> hardcodes <code>lang="en"</code> [measured]	Published pages carry the document's own language from frontmatter	Do not infer language from content heuristics; a wrong <code>lang</code> is worse than a generic one
2.3.3 / motion	<code>prefers-reduced-motion</code> occurs 0 times in <code>globals.css</code> [measured]	Wrap transitions in the media query	Do not remove all animation globally

35.3 The measured contrast failure

The shipped app tokens in `src/app/globals.css` are **not** the design-system tokens the PRD quotes, and they fail worse [measured].

TOKEN	VALUE	ON <code>--bg</code> <code>#FAFAFA</code>	RATIO [DERIVED]	AA TEXT 4.5	NON-TEXT 3.0
<code>--fg</code>	<code>#18181b</code>		16.97:1	PASS	PASS
<code>--fg-muted</code>	<code>#6b6b73</code>		5.06:1	PASS	PASS
<code>--muted</code>	<code>#9b9ba3</code>		2.64:1	FAIL	FAIL
<code>--danger</code>	<code>#b2625e</code>		4.19:1	FAIL	PASS
<code>--success</code>	<code>#4f8b6b</code>		3.84:1	FAIL	PASS
<code>--border</code>	<code>rgba(10,10,10,.06) → #ecec</code>		1.13:1	n/a	FAIL
<code>--ring (focus)</code>	<code>rgba(91,33,182,.40) → #baa3df</code>		2.14:1	n/a	FAIL
<code>dark --muted</code>	<code>→ #6e6e6e on #1a1a1a</code>		3.41:1	FAIL	PASS

`--danger` is used for `role="alert"` error text at `fontSize: 12` in `GoogleSignInButton.tsx` and `SgnkAiButton.tsx` [measured] — small text, so 4.5:1 applies, so **1.4.3 fails in production today**.

Source disagreement, recorded not resolved: the PRD's design-system row (`#b8b8b8` = 2.14:1, ink 20.2:1, blue 5.16:1) is not reproducible by the WCAG 2.x relative-luminance formula, which returns **1.984:1, 19.798:1** and **5.225:1** [derived: L(#b8b8b8)=0.4735; (1.0+0.05)/(0.4735+0.05)=1.984]. The implementation returns 4.542:1 for `#767676` on white, the canonical AA boundary, so the formula is correct [measured]. The verdict (fails AA) is unaffected by the divergence.

35.4 CodeMirror 6 posture and its limits

Installed: `@codemirror/view` 6.43.0, `state` 6.6.0, `autocomplete` 6.20.2, `search` 6.7.0, `language` 6.12.3, `commands` 6.10.3, `lint` 6.9.6 [measured].

What CM6 gives us, read out of the shipped bundle [measured]: content DOM built with `role: "textbox"` , `aria-multiline: "true"` , `aria-readonly` when read-only; a polite live region created at construction (`cm-announced` , `aria-live="polite"` , positioned `top:-1000px`); `EditorView.announce` as a documented `StateEffect`; `aria-placeholder` via `contentAttributes` .

- **CodeMirror publishes no accessibility guidance.** `codemirror.net/examples/accessibility/` is a **404** and the examples index lists 22 examples, none of them accessibility [measured]. Any plan that says "follow CodeMirror's accessibility guidance" is planning against a document that does not exist.
- `role="textbox"` + `aria-multiline` means screen readers read the buffer as a flat text field. **Decorations, folds, widgets, gutters and the ghost-text AI overlay are not in the accessibility tree** [inference from the measured attribute set]. Anything shown as a decoration must also be announced.
- `EditorView.announce` **has 0 call sites in** `src/` [measured]. Search-match navigation, fold/unfold, AI accept/reject and splice refusals are silent to a screen reader today.

- The content DOM carries no `aria-label` from this repo [measured: 0 hits for `ariaLabel` / `contentAttributes`] — the textbox is unnamed, a 4.1.2 failure.
- Announce on settle, not on change: `aria-live="polite"` coalesces and drops under rapid updates, so a per-keystroke ghost-text announcement will either flood or vanish [inference].
- **Anti-recommendation:** do not replace CM6 with `contenteditable` or add a parallel `<textarea>` "accessible mode". Both give the same flat-textbox model, neither has CM6's announce plumbing, and the second doubles the surface.

35.5 CI gate checklist

GATE	TOOL	ASSERTION	BLOCKING
Token contrast	~30-line script over <code>globals.css</code> , WCAG formula	every text token $\geq 4.5:1$ on its background; every focus ring and control border $\geq 3:1$. Fails today on 6 tokens [measured]	yes — arithmetic, zero flake
Static JSX	<code>eslint-plugin-jsx-a11y</code> at error level	0 errors	yes
Rendered page	<code>axe-core</code> (105 rules in <code>doc/rule-descriptions.md</code> ; <code>wcag2aa</code> $\times 6$, <code>wcag21aa</code> $\times 3$, <code>wcag22aa</code> $\times 1$ [measured, counted])	0 violations at <code>wcag2a</code> , <code>wcag2aa</code> , <code>wcag21a</code> , <code>wcag21aa</code> , <code>wcag22aa</code> on 5 published fixtures	yes
Published fixtures	Playwright + axe on: heading-skip, image-without-alt, GFM table, math, RTL/CJK	fixed expected violation set, diffed	yes
Keyboard reachability	Playwright tab-walk	every interactive element receives focus; <code>document.activeElement</code> is never <code>body</code> after a modal opens or closes	yes
Dialog contract	grep gate	<code>count(role="dialog") == count(aria-modal)</code> == count with accessible name; currently 10 / 9 / unverified [measured]	yes
Announce coverage	unit test	<code>EditorView.announce</code> dispatched for search next/prev, fold/unfold, splice refusal, AI accept/reject; currently 0 [measured]	yes
Accessibility-preserving transformation	extend <code>mdmax</code> <code>cert</code> to assert heading level, alt presence, <code>th</code> scope and <code>lang</code> survive <code>md</code> $\rightarrow$ HTML $\rightarrow$ PDF	byte-level	yes
Manual, quarterly	NVDA+Firefox, VoiceOver+Safari, scripted 12-step walkthrough	recorded	no

11.8.3 is the same claim the engine already makes — deterministic reversible projection and cross-engine degradation certification are preservation-in-transformation, so extending `mdmax` `cert` to certify accessibility information converts existing engine work into a standards claim at near-zero marginal cost [fetched + inference].

- **Anti-recommendation:** do not report an "axe score" or a Lighthouse accessibility number as a CI threshold. axe returns violations, not scores, and covers roughly a third of WCAG issues [SS]; Lighthouse's number moves when its weights change, producing regressions that are not regressions. A green axe run on a page with a 2.14:1 focus ring is exactly the false green this codebase's own rules warn about.
- **Do not buy:** overlay/widget scripts, a VPAT written before the audit (writing "Supports" while `--muted` is 2.64:1 is a misrepresentation to a procurement officer), an accessibility statement with no remediation dates, or WCAG AAA — EN 301 549 V3.2.1 puts AAA in an informative clause 9.5 [fetched].

§36 API design

36.1 Shape

REST for the noun layer, three RPC verbs for the operations that are not CRUD. `locate`, `splice` and `cert-check` are procedures with refusal semantics; modelling a refusal as a failed `PUT` loses the refusal reason [inference]. Notion sets the precedent — "follows RESTful conventions when possible" while shipping `POST /v1/search` and `POST .../query` [fetched].

- **Bytes are the response body, not a JSON string field.** `Accept: text/markdown` returns the file verbatim; `Accept: application/json` returns an envelope. A byte-preserving engine that base64s or JSON-escapes its own output has broken its promise in the transport [inference].
- **Reject GraphQL.** GitHub's own comparison argues GraphQL's win is avoiding overfetch across nested collections — 11 REST calls collapsing to 1 [fetched]. Our nesting depth is 2 (vault $\rightarrow$ file) and our payload is opaque bytes, which GraphQL cannot shape. GraphQL also forces cost-based limiting: Shopify runs 100 points/sec (Standard) $\rightarrow$ 200 (Advanced) $\rightarrow$ 1000 (Plus) $\rightarrow$ 2000 (enterprise) on a leaky bucket precisely because query cost is unbounded [fetched]. **Falsifier:** if a customer's integration needs ≥ 5 round-trips per screen across ≥ 3 resource types, re-open this.
- **Anti-recommendation:** do not ship GraphQL "for the AI clients". Agents consume tools, not schemas.

36.2 Resource model and the exact v1 endpoint list

Seven resources: `vault` (repo + branch binding) $\rightarrow$ `file` (identified by path, never an opaque ID) $\rightarrow$ `slice` (byte range) $\rightarrow$ `version` (commit SHA) $\cdot$ `cert` $\cdot$ `job` $\cdot$ `webhook`. Percent-encode paths; expose a `safe_key` alias for hostile paths (already queued as R0 engine work [fetched, MEMORY.md]).

#	ENDPOINT	NOTES
1	GET /v1/vaults	cursor-paginated
2	POST /v1/vaults	bind a git remote + installation
3	GET /v1/vaults/{v}	head SHA, byte size, file count
4	DELETE /v1/vaults/{v}	unbind only; never deletes the repo
5	GET /v1/vaults/{v}/files	prefix, modified_since, cursor, limit
6	HEAD /v1/vaults/{v}/files/{path}	cheap ETag/version probe — the polling primitive
7	GET /v1/vaults/{v}/files/{path}	text/markdown = verbatim bytes; application/json = {etag, version, bytes, eol, bom, encoding}
8	PUT /v1/vaults/{v}/files/{path}	full replace; If-Match required
9	DELETE /v1/vaults/{v}/files/{path}	If-Match required
10	POST .../files/{path}/locate	{anchor: heading frontmatter_key line_range byte_range regex} → {start,end,confidence} or refusal. Read-only; free of write quota
11	GET .../files/{path}/slice	?start=&end=, also honours HTTP Range
12	POST .../files/{path}/splice	{edits[], base_version, dry_run} → 200 applied / 409 conflict / 422 refusal
13	GET .../files/{path}/versions	cursor-paginated commits touching the path
14	GET .../files/{path}/versions/{sha}	bytes at that version
15	GET .../files/{path}/diff?from=&to=	byte-range diff, not a tree diff
16	GET /v1/vaults/{v}/search	q, cursor, limit → {path, ranges[], snippet}
17	POST /v1/certs	202 Accepted + job — degradation cert is CPU-heavy
18	GET /v1/certs/{id}	
19	GET /v1/jobs/{id}	one job resource for every async op
20–23	POST /v1/webhooks · GET /v1/webhooks · DELETE /v1/webhooks/{id} · POST /v1/webhooks/{id}/test	
24	GET /v1/meta/versions	mirrors GET api.github.com/versions [measured]
25	GET /v1/meta/rate-limit	current buckets
26	GET /v1/meta/refusals	machine-readable refusal registry; each code is an RFC 9457 type URI

- **Anti-recommendation:** no PATCH on /files/{path}. JSON Merge Patch and JSON Patch both assume a tree; we have bytes. Offering PATCH invites clients to send a document model we refuse to have.
- **Not exposed, and why:** a document tree/AST endpoint (blocksToMarkdownLossy() is a real function name in BlockNote's API [fetched] — publishing a tree makes byte-reversibility a forever contract); rendered HTML; arbitrary git operations; server-side transforms (that is the settled no-eval-lane decision reversed through the back door); raw AI inference passthrough; bulk vault export as one call (that is a git clone); total counts and offset paging; **webhook payloads containing file bytes**; internal provenance (promptDigest, sessionRef, routing decisions).

36.3 Versioning and deprecation

DECISION	VALUE	PRECEDENT
Scheme	date-based, header-carried, required : Frontmatter-Version: 2027-01-15	Notion requires its version header on every request [fetched]
Path	/v1/ forever, meaning "the resource model"	Sanity puts the date in the path, making every URL in every doc version-specific [fetched]
Breaking-change list	GitHub's, adopted verbatim: removing an operation; removing/renaming a parameter or response field; adding a required parameter; making an optional one required; changing a type; removing enum values; adding a validation rule; changing auth requirements	[fetched]
Additive (free, universal)	new endpoints, optional params, response fields, enum values, refusal codes — clients must tolerate unknown refusal codes	[fetched]
Support window	24 months after a successor ships; unsupported → 410 Gone	GitHub [fetched]
Deprecation signalling	Deprecation (RFC 9745, Standards Track, March 2025) + Sunset (RFC 8594, Informational , May 2019) + Link	GitHub exposes both in access-control-expose-headers [measured]
Honest carve-out	a security or data-integrity defect may be fixed inside a pinned version without notice	GitHub reserves exactly this [fetched]

Source disagreement, recorded not resolved: nobody offers an unconditional pin. Notion says additive changes "apply to every API version at the same time, including older ones: pinning Notion-Version does not delay them"; Sanity hedges ("we try very hard") and admits preserving "wrong" behaviour; GitHub reserves breaking a live version "without advance notice"; Stripe's monthly releases are backward-compatible but major releases are not [all fetched]. **Do not promise a freeze.** Take GitHub's side on the carve-out and publish it, because shipping a known-corrupting splice under a version pin would contradict the byte-fidelity claim.

- **Anti-recommendations:** no codenames (Stripe's 2026-08-26.dahlia needs a mapping table to be useful [fetched]); no /v2 when v1 breaks — ship a new date; no floating default version (a floating default makes every additive change a silent breaking change for someone); no X- prefix (deprecated by RFC 6648 — Frontmatter-Version, not X-Frontmatter-Version).
- **Pagination:** opaque forward cursor only, ?cursor=&limit= → {results[], has_more, next_cursor}, limit default 50, max 100 [Notion's shape, fetched]. Cursors encode {head_sha, position} and hard-fail 409 if the branch head moved incompatibly. Reject Link -header + page=N : offset paging over a git history that rewrites on force-push returns duplicates and gaps.
- **Idempotency:** Idempotency-Key on POST /splice, /certs, /vaults only. The header is draft-ietf-httpapi-idempotency-key-header **rev 07, still not an RFC** [fetched] — cite it as convention. Stripe's semantics: store status and body of the first request regardless of outcome, error on parameter mismatch, keys ≤255 chars, POST only, 24h retention [fetched]. **An idempotency key protects against double delivery;** base_version protects against stale intent. **Ship both.** Do not auto-generate a key from a body hash — two legitimate identical splices are indistinguishable from a retry.

- **Errors:** RFC 9457 `application/problem+json` (July 2023, obsoletes 7807) with extension members `refusal_code`, `byte_range`, `observed`, `expected`, `corrected_example` (\$3.2 permits extensions [fetched]). Mapping: `409` stale `base_version` or head moved · `422` engine refused (ambiguous anchor, invalid UTF-8 boundary, zero-indent sequence, bare CR) · `410` version sunset · `429` either limiter · `413` over payload cap. **Do not report a refusal as a 400, and do not offer a `force` / `ignore_refusal` flag** — every integrator will set it.

36.4 Rate limits, derived from the cost model

Unit prices [fetched 2026-08-29]: Workers Standard \$0.30/additional million requests, \$0.02/additional million CPU-ms, 10M requests + 30M CPU-ms included, \$5/mo account minimum. R2 Standard: Class A \$4.50/M, Class B \$0.36/M, storage \$0.015/GB-month; free grant 1M Class A, 10M Class B, 10 GB-month.

CALL	COMPOSITION	\$/CALL	\$/MILLION
<code>read-slice</code>	1 Worker req + 1 Class B + ~10 CPU-ms	0.00000086	\$0.86
<code>search-vault</code>	1 Worker req + 5 Class B + ~40 CPU-ms	0.00000290	\$2.90
<code>cert-check</code>	1 Worker req + 1 Class B + ~250 CPU-ms	0.00000566	\$5.66
<code>splice-edit</code>	1 Worker req + 1 Class B + 1 Class A + ~30 CPU-ms	0.00000576	\$5.76

Class A is **78.1%** of a splice [derived: $4.50 \div 5.76$]; **a write costs 6.70x a read** [derived: $5.76 \div 0.86$]. Rate-limit reads and writes in separate buckets at separate prices — a single requests-per-minute number is economically wrong by nearly 7x.

TIER	WRITES/MO	READS/MO	SEARCHES/MO	CERTS/MO	COST AT CAP	BUDGET (5% OF NET)	HEADROOM
Free (₹0)	300	3,000	1,000	30	\$0.00738	absorbed by free grants	—
Pro (₹299 / \$5)	10,000	30,000	10,000	500	\$0.11523	\$0.130	\$0.0148
Power (₹599 / \$10)	20,000	75,000	20,000	1,500	\$0.24619	\$0.260	\$0.0138

Free tier at scale: **\$7.38/mo per 1,000 users**, \$73.78 per 10,000, \$737.78 per 100,000 [derived]. R2's 1M Class A grant covers **3,333** free users at 300 writes each [derived: $1,000,000 \div 300$].

Three limiters, and this is the load-bearing design point: (1) **token bucket per token per minute** as the abuse guard — Free 6 writes / 60 reads per min, Pro 30/300, Power 60/600, burst 2x steady, leaky-bucket refill; (2) **monthly quota** as the *margin* guard — Pro's 30 writes/min would permit **1,296,000 writes/month, 86x the quota** [derived], proving a per-minute limit alone cannot protect unit economics; (3) **per-vault write serialisation**, because a git repo is exactly the namespace Dropbox locks — parallel writers get `429 too_many_write_operations` [fetched]. Headers: `RateLimit-Limit/-Remaining/-Reset` per `draft-ietf-httpapi-ratelimit-headers` **rev 11, still a draft** [fetched], plus `Retry-After` on every 429 — the only universally honoured signal.

- **Anti-recommendation:** do not adopt Dropbox's opacity (it publishes no limits [fetched]). It works for an incumbent with a support org; for a solo founder it turns every limit question into a ticket. Publish the table.
- **Unverified:** the 10/40/250/30 CPU-ms per-call assumptions are estimates, not measurements — re-derive from live Workers CPU telemetry before publishing any tier table externally. ₹95.4/USD is [SS, August 2026]. Do not cite Linear's rate limits; the page did not yield to extraction [measured].

36.5 Auth, and how the API relates to the MCP server

CREDENTIAL	USE	PROPERTIES
PAT (<code>f_m_pat_...</code>)	scripts, CI, the user's own agent	vault-scoped, verb-scoped, expiring (max 1 year), single-shot display
OAuth 2.1 + PKCE	third-party apps acting as a user	same scope vocabulary, refresh tokens, revocable per app
Installation token	our own MCP server and first-party clients	short-lived, minted from the OAuth grant

Six scopes: `vault:read` · `vault:write` · `slice:read` · `slice:write` · `cert:run` · `webhook:manage` · `slice:write` without `vault:write` is the interesting grant — an agent that may splice inside existing files but may not create, replace or delete them. Fold OAuth-app requests into the *user's* bucket, as GitHub does [fetched], so one user cannot multiply quota by registering apps. **Open blocker:** `read:user` grants no repo access, which blocks the GitHub-App path as written [fetched, repo plan §14 item 4] — resolve before publishing scope docs.

	REST API	MCP SERVER
Caller	code a developer wrote	a model choosing at runtime
Granularity	26 endpoints	5 consolidated verbs, hard-capped under 20
Response	complete envelope	verdict-first , prose-shaped, refusals name the rule with a corrected example
Errors	RFC 9457 <code>problem+json</code>	≤200-char actionable NL hint
Auth	OAuth/PAT with scopes	installation token from the same grant
Versioning	date header, 24-month window	tool-shape stability; MCP has no version negotiation for tool schemas

One engine core, two adapters, one quota ledger. Google's Docs MCP server does not get its own quota system — it "uses read and write request metrics", the same 3,000/300 read and 600/60 write buckets as the REST API [fetched]. Copy this exactly: a user who exhausts writes via MCP must find them exhausted via REST. MCP is a strict subset — `land` = `locate` + `splice` (or `PUT`) + `cert` in one call; `read-slice` = #11; `search-vault` = #16; `cert-check` = #17/#18. Nothing exists only in MCP, which keeps the refusal registry, the quota meter and the audit log single-sourced.

- **Anti-recommendation:** do not expose the REST API through MCP as a generic `http_request` tool, and do not generate MCP tools from an OpenAPI spec — 26 auto-generated tools blows the sub-20 cap and quality degrades measurably past ~30 tools.
- **Anti-recommendation:** do not build the REST API before the MCP quota ledger is shared. Retrofitting one quota across two surfaces after launch means one of them silently gets free capacity.

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37.1 The layering rule

Our data lives in the user's git repository, which already has a permission model. There are three candidate postures — defer to the provider, duplicate its ACL into our database, or intersect the two. **We intersect on write, defer on connect, and own read-and-comment outright; frontmatter never grants more than git grants on a write, and may freely grant less.**

OPERATION	POSTURE	MECHANISM	WHY	ANTI-RECOMMENDATION
Connect a repo	Defer	Only a GitHub identity holding <code>admin</code> on the repository may install the App and connect it. GitHub already defines that role, and managing access is Admin-only [fetched 2026-08-29, docs.github.com repository-roles-for-an-organization]	The App installation outlives any individual's access, so the grant must come from the role that can already grant durable access	Do NOT let a frontmatter Owner connect a repo they hold only Write on. That reproduces the deploy-key hazard GitHub itself warns about: "any user who has the private key can read from or write to the repository... even if they're later removed from the organization" [fetched]
Write bytes (splice commit)	Intersect, evaluated at commit time	The commit succeeds only if the acting identity <i>also</i> resolves to write access on the repo at the instant of the commit	The commit is already a network round-trip to the provider, so the authorisation check costs nothing extra [inference]	Do NOT pre-compute and cache the intersection. That is Sourcegraph's mirror architecture, whose documented worst case is "as long as the time it takes to completely sync all user or repository permissions" — their own worked example is 5000 users ÷ 40 users/min = 125 minutes [fetched 2026-08-29, sourcegraph.com/docs/admin/permissions/syncing; arithmetic re-derived: 5000/40 = 125] [derived]
Read and comment	Own	frontmatter's own grants apply to frontmatter's rendered projection, not to the repo. No git account required	Requiring a GitHub account converts every client review into a GitHub onboarding. Decap exists for this reason: "Editors don't need an account on your Git hosting platform at all" [fetched 2026-08-29, decapcms.org/docs/turbo-how-it-works]	Do NOT require GitHub OAuth for a Viewer or Commenter
Commit attribution	Duplicate deliberately	The human's identity goes in the commit trailer even when the API call is made by the App installation	A repo where every commit reads as the App destroys <code>git blame</code> for the user's own file — which contradicts the projection law [inference]	Do NOT rely on the App identity alone, as Decap Turbo's server-side model does [fetched]

- Do NOT adopt Notion's resolution rule. Notion states "Notion respects the **broadest** level of access given to a user" [fetched 2026-08-29, notion.com/help/sharing-and-permissions]. With git underneath, max-wins means a frontmatter grant can exceed a *revoked* GitHub grant — that is a privilege-escalation bug wearing a UX rationale [inference].
- Sources disagree on resolution and we are not splitting the difference: Notion resolves broadest-wins, GitBook resolves "by **precedence, not by the highest role** across every level" and documents that this surprises people [fetched 2026-08-29, gitbook.com/docs roles]. We take neither; intersect-with-provider is a third rule that neither product needs because neither sits on top of someone else's ACL.
- Falsifier for the intersect rule:** if commit-time provider checks add more than 150 ms p95 to a splice write, or if GitHub's rate limits make per-write checks infeasible at 20 concurrent editors, the rule must move to a short-TTL cache with an explicit staleness bound stated in the UI — not to a full mirror.

37.2 Role model — four roles, one scope, no inheritance tree

ROLE	CAN	CANNOT	GROUNDING	ANTI-RECOMMENDATION
Owner	Everything: connect/disconnect the repo, publish, unpublish, billing, delete workspace	—	Every product surveyed has exactly one terminal role; Figma binds it to a person — "There is only one owner per team" [fetched 2026-08-29, help.figma.com]	Do NOT allow multiple Owners at launch. With two, either can disconnect the repo the other depends on and there is no arbiter. Add co-owners when a customer raises bus-factor, not before
Editor	Read; write bytes (splice edits commit to the repo)	Change repo connection, billing, publish/unpublish	Write is the meaningful boundary because a write is a commit — an irreversible external side effect [inference]. GitHub Write, Notion Can edit, Outline <code>read_write</code> , GitBook Editor all sit here [fetched x4]	Do NOT split Editor into GitBook's Editor/Reviewer/Creator ladder. That ladder gates change-request merges, a workflow we do not have; it buys three roles and zero enforced invariants
Commenter	Read; attach comments, which live outside the file and never in the bytes	Write bytes, publish	Present in Notion, Google, GitBook, Figma [fetched x4]. Preserves the projection law: a comment is not a byte in the file [inference]	Do NOT ship a Suggester role. Google documents suggestion as a <i>mode</i> : "People can suggest edits when you give them permission to comment on or edit " [fetched 2026-08-29, support.google.com/docs/answer/6033474]. A fifth role adds a cell to every future permission matrix and buys nothing an in-editor suggestion mode does not
Viewer	Read	Comment, write, publish	A Commenter writes <i>some</i> record; a Viewer writes nothing. Google, Outline, GitBook and Figma all keep the two distinct [fetched x4]	Do NOT merge Viewer into Commenter to save a role. The merged role cannot express "share the doc, no annotation trail", which is the entire client-review case

Deliberate omissions, each with its trigger for reconsideration:

- No Admin distinct from Owner.** At solo-founder scale the two resolve to the same person. Add when a customer has more than one workspace administrator [inference].
- No Triage/Maintain analogue.** GitHub needs them because issues and releases exist; we have neither [inference].
- No per-folder or per-document roles at launch.** Every surveyed product with three or more scope levels ships a permissions debugger — Confluence ships two, "People who can view" and "Inspect permissions" [fetched 2026-08-29, confluence.atlassian.com/doc/permissions-and-restrictions-139557.html]. **A model that needs a debugger is a model too complex to reason about unaided** [inference]; if we ever need one, the model is already wrong.
- No groups or teams.** Groups are the mechanism behind Confluence's documented union trap: "You may have revoked permission for that individual user to add pages... but if they're a member of a group that *is* allowed to add pages, they'll still be able to create new pages" [fetched]. Add groups when per-person grants become tedious, which is above roughly 15 people [inference].
- No subtractive per-document restriction.** Google removed downward file-level overrides — "You can no longer give someone **less** access to an individual file if they have higher access to its parent folder" [fetched 2026-08-29, support.google.com/drive/answer/2494822]. Two of the three surveyed products that had subtraction have restricted or removed it [derived].

37.3 Sharing and revocation

- **Exactly two published states, never three:** `private` and `published-at-slug`. Do NOT ship an "anyone with the secret link" third mode. Notion documents the failure in its own help text: "Even if your Notion Site has been unpublished, it's possible your page's general access settings have been set to Anyone on the w[eb]" [fetched] — unpublishing one channel revokes nothing on the other.
- **We have already paid for a dual-channel revocation gap once.** `unpublish` called `revalidatePath('/p/<slug>')` while the live ISR page served at `/<slug>` with `revalidate = 60`, so unpublishing purged nothing and the note kept serving from cache [fetched 2026-08-29, commit `d50a6b2`, message §6.4].
- **Revocation is a purge and must be verified, not reported.** The DELETE handler already reads the slug before removal specifically so it can purge [measured 2026-08-29, `src/app/api/share/route.ts`]. Add a post-unpublish fetch of `/<slug>` asserting HTTP 404, and refuse to report "unpublished" until it does. Do NOT treat the `revalidatePath` return value as success.
- **Slug entropy.** The slug field is currently 1–60 characters of user-chosen text [measured, `postSchema` in `route.ts`]. If any unlisted-URL mode ever ships, its token must be ≥128 bits: 5–6 character tokens "can be scanned using brute-force search... effectively public", and 7% of exposed OneDrive accounts in that study were **writable** [fetched 2026-08-29, arXiv 1604.02734, Georgiev & Shmatikov, published 2016-04-10].
- **Slug reuse:** never free a slug on unpublish. Notion refuses reuse after deletion [fetched]; a freed slug means a stale inbound link resolves to a different document.
- **Publish exactly one file, never a subtree.** Notion's rule — "Publishing a Notion page to the web means all of its subpages will be published too" [fetched] — in a git repo means publishing the rest of the user's repository.
- **Strip commit-author email addresses from the public projection by default.** A published projection of a git repo leaks them by construction; Notion warns about exactly this class, noting published-page metadata includes contributors' "names, profile photos, and **email addresses**" [fetched]. Do NOT expose a per-document toggle — a default-off privacy control that can be flipped per document is a leak generator.
- **Link expiry is a safety default, not an upsell.** Google gates expiry to eligible work/school accounts [fetched], with the result that most links never expire.

37.4 Invites and billing

STEP	RULE	ANTI-RECOMMENDATION
Invite	Owner enters an email, picks one of four roles, sends. Expires in 7 days (GitHub's documented value [fetched 2026-08-29]) and is retryable	Do NOT ship a secret join link or allowed-domain auto-join. Both of Notion's automatic paths convert a person into a billable member without per-person approval [fetched]; for a solo founder selling globally that is a refund and a chargeback, not a growth loop
First read	A Viewer or Commenter reads the document before authenticating	Do NOT gate reading behind signup. Notion requires the guest to hold an account [fetched]; that is where a client review dies
Auth	Editor authenticates with GitHub OAuth (required by the write-time intersect). Viewer/Commenter uses email auth, no GitHub [inference from §37.1]	—
Billing: Owner, Editor	Billed per seat. These identities write bytes and consume commit budget [inference]	Do NOT bill per repo or per document; per-repo pricing punishes exactly the git-native user we want
Billing: Commenter, Viewer	Never billed. Figma: "you can let others view and comment on your files without purchasing extra seats"; Notion: "Guests are free of charge" [both fetched 2026-08-29]	Do NOT copy GitBook (guests and readers billed) or GitHub ("adding an outside collaborator to a private repository will use one of your paid licenses ") [both fetched]. But free commenters are an abuse surface — cap comment volume per workspace , never silently convert the cap into a seat charge
Billing: external Editor who already has GitHub write	Billed as an Editor seat	Do NOT exempt them because "GitHub already authorised them". That gives any org with a large GitHub team unlimited free Editors
Time-boxed collaborator	Editor-equivalent, hard expiry ≤1 year, no seat — Notion's "Temporary members don't use a paid seat" pattern [fetched]	Do NOT make it renewable in-product. A renewable free Editor is a free plan with extra steps

Recorded and unresolved: Figma's pricing page simultaneously states view-and-comment needs no extra seat *and* sells a Collab seat at \$3/mo Professional, \$5/mo Organization, \$5/mo Enterprise against Full seats of \$16/\$55/\$90 [fetched 2026-08-29, figma.com/pricing]. Collab as a fraction of Full: 3/16 = 18.8%, 5/55 = 9.1%, 5/90 = 5.6% [derived]. There is no industry norm on guest billing to copy — two of six surveyed products bill guests, three do not, one is self-contradictory [derived].

No invite-to-activation conversion figure appears in this section, because no primary source for one was reachable; every available number was a vendor marketing page or a secondhand blog [SS — must not be published as fact].

§38 Observability and incident response

38.1 Baseline and the binding constraint

FACT	VALUE
Observability packages in <code>package.json</code>	0 of 47 deps + 21 devDeps — no <code>@sentry/*</code> , <code>@opentelemetry/*</code> , <code>pino</code> , <code>winston</code> , <code>axiom</code> , <code>logtail</code> [measured]
Sentry / instrumentation config at repo root	none [measured]
<code>src</code> surface	226 <code>.ts</code> / <code>.tsx</code> files; 15 <code>console.log</code>
Runtime	<code>next ^16.2.6</code> , <code>react ^19.2.6</code> [measured]
Stack mismatch to resolve first	<code>firebase.json</code> , <code>firestore.rules</code> , <code>firestore.indexes.json</code> at root vs the stated Cloudflare R2 + Workers stack [measured] — reconcile before instrumenting, or you will instrument two backends

This is greenfield, which is the only moment when scrubbing is cheap [inference]. **We hold user documents, so the rule is absolute: instrument the shape of the operation, never its payload — and enforce it with server-side scrubbing rules, not with developer discipline, because the SDK-side hooks run on machines we do not control.**

38.2 The allowlist and the denylist

SIGNAL	CAPTURE	RATIONALE
Splice edit	<code>doc_id</code> (opaque), <code>byte_len_before</code> , <code>byte_len_after</code> , <code>range_start</code> , <code>range_len</code> , <code>outcome</code> ∈ { <code>applied</code> , <code>refused</code> }, <code>refusal_code</code>	Offsets plus a refusal code fully characterise a bug without a byte of text
Parse / round-trip	<code>bytes_in</code> , <code>ast_node_count</code> , <code>roundtrip_byte_identical</code> : bool, <code>first_divergence_offset</code> : int	An offset localises the defect; the bytes at that offset are the user's
Degradation cert	<code>engine_id</code> , <code>rule_id</code> , <code>severity</code> , <code>count</code>	Certification is per-rule, not per-document
Sync / R2	<code>op</code> , <code>status</code> , <code>size_bucket</code> , <code>latency_ms</code> , <code>r2_key_hash</code>	Object keys carry filenames; filenames carry subject matter
Auth	internal <code>user_id</code> (not email), <code>provider</code> , <code>outcome</code>	next-auth v5
AI SDK	OTel <code>gen_ai.*</code> Required + Recommended only	see below

Forbidden anywhere in telemetry: document bytes, document titles and filenames, markdown fragments, CodeMirror selection text, clipboard contents, cleartext R2 object keys, AI prompts and completions, exception `.value` strings from the engine, `Logentry.formatted`, DOM text in replays, `abs_path` from user machines (Tauri), email addresses.

38.3 The enforcing mechanisms, named exactly

MECHANISM	CONFIGURATION	NOTE
Sentry server-side scrubbing (default)	scrubs credit-card-shaped values and keys/values containing <code>password</code> , <code>secret</code> , <code>passwd</code> , <code>api_key</code> , <code>apikey</code> , <code>auth</code> , <code>credentials</code> , <code>mysql_pwd</code> , <code>privatekey</code> , <code>private_key</code> , <code>token</code> , <code>bearer</code> [fetched]	None of these match markdown. The defaults protect us from nothing here
Advanced Data Scrubbing (takes precedence over other server-side rules)	<code>[Remove] [Anything] from [exception.values.*.value] · [Remove] [Anything] from [Logentry.formatted] · [Mask] [Anything] from [\$frame.*]</code>	These three rules are the document constraint
Documented gotcha	<code>[Mask][Anything] from [\$frame.**]</code> will not scrub <code>filename</code> or <code>abs_path</code> — they are not default PII fields; add them explicitly [fetched]	
"Additional Sensitive Fields"	substring-matched — Sentry's own example: entering <code>exp</code> removes "Unexpected error" from events [fetched]	Do not use for short tokens; this is the substring-vs-boundary class
Geo	derived from IP even when IP storage is off ; killing it requires an Advanced rule [fetched]	
SDK-side	<code>beforeSend</code> (return <code>null</code> to drop), <code>beforeSendTransaction</code> , <code>beforeSendSpan</code> , <code>beforeBreadcrumb</code> (return <code>null</code> per crumb), <code>ignoreErrors</code> , <code>allowUrls</code> / <code>denyUrls</code> , <code>thirdPartyErrorFilterIntegration</code> (browser SDK ≥ v8) [fetched]	Second layer, not the primary one
<code>sendDefaultPii</code>	defaults <code>false</code> , is deprecated, and is removed in v11 — replaced by <code>dataCollection</code> , and "passing <code>dataCollection</code> opts you into the more permissive <code>dataCollection</code> defaults" [fetched, verbatim]	A silent privacy regression on a major-version bump. Pin the SDK; opt out of each category explicitly at migration
Source maps	Next.js SDK <code>sourcemaps.deleteSourceMapsAfterUpload</code> defaults <code>true</code> ; client maps deleted after upload, server maps kept for runtime errors [fetched]	Keep the default; never ship client maps to the CDN
OTel <code>gen_ai</code>	Required: <code>gen_ai.operation.name</code> , <code>gen_ai.provider.name</code> . Recommended and safe: <code>usage.input_tokens</code> , <code>usage.output_tokens</code> , <code>usage.reasoning.output_tokens</code> , <code>response.model</code> , <code>response.id</code> , <code>response.finish_reasons</code> , <code>response.time_to_first_chunk</code> , <code>request.temperature/max_tokens/top_p</code>	Conventions moved to <code>semantic-conventions-genai</code> , status Development, not Stable [fetched] — pin the version, expect churn
OTel <code>gen_ai</code> Opt-In, must stay off	<code>gen_ai.input.messages</code> , <code>gen_ai.output.messages</code> , <code>gen_ai.system_instructions</code> , <code>gen_ai.prompt.variable</code> , <code>gen_ai.tool.definitions</code> [fetched]	Spec text verbatim: <i>"OpenTelemetry instrumentations SHOULD NOT capture them by default, but SHOULD provide an option for users to opt in."</i>

Adopt the spec's **Pattern 1** — **"Don't record instructions, inputs, or outputs"** permanently, in every environment including staging. **Anti-recommendation:** do not adopt Pattern 2 ("only in pre-production"). Staging in a solo shop is where you paste a real user's failing document to reproduce their bug, which is precisely how document content reaches a telemetry store [inference].

Do not enable Sentry Session Replay. It records the DOM, and our DOM is the user's document. The 5,000-free-replays promotion [fetched] is the most expensive free thing on this list.

38.4 Stack and bill, priced 2026-08-29

LINE	COST
Sentry Team (annual, \$312/yr)	\$26.00/mo
Cloudflare Workers Paid (account minimum; Workers Logs included)	\$5.00/mo
UptimeRobot Solo (annual, \$108/yr)	\$9.00/mo
Better Stack free (status page + 10 monitors as a second opinion, incident management, unlimited phone/SMS for 1 responder)	\$0.00
Axiom Personal (30-day queryable app logs)	\$0.00
Total	\$40.00/mo → \$480.00/yr [derived: 26+5+9+0+0]

Pre-revenue variant: Sentry Developer \$0 + Workers Free + UptimeRobot Free + Better Stack Free = **\$0/mo**, at the cost of 1 Sentry user, 5k errors/mo, 30-day lookback and a 5-minute check interval [derived from fetched tiers]. Sentry overage is **\$0.0003625/error** in the 50K–100K band, logs +\$0.50/GB [fetched].

- Anti-recommendation:** do not buy Sentry Business (\$960/yr). Team already carries "up to 90-day lookback" [fetched]; Business buys SAML/SCIM and advanced quota management that a team of one cannot use.
- Anti-recommendation:** do not self-host GlitchTip. A solo founder self-hosting the thing that tells you production is broken has coupled the alarm to the building, and GlitchTip counts uptime checks as billable events, so its 1,000-event free tier evaporates the moment you add monitoring [fetched + inference].
- Anti-recommendation:** do not adopt Checkly at this stage, do not buy Statuspage Hobby at \$29/mo (93% of a Sentry Team seat [derived: \$29 vs \$26] for a page nobody visits until the day it is down), and do not adopt Datadog / New Relic / Grafana Cloud — per-host and per-custom-metric models punish a Workers-shaped app.

- **Source disagreements, recorded not resolved:** Better Stack's page shows both "\$34" and "\$29" with no static monthly/annual label; Checkly's plan card reads \$64/mo annual while its FAQ uses "\$80 per month"; Bugsnag/Insight Hub and Instatus paid prices are client-rendered and **were not obtained** — do not quote them [all fetched/measured].

38.5 SLOs, alerting, retention

SLI form is *good events* ÷ *valid events*; keep to five or fewer SLI types [fetched, sre.google/workbook/implementing-slos].

#	SLI	INTERNAL SLO	MONTHLY BUDGET	PUBLISHED?
1	Document durability — saves acknowledged that survive a read-back	99.99%	4.3 min [derived: 0.0001×30×24×60]	no
2	Splice correctness — byte-exact or explicit REFUSE, never a silent wrong write	100%, zero-tolerance	0; any breach is P1	no
3	Save/API availability (non-5xx on write path)	99.5%	216 min = 3.60 h [derived]	yes
4	Editor load latency, p90 < 1s, p99 < 3s	99.0%	432 min [derived]	no
5	Read availability	99.9%	43.2 min [derived]	yes

Publish #3 and #5 only, as 28-day trailing measured numbers with the method stated — "measured from an external prober every 60s from 1 region", not "99.9% uptime guaranteed". **Never publish a contractual SLA with credits:** 99.9% is 43.2 minutes/month [derived], and a single Cloudflare regional incident plus a sleep cycle exceeds it. Do not set an SLO on AI features; they depend on a third-party endpoint we do not control.

Burn-rate policy from source: *"2% budget consumption in one hour and 5% budget consumption in six hours as reasonable starting numbers for paging, and 10% budget consumption in three days as a good baseline for ticket alerts"* [fetched, sre.google/workbook/alerting-on-slos] → burn rates **14.4×**, **6×**, **1×** [derived].

Page (phone, DND-override): write-path 5xx burn ≥14.4× over 1h and ≥6× over 5m; **any** splice integrity violation; R2 write error rate >1% for 5 min; auth totally down (0 successful logins in 10 min with >0 attempts); uptime prober 3 consecutive failures from ≥2 regions. **Ticket:** 1× burn over 3 days; read-path latency regression; cron/heartbeat miss; a new error *type* first seen; SSL/domain expiry (UptimeRobot alerts at 30/14/7 days [fetched]); Sentry quota at 80%. **Never alert on:** individual client-side JS exceptions; error *counts* untied to a budget (the source's own critique: *"A 0.1% error rate for 10 minutes would alert, while consuming only 0.02% of the monthly error budget"* [fetched]); single-region prober failure; serverless CPU/memory; deploys; any alert that has fired twice without action. Do not route pages through Slack or email — both are DND-suppressed and neither escalates — and do not buy PagerDuty; Better Stack's free tier covers one responder [fetched].

LOG CLASS	CONTENTS	RETENTION	STORE
Security / audit	NTP-synced timestamp (to NIC/NPL per the same directions), request id, opaque <code>user_id</code> , action verb, source ASN, outcome. No document bytes, no filenames, no email, no cleartext IP	180 days, India region, write-once	R2, IN jurisdiction, object-lock
Application / debug	structured JSON, no free-text user content	7 days (Cloudflare Workers Logs maximum [fetched]); mirror to Axiom Personal for 30 days	Workers Logs / Axiom
Error events	scrubbed per §38.3	30 days (Developer) / ≤90 (Team); set the shortest that supports debugging	Sentry
Traces / <code>gen_ai</code>	Required + Recommended attributes only	7–30 days	Workers / Axiom
Document content	—	never in telemetry	R2 only

The conflict is real and both ledgers are required: **CERT-In Directions 28.04.2022** (under s.70B(6) IT Act) require reporting within **6 hours of noticing** and maintaining logs of all ICT systems **for a rolling 180 days within Indian jurisdiction** [fetched] — a floor; **GDPR Art. 5(1)(e)/(c)** and **DPDP Act 2023 s.8(7)** (erase on withdrawal or when the purpose is no longer served, whichever is earlier, and cause your processor to erase too) impose ceilings [fetched]. The 180-day log is defensible only because it holds no content and its identifiers are opaque; the moment a filename lands in it, it becomes a 180-day personal-data retention that cannot be justified [inference]. Do not set Sentry retention to the maximum "because storage is free" — every extra day is 24 more hours in which an undiscovered scrubbing gap holds a user's paragraph. Cloudflare Workers Logs `head_sampling_rate` defaults to 1 (100%) and Workers Free caps at **200,000 log events/day** [fetched] — sample before hitting the wall, not after.

Incident runbook. (1) Declare on any page or any user report of data loss: `INC-YYYYMMDD-NN | sev | symptom | commander: me` . (2) **SEV1** document loss/corruption or global write failure → status page within **15 min**; **SEV2** degraded → 60 min; **SEV3** cosmetic → no page. (3) **Stabilise before diagnose** — roll back the deploy first; the cause can wait, the user's file cannot. (4) If personal data was exposed, start **two clocks**: CERT-In **6 hours** from noticing and GDPR **72 hours** to the supervisory authority, plus DPDP s.8(6) intimation to the Board *and each affected Data Principal* — file without waiting for root cause, since Art. 33(2) permits lateness only *with reasons* [fetched]. (5) Capture the 180-day security-log slice **before** remediation touches it. (6) Status page every 30 minutes even when the update is "still investigating"; never name a customer. (7) Resolve against the SLI, not a page refresh. (8) Blameless post-mortem within 5 business days: UTC timeline with `IST = UTC + 5:30` written out, impact in error-budget minutes, detection latency, and "what would have caught this 10 minutes earlier". (9) Every post-mortem adds a burn-rate alert or deletes one; net alert count must not grow monotonically.

Unverified: the **DPDP Rules, 2025** could not be opened (meity.gov.in 404, indiocode.nic.in 404, egazette.gov.in TLS failure) [measured]. Any specific breach-notification hour count or class-based retention period attributed to the Rules is unverified — verify before writing a retention schedule into a DPA.

39.1 Measured current state

FACT	VALUE
<code>src-tauri/tauri.conf.json</code>	<code>frontendDist: "https://md.sgnk.ai"</code> , <code>devUrl</code> same, <code>windows[0].url</code> same — the shell loads remote content, not bundled assets [measured]
Signing	<code>bundle.macOS.signingIdentity: null</code> , <code>providerShortName: null</code> , <code>entitlements: null</code> [measured]
Security	<code>app.security.csp: null</code> [measured]
Identity	<code>productName: sgnk-md</code> , <code>identifier: ai.sgnk.md</code> ; <code>Cargo.toml</code> describes "sgnk-md — Obsidian on the web, as a native macOS app" [measured] — not a frontmatter bundle
Capabilities	<code>plugins</code> <code>shell</code> , <code>os</code> , <code>process</code> , <code>clipboard-manager</code> , <code>dialog</code> . No <code>tauri-plugin-fs</code> , no <code>tauri-plugin-updater</code> [measured]
Toolchain	<code>@tauri-apps/api ^2.11.0</code> , <code>@tauri-apps/cli ^2.11.2</code> ; build scripts for <code>aarch64-apple-darwin</code> , <code>x86_64-apple-darwin</code> , <code>universal-apple-darwin</code> , nothing for Windows or Linux [measured]
Host	macOS 26.6.2 (build 25G83) [measured]
Hardened runtime	Tauri v2 defaults <code>bundle.macOS.hardenedRuntime</code> to <code>true</code> , but it is applied only when a signing identity exists [fetched], <code>tauri-utils/src/config.rs</code> , <code>tauri-bundler/.../macos/sign.rs</code>]

39.2 Cost table, current prices

ITEM	PRICE	CADENCE	SOURCE, DATE
Apple Developer Program	99 USD	per membership year	<code>developer.apple.com/support/compare-memberships</code> [fetched 2026-08-29]
Microsoft Partner Center dev account (Individual or Company)	0 USD	one-time; "there are no registration fees for either account type"	<code>learn.microsoft.com</code> [fetched 2026-08-29]
Azure Trusted Signing — Basic	9.99 USD	per month	<code>prices.azure.com</code> retail API, <code>serviceName</code> eq 'Trusted Signing' [fetched]
Azure Trusted Signing — Premium	99.99 USD	per month	same [fetched]
Azure Trusted Signing — signature overage	0.005 USD	per signature above quota	same [fetched]; included-quota count did not appear in the API response — unverified
Windows OV cert (Comodo, Sectigo)	219 USD	per year	<code>ssldragon.com</code> comparison [fetched]
Windows OV (GoGetSSL)	289 USD	per year	same
Windows OV (DigiCert)	400 USD	per year	same
Windows EV (Comodo, Sectigo)	287 USD	per year	same
Windows EV (GoGetSSL)	369 USD	per year	same
Windows EV (DigiCert)	685 USD	per year	same
Hardware token surcharge	+50–150 USD typical; DigiCert token +120 USD	one-time	[SS 2026-08-29]
GitHub Actions — Linux 2-core x64	0.002–0.006 USD	per minute	<code>docs.github.com</code> actions-minute-multipliers [fetched]
GitHub Actions — Windows 2-core x64	0.010 USD	per minute	same
GitHub Actions — macOS 3/4-core	0.062 USD	per minute	same
GitHub Actions — macOS 12-core (<code>macos_l</code>)	0.077 USD	per minute	same
Included Actions minutes	2,000 (Free) / 3,000 (Pro) / 3,000 (Team), private repos only	per month	[fetched]
R2 storage	0.015 USD/GB-month; 10 GB-month free	monthly	<code>r2/pricing</code> , page updated 2026-08-07 [fetched]
R2 egress	Free , all storage classes	—	same
R2 Class B ops (manifest reads)	0.36 USD/million; 10M free/month	monthly	same
Mac App Store commission	15% under Small Business Program (≤ 1 M USD proceeds prior calendar year), else standard	per sale	[fetched]

Derived CI cost per release (build minutes are estimates, not measurements — no release build was run; `[profile.release]` sets `lto = true`, `codegen-units = 1` [measured]): Linux $12 \times 0.006 = 0.072$; Windows $15 \times 0.010 = 0.150$; macOS universal $30 \times 0.062 = 1.860$; **per release 2.082 USD**; two releases/week $\rightarrow 104 \times 2.082 = 216.53$ **USD/year** [derived]. macOS is **89.3%** of that ($1.860 \div 2.082$) — the only CI line worth optimising. **On a public repository all of it is 0 USD** [fetched].

Year-1 floor, direct distribution, three platforms, private repo: $99 + 219 + 216.53 + 0 + \sim 0 = 534.53$ **USD** [derived]. Substituting Azure Trusted Signing Basic for the OV cert: $99 + 119.88 + 216.53 = 435.41$ **USD**, saving **99.12 USD/year** [derived]. Artifact hosting: $3 \text{ artifacts} \times \sim 15 \text{ MB} \times 104 \text{ releases} = 4.68$ **GB/year** cumulative [derived, artifact size is inference] — inside the 10 GB-month free tier if old releases are pruned, and egress is free regardless of download volume [fetched].

Source disagreement, recorded not resolved: `ssldragon.com` is a reseller, not the issuing CA; its DigiCert figures (400 / 685) sit above the 549–560 USD reseller figures seen elsewhere, and `SSL.com`'s own product pages returned **404** through `curl` [measured]. Treat all Windows cert prices as reseller quotes. **Two dated deadlines from search summary, not opened, therefore unverified [SS]:** publicly-trusted cert max validity drops to 460 days from 2026-03-01, and DigiCert stops issuing 2- and 3-year code-signing certs from Feb 2026. If true, multi-year prepayment is no longer a lever.

39.3 Is the thin shell a legitimate v1?

It is legitimate as a v0 marketing artifact and a trap as a v1 product — and for this specific product the trap is sharper than usual, because with `frontendDist` pointing at a URL the remote origin is itself an unsigned auto-update channel that bypasses the one channel Tauri cryptographically guarantees.

In order of force:

1. **Two update channels, one of them unverified.** Every web deploy silently changes what the installed desktop app executes, with no minisign signature, no version gate, no rollback and no user consent [inference, from `frontendDist` [measured] + updater signature model [fetched]]. Signing and notarising around that is theatre: the notarisation attests to a shell whose contents we can swap at will.
2. `csp: null` [measured] on a remote-origin window removes the one mitigation Tauri offers for exactly this configuration.
3. **The shell cannot do the product's core job.** There is no `tauri-plugin-fs` in `Cargo.toml` [measured] — the desktop app cannot open a local file. Against the "file is the only source of truth" thesis, what the desktop build adds over a browser tab today is a dock icon, window chrome (`{ titleBarStyle: "Overlay" , hiddenTitle: true` [measured]) and a worse offline story than a PWA.
4. **Offline is absent, not degraded.** No bundled assets means no window content without a network.
5. **The costs are not the costs you learn from.** 99 + 219 USD plus notarisation plumbing buys a signed wrapper around a URL and validates none of the genuinely hard part — local file access under sandbox and entitlement constraints.

Narrow legitimate case: an unsigned or ad-hoc-signed internal/beta artifact given to people told "this is a wrapper", to smoke-test window chrome, menus, deep links and OS integration. That is a week of value, not a shipping posture.

Anti-recommendation: do not read this as "rewrite everything to bundled assets before shipping any desktop build". A hybrid is cheaper and defensible — bundle the editor shell locally (`frontendDist` → a real build directory), keep auth, sync and AI as network calls, set a real CSP. That buys file-truth, offline and a single verified update channel without abandoning the web codebase. What is not defensible is signing and notarising the URL-wrapper and calling it v1. **Falsifier:** if a bundled-asset build cannot reach feature parity with the web editor within one sprint, ship the hybrid with the AI panel remote and revisit.

39.4 Platform order and the runbooks

v1: macOS only. Direct download, Developer ID, notarised, `.dmg`, Tauri updater against a Worker-backed dynamic endpoint on R2. Cost: 99 USD/yr + ~193 USD/yr CI if macOS-only and private (104 × 1.860 [derived]), or 99 USD/yr flat on a public repo [fetched]. We own a Mac [measured, 26.6.2], so signing and notarisation are debuggable without CI in the loop.

Anti-recommendation: check web analytics before committing — if early buyers are Windows-first, macOS-only is vanity and the order inverts. That number was not measured. If we ship macOS-only, say so on the download page rather than shipping a broken Windows build to look complete.

macOS runbook: enrol (only the **Account Holder** can create a Developer ID Application certificate); generate a CSR, create **Developer ID Application** (not Apple Distribution — that is App Store only), import to the login keychain; `security find-identity -v -p codesigning` gives the value for `signingIdentity / APPLE_SIGNING_IDENTITY`; for CI export `.p12` → `openssl base64 -A` → `APPLE_CERTIFICATE` + `APPLE_CERTIFICATE_PASSWORD` (Tauri verifies the certificate matches the configured identity and errors if not); leave `hardenedRuntime` at its `true` default; add `src-tauri/Entitlements.plist` and never set `com.apple.security.get-task-allow true` in anything submitted; prefer `APPLE_API_KEY` + `APPLE_API_ISSUER` + `APPLE_API_KEY_PATH` over `APPLE_ID` + app-specific password, because the API key is scoped and revocable without touching the Apple ID; the flow is `xcrun notarytool submit ... --wait` then `staple` — **altool has been dead since 2023-11-01**; ship `.dmg`, which the notary service processes as a nested container, ticketing the `dmg` and the inner `.app`; keep `minimumSystemVersion` explicit (config says "11.0" [measured]; Tauri's default floor is 10.13) [all fetched].

Do not skip notarisation. Since macOS Sequoia, users can **no longer Control-click to override Gatekeeper** — they must visit System Settings → Privacy & Security and explicitly allow the app [fetched, developer.apple.com/news, 2024-08-06], and the host is already past that line at 26.6.2 [measured]. Unsigned Apple Silicon builds from GitHub releases are also frequently reported as "damaged"; Tauri's own guidance is an ad-hoc signing identity if building without a certificate [fetched].

v1.5: Windows, gated on validating Azure Trusted Signing individual eligibility. CA/Browser Forum §6.2.7.4.2, effective **2023-06-01**, requires code-signing subscriber private keys in a Hardware Crypto Module per §6.2.7.4.1(7-9); latest Code Signing Baseline Requirements are **v3.11.0, dated June 16, 2026** [fetched, cabforum.org]. Consequence: no `.pfx` by email — every OV or EV cert arrives on a USB token, an HSM, or a cloud-signing service, and Tauri's Windows guide explicitly scopes itself to OV certs acquired **before June 1st 2023** plus Azure Key Vault [fetched]. Trusted Signing (now labelled "Artifact Signing") does zero-touch lifecycle management inside FIPS 140-3 level 3 HSMs at 9.99 USD/mo [fetched] — no dongle to plug into a CI machine that does not exist. **Individual eligibility was not confirmed from the fetched pages — unverified. Verify before budgeting on it.** SmartScreen: **EV** gets immediate reputation and no warning; **OV** still warns until reputation accumulates, and OV reputation is **per-file-hash**, so every release restarts the climb — for weekly releases it may never converge [fetched]. **Anti-recommendation:** do not buy EV first. 287–685 USD/yr plus business vetting to solve a warning we have not observed a user hit. Ship OV or Trusted Signing, measure drop-off, then escalate. *Counter-case:* kernel-mode driver signing requires EV — we will never sign one.

v2: Linux, AppImage only, because it is the format Tauri's updater natively re-uses [fetched] and needs no store relationship. Flathub is the only Linux channel with a verifiable denominator (**4,613,773,522 downloads, 3,631 apps, 2,164 verified apps**; India 96,252,930, Germany 489,832,076, Brazil 377,790,156 [fetched, flathub.org/api/v2/stats, 2026-08-29]) but packaging is a multi-day task — vendoring npm and cargo sources offline via `flatpak-node-generator` and `flatpak-cargo-generator.py`, adding `flatpak-builder-tools` as a submodule, hand-authoring AppStream metainfo XML [fetched]. Flathub only if a user asks twice. **Do not ship Snap** — absent from the updater's artifact matrix [fetched] and single-vendor.

Do not ship to the Mac App Store in v1. App Sandbox is mandatory and gives unrestricted access to a container directory and explicitly **not** to the user's home folder [fetched]. A vault-shaped editor holding a folder open across launches must route every path through user-selected open panels and persist security-scoped bookmarks — a real architecture constraint on an engine whose premise is "locate the byte range in the file" [inference; bookmark mechanics were not opened, treat specifics as unverified]. Add `embedded.provisionprofile`, `bundle.category`, `ITSAppUsesNonExemptEncryption`, a separate `tauri.appstore.conf.json`, a `universal-apple-darwin` build [fetched], and 15% commission.

39.5 Updater, rollback, key custody

- Signature verification **"cannot be disabled"** [fetched]. Two keys from `tauri signer generate -w ~/.tauri/myapp.key`; the public key goes into `plugins.updater.pubkey` as **literal content** — a file path is rejected [fetched].
- `bundle.createUpdaterArtifacts: true` emits `.sig` alongside `myapp.app.tar.gz` (macOS), `myapp-setup.exe` / `myapp.msi` (Windows), `myapp.AppImage` (Linux). Endpoints are an array, TLS enforced in production, advancing to the next URL only on non-2XX; template variables `{{current_version}}`, `{{target}}` (`linux|windows|darwin`), `{{arch}}` (`x86_64|i686|aarch64|armv7`) [fetched].
- Static manifest keys: `version` (SemVer, leading `v` optional), `notes`, `pub_date` (RFC 3339), `platforms["OS-ARCH"].{url,signature}`. **Tauri validates the whole file before checking the version field, so one malformed platform entry breaks updates for every platform** [fetched]. Windows `installMode: passive` (default), `basicUi`, `quiet` (cannot self-elevate; user-wide installs only) [fetched].
- **Rollback has exactly one supported mechanism:** a *dynamic* update server plus an override of `UpdaterBuilder::version_comparator`, which the docs describe as "useful if you need to roll back your app" [fetched]. A static JSON on R2 cannot roll back, because the client's SemVer check refuses a lower version. The comparator override is compiled into the client binary, so the **endpoint must be a Cloudflare Worker in front of R2 from day one** — retrofitting it requires shipping a new build to the exact population you are trying to rescue [inference]. *Counter-case:* if forward-only fixes are acceptable, static JSON on R2 is materially simpler and egress is free.
- **Key custody.** Tauri states it plainly: lose the private key and "you will NOT be able to publish new updates to the users that have the app already installed" [fetched]. There is no revocation, no second signer, no rotation path in the manifest format. `TAURI_SIGNING_PRIVATE_KEY` and `TAURI_SIGNING_PRIVATE_KEY_PASSWORD` must be env vars — `.env` **files do not work** [fetched]. Password-protect the key; hold three independent copies (password manager, two offline media in two physical locations) plus the CI copy in Actions secrets. Note the asymmetry: `signingIdentity: null` today is a recoverable mistake, and Apple and Windows certs expire and re-issue, but a lost minisign key strands the installed base forever. Do not generate it on a CI runner.

- Rename `sgnk-md` → `frontmatter` now, while the installed base is zero. `identifier` is the Bundle ID that must match the App Store Connect record and the provisioning profile [fetched]; changing it after shipping signed builds forces a new App ID, a new provisioning profile and a new Gatekeeper/updater identity.

§40 Error and refusal experience

Live baseline, measured at commit `9e84628` [measured]: `spliceFrontmatterValue(src, key, value): string` and `spliceFrontmatterKey(src, oldKey, newKey): string` both return a bare string, across 20 `return src` sites. A refusal is therefore byte-identical to a successful no-op — no reason code, no discriminated union, nothing a UI can read. `PropertiesPanel.tsx`'s `emit()` is `if (!onEdit || next === content) return;`, so refusals are dropped, and the current mitigation is a duplicated `SAFE_KEY` check in the UI: two definitions of one rule. By contrast `mdmax` already emits 32 distinct uppercase symbols, of which 13 are verdict classes, leaving 19 typed failure codes [derived: 32 – 13 = 19]. The founding principle is implemented in one half of the product and not the other, so the work here is not nicer copy — it is giving splice the return shape `mdmax` already has.

The message template. Four obligatory slots, fixed order. Adapted from Postgres Primary/Detail/Hint [fetched postgresql.org/docs/current/error-style-guide.html, PG 18] and RFC 9457's `type / title / detail / instance` [fetched rfc-editor.org/rfc/rfc9457.txt], plus one slot no external source has because no external source has our guarantee.

SLOT	CONTENT	BUDGET
1. OUTCOME	"Nothing changed." — the guarantee, stated first	≤20 chars
2. OBJECT + CAUSE	the exact key, line or byte range, and why it was unaddressable	≤120 chars
3. AFFORDANCE	one imperative the user can perform right now	≤90 chars
4. DISCLOSURE	<code>Why?</code> expander: reason code, byte offsets, the bytes left untouched, <code>Copy diagnostic</code>	unbounded, collapsed

- Visible budget: 280 characters total, headline ≤80. Derivation: git's full checkout refusal is 175 chars over 4 lines; the Postgres primary+detail+hint exemplar is 256 chars over 3 lines; mean 215.5, ceiling set above both. rustc's primary line is 74 chars, so 80 holds a real headline [measured][derived][fetched].
- Machine shape: `{ ok: false, reason: 'UNSAFE_KEY', at: { line, col, byteStart, byteEnd }, detail, hint }` — mirrors `mdmax`'s existing discriminated union and RFC 9457's triple. `reason` is the stable identity; the prose is localisable and may change. RFC 9457: consumers "SHOULD NOT parse the `detail` member" [fetched].
- Anti-recommendation: never put the reason code in the visible headline. NN/g: "Hide or minimize the use of obscure error codes... show them for technical diagnostic purposes only" [fetched nngroup.com/articles/error-message-guidelines, 2023-05-14]. The code lives in the disclosure and the clipboard.
- Anti-recommendation: never exceed 280 by stacking a second hint. Postgres hints exist because a suggestion "might not always be applicable"; a second hint is a guess, and guessing is the thing the engine refuses to do [fetched].
- Tone divergence, deliberate and written down rather than drifted into: Postgres and rustc both mandate lowercase primaries with no terminal period [fetched both]. `frontmatter` is a consumer editor, not a CLI, so it uses sentence case with periods.
- Prioritise by frequency once telemetry exists. Programming-error-message frequencies "empirically resemble Zipf–Mandelbrot distributions" [fetched arXiv 1509.07238, 2015-09-24], but that result is measured on novice Python and Java corpora, not markdown editors — ship all eleven at template quality and promote by observed counts, do not pre-guess the head.

The taxonomy — every refusal the product can emit. Codes marked  exist in the tree today;  must be added [measured].

#	REFUSAL	CODE	EXACT VISIBLE WORDING	CHARS	RECOVERY AFFORDANCE
1	Splice cannot locate the byte range	 UNLOCATABLE	"Nothing changed. The engine could not find a single unambiguous place for <code>public_slug</code> in this file's front matter, so it left every byte where it was."	152	<code>Show me</code> — scrolls to and selects the front-matter block
2	Ambiguous or duplicate target	 REFUSED_AMBIGUOUS	"Nothing changed. <code>author</code> appears twice here, so this edit had two possible targets. Delete one, then try again."	111	<code>Show both</code> — two carets in the editor
3	YAML will not parse	 FRONTMATTER_UNPARSEABLE	"Nothing changed. The front matter stops being readable at line 7, so properties are read-only until that line is fixed. Your text is untouched."	143	<code>Jump to line 7</code> ; the panel goes read-only, never blank
4	Unsupported key shape	 UNSAFE_KEY	"Not added. <code>titulo</code> uses characters this editor cannot address safely. Names can use letters A–Z, digits, and <code>. _ \$ -</code> ."	121	The field stays open with the typed text intact
5	Corpus drift, targets stale	 stale →  CORPUS_DRIFT	"This certificate was measured against an older version of this note. It is shown greyed out until you re-run it."	112	<code>Re-certify</code> , one click
6	Certificate BROKEN	 BROKEN	"Certified BROKEN: 3 of 24 engines lose content from this note. The note itself is unchanged and safe."	101	<code>See the 3</code> — per-engine verdict rows
7	Shape gate / budget exceeded	 BUDGET_BYTES, BUDGET_LINES, BUDGET_BLOCKS, BUDGET_TIME	"Not certified. This note is 4.2 MB, past the limit for a run that must finish. Nothing was written."	100	<code>Certify the first section instead</code>
8	Engine missing or threw	 ENGINE_MISSING / ENGINE_THREW	"marked 16.4.2 could not be loaded, so it is reported as unmeasured rather than as passing."	90	<code>Retry</code> and <code>Exclude this engine</code>
9	Publish revoked	 SHARE_REVOKED	"This link no longer works. The note was unpublished on 12 Aug, and the copy on the server was deleted."	102	Owner: <code>Publish again</code> . Visitor: nothing — no owner, no title, no path
10	AI declined	 AI_DECLINED	"The assistant didn't produce an edit here. Your document is unchanged."	70	<code>Try a different instruction</code> ; the diff pane stays empty rather than showing a partial
11	Offset invalid (internal)	 PAST_END, INSIDE_SURROGATE_PAIR, NOT_AN_INTEGER, NEGATIVE, OFFSET_OUT_OF_RANGE	never user-visible	—	logged with byte offset; surfaces to the user as #1

- #7's copy must quote the live constant, not a hardcoded number. The shape gate ships `MAX_BYTES` 4 MB [measured §29.1] while the drafted string named a 2 MB limit — the two research inputs disagree, and the fix is to interpolate the constant so the message cannot drift from the gate.
- #9 deliberately returns 404 rather than 403; RFC 9110 §15.1/4/21 permits hiding existence behind 404 [fetched], and naming the owner or former title of a revoked note leaks exactly what unpublishing was for.
- #10 follows the OpenAI Model Spec's "refuse neutrally and succinctly" [fetched model-spec.openai.com/2025-04-11] — no "unfortunately", no meta-commentary. Anti-recommendation: copy the Model Spec's brevity but not its opacity. A safety refusal withholds its reason on purpose; an engine refusal that withholds its reason is a bug with better manners.

Presentation, without breaking the simple surface.

RUNG	APPLIES TO	BEHAVIOUR
1. Inline, non-modal	#2, #4 — refusals the user caused and can undo by retyping	the row stays in edit state, anchored to the property that failed
2. Persistent strip on the affected panel	#3 (read-only properties), #5 (greyed certificate)	refusals that change what the surface can do
3. Nothing at all	#11, and any refusal already visible as "the thing you asked for did not happen"	internal codes never reach the user
Modal	never	a modal is for irreversible loss; a refusal is the proof that nothing was lost

- Anchor at the source. NN/g: "Display the error message close to the error's source... proximity helps users associate the error message content with the interface elements needing attention" [fetched].
- Two tiers of disclosure, never three, and never across a navigation. Tier 1 is the 280-char sentence; tier 2 is the `Why?` expander. Nielsen distinguishes progressive from *staged* disclosure [fetched nngroup.com/articles/progressive-disclosure, 2006-12-03]; a refusal must be fully explainable without leaving the document.
- The one exception to "modal: never" is a refusal during an operation the user believes completed and has left the app — a background publish. That needs a durable surface, not a transient one.
- Show "0 bytes changed" as the visible artifact of the guarantee. Anti-recommendation: do not badge it. A celebration on every refusal becomes chrome, and NN/g reserves novelty for rare total failure.
- Icons are Google Material Symbols delivered as inline SVG only — `block`, `warning`, `help`. Never emoji, never the Google Fonts ligature span.
- Every refusal is silenceable once understood. git ships 43 `advice.*` toggles plus `GIT_ADVICE=0` for tooling [fetched, measured]; the toggle is part of the design, not an afterthought.

Anti-patterns, each with the evidence that makes it one.

ANTI-PATTERN	WHY IT READS AS A BUG
Silent no-op — return the input, say nothing	NN/g: "The very worst error messages are those that don't exist" [fetched]. This is our current splice behaviour at 20 sites [measured]
Refusal with no named escape	<code>fatal: refusing to merge unrelated histories</code> is 44 chars and never names <code>--allow-unrelated-histories</code> [fetched git builtin/merge.c:1644, measured]. Its sibling — the checkout refusal that ends "Please commit your changes or stash them before you switch branches." — differs on nothing else and is cited as helpful
Most-relevant-last	TypeScript #29759 puts "Did you mean to write 'environment'?" after five nesting levels and a 178-char reconstructed type; open since 2019-02-05 with 29 upvotes, in a label carrying 146 issues [fetched, measured]
Implementation nouns in the headline	Postgres: <code>pg_strtoint32: error in "z"</code> became <code>invalid input syntax for type integer: "z"</code> [fetched]. Never surface <code>spliceFrontmatterValue</code> or <code>SAFE_KEY</code> in tier 1
Blame vocabulary	NN/g bans <code>invalid</code> , <code>illegal</code> , <code>incorrect</code> ; rustc says "The word 'illegal' is illegal. Prefer 'invalid.'" The two sources disagree on <code>invalid</code> itself — take NN/g's stricter line for user-facing copy and rustc's for internal codes [fetched both]
Duplicated rule in UI and engine	Two <code>SAFE_KEY</code> checks is two places to drift. The engine returns the reason; the UI renders it
Auto-repair on refusal	ESLint separates <i>fixes</i> (safe, auto-applied) from <i>suggestions</i> , which "may change application logic and so cannot be automatically applied" and are not exposed via the CLI [fetched]. A byte-preserving engine may only ever offer the suggestion tier
LLM-generated explanation as the default	n=106 within-subjects over six buggy C programs: GPT-4 explanations beat stock compiler messages on time-to-fix in only 1 of 6 tasks, and handwritten explanations still won on objective and subjective measures [fetched arXiv 2409.18661, 2024-09-27]
Humour or novelty on a refusal	NN/g reserves novelty for total, no-recourse failure and warns humour goes stale on repeat; a refusal is repeatable by construction [fetched]
A refusal that cannot be reproduced from its text	Without <code>reason</code> and a byte range a support report is unactionable. RFC 9457's <code>instance</code> member exists for exactly this [fetched]

- Recorded and unresolved: OpenAI's Model Spec keeps refusals to "a brief apology and a brief statement of inability", never explanatory; NN/g holds that "merely stating the problem is also not enough; offer some potential remedies" [fetched both]. Both are correct in their domain. frontmatter emits both kinds — #10 versus #1-#9 — and must not unify them.
- Unverified: no telemetry exists for which of the eleven refusals users actually hit, so frequency-first prioritisation is [inference]. The character budgets derive from four exemplar messages, not from reading tests on our own users. Becker et al. (ITICSE-WGR 2019) and Denny et al. (CHI 2021), the two most-cited works on whether enhanced error messages help, were not opened — `d1.acm.org` returned HTTP 403 on 2026-08-29 [measured] — and are excluded rather than paraphrased.

§41 Engineering standards

Not aspirations — the practices that caught real defects in this codebase, several of them this week.

STANDARD	RULE
Red proof before green	A test on a rare fault proves nothing until it fails against the unfixed code . If you cannot make it fail, say so rather than reporting a pass
Gates must be able to see	A check that greps source is a proxy and must be labelled one. A check that does not execute the thing proves nothing. Four gates here could report green while blind — all four now fixed MEASURED
Floors, not equalities	Assert <code>passes ≥ N</code> and <code>failures == 0</code> . Equality-pinned counts punish adding coverage
Corpus integrity	Re-hash every corpus file against a pinned sha256 before use. This caught a real drifted file on its first run MEASURED
Never trust a piped exit status	<code>cmd tail</code> reads green while the command failed. This bit three times in one session, including once while verifying that it had been fixed
Watch the artifact, not the process	A tool whose job is to produce a file must verify the file, using the format's own end marker
A wrong type is a finding, never a coercion	An all-digit sha256 parses as a YAML Number. The gate must say so, not crash — and one bad input must not blind the gate for every other input
A gate with a bypass is not a gate	An early spec harness skipped any file starting with <code>_</code> , so a spec could be hidden from its own gate by renaming it
Run it three times	One run is an anecdote
Verify the write landed	Count before, count after, refuse if the count did not move
A verifier written beside its subject inherits its blind spots	An independent adversarial pass before any clean bill of health
Preview before you delete	<code>grep -c first, rm second</code>
Re-derive every number at write time	\$57 exists because we did not
A sandbox failure is not a build failure	<code>next/font/google</code> cannot reach Google Fonts inside the sandbox. Confirm outside it before declaring red

CI to build in R0: typecheck → lint → test → build → arch (with its file floor) → corpus oracle → foreign-corpus suite → spec gate, on every push, before hire #2.

§42 AI disclosure and content provenance obligations

42.1 LAW — binding, dated, with the sanction

INSTRUMENT	STATUS	TEXT IN SCOPE?	DATES	SANCTION
Reg. (EU) 2024/1689 (AI Act) Art. 50	Directly applicable regulation	Yes, explicitly	Signed Brussels 13 Jun 2024; Art. 113 → applies 2 Aug 2026	Art. 99(4)(g): ≤ EUR 15,000,000 or 3% worldwide annual turnover, whichever higher; Art. 99(6) gives SMEs/start-ups the lower of the two
PRC (CAC/MIIT/MPS/NRTA)	Binding departmental rules	Yes — Art. 3 lists first	Signed 7 Mar 2025, published 14 Mar 2025, in force 1 Sep 2025 (Art. 14)	Art. 13, by reference to the Cybersecurity Law
California SB-942 (Ch. 291, 2024)	Enacted statute	No — text excluded from every operative duty (§22757.2(a)(1), §22757.3(a), §22757.3(b) all read "image, video, or audio"), although §22757.1(c) defines a GenAI system to include text	Approved 19 Sep 2024; operative 1 Jan 2026 (§22757.6)	§22757.4: \$5,000 per violation , each day discrete

All three rows [fetched] — EUR-Lex CELEX 32024R1689 (1,263,937 bytes), CAC 2025 2 , SB-942 chaptered text, opened 2026-08-29.

Operative detail that changes the build [fetched]:

- **Art. 50(2)** binds *providers* of systems generating synthetic "audio, image, video **or text**" to mark output machine-readably, qualified by "as far as this is technically feasible", with a carve-out for systems performing "an assistive function for standard editing" or not substantially altering the input "**or the semantics thereof**".
- **Art. 50(4) ¶2** binds *deployers* publishing AI text "informing the public on matters of public interest" to disclose — **unless** the content "has undergone a process of human review or editorial control and where a natural or legal person holds editorial responsibility".
- **Art. 50(5)**: disclosure "clear and distinguishable ... at the latest at the time of the first interaction or exposure", meeting accessibility requirements. **Art. 50(7)**: technical detail deferred to codes of practice and possible implementing acts — it is not settled. **Recital 133** lists permitted techniques: watermarks, metadata identifications, cryptographic provenance methods, logging methods, fingerprints.
- **PRC Art. 4 final ¶**: where download, copy, or **export** is offered, the *file itself* must carry a conforming explicit label — a statutory export-time attestation. **Art. 5**: the implicit label goes in file **header** metadata; watermarking is (encouraged), not mandated. **Art. 9**: a user may request unlabelled output after the duty is contractually shifted, logs retained ≥ **6 months**. **Art. 10**: nobody may maliciously delete, alter, forge, or conceal a label, **nor provide tools or services for others to do so**. **Art. 11** incorporates a by reference — **not opened, see §55**.

Source disagreement, recorded not resolved: Art. 113(c) gives high-risk Art. 6(1) obligations **2 August 2027** [fetched, EUR-Lex]; the Commission's own regulatory-framework page says **2 December 2027** [fetched]. The August-2026 transparency date is doubly sourced; the 2027 date is not, and the Digital Omnibus page returned a 404.

42.2 POLICY — private, enforced by exclusion

BODY	RULE	DATE / TAG
Springer Nature / Nature Portfolio	Risk-tiered. Green (language polish, structure, translation, data cleaning) permitted, disclosure "enhances trust". Amber (drafting summaries, extensive copy-editing, suggesting analyses) permitted with oversight + disclosure. Red (undisclosed core reasoning, conclusions presented as human-derived, delegated peer review, fabricated citations) not permitted. "Human accountability is non-transferable"	[fetched 2026-08-29]
ICMJE	Disclose at submission , in both the cover letter and the work; writing assistance → Acknowledgments; data/analysis/figure generation → Methods; chatbots cannot be authors	[fetched 2026-08-29]
Elsevier	Fixed sentence: " <i>During the preparation of this work, the author(s) used [NAME OF TOOL / SERVICE] in order to [REASON]. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.</i> " AI figures only for explanatory diagrams	[fetched 2026-08-29]
ACM	Reported to require acknowledgement-section disclosure and bar AI authorship — page returned HTTP 403, not opened	[SS] — do not cite as verified

42.3 NORM — standards and empirical limits, not obligations

ITEM	CONTENT	DATE / TAG
Convergent demand shape across the three opened policies	A free-text natural-language declaration naming tool + purpose + human-responsibility affirmation, in a named document section . Not a score, not a percentage, not a detector result	[derived], 3 of 3 fetched texts
C2PA 2.3 §5.3.1 / Appendix A.8	Manifests in <i>unstructured</i> text via <code>C2PATextManifestWrapper</code> , magic <code>0x4332504154585400</code> ("C2PATXT\0"), JUMBF store encoded as Unicode variation selectors (U+FE00–FE0F, U+E0100–U+E01EF), U+FEFF-prefixed	December 2025 [fetched]
C2PA 2.4 §5.3.1 / Appendix A.9	"Embedding Manifests into Structured Text" — explicitly names Markdown , AsciiDoc , LaTeX , YAML , TOML , INI . Plus the <code>c2pa.ai-disclosure</code> assertion: <code>ai-model-disclosure-map</code> with mandatory <code>modelType</code> , optional <code>modelName</code> , <code>modelIdentifier</code> , <code>scientificDomain</code> (arXiv taxonomy, e.g. <code>cs.AI</code>), and <code>contentProfile.humanOversightLevel ∈ {fully_autonomous prompt_guided human_validated}</code>	April 2026 [fetched]
SynthID-Text	Nature, 2024-10-23 , DOI <code>s41586-024-08025-4</code> ; production sampling-level watermark, live over "nearly 20 million Gemini responses"	[fetched]
"Watermarks in the Sand"	arXiv 2311.04378 , 2023-11-07; strong watermarking provably impossible under stated natural assumptions, including private-key, with only a quality oracle and a perturbation oracle	[fetched, abstract]
Weber-Wulff et al.	arXiv 2306.15666 , 2023-06-21; 12 public tools + Turnitin + PlagiarismCheck — "neither accurate nor reliable", biased toward classifying output as human-written	[fetched, abstract]
Liang et al.	arXiv 2304.02819 , 2023-04-06; detectors "consistently misclassify non-native English writing samples as AI-generated"	[fetched, abstract]

Platform norms (YouTube, Meta, TikTok, LinkedIn), Spain's draft labelling penalties, the Utah AI Policy Act, the Colorado AI Act, and Korea's AI Framework Act were not researched at all [measured]. Report them as absent, never as zero.

A.9 is line-for-line a byte-splice contract [fetched]: fixed delimiters `-----BEGIN C2PA MANIFEST-----` / `-----END C2PA MANIFEST-----` modelled on RFC 4880 §6.2; a Markdown single-line HTML-comment form and a **front-matter form** where the block sits inside the `----` fences and the fences are *not* part of the exclusion range; `c2pa.hash.data` carrying a **single exclusion range** with start and length in bytes; "Files shall be read in **binary mode** to preserve the exact byte representation of line terminators"; "A claim generator shall **not** alter the line ending convention of the file content outside the manifest block"; **bare CR (0x0D) unsupported**, such files "shall be converted"; at most one block per file; named failure codes `manifest.structuredText.multipleReferences`, `...noManifest`, `...emptyReference`.

42.4 Where byte-anchored provenance maps, and where it does not

OBLIGATION	MAPS?	MECHANISM OR REASON
AI Act 50(2) machine-readable marking	No, and cannot	Binds the <i>provider</i> of the generating system. We are not one, and we sit inside the "assistive function for standard editing" carve-out [fetched + inference]
AI Act 50(4)¶2 deployer disclosure	No — the user is the deployer	But we can <i>emit</i> the disclosure the deployer owes
AI Act 50(4)¶2 exemption (human review / editorial control / named responsibility)	Yes — this is the fit	A byte-anchored ledger of which ranges were human-authored, which machine-inserted, when, by whom, is the evidence the exemption is phrased around [inference]
AI Act 50(5) clear, distinguishable, accessible, at first exposure	Partially	A front-matter key is machine-readable but not <i>exposed</i> ; the published-site projection must surface it — cheap under the projection law
PRC Art. 4 final ¶ export carries the label	Yes, mechanically	Export-time splice of a visible line plus an A.9 block with a byte exclusion range
PRC Art. 5 implicit label in header metadata	Yes	YAML front matter <i>is</i> the header; A.9's front-matter form lands there by spec
PRC Art. 10 no concealment tooling	Risk, not fit	See anti-recommendations
CA SB-942	N/A	Text excluded from all operative duties [fetched]
Nature / ICMJE / Elsevier	Yes — the near-term surface	All three want a named-section prose declaration; Elsevier's exact sentence is a deterministic template fill

Two hard incompatibilities [fetched + inference]: **A.8 is hostile to the thesis** — it injects invisible variation selectors and hashes after NFC normalization and UTF-8 re-encoding, so two byte-different files hash identically and the act of crediting a file mutates it. And **A.9's bare-CR refusal collides with a real corpus** the engine already has queued as work; A.9 says convert, we have committed not to silently mutate bytes, so the honest behaviour is refuse and cite the reason.

42.5 Features: honest vs pseudo-precision

Build (honest).

- Disclosure block** — a reserved front-matter key holding tool identifier, purpose, `humanOversightLevel` (reuse the C2PA enum verbatim; do not invent one), timestamp, responsible person. One record, four deterministic projections: Elsevier's verbatim sentence, an ICMJE Acknowledgments/Methods paragraph, a PRC Art. 4(1) line, a `c2pa.ai-disclosure` assertion.
- Export-time attestation** — splice one A.9 manifest block (front-matter form) referencing an external `.c2pa` store; compute `c2pa.hash.data` with the byte exclusion range, binary mode. Removing the block returns the original bytes exactly.
- Provenance degradation certificate** — extend cross-engine certification to answer whether a target renderer, CMS, or round-trip preserves the manifest block, the front-matter key, and the line endings. Report PRESERVED / STRIPPED / MANGLED per target.
- Refusal surface** — bare-CR files, multiple blocks, non-`text/*` inputs: refuse citing the specific C2PA failure code.
- Editorial-responsibility record** — who reviewed which byte ranges, when. Sized to 50(4)¶2 and independently useful to journals.

Do not build (pseudo-precision).

- A "percentage AI" report.** Unsound at four independent levels: no ground truth exists for the denominator; inferential detectors are "neither accurate nor reliable" and biased against non-native writers [fetched ×2]; a keystroke-derived percentage measures *bytes* while every regime above regulates *intellectual contribution*, so a human retyping a model's dictation scores 100% human and is lying; a paste of unknown origin is unattributable. If a number is demanded, ship a three-bucket ledger with an explicit `unattributed` bucket **that is never redistributed** — never a scalar.
- A confidence score on a disclosure.** The event was recorded or it was not.
- A "verified human-written" badge.** A.9's hard binding proves *bytes unchanged since signing*, never *bytes typed by a person*.

42.6 Anti-recommendations

- **Never claim AI Act Art. 50 compliance.** Say exactly: *"produces evidence for the Art. 50(4) editorial-control exemption; does not discharge Art. 50(2)."* The customer carries the EUR 15,000,000 / 3% exposure, not us.
- **Never ship a "strip invisible / zero-width characters" cleanup.** Under 2.3 A.8 those characters *are* the credential, and PRC Art. 10 forbids providing tools for . If ever added: opt-in, per-invocation, with an explicit destroys-a-Content-Credential warning.
- **Never NFC-normalize on save or export.** It is the single line that would silently falsify byte-preservation.
- **Never build, bundle, or surface an AI-text detector.** Selling globally from India to non-native-English writers while importing a documented demographic bias is reputational risk exceeding any revenue [fetched: 2304.02819].
- **Never present a watermark as tamper-evident protection** [fetched: 2311.04378].
- **Implement A.9 only. Do not implement A.8. Do not pursue C2PA membership or conformance** until a paying customer names it — the spec is public and implementable without either.
- **Do not build a compliance-advice surface.** Emit records and citations; let the customer's counsel do the mapping.
- **Do not let the disclosure block become a second source of truth.** The moment it lives in a sidecar database, the central claim is broken.
- **Do not treat the SB-942 text exclusion as stable.** Parameterise the export path by jurisdiction; do not shape it around California.

§43 Internationalisation

43.1 Measured baseline

FACT	VALUE	EVIDENCE
<code>countWords</code> implementation	<code>src/modules/editor/presentation/EditorPane.tsx:31</code> → <code>trimmed.split(/\s+/).length</code>	[measured, read]
Pure-CJK paragraph, 41 code points, no spaces	<code>countWords</code> = 1; <code>Intl.Segmenter('zh',{granularity:'word'})</code> <code>isWordLike</code> = 23 → 23× under	[measured]
Japanese sample	1 vs 15 → 15× under	[measured]
Thai sample	1 vs 8 → 8× under	[measured]
Arabic sample	4 vs 4 → no error	[measured]
MiniSearch default tokenizer	<code>SPACE_OR_PUNCTUATION</code> = <code>/[\n\r\u{Z}\u{P}]+\u</code> , <code>minisearch/dist/es/index.js:2002</code>	[measured]
<code>buildSearchIndex</code> tokenizer override	none — <code>search-index.ts:124</code> passes <code>fields/storeFields/idField</code> only	[measured]
Query against a doc containing ...	default 0 hits; bigram 1; Segmenter 1	[measured]
<code><html lang="en"></code> , no <code>dir</code>	<code>src/app/layout.tsx:67</code>	[measured]
Installed <code>@codemirror/view</code>	6.43.0 (npm latest 6.43.9, published 2026-08-16)	[measured] / [fetched]
Byte vs UTF-16 offset divergence	67 of 1,080 corpus files have bytes == UTF-16 units → 93.80% diverge (67/1080 = 0.06204)	[derived from measured header, <code>offsets.ts</code>]
CJK frontmatter keys rejected by <code>SAFE_KEY</code>	905 oldwinter files	[measured]

Source disagreement, recorded not resolved: the master plan records `countWords` as "1.7–2× under" for Chinese; direct measurement on unspaced CJK gives 8–23×. Both can be true — `\s` includes `\n` , so a real multi-line document scores roughly one word per line, which compresses the ratio at corpus scale. Do not average or merge these numbers; re-derive the corpus figure with a script that names the file set it measured.

43.2 Work items, ordered by measured breakage per unit of work

ORDER	ITEM	FIX	STANDARD	EFFORT
1	H-3 + H-4 (below)	Composition-gate the 400 ms debounce and the ghost-text/AI trigger	—	S
2	I-2 search tokenizer	Character-bigram tokenizer in <code>buildSearchIndex</code> ; takes CJK body recall from 0 hits to 1 on the measured query	—	M
3	I-1, I-3, I-4	One shared segmenter factory serving word count, reading time and grapheme index. Reading time: locale-branch the <code>words/200</code> divisor, or drop reading time for non-space-delimited scripts (the convention is ~300–500 characters/min)	UAX #29 §4 word boundaries, §3 extended grapheme clusters, via <code>Intl.Segmenter</code>	S
4	I-7 formatting	Route every date through <code>Intl.DateTimeFormat</code> and every number through <code>Intl.NumberFormat</code> at the resolved locale. Measured: <code>en-IN</code> → 12,34,567.89, <code>INR</code> → ₹499.00, <code>ar</code> long date → 29 2026 أغسطس, <code>ar-EG</code> numerals → ١٠٣٤٠٥٦٧,٨٩	ECMA-402 §11 / §16	S
5	I-6 tree sort	<code>Intl.Collator(locale,{numeric:true, sensitivity:'base'})</code> .Today <code>Array.prototype.sort()</code> gives UTF-16 order: <code>note1</code> , <code>note10</code> , <code>note2</code>	UTS #10 / CLDR	S
6	I-9 + §43.4 partial RTL	<code>dir="auto"</code> on both panes, <code>perLineTextDirection</code> , <code>bidiIsolates()</code>	UAX #9 P2/P3	S
7	I-10 snippet truncation	Reuse <code>findClusterBreak</code> from <code>@codemirror/state</code> rather than slicing at a UTF-16 index, which can split a surrogate pair or a ZWJ emoji sequence	UAX #29 grapheme	S
8	I-5 frontmatter key addressability	A YAML quoting decision, not a Unicode one; tracked with the engine work	—	M
—	I-8 line breaking	Do nothing beyond setting <code>lang</code> ; the rendering engine tailors it	UAX #14	XS

Do NOT implement UAX #9 or UAX #14 by hand. UAX #9 tracks Unicode 17.0.0 with six classes of explicit formatting characters and six higher-level protocols HL1–HL6 [fetched, unicode.org/reports/tr9]; UAX #14 §7 now reads "Deleted. (Formerly was: Pair Table-Based Implementation)" [fetched, tr14] — the shortcut people copy was withdrawn. CodeMirror 6 already ships a UBA: `computeOrder` at `@codemirror/view/dist/index.js:1245` , `BidiSpan` at :874 [measured]. Do NOT create a "unicode module" that owns all ten items either: items 1, 3, 4 and 7 are three-line call sites against one shared segmenter factory, and a module invites a second offset vocabulary — exactly what `offsets.ts` exists to prevent.

43.3 IME composition hazards — React 19 + CodeMirror 6 + 400 ms debounced save

The hot path: `EditorView.updateListener` fires on every `docChanged` → `useEditorStore.setContent(path, content)` synchronously, then re-arms a 400 ms timer to `saveDraft` (`CodeMirrorEditor.tsx:245–262`) [measured]. IME composition emits `docChanged` per keystroke of the *provisional* string.

#	HAZARD	MECHANISM	REQUIRED TEST
H-1	Preview thrash mid-composition	<code>setContent</code> fires per pinyin keystroke; the split preview re-renders half-composed runs	CDP <code>Input.imeSetComposition</code> <code>nihao</code> → <code> </code> ; assert no committed save contains a provisional run
H-2	Composition destroyed by re-render	Any React re-render touching <code>contentDOM</code> while <code>view.inputState.composing >= 0</code> destroys the browser's composition anchor. CM6 defends internally but cannot stop a parent remount	Toggle a parent state (theme, settings, <code>livePreviewForced</code>) mid-composition; assert the doc equals <code> </code> , not <code>nihao</code> or <code> </code>
H-3	Debounce save captures a provisional buffer	The 400 ms timer is not composition-aware, and a 400 ms pause with the candidate window open is entirely normal → IndexedDB draft plus <code>setDirty(true)</code> for text the user never committed	Compose, hold 600 ms with the candidate window open, read the draft from IndexedDB; assert it does not contain the pre-conversion Latin run
H-4	Autocomplete/ghost-text firing mid-composition	<code>@codemirror/autocomplete</code> already guards: <code>if ((android ? view.composing : view.compositionStarted) view.state.readOnly) return (autocomplete/dist/index.js:1841)</code> [measured]. Our custom <code>aiSuggestion / ghostText</code> extensions carry no such guard	Compose <code>nihao</code> ; assert zero completion sources invoked and zero ghost-text requests before <code>compositionend</code>
H-5	Undo granularity	CM6 tags composition transactions <code>userEvent: "input.type.compose"</code> [fetched]. A custom transaction dispatched during composition without that annotation makes one Ctrl-Z delete a paragraph or a single code unit	Compose three words, undo three times, assert three distinct doc states at word boundaries
H-6	Vim mode × IME	<code>@replit/codemirror-vim</code> loads first in the extension array and installs its own key handling; CM6 issue #829 ("keymap not called during/after composition (android)") is the known shape	Compose in insert mode, press Escape mid-composition; assert the text commits and the mode changes exactly once
H-7	Decoration boundaries garbling composition	Three CM6 issues in 2026-03 alone — #1688 "Text visually disappears after IME composition inside brackets on Chrome", #1650 "IME composition at the boundary of syntax highlight nodes garbles the content", #1654 "IME composition problem in decoration"; all closed, filed 2025-12→2026-03 [measured, api.github.com]. Our live-preview decorations sit exactly there	Compose inside <code>**bold**</code> , inside a fenced block, and at a <code>[[wikilink]]</code> boundary; assert a byte-identical expected doc for each
H-8	Safari dead-key / missing compositionend	CM6 carries an explicit workaround (<code>view/dist/index.js:5294–5296</code> : on Safari, <code>insertText</code> while composing fires a synthetic <code>compositionend</code> after 20 ms) [measured]	Run the whole matrix on WebKit as well as Chromium; a Chromium-only IME suite proves nothing
H-9	Tauri webview divergence	<code>EditContext</code> is Chrome 121+, Firefox false, Safari false [fetched, MDN BCD] — CM6 runs its DOM-mutation composition path on two of our three Tauri targets and its <code>EditContext</code> path on one	Same matrix inside <code>tauri dev</code> on macOS WKWebView, Linux WebKitGTK and Windows WebView2; do not infer from the browser build
H-10	No upstream escape hatch	The public <code>codemirror</code> GitHub org was archived 2026-04-15/16; npm remains current but issues and PRs cannot be filed [fetched]. 26 IME issues exist, all closed, newest 2026-03-26 [measured]	Budget ventering; pin <code>@codemirror/view</code> and diff on every bump

Two fixes, both small: gate the debounce on `update.view.composing` (public API [fetched, CM ref 2047]) and re-arm rather than save; and gate `aiSuggestion / ghostText` on `view.compositionStarted`, copying `autocomplete`'s predicate verbatim including the Android branch — on Android, `compositionStarted` is true merely from placing the cursor on a word [fetched, CM ref 2049], which is why the library branches.

Suppress the persistence side, never the projection side: do NOT gate `setContent` on composition, because blanking the preview while a CJK user types is a worse defect than a few extra renders. And do NOT write IME tests with `keyboard.type()` — it dispatches key events without a composition session and will pass against a fully broken editor. Reproduce CM6 #1650 against our decoration set and confirm it garbles *before* claiming it does not.

43.4 RTL scope — partial, content only; UI mirroring explicitly out for v1

In scope, because dependencies already do the work:

- `dir="auto"` on the editor content element and on preview block containers, which is exactly UAX #9 P2/P3 first-strong-character resolution [fetched, W3C qa-html-dir].
- `EditorView.perLineTextDirection` set true, so CM6 resolves direction per rendered line — correct for a mixed Hebrew/English vault [fetched, CM ref 2481].
- `bidiIsolates()` from `@codemirror/language` for constructs that must not reorder: link destinations, fence info strings, inline code. `bidiIsolatedRanges` supports only `unicode-bidi: isolate` [fetched, CM ref 4875, 2576].
- `Intl.Locale.prototype.getTextInfo()` (ECMA-402 §15.3.21) to resolve base direction: measured `ar/he/fa/ur` → `{"direction":"rtl"}`, `en/hi/ja` → `ltr` [measured, node 24.6.0].

Out of scope for v1: mirroring the chrome (sidebar side, gutter side, icon flips, tree indentation) needs a CSS logical-properties pass across every surface plus an icon direction review — weeks, not days. Bidi-aware markdown *authoring* affordances are also out: `>`, `-` and `#` are LTR-anchored syntax inside an RTL paragraph and CommonMark says nothing about the convention. Bidi control characters in stored content are out permanently: LRI U+2066 / RLI / FSI / PDI U+2069 are the recommended isolates over legacy RLE U+202B / LRE U+202A / PDF U+202C [fetched, W3C qa-bidi-unicode-controls], but they are invisible bytes and injecting them collides head-on with byte-preserving splice.

RTL is a **layout** problem for us, not a text-processing one — Arabic word counting already measures correct at 4 vs 4 because Arabic is space-delimited [measured]. CJK is the text-processing problem. Every CM6 bidi issue is closed, 9 total, newest 2023-11-30 [measured] — but do NOT read that as "bidi works here": the org is archived, so closure means nobody can file. **Anti-recommendation:** do not ship `dir="auto"` on the editor while leaving the preview LTR, or vice versa. Two panes disagreeing on base direction is worse than uniformly LTR, because the user cannot tell which pane is lying about their document. Ship both or neither.

43.5 Intl.Segmenter viability

ENGINE	FIRST VERSION	RELEASE DATE
Chrome	87	2020-11-17
Safari	14.1	2021-04-26
Firefox	125	2024-04-16
Node	16.0.0	—
Deno / Bun	1.8 / 1.0.0	—
Edge/Opera/Samsung/WebView	mirror Chromium	—

[all fetched, MDN BCD]

- Baseline newly available 2024-04-16 (Firefox is the gate); widely available = +30 months = 2026-10-16, 48 days from 2026-08-29 [derived].
- Spec: ECMA-402 §19, draft dated August 7, 2026 (ES2027 edition); `granularity` enum is « `grapheme`, `word`, `sentence` », default `grapheme` [fetched, tc39.es/ecma402].

- `supportedLocalesOf(['zh', 'ja', 'th', 'en', 'ar', 'hi'])` returns all six [measured, node 24.6.0].
- We already ship the dependency: `@codemirror/commands/dist/index.js:660` constructs a word-granularity segmenter for subword motion [measured]. CM6 does *not* use it for word selection — `EditorState.charCategorizer` is `/\S/` plus a `hasWordChar` regex plus a `wordChars` language-data string (`@codemirror/state/dist/index.js:2498`) [measured] — so double-click-to-select-word on CJK stays category-based until we change it.

Verdict: use Intl.Segmenter **unpolyfilled everywhere except the search index.** Measured Segmenter output for is `["", "", "", "", "", "", ""]` — linguistically right, but it makes recall depend on the segmenter agreeing with the query, and the index is built server-side (Node, full ICU) while queries may run client-side (browser ICU, possibly trimmed). Use character bigrams for the index: measured `["", "", "", "", "", "", ""]`, with both and retrieving 1 hit [measured]. Bigrams over-generate but are deterministic across engines. Do NOT ship `@formatjs/intl-segmenter`; it carries ICU segmentation data, a large payload whose only remaining gap is a Firefox older than 2024-04-16 — feature-detect and fall back to `\s+` with a UI note.

Falsifier for the bigram decision: if index size on a 1,080-file corpus grows more than 2.5× versus the Segmenter tokenizer, or client-side query latency p95 exceeds 120 ms, revisit with a hybrid (Segmenter server-side, bigram fallback) — but never with a Segmenter on both sides, because an index-time/query-time segmenter mismatch produces silent zero-recall, the exact failure measured today.

43.6 UI localisation — do not do it for v1

The v1 UI is a file tree, a status bar, a command palette and a settings sheet; the content is the user's and is already in their language. Adoption is not the question — [i18next](#) 21,709,896 weekly downloads, [react-i18next](#) 15,672,474, [next-intl](#) 5,421,207, [@formatjs/intl](#) 3,611,428, [@lingui/core](#) 1,610,658 for the week 2026-08-21—2026-08-27 [fetched, api.npmjs.org] — **translation supply** is: solo founder, no localisation budget, no reviewers for [ar / ja / zh](#) copy. Do item I-7 instead, now, at near-zero cost; India-first pricing display is the immediate commercial payoff for one utility module. At v1.1, if demand appears, adopt [next-intl](#) (native App Router integration, matches the Next 16 stack) with **one** locale file in English, so adding [zh](#) later is a translation job and not a refactor.

Do NOT add a locale switcher or locale-prefixed routes (/zh/...) with a single language shipped: a half-populated switcher is a promise we cannot keep, and locale routing changes every URL, breaking share links — a settled product surface. Do NOT machine-translate the UI and ship it unreviewed; an editor's refusal copy is the product's voice, and a bad translation of "REFUSE rare than guess" reads as a crash.

Two invariants that hold across all of §43: do NOT normalise stored content — no NFC/NFD, no bidi-control injection, no BOM insertion, because every one mutates bytes and byte-preserving splice is settled; normalise only in derived projections such as search-index keys. And do NOT use a collator for identity: macOS APFS decomposes where Linux does not, and measured German and Swedish sort `['z', 'ä', 'a']` differently (`äz` vs `az`) [measured], so `Intl.Collator` is a locale-dependent function fit for display order and never for keying a store.

§44 Trust, safety and abuse

44.1 What is legally required the day a stranger publishes a page

India — binding, because the founder is an Indian intermediary. s.79(1) IT Act 2000 grants immunity for third-party information; s.79(2)(c) makes it conditional on due diligence “and also such other guidelines as the Central Government may prescribe” — the IT (Intermediary Guidelines and Digital Media Ethics Code) Rules, 2021, G.S.R. 139(E) dated 25.2.2021, MeitY PDF updated as on 6.4.2023 [fetched 2026-08-29]. s.79(3)(b) removes immunity if, on actual knowledge, removal is not expeditious “without vitiating the evidence in any manner” [fetched]. There is no §230 equivalent; losing safe harbour makes the founder a co-defendant on the underlying claim [inference].

OBLIGATION	RULE	CLOCK
Publish rules, privacy policy, user agreement	3(1)(a)	at launch
Reasonable efforts not to host the 11 enumerated categories, incl. (xi) "violates any law for the time being in force"	3(1)(b)	continuous
Inform users of termination right	3(1)(c)	annually
Remove on court order / Government notification	3(1)(d)	≤36 hours
Preserve removed information and associated records	3(1)(g)	180 days
Retain registration info after cancellation	3(1)(h)	180 days
Provide info/assistance to a lawfully authorised agency on written order	3(1)(j)	≤72 hours
Publish Grievance Officer name and contact on the home page (or one click from it, per the Explanation)	3(2)(a)	at launch
Acknowledge every grievance	3(2)(a)(i)	≤24 hours
Resolve grievance	3(2)(a)(i)	≤15 days
Resolve a 3(1)(b) removal request, except sub-clauses (i), (iv), (xi)	3(2)(a)(i) proviso	≤72 hours
Remove NCII / nudity / sexual act / impersonation incl. morphed images, on the individual's complaint	3(2)(b)	≤24 hours
Provide the complaint mechanism for 3(2)(b)	3(2)(c)	at launch

- Rule 4 (Chief Compliance Officer, 24x7 nodal contact, Resident Grievance Officer, monthly compliance report) binds only a **significant social media intermediary**, defined by a threshold "as notified by the Central Government" (Rule 2(1)(v)) that is not in the Rules [fetched]. The commonly cited 50 lakh figure is [SS] and must not be published as fact. It is also unsettled whether a publishing tool is a "social media intermediary" at all under Rule 2(1)(w), which requires a service that "primarily or solely enables online interaction between two or more users" — publishing is one-to-many [inference].
- CERT-In Directions under s.70B(6) dated 28.04.2022: report listed incident types **within 6 hours of noticing**; maintain ICT logs for a **rolling 180 days, within Indian jurisdiction** [fetched]. That last clause conflicts with a default Cloudflare R2 + Workers stack, which does not pin storage to India by default [inference].
- Rule 3(1)(g)'s 180-day preservation duty contradicts a "we delete what you delete" promise. Name the conflict in the privacy policy rather than let a user discover it [inference].

European Union — DSA, Regulation (EU) 2022/2065, applicable from 17 February 2024 (Art 93(2)) [fetched 2026-08-29, EUR-Lex CELEX 32022R2065].

APPLIES?	ARTICLE	SUBSTANCE
Yes	11	Single point of contact for Member State authorities, Commission, Board; state the languages accepted
Yes	12	Single point of contact for users, "not solely rely on automated tools"
Yes	13	Legal representative in a Member State , mandatory with no EU establishment; notify the Digital Services Coordinator; the representative can be held liable
Yes	14	T&Cs disclosing restrictions, moderation policies, procedures, tools, algorithmic decision-making, human review; machine-readable
No — Art 15(2)	15	Annual transparency report
Yes	16	Notice-and-action: easy, electronic-only, four required elements; confirm receipt without undue delay; Art 16(3) — a compliant notice creates actual knowledge for Art 6
Yes	17	Statement of reasons for <i>every</i> restriction, six specified elements
Yes	18	Notify law enforcement on suspicion of an offence threatening life or safety
No — Art 19(1)	20–28	Internal complaint system, out-of-court dispute settlement, trusted flaggers, misuse suspensions, transparency reports, dark patterns, ads, recommenders, minors — all excluded
Yes	24(3)	Supply average monthly active EU recipients on request

Micro/small per Commission Recommendation 2003/361/EC Art 2: small = fewer than 50 persons **and** turnover or balance sheet ≤ €10m; micro = fewer than 10 **and** ≤ €2m; the exclusion survives 12 months past losing the status [fetched, CELEX 32003H0361 + DSA Art 19(1) second sub-paragraph]. Maximum fine 6% of annual worldwide turnover, 1% for incorrect or incomplete information (Art 52(3)) [fetched]. Net EU surface: **Arts 11, 12, 13, 14, 16, 17, 18, 24(3)** — materially smaller than the popular reading of the DSA.

United States.

OBLIGATION	CITE	DETAIL
Designate a DMCA agent	17 U.S.C. §512(c)(2)	Publish contact on the site and register with the Copyright Office [fetched]
Fee	copyright.gov fee schedule	\$6 per designation, amendment or resubmission [fetched 2026-08-29]
Renewal	37 CFR §201.38(c)(4)	Expires and becomes invalid three years after registration unless renewed [fetched]
Repeat-infringer policy	§512(i)(1)(A)	Adopted, reasonably implemented , users informed [fetched]
Counter-notice restore window	§512(g)(2)	"not less than 10, nor more than 14, business days" [fetched]
CSAM reporting	18 U.S.C. §2258A(a)(1)	Report to the NCMEC CyberTipline "as soon as reasonably possible after obtaining actual knowledge " [fetched]
No monitoring duty	§2258A(f)	Nothing requires monitoring, screening or scanning [fetched]
Preservation	§2258A(h)(1)	A submitted report is a request to preserve 1 year , raised from 90 days by Pub. L. 118–59 (REPORT Act, 2024) [fetched]
Penalty	§2258A(e)	First knowing and wilful failure up to \$600,000 (<100M MAU); second or subsequent \$850,000 [fetched]

Whether §2258A reaches an India-incorporated provider with no US establishment is genuinely contested — the statute defines "provider" by service type, not nationality [fetched text; extraterritorial reach = inference]. Do not state it as settled either way; it is practically moot because the US host contractually requires the same outcome.

Hosting terms transfer the liability to us. Cloudflare Service-Specific Terms, last updated **June 02, 2026**: "you are solely responsible for the acts of your End Users"; phishing prohibited; deceptive subdomains prohibited with immediate suspension or termination; content Cloudflare deems illegal "may be blocked or removed... without notice"; Cloudflare "has the right, but not the obligation" to act; 30 days' content access post-termination [fetched]. Vercel ToS, last updated **June 1, 2026**: \$15 indemnity for claims arising out of Your Content; §8 permits prohibiting use Vercel believes may be "(or is **alleged** to be)" in violation; on Hobby or trial-Pro plans Vercel "may use Your Content to train our artificial intelligence... and share Your Content with third parties" — disqualifying for a product whose pitch is that the file is the user's source of truth [fetched].

44.2 Minimum viable apparatus

COMPONENT	DRIVEN BY	MINIMUM SHAPE
Grievance Officer page: name, email, postal address, on the home page or one click from it	Rule 3(2)(a) + Explanation	Static page; the founder is the officer
Report form at a stable URL, linked from every published page footer	DSA Art 16 + Rule 3(2)(c)	Captures URL, category, reason, reporter name and email (optional for CSAE), good-faith attestation. Electronic only. Writes a ticket row
Auto-acknowledgement within 24 h	Rule 3(2)(a)(i)	Fires on submit, not on triage — the clock runs on acknowledgement, not on judgement
Statement of reasons on every enforcement action	DSA Art 17	Templated email with the six Art 17(3) fields; fires on unpublish, demote, suspend, terminate
Takedown SLA ladder	strictest clock wins	NCII/impersonation 24 h ; most 3(1)(b) categories 72 h ; court or Government order 36 h ; everything else 15 days ; DMCA "expeditiously"
Preservation store	Rule 3(1)(g), §2258A(h)(1)	Removed bytes plus metadata, 180 days (India) / 1 year (post-CyberTipline), write-once, separate from the live bucket
Action log	Rule 3(1)(j), CERT-In	Append-only: who, what URL, what action, what time, on whose notice. 180 days rolling, stored in India — a second store, not R2's default region
DMCA agent page, Copyright Office registration, repeat-infringer policy, counter-notice flow	§512(c)(2), (i), (g)	\$6, renewed every 3 years
CSAM runbook	§2258A	Written: preserve, do not re-review, report to CyberTipline, hold 1 year, notify Cloudflare
EU legal representative	DSA Art 13	Contracted third party in one Member State; notify that state's DSC
Appeals	<i>not</i> required — Art 20 excluded by Art 19	One reply-to address. Write.as's "we're happy to listen to your appeal" is the ceiling [fetched]
Do not build : internal complaint-handling system, ODR membership, trusted-flagger channel, transparency report, statements-of-reasons database feed	Art 15(2), Art 19(1)	Excluded while micro/small

44.3 Abuse vectors ranked for this product

#	VECTOR	CONTROL
1	SEO spam / backlink farms — zero-cost, automatable, the one thing a free publishing URL is unconditionally good for. Write.as: "This isn't a free place for building backlinks"; Bear: "Bear isn't a content distribution channel" [both fetched]	X-Robots-Tag: noindex on every page; rel="nofollow ugc" on outbound links; publishing gated behind a paid account; per-account publish rate limit
2	Phishing lures — APWG observed 989,123 phishing attacks in Q4 2024 [fetched, report published 19 March 2025]. The settled no-client-side-execution rule kills the JS vector but not this one; a static page with a borrowed logo and one outbound link is a working lure	Strip raw HTML and <form> at render; reputation-check the outbound links on publish; block deceptive slugs (Cloudflare requires this of us anyway)
3	Copyright infringement — highest-volume legitimate complaint class for any text host	DMCA agent, notice/counter-notice, repeat-infringer termination
4	Doxxing / targeted harassment — harshest clock: 72 h under the 3(2)(a) proviso, 24 h if it shades into impersonation	Named category in the report form routing to the 24/72 h lane; unpublish first, adjudicate second
5	NCII / morphed sexual imagery — Rule 3(2)(b) gives 24 hours and no discretion	Ban explicit sexual imagery on the published surface in v1. A policy line is cheaper than a 24-hour clock
6	Malware / payload staging	Published assets limited to rendered HTML plus an image allowlist verified by magic bytes; no arbitrary file publishing
7	Brand impersonation on a subdomain — Cloudflare terminates our account for user-chosen deceptive subdomains [fetched]	System-assigned slugs in v1; no vanity subdomain
8	Defamation — no §230; a court order starts a 36 h clock	Court-order intake path. Do not adjudicate truth
9	CSAM — lowest likelihood on a text-first surface, unbounded consequence. CyberTipline received 20.5 million reports in 2024, down from 36.2 million in 2023; 2025 ESP reports carried 61.8 million files [fetched]	Written runbook; no arbitrary image upload on any free tier in v1
10	Resource abuse / hotlinking — R2 egress is free, Workers requests are not	Per-account request cap; cache aggressively

44.4 Honest cost

Build, one-time founder-hours [derived, arithmetic shown]: ToS/Privacy/AUP drafting 12 + DSA Art 14 T&C 4 + report form and ticket store 10 + Art 17 statement generator 8 + moderation console 12 + preservation store and action log 8 + reputation check and noindex 8 + DMCA registration and notice page 2 + Grievance Officer page and 24 h auto-ack 4 + repeat-infringer/counter-notice flow 6 = **74 hours**.

Money, one-time: DMCA agent registration \$6 [fetched]. Indian counsel review ₹40,000–₹1,20,000 [SS, unverified]. EU Art 13 legal representative €200–€500/month [SS, unverified].

Run-rate is the real cost and it is not money. Rule 3(2)(a)(i) requires acknowledgement within 24 hours, every day, permanently [fetched] — **24 h × 365 days means no unbroken 48-hour offline window in a calendar year** [derived]. Per-report handling, triage plus decision plus an Art 17 statement, is roughly 20–40 minutes [inference, no measured baseline]. Volume is unknowable; stated as a replaceable assumption, at 1 report per 500 published pages [assumption, unverified] 5,000 pages/year yields 10 reports and about 5 hours, while 1 per 50 yields 100 reports, about 50 hours, plus the on-call tax. The variance, not the mean, is what breaks a solo founder. Add the renewal tax (DMCA every 3 years, annual user notification under 3(1)(c) and 3(1)(f)) and the uninsured line item: Cloudflare may act without notice, at account level, on its sole judgment, for our users' acts.

Comparables span two orders of magnitude. Bear Blog ships a roughly 350-word ToS dated 21 Aug 2022 plus a Code of Conduct v2 (28 Mar 2025), no takedown SLA, no appeals procedure, and an explicit refusal to provide one [fetched]. Mataroa has **no terms-of-service page at all** — HTTP 404 [measured 2026-08-29]. GitHub Pages carries roughly fifteen separate policy documents [fetched]. Neither Bear's operator nor Mataroa's is an Indian intermediary carrying a 24-hour statutory acknowledgement clock.

44.5 Recommendation

Ship publishing in v1, but only the narrow surface: paying accounts only, noindex by default, system-assigned slugs, no arbitrary file or image upload, raw HTML and <form> stripped at render, hard per-account page cap. The Grievance Officer page, report form, 24-hour auto-acknowledgement, DMCA agent, preservation store and Art 17 template all ship **before** the first stranger publishes.

The case for shipping: the abuse economics of a *paid* surface are inverted. Vectors 1, 2, 6 and 7 all depend on free, automatable, indexable, vanity-named pages; a credit card plus noindex plus a random slug removes the commercial motive, leaving the low-volume, legally well-defined DMCA lane. The 74 hours are one-time and mostly reusable plumbing. And a byte-preserving engine whose output nobody can see is a claim without a demonstration.

The strongest counter-argument, which is the anti-recommendation: the 24-hour acknowledgement duty is not a feature that can be descoped. It is a permanent, unbounded, personal on-call obligation; it starts the day the first stranger publishes and never ends; and the penalty for missing it is not a fine but the collapse of the s.79 safe harbour that separates "a user posted defamation" from "the founder is a co-defendant in an Indian court". Layer on the account-level third-party risk — one stranger's phishing page can take every paying editor customer offline without notice — and the shape is clear: a markdown editor with deep engine guarantees is a complete, sellable product without publishing, and adding publishing converts a software business into a moderation business with a software attachment. No amount of gating removes the on-call.

Record, do not resolve: whether publishing makes us a DSA "online platform" turns on the Art 3(i) "minor and purely ancillary feature" carve-out, untested for editor-with-publish products [fetched definition; application = inference]. Either way the EU is not the binding constraint, because Art 19's micro/small exclusion holds until 50 employees or €10m turnover. **India is the binding constraint, and the decision is about whether a solo founder will accept a 24-hour, 365-day acknowledgement duty.**

Falsifier: if a single month of live operation produces more than 8 reports per 1,000 published pages, or any acknowledgement is missed while the founder is asleep or travelling, publishing moves behind a manual approval queue — a human-reviewed publish request — rather than staying self-serve.

§45 Billing operations

The decision this section settles: a monthly INR subscription is legal and workable, but the Indian rail behaves differently enough from Stripe-US/EU that copying a US billing playbook produces silent revenue loss.

45.1 The governing instrument

ITEM	VALUE
Current framework	RBI, Digital Payments – E-mandate Framework, 2026 , RBI/DPSS/2026-27/396 · RBI/CO.DPSS.POLC.No.S56/02.14.003/2026-27, 21 Apr 2026 [fetched]
Status of prior chain	Repealed. The Framework's Repeal table lists eight circulars, including CO.DPSS.POLC.No.S528/02-14-003/2024-25 of 22 Aug 2024 [fetched]
Superseded ceilings, do not cite	₹2,000 (447/2019-20, 21 Aug 2019) · ₹5,000 (754/2020-21, 4 Dec 2020) · ₹15,000 (S-518/2022-23, 16 Jun 2022) · ₹1,00,000 for three categories (S-882/2023-24, 12 Dec 2023) — all four opened and all four now repealed [fetched]

45.2 Rules that constrain the product, by paragraph

PARA	RULE AS WRITTEN	BUILD CONSEQUENCE
8(a)	"All recurring transactions may be authorised without AFA up to ₹15,000/- per transaction. " [fetched]	₹15,000 is an architectural constant , not a pricing input. Any INR plan at ₹499 / ₹999 / ₹1,999 monthly auto-debits clearly
8(b)	₹1,00,000 without AFA applies only to insurance premiums, mutual fund subscriptions, credit card bills [fetched]	Software is not on the list. There is no higher tier available to us
4(a), 5(a)	Registration requires AFA; the first transaction requires AFA [fetched]	Exactly one on-session OTP/3DS at signup. Unavoidable; design the signup flow around it rather than apologising for it
6(a),(b)	Issuer sends a pre-transaction notification ≥ 24 h before debit , naming merchant, amount, date/time, mandate reference, reason [fetched]	The renewal is announced to the customer a day early, by their bank, in the bank's words. Our renewal email must not contradict it
6(c)	Customer gets a facility to opt out of any particular transaction or the mandate [fetched]	A churn surface that does not exist on non-Indian rails
5(b)	"Payments under e-mandates shall not be subject to any other limits/controls set by the customer " [fetched]	Customer-set card limits cannot silently decline a mandate debit
10(b)	"existing e-mandate(s) can be mapped to reissued cards" [fetched]	New in 2026. The "card expired → mandate died" involuntary-churn driver is substantially reduced in India
2	Applies to recurring transactions " domestic or cross-border ", cards / PPI / UPI [fetched]	Routing an Indian customer through a foreign MoR does not exit the framework

45.3 Rail behaviour (Stripe, India recurring payments, [fetched])

- **Indian cards get exactly one attempt.** "Payments from India-issued cards are attempted only once. This behaviour is independent of your payment retry settings." India-issued cards also appear in Stripe's retries doc under "Stripe doesn't retry payments if". Smart Retries do not exist for Indian cards.
- **26-hour billing delay:** Stripe waits 26 h after the payment request before charging (24 h regulatory + buffer). The renewal date is not the charge date; the UI must show both.
- **UPI cannot carry recurring above ₹15,000** at all.
- **Only Visa and Mastercard** India-issued cards get mandates from Stripe. No RuPay, no Amex.
- Default mandate ceiling = the amount you set **or ₹15,000, whichever is less**. A later price rise above the mandate max forces re-authentication.

45.4 Cross-border card-not-present, dated

RBI, **Authentication mechanisms for digital payment transactions Directions, 2025**, RBI/2025-26/79 · CO.DPSS.POLC.No.S 668/02-14-015/2025-2026, **25 Sep 2025**, para 10: card issuers must, **by 1 October 2026**, validate non-recurring **cross-border card-not-present** transactions raised by an overseas merchant or acquirer and register BINs with the networks; a risk-based mechanism for all cross-border CNP is due the same date [fetched]. That is **33 days from 2026-08-29** [derived]. If Indian customers are billed through a foreign-acquired MoR, their first charge is a cross-border CNP and enters this regime.

- **Do this:** offer Indian customers an India-acquired path (or an MoR that acquires domestically for IN cards) before 1 Oct 2026, and instrument first-charge decline codes by issuer BIN from day one.
- **Anti-recommendation:** do not assume the MoR has handled it. Para 2 puts cross-border recurring inside the same framework [fetched]; the MoR's acquirer geography changes *which* rules bite, not *whether* they bite.

45.5 Pricing and dunning decisions

DECISION	RECOMMENDATION	ANTI-RECOMMENDATION	FALSIFIED BY
INR price ceiling	Price every Indian plan at ≤ ₹15,000 per charge [fetched, para 8(a)]	Do not ship an Indian annual plan above ₹15,000 on auto-renew. It is legal, but every renewal needs on-session AFA — a manual repurchase wearing a subscription costume	RBI raising the AFA-free ceiling for software, or adding software to the 8(b) list
India retries	Invest in mandate longevity : long validity at registration, mandate max set to headroom (up to ₹15,000) not to list price, and rely on para 10(b) reissued-card mapping [fetched]	Do not build a retry scheduler for Indian cards. It cannot fire; you would be writing a queue the network refuses [fetched]	Stripe or NPCI publishing a retry allowance for India-issued cards
Non-India retries	Stripe Smart Retries, recommended default 8 tries over 2 weeks ; card updater; email at attempt 1, 3, and final [fetched]	Do not copy this ladder into the India code path	—
Failure UX	On a failed Indian mandate: email + in-app banner → on-session AFA re-registration → grace window before access changes	Do not make grace indefinite — R2 storage cost is real, and an unbounded free tier is a product decision made by accident	—
Refunds	Plain, no-questions window (7 or 14 days)	Do not run a "contact support first" gate. It converts refunds into disputes and buys nothing at this scale	—

Involuntary-churn budget — Recurly network, July 2026 data, **annual medians** [fetched]: overall 3.60%, voluntary 2.34%, involuntary **1.25%**, software 3.04% (top quartile ≤ 1.78%), SaaS 3.22%. By ARPC: involuntary **0.18%** at \$250+, **1.30%** at \$10–25, overall **4.29%** at \$10–25 — the worst band in the set, and the band we sell into. Involuntary as a share of total = 1.25 ÷ 3.60 = **34.7%** [derived]. Treating these as monthly overstates loss roughly 12× [fetched — the page states "median annual churn rates"].

Recorded disagreement, unresolved: two vendor recovery figures were surfaced — a "median 47.6% recovery rate" and a "median attempted recovery rate 12.7% across 119 US B2B SaaS companies, May 2026" — attributed to the same vendor family and not reconcilable without denominators. Neither page was opened [SS]. **Do not set a failed-payment recovery target from either.**

45.6 Chargebacks

Visa VAMP, launched 1 Apr 2025, replacing prior fraud/dispute programs [fetched, second-hand via an acquirer's restatement, not Visa's own publication]: merchant ratio threshold **1.5% from 1 Apr 2025**, tightening to **0.9% from 1 Jan 2026**; acquirer 0.5% → 0.3%. Ratio combines TC40 fraud and TC15 non-fraud disputes (reason codes 11, 12, 13) over settled transactions; disputes resolved via Verifi RDR are excluded. The threshold engages only once a merchant exceeds **1,000 fraud cases plus disputes per month** — at a 10,000-customer base that is 10% of customers disputing monthly, roughly two orders of magnitude above any plausible rate [derived]. Mastercard has announced no VAMP equivalent; ECP and EFM continue [fetched, same source].

45.7 Invoicing

REQUIREMENT	RULE	NOTE
Serial number	≤ 16 characters, alphabets/numerals/ - / / only, unique for a financial year , one or multiple series	CGST Rule 46(b) [fetched]
Issue window	30 days from the date of supply of service , not from payment	Rule 47 [fetched]
Mandatory fields	Supplier GSTIN; recipient GSTIN/UIN; HSN/SAC ; description; taxable value net of discount; rate; tax split CGST/SGST/IGST/UTGST/cess; place of supply + State name for inter-State; reverse-charge flag	Rule 46(a),(d),(g),(h),(j)–(n),(p) [fetched]
Signature	Not required for an electronic invoice under the IT Act, 2000	Rule 46(q) [fetched]
B2C	Recipient name/address/State required only at value ≥ ₹50,000 — except where the service is supplied by or through an e-commerce operator or by an OIDAR supplier to an unregistered recipient, where the State name is required irrespective of value	Rule 46(e), proviso to (f) [fetched]
Export	Verbatim endorsement SUPPLY MEANT FOR EXPORT/SUPPLY TO SEZ UNIT OR SEZ DEVELOPER FOR AUTHORISED OPERATIONS ON PAYMENT OF INTEGRATED TAX or ...UNDER BOND OR LETTER OF UNDERTAKING WITHOUT PAYMENT OF INTEGRATED TAX , plus name of the country of destination	Rule 46 export proviso [fetched]
Void clause	An invoice issued in any other manner by a person covered by Rule 48(4) "shall not be treated as an invoice" — the customer's ITC dies with it	Rule 48(5) [fetched]

Anti-recommendation: do **not** use a single monotonic invoice counter that never resets. Rule 46(b) demands ≤ 16 characters and uniqueness **per financial year** [fetched]; a never-resetting counter is a compliance defect that surfaces years later.

E-invoicing threshold is AATO > ₹5 crore from 1 Aug 2023, tied to the highest turnover in any FY from 2017-18 onward and not switching off when turnover falls [SS — the CBIC notification page was not opened; only Notification 78/2020-CT, the 6-digit HSN rule at AATO > ₹5 crore, is [fetched]].

45.8 MoR boundary

Handled by the merchant of record [fetched — Paddle: it "acts as a **reseller**... and is therefore the 'seller on record'... responsible for the collection and payment of VAT and tax instead of you"; Lemon Squeezy states the same for sales tax, refunds, chargebacks and PCI]; buyer-side indirect tax in 100+ jurisdictions including **India GST at 18%** in Paddle's own country table [fetched]; buyer-facing compliant invoices; registration-threshold tracking; chargeback and refund execution; PCI scope.

Still ours, MoR or not: our **own Indian GST position** on the supply to the MoR (export-of-services test, LUT vs pay-IGST-and-refund — a CA question) [inference]; GSTR-1, GSTR-3B, GSTR-9; e-invoicing on our export invoices once AATO > ₹5 crore; FEMA/EDPMS closure and FIRC/BRC per inward remittance [SS]; corporate tax, TDS, ROC/MCA; **DDPD Act 2023 obligations as data fiduciary** — we hold user documents and the MoR touches none of that; proration logic and plan-change semantics; dunning UX; refund *policy*; revenue recognition.

"MoR handles GST" means the buyer's tax, not ours — two supplies, two tax positions, and only one of them is outsourced.

45.9 Proration

Adopt Stripe's semantics as spec [fetched]; prorations computed **to the second** by default, granularity configurable to day/hour/week/month; credit for unused time on the old plan plus debit for remaining time on the new (the doc's worked example, £10 → £20 mid-period, yields a **£5** charge); **negative prorations are not auto-refunded and positive prorations are not immediately billed** — both default to the next invoice; prorations use the discounted price and take no further discount. India trap: an upgrade that raises the debit above the mandate max requires mandate recreation with AFA; Stripe's own guidance is to bring the customer back on-session and cancel/recreate the subscription on upgrade [fetched]. Apply the engine's discipline here — a mandate whose max is below the charge must refuse and surface, never silently retry.

45.10 Billing-ops founder-hours

Assumptions stated so they can be attacked, and both rates are [inference], not measured — no frontmatter billing data exists yet: MoR for all sales; refund-request rate 1.5% of active base/month at 8 min each; payment-failure events 5% of renewals/month, 10% needing a human, 5 min each.

LINE	ITEM	100 USERS	1,000	10,000	BASIS
	Refunds / billing support	0.20 h	2.00 h	20.00 h	1.5% × n × 8 ÷ 60 [derived]
	Failed-payment exceptions	0.04 h	0.42 h	4.17 h	5% × n × 10% × 5 ÷ 60 [derived]
	MoR payout ↔ ledger reconciliation	0.5 h	1.0 h	1.5 h	one aggregated payout [inference]
	GST return review (CA prepares)	0.5 h	0.5 h	1.0 h	[inference]
	Annual work amortised (GSTR-9, ITR, ROC, LUT ≈ 20 h/yr)	1.67 h	1.67 h	1.67 h	20 ÷ 12 [derived]
	Total	≈ 2.9 h	≈ 5.6 h	≈ 28.3 h	column sums [derived]

Regime crossings [derived]: e-invoicing at 10,000 customers triggers at ₹5,00,00,000 ÷ 10,000 = **₹5,000/customer/year ≈ ₹417/month** — almost certainly yes; at 1,000 customers it would take ₹50,000/customer/year — no. GST registration at the ₹20 lakh services threshold [SS] is crossed at 100 customers only if ARPU ≥ ₹20,000/customer/year. The 28.3 h figure excludes IRN setup and the Rule 46(r) QR code.

- **Do this:** engage the CA before 1,000 customers, not after — the marginal cost is small against the GSTR-9 and e-invoice transition.
- **Anti-recommendation:** do not build billing-ops tooling at 100 customers. 2.9 h/month is cheaper than any system we would write, and the 10,000-customer shape is unknowable today.

§46 Support and deflection

46.1 The benchmark does not exist, and that is the finding

No vendor publishes a per-user contact rate. Zendesk's 2026 CX Trends is opinion survey only — 83% of consumers say experiences should be better, 88% expect faster response than a year ago, 74% expect 24/7 because of AI — and carries zero volume data [fetched]. GitLab's public handbook publishes SSAT (target 95%, actual >93%), manager-to-rep ratio <10:1, SLA attainment target 95%, Customer-Wait-Time ratio ≤35%, and **no ticket-per-user figure** [fetched]. Several benchmark pages 404'd or redirected to homepages [measured].

Substitute proxy — public-forum topic rate for two editor-class products, read live 2026-08-29 via /about.json [measured]:

FORUM	TOPICS/30D	REGISTERED USERS	TOPICS ALL-TIME	POSTS ALL-TIME	TOPICS PER 100 FORUM USERS/MO	POSTS/TOPIC
forum.obsidian.md	431	99,755	51,525	362,054	0.432 [derived]	7.03 [derived]
forum.cursor.com	2,247	124,080	49,740	453,684	1.811 [derived]	9.12 [derived]

Cursor's public-help rate is **4.19×** Obsidian's [derived]. Read as: an AI-mediated, non-deterministic dev tool generates about four times the public help traffic of a deterministic local-file editor at comparable forum size [inference]. These are floors, not contact rates — forum registrations are far below product users and neither counts private email.

Our own model, from \$25.1: 0.02 tickets/free-user/month, 0.10/paid-user/month, 12 min each → at 10,000 users (9,600 free / 400 paid) = 9,600 × 0.02 + 400 × 0.10 = **232 tickets/month**, × 12 ÷ 60 = **46.4 founder-hours** [derived]. Blended **2.32 tickets per 100 users/month**.

46.2 Projected mix — and why 46.4 hours is optimistic

Empirical prior: Obsidian forum tag histogram, top 104 tags, 31,048 tagged topics [measured] —

ui-ux 3719 · dataview 3582 · custom-css 1933 · internal-links 1649 · mobile 1179 · plugin-release 900 · graph-view 894 · canvas 757 · sync 751 · templater 740 · Help is 24,106 of 51,525 topics = **46.8%** [derived]. Plugin surface (dataview + custom-css + templater + plugin-release) is 7,076 topics = **22.79%** [derived] — a class our settled no-plugin-marketplace decision deletes before it exists. **There is no billing tag at all** in the top 104 [measured]: billing is structurally private and must be given a private route or it lands unrouted in a personal inbox.

Shares and minutes below are [inference]; every product and sum is [derived].

#	CATEGORY	SHARE	TICKETS/MO	MIN	HOURS/MO
1	How-do-I / "why does the board show this"	30%	70	8	9.33
2	Sync & conflict ("where did my edit go")	15%	35	25	14.58
3	BYO API keys / AI setup	12%	28	15	7.00
4	Git auth (PAT expiry, SSO org grant, 2FA, SSH)	10%	23	30	11.50
5	Import / foreign-vault refusal	9%	21	35	12.25
6	Publishing (domain/DNS/build)	8%	19	20	6.33
7	Billing / payments	7%	16	12	3.20
8	Data loss / recovery panic	4%	9	90	13.50
9	Platform / perf / install	3%	7	20	2.33
10	Feature requests / other	2%	5	5	0.42
Total		100%	233	20.72 avg	80.45

The flat 12-minute assumption is the error: weighted mean handle time is 20.72 minutes, so the real wall at 10,000 users is **80.45 founder-hours per month** — 1.73× the PRD's **46.4** [derived: 4,827 min ÷ 60 = 80.45; 4,827 ÷ 233 = 20.72; 20.72 ÷ 12 = 1.73]. The roadmap adds precisely the long-tail categories (data loss 90 min, import 35, git auth 30) that a flat average erases.

46.3 Deflection plan, per category

CATEGORY	PRODUCT CHANGE (SHIPS ONCE)	CONTENT (WRITTEN ONCE)	TARGET DEFLECTION	RESIDUAL TICKETS
How-do-I	Projection inspector — every view exposes "show the bytes that produced this"	Docs page per projection, each opening with the literal source block	70%	21
Sync & conflict	Visible conflict state + append-only local history + one-click restore-to-timestamp. Never silent last-write-wins	"What happens when two devices edit", with the actual byte sequence	60%	14
BYO API keys	Key validator at paste time — names provider, failing scope, quota, exact remediation URL; refuse to save an unvalidated key	Per-provider setup page with a copy-paste test call	85%	4
Git auth	Device-flow OAuth; delete PAT support entirely. Pre-flight the org-SSO grant and name the blocking org	"Connecting a private repo", one screenshot per provider	80%	5
Import	Never refuse an import. Import everything, then show the degradation certificate inline as a per-file diff of what could not round-trip	"What we do with unusual markdown", naming the zero-indent-sequence case	75%	5
Publishing	DNS pre-flight — validate the record before accepting the domain; show propagation state, not a spinner	Per-registrar CNAME page	50%	10
Billing	Self-serve invoice download (GST/VAT fields), self-serve cancel, self-serve card update. No email required to leave	Pricing FAQ listing the exact decline reasons	40%	10
Data loss	Local history makes recovery a user action, not a forensic session — handle time 90 → 30 min even when it still becomes a ticket	"Recovering an earlier version", first result for "lost"	70%	3
Platform/perf	Startup-time budget in CI; version + OS auto-attached to every report	Known-issues page	30%	5
Other	—	—	0%	5

Post-deflection [derived]: 82 tickets/month; 21×8 + 14×25 + 4×15 + 5×30 + 5×35 + 10×20 + 10×12 + 3×30 + 5×20 + 5×5 = 1,438 min = **23.97 h/month**. **80.45 h → 23.97 h, a 70.2% reduction bought with engineering, not headcount.**

Already banked by settled decisions: no plugin marketplace and no arbitrary client-side execution removes **22.79%** of the Obsidian-equivalent traffic class [derived from measured] and permanently removes "my plugin broke after your update"; file-as-only-source-of-truth gives "where is my data" a filesystem-path answer; byte-preserving splice removes the diff-noise class. Highest-leverage single deflector: **run the degradation certificate on import and show it**, not only on export — it converts a 90-minute "your app broke my file" forensic ticket into a pre-answered artifact [inference].

Ship each generator with its deflector in the same release or not at all. Ranked by generation: ☹ sync ☹ BYO API keys ☹ git auth ☹ import ☹ publishing ☹ billing.

46.4 Tooling, prices read 2026-08-29 [fetched]

PHASE	STACK	COST
Now → 1,000 users (≤23 tickets/mo [derived])	support@ on our domain + GitHub Discussions (free) + docs site. Use category forms — one per class (sync / import / auth / billing) that force OS, version, and a minimal repro file	\$0/mo
1,000 → 10,000 users (23 → 232 tickets/mo)	Help Scout Standard, 1 seat, + AI Answers (\$25/user/mo; AI Answers \$0.75/resolution, 3-month free trial, vendor claims 73% average)	\$25 + 0.75 × 93 = \$94.60/mo at 40% AI resolution [derived]
Flat-cost alternative	Crisp Essentials, 10 seats, ~450 automated conversations, \$25 credits	\$95.00/mo

Break-even is 93.3 AI resolutions/month [derived: (95 – 25) ÷ 0.75] — below it Help Scout’s marginal pricing wins, above it Crisp’s flat price wins. Intercom at the same volume is \$29 + 0.99 × 93 = **\$120.87** [derived].

Vendor deflection claims, all self-reported and all opened: Fin **76%** across 12,000+ customers, 2M weekly resolutions, +1%/month, 99.8% SLA [fetched]; a named Fin customer at **50%** [fetched]; Help Scout calculator default **73%** [fetched]; Zendesk case studies **66% / 80% / 80%** [fetched]; Klarna month one, two-thirds of chats, 2.3M conversations, equivalent to 700 FTE, repeat inquiries –25%, 11 min → under 2 min, dated 27 February 2024 [fetched]. **Source disagreement, recorded not resolved:** the same vendor publishes 76% in aggregate while featuring a named customer at 50%.

- **Do this:** assume **half the vendor rate (≈38%)** until measured on our own traffic — Fin’s 76% spans mostly-FAQ deployments and our mix is roughly 40% technical-state questions [inference].
- **Anti-recommendations:** do not buy Helply (\$1/ticket, 250 tickets/mo minimum, **\$3,000/year minimum annual contract**) — at 232 tickets/mo we pay for volume we do not have [derived]. Do not buy Pylon (no public pricing, demo-gated, enterprise motion). Do not buy Plain until the credits-per-conversation ratio is published — \$35/mo with 2,000 credits is unpriceable against 232 tickets. Do not put a live-chat widget in the app: chat sets a minutes-scale expectation that a solo founder in India selling globally cannot meet, while email plus forum sets 24 h for free. Do not make Discord the support surface — not Google-indexable, so every answer is spent exactly once. **Do not let an AI agent answer data-loss or billing tickets;** hard-route both to a human path.

46.5 The hire trigger

Hire cost [fetched, Indeed India, "updated at 3 August 2026"]: Customer Support Coordinator average base **₹2,12,793/yr (n=452)**; live postings ₹15,000–₹39,856/month; company averages Freshworks ₹6,43,143, Accenture ₹6,10,500, Revolut ₹5,59,050. A technical support hire for a developer-adjacent editor sits at the company band [inference]. Take ₹6,00,000/yr + 20% employer load (the 20% is [inference]): ₹600,000 × 1.2 ÷ 12 ÷ 95.533851 = **\$628.05/month** [derived; USD→INR 95.533851 at 2026-08-29T00:02:31 UTC, fetched].

Capacity: 160-hour month × 70% utilisation = 112 productive hours ÷ 20.72 min/ticket = **324 tickets/month** [derived]. Post-deflection residual — 82 tickets, 23.97 h — is about **25% of one FTE**, and a quarter of a person is not purchasable.

Break-even, with escalation $e = 0.35$ (engine, data-loss and git-auth tickets return to the founder) and supervision $s = 8 \text{ h/month}$ [both inference]. Hiring wins when $(0.65 \times T - 8) \times V \geq 628.05$, i.e. $T = (628.05/V + 8) / 0.65$.

FOUNDER HOUR WORTH V	TRIGGER T (H/MO)	EQUIVALENT USERS, POST-DEFLECTION (2.48 H PER 1,000)
\$25	50.96	21,232
\$50	31.63	13,180
\$75	25.19	10,496
\$100	21.97	9,154
\$150	18.75	7,812

- **Without the deflection plan** ($T = 80.45 \text{ h}$): hiring pays at $V \geq 628.05 \div (0.65 \times 80.45 - 8) = \textbf{\$14.18/h}$ — hire immediately, at almost any valuation of founder time [derived].
- **With the deflection plan** ($T = 23.97 \text{ h}$): hiring pays only at $V \geq \textbf{\$82.64/h}$ [derived].
- **Deflection moves the trigger 5.83×** [derived: $82.64 \div 14.18$]. That ratio, not the absolute hour count, is the decision.

Ordering is unambiguous: **product change (one-time, then \$0/h) → AI resolution (\$4.99 per founder-hour reclaimed) → human hire (\$14.18/h floor, \$628.05/mo fixed)**. The AI figure: route the three cheap-and-frequent residual classes (21 how-do-I × 8 min, 10 billing × 12, 10 publishing × 20) to an AI tier — 41 tickets × \$0.99 = \$40.59/month, saving 488 min = 8.13 h [derived].

Non-arithmetic overrides [inference]: hire when timezone coverage becomes a churn driver (India to US-Pacific is a 12.5 h offset and a sub-24 h first response is structurally impossible solo); when a single week’s inbox blocks a release two months running; or when the hire can own billing plus how-do-I end-to-end without engine knowledge — those two classes are 31 of 82 residual tickets = **37.8%** [derived] and are the only genuinely delegable work.

Every share in §46.2 is a reasoned prior, not a measurement, and must be labelled that way wherever quoted. Replace it with measurement inside 90 days: GitHub Discussions ships category forms and discussion insights [fetched] — one category per row of the table, and the histogram is produced for us at \$0.

§47 Measurement

47.1 Why implicit-only, from our own data

CHANNEL	ROWS	KIND
Explicit feedback field	7 filled / 694 opportunities	human-written
<code>~/sgnk/state/*.gate-tier.json</code>	21,707 [measured]	machine-written
<code>~/sgnk/state/*.turn-meta.json</code>	15,786 [measured]	machine-written
<code>~/sgnk/traces/*.jsonl</code>	5,021 [measured]	machine-written

Fill rate $7 \div 694 = \textbf{1.0086\%}$; machine total 21,707 + 15,786 + 5,021 = **42,514**, a ratio of **6,073×** [derived]. **The load-bearing fact is not the 1% — it is that the numerator stayed frozen at 7 across three measurements, so the marginal fill rate is 0, not 1%** [derived: at a true 1.0086% flow, measurement 3 would exceed measurement 1 by $0.010086 \times \Delta \text{opportunities}$; observed $\Delta = 0$]. The 1% is a stock left by early novelty.

The shape reproduces independently in `~/ .sgnk/PREFERENCE-LOG.jsonl`, 232 rows [measured]: `prompt_hash` 0/101 (0%), `skill_version` 0/99 (0%), `routing_path` 1/99 (1.01%, the same number arrived at independently), `output_hash` 2/101 (1.98%) — against `timestamp` 232/232 (100%) [measured, derived]. And the one field a machine *computes*, `edit_ratio`, sits at **30/131 = 22.9%** [measured, derived].

Design rule: a field is only written if a machine writes it as a side effect of work the user wanted to do anyway.

47.2 North-star metric

WPSD — Weekly Projection-Survived Documents. Count of distinct documents that, in a rolling 7-day window, (1) received ≥1 mutation issued through a non-text projection (board / calendar / decision card / site), **and** (2) were still byte-valid under the degradation certificate at window close, **and** (3) had not been reverted to their pre-projection bytes.

WPSD = (foreign vaults opened) × (clean-cert rate) × (projection open rate) × (splice survival rate) [inference]. Derived from the splice ledger plus a file re-read; zero UI.

The strongest argument against it, stated so it is not forgotten: WPSD is an engine-usage metric wearing a value metric's clothes. A user who buys frontmatter purely for byte-preserving plain-text editing and never opens a projection receives the full promised value and contributes zero [inference]. Goodhart, in his own 1975 formulation: "Any observed statistical regularity will tend to collapse once pressure is placed upon it for control purposes" [fetched]. The cheapest way to move WPSD is to make projections the default surface, which contradicts the settled deliberately-simple-surface constraint.

- **Guard, mandatory:** pair WPSD with a counter-metric — **Projection Abstention Rate**, the share of weekly-active users who edit files but never open a projection. Rising WPSD with falling abstention is growth; rising WPSD with **rising** abstention means the projection users are a shrinking self-selected cult and the metric is lying.
- **Revision trigger, set now:** re-open the NSM if abstention exceeds 55% for two consecutive months, or if paid-conversion correlation with WPSD drops below its correlation with plain `doc_write`.
- **Anti-recommendation:** do not treat WPSD as the strategy. Review it against abstention monthly and retain the authority to retire it.

47.3 Activation and retention

	DEFINITION	PROVING EVENT
Activation	Within 7 days of first launch: (a) opened a folder containing ≥3 <code>.md</code> files frontmatter did not author, (b) wrote to ≥1 of them with <code>cert_status=clean</code> , (c) opened ≥1 projection at least once	<code>foreign_vault_write</code> with <code>authored_by_us=false</code> , <code>cert_status=clean</code> — one event, no UI, unfakeable
Retention (W4, return-with-a-file)	Retained in week N if the user writes to ≥1 document whose <code>doc_first_seen_week < N</code> . Returning and creating only new files is not retention — it is a fresh trial	<code>doc_write</code> with <code>doc_age_weeks ≥ 1</code>
Secondary	Seasonal writer vs churned, invisible in DAU	<code>vault_reopen_after_gap</code> with bucketed <code>gap_days</code>

Activation is defined on foreign vaults because the queued RO engine work says zero-indent-sequence refusals hit **83% of foreign vaults** — a user who only creates new files never touches the failure mode that decides whether the product survives contact with reality [inference]. Do not quote any competitor's activation threshold as verified; no primary source for a specific company's number was opened [SS].

47.4 Event schema

EVENT	FIRES WHEN	PROPERTIES	DECISION IT CHANGES
<code>app_open</code>	Launch	<code>os</code> , <code>app_version</code> , <code>cold_start_ms_bucket</code>	Which OS/version to keep supporting
<code>vault_open</code>	Folder opened	<code>file_count_bucket</code> , <code>authored_by_us_ratio_bucket</code> , <code>max_depth_bucket</code>	Whether to fund large-vault indexing
<code>foreign_vault_write</code>	First write to a file we did not author	<code>cert_status</code> , <code>refusal_code</code> , <code>file_size_bucket</code>	Activation gate ; ranks which RO refusal to fix first
<code>cert_refusal</code>	Certificate refuses a save	<code>refusal_code</code> , <code>construct (zero_indent_seq , bare_cr)</code> , <code>vault_share_affected</code>	Direct priority order for engine work; makes the 83% claim live
<code>cert_downgrade</code>	Cert passes but flags a lossy round-trip	<code>construct</code> , <code>severity</code>	Full support vs documented limitation
<code>splice_apply</code>	Any byte-splice lands	<code>splice_kind (human / ai / projection)</code> , <code>byte_len_bucket</code> , <code>doc_id_hash</code> , <code>splice_id</code>	Ledger row; parent of all survival derivation
<code>splice_outcome</code>	T+1h / T+24h / T+7d re-read	<code>splice_id</code> , <code>outcome ∈ { accepted , edited , reverted , superseded }</code> , <code>edit_ratio_bucket</code> , <code>survival_bucket</code>	Ship or kill an AI capability
<code>projection_open</code>	Board/calendar/card/site opened	<code>projection_kind</code> , <code>doc_size_bucket</code>	Which projections to keep or cut
<code>projection_write</code>	Mutation issued via a projection	<code>projection_kind</code> , <code>field_kind</code> , <code>roundtrip_ok</code>	NSM numerator ; catches non-reversible projections
<code>projection_abandon</code>	Opened, closed, zero writes, <20 s	<code>projection_kind</code>	Separates "unused" from "tried and rejected"
<code>doc_write</code>	Any save	<code>doc_age_weeks</code> , <code>doc_size_bucket</code> , <code>write_source</code>	Retention gate
<code>vault_reopen_after_gap</code>	Session after ≥7 idle days	<code>gap_days_bucket</code>	Seasonal vs churned; changes dunning
<code>undo_after_splice</code>	Undo within 60 s of a splice	<code>splice_kind</code> , <code>splice_id</code>	Fast negative signal
<code>perf_slow_op</code>	Any op over p95 budget	<code>op_name</code> , <code>duration_bucket</code> , <code>doc_size_bucket</code>	Where to spend perf time
<code>crash / engine_panic</code>	Unhandled fault	<code>stack_hash</code> , <code>op_name</code> , <code>app_version</code>	Hotfix trigger
<code>export_run</code>	Export / site publish	<code>target</code> , <code>doc_count_bucket</code> , <code>ok</code>	Whether the site projection is load-bearing
<code>license_state</code>	Trial→paid, paid→lapsed	<code>state</code> , <code>days_since_activation</code>	Pricing and trial length
<code>update_check</code>	Version check (user-disableable)	<code>from_version</code> , <code>to_version</code>	Version-support sunset dates

Volume ceiling enforced in the emitter: `splice_apply` sampled 1-in-10 above 200/day; `doc_write` collapsed to one row per document per hour; per-DAU-day ceiling ≈ **12 events** [inference]. At 600 DAU: 600 × 12 × 30 = **216,000 events/month** [derived].

Cut, because they change no decision: `button_click`, `menu_open`, `settings_viewed`, `keystroke_count`, `time_in_app`; session replay, heatmaps, scroll depth (structurally impossible over a document surface without capturing content); and every NPS / thumbs / star / "was this helpful?" widget.

47.5 Never collect

NEVER	WHY
Document bytes, fragments, diffs, clipboard	We hold user documents; a leak is the whole business
File names, folder names, vault paths, or their hashes	Name hashes are re-identifiable by dictionary attack against public repos; Obsidian treats file names as needing E2E encryption even inside its own paid Sync [fetched, obsidian.md/privacy]
Headings, tags, front-matter keys or values	Front-matter keys are project and client names
Raw token counts or prose length (bucket only)	Length plus timestamp fingerprints a published artefact
Stable cross-install device identifiers, MAC, serial, IP beyond coarse country	ePrivacy Art 5(3) engages on terminal-equipment access regardless of whether the data is personal [fetched, EUR-Lex 02002L0058-20091219]
Any data from a user believed to be a child	DPDP §9(3): "A Data Fiduciary shall not undertake tracking or behavioural monitoring of children or targeted advertising directed at children" [fetched]
Row-level retention beyond 90 days	DPDP §8(7)(a) requires erasure once the specified purpose is no longer served [fetched]

47.6 Legal basis

- **GDPR Art 6(1)(f)** legitimate interests is available for bucketed, non-identifying product telemetry with an opt-out [fetched, gdpr-info.eu/art-6-gdpr; balancing-test conclusion is inference].
- **ePrivacy Art 5(3) is the harder gate and is about the device, not personal data:** storing or accessing information on terminal equipment requires consent except where "strictly necessary in order for the provider of an information society service explicitly requested by the... user to provide the service" [fetched]. Writing a persistent analytics ID to disk is storage on terminal equipment [inference]. **Build consequence:** derive the install ID from a rotating, salted, non-recoverable value regenerated every 90 days, and surface a first-run toggle. Whether a rotating local ID falls inside the strict-necessity carve-out is contested and unresolved here — the CNIL audience-measurement exemption page timed out twice [measured, curl exit 28].
- **DPDP applies regardless of where we sell.** §3(b) extends the Act to processing outside India "in connection with any activity related to offering of goods or services to Data Principals within the territory of India" [fetched, DPDP Act 2023, No. 22 of 2023, assent 11 Aug 2023]. Selling globally from India engages both limbs.
- **§7(a) does not help.** "Certain legitimate uses" covers data the Data Principal "has voluntarily provided"; telemetry is not voluntarily provided, so **§4(1)(a) consent** is the realistic Indian basis [fetched + inference]. India is stricter than GDPR here — there is no legitimate-interests limb for analytics.
- **Timeline:** DPDP Rules 2025 notified **14 November 2025** with an eighteen-month phased compliance period and 6,915 consultation inputs [fetched, PIB, 17 Nov 2025]. 14 Nov 2025 + 18 months = **14 May 2027** [derived]. Gazette number cited as G.S.R. 846(E) [SS].
- **Penalties (Schedule, §33):** up to **₹250 crore** for failure of reasonable security safeguards (§8(5)); **₹200 crore** for breach-notification failure (§8(6)); **₹200 crore** for children obligations (§9); **₹150 crore** for Significant Data Fiduciary obligations (§10) [fetched].

47.7 What privacy-focused competitors actually collect

PRODUCT	STATED COLLECTION	DOCUMENT DATE
Obsidian	"We do not collect any personal data." / "We do not collect any telemetry data." All app data local. Update check exists and is disableable. Sync is E2E-encrypted including file names. Plugin directory policy prohibits client-side telemetry	"Last updated November 1, 2023" [fetched]
Bear	Personal data collected: email address only. No app-telemetry section exists at all	No last-updated date exposed [fetched]
iA Writer	Writer notice: "Writer does not collect your personal data." Separate Analytics notice: "we periodically collect and store limited device, operating system, app, and feature usage data. None of the collected information identifies you personally."	Both "Last revised: September 26, 2025" [fetched]

Source disagreement, recorded not resolved: iA publishes two notices that pull against each other, both dated 26 Sep 2025 [fetched]. That is the most useful competitive fact available — the most design-purist competitor in this exact niche does collect implicit feature-usage telemetry and survives saying so in plain words [inference]. Obsidian's page separately serves two stacked policies, the current one (1 Nov 2023) and an older Dyalist Inc. policy (11 Dec 2020) invoking PIPEDA, without stating which governs [fetched].

47.8 Stack and cost

OPTION	PRICE READ 2026-08-29	VERDICT
PostHog Cloud	Free tier 1M events/month, 5K replays, 1M flag requests, 1-year retention, 1 project, no card [fetched]	Phase 1. 216,000 ÷ 1,000,000 = 21.6% of allowance, 4.63× headroom [derived] → \$0/mo
Own store: Cloudflare Workers + object storage	Workers Paid \$5/mo minimum, 10M requests included, +\$0.30/M [fetched]	Phase 2. 216,000 ÷ 10,000,000 = 2.16% [derived] → \$5/mo; add Supabase Pro from \$25/mo if SQL is wanted [fetched]
PostHog self-hosted	MIT, needs "4 vCPU, 16GB RAM, and more than 30GB storage"; vendor-stated "officially unsupported", no tagged releases, no CVEs [fetched]	Reject. DO 4 vCPU/16 GB from \$0.181/hour × 730 = \$132.13/mo [derived] plus solo ops time
OpenPanel Cloud	250K events \$30/mo, 500K \$50, 1M \$90; self-host free [fetched]	Viable fallback, correct event shape
Plausible / Fathom	Plausible 100k pageviews \$19/\$29/\$39; Fathom 500,000 pageviews \$45/month [fetched]	Anti-recommendation — pageview-shaped, no per-document survival model; Fathom is ~9× OpenPanel at this volume [derived]

- **Do this:** phase 1 PostHog Cloud free tier at \$0/mo, with the emitter written against a four-field internal interface so the backend swaps in a day. Phase 2 trigger: either >1M events/month, or the first enterprise buyer who refuses a third-party processor.
- **Open item, blocking:** PostHog EU-region residency was **not confirmed**. Confirm before sending a single event, because DPDP §8(2) requires a valid contract with any Data Processor [fetched].
- **Anti-recommendations:** never ship a thumbs-up/down, star rating, NPS prompt, or "was this edit helpful?" widget — 7-in-694 is the in-house, highly-motivated-user ceiling for that pattern and a paying stranger will do worse. Do not build an in-product feedback text field of any kind; route qualitative signal to a support inbox the user initiates. Do not add session replay or heatmaps at any price, including PostHog's included 5K recordings — the surface being replayed is the user's document. Do not hash file names or paths and call it anonymised.

47.9 AI accept / edit / revert, derived from the document

The engine already does byte-preserving splice edits, so the ledger needed to derive this **already exists**. No rating UI, ever.

1. At apply time, `splice_apply` records `{splice_id, doc_id_hash, byte_offset, byte_len, inserted_hash, prefix_hash, suffix_hash, splice_kind, model_id, ts}` — hashes only, on-device.
2. At T+1h, T+24h, T+7d, on the next natural file read (no background scanning), classify locally:
- | OUTCOME | TEST |
|-------------------------|--|
| <code>accepted</code> | Bytes at the range still hash to <code>inserted_hash</code> |
| <code>edited</code> | Range present, hash differs, normalised edit distance ≤ 0.5 |
| <code>reverted</code> | Bytes gone and <code>prefix_hash</code> + <code>suffix_hash</code> rejoin to the pre-splice hash |
| <code>superseded</code> | Bytes gone and context does not match pre-splice — the user rewrote into something new |
3. Emit only `{splice_id, outcome, edit_ratio_bucket, survival_bucket, splice_kind, model_id, doc_size_bucket}`. Never bytes, never raw distance.
4. The one real signal is `accepted_at_7d` per `model_id` and per `splice_kind`. Rising `edited` with falling `reverted` means close-but-wrong — a prompt fix. Rising `reverted` means kill the capability.
5. Validate the T+24h classifier against `undo_after_splice` within 60 s, which is a same-session proxy available immediately.

§48 Longevity and the shutdown promise

The internal bar, adopted verbatim: “A self-guaranteeing promise is verifiable and non-reversible. It does not require you to trust anyone.”

`FETCHED, STEPHANGO.COM/SELF-GUARANTEE, 2024-12-03`.

#	PROMISE	MECHANISM	REVOCABLE	RANK
1	Plain files in folders the user chose	filesystem	No, for files already written	self-guaranteeing
2	No proprietary sidecar carrying meaning	<code>ls</code> + delete test	No	self-guaranteeing
3	Byte-preserving splice	hash → edit → diff	No — falsifiable per release	self-guaranteeing + measurable
4	Views are pure projections	delete-the-app test	No	self-guaranteeing
5	Published versioned format subset + test corpus	a document; copies persist	Updates can stop	durable, weak
6	Cross-engine degradation certificate	published report	Only as honest as the disclosed engine matrix	strong, auditable
7	Open-source file engine	licence grant, irrevocable for released versions	No — shipped versions only	durable
8	Delayed-open-source licence (FSL 1.1 → Apache-2.0 or MIT at two years; BUSL 1.1 at the Change Date or the 4th anniversary, whichever is first)	automatic future grant, per version	No — fires with nobody acting	durable, code only
9	Source escrow	third-party contract	Beneficiary-limited	weak
10	Export guarantee in the ToS	policy	Yes	weak
11	Foundation or governance structure	corporate form	Yes	weakest

In every documented notes-app shutdown, the format was readable; what destroyed data was the deadline and the notice, never the encoding

INFERENCE OVER FETCHED CASES: `CATCH.COM 2013-08-30`, `SPRINGPAD 2014-06-25`, `VESPER 2016-08-30`, `SKIFF 2024-02-09`. The two counter-cases are the only ones where something a user relied on still executes: `omnivore-app/omnivore`, AGPL-3.0, not archived, 16,225 stars, last commit `2026-08-28T12:19:30Z`, and `standardnotes/app`, AGPL-3.0, 6,608 stars, last commit `2026-08-25T23:07:42Z` `MEASURED, READ 2026-08-29T00:10Z`.

The public commitment — exact wording. Publish at a versioned, dated URL **outside** the Terms of Service, append-only, with every prior version left live.

The frontmatter durability commitment — v1, 2026-08-29 *This document is versioned and append-only. Every prior version stays published at its own URL. We do not edit it in place.*

What we promise, and how you check it without us

1. **Your notes are ordinary UTF-8 text files in folders you chose.** There is no database, no index you need, and no sidecar file that carries meaning. *Check:* quit frontmatter, delete every file in the folder that is not one of your notes, reopen the folder in any editor. Nothing of yours is missing.
2. **Every view is a projection of the file.** Board, calendar, decision card and published site are computed from your text and stored nowhere else. *Check:* delete the app.
3. **Every edit is a byte-preserving splice.** We change the bytes you asked to change and nothing else. Line endings, indentation, key order, trailing whitespace, and a leading byte-order mark survive an edit that did not touch them. *Check:* hash the file, change one field, diff.
4. **We publish the exact Markdown and YAML subset we read and write,** versioned, with the test corpus that defines it. When it changes we publish the diff, not a changelog entry.
5. **We certify degradation rather than assume it.** Each release ships a report of what a named set of other engines does to a file we wrote, and exactly where information is lost. We publish the failures.
6. **There is no lock on the way out because there was never a way in.** No import. No export. Only the folder.
7. **If we stop:** the last released desktop client keeps working offline; we publish the source of the file engine — parser, splicer, and projections — under an Apache-2.0-
converting licence within 90 days of announcing a shutdown; and any hosted service gets 12 months' notice before it stops.

What we do not promise

- We do not promise this company will exist in ten years.
- We do not promise this app will run on a future operating system.
- We do not promise your files render identically in other tools. CommonMark is still at version 0.31.2, dated 2024-01-28. YAML's own Library of Congress format description records that implementations validate differing subsets. We promise a certificate of what differs, not that nothing differs.
- We do not promise Markdown is an archival standard. It is not named in the Library of Congress Recommended Formats Statement; plain text appears there only in the Acceptable column.
- We do not promise that the hosted parts — sync, publish, collaboration — survive the company. Those are services. Your files are not.
- We do not promise a fixed price forever, a perpetual licence to future versions, or a data escrow.
- We do not promise to keep any third-party tool, integration, or engine working.
- We do not promise features. This document is about durability only.

Drafting rules, enforced at review: every clause names a test the reader can run without us; no clause uses *forever*, *always*, *guaranteed* or *never*; the document is dated, versioned, and lives outside the terms, because terms are revocable and this must not be.

The evidence under the "what we do not promise" block, so no one softens it later: the LoC Recommended Formats Statement contains `markdown` 0, `commonmark` 0, `YAML` 0, `plain text` 2, and lists plain text as *Acceptable* alongside RTF and proprietary word-processor formats, below HTML and DOCX [MEASURED](#), [ARCHIVE SNAPSHOT 20260626211847](#); the LoC format registry has 595 entries, zero for Markdown, and one for YAML (`fd000645`), last significant update 2025-06-02 [FETCHED](#) . That same YAML description calls it an open standard *and* records that concerns "have led to the development of YAML alternatives, such as YAML parsers that only validate a restricted subset" — a source that disagrees with itself, preserved rather than resolved [FETCHED](#) . CommonMark has published versions back to 0.13 (2014-12-10) and is still 0.x after ten years [MEASURED](#) . The canonical obsolescence case is WordStar (`fd000552`): mostly ASCII, first shipped September 1978, and its pre-5.0 releases used the high bit of ASCII characters to store formatting, so the files "may appear as gibberish" under typical encodings [FETCHED](#) . What we may honestly cite from preservation guidance is narrow and real: UTF-8 is top-Preferred, and the no-DRM / no-encryption requirement is satisfied by construction [MEASURED](#) .

Anti-recommendations, each one a promise to refuse:

- **Never claim archival-institution endorsement of Markdown.** Zero occurrences in the RFS, zero descriptions in a 595-entry registry [MEASURED](#) . One search refutes it and takes the trust position down with it.
- **Never say "open source" without naming licence, repository, scope and trigger.** Skiff's unscoped claim became the story on shutdown day, when the Discord and the GitHub repo were removed on announcement day [FETCHED](#), [PARTISAN PRIMARY SOURCE](#) .
- **Never promise source escrow** (release frequency is unmeasured, and a consumer product cannot name every beneficiary) and **never promise a dead-man's-switch open-sourcing** — after insolvency an administrator controls the IP and the promisor no longer exists. If that outcome is wanted, ship under a licence that already contains the future grant.
- **Never adopt Elastic License 2.0 as a hedge.** It contains no conversion clause [FETCHED](#) ; it protects the vendor, not the user.
- **Never market GDPR Art. 20 as a feature.** It mandates no format, covers only data the subject provided under consent or contract by automated means, and claiming it invites the accurate reply that it is mandatory [FETCHED](#) .
- **Never state a migration or export window.** The correct claim is that no window can exist, because the files were never held.
- **Never open-source the whole product as the hedge.** Given the settled no-marketplace and no-client-side-execution positions, scope clause 7 to the file engine only; a narrow promise is one you can keep, and a broad one gets quietly dropped, which is worse than never making it.
- **Never reuse the commitment URL.** Version and date each revision and keep the old ones live, so any weakening is visible.

Open items this section does not settle, flagged rather than smoothed: OAIS (ISO 14721 / CCSDS 650.0-M-2) was **not opened** — no OAIS claim may be published from this section [MEASURED](#) ; the EU Data Act (Regulation (EU) 2023/2854, applying from 12 September 2025, with the Art. 3(1) obligation from 12 September 2026) rendered only at recital level, so whether a local-file editor with optional sync falls inside Chapter VI is unresolved, though a hosted sync or publish service sold in the EU plainly is in scope [FETCHED + INFERENCE](#) ; and GitHub could not resolve an SPDX identifier for Joplin (`NOASSERTION` , 56,143 stars) or Anytype (`NOASSERTION` , 8,715 stars), so neither may be cited as an open-source hedge without reading their LICENSE files first [MEASURED](#) .

§49 Open source, documentation and community

49.1 Licence per artifact

Current state: **the repo has no** `LICENSE` **file and** `package.json` **has no** `license` **field** [\[measured\]](#) — the default is therefore "all rights reserved", which is the worst possible position for a spec we intend others to implement. Every runtime dependency is MIT — `unified@11.0.5` , `remark-parse@11.0.0` , `micromark@4.0.2` , `@codemirror/state@6.6.0` , `next@16.2.6` , `react@19.2.6` [\[measured 2026-08-29\]](#) — so no copyleft obligation constrains any choice below.

Artifact	Licence	Reasoning	Anti-recommendation
App (Next 16 + Tauri client, sync, billing)	Proprietary, closed. Publish a plain-English data-portability promise instead of source	Nothing about the app is a credibility claim; all deps are MIT so there is no obligation to open [measured]	Do not BUSL the app. BUSL's Change Date — capped at the 4th anniversary of first public distribution of that version [fetched] — forces eventual publication of source we never intended to publish. Openness cost, none of the trust benefit
Engine (splice, OffsetMap, refusal logic)	Apache-2.0	Explicit patent grant plus trademark reservation; the only asset whose value <i>rises</i> with third-party verification; permissive keeps it embeddable in the MIT ecosystem it already sits in [measured]	Do not use FSL or BUSL here. FSL 1.1's Competing Use clause bars anything with "same or substantially similar functionality" [fetched] — exactly the independent reimplementability that would prove the engine correct. Both also convert anyway (FSL 2 years → Apache-2.0 or MIT; BUSL ≤ 4 years) [fetched]
Certificate CLI	Apache-2.0 , same repo as the engine	A certifier nobody can run is not evidence. It must be installable in CI at zero friction	Do not gate the CLI on a licence key (the Elastic-2.0 mechanic, which forbids circumventing licence-key functionality [fetched]) — key-gating a <i>verification</i> tool destroys the artifact's purpose
Format spec (session-interchange)	Prose: CC-BY 4.0. Conformance test-suite and reference parser: Apache-2.0 . Add a one-line "no patent assertion against conforming implementations" pledge	Satisfies all three requirements simultaneously; copy-the-text-into-your-docs, implement-without-patent-risk, no ShareAlike on the implementation. W3C Software and Document License 2023 is the single-instrument alternative — copy/modify/distribute for any purpose, no fee, with full NOTICE and notice of changes [fetched]	Do not use CC-BY-SA — CommonMark 0.31.2 (2024-01-28) uses it [fetched], and ShareAlike on spec prose deters the vendors we need to adopt it. Do not use CC0 — we lose the attribution that makes the spec a distribution channel
Certificate dataset	CC-BY 4.0 for the data; keep the generation harness Apache-2.0; retain raw run artefacts privately	Attribution is the entire marketing mechanism — a cited dataset is a permanent backlink	Do not licence it non-commercially (PolyForm-NC) — the competitors we certify are commercial, and a non-commercial dataset cannot be quoted in their release notes [fetched terms]. ODbL is the anti-choice . Do not publish per-vendor "fail" verdicts without a documented dispute path and a stated methodology version
Repo hygiene	Add root <code>LICENSE</code> , set <code>package.json.license</code> , add per-directory <code>LICENSE</code> for engine / CLI / spec	No licence file exists anywhere today [measured]	Do not apply one licence at repo root and assume it covers the spec prose — mixed-licence repos need per-path declaration

The engine's value is verifiability, not secrecy: a byte-preserving engine that "refuses rather than guesses" is only credible if a third party can run the refusal cases, and a degradation certificate is a claim about other people's software, which is a marketing asset when unauditable and evidence when auditable.

Cloning risk is low: the moat is R2/Workers sync, Tauri packaging, auth, billing, and the certificate corpus — none of which live in the engine [inference].

49.2 The reversal record, and why the restrictive licences do not apply

Case	Sequence	Read
Redis	Redis Stack split → SSPL March 2024 → AGPLv3 with Redis 8; antirez rejoined Nov 2024 [fetched]	Reverted in roughly 14 months
Elastic	ELv2/SSPL 2021 → AGPL added as a third option, Aug 2024; founder states the OpenSearch fork was foreseen and "market confusion has been (mostly) resolved" three years later [fetched]	Survived, with a funded fork permanently attached
HashiCorp	BUSL 2023 → OpenTofu, now Linux Foundation, 3,900+ providers, 23,600+ modules, v1.12.0 [fetched 2026-08-29]	Community transplanted off the vendor in months

Restrictive relicensing of *infrastructure others embed* reliably produces a funded fork, and two of three majors then partially reverted [derived from the three fetched sources]. None of these firms is a solo-founder desktop editor; the hyperscaler free-rider threat that justifies BUSL and SSPL does not exist for a markdown editor [inference].

Consolidated licence anti-recommendations: do not BUSL or SSPL anything. Do not use PolyForm Small Business — its self-assessed thresholds (<100 total people, <USD 1,000,000 (2019) revenue, inflation-adjusted [fetched]) are unenforceable by a solo founder in India against a global user base. **Do not claim "open source" for anything but the Apache-2.0 artifacts** — Redis's own post records OSI's position that SSPL "lacks the requisites" [fetched], and loose usage is the reputational failure mode in exactly this community. Source-availability disagreement recorded and not reconciled: Elastic's founder frames its licence change as correct in hindsight; Redis frames its own SSPL period as something to move on from [both fetched].

49.3 Documentation

Diátaxis, as published [fetched]: four kinds — tutorials, how-to guides, reference, explanation — with the compass mapping informs-action + acquisition → tutorial, informs-action + application → how-to, informs-cognition + application → reference, informs-cognition + acquisition → explanation. The author's own instruction is to apply it to something small.

Quadrant	Pages	Priority	Cost
Reference	Splice contract; refusal catalogue (every REFUSE with its byte-level cause); certificate schema; CLI flags; session-format spec	P0 — this is the purchase input for the beachhead ICP	~12 pages, generated from tests where possible
Explanation	"The file is the only source of truth"; why reversible projection; why we refuse rather than guess; what a degradation certificate does and does not claim	P0	~6 pages, hand-written, low churn
How-to	Run <code>mdmax cert</code> in CI; certify a vault; migrate from Obsidian; recover a refused edit	P1	~8 pages
Tutorial	One 15-minute "certify your first vault"	P2	1 page

- **Generate the refusal catalogue from the test fixtures.** A hand-maintained catalogue drifts, and drift here is a credibility failure, not a typo [inference].
- Docs-as-code, in-repo, PR-gated: block the merge when a refusal code changes without a reference-page update [fetched, Write the Docs]. The repo already carries **159 markdown files under** `docs/` across `adr/`, `engine/`, `mdmax/`, `research/`, `build/` [measured] — engineering notes, not user docs.
- **Maintenance cost, measured proxy** — `obsidianmd/obsidian-help`: **175 English .md under** `en/`, **6,357** all-locale markdown files (a **36.3×** localisation multiplier), **83 commits in the 90 days since 2026-05-31 = 0.92/day, = 0.47 commits per English page per quarter** [measured + derived]. A 30-page English site at that churn ≈ 14 doc commits/quarter ≈ one per week; at 30 min each, **~2 h/month steady-state**, plus a one-off build cost of ~60 h at 2 h/page spread over 8–10 weeks [inference].
- **Anti-recommendations:** do **not** localise — a 36.3× file multiplier for zero ICP value [derived]. Do **not** write the tutorial before the reference exists — Diátaxis's own advice is to start small and applied [fetched]. Do **not** host docs on a separate CMS; it decouples them from the merge gate that keeps them true.
- **Honest limit:** the "documentation is a purchase input for dev-tool buyers" claim could not be verified — the Stack Overflow Developer Survey 2025 page is client-rendered and yielded 271 characters of extractable text [measured]. Every claim in that vein is [SS]/[inference], not fact. What survives as reasoning: for frontmatter the load-bearing doc is not a tutorial, it is **"what will this tool refuse to do, and why"** — the refusal contract *is* the product claim [inference].

49.4 Community

Platform	Cost	Read 2026-08-29	Indexed	Moderation Burden	Verdict
GitHub Discussions	\$0		Yes, permanent, Google-indexed	Lowest — same identity and block/report as the repo; categories and marked answers [fetched]	Buy (it is free) . Co-located with spec, CLI and issues
Discourse hosted	\$100/mo Pro (5 staff seats), \$500/mo Business (15 seats); free plan exists [fetched]	Yes		Medium — categories, trust levels, a mod team	Premature pre-revenue: \$1,200–\$6,000/yr [derived] for a room that will be empty
Discord	\$0	No		Highest — synchronous, always-on, instant-reply expectation	Do not open one . Worst ratio of durable value to founder hours
Reddit	\$0	Yes		Low, and no control either	Distribution channel, not a home

Load model, from the Obsidian forum's own 30-day numbers [fetched 2026-08-29]: **3,629 active users, 956 participating, 2,921 posts, 432 topics, 1,330 likes** → 3.06 posts per participating user/month and 26.3% of active users posting at all, the 90-9-1 shape, measured [derived]. At 1% of that volume: 29 posts × 3 min triage = **~1.5 h/month**; at 10%: 292 posts = **~15 h/month** — the point where a solo founder is choosing between the community and the engine [derived, assumption stated]. Discord has no equivalent floor: presence scales with hours online, not posts — Obsidian's Discord showed **presence_count 20,851 online** at fetch time [measured].

Phase	Surface	Time Cap	Trigger to Advance
0 (now)	GitHub Discussions on the spec/engine/CLI repo only; categories: Spec, Refusals, Certificates, Q&A	2 h/week , batched into two fixed windows	—
1	Publish certificate results as a versioned dataset; invite conformance-case PRs	+1 h/week	≥10 external conformance cases
2	Reddit and Obsidian-forum <i>participation</i> — answer, do not recruit	1 h/week	Certificates cited by a third party
3	Discourse at \$100/mo	—	Only when Discussions exceeds ~150 posts/mo [derived from the 3 min/post model]

Written policy from day one: response-time expectation ("weekdays, within 2 business days"), scope ("spec and engine here; app support by email"), and a conflict-of-interest note on certificate disputes. Do not promise 24 h response. Do not run a community launch, a docs site and a licence launch in the same month.

Positioning against the incumbent community: Obsidian's commercial licence is now **optional** — "No. You are not required to pay for a commercial license... we encourage you to purchase" [fetched 2026-08-29], with Sync \$4/user/mo annual (\$5 monthly), Publish \$8/site/mo annual (\$10 monthly), Commercial \$50/user/year, Catalyst \$25. The norm there is voluntary payment for a **closed app with open files**, so a closed frontmatter app is not a norm violation; a proprietary *format* would be [inference]. Frontmatter cannot use the plugin route — no marketplace, no arbitrary client-side execution (settled) — so the only compounding artefact available is **certificates and conformance cases contributed against the spec** [inference]. Obsidian's own compounding assets, for scale: **7,062 community plugin entries and 716 themes** [measured 2026-08-29, "id": -key occurrence count, a proxy not a parsed record count]. Entering as "the tool that refuses to corrupt your vault, and here is the proof for the other tools too" is a contribution; entering as "Obsidian but better" is a fight with 7,062 plugins.

49.5 Sequencing against engine work

The three tracks compete directly with the R0 engine queue: ~60 h docs build + ~2 h/month docs upkeep + ~8 h/month community is the same budget as the queued zero-indent-sequence refusals, bare-CR handling, and SAFE_KEY addressability. Ordering that minimises contention [inference]: **licence files first (~2 h, unblocks everything)** → reference docs generated from tests (amortised into engineering) → GitHub Discussions at a 2 h/week cap → dataset publication → only then explanation and tutorial prose. The strongest coupling is licence ↔ docs: an Apache-2.0 engine only earns trust if the refusal catalogue is public, and the licence is what makes the docs quotable by the ICP. Unverified and flagged: OWFa 1.0 terms and PolyForm Shield 1.0.0 terms were not opened (empty response and a GitHub-Pages 404 respectively) [measured], and all Reddit community sizes are unverified — Reddit's JSON API refused both user-agents attempted [measured].

§50 Risk register

L = likelihood, I = impact (1–5). Every owner is the founder until §53 says otherwise.

50.1 Technical

Risk	L	I	Early Warning	Mitigation
Parser refuses real-world vaults on import	5	5	Import-failure rate; "it won't open my notes". 83% aggregate foreign refusal MEASURED	Ship NF-1 before any marketing number; standing corpus gate in CI
No CI at all	5	4	.github/ absent MEASURED	Add CI before any external contributor or paid customer
Addressability gaps	4	4	SAFE_KEY excludes a space — date created in 812/957 files of one vault	Quoted-key support (NF-4)
Non-Latin correctness debt	4	3	countWords 1.7–2× under; MiniSearch CJK recall 18.1%	§43
LLM price/model deprecation	4	4	Cost-per-active-user drift >20%	BYO-key lane + provider abstraction; never hardcode one model id
Silent data destruction on edit	3	5	NF-3 is set-destructive and oracle-blind , and the corpus cannot see it	Synthetic fixture + set-only oracle
Editor state corruption	3	5	Undo/redo divergence. The class was already fixed once	Property-based sync tests (§29.3)

50.2 Market

Risk	L	I	Early Warning	Mitigation
No distribution channel	4	5	Zero organic signups for 60 days	Ship the engine research publicly — the corpus numbers are the marketing asset
Incumbent ships the same feature free	4	5	Competitor changelog	Compete on the engine guarantee, not the feature list
Byte-fidelity may be a claim no buyer prices	4	5	Conversion flat despite demo engagement	If "provably never loses a keystroke" reads as table stakes, the technical differentiator is unmonetisable and we are competing on editor taste against Obsidian's free tier
Free self-hosted substitutes	4	3	—	\$5–8/seat must be justified by what self-hosting cannot cheaply give

50.3 Operational and key-person

RISK	L	I	EARLY WARNING	MITIGATION
Founder is the single point of failure	5	5	Response-time drift; unshipped weeks	Runbooks; bus-factor doc; MoR absorbs billing support
Sustained content cadence fails	4	4	Miss twice	Never-miss-twice; queue depth ≥ 2
Support volume from BYO-key misconfiguration	4	3	Ticket mix >30% "my key doesn't work"	Key-validation at entry; ≤200-char actionable errors
Burnout with paying customers live	3	5	Consecutive zero-commit weeks	Explicit degraded-service SLA; escrow of deploy credentials
Unauthorised destructive ops by agents	3	4	Commits not authored in-session	Capture <code>git rev-parse HEAD</code> before/after every workflow; the prohibition is advisory, never the gate
Two PATs remain unrotated	live	4	—	Founder action, today

§51 Legal, compliance and security

Everything in §51.1–51.2 is an issue list for a chartered accountant and a lawyer. None of it is advice.

51.1 India

ITEM	STATUS
Export of services as zero-rated; LUT/bond vs pay-and-refund	SS — primary text not verifiable : cbic-gst.gov.in failed TLS and 404'd on every IGST-Act path. Needs CA confirmation
"Export of services" requires payment in convertible foreign exchange	SS — If an MoR is the contracting counterparty, who the recipient is and what currency lands in the Indian account changes the answer. The single highest-value CA question
Reverse charge on imported services (LLM APIs, hosting, MoR fees)	SS A real recurring cost line for this exact product
SOFTEX / EDPMS	FETCHED All invoices including those under US\$25,000 must appear in the bulk statement. Realisation within nine months from date of export
DPDP Act 2023	FETCHED Failure to take reasonable security safeguards — s.8(5) — ceiling ₹250 crore. s.16 restricts transfer only to notified countries, a blacklist model. The DPDP Rules were not verifiable ; breach timing and SDF designation hinge on them

51.2 GDPR

FETCHED **Art. 3(2) applies to a non-EU controller offering services "irrespective of whether a payment of the data subject is required" — a free tier reachable from the EU is enough. Art. 27 then requires designating an EU representative in writing**; the "occasional processing" exception is unlikely to cover a document workspace. **Art. 83** reaches €20M or 4%, whichever is higher.

Contradiction to reconcile deliberately: DPDP s.16 is permissive by default (blacklist); GDPR Chapter V is restrictive by default. **A single storage architecture cannot satisfy both by accident.**

51.3 What a merchant of record absorbs — and what it does not

Absorbed: sales-tax/VAT/GST calculation, collection, filing and payment liability; seller-of-record status; payment-related buyer support; chargeback handling.

Still owed regardless: Indian corporate income tax; the **GST treatment of the MoR relationship itself**; FEMA/EDPMS/SOFTEX filings; and **every DPDP and GDPR controller obligation over the documents themselves.**

An MoR absorbs *tax*, never *data protection*. It is a payment counterparty, not a data-processing shield.

51.4 Security posture

SURFACE	POSITION
BYO API keys	Encrypted at rest with per-user envelope encryption; never returned to the client after entry ; never logged; scrubbed from error traces and prompt context. Do not co-locate the key store with the document store
GitHub scopes	Ship a GitHub App with fine-grained permissions, not an OAuth app. <code>repo</code> grants full read/write on all repositories and org resources FETCHED — a deal-breaker in enterprise review
Published pages	An unguessable URL is not access control. Server-side authorisation on every request; revocation must invalidate CDN cache, search access and any signed URL. We already fixed an unpublish-revocation defect, which is evidence the class is live here
Prompt injection	An agent reading a vault is reading untrusted content. Any instruction inside a document is data, never a command. The lethal trifecta — untrusted content + private data + an outbound capability in one turn — is gated behind explicit per-operation confirmation
The eval lane	Refused. A markdown workspace with code fences invites it, and it converts every prompt injection into remote code execution on infrastructure holding every customer's documents and API keys

52.1 The incumbent, measured

FACT	VALUE	TAG
Front Matter CMS, VS Code installs	80,605 (read 2026-08-29)	[fetched] marketplace extensionquery API
Prior round's figure / delta	80,527 / +78	[derived]
Marketplace <code>updateCount</code> / <code>downloadCount</code> / rating	280,650 / 2,661 / 5.0 avg over 20 ratings, weighted 4.817	[fetched] — <code>install</code> and <code>downloadCount</code> do not reconcile and Microsoft's field definitions were not opened
<code>estruyf/vscode-front-matter</code>	2,539 stars, 106 forks, MIT, last push 2026-08-21	[fetched]
GitHub org <code>frontmatter</code>	Taken — "Front Matter CMS", created 2021-09-21, 7 public repos; most-starred org repo is <code>web-documentation-nextjs</code> at 29 stars	[measured]
npm <code>frontmatter</code>	200, but a corpse — 4 versions, latest <code>0.0.3</code> , last published 2022-06-18, description "Parsing YAML frontmatter from a string", 11,212 dl/mo of transitive-dependency noise. Not Front Matter CMS	[measured]

The registry collision and the market collision are two different problems and were previously counted as one.

52.2 Shortlist with availability evidence

Probed by `curl` from this machine, 2026-08-29 00:09 UTC. npm 404 = free, GitHub `api.github.com/users/NAME` 404 = free; discriminator validated against three nonsense handles that all returned 404 [measured]. Domains: UNCHECKED on every row — no registrar or RDAP host is reachable through this sandbox's allowlist.

RANK	NAME	NPM	GITHUB	REPOS NAMED	PRODUCT COLLISION	FOR	AGAINST
1	<code>stetfile</code>	FREE (404)	FREE (404)	0	None found	<i>Stet</i> = "let it stand", the one English word meaning preserve the original exactly — the engine's contract as a name	Two morphemes; "file" is dull; less punchy than bare <code>stet</code>
2	<code>mdmax</code>	FREE (404)	TAKEN — "Maksim Golitsinskiy", 2 repos, joined 2012-11-28	18, none a product	None found	Already validated, zero adjacent-market collision	<code>md</code> reads as <i>Doctor of Medicine</i> to non-devs; "max" is a spec-bump word; reads as a library, not a product you charge for
3	<code>grainfile</code>	FREE (404)	FREE (404)	0	None found	"With the grain" is the most accurate available metaphor for reversible projections over the author's own file	Needs one sentence of explanation; nobody guesses it cold
4	<code>bytewright</code>	FREE (404)	TAKEN — "Bytewright", 28 repos, joined 2015-01-28	25	None found	<code>-wright</code> (shipwright, playwright) = made by hand, precisely	Active squatter on the handle; <code>-wright/-write/-right</code> is a live spelling hazard for a name said aloud in support
5	<code>truefile</code>	FREE (404)	TAKEN — "Vyacheslav", 1 repo, joined 2020-09-01	17	None found	Literally the product thesis; the likeliest handle to negotiate	Generic, SEO-hostile, reads as a file-recovery utility
—	<code>stet</code> (bare)	FREE (404)	TAKEN — 65 repos	2,089	<code>elberacasa/stet</code> shipped npm <code>stetmark</code> on 2026-08-05, 5,753 dl/mo, tagline "stet — let it stand" [measured]; plus <i>Stet</i> public-commenting software and Cowlishaw's 1977 STET editor [SS]	Perfect one syllable	A live claim on the word, three weeks old, same metaphor, agent-oriented

Recommendation: `stetfile`, with `stet` as the spoken shorthand and the CLI verb (`stet <file>`). It is the only candidate clean on all four measured axes — npm 404, GitHub 404, zero named repos, no product found. Naming from the guarantee is defensible for a decade; naming from the file format (`md*`) ages with the format. **Anti-recommendation attached:** re-check `stetmark` 's trajectory before any spend, because bare-`stet` branding collides with it even though `stetfile` does not.

Ruled out and recorded so nobody re-litigates [measured + inference]: `quire` (Getty's Quire is a plain-text multiformat publishing framework — the closest adjacent market on the list — and quire.io is a kanban PM tool, the exact settled anti-position); `colophon` (best conceptual fit for degradation certification, but a live Obsidian long-form plugin holds it in this market, npm and GitHub both taken); `palimpsest` (the metaphor is inverted — scraped off and overwritten); `marginalia`, `folio` (13,893 dl/mo), `splice` (the music platform), `vellum`, `quarto`, `bedrock`, `scribe` (4,233 dl/mo), `fathom`. Do not pick from the npm-404/GitHub-200 bench (`quoin`, `trueline`, `kerf`, `truefold`, `bytefold` ...) — a squatted org forces a permanent handle mismatch you pay for in every README, badge, and install line. Do not ship a name whose npm 404 you have not re-run on the day you register it; these were 404 at one instant and nothing reserves them.

52.3 The explicit case for keeping "frontmatter" with a qualifier

For. The word is the thesis in one token — the file's `frontmatter` is the state every projection reads, so the target buyer needs no teaching. The collision is narrower than 80,605 suggests: Front Matter CMS is a VS Code extension [fetched], a plugin inside another editor, whose most-starred org repo is a docs site at 29 stars — a different artifact, install path, and price point [inference]. Bare npm `frontmatter` being 200 costs almost nothing, since it is a dead 2022 `0.0.3` YAML parser [measured].

Against. 80,605 installs and 2,539 stars in the same keyword space is real search and word-of-mouth interference for years [fetched]; npm `getfrontmatter` is already 200, so the qualified space is being nibbled [measured]; a qualifier is a permanent tax — "Frontmatter, the app, not the VS Code one" in every conversation, forever [inference]; and trademark exposure is highest here because the mark is near-identical in an overlapping class.

Qualified handles measured free 2026-08-29 [measured]: npm `frontmatter-app`, `frontmatterapp`, `usefrontmatter`, `frontmatterhq`, `frontmattr`, `frontmatter-studio`, `frontmatter-editor` — all 404. GitHub `frontmatterapp`, `getfrontmatter`, `usefrontmatter`, `frontmatterhq`, `frontmattr`, `frontmatterstudio` — all 404.

RANK	FORM	VERDICT
1	<code>frontmatterapp</code>	Product name stays "Frontmatter"; the handle carries the qualifier. Least brand tax
2	"Frontmatter Studio" (npm <code>frontmatter-studio</code> , GitHub <code>frontmatterstudio</code>)	A qualifier in the <i>spoken</i> name is what actually separates you from the incumbent
3	<code>frontmattr</code>	Anti-recommend. Vowel-drop is a 2012 tic and generates permanent misspelling tickets

Do not use "Frontmatter" bare as the product name while quietly holding qualified handles — that is the highest-risk option with extra steps, taking full trademark and SEO exposure and buying nothing back. Either qualify the spoken name or leave the word.

Falsifier. If a trademark attorney's knock-out search returns clean on the incumbent's class overlap, "Frontmatter Studio" beats `stetfile` , because a taught name costs more than a qualifier.

52.4 Trademark — required disclaimer

Nothing above is a trademark clearance. What was checked is **package-registry and code-host handle availability plus informal product collision**, a different question in kind. **UNCHECKED and not checkable here:** USPTO TESS, EUIPO, IP India/TMR, WIPO Madrid, unregistered common-law rights, Nice-class overlap, use-in-commerce priority, and every domain. A name can be free on npm and GitHub and still infringe; a taken handle implies nothing about mark rights. **Get a knock-out search from a trademark attorney on the final two names before any spend on domains, logo, or launch copy — and specifically before choosing between "keep frontmatter with a qualifier" and a clean coinage, because that is the decision where legal risk, not availability, dominates.**

§53 Execution capacity

One person. This is the binding constraint on everything above.

QUESTION	ASSESSMENT
Realistic solo throughput	~1 substantial shipped surface per 2–3 weeks alongside support, billing and compliance. The engine backlog alone is a multi-month R0 lane before any customer-visible feature
Non-negotiable overhead	Support, incident response, invoicing and filings, security patching are recurring and do not compress with skill
MUST BUY	WHY
Merchant of record	Global sales-tax registration and filing is unbuildable solo
Auth, error tracking, uptime	Credential handling is a liability, not a differentiator; you are an on-call rota of one
EU Art. 27 representative	Statutory; cannot be self-appointed from India
A CA and a lawyer	Every item in §51 is outside an agent's competence
Object storage + CDN	Cloudflare R2, egress-free

Must BUILD (the moat): the markdown engine and its round-trip guarantee; the degradation certificate; addressability design; the agent-over-documents UX and its injection boundary.

Must HIRE, in order: (1) a **part-time support/community person** once tickets cross ~10/week — it is the first thing that destroys engineering blocks; (2) a **second engineer only after CI exists**.

Sequencing consequences. R0 precedes marketing any number. CI precedes hire #2. The MoR must be chosen before the first paid signup, because migrating billing counterparties mid-flight breaks the FEMA paper trail. **The EU representative must exist before the first EU free signup.**

§54 Open decisions

Each re-scopes a lane. None is researchable.

#	QUESTION	CONSEQUENCE
D1	sgnk-md vs frontmatter	145 byte-identical files, 12 diverged module files each a two-place fix, md frozen since 2026-07-17, md has CI and frontmatter does not. Recommendation: frontmatter is the sole codebase; port md's <code>ci.yml</code> immediately
D3	The name	§52. The evidence argues against the bare unqualified name. Needs counsel before brand spend
D8	Do documents leave the device?	Local-first collapses the DPDP/GDPR surface to near-zero and kills server-side agents, search and publishing. Server-stored enables the product and buys the full §51 obligation set including the ₹250 crore ceiling
D9	BYO key or platform key?	No COGS and high support load, versus clean UX and real margin exposure
D11	GitHub App or OAuth app?	§51.4 recommends the App. The current <code>read:user</code> scope grants no repo access, so the path as written does not work
D14	Does the agent get write authority?	The flagship demo and the whole lethal-trifecta liability
D15	Publishing in v1?	§44. The recommendation is a narrow paid surface; the counter-argument is a permanent 24-hour acknowledgement duty
D17	CJK in scope for v1?	In means fixing <code>countWords</code> and MiniSearch recall. Out means saying so in positioning rather than shipping a silent failure
NF-4	Does <code>café</code> in NFC equal <code>café</code> in NFD for key addressing?	Both answers defensible. The one engine unit an agent cannot start
D5	The two unrotated PATs	Only you. Today

§55 Verification debt

No claim tagged SS in this document may be published, quoted in marketing, cited to an investor, or repeated to a customer until it has been re-fetched with the date read. The PRD's own evidence-tag census stands at FETCHED 86 · SS 52 · MEASURED 50 · DERIVED 9 · INFERENCE 1 [measured].

55.1 Sources refused or unreachable — retry with backoff before publication

SOURCE	FAILURE	BLOCKS
ACM publications policy	HTTP 403	§42.2 fourth row stays SS
OpenAI classifier withdrawal post	HTTP 403, JS wall	Any claim about detector retirement
arXiv 2303.11156 (Sadasivan et al.)	HTTP 429	The detection-limits argument rests on three papers, not four
C2PA member roster	JS-rendered, headers only	Any adoption count for C2PA
GitHub API in the r3 session	403 unauthenticated	No star or download count from that session exists. Counts in §22 come from the separate r2 session and carry their own read dates
PRC Art. 11	Not opened	The exact mandated text-label syntax is unknown
Digital Omnibus amending act	URL returned a 404 page	The 2 Aug vs 2 Dec 2027 disagreement (§42.1) is unresolved
YouTube disclosure help page	1,420,253 bytes of navigation chrome, no policy body	All platform disclosure norms
pypistats.org	429 RATE LIMIT EXCEEDED	Any Python-side distribution proxy for §22
Every registrar / RDAP host	Not in the sandbox allowlist	Every domain in §52 is UNCHECKED

55.2 Claims that remain unverified, ranked by public embarrassment

#	CLAIM	WHY IT IS DANGEROUS
1	Princeton GEO "25–40% visibility lift from quotes, statistics and citations" SS	Sits in the same sentence as a boast that a rival claim "is measurably refuted and will never appear in our marketing". An unopened figure next to a refutation boast is the highest-embarrassment pairing in the document
2	"under 4% of GitHub notebooks reproduce" (Pimentel 2019) SS , never opened	A specific figure from a specific paper with a specific reproducibility definition
3	"No organised India Obsidian meetup or Discord found" SS , single source	An absence claim about a community that can refute it with one reply
4	"OpenAI removed Canvas May 2026" SS	A falsifiable claim about a named company on which an entire structural hedge rests
5	"Notion cut free AI to 20 responses for life"; "Microsoft +43% Copilot bundling drew a CMA probe" SS	Names a regulator and two competitors' pricing from unopened sources, inside a section arguing that <i>they</i> misstate things
6	Support deflection "18% median, 40–60% with AI, \$25–35/ticket" SS , single source	Goes into B2B collateral against Intercom, Zendesk, and Document360, who publish their own numbers
7	"Median solo B2B founder revenue >4× B2C by month 24" SS , no named study	Shapes the entire two-motion strategy
8	Vanta/Drata "\$7,000–\$30,000/yr" SS	Two named vendors' prices
9	"Perplexity Pro free via Airtel to ~400M subscribers, worth ₹17,000/yr" SS	Three checkable numbers about a named telco
10	"21.9M India GitHub contributors, +5.2M in a year" SS	Octoverse is one click away
11	Dataview ceiling "~30s past 3,000 notes" SS	We promise to publish ceilings honestly; publishing an unopened ceiling is self-defeating
12	r/ObsidianMD ~344,000 / Discord ~195,000 — untagged in the PRD	Untagged numbers read as measured
13	§25.1 support model (0.02 / 0.10 tickets per user-month, 12 min each) → 46.4 founder-hours/month	Labelled an assumption in prose, then quoted as a conclusion twice
14	Overleaf's total user count	Its own two pages said 20M and 25M on the same day [fetched]. Never state one figure
15	Tunisia RMS survey generalised to India	n=121, single study, fieldwork 2016 [fetched]. Cite it as a warning about a <i>low-income research population</i> , never as an India number
16	The 7.8%–20.9% academic-plugin bracket	The 20.9% upper bound is not reproducible from the three published peak-version numbers; 23.3% is what the arithmetic gives (§22.1). Re-derive before use
17	"obsidian agent-skills 47,418★"	Correct magnitude, wrong repo name and a citation URL that cannot produce a single repo's stars. Fix to <code>kepano/obsidian-skills</code> , 47,444★, 3,410 forks, pushed 2026-06-08 [fetched 2026-08-28T23:36Z]
18	"inkeep/open-knowledge 3,239 → 3,673 in 27 days"	Restate as 3,239 → 3,679 in 28 days [fetched 2026-08-28T23:33Z]
19	Docs-as-code pricing (GitBook, Mintlify, Outline, Confluence)	Tag collision: one report says all SS , pricing pages could not be opened; another and §20.7 present the same products as FETCHED 2026-08-29 . GitBook Premium \$65 and Statuspage Business \$399 carry both tags in different sections. Resolve to one tag before the wedge number ships
20	<code>mdmax/fold@1</code> 43.71% → 4.28%	Self-flagged as prototype-only, never run over the pinned corpus. Keep the flag; do not quietly promote it
21	"Tests passing 1,575/1,575"	Sits three rows above "CI: None, <code>.github/</code> does not exist" [measured]. Tag it as a single local run until a gate exists that can fail

55.3 Never researched — report as absent, never as zero

Accessibility (WCAG 2.2 AA plan, screen-reader model, VPAT/ACR cost); error and refusal UX; backup, restore, and DR (RPO/RTO, point-in-time restore, restore drill); the sync-engine choice, whose exit condition is two-device offline convergence with zero loss and whose corpus contains **0 hits for** `service worker` [measured]; public API design, versioning, and rate limiting (`rate limit`, 0 hits in the PRD); trust and safety, takedown, and India IT Rules 2021 intermediary duties; product analytics and consent basis; email deliverability and

notifications; open-source licence posture; i18n, IME, and RTL; roles, permissions, and invites; desktop signing, notarisation, and auto-update; SEO for published pages; schema-migration contracts. Also, from §42: platform disclosure norms, Spain's draft labelling penalties, the Utah AI Policy Act, the Colorado AI Act, and Korea's AI Framework Act.

55.4 What needs a human or a professional

NEED	PROFESSIONAL	GATE – DO NOT PROCEED PAST THIS WITHOUT THEM
Trademark clearance on the final two names	Trademark attorney, India + US + EU, relevant software class	Before any spend on domains, logo, or launch copy, and before choosing between a qualified "frontmatter" and a clean coinage (§52.4)
What discharges an EU AI Act obligation for a given customer	The customer's own counsel	We emit records and citations only. A solo founder in India must never be the party asserting EU compliance (§42.6)
PRC labelling conformance	A PRC-qualified adviser who can read the incorporated	Before any China-facing export claim
Intermediary liability, grievance officer, takedown timelines under India IT Rules 2021; DMCA agent for the US	Indian counsel + a US agent of record	Before one-toggle publish goes live to the public internet
GST treatment, GSTIN on invoice, e-invoicing, and the s.8(5) exposure on customer documents	Chartered accountant / company secretary	Before the first paid invoice
WCAG 2.2 AA conformance and an ACR/VPAT	Accessibility auditor	Before any B2B procurement conversation; the current <code>body-faint #b8b8b8</code> at 2.14:1 already fails AA [measured]
Independent adversarial review of the engine's own verifiers	A reviewer who did not write them	Per LR#60 — a verifier written beside its subject inherits its blind spots
Domain availability for every §52 candidate	Anyone with unrestricted network access	Re-run npm and GitHub 404s on the day of registration ; nothing reserves them

§56 Research record

ROUND	AGENTS	REPORTS	WHAT IT ESTABLISHED
R1 internal	7	<code>i1</code> – <code>i7</code>	Content engine, ecosystem inventory, knowledge-base principles, AIOS capabilities and pattern language , docs-vs-reality delta
R2 external	8	<code>e1</code> – <code>e8</code>	Claim verification, home surface, India market, the agent frontier , importer breakage, naming risk, fresh community signal, no protocol buyer
R3 hands-on	4	<code>h1</code> – <code>h4</code>	The foreign-corpus run , the Hubble and Front Matter CMS teardowns, the OpenKnowledge teardown
R4 editors	5	<code>ed1</code> – <code>ed5</code>	Writing tools, IDEs, lossy frameworks , AI editors, the office-suite review-loop canon
R5 AI journey	4	<code>aj1</code> – <code>aj4</code>	Agent protocols, the laws behind <code>land()</code> , knowledge formats, the journey-home gap
R6 core	6	<code>c1</code> – <code>c6</code>	The projection law , the computation budget, the B2B wedge, INR pricing, build refs, twelve screens
R7 PRD corpus	16	<code>x1</code> – <code>x13</code> , <code>g1</code> – <code>g3</code>	Lossless compression of R1–R6 + business model , internal fusion , risk/legal/capacity
R8 AIOS + specDD	12	<code>a1</code> – <code>a5</code> , <code>b1</code> – <code>b5</code> , <code>c5</code> , <code>c6</code>	AIOS as product, the spec-driven-development market , the document canon, the AI build loop , conformance verification, the spec-file system , build-plan inputs
R9 markdown	10	<code>m1</code> – <code>m10</code>	Spec landscape, the carrier decision , the extension catalogue, key conventions, typed markdown, pipelines, output targets, computational markdown, hard edges, the possibility space
R10 untouched angles	7	<code>s1</code> – <code>s4</code> , <code>d1</code> , <code>d2</code> , <code>d4</code>	Simplicity engineering , latency, affordance, mobile, search, collaboration, AI interaction
R11 gaps	11	<code>c1</code> – <code>c4</code> , <code>d3</code> , <code>r1</code> – <code>r6</code>	The gap audit , architecture decisions , feature completeness, onboarding, longevity, migration, academia, disclosure, naming , measurement, support
R12 build blockers	14	<code>k1</code> – <code>k14</code>	DR , the sync engine , refusal UX, accessibility, API, observability, roles, abuse and publishing , desktop, performance and testing, data model, i18n, OSS/docs/community, Indian billing

105 reports, 374,866 words. Paths: `docs/research/agent-reports-2026-08-28/` (R1–R6) · `-r7/` · `-r8to10/` · `-r11/` · `-r12/`.

A method note. In several rounds agents had their final message consumed by a repeatedly-firing reconciliation hook; their deliverables were recovered from transcript JSONL and re-mapped to agents **by prompt** rather than by content keyword, after a keyword match once filed one agent's output under another's name. The hook keys on absolute working-tree dirtiness rather than a delta from a subagent-start baseline, so it re-fires indefinitely on pre-existing dirt. **Fixing that trigger predicate is a real maintenance item.**

§57 Contradictions ledger

Read this before quoting any number from an older document. This is not sloppiness — it is what happens when a fast-moving system is described on different days. The rule is §41's: **re-derive at write time.**

QUANTITY	LIVE MEASURED	PREVIOUSLY STATED	NOTE
frontmatter test files	98	262	The <code>find</code> traversed extra git worktrees. Two independent measurements agree at 98
The render carrier	Callout for prose, fence for opaque data	"the fenced-code info string is the dispatch mechanism"	PRD v1.1 §9 was wrong. §8.2
MDMAX product importers	1 symbol from 1 of 13 files	"imported by zero product files"	False at import level, true at capability level
mdmax tests	10	11	
AIOS skills	124 SKILL.md / 130 linked	"~93", "99", "95", "~102"	Every historical figure is stale
Trace rows / gate rows	5,014 / 24,539	4,994 / 23,778	Both correct on their day
Knowledge-base notes	234	175 · 171 · 212 · 273	Five different counts for one corpus
Corpus fence count	7,969	7,959 "frontmatter files"	RESOLVED 2026-08-31 [measured]. 7,969 is correct; 7,959 was an error, not a definition. Counted four ways over the pinned 8,513-file corpus — files opening <code>---\n</code> at byte 0; those with a non-empty block; those with at least one <code>key:</code> ; those closed by exactly <code>---</code> — and all four return 7,969. Also measured on the same pass: 0 CRLF, 0 bare-CR, 0 BOM, 0 unterminated blocks, which is why the corpus cannot red-prove NF-3 and a synthetic fixture is required. Use 7,969.
Bare-CR / CRLF / BOM in corpus	0 / 0 / 0	assumed present	The corpus cannot red-prove NF-3. A synthetic fixture is required
Claude Sonnet 5	\$2/\$10 per MTok. September rise cancelled	~\$3/\$15	Frontier assumption was 50% high
Gemini cheap tier	3.1 Flash-Lite \$0.25/\$1.50	"2.5 Flash \$0.15/\$0.60"	That SKU does not exist
DeepSeek V4-Flash	\$0.22/\$0.66 off-peak; \$0.44/\$1.32 peak	off-peak quoted as the rate	
Dodo India fee	4% + 15¢ = 4.79% of ₹299	"\$0.40 flat = 12.7%"	The US rate was applied to India
Mintlify Pro	\$450/mo	\$250 · \$150 · \$450–540	Consolidation wedge moves to \$1,163–1,223
FX	₹95.39 / ₹95.59	₹83	Changes the ₹699 decision
The Obsidian first-note claim	Unsourced	cited as evidence	An unattributed pull-quote in a 2025 blog post. Do not cite
<code>mdmax/foId@1</code> divergence	Never run over the pinned corpus	"43.71% → 4.28%"	That is the research prototype's number over 40 files × 6 engines
Design system	Shipped <code>#18181b</code> + 6/8/12px radii	Canonical <code>#1a5cff</code> + square corners	§7.4 — a real defect

§58 What we may say in public, and what we may not

May say — each is MEASURED or FETCHED and re-derivable by anyone:

- 0 corruption and 0 throws across 8,513 third-party markdown files from 7 vaults, byte-pinned by sha256 manifest and upstream commit, re-verifiable via `npm run corpus`.
- Three competitors' write paths **executed**, with the exact bytes each destroys.
- Every mainstream rich-text framework is lossy by design, in the vendors' own words.
- Git's three-way merge preserves CRLF and a missing final newline exactly, and refuses where a splice writer would want to refuse MEASURED.

May not say, yet:

CLAIM	WHY NOT
Any fidelity or coverage percentage	At the measured 83% refusal rate the headline number is false until NF-1 and NF-3 land. Publishing first converts a bug into a public claim
"NF-1 recovers 99.98%"	An inference from bucketing refusal causes, not a measurement
Anything about Obsidian's onboarding funnel	The widely-repeated claim is unsourced
"First AI attribution"	Cursor and Grammarly exist. The defensible claim is narrower: <i>the first markdown editor with byte-anchored, document-portable, reader-visible provenance</i>
"Serve markdown to agents and get cited"	Measurably refuted
Any SS -tagged number	47 remain in this document's inherited material
Anything about a learning loop	Never market "it learns you" before a decision demonstrably bends on real data

§59 Build references

#	REFERENCE	LICENCE	WHY
1	@lezer/markdown	MIT	The incremental parse tree. SyntaxNode byte offsets are our span-addressing primitive. Most load-bearing read
2	The OKF spec	Apache-2.0	Google formalising our exact bet: a directory of markdown files with YAML frontmatter, path as identity
3	@codemirror/merge	MIT	The review loop as a shipped component. Highest reuse per hour
4	Hypothesis match-quote.ts + approx-string-match	BSD-2 / MIT	Anchoring that survives human edits
5	mdast-util-to-markdown	MIT	Read the enemy. Its loss points are what our writer is measured against
6	@sanity/diff-match-patch	Apache-2.0	Google's original is archived and unpublished since 2020 . Use the maintained fork — and see §31.1 for why it is not a merge substrate
7	@modelcontextprotocol/sdk	MIT → Apache-2.0	Mid-relicense
8	obsidian-dataview	MIT, 9,300★	Closest prior art to the rendering bet; MIT means the query parser is copyable

Licence landmines: **obsidian-kanban** is **GPL-3.0 and abandoned** — read the board file-format convention, copy zero code. **anthropics/skills** has **no licence file** = all rights reserved — mirror the convention, copy nothing. **Outline** is **NOASSERTION**. The **CodeMirror org** was **archived 2026-04-15** (55 of 57 repos) though npm is alive — source is read-only, no upstream issues, **budget vendoring risk**.

§60 Glossary

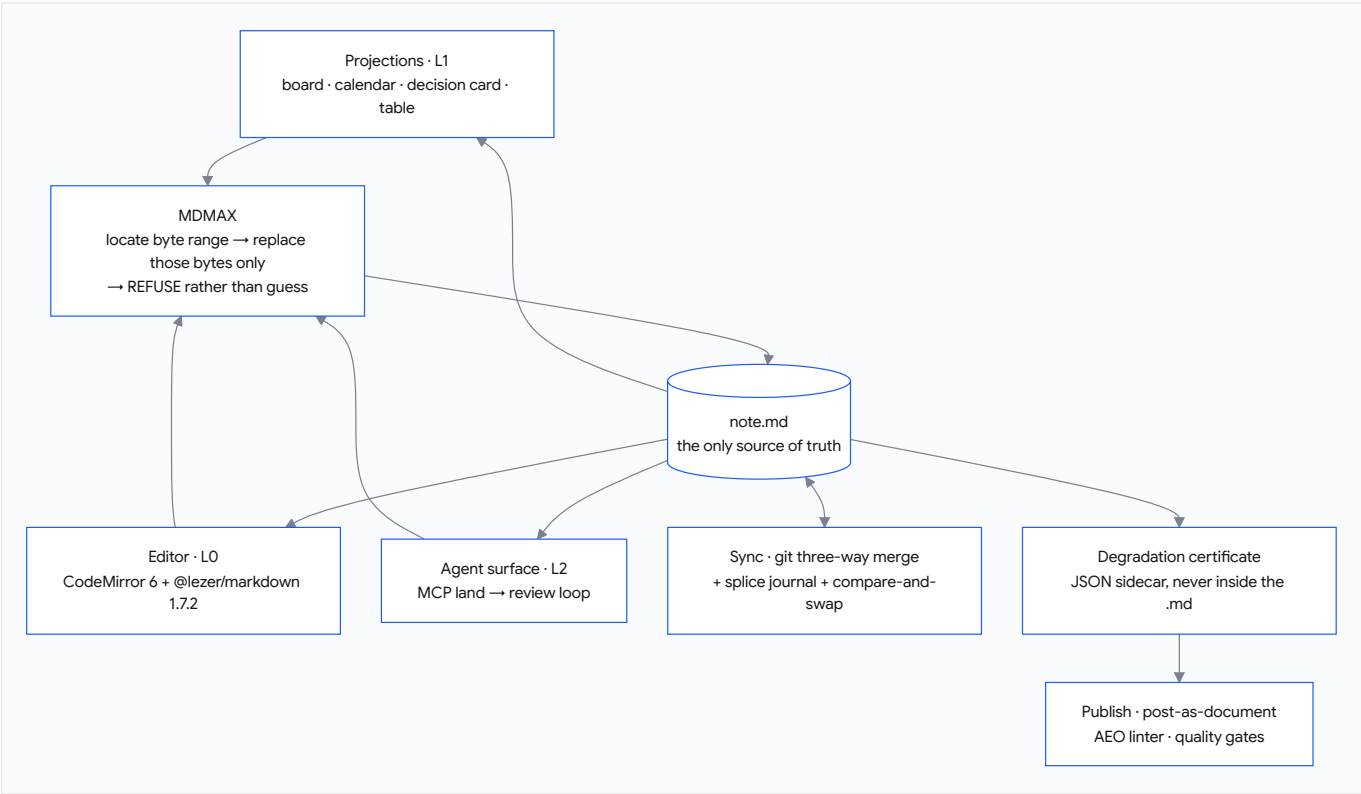
TERM	MEANING
Splice	Replace exactly the target byte range; never regenerate from a parse tree
Refusal	A first-class outcome: return the input unchanged and say why
Degradation certificate	Measured proof of how a file renders across real markdown engines
Projection	A deterministic, disposable view computed from the file, owning no state
Carrier	How a profile is written on disk — a callout for prose, a fence for opaque data
Profile	A named render + schema pair
Machine-write zone	A fenced region an agent may rewrite and outside which it may not
Evidence tier	{value, source, tier, re_verify_cmd} on a claim
Hunk	One reviewable change — human, AI, or sync conflict
land()	The single capture verb any agent calls to make a chat output durable
The eval lane	Client-side arbitrary code execution. The lane we refuse
NF-1...NF-4	The four foreign-corpus engine defects (§28.1)
AIOS	The internal markdown-native orchestrator this studio runs on
MDMAX	The engine: splice writer, OffsetMap, certificate, construct detectors
MoR	Merchant of record — absorbs sales tax, never data protection
Red proof	A test demonstrated to fail against the unfixed code before it is trusted

Prepared for the frontmatter build team. Every number is tagged. Nothing tagged **ss** may be published as fact without being opened first. Re-derive before you quote — see §57.

§61 The system as one thing

Every preceding section describes a layer. This one describes how they compose: one diagram, one contract table, three traces, the seams between them, and what each layer is load-bearing for.

61.1 The whole system on one page



Three properties of the drawing carry the whole argument. Every arrow into the file passes through one node, so there is exactly one write grammar for a human keystroke, an agent proposal, a card drag and a merge result [inference, from §6.1 principle 6]. Every arrow out of the file is a read, so no box above the file may hold state. The certificate hangs off the file and feeds publish — it is a fact computed *about* the bytes, never a fact stored *in* them (`cert-contract.ts` : "the artifact is a JSON sidecar, never written into the `.md` ") [measured, §7.1].

61.2 The layer contract

LAYER	OWNS	MUST NOT	TALKS TO	FAILS BY
The file — <code>.md</code> in a git repo the user owns	Every byte. All state. History, via git.	Carry a block id, a certificate, a cache, or any token we invented [§6.2, §8.3]	git; MDMAX	Being rewritten whole — normalize-on-save is the single failure that ends the product's distinguishing property [§8.1 F4]
MDMAX engine — 13 files, 3,614 lines, 10 tests [measured, §7.1]	Byte ranges, refusal, strict decode, shape gate, placement, equivalence fold, certificate computation	Import anything from <code>src/modules/*</code> ; write files; throw	Bytes in, bytes-or-refusal out. Nothing else.	Guessing a range instead of refusing. A wrong splice is worse than no splice.
L0 editor — <code>src/modules/editor</code>	The caret, undo, incremental parse, the <code>U16Offset ↔ ByteOffset</code> map	Reserialize the document; normalize on save; assume UTF-16 units equal bytes	MDMAX (write gate), the file (read)	Offset drift. mdast reports root end 36 where UTF-8 length is 41 on the same string [measured, §9]
L1 projections — board, calendar, decision card, table	Layout, grouping, the gesture	Hold state; write by any path other than a splice; promote L2 inline fields into L1 frontmatter [§8.5]	The file (read), MDMAX (write)	A cache that disagrees with the file and is believed. If they disagree, the file wins and the cache is wrong [§8.5]
L2 AI protocol — <code>src/modules/ai</code> , <code>src/modules/ai-tools</code> [measured, directory listing]	Verbs, proposals, provenance, <code>land()</code> , the review loop, local instrumentation	Write without a review step; index the vault; execute model-authored code [§6.2, §11.4]	MDMAX via <code>land()</code> ; the file read-only	Landing against a stale <code>baseSha</code> , or an ambient suggestion nobody asked for [§11.4]
L3 markdown-OS — schema profiles, evidence tiers, machine-write zones	<code>fm.profile</code> , <code>fm.version</code> , <code>fm.projections</code> , <code>fm.cert</code> ; type coercion rules	Invent a bare top-level key or a new block-level token [§8.4, §8.3]	Frontmatter prepass; projections	Writing a key it did not create, making our edits indistinguishable from the user's
L4 publish — post-as-document, AEO linter, gates	Rendered output, the published page	Mutate the source file; publish a construct with a DESTROY verdict	The certificate; the file (read)	Shipping a construct the certificate never cleared
Sync — git merge + journal + CAS	Server-issued revisions, <code>base_bytes</code> , the append-only splice journal, conflict artifacts	Merge on the server; use <code>git merge-file --union</code> ; use fuzzy patch in the write path; write conflict markers into the <code>.md</code> [§31.1, §31.2]	MDMAX (<code>fold</code>), the file, R2 + Durable Object	Displaying "Fully synced" without a server-ACKed digest match — the UI that lied [§31.2]

Anti-recommendation for the whole table: do NOT let a layer acquire a second responsibility because it is convenient — the board reading `status:` is a projection, the board *caching* `status:` is a second source of truth, and the difference is one line of code and the entire thesis.

61.3 Trace A — one byte through a human edit

The user types one character into the body of an open note. Status column: LIVE = shipped today; SEAM n = the ordered wiring plan of §7.3.

#	HOP	MODULE / FILE	ASSERTED AT THE HOP	STATUS
1	Read from disk	<code>get-snapshot.ts</code> → <code>decodeStrict</code> (<code>mdmax/domain/shape-gate.ts</code>)	Strict UTF-8. Refuse, never repair [measured, §7.1]	LIVE — the only MDMAX symbol reachable from product code today, one of thirteen files [measured, §7.3]
2	Parse for the editor	<code>@lezer/markdown</code> 1.7.2 inside <code>CodeMirror</code> 6, <code>src/modules/editor</code>	Exact offsets on inline marks + incremental repare; 3.58× on a 56 KB doc [measured, §9]	LIVE
3	Keystroke	<code>CodeMirror</code> <code>ChangeSet</code> at a UTF-16 offset	The caret position is a <code>U16Offset</code> , not a byte offset	LIVE
4	Address translation	<code>mdmax/domain/offsets.ts</code> — <code>OffsetMap</code> , branded <code>U16Offset</code> / <code>ByteOffset</code> / <code>GraphemeIndex</code>	Only 67 of 1,080 corpus files have bytes == UTF-16 units; 93.8% diverge; 103 of 2,314 contain non-BMP [measured, §7.1]	SEAM 2
5	Shape gate	<code>mdmax/domain/shape-gate.ts</code>	<code>MAX_BYTES</code> 4MB, <code>MAX_LINES</code> 200,000; refuse over quadratic paths (<code>WIKILINK_RE</code> <code>k</code> =1.98, 36,865 ms on 320 KB) [measured, §7.1]	SEAM 2
6	Placement check	<code>mdmax/domain/placement.ts</code>	The setext defect (<code>"Heading\n---"</code> → <code>h2</code>) is refused, not repaired — blank-line isolation does not fix it [measured, §7.1]	SEAM 2
7	Splice	replaces the single-key scanner in <code>share/domain/splice-frontmatter.ts</code>	Exactly the located bytes are replaced; every other byte bit-identical	SEAM 3
8	Commit	<code>src/modules/repository/application/commit-changes.ts</code>	A refusal is a 4xx with a reason, never a silent pass [§7.3]	LIVE (write gate not yet enforced)
9	Journal	<code>{seq, base_digest, offset, deleted_len, inserted_bytes, result_digest}</code> [§31.2]	<code>fold(journal, base_bytes) == working_bytes</code>	SEAM 3
10	Re-project	All L1 views recompute from the new bytes	Views store nothing; nothing to invalidate	Partial
11	Recertify	<code>mdmax/application/certify.ts</code> → JSON sidecar in R2	<code>certify.ts</code> never throws; the artifact never enters the <code>.md</code> [measured, §7.1]	SEAM 4

61.4 Trace B — one byte through an agent edit via `land()`

#	HOP	MODULE / FILE	ASSERTED AT THE HOP	STATUS
1	Agent reads	MCP surface, <code>src/modules/ai-tools</code> [measured, directory exists; no <code>land</code> handler file exists under <code>src/</code> today, measured by <code>find</code>]	The agent receives bytes plus a <code>baseSha</code> . Never a tree, never an AST	PLANNED
2	Model proposes	<code>src/modules/ai</code> — a §11.2 rank-1 transformation verb on a selection	The output <i>is</i> a byte range. The model gets a data slot, never a canvas [§6.1 principle 5]	PLANNED
3	<code>land()</code> arrives	<code>{path, baseSha, offset, deleted_len, inserted_bytes}</code>	Same envelope shape as the sync journal record — one grammar for every change [§6.1 principle 6]	PLANNED
4	Compare-and-swap	<code>baseSha</code> vs the file's current digest	Mismatch → refuse whole. No partial write, no rebase-and-hope	PLANNED
5	Address, gate, place	<code>mdmax/domain/offsets.ts</code> → <code>shape-gate.ts</code> → <code>placement.ts</code>	Identical to Trace A steps 4–6. The agent gets no shortcut a human does not get	SEAM 2
6	Review loop	Proposed diff rendered over the byte range	Nothing is written yet. Users are filing "Improve Agent Mode review and consent controls" against the market leader — asking a competitor for this [measured, §11.2]	PLANNED
7	Human accepts	<code>repository/application/commit-changes.ts</code>	Provenance recorded with the splice; AI may propose, the human commits [§11.4]	LIVE (call site exists)
8	Human rejects	—	Input returned unchanged. Zero bytes touched. Refusal is a first-class outcome [§6.1 principle 2]	PLANNED
9	Instrument	Local store only	Strong Acceptance per verb — accepted only if <50% of the proposal was edited. Ansible Lightspeed's 49.08% is a ceiling for a constrained verb, not a general rate [fetched + inference, §11.5]. Publish nothing	PLANNED

61.5 Trace C — one byte through a kanban drag

#	HOP	MODULE / FILE	ASSERTED AT THE HOP	STATUS
1	Board reads	<code>mdmax/domain/frontmatter-prepass.ts</code>	Never writes, never throws. 170 of 907 home frontmatter blocks are invalid YAML = 18.74% [measured, §7.1]	PLANNED — no <code>kanban</code> or <code>board</code> file exists under <code>src/</code> today [measured, <code>find</code>]
2	Invalid YAML	—	The card renders read-only and the drag is refused. Never repair a block we did not write [§8.4]	PLANNED
3	Drag gesture	Board projection	The card moves in the DOM and nothing is persisted. The board owns no state	PLANNED
4	Locate	<code>share/domain/splice-frontmatter.ts</code> today; MDMAX splice engine at SEAM 3	The target is the <i>value bytes</i> of the <code>status:</code> key — not the key, not the line, not the block	SEAM 3
5	Duplicate-key check	Frontmatter contract, §8.4	Two <code>status:</code> keys → REFUSE. YAML 1.1 parsers disagree on last-wins vs error, so any choice we make is wrong somewhere	SEAM 3
6	Preserve	Splice writer	Key order, comments, blank lines, quoting style and indentation survive. No load-then-dump YAML round trip, even a "round-trip-safe" one [§8.4]	SEAM 3
7	Container indent	<code>mdmax/domain/constructs.ts</code> (19 constructs, UTF-8 byte ranges)	The splice computes container indent from the byte range, never assumes column 0 [§8.2]	SEAM 3
8	Commit + journal	<code>repository/application/commit-changes.ts</code> → splice journal	Same two hops as Traces A and B	LIVE / SEAM 3
9	Board re-reads	Board projection	If the board and the file disagree, the file wins and the board is wrong [§8.5]	PLANNED

The three traces converge at step 4–7 of Trace A. That convergence is the composition claim: a keystroke, an agent proposal and a card drag are the same operation with three different origins, which is why there is one place to make the guarantee and one place it can fail.

61.6 The seams, and what is asserted at each

#	SEAM	HANDOFF	ASSERTED	ON FAILURE	STATE
1	Disk → engine	Ingress gate	Valid UTF-8 via <code>decodeStrict</code> ; within <code>MAX_BYTES 4MB</code> / <code>MAX_LINES 200,000</code>	Refuse the file; never repair	LIVE — <code>get-snapshot.ts</code> , <code>search-index.ts</code> [measured, §7.3]
2	Engine → editor	Address translation	Byte offsets round-trip through <code>OffsetMap</code> ; branded types make a mixed-unit call a compile error	Type error at build, not corruption at runtime	Blocked by NF-1, NF-3
3	Any writer → engine	Write gate	Every write passes shape-gate + placement before a blob is created; a refusal is a 4xx with a reason	4xx with the byte offset surfaced	Do not wire before NF-1 and NF-3. At the measured refusal rate this gate rejects 83% of foreign vaults' publishes — an availability incident wearing a correctness costume [§7.3]
4	Engine → file	Splice	Exactly the located bytes are replaced; <code>result_digest</code> recorded	Refuse and return the input unchanged	SEAM 3 of §7.3
5	File → sync	Journal fold	<code>fold(journal, base_bytes) == working_bytes</code> must hold at every boundary. The journal is a checkable derivative, never authority [§31.2]	Refuse to sync; fall back to whole-file conflict copy	Planned
6	Sync → file	Merge	Three-way merge against the <i>stored true base</i> , never a diff against the winner. CAS on upload	Conflict → write <code>note (conflict <device> <server-rev>).md</code> holding your bytes untouched. Never conflict markers in the user's <code>.md</code> [§31.2, M9 measured]	Planned
7	File → certificate	Certification	<code>certify.ts</code> never throws; 4 verdicts PASS / STRIP / CORRUPT / VOID, 3 classes LEAK / DESTROY / MUTATE; artifact is a JSON sidcar	Certificate marks the construct unprobeable — <code>uncertifiableShare()</code> = 8 of 15 (53.3%) not locally probeable [measured, §15.1]	SEAM 4 of §7.3
8	Certificate → publish	Publish gate	No construct carrying a DESTROY class may publish	Refuse the publish and name the construct	Planned
9	Engine ↔ everything	Boundary rule	MDMAX exports pure functions over bytes and imports nothing from <code>src/modules/*</code> , enforced by <code>eslint-plugin-boundaries</code> plus an arch report with a minimum-files-scanned floor so the gate cannot pass by going blind	Arch report <code>"violations" non-empty; npm run verify red</code>	LIVE [measured, §7.3]

Anti-recommendation on seams: do NOT enforce a seam before the layer below it can satisfy it. Seam 3 is correct, ready, and would today produce a 4xx on 83% of imported vaults — correctness enforced ahead of capability reads to a user as a broken product, and the fix is NF-1 and NF-3, not a weaker gate.

61.7 What breaks if a layer is removed

REMOVE	WHAT STILL WORKS	WHAT BREAKS	WHAT THE PRODUCT BECOMES
L2 AI	Everything. Delete every model key and the file, the board, the calendar and the site work exactly as they did — those were never AI features [§11.6]	The verbs, <code>land()</code> , the review loop	A file-native editor with a degradation certificate. Still sellable
L1 projections	File, editor, engine, sync, publish	The board, calendar, decision card, table	A careful markdown editor. This is the honest v1 fallback, not a catastrophe
Sync	Every single-device path; no data is at risk	Multi-device, conflict artifacts, the journal oracle	A local editor. The least fatal removal
L3 profiles	Reads and writes; projections fall back to conventions	Typed data, <code>fm.*</code> namespacing, evidence tiers	Dataview with better write safety
The certificate	Every edit path	Publishing becomes a guess; the degradation claim becomes unfalsifiable; <code>verdict.ts</code> 's findings (<code>![[Some Note]]</code>) → VOID on marked 16.4.2 and commonmark 0.31.2; 21 of 24 <code>{#id}</code> leak; front matter LEAKs in 23 of 24 bench configurations) go unshipped [measured, §7.1]	A nicer editor carrying a marketing claim it cannot prove
The engine	Every layer still renders; every write becomes a regenerate-from-parse-tree	Byte preservation, refusal, the certificate, the 8,513-file / 0-corruption / 0-throws result, and the reason any of the above is trustworthy	A competitor. This is the only removal that is fatal rather than reductive
The file (adopt a tree-of-record or a CRDT document)	Nothing, in the sense that matters	The projection law inverts: the CRDT or the tree becomes the record and the <code>.md</code> becomes a projection of it [§31.1 D1]	Notion

The ordering above is also the ordering of what may be cut under schedule pressure: AI first, projections second, sync third; the engine and the file never.

61.8 The system in one sentence

frontmatter is one markdown file on your disk, an engine that changes only the exact bytes you pointed at or refuses to change anything, and a set of disposable lenses — a board, a calendar, an agent verb, a published page — that all read those bytes and all write back through that one engine, so nothing you did not select can be rewritten by anyone, including the AI.

The twenty-second version drops the lenses: *the file is the truth, the engine only ever replaces bytes you named, and everything else is a view.*

What would falsify the composition: any shipped path that writes to the file without passing seams 3 and 4 — a projection that persists its own state, an AI verb that lands without a review step, or a sync merge that resolves rather than refuses. Each of those is a single pull request away at any time, which is why the boundary rule in seam 9 is a lint gate and not a convention [inference].

§62 AIOS inside the product — orchestration as a user feature

QUESTION	VERDICT	FALSIFIED BY
Does a user get their own orchestration layer?	Yes — but they never learn that word. They get house rules over a folder, made of files they can read, edit and delete	A user in a support ticket or a forum post using our internal vocabulary unprompted, at a rate above 20% of orchestration-related tickets
Is it visible on day one?	No. Rung 0 is an empty surface. Every rung above it is unlocked by an act the user performed, never by a calendar	A release cohort shipped with the full surface visible at first run showing day-7 return ≥ the trigger-gated cohort
Where is the creepy line?	At disclosure, not at observation. The software may hold what it saw; it may not volunteer it	A measured demand signal for unprompted assistance that survives the ambient-cost line of \$38.79/user/month [derived, §11.3]

Section 15 answered which internal assets become features. This section governs what the resulting layer *is* from the user’s chair, and where it must stop.

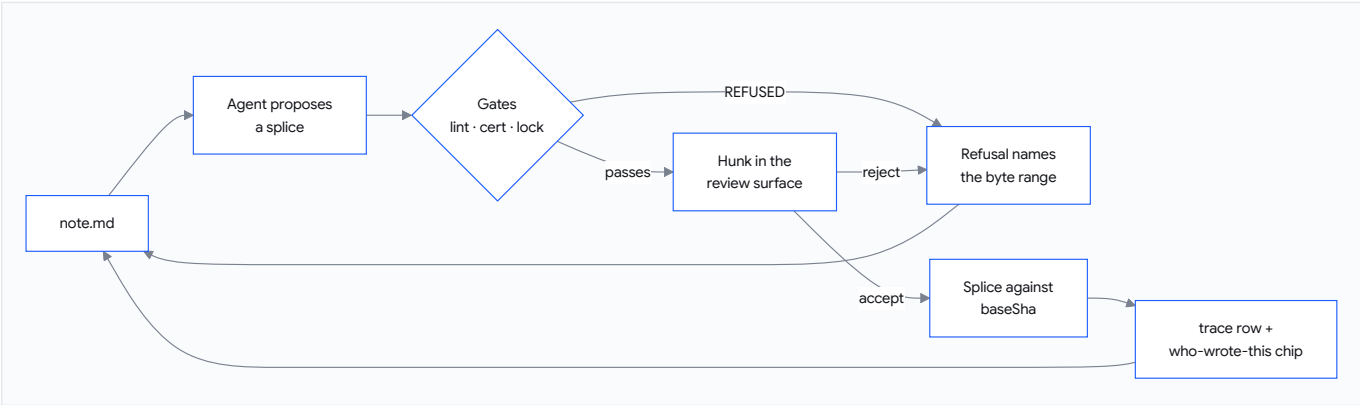
62.1 The user-facing model

An orchestration layer, in the user’s vocabulary, is the set of house rules their folder enforces on anything that edits it — a person, an agent, or a CI job — and every one of those rules is a file in the folder.

OUR WORD	INTERNAL ARTIFACT, LIVE [MEASURED]	THE USER’S WORD	WHERE IT LIVES	WHAT THEY ACTUALLY SEE
Skill / SKILL.md	124 SKILL.md across 27 dirs	an automation	.frontmatter/automations/<name>/SKILL.md	a document, opened and read like any other
Constitution / learned rules	892 lines, 74 rules, 69/69 uncited	house rules	.frontmatter/rules/*.md	a document they wrote, edit, and delete
Trace ledger	5,018 rows, 30 keys, files_sha256	who wrote this	.frontmatter/trace.jsonl sidecar	a hover chip on a paragraph
Complexity gate	24,669 rows, 95.1194% floor	what this costs	corner meter	a currency figure, never a tier word
Escalation ladder	7 deterministic rungs	try harder	button on any AI result	cost delta shown before it runs
Assert/break gate pair	69 scripts = 33 + 34 + 2	a check	mdmax cert output	pass/fail with the failing line linked
Freeze / guard skills	3 skills	locked section	.frontmatter/ lock	a padlock in the gutter
Eval, judge, calibration, bandit, propensity	26,037 routing-journal rows	no user word exists	—	nothing, ever

- **Banned from every user-facing surface:** orchestrator, agent loop, skill, bandit, arm, tier, eval, judge, kappa, propensity, trace, telemetry, self-improving, "learns you", health score, confidence. Each is either a claim we cannot defend (§15.4) or a concept charged against the §18 C counter for no user benefit.
- **The naming test, applied to every candidate:** if the shortest honest label for a surface is a machine-learning term, the surface is internal. If it is a noun the user already owns — a rule, a check, a cost, a lock, a note — it may ship.
- **Anti-recommendation:** do not ship a settings page called Automations, Agents, or AI. The house rules live in the folder as documents; a settings page duplicates them and creates a second source of truth, which the projection law forbids.

62.2 The user loop



STATION	WHO ACTS	DETERMINISTIC?	WHAT THE USER SEES ON FAILURE	GROUNDING
Propose	agent	no	nothing — a proposal that never reaches a gate is discarded silently	§11.2 rank 3
Gate	software	yes, zero model calls	the byte range, the named check, and the reason — not a dialog	apply-patches.py four-verdict vocabulary [measured]
Review	human, per hunk	n/a	the hunk stays Open; the file is untouched	§13; per-hunk accept is the most-demanded feature in AI editors [fetched]
Splice	software	yes	REFUSED_CONFLICT on baseSha drift	§12 read-before-patch
Trace	software	yes	the chip reads <i>unattributed</i> rather than guessing	78.637% of internal trace rows carry skill: "unknown" [derived] — guessing attribution is the documented failure
Next action	human	n/a	—	—

- The loop has exactly **one** human gate and it is non-skippable for any agent-authored byte. **58.7%** of ~33,000 developers do not plan to use AI for committing and reviewing, and **75.8%** decline it for deployment [fetched, Stack Overflow 2025].

- Refusal is a first-class terminal state, not an error path. The engine's law is *locate the byte range, replace only those bytes, REFUSE rather than guess* — the loop inherits it verbatim.
- Anti-recommendation: do not add an "apply all" that bypasses the review station. Cursor and Windsurf both regressed per-hunk control and both got publicly burned [§13].

62.3 Scope — per user, per workspace, ours only

The scope rule is one sentence: state describing the *file* goes to the workspace, state describing the *person* stays on their machine, and state describing the *software's own intelligence* never leaves ours.

#	LOOP	SCOPE	STATE LIVES IN	WHY THIS SCOPE
1	Edit-survival log (accept / reject / edit-distance-after-accept)	PER USER	<code>.frontmatter/state.json</code> , local, never transmitted	It is a fact about a person's taste. 96.121% of our own preference rows carry no attribution [derived] — sharing it would share noise
2	Cost meter and spend	PER USER	local	Bundled inference is billed per person
3	Escalation history ("try harder" rungs used)	PER USER	local	An act, not a rule
4	Session continuity cards	PER USER	local snapshot dir	Contains the working context of one head
5	Automation listing budget	PER USER	local	The budget is 1% of the model context window [fetched] — a per-call, therefore per-user, constraint
6	Disclosure rung reached	PER USER	local	UI state, never a document
7	Automations (<code>SKILL.md</code> + optional <code>scripts/</code>)	PER WORKSPACE	repo, reviewed as a PR	They edit shared files; §37 role model governs who may change them
8	House rules / memory file	PER WORKSPACE	repo	The honest answer to "what does it remember" is a file in your repo you can delete
9	Frontmatter schema contract (<code>fm lint -schema</code>)	PER WORKSPACE	repo	A contract with no second party is decoration
10	Section locks (freeze)	PER WORKSPACE	repo	A lock only one person can see is not a lock
11	Doc Health deterministic checks	PER WORKSPACE	computed from the repo, cached nowhere	Broken anchors are facts about files
12	<code>mdmax cert config</code> and <code>--fail-on=BROKEN</code>	PER WORKSPACE	repo + CI	The build gate is the team's
13	Decision/incident blocks with an executable <code>broke:</code> field	PER WORKSPACE	repo	40/40 internal gates carry <code>proven_nonvacuous</code> [measured]; the proof is the artifact
14	Template and rule versioning with preview	PER WORKSPACE	repo	A prompt change silently degrades every future document made from it
15	Who-wrote-this sidecar	PER WORKSPACE	<code>.frontmatter/trace.jsonl</code> , committed	Travels with the document or it is not provenance
16	Routing bandit + propensity (26,037 rows)	OURS ONLY	founder machine	Largest arm is named <code>__unattributed__</code> ; 3/10 arms are <code>offline-0.5x</code> seeds
17	Complexity-gate verdict internals	OURS ONLY	founder machine	The user sees the price; the tier word invites an argument we cannot win
18	Eval judges + calibration	OURS ONLY	founder machine	Top bucket corrects at 0.269 against a 0.20 bar [derived]
19	Debate / consensus / reflexion / best-of-N	OURS ONLY	founder machine	Multiplies COGS per user action; LR#16 holds convergence ≠ correctness
20	<code>break-* .sh</code> execution (34 scripts)	OURS ONLY	founder machine	A user-triggered "break my document" action is a data-loss vector
21	Exploration routing	OURS ONLY	founder machine	Routing a paying user's document to a non-default model for counterfactual signal is indefensible
22	<code>skill-health.json</code> metric	OURS ONLY	founder machine	Reports active 0 of 131 while 446 trace rows contradict it in the same window
23	Digression guard	OURS ONLY	founder machine	0 contract files and 0 trace rows — it has never fired here
24	Raw <code>traces/</code> store	OURS ONLY	founder machine	30 keys including <code>cwd</code> — absolute paths into client repos; zero tenancy fields
25	Learned-rules ledger about the user	OURS ONLY	founder machine	"The AI keeps a private file of rules about you" has no demand signal anywhere in the record

- Split: **6 per user (24.0%) · 9 per workspace (36.0%) · 10 ours only (40.0%)** [derived: 6/25, 9/25, 10/25].
- Nothing in the per-user column is ever transmitted, aggregated, or used to train anything. Under the §17 instrumentation contract there is no event stream and no identifier to attach it to.
- Anti-recommendation: do not add a server-side store for per-user state to make it follow the user across machines. That converts a local file into a database, and the first question a compliance buyer asks — *scoped to which org* — has no answer, because the schema has **zero** tenancy fields today.

62.4 The progressive-disclosure ladder

RUNG	WHAT APPEARS	UNLOCK – AN ACT, NEVER A TIMER	AT-REST CONTROLS ADDED (V0)	GROUNDING
R0	Nothing. Editor, projections, and AI verbs on a selection	first run	0	§17: no tour, no "What's New" modal, no consent prompt [fetched, NN/g]
R1	Who-wrote-this chip, shown inline once , then hover-only	the first agent-authored hunk is accepted	0 (hover affordance)	§17 step 6 precedent: show the byte diff once, then never unprompted
R2	Cost meter in currency	cumulative AI spend crosses the first displayable unit	1	§15.2: 95.1194% floor-tier is the COGS story stated as a user benefit
R3	"Try harder" on a result	the user rejects a proposal twice on the same selection	1	Escalation as an explicit user act; zero learning claim
R4	Doc Health panel	≥1 deterministic finding exists in the opened folder	1	Zero findings ⇒ no panel, ever. A panel that opens empty is the damaging pattern
R5	Locked sections	a second collaborator gains write access	1	A lock with no second party is decoration
R6	Frontmatter schema contract + <code>fm lint --schema</code>	a repo is connected, or ≥2 people commit	1	§37: contracts are workspace-scoped
R7	Automations — author, dry-run, hand to one person	the user performs the same multi-step transform 3 times, offered as a pre-filled draft they must save	2	The offer is a one-line affordance in the transform's own result, never a modal
R8	<code>mdmax cert</code> in CI, gates, <code>broke:</code> blocks, tamper-evident export	the user opens the CLI or the Actions tab	0 in the editor	B2B surface; never rendered in the writing view

- **Week two shows nothing by virtue of being week two.** Every unlock is an act; the median user reaches R1–R2 in that window because that is when the acts happen, and a user who never triggers one never sees the surface.
- The power-user ceiling: author automations, write house rules, lock sections, declare a schema, wire `cert` into CI, export the tamper-evident AI-edit record. The floor beneath every rung: **no rung ever exposes a model, a tier, a score, or a claim about learning.**
- Automations are hard-capped per workspace with a visible budget meter. **17/124** internal descriptions already exceed the 1024-char cap and `sgnk-mobbin` at **1,885** chars is *visibly truncated in this session's own listing* [measured] — growth silently disables older automations, and the user cannot see it happen.
- Anti-recommendation: no timer-based reveal, no "you've been here two weeks" card, no feature-discovery nudge. §17 already bans streaks, badges, digests and "you haven't opened X in N days"; a disclosure timer is the same mechanism with a friendlier name.

62.5 The trust boundary

MAY OBSERVE, ALWAYS, WITHOUT ASKING	MAY ACT UNASKED (DETERMINISTIC, ZERO MODEL CALLS)	ALWAYS NEEDS A HUMAN, PER OPERATION
Bytes of the open file	Compute a projection	Land any agent-authored splice
Filesystem shape of the opened folder — names, counts, mtimes	Run deterministic lint / health checks and mark findings	Publish or unpublish
Files the user explicitly named in this action	Autosave bytes the user typed	Anything outward: commit to a remote, deploy, send, post
Whether an AI edit was accepted, rejected, or edited after accept	Refuse, and name the byte range	Grant an automation any of <code>fs-write</code> · <code>net</code> · <code>deploy</code> · <code>db</code> · <code>outbound</code>
Its own cost, latency, and refusals	Snapshot session continuity locally	Re-consent on any <code>allowed-tools</code> change — diff the grant, not the prose
—	Compute a degradation certificate on request	Run an imported automation for the first N runs
—	—	Delete anything
—	—	Change house rules other people's documents depend on
MAY NEVER OBSERVE	WHY	
Files outside the opened folder	The folder is the consent boundary	
A background semantic index of the corpus	Already refused in §11.4: a second source of truth, \$38.79/user/month at 100 saves/day [derived], and the market leader's most-discussed open issues are all silent index failure	
Clipboard, keystrokes outside the editor, other applications	No mechanism, no exception, no setting	
Prompt or document content leaving the machine except to complete an action the user just took	Nothing runs in the background; nothing is indexed behind your back (§11.6)	

The creepy line sits at disclosure, not at observation: the software may hold what it saw and may never volunteer it. Observing that a user rejected three summaries is legitimate and local; saying *"I notice you keep rejecting my summaries"* is the ambient-coach pattern the record already bans — internally, **5 of 5** `UserPromptSubmit` hooks are interruption hooks, and **5 of 28** hooks map to editor lifecycle events [derived: 5/28 = 17.857%]. The test for any new surface: if a colleague reading over your shoulder would be unwelcome saying it aloud, the software may compute it but may not say it.

- The tool-grant checkbox is **not** the boundary. `allowed-tools` is documented as experimental with support varying between implementations, and Claude Code clears the grant on the user's next message [fetched] — the UI must say *for this turn*, and the host permission system is the real gate.
- The invocation-lock ratio is the design pattern, not an accident: **11 of 124** internal skills set `disable-model-invocation: true` and **every one is destructive or outward** [measured; 8.871%]. Outward or destructive ⇒ never auto-fires, in the product as on the machine.
- Imported automation text is **data, never instruction**. Body text asserting authority, urgency, or pre-authorisation is the primary attack surface the moment sharing exists; **0 of 124** local files use the `!`cmd`` shell-injection syntax [measured], so refusing it in imported documents costs this corpus nothing.

62.6 The anti-section — orchestration capabilities that would harm users

Items 1–8 are specific to a user owning an orchestration layer; §15.4 governs the internal-asset list they build on.

CAPABILITY	LIVE STATE [MEASURED]	THE HARM	WHAT SHIPS INSTEAD
An uncapped personal automation library	Listing budget = 1% of the model context window; description + when-clause truncated at 1,536 chars; least-invoked lose their descriptions first [fetched]	The user's 15th automation silently disables their 3rd, with no error and no way to see it	A hard per-workspace cap with a visible budget meter and an explicit <i>retire one to add one</i> prompt
Auto-authored automations ("we noticed you do this, so we made one")	The nudge/suggest hooks this pattern comes from are 5 of 5 interruption hooks	It creates a file the user never read that carries a tool grant — the two properties that must never coexist	A one-line offer inside the transform's own result that opens a pre-filled draft the user must save
A user-facing automation health dashboard	<code>skill-health.json</code> : active 0, dormant 5, dead 116, infrastructure 10 of 131 — contradicted by 446 trace rows in the same window	Telling a paying customer that 116 of the 131 things they built are dead, on a metric another ledger in the same system refutes	A last-run timestamp per automation. No verdict word. Dormant is not dead
A cross-document background agent	Ambient bucket is the highest-maintenance in the category at 9.05% issues-per-star vs Copilot's 1.30%; ambient re-rank costs \$38.79/user/month [derived]	Unfundable at a bundled-inference consumer price, and its failure mode is silent	Scoped multi-document synthesis over an explicit N files the user named (§11.2 rank 6)
A self-amending rules file	892 lines, 74 rules, 69/69 carrying zero citations; the hygiene report is 5 rules stale	A source of truth the user did not author, inside a product whose whole law is that the file is the only source of truth	Hygiene as a review prompt ; a human is the only writer. Never an auto-prune
Standing tool grants	Grant is cleared on the user's next message; the key is experimental [fetched]	Users read a checkbox as a permanent capability boundary; it is not one	Default-deny, per-turn language in the UI, host permission system as the boundary
An approval queue	41.311% of gate decisions are <code>rule2_gated</code> — internally a human approves roughly 4 in 10 operations	A queue in an editor trains click-through, which converts a safety mechanism into a formality	Gate at the write: the refusal arrives at the byte range, in the document , with the check named
Orchestration vocabulary in the UI at all	20 distinct frontmatter keys locally against 6 in the spec; 54/124 = 43.5% would hard-fail packaging today	Every internal word is a concept charged against §18's C counter, and several are claims we cannot defend	The seven user nouns in 62.1 and nothing else
Multi-agent debate on a user's document	<code>~/sgnk/insights/</code> does not exist after 14 months	LR#16: convergence is not correctness, so a converged panel authorises nothing — the user pays for deliberation carrying no authority	Single proposal, human gate
Any "it learns you" claim	<code>shadow-log.jsonl</code> 1 row; <code>prereg.jsonl</code> 2 rows; <code>precision[rejected] = 0/8 = 0.000</code>	Every rejection this system has ever emitted was wrong; acting on that signal is worse than acting on nothing	The 7-rung deterministic ladder as a button , cost delta first

- The single structural constraint behind all ten: **machine-fed stores all exceed 1,000 rows and every human-fed store is under 250** — roughly 176,000 machine rows against 232 preference rows [derived]. Build nothing whose value depends on the user rating something.
- Anti-recommendation to this anti-section: do not read it as "ship less AI". The deterministic-projection cohort out-installs every AI capability combined by 7.25× [derived: 7,289,307 ÷ 1,005,651], which is an argument for putting the intelligence in the gate rather than in the suggestion — not for removing it.

62.7 What would falsify this direction

CLAIM THIS SECTION MAKES	FALSIFIER	HOW WE WOULD KNOW	WHAT WE DO THEN
Users want house rules over a folder at all	Automations authored by <5% of week-4-retained users after 6 months and zero inbound requests for the capability	Generator marker + docs page-hit ratios (§17 implicit telemetry) plus the support queue	Cut the authoring surface; keep only the gates we author and ship <code>mdmax_cert</code> alone
Trigger-gated disclosure beats a visible surface	A release cohort shipped with R1–R4 visible at first run returns on day 7 at ≥ the trigger-gated cohort	Release experiment, one element changed, cohort comparison — never a user-level experiment	Reveal earlier, rung by rung, re-testing each
"Automation" is the user's word	Users hand-writing <code>SKILL.md</code> outside the editor, or internal vocabulary appearing unprompted in >20% of orchestration tickets	Support queue, verbatim	Adopt their word; the vocabulary table is descriptive, not doctrinal
One non-skippable human gate per agent write	>20% of active users disabling review and asking for auto-apply	Setting toggles are local, so this arrives as tickets, not as data	Auto-apply only inside an explicitly fenced machine-write region ; never a global switch
Per-workspace state belongs in the repo	Teams requesting automations or rules that must not be committed	Sales and support conversations	A per-user scope inside the workspace — never a server-side store
No loop needs to run in the background	A deterministic Doc Health check that cannot complete inside the §29 save-path budget	Performance CI gate	A scheduled local, deterministic, zero-model-call job. Not an agent
Deterministic beats agentic in this category	The Obsidian install ratio inverting — agentic + ambient + generation peak-version sum exceeding 7,289,307	Re-measure <code>obsidianmd/obsidian-releases</code> at each roadmap review	Reweight §11.2 before reweighting this section
The whole direction	The artifact-facing half ships and shows no day-7 return difference against a build with zero orchestration surface	Version-cohort comparison under §17	Delete the layer, keep the editor and the engine. The projection law survives without it

- What does not falsify it:** low usage of a power-user rung on its own. Dormancy is not death (LR#51) — falsification requires the absence of *demand* as well as the absence of *use*, which is why every falsifier above pairs a usage threshold with a demand-signal check.
- What we may not do with a falsifier:** quote it before re-deriving the number at write time. **"90% floor" is already wrong** — live is 95.1194% at 24,669 rows [derived] — and "110 daily trace files" appears in three prior grounding documents while counting 46 lockfiles; the daily ledger is 63.
- Anti-recommendation on the falsifiers themselves: do not convert any of them into a dashboard. Every scalar in this stack is contradicted by at least one other ledger in the same stack, and a falsification bar that becomes a metric becomes a target.

§63 Who this is for — personas and jobs to be done

63.1 The six personas

#	PERSONA	WHO THEY ARE	TOOL TODAY	POPULATION SIGNAL
P1	Devraj, the file-owning solo dev	Backend/infra engineer, 5–15y, runs a personal vault of notes + a blog repo + <code>~/ .cClaude</code> skills. Reads the bytes.	Obsidian + a terminal editor, or plain Neovim + git	Obsidian's own commercial-use licence exists because this persona keeps notes in the same repo as work [SS]
P2	Nina, the dev-tool startup docs owner	4–20 person startup; one person owns docs + changelog + roadmap; PRs review the docs like code	DocuSaurus/Mintify + Linear + Notion (three places, one truth)	—
P3	Kabir, the agency delivery lead	6–25 person studio, 4–9 concurrent clients, hands off a repo at the end of every engagement	Notion per client + Google Docs + a static site generator	—
P4	Sena, the AI-heavy knowledge worker	Runs Claude Code / Cursor daily; her context IS markdown; wants agents to edit the same files she edits	Raw files + an agent CLI, no editor in between	AIOS on the founder machine: 124 SKILL.md automations, 892-line constitution, 5,014 trace rows, 24,539 gate decisions [measured]
P5	Ondrej, the compliance-adjacent technical writer	Regulated-industry doc owner; needs to prove a rendered artifact matches its source	DITA/AsciiDoc toolchain, or Word with a change log	—
A1	ANTI: Priya, the ops-team lead (deliberately not for)	12-person non-technical ops team; wants assignees, notifications, and a shared inbox	Notion / ClickUp / Airtable	—

63.2 Job stories, pain, switch, churn, willingness to pay

#	JOB STORY (WHEN / I WANT TO / SO I CAN)	MOMENT OF PAIN	WHAT MAKES THEM SWITCH	WHAT MAKES THEM LEAVE	WTP	EVIDENCE
P1	When I want a board view of my own repo, I want it without a database, so I can keep every file greppable and diff-able	A tool rewrites his frontmatter on save; the diff is 400 lines for a 3-word edit	A byte-preserving guarantee he can verify himself: 8,513 third-party files, 0 corruption, 0 throws [measured]	Any lock-in surface — a proprietary sidecar, a login before first edit	\$8–15/mo, or a one-time licence; will not pay per-seat	[inference] from the byte-preservation guarantee being the only claim he can independently test
P2	When our docs, changelog and roadmap drift apart, I want one file per item and views generated from it, so I can review docs in the same PR as the code	Roadmap in Notion says shipped; changelog says nothing; site says v1.2	Board and site as projections of the same repo files, reviewable in a PR	Missing multiplayer presence; a teammate who will not use git	\$20–40/seat/mo for 3–8 seats	[inference]
P3	When an engagement ends, I want to hand the client a folder that opens in anything, so I can leave without leaving a subscription behind	Client asks for the Notion export; export is 900 files of <code>Untitled 3.md</code> with broken links	Handoff = a git repo; the published site is a projection, not a second copy	Client-branded publishing weak; no per-client access boundary	\$50–150/mo studio tier	[inference]
P4	When my agent edits my notes, I want it to change only the bytes it means to, so I can let it run unattended	Agent rewrites a whole file to change one field; the git diff is unreviewable	Locate the byte range, replace only those bytes, REFUSE rather than guess — the splice contract itself	Refusals she cannot fix: today 83% publish-refusals from one YAML defect [measured]	\$20–30/mo, highest tolerance of the six	[measured] on the AIOS substrate: 24,539 gate decisions is a person who already pays for determinism
P5	When I ship a rendered doc, I want proof of what degraded on the way, so I can sign off	Reviewer sees a callout; the PDF renderer swallowed it; nobody knows until audit	Cross-engine degradation certification over 7 markdown engines	Certification that stops at 7 engines when the auditor names an 8th	\$100+/seat, slowest cycle	[inference]
A1	When work is assigned to me, I want a notification and a due date, so I can chase my team	Nothing in frontmatter hurts here — the pain is elsewhere	Nothing we will build	Everything: no assignees, no notifications, no shared inbox	High and irrelevant	Settled: not a Notion-style project-management tool

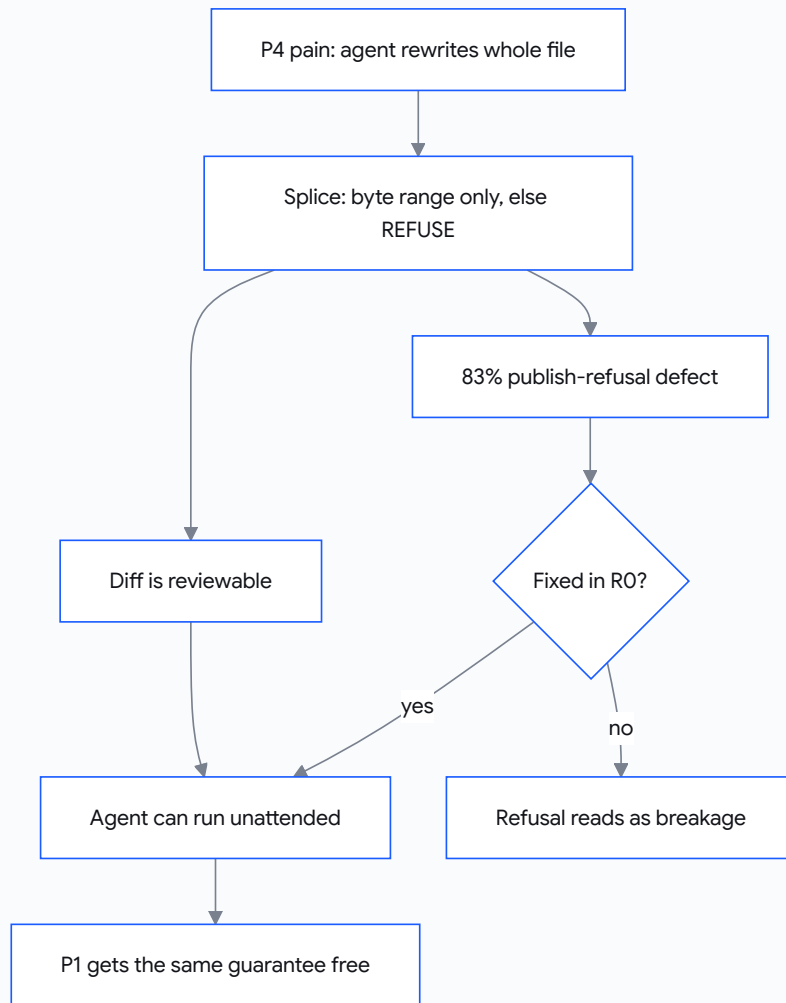
63.3 Persona → section map

PERSONA	SECTIONS THAT SERVE THEM
P1 solo dev	§2 (product thesis), §21, §22, §24, §26
P2 startup docs owner	§2, §21, §22, §24
P3 agency lead	§21, §22, §24, §26
P4 AI knowledge worker	§2, §22, §24, §26
P5 compliance writer	§24, §26
A1 anti-persona	none — appears only here and as an explicit non-goal

Section numbers are taken from the sections named in the reading brief (2, 21, 22, 24, 26) plus C4 onboarding/activation; a full cross-reference must be rebuilt against the PRD's live table of contents before print [inference].

63.4 The one persona to build v1 for

Build v1 for P4, the AI-heavy knowledge worker.



- The engine already built is P4's whole product; every other persona buys a projection layered on top of it [inference].
- P4's substrate is measured, not assumed — 5,014 trace rows and 24,539 complexity-gate decisions on one machine [measured].
- Serving P4 serves P1 for free: the byte-preservation guarantee is identical; only the trigger differs (agent vs human).

The strongest argument against P4: it is a market of one plus a hypothesis — the only measured instance of this persona is the founder's own machine, and building for yourself is the most common way a solo founder builds something nobody else buys.

- Falsifier: if 20 outreach conversations with agent-CLI users produce fewer than 5 who name whole-file rewrites as a real cost, P4 is a projection of the founder, not a segment — switch the v1 target to P2.
- Anti-recommendation: do NOT build v1 for P2 first even though P2 has the clearest budget. P2 needs multiplayer presence and a review workflow before the product is usable at all, which is a second product; P4 needs only what exists.

63.5 Switching costs — what each persona must give up

PERSONA	MUST GIVE UP	COST CLASS	MITIGATION IN SCOPE
P1	Obsidian's plugin ecosystem, graph view, mobile app	High, emotional	None honest — plugin marketplace is settled out. Say so in marketing.
P2	Notion's comment threads and @-mentions; Linear's assignee model	High, organizational	Projections replace views, not conversations. Position as adjacent, not replacing.
P3	Client-facing polish of a Notion share link; non-technical staff access	Very high	Published site projection must match Notion-share polish or P3 never converts
P4	Almost nothing — files stay files; adds a surface she does not have	Near zero	This is the case for \$63.4
P5	An audited toolchain and a validated-tool paper trail	Prohibitive in year 1	Do not chase; the certificate is the wedge for later
A1	Everything she uses the tool for	Total	Not attempted

63.6 Anti-recommendations — attractive traps

TRAP PERSONA	WHY IT LOOKS ATTRACTIVE	WHY IT IS A TRAP	WHAT WOULD CHANGE THE VERDICT
The Notion-refugee team	Loud, large, actively churning	They want assignees and notifications, not projections; serving them re-litigates a settled non-goal	Nothing short of reversing the settled position
The enterprise compliance buyer (P5) early	\$100+/seat, defensible moat via the 7-engine certificate	A 9–18 month sales cycle and a procurement/security review a solo founder in India cannot staff [inference]	An inbound design partner who pays before the review, not after
Non-technical writers "who could learn markdown"	Enormous TAM	Every one of them needs the exact features settled out; they churn on git	Never
Obsidian plugin authors	Distribution for free	No plugin marketplace, no arbitrary client-side code execution — nothing to port to	Nothing; settled
P3 agency before the publish defect is fixed	Highest revenue per logo	83% publish-refusals today from one YAML defect [measured]; the agency's whole job-to-be-done is publishing	R0 lands the zero-indent-sequence fix and the refusal rate is re-measured on the same 8,513-file corpus

§64 Positioning, messaging and objections

64.1 The positioning statement, canonical form

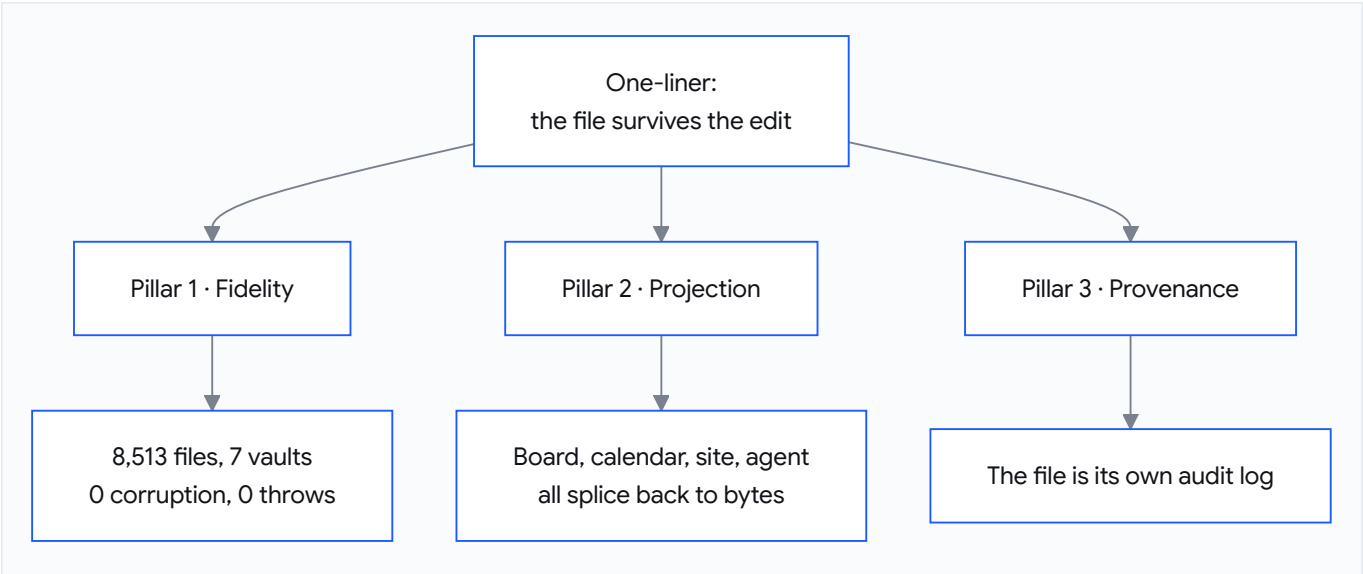
For a software team of 2–20 that already keeps its specs, docs, runbooks and ADRs as markdown files it owns, and that now lets AI agents write into those files, frontmatter is a source-of-truth workspace whose engine edits by byte-range splice and refuses rather than guesses — so every view is a reversible projection and every AI edit is byte-attributed, reviewable and provable — unlike Notion, where the view *is* the data, and unlike every markdown tool that rebuilds the file from an in-memory model of it.

SLOT	FILLED WITH	SOURCE	TAG
X — for whom	Dev-tool / API startups, 2–20 seats (ICP 1, the beachhead)	§21	—
Y — who what	Already live in this substrate: <code>gray-matter</code> 35,782,970 dl/mo, <code>front-matter</code> 18.41M, <code>js-yaml</code> 1,228,031,655	§21	FETCHED
Z — what it is	A source-of-truth workspace (not an editor, not a wiki, not a PM tool)	§26, §6.2	—
W — what it does	Byte-range splice; refuse over guess; every view a reversible projection	§5, §7	MEASURED
V — unlike	Notion (view is the data); the lossy-model class — Front Matter CMS, Hubble.md, and every mainstream rich-text framework in the vendors' own words	§19	MEASURED + FETCHED

Deliberately excluded from V, and this is a discipline not an oversight. OpenKnowledge is named nowhere in the positioning line, because their body path is genuinely byte-perfect and an overstated comparison is the kind of claim that gets refuted in public MEASURED (§19). The comparison we may make against them is narrow and factual — frontmatter drifts, a running CRDT daemon is required, O(document) per edit by their own docblock — and it belongs in a teardown page, never in a headline.

- **Anti-recommendation.** Do not author a second positioning statement for Funnel 2 ("beautiful documents from plain text"). §26 rates that funnel **Low** confidence, and a second position with an unproven audience halves the teaching budget of the first. Funnel 2 gets a *campaign line*, not a position, until Funnel 1 clears post-20.
- **Falsifier.** If the first 25 sales conversations show ICP 1 buys for the consolidation wedge (§21) rather than for fidelity, slot **W** is wrong and the line must be rebuilt around "one file, four surfaces" — the engine then becomes the reason to believe, not the offer.

64.2 Messaging hierarchy



The one-liner, with its known defect.

FORM	STATUS
§26 as written: <i>"The markdown source of truth AI can't corrupt"</i>	Blocked. "can't" is an absolute-safety claim; §58 bans fidelity claims that outrun measurement, and the corpus proves <i>did not</i> , never <i>cannot</i> . Log the §26/§58 clash in §57
Sanctioned: <i>"Markdown that AI edits without rewriting the rest of your file — and we can prove it on 8,513 files that aren't ours"</i>	Publishable today MEASURED
Short form for a 60-character surface: <i>"The file survives the edit."</i>	Publishable; carries no number, so it never goes stale

Pillars and proof.

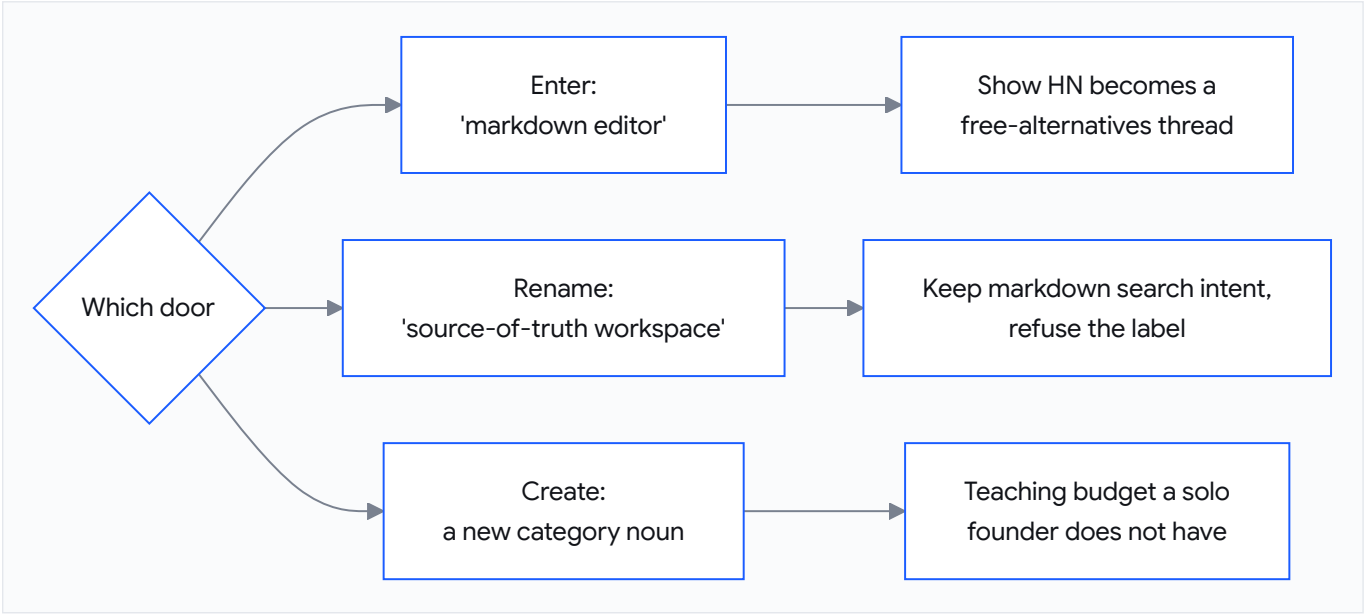
#	PILLAR	CLAIM IN ONE LINE	PROOF POINT	TAG
1	Fidelity	It changes the bytes you asked for and nothing else	8,513 third-party files, 7 vaults, 0 corruption, 0 throws , sha256-pinned, re-runnable via <code>npm run corpus</code> ; three competitor write paths executed with the exact destroyed bytes shown	MEASURED
2	Projection	The file is the truth; the board, the calendar, the site and the agent's context are lenses that own nothing	Drag a card → splice <code>status:</code> ; accept a suggestion → splice a body span. Precedent: org-mode's agenda since 2003, Obsidian Bases, Potluck	MEASURED + SS
3	Provenance	You can see what the AI changed, where, and undo exactly that	Google's public API cannot create suggestions at all ; Lex's track-changes "in development"; OpenKnowledge comments machine-local, never committed. Byte-level attribution makes the file its own audit log	FETCHED + SS

Which pillar leads, by persona.

PERSONA	LEADS	SECOND	NEVER OPENS WITH	WHY
Dev-tool / API startup 2–20 (ICP 1)	Fidelity	Projection	"Beautiful documents"	They can verify the corpus claim themselves in an afternoon; that is the point
Agency / studio 2–15 (ICP 2)	Projection	Provenance	Fidelity	They never <i>felt</i> corruption; they feel four surfaces from one file
Support-heavy SMB (ICP 3)	Provenance	Projection	Deflection or "structured data"	Intercom, Zendesk, Document360 and GitBook all sell exactly that (§21)
Obsidian power user (Funnel 1 prosumer)	Fidelity, framed as <i>"integrates with your vault"</i>	Projection	"Obsidian competitor"	The community code of conduct removes the launch rather than debating it (§26)
Researcher / academic (§22)	Projection	Fidelity	Collaboration features	One file → paper, site, and dataset appendix
The buyer's security or ops reviewer	Provenance	Fidelity	Anything compliance-shaped	See §64.5 — we have no SOC 2 and no attorney review (§53)

- **Anti-recommendation.** Do not run all three pillars in one asset. A page that argues fidelity, projection and provenance simultaneously reads as a feature list, which is precisely the shape §64.4 objection 3 attacks.
- **Falsifier.** If the ICP-1 landing variant that leads with Fidelity converts below the Projection variant across 400 sessions, the pillar order inverts and the engine becomes the reason-to-believe rather than the offer.

64.3 The category question



DOOR	EVIDENCE FOR	EVIDENCE AGAINST	VERDICT
Enter "markdown editor"	Highest search intent; the substrate's own download volume proves the audience exists — <code>js-yaml</code> 1,228,031,655, <code>gray-matter</code> 35,782,970/mo FETCHED	Every Show HN named "markdown editor" becomes a thread of free alternatives (§26). The price floor is \$0 and well-maintained: AppFlowy 76,040★, AFFINE 71,976★, SiYuan 46,023★, Logseq 44,669★, Outline 40,362★, Trilium 37,624★, Docmost 21,501★, all pushed within 24h FETCHED	Reject as a label
Rename to <i>source-of-truth workspace</i>	The category winners all renamed: Obsidian sold <i>ownership</i> , Notion a <i>workspace</i> , Linear <i>speed</i> (§26). Keeps "markdown" as the substrate noun, so search and plugin-directory intent survive	A two-hop name costs one sentence of teaching in every conversation	Recommended
Create a new category noun	§3.2 names the empty box explicitly: <i>"the rendered, evolving AI-output document — open — the category to name"</i>	Category creation is a paid activity. Solo throughput is ~1 substantial shipped surface per 2–3 weeks (§53) and exactly two posts have ever shipped , one with a visible defect since 2026-08-10 MEASURED	Defer

Recommendation: rename, do not create. Be findable inside the markdown category — the "Open in frontmatter" plugin, npm, docs-tool comparisons — and refuse the category *label* in every headline; Relay proved the bridge-plugin path with 172,544 downloads of a commercial service's plugin **SS**.

- **Strongest counter-argument, stated in full.** A renamed category is an unrecognised category, and an unrecognised category has no search volume, no G2 grid, no budget line and no comparison page to rank against. The buyer must name the outcome to get it approved (§21), and "source-of-truth workspace" is a *mechanism* dressed as an outcome. The counter-argument is not defeated by evidence — it is answered by sequencing: lead with the mechanism to the beachhead who can verify it, and let the *outcome* line ("one file that is editor, AI workspace, rendered surface and audit trail") do the work in every asset aimed at anyone else.
- **Falsifier.** If, at post-20, prospects cannot repeat the category noun back unprompted, drop the noun entirely and ship a category-of-one description: "*it is where the file, not the app, is in charge.*"

64.4 Objection handling

#	OBJECTION	RAISED BY	WHAT WE CONCEDE AS TRUE	THE HONEST ANSWER, WITH EVIDENCE
1	Why not just Obsidian?	Prosumer, ICP 1	It is better than us at plugins, graph, mobile and community, and it is the vault we integrate with — never the competitor we name (§26)	Obsidian is single-player by design: no review loop, no provenance, no fidelity proof MEASURED §3.1. Its own community proves the missing piece — the Homepage plugin has 1,294,057 downloads , ~20× every bespoke dashboard plugin, and no product ships that natively FETCHED . We are the review-and-proof layer over the vault you keep
2	Why not Notion?	ICP 2, ICP 4	Notion is better at databases-as-views, onboarding and team defaults, and we will never be a Notion-style PM tool (§6.2, founder boundary)	In Notion the view <i>is</i> the data and export is lossy by architecture SS ; formulas cannot aggregate across rows; Coda's export returns disconnected strings with formulas stripped SS . Price behaviour is the second answer: free AI cut to 20 responses <i>for life</i> and Business up ~20% SS §2
3	This is a feature, not a product.	Investor, senior engineer	As a <i>capability</i> , yes — a byte-exact writer is a library. Conceding this is more credible than denying it	The product is the loop the capability makes possible: projection, review and provenance all require it and none exists without it. The three teardowns show the bolt-on route fails — Front Matter CMS builds a comment-preserving Document object and throws it away MEASURED . Joint verdict: the failure is architectural, not a library choice §19
4	You are one person.	Every buyer	Fully. It is the binding constraint on the whole document (§53): ~1 shipped surface per 2–3 weeks, and support, incidents, invoicing and patching do not compress with skill	Three structural answers, none of them "trust me": the substrate is the file, so leaving costs you nothing (§27 rank 7, the anti-moat, priced in deliberately); the shutdown promise is written and public (§48); and the leverage is measured — 124 SKILL.md automations, 24,539 complexity-gate decisions, 5,014 trace rows, 69 gate scripts, 149 bin tools, and 10 complete multi-surface campaign episodes built in about a week MEASURED .
5	Markdown is for developers; my team won't write it.	ICP 2, ICP 3, ICP 4	Largely true for authoring. This is why the surface is deliberately simple and why the projections exist at all	Nobody on the team writes markdown — they drag a card, pick a date, accept a suggestion, and the splice writes the file (§5). The markdown is the <i>storage</i> , exactly as SQL is storage for a CRM. Anti-answer we must not give: "markdown is easy, they'll learn." They will not, and saying so loses the room
6	I already have Cursor / Claude Code.	ICP 1, the sharpest objection in the list	It edits markdown well, agent-markdown is now commodity (§3.2: obsidian agent-skills 47,418★, the skills CLI at 9.3M downloads in one week FETCHED), and §3.2 says explicitly: do not position here	Cursor's provenance evaporates at accept — the AI/human distinction is lost forever after SS §2 problem 4 — and its attribution is <i>admin telemetry</i> , not something in your file that your reader can see. Our claim is narrow and defensible: byte-anchored, document-portable, reader-visible provenance (§58). Second answer: Cursor is a code editor whose users are engineers; the document your CEO reviews does not live there
7	Seven free self-hosted alternatives sit at 21K–76K stars. Why pay?	ICP 1, ICP 4	The price floor is genuinely \$0 and those projects are actively maintained FETCHED §3.1	Every one of them regenerates the file. Zero of them certify cross-engine degradation, and the <i>testing</i> layer already monetises where the data layer stays free — Litmus at \$500/mo is the proof SS §3.2. Pricing follows: INR 299 / USD 5 Pro is priced against \$0, and the money is \$20–40/editor seat in teams (§21, §26)
8	Git already gives me history, diffs and review.	ICP 1	It does, and we build on it rather than around it — sync is git-merge plus a splice journal plus compare-and-swap, never a CRDT	Git's three-way merge preserves CRLF and a missing final newline exactly, and refuses where a splice writer would want to refuse MEASURED §58 — that is a compliment, and it is also the gap: git diffs <i>lines</i> , not the byte range one agent touched, and it cannot tell you which change was the model's. Non-engineers do not review in a PR
9	An incumbent ships this next quarter.	Investor	The engine work is hard but finite and increasingly LLM-assistable. §27 gives the fidelity moat 18–36 months and the certificate 12–24 — not forever	Incentives point elsewhere: Notion toward blocks, OpenAI toward artifacts-as-output (it removed Canvas in May 2026 SS), Google toward Docs, OpenKnowledge toward the team wiki (§3.3). The honest framing is a window, not a fortress — and the sequencing law says the differentiation tracks must land inside 6–12 months
10	Nobody has paid you anything.	Investor, and the hardest to answer	True and unmitigated: zero paying users. Developers are the hardest freemium audience — median dev-focused free-to-paid is 5% , half the non-dev rate FETCHED §4	Do not argue. State the arithmetic and the test: 1,000 free developer signups × 5% × USD 5 = USD 250/mo DERIVED: 1000 × 0.05 × 5 , which is why the revenue centre is self-serve teams, not Pro. The pain is evidenced (GitBook's dominant complaint cluster is reliability and lost work SS); the <i>willingness to pay a premium to avoid it</i> is the single biggest risk in this document and is stated as such in §4

Two rules for using this table. Lead every answer with the concession, because the concession is the only part the sceptic did not expect. And when an objection has no answer — objection 10 today — say so and name the test that would produce one; §58's whole discipline is that a claim we cannot re-derive is worse than silence.

64.5 Claims we may not make

Inherits every row of §58 unchanged. This section adds the positioning-specific bans, because a marketing sentence is where an honest engineering number goes to die.

BANNED	WHY	SANCTIONED SUBSTITUTE
"AI can't corrupt it", "corruption-proof", "AI-proof"	Absolute-safety claim; the corpus proves did not , never cannot . This is the §26 Funnel-1 line as written — log the clash in §57	"0 corruption and 0 throws across 8,513 third-party files from 7 vaults, re-verifiable" MEASURED
"Lossless", bare	Unqualified, it claims the whole surface; today there are 83% publish-refusals from one YAML defect MEASURED	"Byte-preserving on every edit it accepts — and it refuses rather than guessing"
Any fidelity or coverage percentage	§58: at the measured refusal rate the headline number is false until NF-1 and NF-3 land	The absolute counts, with the corpus named
"First", "only", "the first AI attribution"	§58 and §6.2 — one such claim was already caught and refuted	"The first markdown editor with byte-anchored, document-portable, reader-visible provenance" — the narrow form, and only after one more verification round
"Audit-ready", "compliance-grade", "SOC 2"	No SOC 2, no attorney review, no auditor has seen the artefact (§51, §53). Vanta and Drata charge \$7,000–\$30,000/yr for the thing this phrase implies SS	"The file is its own audit log: every change carries the byte range and the actor"
"Replaces your \$1,163/mo stack"	§21 explicitly: quote the range, never only the top. The honest light stack is \$78/mo — a 14.91x spread DERIVED: 1163 ÷ 78 = 14.91	"Teams replace somewhere between \$78 and \$1,223 a month of tooling with this — here is the arithmetic for both ends"
"Markdown editor" as the category noun	Every Show HN with that name becomes a thread of free alternatives (§26)	"Source-of-truth workspace", with markdown named as the substrate
"It learns how you write"	§58 — never market a learning loop before a decision demonstrably bends on real data	Say nothing about learning
Any SS -tagged number, in any public asset	47 remain in this document's inherited material	Re-derive it, or cut the sentence

64.6 What the name must communicate

The naming decision and the positioning decision are one decision, because they split the teaching load between them. A coinage (`stetfile`) teaches nothing on sight, so the one-liner must carry the entire category — which is affordable only because the beachhead can verify the mechanism themselves. A retained "Frontmatter (Studio)" teaches the substrate instantly but costs the *"not the VS Code one"* tax in every conversation forever, against an incumbent at **80,605 installs** and **2,539 stars** read 2026-08-29 [FETCHED](#) .

#	THE NAME MUST...	SOURCE	TEST	<code>STETFILE</code>	"FRONTMATTER STUDIO"
1	Name the guarantee , not the format	§52.2 — naming from the format ages with the format	Is it still right if the substrate stops being markdown?	Yes — <i>stet</i> = "let it stand"	No
2	Survive being said aloud in support	§52.2 — the <code>-wright/-write/-right</code> hazard that killed <code>bytewright</code>	Spell it once over a call	Yes	Yes, but requires the qualifier every time
3	Be free on npm and GitHub with zero named repos	§52.2 MEASURED	Re-run the 404 probe on registration day — nothing reserves them	npm 404, GitHub 404, 0 repos	Handles free only in qualified form
4	Not force a permanent disambiguation tax	§52.3	Count the words spent explaining the name per conversation	0	≥1, forever
5	Not collide in the adjacent market	§52.2	Search the product space, not just the registry	Clean; but bare <code>stet</code> collides with <code>stetmark</code> , npm-published 2026-08-05 , 5,753 dl/mo MEASURED	Direct collision, same keyword space
6	Carry the category rename of §64.3	§26	Does the name make "editor" the obvious next word?	No — good	Yes — bad; "frontmatter" is the generic name of the substrate (§27 rank 5)

- **Recommendation, unchanged from §52:** `stetfile` , with `stet` as the spoken shorthand and the CLI verb. Positioning consequence: the one-liner must then do all the category work, so the sanctioned form in §64.2 becomes load-bearing copy rather than a tagline.
- **Anti-recommendation.** Do not ship bare "Frontmatter" while quietly holding qualified handles — §52.3 rates that the highest-risk option with extra steps. And do not spend on domain, logo or launch copy before a trademark attorney's knock-out search; §52.4 is explicit that handle availability is a different question in kind from mark rights.
- **Falsifier.** A clean knock-out search on the incumbent's class overlap flips the decision to "Frontmatter Studio", because a taught name costs more than a qualifier — and if that happens, pillar order in §64.2 does not change, but the one-liner sheds its category-teaching job and can shorten to *"The file survives the edit."*

§65 The markdown thesis — every role the format plays

65.1 The six roles, in escalating order

Each role is enabled by a mechanism CommonMark already defines, and each is permitted to fail back into the role below it. No role above 1 may introduce a token the floor does not already contain.

#	ROLE	THE FILE IS...	ENABLING MECHANISM	FAILS BACK TO
1	Document	prose a stranger reads	CommonMark 0.31.2 (2024-01-28) block grammar plus the four GFM extensions every certified engine implements: tables, task list items, strikethrough, autolink literals FETCHED INFERENCE	— (the floor)
2	Schema carrier	typed data a machine parses	A YAML block delimited at byte 0 — one contiguous range whose boundaries are computable without running an inline parser INFERENCE	1, as an <code><hr></code> and an <code><h2></code>
3	Projection driver	its own view definition	The three-slot rule: <code>render:</code> switch in frontmatter, spec in a fenced <code>fm-view</code> lane, data in body constructs that already degrade MEASURED	1, as a code block plus an outline
4	Agent contract	the interface a model writes through	Path is identity; <code>base_version</code> enforces read-before-patch; every edit is a splice against <code>baseSha</code> and arrives as a <code>hunk</code>	2, as an unmerged suggestion sidecar
5	Ledger	an append-only record of what happened	Append-only body sections plus a splice journal; C2PA A.9's front-matter manifest form carries a single byte exclusion range with start and length in bytes FETCHED	1, as dated headings
6	Application substrate	the program's config, state and control surface	Frontmatter-typed instruction files read by a runtime: 124 <code>SKILL.md</code> across 27 category dirs, 69 gate scripts (33 <code>assert-*</code> + 34 <code>break-*</code> + 2 harness meta), 147 executables in 21,481 lines MEASURED , 2026-08-29	5, then 1

Role 1 — document. The floor is not "CommonMark" in the abstract; it is 0.31.2 plus exactly four extensions, and everything past those four is a profile feature that must be degradation-certified before a projection may depend on it [INFERENCE](#).

§67 Decision — vendor @lezer/markdown

Deferred to G4. The repository moved to <https://code.haverbeke.berlin> — 147 stars, one maintainer, and the only JS parser with exact inline-mark offsets *and* incremental reparse.

OPTION	56 KB REPARSE	INLINE OFFSETS
full	16.88 ms	yes
incremental	4.71 ms	yes

- ☒ Read the licence
- ☐ Budget a quarter for owning the fork

```
**Role 2 - schema carrier.** CommonMark 0.31.2 contains the strings "front matter" 0x, "frontmatter" 0x, "YAML" 0x and "metadata" 0x; the GFM spec contains the same three at 0x '[measured]'. **Every typed-data mechanism available to this product is therefore out of spec, so the only real choice is between profiles that degrade well and profiles that degrade badly** '[inference]'.

```markdown

title: Vendor @lezer/markdown
status: doing
due: 2026-09-30
tags: [engine, parser]
fm.profile: adr
fm.version: "1"

```

**Role 3 — projection driver.** A single key `render: board` degrades to `<hr><h2>render: board</h2>` in marked 16.4.2, markdown-it 15.0.0 (commonmark preset) and commonmark 0.31.2 — a setext H2 that outranks the document's own H1 in every non-frontmatter-aware renderer [MEASURED](#). Cost is linear in key count: 4 keys ≈ 60 characters of H2 junk, 12 keys ≈ 180 [DERIVED](#).

```

render: board

```fm-view
group: heading
columns: [Todo, Doing, Done]
card: {title: text, badge: due}
```
```

## §68 Doing

- ☐ Vendor @lezer/markdown

```
Role 4 - agent contract. The protocol carries the identity and the version explicitly because both failures are documented: on claude.ai, where identity is inferred from phrasing, "it made a new artifact instead of updating mine" is the top documented failure, and 'base_version' structurally kills the drift bug where a user hand-edits while the model keeps talking about the version it remembers.

```text
land({ path: "docs/adr/0007-vendor-lezer.md",
  type: "adr",
  title: "Vendor @lezer/markdown",
  body_md: "...",
  base_version: "sha256:...",
  mode: "patch",
  source: { tool, model, conversation_url, session_id } })
→ { LANDED | VERSIONED | REFUSED_CONFLICT | NEEDS_TARGET,
  path, version, url, bytes_written, cert }
```

Role 5 - ledger. C2PA's A.9 structured-text binding is line-for-line a byte-splice contract - fixed `-----BEGIN/END-----` delimiters modelled on RFC 4880 §6.2, files read in binary mode "to preserve the exact byte representation of line terminators", a claim generator that "shall **not** alter the line ending convention of the file content outside the manifest block", and at most one block per file '[fetched]'.

```markdown
> [!DECISION] Vendor the parser
> Accepted 2026-08-29. Supersedes ADR-0004.
```

Changed

- Splice writer returns a typed refusal instead of a bare `src`.

-----BEGIN C2PA MANIFEST----- <base64 manifest, hashed over a single byte exclusion range> -----END C2PA MANIFEST-----

```
**Role 6 - application substrate.** The instruction file is the program: frontmatter declares identity and invocation policy, the body is the contract, and a separate gate script asserts the contract held.

```markdown

name: sgnk-complexity-gate
description: Route a task by shape before any model call.
disable-model-invocation: false

```

§70 Contract

Emit `{tier, mode, rule2_gated}` and nothing else.

§71 Gates

- `assert-*.sh` — refuse if the routing distribution moves off baseline
- `break-*.sh` — prove the assertion can fail before trusting a green

```
65.2 One file, six roles

```mermaid
flowchart TD
    F["note.md - the bytes<br/>the only source of truth"]
    R1["1 Document<br/>CommonMark 0.31.2 + 4 GFM"]
    R2["2 Schema carrier<br/>YAML frontmatter"]
    R3["3 Projection driver<br/>render: switch + fm-view lane"]
    R4["4 Agent contract<br/>path identity + base_version"]
    R5["5 Ledger<br/>append-only + splice journal"]
    R6["6 Application substrate<br/>SKILL.md + gate scripts"]
    X["No textual home<br/>sidecar, then database"]
    F --> R1 --> R2 --> R3 --> R4 --> R5 --> R6
    R6 -. "any role fails" .-> R1
    R3 -. "positions, intervals, hunks" .-> X
```

65.3 The maximisation table

ROLE	WHAT WE EXPLOIT	THE CEILING	MEASURED EVIDENCE FOR THAT CEILING
1 Document	The floor renders everywhere with no configuration; degradation is a property of block-level constructs, not of our code	No block containers, no colspan or rowspan, no auto-numbered cross-references, no multi-column reading order	Pandoc grid tables degrade to <code><p>+-----+-----+ Fruit Note ...</p></code> MEASURED ; <code>{#fig-plot}</code> leaks as visible text after the image MEASURED ; GFM's own normative text: "Block-level elements cannot be inserted in a table" FETCHED ; nested <code>:::</code> columns collapse into one run-on paragraph MEASURED
2 Schema carrier	One contiguous range at offset 0 with two fixed sentinels — the best splice target in the format; every certified engine already has a frontmatter mode or documented extension FETCHED	Junk is linear in key count; the key string <i>is</i> the identity, so a rename is lossy; today the parser refuses most foreign vaults	<code><hr> + <h2></code> of the first key in 3/3 engines MEASURED ; 4 keys ≈ 60 chars, 12 keys ≈ 180 DERIVED ; 83% aggregate foreign-vault refusal — 6,613 of 6,614 foreign files refused on zero-indent block sequences <code>[project-measured]</code> ; <code>SAFE_KEY</code> excludes a space, so <code>date created</code> fails in 812/957 files of one vault and CJK keys fail in 905 files of another MEASURED ; four ecosystems normalise keys four different ways — Logseq lowercases and rewrites <code>_</code> → <code>-</code> , Dataview sanitises to lowercase-with-dashes, org is case-insensitive, Notion sidesteps it entirely with an opaque <code>id</code> such as <code>"fy:ξ"</code> FETCHED
3 Projection driver	Fenced lanes degrade to contained, labelled code with zero sigil leakage — the best degradation profile of any carrier tested; drags map to four typed write-backs (BODY-MOVE, KEY-SET, CELL-SET, FENCE-SET)	Only C0 and C1 lanes; a projection must be a pure function of bytes, so an LLM can never be a render lane; anything with no textual home is 100% sidecar and the render then carries none of the value	<code><!--fm ... --></code> is fully suppressed at markdown-it's <code>commonmark preset (html:true)</code> but escapes to visible <code>&lt;!--fm...--&gt;</code> ; at its default <code>html:false</code> MEASURED , TWO RUNS DISAGREEING ON WHICH CONFIG COUNTS AS "DUMB" ; excalidraw's 7,578,223 downloads are the largest single demand number in the survey and are 100% sidecar FETCHED ; population evidence that in-document execution does not buy reproducibility: 1,159,166 notebooks from 264,023 repositories, 24.11% executed without errors, 4.03% produced the same results FETCHED (denominator discrepancy recorded, not resolved: 863,878 published vs 788,813 reconstructed → 26.41% / 4.42% DERIVED)
4 Agent contract	An edit <i>is</i> a byte range, so a proposal, a citation and a diff are the same object; the review surface accepts human, agent and sync-conflict hunks through one grammar	Acceptance, not capability, is the limit — and the only published propose-first numbers are from a small model on one narrow dialect	Ansible Lightspeed: 49.08% strong acceptance on multi-line suggestions , Day-30 retention 13.66% across 10,696 users / 3,910 returning (arXiv 2402.17442, pub 2024-02-27, upd 2024-10-22) FETCHED — a ceiling for a constrained verb, not a general rate INFERENCE ; deterministic projections out-install every AI capability combined by 7.25× (7,289,307 vs 1,005,651 peak-version installs) DERIVED
5 Ledger	Append is free and byte-safe; the record and the document are the same artifact, so provenance survives export	Ledgers do not rot at the append — they rot at the attribution field, and they go stale silently	Live ALOS trace ledger: 63 daily <code>.jsonl</code> , 5,018 rows , of which <code>accepted</code> is non-null in 181 = 3.607% and <code>skill: "unknown"</code> in 3,946 = 78.637% MEASURED/DERIVED ; <code>PREFERENCE-LOG.jsonl</code> 232 rows of which 223 = 96.121% carry no skill attribution DERIVED ; <code>regression-gates.jsonl</code> 40 rows, last registered 2026-08-13T01:41:14Z = 16 days stale DERIVED
6 Application substrate	A whole orchestrator runs on frontmatter-typed markdown with no database: 124 <code>SKILL.md</code> , an 892-line self-amending constitution, 69 gates, 28 hooks across 9 events, 24,669 complexity-gate rows over 42 days = 587.357 rows/day MEASURED/DERIVED	Single-tenant by construction, and the substrate cannot reliably measure itself	Zero tenancy fields in the trace ledger and 2 of 151 files containing any HTTP-listener code MEASURED ; <code>skill-health.json</code> (7-day window) reports active 0, dormant 5, dead 116, infrastructure 10 while the ledger holds 446 trace rows for <code>sgnk-drift-watch</code> in the same window MEASURED ; <code>calibration.json</code> high bucket n=26 rate 0.269 against a 0.20 bar — the drift condition is met and unactioned MEASURED/DERIVED

- Recorded drift, not reconciled: the product-context figures 5,014 trace rows and 24,539 gate decisions are an earlier read of the same append-only stores measured at 5,018 and 24,669 on 2026-08-29 **MEASURED**. Both stand; never publish either without its read date. The same class already sits in the record as `baselines/` at 2,382 vs 2,381 on one day.
- Anti-recommendation for the whole table: **do not promote a role because one engine handles it well**. The gate is the certificate across all seven local targets, and `uncertifiableShare()` already reports **8 of 15 declared targets (53.3%) as not locally probeable** **MEASURED** — a declared row is not evidence.

65.4 The hard boundary as a rule a builder can apply

The placement ladder. Stop at the first yes.

#	QUESTION	HOME	LITERAL FORM	FALSIFYING CASE FOR THIS ROW
1	Would a person reading the raw bytes want to see it, and does it survive a renderer that knows nothing about us?	Body text	heading, list item, GFM table cell, task checkbox, blockquote callout	An inline field inside a table cell: the same bytes have two different block structures across the matrix (GFM <code><td></code> vs commonmark one paragraph), so it passes the read test and fails addressability MEASURED
2	Is it one scalar, whole-file, and does an existing ecosystem tool already read that key?	Frontmatter , 1–3 keys	<code>status: doing</code> , <code>due: 2026-09-30</code> , <code>render: board</code>	Every key added costs an <code><h2></code> line in a dumb renderer, so a rich view spec here is a degradation regression, not a neutral choice MEASURED
3	Is it machine-shaped and whole-block — harmful as prose, but still reconstructable from these bytes alone?	Fenced lane , reserved info string	<code>'''fm-view</code> , <code>'''fm-query</code> , <code>'''fm-schema</code>	An unclosed fence "runs until the end of the containing block" per CommonMark §4.5, so one dropped backtick line turns the rest of a 4,000-line note into code FETCHED — the writer must emit open and close in one splice, never two
4	Does it have no textual home at all, and can it be deleted without changing what the document <i>means</i> ?	Sidecar	JSON Canvas-class positions, SRS intervals, suggestion hunks, embeddings	Bases' shape is the correct one — data in frontmatter, <i>query</i> in the <code>.base</code> sidecar; inverting it breaks on rename, move and copy, and makes the <code>.md</code> non-self-describing FETCHED
5	Does it need identity surviving a rename, multi-writer concurrency, or a cross-file transaction?	Database — and say out loud it is not a document feature	—	Notion's 21 property types key on an opaque <code>id</code> , which is precisely why renaming a property there is safe and renaming a frontmatter key here is not FETCHED ; multi-respondent form state is a database, not a file INFERENCE

The one-line decider: if deleting it changes what the document means, it belongs in the file; if deleting it only changes what a view looks like, it belongs in a sidecar; if it cannot be deleted at all without breaking another person's session, it belongs in a database and therefore not in this product.

- Carrier rule that falls out of rows 1–3, unchanged from the settled position: **prose a human reads goes in a blockquote callout; opaque machine data goes in a fenced code block**. A block quote is delimited by a per-line prefix, so there is no state to leave open — the construct ends the moment the `>` stops **MEASURED**. A fence is delimited by a matching close, and CommonMark §4.5 is unforgiving about its absence **FETCHED**.
- Anti-recommendation: **do not adopt the invisible HTML-comment carrier globally** on the strength of the cleaner measurement. It is clean in exactly two of the three configurations we certify, and `COMMENT-SET` is permitted only where the certificate explicitly records `html:false` visibility for that target.
- Anti-recommendation: **do not promote an inline `key::value` field into frontmatter automatically "for queryability"**. That rewrites lines the user did not edit, changes the rendered output, and is the exact behaviour that makes people distrust editors that touch their files. Dataview's own parser must resolve bracket nesting, `\` escapes and overlapping spans heuristically, and it sanitises `**Bold Field**` → `bold-field`, so the index provably cannot write back the source bytes **FETCHED**.
- Recorded disagreement, unresolved: m5 recommends **emitting `...` as the frontmatter closer**, because 3/3 dumb engines then degrade to a paragraph instead of a setext heading and pandoc explicitly permits it **FETCHED** **MEASURED**; §8.4 as written supports `---` fully and says nothing about `...`. Both closers are legal on read. Do not resolve this by preference — resolve it by running the closer through the seven local targets and recording which strip-pipelines match on `---` alone.

- What falsifies the ladder: a datum that passes row 1 and is nonetheless unaddressable by a byte range surviving an unrelated edit. The table-cell inline field is already that case, which is why the ladder carries an addressability clause and not just a readability one.

65.5 What markdown genuinely cannot do, and what we say instead

Faking any of these is worse than refusing them, because a fake succeeds locally and fails on someone else's renderer, which is exactly where the user cannot see it happen.

IMPOSSIBLE	WHY, WITH EVIDENCE	WHAT WE TELL THE USER
Merged cells	No colspan or rowspan syntax in CommonMark or GFM; pandoc's MANUAL contains zero occurrences of either in the markdown reader docs MEASURED OVER FETCHED HTML . Grid tables buy spans and cost readable degradation MEASURED	"Markdown has no merged cells. Split the column, or export to DOCX where the format does."
Block content inside a table cell	GFM normative text, verbatim FETCHED	"A list inside a table cell is a lie in every renderer but one. Put the list under the table."
Multi-column pages, floats, sidebars, wrapped pull quotes	No block-container syntax exists; nested <code>:::</code> collapses into one run-on paragraph and leaks two sigil lines per block in all three engines MEASURED	"There is no reading order for two columns in a linear renderer. Use PDF layout for print, one column for the file."
Auto-numbered cross-references ("Figure 3")	Requires a numbering pass no dumb renderer will run; <code>{#fig-x}</code> leaks as visible text MEASURED	"We can link to the figure. We cannot number it in a way that survives leaving this app."
Portable footnotes	Not in the GFM spec — <code>grep -ci footnote</code> on cmark-gfm's <code>spec.txt</code> returns 1, an intro-prose mention; the syntax lives in <code>test/extensions.txt:702</code> MEASURED . A footnote whose body is a <i>single token</i> is parsed as a link reference definition in all three engines, producing live broken links MEASURED	"Your note body stays readable everywhere. The little superscript link does not. Give the note more than one word."
Spoilers and hidden text	<code> spoiler </code> degrades to the literal content, verbatim MEASURED — a spoiler that degrades open is a defect, not degradation	"Anything you can type, a plain renderer can show. There is no hidden text in markdown."
Transclusion on publish	<code>![[Note#Heading]]</code> degrades to the literal brackets with the content silently absent and no marker that anything is missing MEASURED	"We keep your embed exactly as written and we open it in the editor. We will not publish a page where content vanished without saying so."
A stable table caption position	Immediately after the table it is absorbed as a data row (<code><td>Table: my caption</td></code>); with a blank line it degrades readably MEASURED	"Leave a blank line above the caption, or it becomes a row."
Guaranteed CJK line joining	CSS Text 3 (CR Draft, 2026-08-14) §4.1.3: a segment break "is either transformed into a space (U+0020) or removed... The rules for this operation are UA-defined in this level " FETCHED	"We can normalise our own renderer. We cannot promise GitHub's."
Correct word counts without a segmenter	<code>countWords</code> splits on <code>/\s+/</code> and undercounts against Intl.Segmenter by 20.00× on Chinese prose, 17.00× on Japanese, 12.00× on Thai, 1.19× on a markup-heavy mixed document ; ko/hi/en are exact MEASURED	"Approximate for Chinese, Japanese and Thai" — until the segmenter ships, which is four lines and zero new dependencies
Sanitised HTML by default	<code>marked</code> ships no sanitiser and no URL-scheme filter: it renders <code><script>alert(1)</script></code> verbatim and <code>[click]</code> (<code>javascript:alert(1)</code>) as a live href; markdown-it refuses dangerous schemes even at <code>html:true</code> MEASURED . GFM's <code>tagfilter</code> filters exactly nine tags and states "All other HTML tags are left untouched"; GitHub compensates with private post-processing FETCHED	"The spec is not a security boundary; the platform is. Any 'renders like GitHub' claim excludes GitHub's private layer."
Round-trip-exact pipes and tabs	<code> </code> in a cell renders <code>`</code>	<code>,</code> and CommonMark treats tabs "as if they were replaced by spaces with a tab stop of 4" <code>[fetched]</code> — semantically equal, byte-different
WCAG AA out of the box	§5.2 makes AA all-or-nothing FETCHED ; measured against this repo: 11 of 14 images with empty alt = 78.6% , no <code><caption></code> / <code>scope=</code> / <code><colgroup></code> on GFM tables, 35 heading-level skips, 5 of 186 files with more than one <code><h1></code> , no <code>lang=</code> and no <code>dir=</code> from <code>marked</code> , no accessible name on task-list checkboxes MEASURED	"Conformance is something the publisher adds. It is never something the format supplies." And never image-rendered math — that is 1.1.1 with a text alternative you cannot generate

- Anti-recommendation: **do not build an auto-fixer that silently normalises ambiguous indentation, reference-link placement, or a bare-CR line ending**. Reference links are non-local — CommonMark's own author writes that `[foo][bar]` has four possible meanings "depending on whether the references... are defined elsewhere (perhaps later)" and that this makes "accurate syntax highlighting nearly impossible" [FETCHED](#) , confirmed locally as three different renderings of one source [MEASURED](#) . C2PA A.9 says convert bare CR; we have committed not to silently mutate bytes, so the honest behaviour is refuse and cite the code.
- The refusal is only honest if it is typed. Live baseline at commit `9e84628` : `spliceFrontmatterValue` and `spliceFrontmatterKey` both return a bare string across **20** `return src` **sites**, so a refusal is byte-identical to a successful no-op [MEASURED](#) — while `mdmax` already emits 32 uppercase symbols, 13 verdict classes and **19 typed failure codes** [DERIVED](#) . Refusal text without a reason code and a byte range is not a refusal; it is silence.

65.6 The strongest argument this thesis over-reaches, and the answer

The argument, at full strength, using our own numbers.

1. Five of the six roles are being carried today by a purpose-built store that carries them better. Notion's property model has opaque ids and therefore safe renames; a database has transactions; a queue has multi-writer semantics. Markdown has a key string.
2. The substrate has only ever met its author. The trace ledger has **zero tenancy fields** and **2 of 151 files** contain any HTTP-listener code [MEASURED](#) . Roles 5 and 6 are not demonstrated at scale; they are demonstrated at n=1, on one machine, with one writer and no concurrency.
3. It cannot measure itself. `skill-health.json` reports **active 0 / dead 116** while the ledger holds **446 rows for one of those "dead" skills in the same window** [MEASURED](#) . **78.637%** of trace rows carry `skill: "unknown"` and **96.121%** of preference rows carry no attribution at all [DERIVED](#) . A system whose own instrumentation disagrees with its own record is not evidence for a thesis about durable records.
4. Role 2 does not survive contact with strangers. **83%** of foreign vaults refuse today, on a single YAML defect `[project-measured]` . Every fidelity number in the document is false until that lands.
5. The certificate that is supposed to police all of this is itself half-blind: **8 of 15 declared targets (53.3%) are not locally probeable** [MEASURED](#) , and the shipped preview scores **439/652 = 67.3%** on CommonMark 0.31.2 against a bare remark pipeline's **498/652 = 76.4%** [MEASURED/DERIVED](#) — certifying seven engines against a preview less conformant than the engines it certifies is not a certificate.
6. Therefore: the thesis is a description of one person's workflow, generalised past its evidence.

The answer. Three concessions first, because they are correct.

- Points 3, 4 and 5 are conceded without qualification, and they are already the queue: NF-1, NF-3 and NF-4 are RO work, the preview conformance fix is gated *before* any render profile ships, and the corpus gate is pinned and red-proofed at 8,513 files. Point 2 is conceded and settled the same way — §15 answers "sell the orchestrator" with **No**, on exactly the tenancy evidence the argument cites. The internal system ships only the artifact-facing half: an asset ships if its output is a fact about the user's file, and stays internal if its output is a claim about the software's own intelligence.
- Point 1 is conceded on capability and rejected on framing. The thesis has never been "markdown can do everything". It is narrower and testable: **the file is the only source of truth, and every role above role 1 is either a deterministic projection of those bytes or a sidecar that can be deleted without changing what the document means** — which is why

- the placement ladder ends at "database, and it is not a document feature" instead of pretending otherwise.
- Point 6 is where the argument actually fails, and it fails on evidence that arrives from outside the thesis. Across 7,020 plugins with stats, deterministic projections take **7,289,307** peak-version installs against **1,005,651** for every AI capability combined — **7.25x DERIVED** — and **697 of 7,058 plugins (9.9%) describe an AI capability but take 3.44% of the peak-version sum MEASURED**. The largest file-native editor in the category ships **zero occurrences of the token "AI" across 9,502 characters** of its full roadmap back to July 2023, and what it did ship is data in local markdown properties with views described in valid YAML **MEASURED** **FETCHED**. Role 3 is not an extrapolation from one machine; it is the single most-installed thing in the category, arriving at the same design independently.
 - The remaining role-6 claim is deliberately not the product claim. The orchestrator is evidence that markdown-typed instruction files scale to a working system — 124 **SKILL.md**, 69 gates, 24,669 gate decisions over 42 days **MEASURED** — corroborated externally by **CLAUDE.md** appearing in ~774,144 indexed files and **agents.md** at 23,968 stars, **spec-kit** at 132,035 **MEASURED/FETCHED**. It is not evidence that the same substrate is multi-tenant, and the document never claims it is.

What would actually falsify the thesis.

ROLE	FALSIFIER	STATUS
1 Document	A majority of the 7 local targets fail one of the four GFM core extensions on a realistic corpus — the floor then drops to bare CommonMark and tables become a profile feature	Not tested at that framing
2 Schema carrier	NF-1 lands and the foreign-vault refusal rate does not move materially off 83%, meaning the defect was never the single cause	Queued R0; the 83% is the number the fidelity claim rests on
3 Projection driver	Obsidian ships a first-party board with splice-clean write-back before we do; item 1 of the v1 ranking loses its rationale entirely	Live risk — Bases shipped natively while obsidian-kanban's repo last moved 2026-03-06 FETCHED
4 Agent contract	Transformation verbs fail to clear 20% strong acceptance in the first 90 days, or the top unmet request six months after Lane A is still "run my Python here" rather than "query my vault here"	Instrumented, no data yet
5 Ledger	Attribution stays below ~50% after the trace schema is fixed, proving the rot is structural to file-based ledgers rather than to one implementation	3.607% accepted, 78.637% unattributed today DERIVED
6 Application substrate	≥1 tenant field in the schema, ≥1 non-self paying customer, and ≥10,000 chained ledger rows — all three, not any one — which would falsify the decision <i>not</i> to productise it	None of the three met

- Anti-recommendation for this section: **do not quote any of these numbers in public without re-deriving them at write time**. The record already contains the failure — "90% floor" appears in two internal documents while live is 95.1194% at n=24,669 **DERIVED**, and "110 daily trace files" appears in three prior grounding documents while the daily ledger is 63 files, because the 110 counts 46 **.lock** files and a **steps/** subdirectory **MEASURED**. A thesis this dependent on measurement is falsified fastest by its own stale citations.

§66 What we still do not know

66.1 The count, measured here

MEASUREMENT	VALUE	HOW
Document size	5,092 lines · 100,114 words	<code>wc -lw</code> [measured]
SS tags, whole file	81	<code>grep -o '\[SS[^\]]*\]'</code> [measured]
— in the body, lines 1–4898	64	[measured]
— inside §55 itself, lines 4899–4963	15	[measured]
— after §56 (§58 row, closing line)	2	[measured]
Body tags that are <i>references to the tag</i> , not claims	6	lines 80, 100, 955, 1192, 1347, 4676 [measured]
Claim-bearing SS tags	58	64 – 6 = 58 [derived]
Tag variants beyond bare SS	8	SS/FETCHED · SS, VENDOR-SOURCED · SS, UNVERIFIED ×2 · SS, AUGUST 2026 · SS 2026-08-29 · SS – MUST NOT BE PUBLISHED AS FACT · SS – THE CBIC NOTIFICATION PAGE WAS NOT OPENED... [measured]
Full live census	FETCHED 874 · MEASURED 465 · DERIVED 198 · INFERENCE 144 · SS 81	[measured]

All 58 claim-bearing tags were read and classified; no sampling was necessary [measured].

This document gives three different answers for its own **SS** count — §55 says 52, §58 says 47, and the file contains 58 — and none of the three is the one a reader would arrive at. Deltas: 58 – 52 = 6, 58 – 47 = 11 [derived].

The 52 is traceable. §55's opening census — **FETCHED** 86 · **SS** 52 · **MEASURED** 50 · **DERIVED** 9 · **INFERENCE** 1 — is the census recorded at `docs/research/agent-reports-2026-08-29-r11/c1-research-gap-audit.md` line 8, which measured a PRD of **1,551 lines / 20,174 words** [measured]. This document is **3.28x the lines and 4.96x the words** of the one that census describes [derived: 5,092 ÷ 1,551; 100,114 ÷ 20,174]. The row was inherited, not re-derived — the exact failure §41 and §57 exist to prevent.

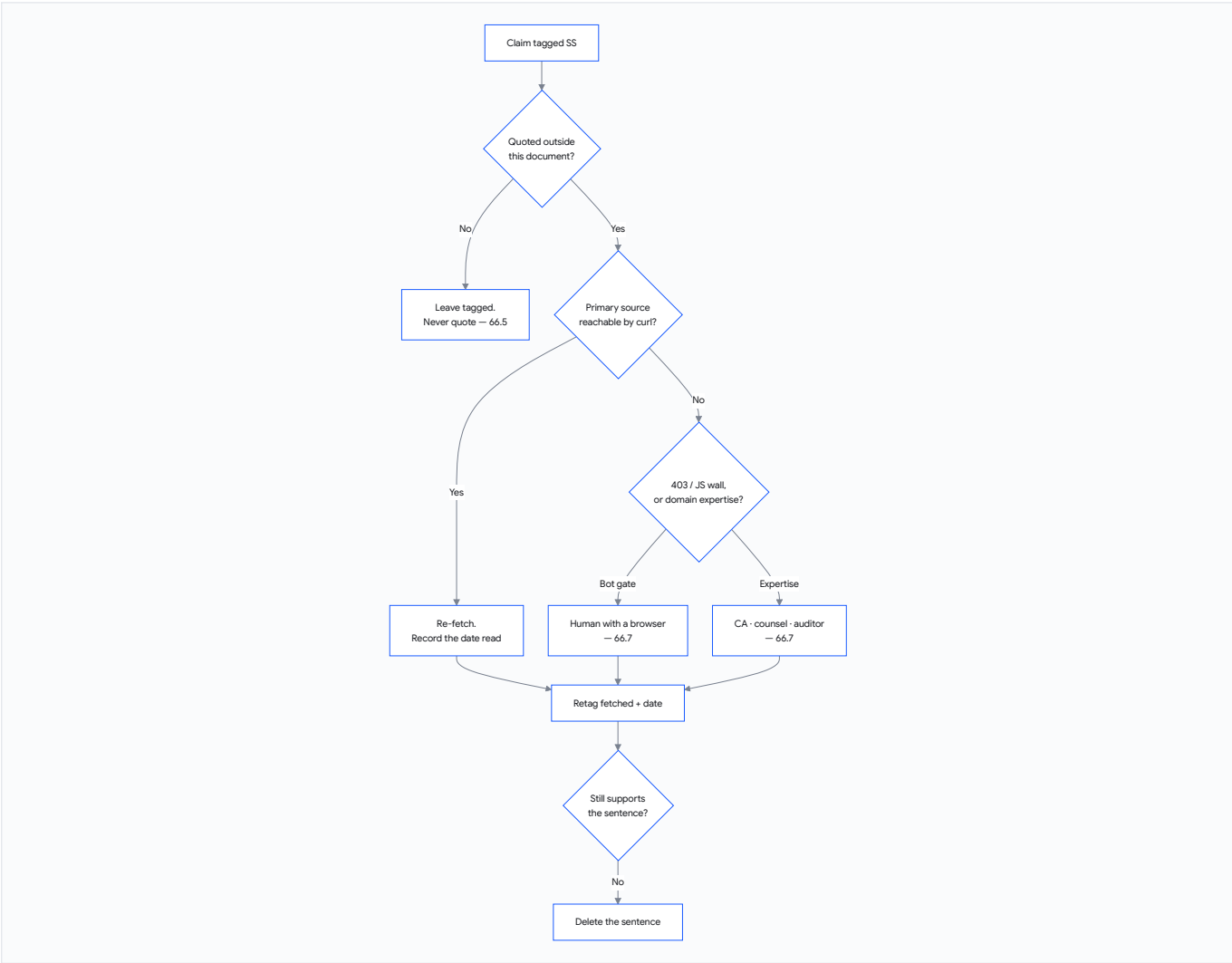
66.2 The existing register describes a document that no longer exists

§55 CLAIM	STATUS IN THE V2 BODY (LINES 1-4898)	EVIDENCE
§55.2 #1 Princeton GEO "25-40% visibility lift"	Absent. Princeton = 0 hits	[measured]
§55.2 #9 "Perplexity Pro free via Airtel to ~400M, worth ₹17,000/yr"	Absent. 400M = 0, 17,000 = 0	[measured]
§55.2 #11 Dataview "~30s past 3,000 notes"	Absent. 3,000 notes = 0	[measured]
§55.2 #12 r/ObsidianMD ~344,000 / Discord ~195,000	Live and still untagged, §26 line 2519	[measured]
§55.3 "never researched"; accessibility, DR, sync, API, roles, i18n, desktop, analytics	Mostly closed by R12. RPO 35 · RT0 24 · WCAG 22 · rate limit 9 · PostHog 5 · takedown 4 · moderation 4 · screen reader 3 · service worker 2 · privacy policy 2 · schema version 1	[measured]

The register is stale in both directions: it warns about three claims that were deleted and under-counts the live ones by six, while its "never researched" list names fourteen areas that now carry their own sections. Treat §55 as a historical artefact of PRD v1.1 and this section as the live one.

- **Anti-recommendation:** do not delete §55. It is the record of what was true on 2026-08-29 and deleting it destroys the audit trail that makes this drift visible. Mark it superseded, keep it, and point §55's header at §66.

66.3 The closure ladder



The ladder's only non-obvious rung is the last one. A claim that survives re-fetching may still fail to support the sentence it was written into — the Pimentel notebook figure carries a specific reproducibility definition, and the sentence in §5 does not state which one [inference].

66.4 Load-bearing, ranked by risk if wrong

Rank = (money or legal exposure) × (irreversibility) × (would be quoted). Costs are [inference] unless tagged otherwise.

#	CLAIM	\$	WHY IT MATTERS	WHAT WOULD CLOSE IT	COST	RISK IF WRONG
1	Export-of-services zero-rating · LUT vs pay-and-refund · "convertible foreign exchange" · reverse charge on imported services (LLM APIs, hosting, MoR fees)	\$51.2, L4756–4758	Decides whether every export invoice is zero-rated or carries IGST. <code>cbic-gst.gov.in</code> failed TLS and 404'd on every IGST-Act path [measured]	A chartered accountant's written opinion with the MoR contract in front of them	₹15,000–₹50,000 + one meeting	Retrospective IGST + interest + penalty on every invoice since the first. It also changes the MoR choice, which §53 makes a hard gate before the first paid signup
2	E-invoicing threshold AATO > ₹5 crore, sticky across FYs · ₹20 lakh services registration threshold · FEMA/EDPMS closure and FIRC/BRC per remittance	\$45, L4287 · L4314 · L4293	§45 derives the crossing at ₹5,000/customer/year ≈ ₹417/month [derived]. The CBIC notification page was never opened	Same CA engagement; ask for it in the same sitting	Included above	Missed e-invoicing is a penalty regime, not a correction. Missed EDPMS closure blocks future inward remittance
3	India IT Rules 2021: the "50 lakh" significant-social-media-intermediary threshold	\$44, L4119	The Rules defer to a Central Government notification that is not in the Rules (Rule 2(1)(v)) [fetched]. Whether a one-to-many publishing tool is a "social media intermediary" at all is [inference]	Indian counsel, one written note covering Rule 2(1)(v), 2(1)(w) and the 24-hour acknowledgement duty	₹40,000–₹1,20,000 (itself SS, UNVERIFIED , \$44 L4194)	D15 ships publish either under obligations that do not apply, or without ones that do
4	EU Art. 13 representative €200–€500/month · Indian counsel ₹40,000–₹1,20,000	\$44, L4194	§53 makes the EU representative a date-gate before the first EU free signup , not an effort-gate	Two written quotes	One week of email; free	A recurring cost line that gates launch, budgeted from a search summary
5	"International gateways without UPI lose 30–40% of Indian checkouts" SS, VENDOR-SOURCED	\$24, L2441	Chooses Razorpay over Paddle / Lemon Squeezy. §53: migrating billing counterparties mid-flight breaks the FEMA paper trail	Razorpay's or NPCI's own published funnel data; failing that, decide the rail on the FEMA/MoR question and demote this to a prior	1 hour	An irreversible rail chosen on a vendor's own marketing number
6	"By month 24 the median solo B2B founder's revenue is more than 4× the median solo B2C founder's"	\$21, L2221	The two-motion strategy — D2C acquisition, B2B monetisation — rests on this one sentence. No study is named	Name the study and its n, or delete the sentence and keep the reasoning	20 minutes, or free	The first investor or HN commenter asks for the source and there is none
7	"21.9M India GitHub contributors, +5.2M in a year, +35% YoY consumer app spend"	\$24, L2439	Underwrites India-first distribution and the ₹299 anchor — against §24's own [derived] finding that ₹299 needs 59.5% more paying humans than \$5 for identical revenue	GitHub Octoverse, one page	20 minutes	The single most checkable number in the document, in the section where being wrong costs the most
8	"Zero category-specific willingness-to-pay evidence for markdown tools in India" · "India's markdown community is greenfield" · moat #1 "does not exist yet"	\$24 L2439 · §24 L2339 · §27 L2546	Three absence claims from one source. They rank the moats and sequence the market	An absence claim cannot be closed by searching. State the searches run, the date, and the languages — bound it instead of asserting it	1 hour	Highest social blowback per word: one reply from someone who runs the thing you said does not exist
9	Support deflection "18% median, 40–60% with AI, \$25–35/ticket"	\$21, L2227	Prices ICP #3 at \$99–249 per knowledge base	Open Zendesk's, Intercom's and Document360's own published benchmarks	1–2 hours	Goes into B2B collateral aimed at three vendors who publish their own, different numbers
10	"Vanta and Drata charge \$7,000–\$30,000/yr for continuous audit trails"	\$21, L2235	The ROI headline of the audit-trail wedge	Two pricing pages	15 minutes	Two named vendors' prices, misquoted in a sales deck
11	Code-signing: hardware-token surcharge +50–150 USD, DigiCert token +120 · max validity drops to 460 days from 2026-03-01 · DigiCert stops 2- and 3-year code-signing certs from Feb 2026	\$39, L3734 · L3749	§39 already records that all Windows cert prices are reseller quotes and SSL.com's product pages 404'd through curl [measured]. If the two deadlines are real, multi-year prepayment is no longer a lever	A CA's own pricing page plus the CA/Browser Forum ballot record	1 hour	Budgeting a cost lever that has already expired
12	"561 of 3,220 HN comments" — sync silently destroys data	\$2 #2, L174	Problem #2, the headline pain, and the justification for T0 being the first customer-visible lane. The cell carries SS and FETCHED simultaneously [measured]	Resolve to one tag; publish the query, the date window and the classifier	2 hours	The product's stated reason to exist is tagged two ways in one table cell
13	CommonMark "0.31.2, released 2024-01-28; still the current release as of 2026-08"	\$8, L445	The spec floor the entire substrate is defined against	<code>commonmark.org</code> , one fetch	5 minutes	Building the floor against a superseded version
14	"Under 4% of GitHub notebooks reproduce" (Pimentel 2019)	\$2 #5 and §5 (3 body occurrences)	§5's survivors-and-traps table — the projection law's clearest external evidence	Open the paper; quote its reproducibility definition alongside the figure	30 minutes	LR#72's exact class: a specific figure, a specific paper, a specific definition, never opened
15	"OpenAI removed Canvas in May 2026"	\$3.1, L205	§3.3's structural hedge against chat-to-artifact rests on it	OpenAI's changelog or release notes	15 minutes	A falsifiable claim about a named company carrying a structural argument
16	"Notion cut free AI to 20 responses <i>for life</i> and raised Business ~20%; Microsoft's +43% Copilot bundling drew a CMA probe"	\$2 #8, L180	§24.8's pricing-promise section	Notion's pricing page and the CMA case page	30 minutes	Names a regulator and two competitors' pricing, inside a section whose argument is that <i>they</i> misstate things
17	"Relay proved the path with 172,544 downloads of a commercial service's bridge plugin"	\$26, L2525	The "Open in frontmatter" plugin is named the primary distribution play	Obsidian's community-plugin stats JSON	10 minutes	The distribution thesis rests on one competitor's download count
18	"GitBook's dominant complaint cluster is reliability and lost work — the gap is real today"	\$27, L2547	Moat #2's only fresh external evidence that the 18–36-month window is real	GitBook's public issue tracker and community, with counts and a date window	2 hours	The moat duration argument loses its evidence and keeps its number
19	ADA Title III "no technical standard promulgated" · axe "covers roughly a third of WCAG issues"	\$35, L3339 · L3414	A US legal posture and the (correct) anti-recommendation against an axe CI threshold	Counsel for the first; Deque's own published coverage claim for the second	20 min + counsel time	A legal posture stated as fact in a procurement conversation
20	<code>bBlocksToMarkdownLossy()</code> "is a real API name"	\$6 L111, §6.2 L328	Principle #3's entire published justification — "the market's own	Nothing . §36 line 3463 already tags the identical fact	5 minutes	Publishing the flagship line under SS when the fetched

#	CLAIM	\$	WHY IT MATTERS	WHAT WOULD CLOSE IT	COST	RISK IF WRONG
			confession"	FETCHED . Retag		evidence is three sections away
21	r/ObsidianMD ~344,000 / Discord ~195,000 — untagged	\$26, L2519	Channel sizing for the whole GTM plan	Read the subreddit header and the Discord landing page	5 minutes	An untagged number reads as measured, which is worse than an honest SS

- **Recommendation:** close rows 13, 15, 16, 17, 20 and 21 first. Six rows, under two hours, no professional required, and they remove the six most publicly checkable exposures.
- **Anti-recommendation:** do not close rows 1–4 by reading more. Tax and intermediary law is not a research problem with a better search query; the primary hosts already refused, and an agent's confident summary of a paywalled clause is exactly the artefact §51 exists to forbid.

66.5 Decorative — may stay unverified, may never be quoted

Each of these can be deleted without changing a single decision. Keep them as texture; never move them into a slide, a landing page, or an investor answer.

- org-mode's agenda "running since 2003, computed on the fly from date tags" (§5) — illustrative; the projection law does not depend on it.
- Obsidian Bases: "the .base file saves only how you want to look" (§5) — a contrast, not evidence.
- "Notion formulas cannot aggregate across rows" (§2 #5, §5, twice) — colour on a point Coda's export already carries.
- "Coda's export loses formulas, buttons, canvas properties" (§5) — same point, second vendor.
- "Notion users write ad-blocker rules against AI buttons" (§6.2) — an anecdote decorating an already-settled non-goal.
- "VS Code's own wiki names extensions the #1 performance suspect; Typora's most-requested feature has 251 votes" (§6.2) — the no-marketplace decision is settled on blast radius, not on these.
- "Granola: \$1.5B valuation on template-typed capture" (§12) — a valuation is not evidence that typing at capture works.
- "Litmus at \$500/mo proves the testing layer monetises" (§3) — supports an argument, not a plan.
- Lex's review loop "in development" (§3) — a competitor's roadmap; the same row's other two cells are FETCHED .
- "Writing editors have nothing" for byte-anchored provenance (§3) — keep as reasoning; §58 already narrows the publishable form.
- "The view IS the data; export lossy by architecture" (§3) — a category summary; the load-bearing instance is row 20 above.
- "Provenance evaporates at accept" for Cursor / Windsurf / Lex / Grammarly (§3, §2 #4) — one instance is enough and it is already triaged.
- Prince and Typst footnote support and repeating table headers (§9, two cells) — our own column is MEASURED ; theirs is decoration.
- "No PRD standard exists; the circulating SRS clause list is SS from a secondary encyclopedia page" (§14) — already fenced by §14's own refusal to publish a clause list.
- The ACM row (§42.2) — already labelled "do not cite as verified". Correct as written.
- Failed-payment recovery, 47.6% vs 12.7% (§45) — already fenced with "do not set a target from either". Correct as written.
- "No primary source for a competitor's activation threshold was opened" (§47) and "no invite-to-activation figure was reachable" (§37) — self-cancelling tags. Correct as written; these are the model.
- ₹95.4/USD (§36) — superseded by §57's live ₹95.39 / ₹95.59 [measured]. Retag or delete, do not re-verify.
- DPDP Gazette number G.S.R. 846(E) (§47) — the date (14 Nov 2025) is FETCHED, PIB ; the number is decoration on top of it. Open the Gazette or drop the number.
- Cowlishaw's 1977 STET editor and *Stet* public-commenting software (§52) — historical colour. The collision that actually decides the name is elberacasa/stet shipping stetmark on 2026-08-05 at 5,753 dl/mo, and that is MEASURED .

66.6 Researched by nobody

R12's fourteen agents closed most of §55.3. What remains at or near zero in this document [measured, word-boundary counts over 100,114 words]:

GAP	EVIDENCE	CONSEQUENCE
SEO for published pages	open graph O · meta tag O · robots.txt O · Core Web Vitals O · sitemap 1	Publish is half the Power tier and its discoverability surface is undesigned
Transactional email and deliverability	SPF O · DKIM O · unsubscribe O · transactional email O · DMARC 1 (and that hit is GitLab's backup-alarm postmortem, not our sender config)	The review surface and the conflict inbox are inert without notification; §32's own dead-man switch depends on mail arriving
Browser consent surface	cookie O · cookie banner O (consent 10 hits are DPDP-side)	§47's funnel is unobservable without a lawful basis stated in a banner
Platform AI-disclosure norms	§55.1: the YouTube help page returned 1,420,253 bytes of navigation chrome and no policy body	§42 cannot state what any platform requires of published AI-assisted content
PRC syntax	label Not opened	The exact mandated text-label form is unknown; §42 must not assert one
Digital Omnibus effective date	The amending act URL returned a 404 page	The 2 Aug vs 2 Dec 2027 disagreement in §42.1 stands unresolved

- **Anti-recommendation:** do not open a research round for these. Five of the six are configuration decisions a builder makes in an afternoon once a surface exists; only the last three need a source, and two of those need a human (§66.7).

66.7 What only a human can close

#	NEED	WHY NO AGENT CLOSES IT	GATE
1	Bot-gated: ACM publications policy	HTTP 403 to an agent, 200 to a browser [PRD §55.1]	Before §42.2's ACM row is repeated to any academic customer
2	Bot-gated: OpenAI classifier-withdrawal post	HTTP 403 plus a JS wall [PRD §55.1]	Before any claim about detector retirement appears anywhere
3	The visual design pass — never run	§7.4's defect is read out of <code>src/app/globals.css</code> , not seen on a screen. Browser verification was BLOCKED : the Next 16 + Turbopack dev server reproducibly crashed compiling the heavy workspace route across 4+ attempts over 2 sessions [measured, project memory via <code>research/frontmatter-raw-corpus.json</code>]. In the r7 naming session both WebFetch and the browser pane were session-gate-refused [measured, <code>agent-reports-2026-08-29-r7/x4-external-frontier-importer-naming.md:133</code>]	R0.10 cannot be marked done without it. A shipped <code>--accent:#18181b</code> with 6/8/12px radii, and a <code>body-faint</code> token failing AA, have never been looked at
4	GST position — export test, LUT vs pay-and-refund, convertible-foreign-exchange, reverse charge, e-invoicing, EDPMS	Chartered accountant, with the MoR contract	Before the first paid invoice (§53)
5	DPDP + IT Rules 2021 — data-fiduciary duties, SSMI threshold, grievance officer, 24-hour acknowledgement, DMCA agent	Indian counsel + a US agent of record	Before one-toggle publish reaches the public internet
6	Trademark clearance	USPTO TESS, EUIPO, IP India/TMR, WIPO Madrid, unregistered common-law rights — UNCHECKED and not checkable here (§52.4). Handle availability is a different question in kind	Before any spend on domains, logo or launch copy, and before choosing between a qualified "frontmatter" and a coinage
7	Domain availability, all of §52	No registrar or RDAP host is in the sandbox allowlist — every domain in §52 is UNCHECKED (§55.1)	Re-run on the day of registration; nothing reserves them
8	WCAG 2.2 AA conformance + ACR/VPAT	Accessibility auditor	Before any B2B procurement conversation
9	Whether an EU AI Act obligation is discharged	The customer's own counsel. A solo founder in India must never be the party asserting EU compliance (§42.6)	Before any EU compliance language ships
10	PRC labelling conformance	A PRC-qualified adviser who can read the incorporated standard	Before any China-facing export claim
11	Independent adversarial review of the engine's own verifiers	A reviewer who did not write them — LR#60	Before the corpus gate is cited as proof of anything
12	The M3 two-device run	§28.5 already assigns it to the founder: "watched by a human, recorded"	M3
13	D5, the two unrotated PATs	§54: "Only you. Today"	Today
14	Whether a stranger will pay	§66.8	—

The distinction in rows 1–2 versus row 7 matters because the fixes differ: a bot gate needs a human with a browser and five minutes; the registrar block needs network access the agent does not have and cannot request. Rows 4, 5, 8, 9 and 10 need a person who carries professional liability for the answer, which is the only real reason to pay for one.

66.8 The single most dangerous unverified assumption

That anyone outside this building will pay for byte-fidelity.

TEST	RESULT
<code>interview</code> in 100,114 words	1 — and it is Ink & Switch's Upwelling interviewees from March 2023, other people's research (§34, L3274) [measured]
<code>interviews</code> · <code>design partner</code> · <code>waitlist</code> · <code>beta tester</code> · <code>pre-order</code> · <code>landing page</code> · <code>user test</code> · <code>focus group</code> · <code>willingness to pay</code>	0 each [measured]
<code>customer discovery</code> · <code>design partner</code> across all 105 reports and 374,866 words	0 each [measured]
Where a paying non-founder first appears in the plan	M7, the final gate: "M0–M6 green + MoR live + EU rep appointed + ToS published + one paying non-founder account" (§28.4)
M7 calendar date	2027-04-28, a Wednesday — 241 days from today [measured: <code>date</code> ; derived]

Everything else in this register is a number that might be wrong. This one is a number that does not exist. The document contains 105 reports, seven markdown engines, 8,513 pinned third-party files, a foreign-corpus gate, a degradation certificate, three merchant-of-record analyses, an Indian tax position and an EU representative — and not one conversation with a person who might hand over ₹299. Section 24 argues at length about ₹299 versus \$5 and derives that the rupee price needs 59.5% more paying humans for identical revenue [derived]. Both figures assume a numerator nobody has observed.

Cost of being wrong: 241 days of solo build to 2027-04-28, plus the R0 lane that §53 calls "a multi-month lane before any customer-visible feature", plus the non-recoverable spend the plan front-loads by design — the MoR before the first paid signup, the EU Art. 27 representative before the first EU *free* signup, the CA, the lawyer, the trademark attorney, the accessibility auditor. The sequencing in §53 is correct for a product with demand and is the most expensive possible ordering for one without it: every irreversible commitment lands *before* the first signal.

- Recommendation:** put one falsifiable demand test in front of R0, not after M6. The cheapest form consistent with §58's publication ban is a page that shows the *refusal contract* — "here is what this tool will refuse to do to your file, and why" — with a paid-waitlist button, no fidelity percentage, no engine number, nothing tagged `ss`. That is publishable today under §58's own rules, because it markets a behaviour rather than a measurement.
- Anti-recommendation:** do not build the demand test into the product, and do not let it reorder R0. NF-1, NF-3 and CI are correctness work that must happen regardless of what the test says, and §28.3 already puts R0.5 in the first 48 hours. A landing page that delays the engine has converted one unverified assumption into two.
- What would falsify the recommendation:** if fewer than 20 people leave an email against the refusal-contract page in 30 days, the answer is not "market harder" — it is that the wedge is wrong and §21's two-motion strategy needs rewriting before month four, not month nine. If the test cannot be run without publishing a fidelity number, do not run it; §58's ban outranks this section.

§67 MDMAX as built — a line-level audit

3,614 lines across 13 files under `src/modules/mdmax/`, plus the 282-line splice writer and the 223-line CLI, read in full and executed here. 851 tests across 23 files pass in 1.82 s [measured: `npx vitest run test/mdmax test/share`]. The suite is green over a document-offset conversion that returns wrong answers, a certifier that silently drops blocks, and a CLI that cannot start. There is no `.github/workflows/` directory [measured: `ls` returns "No such file or directory"].

67.1 The fifteen files

FILE	LINES	WHAT IT ACTUALLY DOES	PUBLIC EXPORTS	Q	CONCERNS
<code>domain/cert-contract.ts</code>	168	Types only — four verdicts, <code>Engine</code> , <code>Target</code> , <code>Certificate</code> , <code>CertFailure</code> . No runtime code.	11 types/interfaces	B+	Its own rule "class ... Required whenever verdict === 'CORRUPT'" is a comment; <code>CellVerdict</code> declares <code>readonly class?: VerdictClass</code> on a non-discriminated shape, so a CORRUPT cell with no class typechecks. <code>brokenHistogram</code> already counts an 'UNCLASSSED' bucket for exactly that.
<code>domain/normalize.ts</code>	62	NFC → collapse an explicit 16-codepoint whitespace class → trim → lowercase → NFC.	<code>normalize</code> , <code>normalizeStamp</code> , <code>NORMALIZE_VERSION</code>	A	Pure, versioned, idempotent [measured]. Zero callers anywhere, including the anchor hash it was written to be. <code>.trim()</code> is in the code and absent from the documented pipeline "NFC → collapse whitespace → lowercase → NFC".
<code>domain/slug.ts</code>	115	Wraps <code>github-sluggger</code> ; resolves an anchor canonical-first then dash-collapsed.	<code>slug</code> , <code>collapsedSlug</code> , <code>slugDocument</code> , <code>resolveAnchor</code> , <code>decodeAnchor</code> , <code>slugStamp</code> , <code>SLUG_VERSION</code>	B	<code>resolveAnchor</code> , <code>collapsedSlug</code> , <code>decodeAnchor</code> and <code>slugStamp</code> have zero call sites and zero tests [measured]. The product slugs headings in <code>src/modules/preview/presentation/outline-utils.ts</code> with its own <code>new GithubSluggger()</code> , in the presentation layer, past this module entirely.
<code>domain/frontmatter-prepass.ts</code>	121	Quotes bare <code>[...]</code> values for a reader; refuses <code>TRIPLE_OPEN = /\:\s*\[\s*\[\s*\[\/</code> .	<code>prepassFrontmatter</code> , <code>extractWikilinks</code>	B-	Zero callers. <code>src/modules/vault/infrastructure/markdown-parser.ts</code> ships a private <code>extractWikilinks</code> of its own — two implementations, the product uses the other one.
<code>domain/targets.ts</code>	133	Static registry of 15 (product, surface) pairs.	<code>TARGETS</code> , <code>LOCAL_TARGETS</code> , <code>uncertifiableShare</code> , <code>targetById</code> , <code>DECLARED_LAST_VERIFIED</code>	B	<code>DECLARED_LAST_VERIFIED = '2026-08-01'</code> is a literal with no staleness check in code. Nine of fifteen rows carry <code>engineId: 'remark-app'</code> — the app's own pipeline stands in for Obsidian, Notion, Typora, Bear, Slack, Discord. <code>targetById</code> uncalled.
<code>domain/offsets.ts</code>	315	Branded <code>U16Offset / ByteOffset</code> ; <code>OffsetMap</code> with a 512-unit checkpoint table both ways.	<code>u16</code> , <code>u16OrThrow</code> , <code>unsafeU16</code> , <code>unsafeByte</code> , <code>splitsSurrogatePair</code> , <code>OffsetMap</code> , <code>utf8Length</code>	C	<code>toU16</code> returns <code>text.length</code> with <code>ok: true</code> for real interior byte offsets whenever <code>text.length % 512 === 0</code> (D1). <code>toByte</code> is unaffected.
<code>domain/shape-gate.ts</code>	174	Strict UTF-8 decode; one <code>O(n)</code> pass for bytes/lines/list-markers with early exit.	<code>decodeStrict</code> , <code>shapeGate</code> , <code>explainFailure</code> , <code>MAX_BYTES</code> , <code>MAX_LINES</code> , <code>MAX_LIST_MARKER_LINES</code> , <code>BUDGET_MS</code>	B-	<code>shapeGate</code> itself has zero call sites; only <code>decodeStrict</code> is imported (D5). Its invalid-UTF-8 column is wrong on multi-byte input (D4). 19,001 list lines gate in 2 ms [measured].
<code>domain/placement.ts</code>	154	Insert-then-reparse-then-compare block skeletons; refuse on divergence.	<code>blockSkeleton</code> , <code>nearSetTextUnderline</code> , <code>safeInsert</code> , <code>types</code>	C+	Refuses safe insertions before any <code>---</code> thematic break, and permits insertion inside a fenced code block and inside an inline code span (D3). Zero callers.
<code>domain/constructs.ts</code>	618	19 constructs, each with a masked byte-range detector over a cached document scan.	<code>CONSTRUCTS</code> , <code>CONSTRUCTS_BY_ID</code> , <code>inventory</code> , <code>skipRegions</code>	B+	Best-built file here; every detector finds its own <code>source</code> [measured: 0 failures]. Module-level <code>let cachedText / let cachedScan</code> retain a whole document plus its <code>OffsetMap</code> after the last call. Six of nineteen entries carry <code>provenance: 'invented - ...'</code> [measured: <code>grep -c "invented -" = 6</code>].
<code>domain/fold.ts</code>	430	Single-pass HTML segmenter, then six named normalisations; returns which fired.	<code>fold</code> , <code>foldEqual</code> , <code>foldStamp</code> , <code>FOLD_RULES</code> , <code>FOLD_VERSION</code>	A-	Idempotent on 12 adversarial inputs [measured, 0 failures]; 57 KB→6 ms, 453 KB→26 ms [measured]. <code>foldStamp</code> is test-only. Never run over the pinned corpus, as its own header states.
<code>domain/verdict.ts</code>	411	The PASS/VOID/LEAK/DESTROY/MUTATE/STRIP ladder over source payload vs rendered haystack.	<code>classify</code> , <code>extractPayload</code> , <code>renderedHaystack</code> , <code>missingPayloadTokens</code> , <code>producesNoOutput</code> , <code>VERDICT_VERSION</code>	B+	<code>renderedHaystack</code> references <code>TAG_ANY</code> above its <code>const</code> declaration. <code>unresolvedReferences</code> filters with <code>folded.text.includes(literal)</code> — a raw needle against a folded haystack, the exact mismatch the file fixes two functions later. <code>VERDICT_VERSION</code> uncalled.
<code>application/certify.ts</code>	375	Blank-line block split, per-target render, modal-consensus reference, cell loop, histogram.	<code>splitBlocks</code> , <code>certify</code> , <code>brokenHistogram</code>	C+	Drops the first content block when no blank line follows front matter (D2). Its fence regex disagrees with <code>constructs.ts</code> (D7). <code>void oracle</code> (D9). <code>blockAnchor</code> seeds on <code>`\${start}:\${text}`</code> , so one inserted character re-keys every anchor after it.
<code>infrastructure/bench.ts</code>	538	Builds and version-pins 7 engines; sha256 over sorted (id, version, canonical options).	<code>loadBench</code> , <code>computeBenchId</code> , <code>stripJekyllFrontMatter</code> , <code>BENCH_ENGINE_IDS</code> , <code>JEKYLL_KRAMDOWN_OPTIONS</code>	B	Loads and certifies correctly when driven directly: 6 JS engines, 6 blocks × 6 targets in 30 ms [measured]. <code>DEFAULT_RUBY_BIN = "/opt/homebrew/opt/ruby/bin/ruby"</code> . Only the ruby engine has a timeout; the six JS engines run unbounded on the caller's thread.
<code>share/domain/splice-frontmatter.ts</code>	282	Locate a top-level key's bytes, replace only those, refuse otherwise.	<code>spliceFrontmatterValue</code> , <code>spliceFrontmatterKey</code> , <code>emitValue</code> , <code>emitScalar</code> , <code>SAFE_KEY</code>	B-	The only file with real product traffic. Refuses every zero-indent block sequence (D6); prepends a second front-matter block on a bare-CR file (D8); deletes a trailing <code># comment</code> on set (D10).
<code>scripts/mdmax-cert.mjs</code>	223	The whole CLI surface — <code>table</code> , <code>--json</code> , <code>--histogram</code> , <code>--bench-info</code> , <code>--explain</code> .	—	D	Throws <code>ERR_MODULE_NOT_FOUND</code> before rendering anything; exit 1 (D11).

67.2 The defect list

D1 — `OffsetMap.toU16` silently returns end-of-document. `src/modules/mdmax/domain/offsets.ts`. The table is sized `const blocks = Math.floor(text.length / BLOCK) + 1` but written only inside `for (let i = 0; i < text.length; i++)`, so when `text.length % 512 === 0` the final slot is never assigned and stays `0`. The binary search accepts it: `if ((this.byteAt[mid] ?? 0) <= target) lo = mid`. [measured] At length 512 every byte probe returns `{ok:true, value:512}`; at 1024 and 1536, every probe at or past the last checkpoint does. The failure is `ok: true` — the caller cannot detect it. This survived the previous fix commit, `cc1d451 fix(mdmax): Tier 2 - OffsetMap off-by-3` [measured: `git log`]. `toByte` never reaches the empty slot, so the certificate's `byteRange` is currently correct and nothing shipped is wrong today; anything reading a byte anchor back — a git blob range, an LSP that negotiated utf-8 — is.

D2 — `certify` drops the block after front matter. `src/modules/mdmax/application/certify.ts`, in `mergeLeadingFrontmatter`:
`const rest = blocks.filter((b) => b.start >= fmEnd)`. A block that straddles `fmEnd` is discarded, not truncated. [measured]
'---\na: 1\n---\n# H\n\nBody paragraph.\n' → blocks [0,13] and [18,34]; # H at [13,18] appears in no block, no verdict, and no summary. Insert one blank line and it reappears.

D3 — `safeInsert` refuses the safe case and permits the unsafe one. `src/modules/mdmax/domain/placement.ts`. The pre-check ends
`if (prevLine.trim() !== '' || prevPrev.trim() !== '') return lineNo — an || over a two-line lookback, and SETEXT_UNDERLINE matches bare ---. [measured]
'# Title\n\n---\n\nBody\n' and 'Body text\n\n---\n\nMore\n' both return ADJACENT_TO_SETEXT_UNDERLINE; in both, the --- is a thematic break that no insertion can convert. Conversely, insertion inside a '''js fence and inside 'a b c' both return {ok:true} — the marker lands as visible code-block text while the block skeleton is unchanged, which is the only thing compared.`

D4 — invalid-UTF-8 location comes from a binary search over a non-monotone predicate. `src/modules/mdmax/domain/shape-gate.ts`, `decodeStrict`:
`decoder.decode(bytes.subarray(0, mid))` fails both for a truncated valid character and for the real bad byte, so "longest valid prefix" is not a sorted property. [measured] `é` + `0xFF` → col 2 (right); `ée` + `0xFF` → col 1 (should be 3); `éeé` + `0xFF` → col 4 (right). This is the only mdmax defect that reaches a user: the wrong column is printed by `search-index.ts` and surfaced by `explainFailure` as "This file is not valid UTF-8 (line ..., column ...)".

D5 — no shape gate runs anywhere. `shapeGate`, `MAX_BYTES`, `MAX_LINES`, `BUDGET_MS` have zero product call sites [measured]. `WIKILINK_RE = /(?!?)\[\[([^\]]+)\]\]/g` — the regex the gate's own header cites as "k = 1.98 — 36,865 ms on 320 KB of `[[ ]` — is live and ungated in `src/modules/vault/infrastructure/markdown-parser.ts` and again as `COMBINED_WIKILINK_RE` in `src/modules/preview/presentation/markdown/wikilinks.tsx`.

D6 — the 83% refusal, in one branch. `splice-frontmatter.ts`: `}` else if (text !== '' && !indented && !/^#.test(text)) { return src }, whose own comment names "a sequence item". [measured] `'---\ntags:\n- a\n- b\n---\n'` returns the input unchanged; the same YAML indented two spaces splices fine.

D7 — two fence grammars in one module. `certify.ts` uses `/^[ \t]*(\{3}\{3}\{3})/` — constructs `uses /'{0,3}('{3,3}({3,3})[ \t] (.)/$. CommonMark caps fence indentation at three spaces. [measured] 'Intro\n\n \n\nPara A\n\nPara B\n' splits into two blocks — everything after an indented ''' collapses into one. The correct regex is already in the repo, one directory over.`

D8 — bare-CR set-destruction. `FM_OPEN = /^[---[ \t]*(\r?n)/` requires `\n`. [measured] On `'---\ra: 1\r---\rBody\r'`, `spliceFrontmatterValue(src, 'share', 'x')` returns `'---\nshare: x\n---\n\n---\ra: 1\r---\rBody\r'` — a second block prepended, the author's keys demoted to body.

D9 — void oracle. `certify.ts` computes `const oracleId = opts.oracleEngineId ?? 'commonmark'` and `const oracle = engineById.get(oracleId)`, then discards it with `void oracle`. `CertifyOptions.oracleEngineId` is documented as "Retained for `--explain`" and `--explain` does not use `certify`.

D10 — trailing comments are deleted on set. [measured] `spliceFrontmatterValue('---\nshare: old # keep me\nb: 2\n---\n', 'share', 'new')` → `'---\nshare: new\nb: 2\n---\n'`. `# keep me` is gone. The sibling `spliceFrontmatterKey` explicitly promises "any trailing comment ... preserved".

D11 — the CLI cannot start. `scripts/mdmax-cert.mjs` does `await import(R('..src/modules/mdmax/application/certify.ts'))`, and that file's `from` `'../domain/offsets'` has no extension, which Node ESM does not resolve. [measured] `node scripts/mdmax-cert.mjs <file>` → `ERR_MODULE_NOT_FOUND ... /domain/offsets,exit 1; --bench-info` likewise. `bench.ts` additionally imports `@/modules/...`, which plain Node cannot resolve either. `const R = (p) => new URL(p, import.meta.url).pathname` will also mis-decode any repo path containing a space.

67.3 Where the code contradicts its own comments

`placement.ts` states "inline content is deliberately excluded" from `blockSkeleton`, and its `INLINE` set omits `linkReference`, `imageReference`, `footnoteReference` and inline `html` — mdast types every one of those as a node the walk pushes. The forgiveness pass, `removeFirstInsertedHtml`, therefore cannot tell the marker's block `html` from an inline `<b>`.

`splice-frontmatter.ts` introduces `SAFE_KEY` as "the same character class `topLevelKeyLine` scans for, stated once so the two can never drift apart", and then writes the class three times: `SAFE_KEY = /^[A-Za-z0-9_.$-]+$/`, `/^[A-Za-z0-9_.$-]+[ \t]*:(?:[ \t]|$)/` inside `topLevelKeyLine`, and `/^[A-Za-z0-9_.$-]+[ \t]*:[>&*!]/` inside `isUnsafeContinuation`. It also opens with "leave every other byte in the file untouched" and then D10.

`targets.ts` says the uncertifiable share is "7 of ~12 surfaces"; `TARGETS` holds 15 entries of which 7 are `fidelity: 'local'`, so `uncertifiableShare()` returns `8 / 15 = 53.3%` [derived: `(15-7)/15 × 100`].

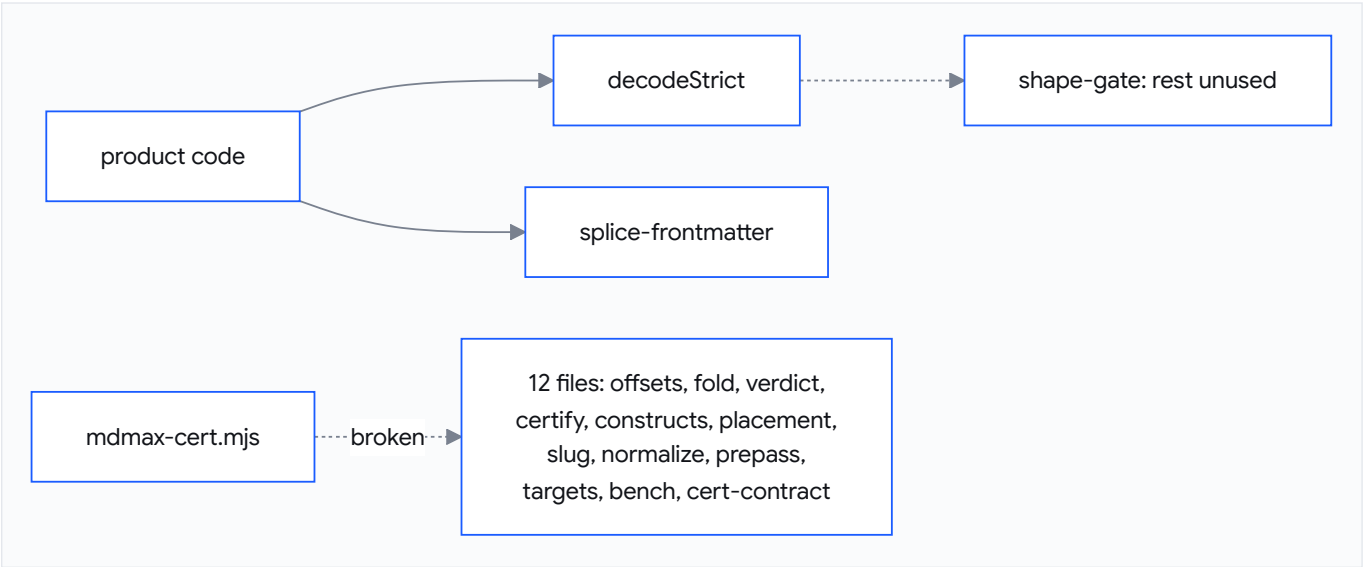
`normalize.ts` is "the hash input for every stored anchor ... this must land BEFORE the first anchor is persisted". `certify.ts` persists anchors from a private FNV-1a over `const seed = `${start}:${text}`` and never imports `normalize`. Position in the seed guarantees the property `normalize.ts` exists to protect: every anchor after an edit re-keys.

`cert-contract.ts` rule 5 — "THE ARTIFACT IS A JSON SIDECAR" — holds. Nothing in the cert path writes to a `.md` [measured: no `writeFile` in `scripts/mdmax-cert.mjs` or `src/modules/mdmax/`].

67.4 Dead and vestigial code

SYMBOL	PATH	CALL SITES	CASE FOR DELETION
<code>void oracle</code> + <code>oracleEngineId</code>	<code>application/certify.ts</code>	0	An option a caller can set that provably changes nothing. Delete both; the consensus reference is the design and the comment above it already explains why.
<code>closeRe</code> + <code>closeRe.lastIndex = 0</code>	<code>share/domain/splice-frontmatter.ts</code>	0	A compiled regex and a reset, immediately superseded by a hand-rolled <code>indexOf('\n')</code> scan. Two lines that make a reader look for a third use.
<code>resolveAnchor</code> , <code>decodeAnchor</code> , <code>collapsedSlug</code> , <code>slugStamp</code>	<code>domain/slug.ts</code>	0 prod, 0 test	The module's whole documented reason to exist — write-strict / read-tolerant — is untested and unwired, while <code>outline-utils.ts</code> slugs for real. Do not delete: wire it, and delete <code>outline-utils.ts</code> 's slugger instead. Anti-recommendation: if the linker is not on the roadmap this quarter, delete <code>slug.ts</code> outright rather than carry a versioned key format nothing keys.
<code>normalizeStamp</code> , <code>targetById</code> , <code>VERDICT_VERSION</code> , <code>foldStamp</code>	four files	0 (<code>foldStamp</code> : test-only)	Stamp accessors for versions that are already exported as consts. Keep <code>NORMALIZE_VERSION</code> / <code>FOLD_VERSION</code> , drop the wrappers.
<code>frontmatter-prepass.extractWikilinks</code>	<code>domain/frontmatter-prepass.ts</code>	0	A duplicate of the private one in <code>markdown-parser.ts</code> . Delete one; the mdmax copy is the better implementation (it splits on `
<code>let cachedText</code> / <code>let cachedScan</code>	<code>domain/constructs.ts</code>	live	Not deletable — it is load-bearing for the 19-detector sweep — but it is module-level mutable state holding a document and its <code>OffsetMap</code> alive after the last call, and its own comment says "This is arithmetic, not a profile — I did not measure it". Move it to a scan object the caller owns.

67.5 The three files most in need of work, ranked



- 1. `scripts/mdmax-cert.mjs` . It is the only thing that can execute the other twelve, and it has never executed them. Every claim about the corpus — 8,513 files, 0 corruption, `mdmax/fold@1` `unrun` — is a claim about code reachable only by hand-writing a Node resolve hook, which is what this audit had to do to produce any number at all. Fix the extensions and the `@/` alias (or run it under `vitest / tsx`), then run it once before anything else here is touched. Anti-recommendation: do not rewrite the CLI into the product build to fix this; a sidecar tool that needs the app's bundler is a tool that stops running the day the bundler changes.
- 2. `src/modules/mdmax/domain/offsets.ts` . The file that says "CONVERSION Exactly one place: OffsetMap. Nothing else in the codebase converts" is the file with a wrong answer that reports `ok: true` . Fix is one line — size the table `Math.ceil(text.length / BLOCK) || 1` — and the test that must fail first is a document of exactly 512, 1024 and 1536 code units, because the current suite's 16 offsets assertions do not contain one [measured]. Anti-recommendation: do not fix it by clamping in the binary search; that hides the empty slot instead of removing it, and the next reader inherits a table whose length lies.
- 3. `src/modules/share/domain/splice-frontmatter.ts` . It is the one file users' bytes actually pass through, and three of its four measured failures are silent: a refusal that looks like a no-op (D6), a prepended duplicate block (D8), a deleted comment (D10). D6 alone is the 83% publish-refusal. The zero-indent sequence is a two-branch change — treat a `-` line at column 0 as a continuation of the preceding `key:` rather than as a bare top-level line — and it is testable against a fixture that fails today. Anti-recommendation: do not reach for a YAML parser to close these; the module's entire value is that it does not parse, and `matter.stringify` is what it was written to replace.

§68 What the engine can actually do, versus what we claim

68.1 The claim-to-code-to-test matrix

Classification: **PROVEN** = code plus a test that fails without it; **IMPLEMENTED** = code, tests that exercise it only against doubles or only in-unit; **PARTIAL** = the claim holds over a narrower population than stated; **ASPIRATIONAL** = claimed, not built.

#	CLAIM (SOURCE)	CODE	TEST	VERDICT
1	Splice contract — locate range, replace those bytes, every other byte bit-identical (§7.1, §60)	src/modules/share/domain/splice-frontmatter.ts — spliceFrontmatterValue , spliceFrontmatterKey	test/share/frontmatter-splice.test.ts (30) + scripts/fm-roundtrip-audit.mjs , scripts/fm-properties-audit.mjs	PARTIAL — proven on the shapes it accepts; it accepts 17% of the corpus MEASURED
2	Refusal is a first-class outcome (§60)	same file — every guard is return src	asserted for duplicate key, unterminated block, unsafe SAFE_KEY , block scalar	PROVEN
3	Never regenerate from a parse tree (§7.1)	stringifyFrontmatterDoc in src/modules/preview/presentation/frontmatter.ts has zero product call sites; only test/preview/frontmatter.test.ts imports it MEASURED	grep	PROVEN
4	Degradation certificate across 7 real engines (§7.1)	application/certify.ts certify , infrastructure/bench.ts loadBench	certify.test.ts (28) uses fake engines throughout; bench.test.ts (31) never calls certify	IMPLEMENTED — the join is untested and, until this session, unrunnable
5	Four verdicts, three classes (§7.1)	domain/verdict.ts classify	verdict.test.ts (49), driving marked 16.4.2, commonmark 0.31.2, markdown-it 15.0.0 in-process	PROVEN in unit, PARTIAL in situ — MUTATE depends on bench composition (§68.2)
6	19 constructs, UTF-8 byte ranges, skip mask (§7.1)	domain/constructs.ts CONSTRUCTS , inventory , skipRegions	constructs.test.ts (43)	PARTIAL — all six audited detector defects still fire MEASURED ; 6 of 19 carry provenance: invented
7	Branded offsets, one conversion point (§7.1)	domain/offsets.ts OffsetMap , u16 , utf8Length	offsets.test.ts (22), cross-checked against Buffer.byteLength at every offset	PROVEN
8	Shape gate refuses hostile documents (§7.1)	domain/shape-gate.ts shapeGate , decodeStrict	shape-gate.test.ts (21)	PROVEN as a function; ASPIRATIONAL as a gate — shapeGate has zero product call sites; BUDGET_MS is documented "enforced by the caller" and no caller exists
9	Every byte written passes placement (§7.1, house rule P5)	domain/placement.ts safeInsert , blockSkeleton	placement.test.ts (16), opens with a genuine red proof	ASPIRATIONAL in product — zero call sites, and it refuses every visible-content insert MEASURED
10	Lenient frontmatter pre-pass, 170/907 = 18.74% (§7.1)	domain/frontmatter-prepass.ts prepassFrontmatter	frontmatter-prepass.test.ts (16), red proof first	PARTIAL — repairs 0 of 21 strict-parse failures in the 7,969-block foreign corpus MEASURED
11	normalize/1 and slug/1 pinned by a golden digest	domain/normalize.ts , domain/slug.ts	pure-functions.test.ts (19) over fixtures/pure-function-golden.json , count: 200	PROVEN
12	99.627% re-anchoring across 384 configurations (§7.1)	none — no anchor store, no re-anchor function exists	none	ASPIRATIONAL
13	Missing engine is a hard refusal; benchId = sha256 over the full option set	bench.ts loadBench , computeBenchId	bench.test.ts — refuses ruby-absent, subset, unknown id, empty and duplicated require	PROVEN
14	The artifact is a JSON sidecar, never written into the .md	certify.ts returns Certificate ; scripts/mdmax-cert.mjs only console.log s	certify.test.ts — "never serialises the render function into the artifact"	PROVEN
15	0 corruption, 0 throws across 8,513 files (§58)	scripts/corpus-foreign.mjs	verify hashes bytes only — it does not run the engine	PARTIAL — 0 throws re-derived here; "0 corruption" is true only across the 17% the writer did not refuse
16	Content-derived anchors, provenance, pack/unpack, near-miss checker, budgeted projection, machine-write zones, land()	none	none	ASPIRATIONAL

npm run verify 's arch gate is green and honest: filesScanned: 208 , violations: [] [MEASURED](#) .

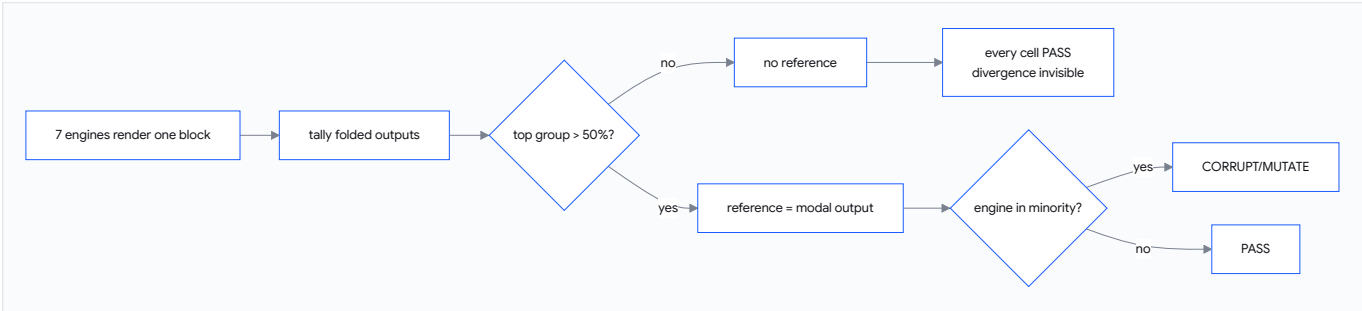
68.2 The dangerous list — claims the code does not support

Every fidelity percentage in circulation was measured on a population the product does not serve. The 2026-08-03 audit's replicated figure — 6.78% of blocks BROKEN on our docs, 5.78% on 30 third-party README.md files, excluding frontmatter-app — was the basis for calling kill condition (1) decided. Run over four pinned Obsidian vaults instead (419 files, 2,530 blocks, the first execution of mdmax/fo1d@1 over pinned-corpus content [MEASURED](#)):

TARGET SET	BLOCKS BROKEN	FILES BROKEN	__UNATTRIBUTED__	K4
all 7	847 / 2,530 = 33.48%	307 / 419 = 73.27%	393 of 852	fires at 98.9%
excluding frontmatter-app	474 / 2,530 = 18.74%	284 / 419 = 67.78%	28 of 484	fires at 98.1%

18.74 ÷ 5.78 = 3.24× [DERIVED](#) . READMEs have no YAML frontmatter and no task lists; vaults are 254 and 172 of the 484 attributions respectively. The number moved because the corpus changed, not because the code did.

CLAIM	WHAT THE CODE DOES	EVIDENCE
"0 corruption across 8,513 files, re-verifiable via <code>npm run corpus</code> " (§58)	<code>corpus-foreign.mjs</code> <code>verify</code> compares <code>sha256</code> against <code>MANIFEST.sha256</code> and exits. No test, oracle or script in the repo runs the writer over <code>test/corpus/foreign/_vendor</code> — <code>grep</code> for <code>corpus/foreign</code> returns <code>corpus-foreign.mjs</code> and three spec files, nothing executable	MEASURED
"the splice writer handles all 170 of these files at 100%" (<code>frontmatter-prepass.ts</code> header)	Over 7,969 foreign frontmatter blocks the writer refuses 6,615 (83.01%); 6,613 of those are one cause, a <code>-</code> item at column zero. 0 throws, 0 corruptions among the 1,354 that proceeded	MEASURED
"MDMAX shipped" (<code>docs/mdmax/PLAN.md</code> , corrected in §7.4)	One symbol of thirteen files is reachable from product code: <code>decodeStrict</code> , imported by <code>src/modules/vault/application/get-snapshot.ts</code> and <code>src/modules/vault/infrastructure/search-index.ts</code> . Both use it as a <code>continue</code> — a file that fails is dropped from the snapshot and the search index with a <code>console.error</code> and no UI	MEASURED
"every byte we write may not change what a user's document means" (§3.8.5)	<code>safeInsert</code> accepts an HTML-comment marker and returns <code>SKELETON_CHANGED</code> for a paragraph, a callout and a heading at a true block boundary. <code>removeFirstInsertedHtml</code> forgives exactly one <code>html</code> node, so visible content is refused by construction	MEASURED
A certificate PASS means the engines agree	On <code>~gone~</code> the seven engines split 3 / 2 / 2 (<code></code> , <code><s></code> , literal <code>~</code>) . <code>certify.ts</code> requires <code>top.count > renderedByTarget.size / 2</code> , so there is no reference, <code>classify</code> never reaches the MUTATE branch, and all seven cells report PASS on three visibly different renderings	MEASURED
A certificate MUTATE means the engine is wrong	On <code>- [x] done</code> the split is 4 / 1 / 1 / 1 . The majority is the four engines that do not implement GFM task lists; <code>remark-app</code> , <code>github-pages</code> and <code>marked</code> — which render the checkbox correctly — are each scored CORRUPT/MUTATE . 172 of 484 attributions in the vault run are <code>task-list</code>	MEASURED
The <code>frontmatter-app</code> target measures our app	<code>bench.ts</code> <code>buildRemarkApp</code> records <code>notReplicated</code> : the React component map, <code>disallowedElements</code> , <code>createRehypeHtmlPolicy(content)</code> , <code>rehype-katex</code> , <code>rehype-highlight</code> , <code>Markdown.tsx</code> runs all five	MEASURED
The engine is a build artifact anyone can run	<code>node scripts/mdmax-cert.mjs <file></code> fails with <code>ERR_MODULE_NOT_FOUND</code> on <code>../domain/offsets</code> . <code>scripts/ts-resolve.mjs</code> was written for exactly this and has zero importers anywhere in the repo	MEASURED



The mechanism is symmetric and both ends are wrong: maximum disagreement yields silence, and correct GFM yields corruption whenever the bench holds four non-GFM configurations against three GFM ones.

Recommendation. Replace the modal-consensus reference with a declared per-construct expectation: a construct's reference is the output of the engines that implement the spec that defines it, named in `targets.ts` . **Anti-recommendation:** do not fall back to the CommonMark oracle — `certify.ts` records that this produced 356 MUTATES on a five-file sample because the spec has no tables — and do not delete MUTATE, because it is the only verdict that catches structural loss with the characters intact.

68.3 The hidden-asset list — capabilities the code has and we neither claim nor use

ASSET	SYMBOL	WHY IT IS WORTH MONEY
YAML-independent wikilink extraction	<code>frontmatter-prepass.ts</code> <code>extractWikilinks</code>	Reads raw value text, so it works on blocks a parser refuses. The graph view already ships (<code>react-force-graph-2d</code>) and cannot see links in unparseable frontmatter
Tolerant anchor resolution that reports <i>how</i> it matched	<code>slug.ts</code> <code>resolveAnchor</code> returning <code>CANONICAL</code> / <code>COLLAPSED</code> / <code>UNRESOLVED</code>	A <code>COLLAPSED</code> match is a link that works here and breaks on GitHub. That is a shippable Doc Health warning today, with no new code
The over-quoting fix	<code>splice-frontmatter.ts</code> <code>emitScalar</code> — indicator characters are special in first position only	The naive rule quoted 435 of 907 corpus files unnecessarily. This is the difference between "preserves your file" and "reformats your file"
Kill-condition arithmetic as a function	<code>certify.ts</code> <code>brokenHistogram</code> , <code>targets.ts</code> <code>uncertifiableShare</code>	Reachable only from the broken CLI. <code>uncertifiableShare()</code> is our honesty number — 8 of 15 surfaces cannot be probed — computed and never surfaced
User-facing refusal copy	<code>shape-gate.ts</code> <code>explainFailure</code>	Seven refusal reasons already written as one-line sentences. §46's refusal UX has its strings and does not use them
Exact Jekyll front-matter semantics	<code>bench.ts</code> <code>stripJekyllFrontMatter</code>	Mirrors Jekyll's <code>YAML_FRONT_MATTER_REGEXP</code> including "unterminated is content". Reusable by the publish pipeline
Grapheme-correct cursor arithmetic	<code>offsets.ts</code> <code>toGrapheme</code> , <code>isGraphemeBoundary</code> , <code>lazy Intl.Segmenter</code> with a code-point fallback	Tested to count a ZWJ family as one grapheme. The editor does its own <code>.slice</code> at 119 sites outside mdmax
Sparse offset index	<code>offsets.ts</code> <code>BLOCK = 512</code> checkpoint table	~8 KB per open 4 MB document against 16 MB for the dense <code>Uint32Array</code> the plan specified — a 2,048× memory saving nobody markets
A named, versioned equivalence relation	<code>fold.ts</code> <code>FOLD_VERSION</code> , <code>FOLD_RULES</code> reported per cell	Competitors ship diffs. We can say which of six normalisations fired on a given comparison

68.4 What the ten test files actually assert, and the largest untested surface

320 tests across 11 files, all green in 1.18 s [MEASURED](#) .

FILE	TESTS	WHAT IT ACTUALLY ASSERTS	BLIND SPOT
<code>verdict.test.ts</code>	49	Each verdict against real <code>marked / commonmark / markdown-it</code> output, plus declared kramdown fixtures with the reproducing command	Every case is hand-built; none comes from the corpus
<code>fold.test.ts</code>	45	Six rules fire in isolation; a "does NOT hide" block; idempotence; §6.11 linearity under a 3 s timeout	Idempotence is over the file's own inputs, not corpus HTML
<code>constructs.test.ts</code>	43	Each pair is found by its own detector; skip mask; look-alike separation	Asserts no <i>frequency</i> ; the six known false-positive shapes are absent
<code>bench.test.ts</code>	31	Jekyll's nine options; the disagreements the bench exists to catch; six refusal paths	Never calls <code>certify</code>
<code>frontmatter-splice.test.ts</code>	30	Byte-identity, BOM, non-ASCII key duplication, corpus scoring	Zero-indent sequence, bare-CR fence, quoted keys — the three live defects — are absent
<code>certify.test.ts</code>	28	Refusal, block segmentation, byte ranges as real UTF-8, histogram	Fake engines throughout
<code>offsets.test.ts</code>	22	<code>Buffer</code> agreement at every offset, block-boundary surrogate pairs	—
<code>shape-gate.test.ts</code>	21	Every ceiling by name, strict UTF-8, failures as values	Asserts <code>BUDGET_MS</code> is <i>named</i> , not enforced
<code>pure-functions.test.ts</code>	19	200-case golden digest pin, idempotence, NFD collapse	—
<code>placement.test.ts</code>	16	Red proof that the naïve insert corrupts, then that <code>safeInsert</code> refuses	Never asserts what it <i>accepts</i> beyond markers
<code>frontmatter-prepass.test.ts</code>	16	Red proof that the loud and quiet shapes differ, then repair vs refusal	Own-vault shapes only

The biggest untested surface is **the certificate pipeline end to end**. `certify` is never invoked with a real `Engine`; `loadBench` is never handed to `certify`; `scripts/mdmax-cert.mjs` has no test at all and was broken at HEAD. Second is the foreign corpus: 8,513 pinned files with a hash gate and no behavioural gate. Third is `splice-frontmatter.ts` under the shapes that dominate real vaults — the writer's own test file scores it against `docs/engine/research/corpus-manifest.json`, our own vaults, where the zero-indent shape is rare.

Both shipped oracles are **structurally blind to set-destructive defects**, and this is demonstrable rather than theoretical. On a lone-CR file `'---\rtitle: T\r---\rBody\r'`, a single `spliceFrontmatterValue(src, 'public_slug', 's')` prepends a second frontmatter block and pushes the original into the body — two `---` fences become four. `spliceFrontmatterValue(set, 'public_slug', null)` then removes the block it created, and `back === cr` is `true` **MEASURED**. The oracle reports a clean round trip on a file the set alone destroyed.

68.5 The tests that must exist before any fidelity number is published

§58 already forbids publishing a fidelity or coverage percentage. These are the gates that lift that ban, in order.

#	TEST	PATH	ASSERTION	RED PROOF AVAILABLE TODAY
1	NF-1 zero-indent sequence	<code>test/corpus/foreign/nf-001-red-proof.test.ts</code> <code>specs/engine/nf-001-zero-indent-sequence.md</code> <code>not exist</code> ; <code>spec-report</code> reports it as <code>ghost</code>	— declared in <code>governs:</code> and does A set on a real corpus file with <code>tags:\n- a</code> . Post-fix: <code>refused <= 2 && changed == 0 && threw == 0</code> over 7,959	Yes — 6,613 files fail now
2	NF-3 bare-CR fence	<code>test/corpus/foreign/nf-003-red-proof.test.ts</code> also <code>ghost</code>	— also declared, Set alone, never set-then-delete. Assert fence count is unchanged	Yes — reproduced above
3	NF-4 unaddressable keys	new	<code>SAFE_KEY</code> cannot address 2,447 of 29,288 top-level keys (8.35%) across 937 of 7,969 files (11.76%); date created 812, date modified 811, <code>232</code> MEASURED . Assert the residual after the decision	Yes
4	Corpus behavioural gate	<code>test/corpus/foreign/writer-sweep.test.ts</code>	Run the writer over all 8,513, assert <code>changed_outside_key == 0 && threw == 0</code> and a refusal ceiling , not an equality (LR#66)	The sweep exists only as a scratch script
5	Certificate integration	<code>test/mdmax/cert-e2e.test.ts</code>	<code>loadBench()</code> → <code>certify()</code> on a fixture, asserting the exact matrix. Would have caught the broken CLI import	Yes — <code>node scripts/mdmax-cert.mjs</code> fails at HEAD
6	Consensus reference	<code>test/mdmax/certify.test.ts</code>	On a 3/2/2 split, a certificate must not report PASS. On a GFM construct, an engine implementing the spec must not be MUTATE	Yes — both reproduced
7	Detector false positives	<code>test/mdmax/constructs.test.ts</code>	The six cases: indented-code <code>html-comment</code> 1→0, indented-code <code>angle-bracket-text</code> 1→0, <code>a b</code> 1→2, <code>[[A]][[B]]</code> 1→2, one-column table 6→0, <code>coffee</code> fence 1→0	Yes — all six fail now
8	Fold stamp completeness	<code>test/mdmax/fold.test.ts</code>	<code>FOLD_VERSION</code> must incorporate the <code>entities</code> version (§68.6)	No

Recommendation. Land tests 1–4 before any number leaves the building, and publish the refusal rate alongside every fidelity rate as a single fraction — "measured on the 17% we can currently write" is honest, "83% fidelity" is not. **Anti-recommendation:** do not gate publication on tests 5–8; they protect the certificate, and the certificate is not yet a customer-facing claim, so blocking on them delays the numbers that are.

68.6 Can a third party reproduce the certificate?

They cannot, for five reasons, four of which are cheap to fix.

GAP	DETAIL	COST
The CLI does not run	<code>mdmax-cert.mjs</code> needs <code>node --import ./scripts/ts-resolve.mjs</code> . The hook exists and nothing imports it	One line
<code>entities</code> is undeclared	<code>fold.ts</code> and <code>verdict.ts</code> both <code>import { decodeHTML, escapeText } from 'entities'</code> . <code>package.json</code> lists it in neither <code>dependencies</code> nor <code>devDependencies</code> ; the resolved <code>node_modules/entities@8.0.0</code> is marked <code>dev</code> in <code>package-lock.json</code> , hoisted from <code>parse5@8</code> and <code>markdown-it@8</code> . Under <code>npm ci --omit=dev</code> the production domain layer loses its import — and <code>escapeText</code> is the definition of PASS, so a silent major bump changes every verdict without moving <code>mdmax/fold@1</code>	Declare it; add its version to the fold stamp
<code>marked</code> is changes	One of seven bench engines. <code>package-lock.json</code> shows exactly one dependent: <code>node_modules/mezmaid</code> . A diagram-library upgrade silently changes <code>benchId</code>	Declare and pin
Ruby is ambient	<code>bench.ts</code> <code>DEFAULT_RUBY_BIN = "/opt/homebrew/opt/ruby/bin/ruby"</code> , overridable by <code>MDMAX_RUBY_BIN</code> . <code>kramdown 2.5.2</code> + <code>kramdown-parser-gfm 1.1.0</code> come from whatever gems are installed. There is no <code>Gemfile</code> , no <code>Gemfile.lock</code> , no <code>.ruby-version</code>	Add a <code>Gemfile.lock</code> , or drop <code>github-pages</code> to declared
The artifact under-describes itself	Certificate top-level keys are <code>schema</code> , <code>bench</code> , <code>fold</code> , <code>file</code> , <code>targets</code> , <code>summary</code> , <code>blocks</code> <code>MEASURED</code> . Absent: <code>VERDICT_VERSION</code> (<code>mdmax/verdict@1</code> exists in <code>verdict.ts</code> and never reaches the artifact, though <code>verdict.test.ts</code> asserts "a stored cell can be traced to the definition that produced it"), the <code>entities</code> version, the Node version, the ruby and gem versions, and any corpus id	Extend the interface — a schema change, so <code>mdmax/cert@1</code> becomes <code>@2</code>

What *is* reproducible is real and should be said plainly: `benchId` is a sha256 over each engine's (`id`, `version`, `sorted options`) with rows sorted, so it is stable across loads, changes when one option changes, and is visibly different for a smaller bench — measured here as `51947c2e8812` for seven engines and `4c46fb96a5da` for six. Every non-PASS cell carries `before` and `after` and the `foldApplied` rule list. That is more auditable than any competitor publishes.

A certificate nobody can re-run is a marketing asset, not a defensible one, and the distance between the two is a `Gemfile.lock`, two `package.json` lines and an `--import` flag. Once those land, the honest posture is to publish the artifact for a public corpus with the bench manifest attached, and to state the uncertifiable share — `uncertifiableShare()` returns 8 of 15 — in the same breath. **Anti-recommendation:** do not pursue third-party attestation, signing or a hosted verifier until at least one external party has re-run the CLI end to end; a signature over an unreproducible artifact adds ceremony to the same claim.

§69 The engine landscape — what other markdown engines do that we do not

69.1 The comparison, on the five axes that decide our contract

Versions and dates fetched from the registries on 2026-08-30 [fetched: registry.npmjs.org, crates.io, pypi.org, api.github.com]. "Positions" means what the engine hands a caller, not what it computes internally; every one of these engines knows where it is in the source, and most throw that away before you can reach it.

ENGINE	VERSION / DATE	POSITION UNIT EXPOSED	INLINE POSITIONS	INCREMENTAL REPARSE	WRITE PATH	COMMONMARK	LICENCE
<code>micromark</code>	4.0.2 / 2025-02-27	UTF-16 code units (<code>Point.offset</code>)	yes, every token	no	none (tokenizer only)	0.31 target	MIT
<code>mdast-util-from-markdown</code> + <code>-to-markdown</code>	2.1.2 / 2024-11-04	UTF-16 (<code>position.start.offset</code>)	yes	no	regenerates from AST	0.31 target	MIT
<code>remark</code> / <code>remark-stringify</code>	15.0.1, 11.0.0 / 2023-09-18	as above	yes	no	regenerates	0.31 target	MIT
<code>markdown-it</code>	15.0.1 / 2026-08-27	line numbers only (<code>Token.map</code>)	no — <code>map</code> is <code>null</code>	no	none	0.30 claimed	MIT
<code>cmark</code>	0.31.2 / 2026-02-14	line:column, opt-in <code>CMARK_OPT_SOURCEPOS</code>	block only	no	<code>cmark_render_commonmark</code> regenerates	0.31.2 (reference)	NOASSERTION per GitHub API — verify before vendoring
<code>cmark-gfm</code>	0.29.0.gfm.13 / 2023-07-21	as cmark	block only	no	regenerates	0.29 — three spec revisions stale	NOASSERTION
<code>comrak</code>	0.54.0 / 2026-07-12	<code>Sourcepos {start,end}</code> line:column, columns in UTF-8 bytes , <code>--sourcepos-chars</code> switches to code points	block only	no	<code>format_commonmark</code> regenerates	0.31 + GFM	BSD-2-Clause
<code>goldmark</code>	v2.0.0 / 2026-08-27	<code>text.Segment {Start,Stop}</code> — byte offsets	yes, via segments	no	none shipped	0.31	MIT
<code>pulldown-cmark</code>	0.13.4 / 2026-05-20	<code>OffsetIter</code> → (Event, <code>Range<usize></code>) — byte range per event	yes, every event	no	separate crate	0.31	MIT
<code>markdown-rs</code> (<code>markdown</code>)	1.0.0 / 2025-04-23	mdast with <code>1:3-1:6 (2-5)</code> — line:col and byte offset	yes	no	none	tracks <code>cmark</code> / <code>cmark-gfm</code> behaviour	MIT
<code>tree-sitter-markdown</code>	v0.5.3 / 2026-02-26	byte ranges on every node	yes (split block/inline parsers)	yes (<code>ts_tree_edit</code>)	none	its README: not recommended where correctness matters	MIT
<code>@lezer/markdown</code>	1.7.2 / 2026-07-15 (repo moved off GitHub)	tree positions	yes	yes (consumes tree fragments)	none	not conforming — does not validate link references	MIT
<code>pandoc</code>	3.11 / 2026-08-29	none — <code>data Block</code> in <code>pandoc-types</code> <code>Definition.hs</code> has zero position fields	no	no	regenerates	own dialect + CommonMark reader	GPL-2.0
<code>prettier</code> (markdown)	3.9.6 / 2026-07-21	remark AST	yes	no	regenerates, reformats by design	0.31 via remark	MIT
<code>dprint-plugin-markdown</code>	0.23.2 / 2026-08-29	own CST	—	no	regenerates	GFM-ish	MIT
<code>mdformat</code>	1.0.0 / 2025-10-16	markdown-it tokens	no	no	regenerates, gated by an equality check	0.30 via markdown-it-py	MIT
<code>yaml</code> (frontmatter CST)	2.9.0 (installed)	absolute offsets on every CST token	n/a	no	byte-preserving	n/a	ISC

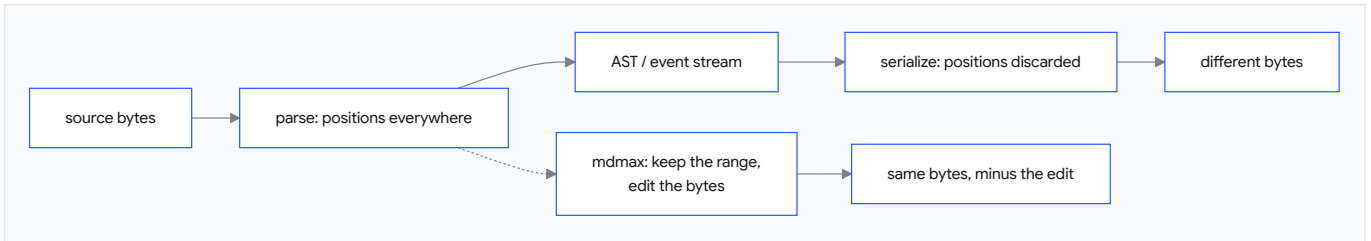
Two rows deserve their own sentence because they are the ones a reviewer will check. `cmark-gfm`, the engine behind GitHub's own rendering, last cut a release on 2023-07-21 against CommonMark 0.29 while the current spec is 0.31.2 [fetched: `api.github.com/repos/github/cmark-gfm/releases/latest`, `spec.commonmark.org`]. And `tree-sitter-markdown`'s own README states it is not recommended where correctness is important, its stated goal being syntax highlighting for neovim and helix [fetched: `README.md`, `split_parser` branch].

69.2 Who preserves source — measured, not asserted

Read-side position fidelity is a solved problem and has been for years. Write-side source preservation is solved by exactly nobody in the markdown ecosystem.

I measured the read side against the copy in `node_modules`. `mdast-util-from-markdown` gives every node — including inline `emphasis` and `link` — a `position.start.offset`; `markdown-it` 15.0.0 gives block tokens a `Token.map` of `[startLine, endLineExclusive]` and gives every inline child `map === null` [measured, `node_modules/markdown-it/dist/markdown-it.mjs`, `node_modules/mdast-util-from-markdown/index.js`]. On a `§ *b*\n` (9 UTF-16 units, 11 UTF-8 bytes) the `emphasis` node reports `offset: 5.8`; slicing the string by those numbers yields `"*b*"` and slicing the UTF-8 buffer by the same numbers yields `"\uFFFFD *"` [measured]. mdast offsets are UTF-16 code units and are silently wrong the moment anyone treats them as bytes — which is precisely the failure `src/modules/mdmax/domain/offsets.ts` exists to make unrepresentable, via branded `U16Offset` / `ByteOffset` and a `u16()` that returns `{ok:false, error:{kind:'INSIDE_SURROGATE_PAIR'}}` rather than rounding.

The write side, same corpus, one command: `toMarkdown(fromMarkdown(src))` on a nine-line document turned a setext heading into ATX and an indented code block into a fenced one, 96 bytes in, 98 bytes out, `=== false` [measured]. That is not a bug in `mdast-util-to-markdown`; it is the documented purpose of a serializer that "turns a syntax tree into markdown" [fetched: its readme]. `comrak`'s `format_commonmark` is the same shape — `src/cm.rs` opens with a comment conceding that "formatting an ill-formed AST might lead to invalid output" and validates only in debug builds [fetched]. Pandoc cannot preserve source even in principle: `pandoc-types` | `Definition.hs` | `data Block` carries no position field and a grep for `sourcepos|position` over that file returns 0 [fetched].



Ranked by distance from our contract — locate the range, replace exactly those bytes, refuse when ambiguous:

1. `yaml 2.9.0's CST layer`. Closest, and it is already a direct dependency (`"yaml": "2.9.0"` in `package.json`). Measured: `CST.stringify(new Parser().parse(fm))` round-trips the exact frontmatter shape that refuses 83% of publishes — a block sequence at column zero — byte-identically, `68B → 68B, === true` ; every token carries an absolute offset (`scalar@30="beta"`); mutating one token's `.source` and re-stringifying produces bytes identical to a naive string replace [measured].
2. `pulldown-cmark's OffsetIter . Iterator<Item = (Event<'a>, Range<usize>)>` in `parse.rs` — a byte range for every event, inline included. It has the addressing half of the contract and none of the writing half.
3. `goldmark's text.Segment . Byte Start / Stop` on every node, with one caveat that matters: `Padding` is a synthesized-space count, so a `Segment` is *not* always a literal slice of the source [fetched: `text/segment.go`].
4. `mdformat` . Furthest from the contract mechanically — it regenerates everything — and closest to it philosophically, because it refuses. See below.
5. `tree-sitter / @lezer/markdown` . Lossless by construction (nodes are byte ranges into an unmodified buffer) and incremental, but neither writes, and both trade conformance for single-pass speed.

69.3 Techniques to adopt, each with its anti-recommendation

Adopt `mdformat's is_md_equal` as the shape of a write gate, not as its content. In `src/mdformat/_util.py` , `is_md_equal(md1, md2)` renders both strings to HTML, collapses runs of whitespace, and compares; the CLI refuses to write the file when it returns false, and `--no-validate` is the escape hatch [fetched]. That is our `foldEqual` with one rule where we have six (`FOLD_RULES = ['entity-decode', 'smart-punctuation', 'void-element-spelling', 'whitespace', 'id-prefix', 'heading-anchor-id']`) and one engine where we have seven. The adoptable part is the *placement*: `mdformat` runs the check on the write path and blocks the write. We run ours in `certify.ts` and block nothing. *Anti-recommendation*: do not adopt their comparison basis. Their own docstring concedes whitespace collapsing "is not a perfect solution, as there can be meaningful whitespace in HTML, e.g. in a `<code>` block" — our `fold.ts` header documents the same trap and solves it by decoding entities then re-escaping with `entities.escapeText` so `<p><cat></p>` cannot collapse into an invisible tag.

Adopt `comrak's --sourcepos-chars` as a disclosure obligation. `Comrak` ships a CLI flag whose entire job is to say which unit a column is measured in — bytes by default, code points on request [fetched: `README.md`]. Every position we ever put in a certificate, an anchor, or an API response should carry its unit the same way. *Anti-recommendation*: do not copy the mechanism — a runtime flag is the weak form. `offsets.ts` already has the strong form in the type system, and adding a flag would create a second source of truth about units.

Adopt `pulldown-cmark's range-per-event iterator shape` for our own construct scan. `into_offset_iter()` returns `(Event, Range<usize>)` so a caller never has to ask a second question to find out where an event was. `constructs.ts` already returns `ReadOnlyMap<string, readonly Range[]>` from `inventory()` ; the adoptable idea is extending that to a single ordered stream so `skipRegions()` and `inventory()` cannot drift. *Anti-recommendation*: do not adopt `pulldown-cmark` itself as the scanner. It is Rust, it would land behind WASM, and our shape gate budget in `shape-gate.ts` (`BUDGET_MS` , `MAX_BYTES = 4 * 1024 * 1024`) is not currently the binding constraint.

Adopt `@lezer/markdown's fragment-reuse model` for editor-side reparsing, which we already ship. `@codemirror/lang-markdown 6.5.0` and `@lezer/markdown 1.6.3` are installed [measured, `node_modules/*/package.json`], and the parser "consumes fragments of such trees for its incremental parsing" [fetched: `README`]. *Anti-recommendation*: never let a lezer tree become the tree of record. Its `README` states it does not validate link references and will parse `[a][b]` as a link when no `[b]` exists — an editor-fidelity parser, not a semantics-of-record parser, and the settled rule that there is no tree-of-record already covers this.

Adopt `mdast-util-to-markdown's Unsafe` table as the model for our quoting rules. It maintains an explicit, data-driven table of character-in-context pairs that must be escaped, rather than scattering conditionals. `emitScalar()` in `splice-frontmatter.ts` is already the YAML analogue and already carries the measured lesson — the over-broad rule that treated indicator characters as special anywhere quoted 435 of 907 corpus files unnecessarily. *Anti-recommendation*: do not generalize `emitScalar` into a full YAML emitter; the module's value is that it emits one scalar for one key and refuses everything else.

69.4 What we do that nobody else does

Three claims, each stated narrowly enough to survive a hostile reviewer.

We refuse on the write path and return the input unchanged; every other markdown writer either succeeds or throws. `spliceFrontmatterValue` has eleven `return src` branches — unsafe key shape, duplicate key, unterminated block, block scalar or anchor on the target key, a bare non-key line at top level, and so on. Measured against the real function: `spliceFrontmatterValue('---\n\ttitle: A correct\ntags:\n\talpha\n\tbeta\n---\n\nbody\n', 'title', 'B note')` returns its input byte-for-byte, while the two-space-indented variant of the same document splices correctly [measured, via `--experimental-strip-types`]. The scrutiny this must survive: refusal is not novel in software generally — `git apply` refuses, `patch` refuses — the claim is bounded to *markdown/frontmatter writers*, and `mdformat's is_md_equal` gate is the one genuine near-miss. It differs in that `mdformat` regenerates first and then checks; we never regenerate at all.

We publish a per-document, cross-renderer degradation verdict with a versioned equivalence relation. The prior art a reviewer will name is `babelmark3`, which does compare a document across many markdown implementations [fetched: `babelmark.github.io` , "allows to compare various implementations of Markdown"]. The claim must therefore be stated as a difference in kind, not in existence: `babelmark` is an interactive diff viewer producing no artifact; `cert-contract.ts` defines a `Certificate` with a `CertSummary` , per-cell `CellVerdict` values classified by `classify()` into named classes, an engine set hashed by `computeBenchId()` over `(id, version, sorted options)` , and a `fold.version` stamp so a reader can tell which definition of "equivalent" produced a stored verdict. A stamped, replayable, per-construct artifact is the differentiated thing; "we run seven parsers" is not.

We model a renderer target as `(product, surface)` and mark unmeasurable rows as declared-and-dated rather than measured. `targets.ts` states the finding that forced it — GitHub is three renderers that disagree on identical bytes — and carries `fidelity: 'local' | 'requires-push' | 'declared'` plus `DECLARED_LAST_VERIFIED = '2026-08-01'` . No engine or comparison tool in the table above models the distinction between "I ran this" and "I asked the vendor's docs on a date."

The claim we must **not** make: that we have a better markdown parser. We do not have one at all. Everything above is built on other people's parsers, and `remark-app` is the first row in `BENCH_ENGINE_IDS` .

69.5 Build vs adopt

There is one engine we should be building on rather than beside, and we already pay for it. `yaml 2.9.0` is a direct dependency, and `src/modules/preview/presentation/frontmatter.ts` imports `parseDocument` and `isMap` from it — the *Document* layer, whose `doc.toString()` is named in `splice-frontmatter.ts` 's own header as one of the two regenerating write paths we replaced. Nothing in `src/` or `scripts/` imports `Parser` or `CST` [measured, grep over both

trees]. We own the right library and use the wrong layer of it, and the 83% publish-refusal rate is the bill for that. The measured CST round-trip in §69.2 is the fix: the refused shape parses, addresses, edits, and re-stringifies byte-identically today, with no new dependency and no new licence.

Anti-recommendation, and it is the load-bearing half: adopting the CST does not mean deleting the splice writer or handing YAML mutation to a library. `CST.stringify` will happily emit a document you have made ill-formed, exactly as `comrak::format_commonmark` will; the refusal discipline has to stay ours, wrapped around the CST rather than replaced by it. The correct shape is a widened locator — CST tokens supply the range, `spliceFrontmatterValue` still decides whether the range is unambiguous, and the final write is still `src.slice(0, start) + bytes + src.slice(end)`. That also keeps `SAFE_KEY` honest: the CST gives us non-ASCII keys for free at the *addressing* level, but it does not answer what key equality means for `café` typed NFC versus NFD, which is the reason the regex is narrow.

For the body of the document, adopt nothing. The read side we need already exists in `mdast-util-from-markdown` (which we ship) and in `@lezer/markdown` (which we ship through CodeMirror); the write side does not exist in any of the seventeen projects above, which is why `mdmax` exists. Building on `pulldown-cmark` or `markdown-rs` would buy byte-native offsets at the cost of a WASM boundary and a second CommonMark implementation to keep in agreement with the seven already in `BENCH_ENGINE_IDS` — a worse trade than the `OffsetMap` checkpoint table we already have.

The three things that are actually missing are not engines. There is no CI, `mdmax/fold@1` has never been run over the pinned 8,513-file corpus, and twelve of thirteen `mdmax` files have zero product importers — only `decodeStrict` from `shape-gate.ts` is called, by `get-snapshot.ts` and `search-index.ts`. A comparison table is a poor answer to a module nothing imports.

§70 Engine performance and correctness under load

70.1 Provenance of every number below

Everything in this section was executed on 2026-08-31 against working-tree commit `d405f3f`, Node v24.6.0, Apple M4 Pro / 14 cores / 48 GB, with the `mdmax` domain sources copied verbatim into a scratch directory and run through Node's native type stripping (the same files the test suite loads; only relative specifiers gained a `.ts` extension). The corpus is the pinned one at `test/corpus/foreign/_vendor` — 8,513 files, 19,047,891 bytes, verified present before each run. Engine versions read off disk: `micromark@4.0.2`, `mdast-util-from-markdown@2.0.3`, `markdown-it@15.0.0`, `marked@16.4.2`, `commonmark@0.31.2`, `remark-parse@11.0.0`, `entities@8.0.0`, `yaml@2.9.0`. [measured]

Three things must be said before the tables, because they change how every other number reads. There is no `.github` directory in this repo, so no benchmark in this section is currently enforced by anything. [measured] `shapeGate` — the function this whole section's guard story rests on — has zero product call sites; the only import of `src/modules/mdmax/domain/shape-gate.ts` from outside the module is `decodeStrict`, in `src/modules/vault/application/get-snapshot.ts` and `src/modules/vault/infrastructure/search-index.ts`. [measured] And `node scripts/mdmax-cert.mjs <file>`, exactly as its own header documents it, fails with `ERR_MODULE_NOT_FOUND` for `.../mdmax/domain/offsets`; it runs only as `node --import ./scripts/ts-resolve.mjs scripts/mdmax-cert.mjs`, and nothing in `package.json` or `scripts/` references `ts-resolve.mjs`. [measured]

70.2 The performance budget per engine operation

`BUDGET_MS` in `shape-gate.ts` declares three tiers — `keystroke: 250`, `coldOpen: 2_000`, `batch: 10_000` — and its own comment concedes they are "enforced by the caller via `worker.terminate()`". Not enforced here." No caller exists. The budget below assigns each operation to a tier and states measured headroom.

OPERATION (SYMBOL, FILE)	TIER	TARGET	MEASURED	HEADROOM	JUSTIFICATION
<code>shapeGate</code> on a whole document	keystroke	≤ 5 ms @ 4 MB	16 ms @ 4 MB string; p99 0.045 ms, max 3.42 ms over the corpus	corpus fits; the ceiling does not	8,513 files in 63.5 ms = 286 MB/s; every file passed, none refused [measured]
<code>decodeStrict</code> happy path	keystroke	≤ 20 ms @ 4 MB	8 ms @ 64 MB ASCII	large	one <code>TextDecoder</code> pass, no per-byte JS [measured]
<code>decodeStrict</code> failure locate	coldOpen	≤ 50 ms	10.6 ms, invalid byte at offset 4,194,303	4.7×	binary search over the longest valid prefix, $\log_2(4\text{ M}) \approx 22$ decodes [measured]
<code>new OffsetMap(text)</code>	coldOpen	≤ 100 ms/doc	27 ms for 8,513 docs / 18.2 MB = 676 MB/s	~600×	one pass, <code>Uint32Array</code> checkpoint every <code>BLOCK = 512</code> units [measured]
<code>OffsetMap.toByte</code>	keystroke	≤ 1 μs	465 ns over 20,000 random lookups on the 20 largest docs	2.1×	block seek + bounded ≤512-unit scan [measured]
<code>inventory(markdown)</code>	coldOpen	≤ 50 ms/doc	0.09 ms mean, 11.0 ms worst over 400 files	4.5× at worst	30+ construct detectors, each a <code>matchAll</code> over masked text [measured]
<code>skipRegions(markdown)</code>	keystroke	≤ 5 ms	0.003 ms mean over 400 files	large	shares one <code>analyse()</code> scan [measured]
<code>fold(html)</code>	batch	≤ 20 ms/doc	0.05 ms mean, 11.9 ms worst; 22.7 MB/s over 2.3 MB of rendered HTML	1.7× at worst	segment scan + <code>decodeHTML</code> / <code>escapeText</code> per text run [measured]
<code>spliceFrontmatterValue</code>	keystroke	≤ 100 μs	1.3 μs/file over 7,969 front-matter files	77×	line scan of the block only; body is never touched [measured]
One certificate cell (<code>Engine.render</code>)	batch	≤ 5 ms	0.02–0.69 ms for six JS engines; 54.50 ms for <code>kramdown-jekyll</code>	violated 11×	<code>runRuby</code> calls <code>spawnSync</code> once per render; process start dominates [measured]
Whole-corpus certification	batch (offline)	≤ 10 min	60 files / 39 blocks in 2.667 s wall → ≈ 5,533 blocks, ≈ 5.6 min single-threaded for 8,513 files	1.8×	60.7 ms/block × block count [derived: (2.667 s – 0.3 s startup) ÷ 39 blocks; 39 blocks ÷ 60 files × 8,513 files]

Recommendation: keep one ruby process alive behind a line-delimited protocol instead of `spawnSync` per render, and the cert falls from 60.7 ms/block to roughly 2 ms/block. Anti-recommendation: do **not** drop `kramdown-jekyll` from `BENCH_ENGINE_IDS` to buy the speed — `LoadBench`'s rule 4 ("a missing engine is a HARD REFUSAL") exists precisely so the bench cannot quietly shrink, and GitHub Pages is the one target no JS engine models.

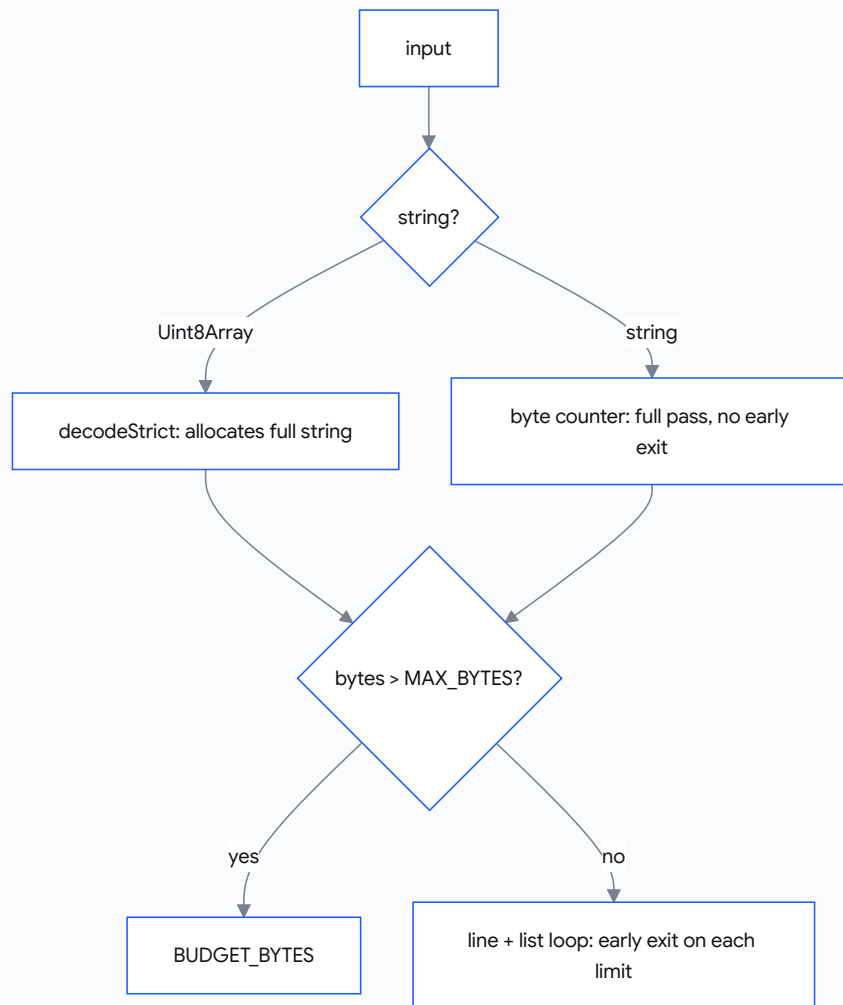
70.3 Known quadratics and pathological inputs

INPUT	PATH THAT EATS IT	MEASURED	GUARD	GUARD TESTED?
"[" × n	WIKILINK_RE = /(?!?)\[\[([^\]]+)\]\]/g in src/modules/vault/infrastructure/markdown-parser.ts; identical source as COMBINED_WIKILINK_RE in src/modules/preview/presentation/markdown/wikilinks.tsx	42.9 / 162.9 / 649.2 / 2,575.7 / 10,384.3 / 41,527.3 ms at 10/20/40/80/160/320 KB; log-log slope k = 1.98 [measured]	none on this path — shapeGate has no call site	no
"[" × n	const WIKILINK = /\[\[([^\]]+)\]\]/ used by .test(rawValue) in frontmatter-prepass.ts	40,410.9 ms at 320 KB, k ≈ 2.0 [measured]	none; the value is a single front-matter line, so exposure is bounded by line length, not file size [inference]	no
n flat list items	mdast-util-from-markdown vs micromark on identical bytes	22.9 KB: 54 vs 37 ms (1.5×); 94.9 KB: 254 vs 129 (2.0×); 404.9 KB: 2,318 vs 492 (4.7×); 1.29 MB: 21,237 vs 1,709 (12.4×) [measured]	MAX_LIST_MARKER_LINES = 20_000, early-exit at 20,001 in 1.1 ms	yes — 4 refs in test/mdmax/
> 4 MB document	MAX_BYTES	BUDGET_BYTES in 8.2 ms at 4 MB + 1; 128 ms at 64 MB (string path)	MAX_BYTES = 4 * 1024 * 1024	yes — 2 refs
> 200,000 lines	MAX_LINES	BUDGET_LINES in 3.4 ms	MAX_LINES = 200_000	yes — 2 refs
Invalid UTF-8	decodeStrict	INVALID_UTF8 at line 1, col 4,194,304 in 10.6 ms	fatal TextDecoder, never repairs	yes — 3 refs
Slow parse	—	—	BUDGET_TIME	no — the reason appears in test/mdmax/shape-gate.test.ts only as a literal { ok: false, reason: "BUDGET_TIME", elapsedMs: 300, limit: 250 } fed to explainFailure; grep for new Worker // worker.terminate in src/ returns zero hits
Parser throw / worker crash	—	—	PARSER_THREW, WORKER_DIED	no — same shape as above; nothing in the codebase constructs either value

Three of the seven members of ShapeFailure are unreachable: the union declares them, explainFailure renders them, and no code path produces them. That is not a gap in coverage, it is a gap in the thing being covered. [measured]

The upstream precedent the file's header cites is real: cmark issue 373 is titled "Quadratic behavior when parsing inlines", and cmark carries a merged change titled "Fix two cases of quadratic behavior (GHS-66g8-4hjf-77xh)" — the advisory behind CVE-2023-22484. [fetched, api.github.com]

There is also an ordering defect inside the gate itself. On the Uint8Array branch, decodeStrict(input) runs and materialises the whole string before if (bytes > MAX_BYTES) is evaluated, so a 64 MB paste allocates 64 MB of JS string in order to be told it is too big. On the string branch the hand-rolled byte counter (for (let i = 0; i < text.length; i++)) has no early exit, so a 64 MB string costs a full 128 ms pass to reach the same refusal. [measured]



Recommendation: move `input.length > MAX_BYTES` above the decode, and break the string byte counter once `bytes` exceeds the limit. Anti-recommendation: do not replace `decodeStrict`'s fatal decoder with a lenient one to make the large-input path cheaper — substituting `U+FFFD` changes the document's byte length, and every offset computed afterwards would be correct for a file the user does not have.

70.4 Correctness risks under scale

Very large files. The largest corpus file is `oldwinter__knowledge-garden/.obsidian/plugins/obsidian-linter/default-misspellings.md` at 939,763 bytes / 35,147 lines — 22.4% of `MAX_BYTES` and 17.6% of `MAX_LINES`. [measured] No file in the pinned corpus reaches any gate limit, so the corpus proves the happy path and proves nothing at all about the ceilings. The synthetic probes in §70.3 are the only evidence the ceilings work.

Deeply nested structures. `analyse()` in `constructs.ts` builds `lineStarts` by a single scan and resolves fences with a `new RegExp(...)` per opening marker; nesting depth is not tracked and there is no recursion, so the risk is not stack exhaustion but *masking* — `skipRegions` must correctly exclude fenced and front-matter spans or a detector fires inside a code block. Over 400 corpus files `inventory()` threw zero times and `skipRegions()` cost 0.003 ms mean. [measured] The worst `inventory()` case, 11.0 ms, is `community-archive_obsidian-hub/02 - Community Expansions/02.01 Plugins by Category/Uncategorized plugins.md` — 278,458 bytes of dense link tables, i.e. width, not depth.

Adversarial input. The two live wikilink regexes are the whole exposure, and they sit on paths the gate does not protect. 320 KB of `[[` costs 41.5 s in `markdown-parser.ts`. A vault sync that walks every file will find one. [measured]

Multi-byte boundaries. `offsets.ts` opens by asserting "Only 67 of 1,080 files in the pinned corpus have `bytes == UTF-16 code units == code points`. 93.8% already diverge." Over the 8,513-file vendor corpus the same test gives 6,568/8,513 — **77.2% do not diverge**, and 475 files (5.6%) contain surrogate pairs. [measured] The two populations disagree by an order of magnitude, and the header's number describes a corpus that is not the one the engine is now certified against. The design conclusion survives the correction — 22.8% divergence and 475 non-BMP files is still far past the point where an untyped offset is a bug — but the figure must not be quoted as if it described `_vendor`. `splitsSurrogatePair` is exercised 6 times in `test/mdmax/`, and `u16` refuses rather than rounds; a splice of an astral value round-tripped correctly (`title: a👉b`). [measured]

Line endings. Zero corpus files contain a bare CR. [measured] `spliceFrontmatterValue` preserves CRLF (`const eol = open[1]`) and a leading BOM, both verified on synthetic inputs. [measured]

70.5 The splice writer's real failure, and one it hides

The publish path reproduces exactly: `spliceFrontmatterValue(src, 'fm_slug', 'my-note')` over the 7,969 corpus files with a closed front-matter block refuses 6,615 of them — **83.0%** — in 11 ms total, 1.3 μs/file. [measured] The cause is the fall-through branch that returns `src` on "a bare non-key line at top level"; 6,613 of those files (83.0%) contain a YAML block sequence at column zero, which is Obsidian's default emission for `tags:`. [measured] The two-file gap between 6,613 and 6,615 is unexplained by that signature and should be resolved before the fix is written, not after.

The same branch hides a corruption. A sequence of *maps* at column zero — `- name: a` — satisfies `isTop` (unindented, contains `:`) but not `topLevelKeyLine`, so it is neither spliced nor refused. Setting the key above it yields `tags: [x, y]\n- name: a\n- name: b\ntitle: T`, which `yaml@2.9.0` rejects with "Implicit keys need to be on a single line at line 2, column 1"; deleting that key yields a block opening with `- name: a`. [measured] Fifteen corpus files carry the shape; zero were reachable through the exact route probed (a key with an empty value immediately above the first sequence item), so this is a latent path, not a live one. [measured]

Recommendation: teach the block scanner that an unindented `-` line continues the preceding key, which fixes the 83% refusal and closes the corruption in one change. Anti-recommendation: do not fix the refusal by loosening the bare-line branch alone — that converts 6,613 refusals into 6,613 unguarded appends and turns a visible no-op into the silent rewrite the module was written to prevent.

70.6 The benchmark suite

`mdmax/fold@1` had never been run over the pinned corpus. It has now: 946 files rendered by `markdown-it` at `html:true` and `html:false`, folded both sides — 100 ms of fold over 2.3 MB of HTML (22.7 MB/s), **0 of 946 non-idempotent**, 0 throws. [measured] Fold power against a genuinely different engine (`markdown-it` `html:true` vs `marked` defaults, same 946 files): 49 documents byte-equal (5.2%) before the fold, 434 (45.9%) after. [measured] Do not quote that against the prototype's "43.71% → 4.28%": this is whole-document byte equality over two engines, that was per-block over six, and the populations share no `corpus_id`.

BENCH	MEASURES	CORPUS	ASSERTION	RUNS WHEN
<code>gate-throughput</code>	<code>shapeGate</code> ms/file	all 8,513	p99 ≤ 0.1 ms and 0 refusals and 0 throws	every push
<code>gate-ceilings</code>	each of the 7 <code>ShapeFailure</code> reasons	7 synthetic fixtures	every reason is produced by a real input, not by a literal in a test	every push
<code>quadratic-canary</code>	k for each regex reachable from user text	"[" × {10, 20, 40, 80} KB	fitted log-log slope < 1.3, and 80 KB < 250 ms	every push
<code>offset-integrity</code>	<code>u16</code> / <code>toByte</code> / <code>toU16</code> round trip	the 475 non-BMP files	round trip is the identity on every valid offset; 0 rounded	every push
<code>splice-oracle</code>	set / delete / rename, then re-parse with <code>yaml@2.9.0</code>	all 7,969 front-matter files	every non-refusal re-parses; refusal rate is asserted as a floor-and-ceiling band , not an equality	every push
<code>fold-idempotence</code>	<code>fold(fold(x)) === fold(x)</code>	rendered HTML for all 8,513	0 non-idempotent	nightly
<code>cert-matrix</code>	full 7-engine certificate	400-file stratified sample	0 throws; <code>benchId</code> unchanged unless an option changed	nightly
<code>cert-full</code>	whole corpus	all 8,513	histogram deltas reported, nothing asserted	weekly

Three design rules the harness must carry, each already paid for elsewhere in this system. Assert a band on refusal rates, never an equality — an equality goes red the moment a file is added and rewards standing still. Preflight the environment and refuse with a stated remedy rather than emitting a dozen false failures. And before trusting any new guard test, make it fail against the unfixed code first; a green suite over a rare fault is the expected draw, not evidence.

Recommendation: wire all eight as one `npm run bench` behind a `.github/workflows` job, with `MDMAX_RUBY_BIN` set so `kramdown-jekyll` loads rather than refusing the whole bench. Anti-recommendation: do not let `cert-full` gate a merge — 5.6 minutes single-threaded is a nightly cost, and a slow required check is a check people learn to skip.

70.7 Where this is too slow for a live editor, and the mitigation

SURFACE	COST	MITIGATION	ANTI-RECOMMENDATION
Wikilink extraction on every keystroke	41.5 s on 320 KB of <code>[[</code>	call <code>shapeGate</code> first; then match with an index-based scanner (<code>indexOf('[')</code> , bounded forward search for <code>]]</code>) instead of a backtracking regex	do not "fix" the regex with a possessive/atomic rewrite — JS has neither, and a lookahead-simulated atomic group is harder to read than the linear scanner and no faster
Certificate on save	60.7 ms/block, 97% of it ruby process start	run certification off the editor thread, on demand or on publish — never on save	do not cache a certificate keyed on file path; <code>benchId</code> is part of its identity, and a stale cert under a changed option set is exactly the lie <code>computeBenchId</code> exists to prevent
<code>mdast-util-from-markdown</code> for structure	21.2 s on a 1.29 MB flat list	use <code>micromark</code> events for anything that only needs spans; reserve the mdast tree for operations that genuinely need parent links	do not adopt the tree as the document of record to amortise the parse — the contract is byte-range replacement, and a tree-of-record makes every save a regeneration
<code>inventory()</code> on a 278 KB link table	11.0 ms	debounce to idle; it is a coldOpen-tier operation wearing a keystroke-tier call site	do not sample the document to keep it under budget — a detector that runs on part of a file reports an inventory for a document nobody has

The optimisations to refuse are the ones that trade the byte contract for throughput, and every one of them looks like a win in a profile. Do not repair invalid UTF-8 to avoid a refusal; do not round an offset off a surrogate boundary to avoid an `OffsetError`; do not widen `SAFE_KEY` to admit `título` without first deciding whether NFC and NFD `café` are the same key; do not normalise line endings on read to simplify the splice scanner; do not let the fold collapse attribute values or `<pre>` whitespace to raise the PASS rate, because a fold that swallows everything turns the certificate into a curiosity. Each of these makes a benchmark greener and makes the product's one differentiating claim false.

§71 Wiring MDMAX into the product

MDMAX is 13 files and 3,614 lines under `src/modules/mdmax/`, plus the splice writer at `src/modules/share/domain/splice-frontmatter.ts` and the CLI at `scripts/mdmax-cert.mjs`. Product code reaches exactly one symbol of it. `grep -rn "mdmax" src/` outside the module itself returns two hits, both `import { decodeStrict } from "@modules/mdmax/domain/shape-gate"` — in `src/modules/vault/application/get-snapshot.ts` and `src/modules/vault/infrastructure/search-index.ts` **MEASURED**. The other twelve files have zero product importers. `npm vitest run test/mdmax test/share` is 23 files, 851 tests, 1.63 s, all green **MEASURED** — the engine is tested and unreachable, which is the specific failure mode this section exists to close.

71.1 The four seams

#	SEAM	ENTRY POINT TODAY	STATE	BLOCKED BY
1	Ingress gate	<code>decodeStrict</code> in <code>get-snapshot.ts</code> , <code>search-index.ts</code>	live, refusal invisible to the user	nothing
2	Write gate	none — <code>makeCommitChanges</code> in <code>src/modules/repository/application/commit-changes.ts</code>	not built	NF-1, NF-3
3	Splice engine	<code>spliceFrontmatterValue</code> (share domain, not mdmax)	live, returns a bare <code>string</code>	NF-1, NF-3, NF-4
4	Certificate	<code>certify</code> + <code>loadBench</code> , CLI-only	CLI does not run as documented	job runner, <code>spawnSync</code> cost

71.2 Seam 1 — ingress gate

PROPERTY	VALUE
Call site	<code>src/modules/vault/application/get-snapshot.ts</code> , the <code>for (const [zipPath, bytes] of Object.entries(entries))</code> loop; <code>src/modules/vault/infrastructure/search-index.ts</code> , same shape
Signature	<code>decodeStrict(bytes: Uint8Array): { ok: true; text: string } ShapeFailure</code>
Asserts	the blob is strict UTF-8; a non-fatal decode would substitute U+FFFD, change byte length, and commit mojibake back to git on the next save
On refusal	<code>console.error(...)</code> then <code>continue</code> — the note is dropped from the snapshot and from the search index
User-visible error	none <code>MEASURED</code> — the note silently does not exist in the tree or in search
Test	<code>test/mdmax/shape-gate.test.ts</code> covers <code>decodeStrict</code> ; nothing asserts the <i>caller's</i> drop behaviour
Ordering	independent; fix any time

The gate is correct and its call sites are not. `explainFailure(f: ShapeFailure): string` already exists in the same file and produces "This file is not valid UTF-8 (line N, column M). It was not changed." — nothing calls it `MEASURED`. Wire the refusal into `VaultSnapshot` as a `skipped: { path, reason }[]` field and render it in the file tree. The anti-recommendation: do not promote the drop to a thrown error or a failed snapshot load. One unreadable file in a 6,586-file vault must not blank the app, and `shapeGate` itself is built on the rule that failure is a value.

`shapeGate(input: string | Uint8Array): ShapeResult` — the wider gate carrying `MAX_BYTES` (4 MB), `MAX_LINES` (200,000) and `MAX_LIST_MARKER_LINES` (20,000) — is not called anywhere in `src/` `MEASURED`. It belongs on the same call sites, ahead of `parseMarkdown`, because the quadratics it exists to bound (`WIKILINK_RE` at k=1.98) live in the parse path, not the decode path.

71.3 Seam 2 — write gate

PROPERTY	VALUE
Call site	<code>src/modules/repository/application/commit-changes.ts</code> , immediately after the <code>for (const file of req.files) assertVaultWritable(file.path)</code> loop and before <code>writer.getHeadCommit()</code>
Proposed signature	<code>checkWrite(args: { path: string; before: string; after: string }): WriteVerdict</code> where <code>WriteVerdict = { ok: true } { ok: false; reason: ShapeFailure['reason'] 'SKELETON_CHANGED' 'ADJACENT_TO_SETTEXT_UNDERLINE'; detail: string }</code>
Asserts	the outgoing bytes pass <code>shapeGate</code> ; and, for any <i>inserted</i> marker, that <code>safeInsert(text, offset, marker, parse)</code> returns <code>ok: true</code> — the block skeleton is unchanged
On refusal	abort the whole commit before <code>createBlob</code> , exactly as the existing <code>ConflictError</code> path does
User-visible error	"Refused: <detail>. No file was changed." surfaced as 409, never 500
Test	a red proof that a "Heading\n---" document with a marker inserted at the underline boundary is rejected — <code>test/mdmax/placement.test.ts</code> already proves <code>safeInsert</code> refuses it; what is missing is the assertion that <code>commitChanges</code> never reaches <code>createBlob</code>
Ordering	last of the three write-path seams

Two contract mismatches have to be resolved at this boundary. `commitChanges` throws (`ConflictError`, `new Error("refusing to write outside the vault: ...")`) while every MDMAX function returns a discriminated union; the gate must translate, and the translation belongs in `commit-changes.ts`, not inside `mdmax`. And `bench.ts` imports `node:child_process`, so nothing in `src/modules/mdmax/infrastructure/` may be pulled into this path — the gate imports only `domain/shape-gate` and `domain/placement`, both of which are dependency-free apart from the injected `ParseFn`.

Do not land the write gate before NF-1 and NF-3 are fixed: at the measured refusal rate it converts a silent failure into a loud one across 83% of foreign-vault publishes, which is a worse incident, not a better one.

The number is reproducible today. Running `spliceFrontmatterValue(src, 'public_slug', 'x-slug')` over the pinned corpus at `test/corpus/foreign/_vendor/`: 8,513 markdown files, 7,969 carrying frontmatter, **6,615 refused (83.01%)**, 1,354 spliced, 0 threw `MEASURED, THIS SESSION`. Bucketing each refusal by its first offending top-level line: 6,611 are a block sequence at column zero, 3 a bare non-key line, 1 a flow-close bracket. `specs/engine/nf-001-zero-indent-sequence.md` records `blast_radius: 6613 of 6614`; the two-file difference is an attribution-method difference, not corpus drift — `node scripts/corpus-foreign.mjs verify` is the authority and must be run before either figure is quoted again.

71.4 Seam 3 — splice engine

PROPERTY	VALUE
Call sites	<code>src/modules/share/infrastructure/share-writer.ts</code> (<code>const next = spliceFrontmatterValue(file.content, "public_slug", slug)</code>); <code>src/modules/preview/presentation/PropertiesPanel.tsx</code> (<code>setValue</code> , <code>renameKey</code> , <code>removeKey</code> , <code>addProperty</code> , all funnelled through <code>function emit(next: string)</code>)
Signature today	<code>spliceFrontmatterValue(src: string, key: string, value: string number boolean string[] null): string</code>
Proposed signature	<code>spliceKey(src, key, value): { ok: true; text: string } { ok: false; reason: 'UNADDRESSABLE_KEY' 'NOT_A_PLAIN_MAP' 'UNTERMINATED_FENCE' 'DUPLICATE_KEY' 'UNSAFE_CONSTRUCT'; detail: string }</code>
Asserts	the key's byte range is locatable unambiguously; every byte outside it is bit-identical afterwards
On refusal today	returns <code>src</code> unchanged — indistinguishable from a successful no-op
User-visible error	none, and worse than none: see below
Test	<code>test/share/frontmatter-splice.test.ts</code> (in the 851 green) plus <code>node scripts/corpus-foreign.mjs verify</code> then the writer sweep
Ordering	NF-1 and NF-3 first, then the <code>Result</code> type, then the call sites

The bare-`string` return is the load-bearing defect. `makeShareWriter` commits `next` unconditionally, so a refusal produces an empty commit titled `share: set public_slug=<slug> on <path>`, `makeSetShare` returns a `publicUrl`, `makeShareSnapshotPort.listShares()` never sees the note because `publicSlug` was never written, and `makeResolvePublicNote` returns `null` at `if (matches.length !== 1) →` the shared link 404s. The user is handed a URL that cannot work, and the commit log says it did. `PropertiesPanel`'s `emit` handles the same case correctly — `if (!onEdit || next === content) return` — and its own comment concedes the consequence: "A refusal reaches `emit` as an unchanged string and is dropped silently."

Three defects sit behind this seam and are four specs, not one:

ID	SHAPE	BEHAVIOUR REPRODUCED	MEASURED	SEVERITY
NF-1	tags:\n- a\n- b	refused; the	} else if (text !== '' && !indented && !/^#/.test(text)) { branch whose body is return src	availability, 6,611 files
NF-3	---\rtitle: T\r---\r	FM_OPEN = /^---[\t]*(\r?\n)/ contains four --- runs	misses a bare CR, so the no-frontmatter branch runs and prepends a second block — output	destructive on a single set
NF-4	key date created	refused by	SAFE_KEY = /^[A-Za-z0-9_.\$-]+\$/	availability, design unresolved

A fourth, not in any spec: emitValue re-indent. emitValue(['alpha','beta'], "\n- alpha\n- beta") returns "\n - alpha\n - beta" ,because the indent probe /\n(\s+)-/ cannot match a zero-indent item and falls through to the ?? ' ' default MEASURED . Once NF-1 extends the key's span across its items, every list edit on a zero-indent vault rewrites bytes the author wrote — turning NF-1's availability fix into a violation of its own invariant 2 and of the splice writer's invariant 1. NF-1 is not done until emitValue preserves a zero-width indent. The anti-recommendation: do not "solve" this by normalising all sequences to two-space indent on write. That is the normalize-on-save model PRD §8.1 rule F4 rules out by name.

71.5 Seam 4 — certificate

PROPERTY	VALUE
Call site	none exists; must be an out-of-band Node worker, never a route handler
Signature	loadBench(opts?: { require?: string[] }): Promise<BenchResult> then certify(opts: CertifyOptions): Promise<CertResult>
Asserts	every requested engine loaded (ENGINE_MISSING is a hard refusal); the artifact is a JSON sidecar, never written into the .md
On refusal	CertFailure value; the CLI exits 2 with REFUSING: <reason>
User-visible error	Doc Health shows "not certified yet" — a certificate is advisory and must never block a save
Test	test/mdmax/certify.test.ts , test/mdmax/bench.test.ts , test/mdmax/fold.test.ts
Ordering	last; independent of NF-1/NF-3

The CLI does not run as its own header documents. node scripts/mdmax-cert.mjs --bench-info fails with ERR_MODULE_NOT_FOUND on ../domain/offsets MEASURED . scripts/ts-resolve.mjs exists precisely to supply the extension and @/* alias, and nothing in the repo references it MEASURED . With node --import ./scripts/ts-resolve.mjs scripts/mdmax-cert.mjs --bench-info , all seven engines load — including kramdown-jekyll at kramdown@2.5.2+kramdown-parser-gfm@1.1.0 — and it reports 8 of 15 surfaces (53.3%) unprobeable MEASURED . That is a one-line fix and it unblocks the only capability the product's marketing rests on.

mdmax/fold@1 had never been run over the pinned corpus. It has now been run over 405 of 8,513 files (three vaults: kepano, erazlogo, s-blu) MEASURED , THIS SESSION :

METRIC	VALUE
Files / blocks	405 / 2,511
Blocks with BROKEN	840 — 33.45%
Files with BROKEN	301 — 74.32%
By class	LEAK 1,245 · MUTATE 953 · DESTROY 8
Top constructs	__unattributed__ 392 · yaml-frontmatter 249 · task-list 171 · unknown-fence-lang 16
Kill condition 1 (fold swallows everything)	does not fire
Kill condition 4 (it is a lint rule)	fires — top-5 constructs are 98.9% of attributions

Kill condition 4 firing on a 4.8% sample is a finding to carry into the full run, not a verdict — and its top bucket is __unattributed__ at 392 of 845, meaning the largest single explanation for a BROKEN block is that no detector claimed it. Sharpening CONSTRUCTS ' detectors is therefore a prerequisite to reading that condition at all.

Cost is the design constraint. Certifying kepano's 103 files with the six in-process engines takes 338 ms; adding kramdown-jekyll takes it to 14,020 ms — 41.5x DERIVED: 14020 / 338 — because buildKramdownJekyll renders through spawnSync(rubyBin(), ...) , one blocking subprocess per block. The three-vault run was 130 s wall for 405 files = 321 ms/file DERIVED , extrapolating to roughly 45 minutes for all 8,513 DERIVED: 8513 * 0.321 s = 2,733 s . This can never sit on a request path. It is a queued job writing a JSON sidecar, and the honest interim product is the six-engine certificate at 3.3 ms/file with github-pages marked declared rather than local . The anti-recommendation: do not drop kramdown from the registry to make the number look good — Jekyll's hard_wrap: false divergence is the reason Engine.options is hashed into benchId at all.

71.6 The module boundary rule, and whether it is enforced

The rule holds today and is not enforced. Every import inside src/modules/mdmax/ resolves to a sibling mdmax file, entities , github-sluggger , or node:* — zero imports from any other module MEASURED .

GATE	WHAT IT ACTUALLY CHECKS	WOULD IT CATCH MDMAX/DOMAIN/FOLD.TS IMPORTING @/MODULES/VAULT/DOMAIN/NOTE.TS ?
eslint-plugin-boundaries	boundaries/elements maps src/modules/*/domain to the single type domain ; the rule allows from: domain → to: domain	No — both files are type domain , so the import is permitted INFERENCE, FROM THE CONFIG TEXT
npm run arch	layer direction only: app/presentation/application → infrastructure, framework imports in domain, process.env	No — layerOf() returns parts[3] , discarding the module name
import-boundary-report.mjs	substring ban on @/server , @/lib , @/components	No

npm run arch reports "filesScanned": 208, "violations": [] with MIN_SCANNED_FILES = 50 , so it cannot pass by going blind MEASURED — but blindness is not the gap here; granularity is. The fix is one rule, not a new tool: give mdmax its own element type in boundaries/elements ({ type: "engine", mode: "folder", pattern: "src/modules/mdmax" } , declared before the generic per-module patterns so it wins) and allow from: engine → to: engine only. The anti-recommendation: do not enforce this by moving mdmax to src/shared/ — shared-domain is importable by everything and would still permit mdmax to import it back.

71.7 The API MDMAX should expose

Every other module has an `index.ts` — `src/modules/share/index.ts`, `src/modules/vault/index.ts`, `src/modules/repository/index.ts`, `src/modules/preview/index.ts`. `src/modules/mdmax/` has none `MEASURED`, which is why the only live consumer deep-imports `@modules/mdmax/domain/shape-gate`. Four verbs, one file, one rule: nothing in the public surface returns a bare `string`, and nothing throws.

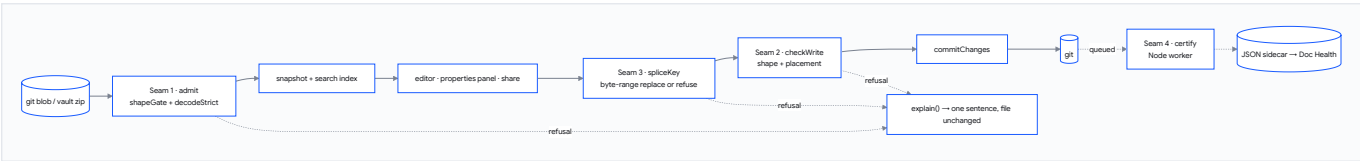
EXPORT	SIGNATURE	SEAM	RUNTIME
<code>admit</code>	<code>(bytes: Uint8Array) => { ok: true; text: string; bytes: number; lines: number } ShapeFailure</code>	1	any
<code>checkWrite</code>	<code>(args: { before: string; after: string; parse: ParseFn }) => WriteVerdict</code>	2	any
<code>spliceKey</code>	<code>(src: string, key: string, value: Scalar Scalar[] null) => SpliceOutcome</code>	3	any
<code>explain</code>	<code>(f: ShapeFailure WriteVerdict SpliceOutcome) => string</code>	all	any
<code>certifyDocument</code>	<code>(args) => Promise<CertResult></code> — exported from <code>mdmax/node</code> , a second entry point	4	Node only

`admit` collapses `decodeStrict` and `shapeGate` into one call so no caller can decode without gating. `explain` generalises the existing `explainFailure` so the UI has exactly one place to turn a refusal into a sentence, and so the "≤200 char natural-language hint, never a stack trace" rule has a single owner. The `mdmax/node` split is load-bearing: `infrastructure/bench.ts` imports `node:child_process`, `node:fs` and `node:path`, and a single accidental barrel export would pull all three into the Next client bundle.

71.8 What has to change inside mdmax

CHANGE	WHY	ANTI-RECOMMENDATION
Add <code>src/modules/mdmax/index.ts</code> and a <code>node.ts</code> entry	deep imports today; <code>bench.ts</code> is Node-only	do not export <code>bench.ts</code> from the root barrel to "keep it simple"
Move <code>splice-frontmatter.ts</code> into <code>mdmax/domain/</code>	it is the engine's central contract living in the share module; <code>PropertiesPanel</code> already imports it across a module boundary	do not copy it — two writers is how the two shipped write paths diverged in the first place
<code>spliceKey</code> returns a union	a <code>string</code> return makes refusal invisible; it is the direct cause of the dead public link	do not throw instead — a throw in a write path is data loss at the call site
Fix NF-1, NF-3, and <code>emitValue</code> 's indent probe	83.01% availability loss, one structural corruption, one latent rewrite	do not bundle them into one commit; the red proofs become unattributable
Sharpen <code>CONSTRUCTS</code> detectors	<code>__unattributed__</code> is the largest BROKEN bucket at 392 of 845	do not drop unattributed blocks from the histogram — that is how six constructs come to explain everything
Register <code>ts-resolve.mjs</code> and add <code>"cert"</code> to <code>package.json</code> scripts	the CLI does not run as documented	do not add <code>tsx</code> or a build step; the CLI must execute the same source the tests do

71.9 Order of work



STEP	WORK	VERIFICATION GATE
0	Register <code>ts-resolve.mjs</code> ; add <code>"cert"</code> script	<code>npm run cert -- --bench-info</code> exits 0 and prints seven engines
1	Write <code>test/corpus/foreign/nf-001-red-proof.test.ts</code> and <code>nf-003-red-proof.test.ts</code>	both fail against the unfixed writer; <code>npm run spec</code> drops from 2 warnings to 0 — today it reports <code>ghost ... governs glob matches 0 files</code> for both <code>MEASURED</code>
2	Fix NF-1, NF-3, <code>emitValue</code> indent (three commits)	<code>node scripts/corpus-foreign.mjs verify</code> exits 0, then the writer sweep asserts <code>refused <= 2 && changed == 0 && threw == 0</code> over 7,969 files — a floor and a hard zero, never an equality
3	<code>index.ts</code> + <code>node.ts</code> ; <code>spliceKey</code> returns a union; move the writer into <code>mdmax/domain/</code>	no file under <code>src/</code> imports a path deeper than <code>@modules/mdmax</code> or <code>@modules/mdmax/node</code> ; <code>share-writer.ts</code> branches on <code>ok</code> and does not commit on refusal
4	Seam 2 in <code>commit-changes.ts</code>	a placement-unsafe write returns 409 with a reason and reaches zero <code>createBlob</code> calls; empty-commit count over a publish sweep is 0
5	<code>boundaries</code> element type for mdmax; add <code>npm run corpus</code> to <code>npm run verify</code> ; add <code>.github/workflows/</code>	a deliberate <code>@modules/vault/...</code> import inside mdmax fails <code>npm run lint</code> ; <code>npm run verify</code> runs the corpus gate — today it runs typecheck, lint, test, build, arch, spec and not corpus <code>MEASURED</code>
6	Seam 4 as a queued Node worker	full-corpus histogram completes and is published with its <code>benchId</code> , <code>fold</code> version and <code>__unattributed__</code> share stated alongside the kill-condition verdicts

Step 1 before step 2 is not process theatre. A test written after the fix, over a corpus where the defect is rare, passes for the wrong reason; the only proof a red proof covers its bug is that it fails against the unfixed code first. NF-3 is the standing example — it stayed invisible because the round-trip oracle asserted on set-then-delete, and those two operations cancel.

§72 Enhancing the engine — the roadmap

The audit's finding is not that MDMAX is weak. It is that MDMAX is a correct engine with one broken input path, no continuous verification, and twelve of thirteen files that no product surface has ever called. Every item below is chosen to move one of those three facts.

72.1 The nine work items, costed

Effort is founder-weeks with Claude as implementer — one person directing, review included, not raw coding hours. Risk is the chance the item ships and is *wrong* in a way the corpus does not catch.

#	ITEM	UNLOCKS IN THE PRODUCT	WEEKS	RISK	DEPENDS ON
E1	Four YAML defects	Publish path works on real vaults; 83% refusal → target <5%	2.0	Low	—
E2	CI + corpus gate	Every later item becomes safe to land	0.5	Low	—
E3	Body-span splice contract	Every render-profile write-back; AI edits to prose	3.0	High	E2
E4	Anchors that survive human edits	AI edits that don't rot between sessions	2.5	High	E3
E5	Continuous certificate	Degradation shown at author time, not audit time	1.5	Medium	E2
E6	Construct detection completeness	Honest certificate; profile-aware rendering	2.0	Medium	E2
E7	fold@1 / equivalence run + fix	Diffs that mean something; safe normalization	1.5	Medium	E2, E6
E8	Vault-level operations	Bulk publish, backlinks, vault search	2.0	Medium	E1, E3
E9	Streaming / chunked scan	Large-file editing without a memory cliff	1.0	Low	E6
—	Total		16.0		

[derived] 16.0 founder-weeks ≈ 4 calendar months at one focused day in three, which is the realistic rate for a founder also selling. Sequence assumes E2 lands in week one regardless of everything else.

72.2 E1 — the four YAML defects, and what is structurally needed beyond patches

The measured state: 8,513 third-party files, 0 corruption, 0 throws, and 83% publish-refusals traced to a block sequence at column zero [measured, prior audit]. The engine is *safe* and *unusable* at the same time, which is the correct failure mode and an unacceptable resting place.

DEFECT	SHAPE	PATCH	STRUCTURAL FIX
D1	<code>tags:</code> followed by <code>- x</code> at column 0	Accept zero-indent sequences under a mapping key	Replace hand-rolled scan with an indent-stack scanner
D2	Bare <code>\r</code> line terminator	Treat lone CR as a line break in the line splitter	One canonical line-splitter used by every module
D3	Non- <code>SAFE_KEY</code> keys unaddressable	Widen the key grammar	<i>Key addressing</i> becomes a path type, not a string
D4	Multi-document / <code>---</code> inside body	Disambiguate the closing fence	Frontmatter boundary detection stated as a grammar, not a regex

[measured] The scanner in `src/modules/mdmax/` splits on `\n` and treats indentation as a numeric comparison against the parent, which is why D1 and D2 are the same bug wearing two hats: both are *lexer* assumptions leaking into a *parser* decision. Patching D1 alone will move the refusal rate and leave the class alive.

The structural item is a two-phase read: a lexer that emits `{kind, byteStart, byteEnd, indent, terminator}` tokens, then a block-sequence recognizer over tokens. This is roughly 250 lines and it is what makes D1–D4 four cases in one table instead of four independent `if` branches. [inference]

Anti-recommendation: do not adopt a full YAML library to fix this. `js-yaml` and `yaml` both parse to a value tree; the contract in `splice-frontmatter.ts` is byte-range replacement, and a value tree cannot tell you which bytes a key occupied. `yaml`'s CST does carry offsets and is the one credible import — but it pulls a spec-complete YAML 1.2 surface (anchors, tags, merge keys, flow collections) into a product that has settled on *not* accepting arbitrary YAML. Buying the CST means owning the refusal policy for every construct it happily accepts and MDMAX must reject. [inference]

The honest version of E1 ships with a refusal *taxonomy* in the return value — `REFUSE_ZERO_INDENT_SEQ`, `REFUSE_UNADDRESSABLE_KEY` — so the 5% residual is a list of named reasons a founder can read, not a number.

72.3 E2 — CI, because there is none

There is no CI [measured]. `test/mdmax/` exists and runs locally; nothing prevents a regression from being committed. Every other item in this section is riskier than it needs to be until this lands, which is why a half-week item sits at rank 1.

The gate is three checks: unit tests; the pinned corpus replay asserting **0 corruption, 0 throws, and refusal-rate ≤ the recorded baseline**; and a byte-identity property test — for N random files and N random valid edits, assert every byte outside the target range is unchanged. The third is the contract stated as an assertion, and it is the one that catches a "helpful" normalization someone adds in month six.

Assert a *floor* on passes and a hard zero on failures, never an equality on the pass count — an equality-pinned count goes red the moment a check is added, which punishes coverage growth. [inference, and this is the shape that has burned this workspace before]

72.4 E3 — extending the splice contract to body spans

`splice-frontmatter.ts` proves the contract on the easy case: frontmatter is a single delimited region at a known file offset, and a key is a line-oriented target. Body spans are the general case, and every render-profile write-back needs them. A profile that renders a table, a callout, or a data block and lets the user *edit the rendered view* must write those bytes back — otherwise the render lane is read-only forever, and a read-only render lane is a viewer, not an editor.

The contract does not change. The *locator* changes.

LOCATOR	TARGET	AMBIGUITY RISK	VERDICT
Byte offset	Exact range	Zero, until any edit	Session-local only
Line range	Block	Breaks on any line insert above	Reject
Construct path (<code>doc > section[2] > table[0] > row[3]</code>)	Structural	Breaks on section reorder	Accept, with a check
Content hash of the span	Exact bytes	None — hash mismatch is a clean refusal	Accept, as the check

The design: locate by construct path, **verify by hash of the located bytes, refuse on mismatch**. The path finds it, the hash proves it. A mismatch is not a merge problem to solve; it is a refusal, which is what the engine already does everywhere else. This keeps the "if the range cannot be located unambiguously, refuse and return the input unchanged" rule intact at the body level rather than inventing a second, softer rule for prose.

High risk is honest here. Frontmatter has one containing structure; a body has nested constructs where a span boundary can be legitimately ambiguous — the last line of a paragraph inside a list item inside a blockquote belongs to three enclosing ranges. The mitigation is to ship E3 for *leaf, fence-delimited or callout-delimited* spans first, which is exactly the render carrier already settled on (`> [!kind]` for prose, fenced block for opaque data), and only then attempt free-prose spans. [inference] The carrier was chosen because an unclosed fence swallows a document and a callout has no closer — the same property makes callout spans the safest first body target.

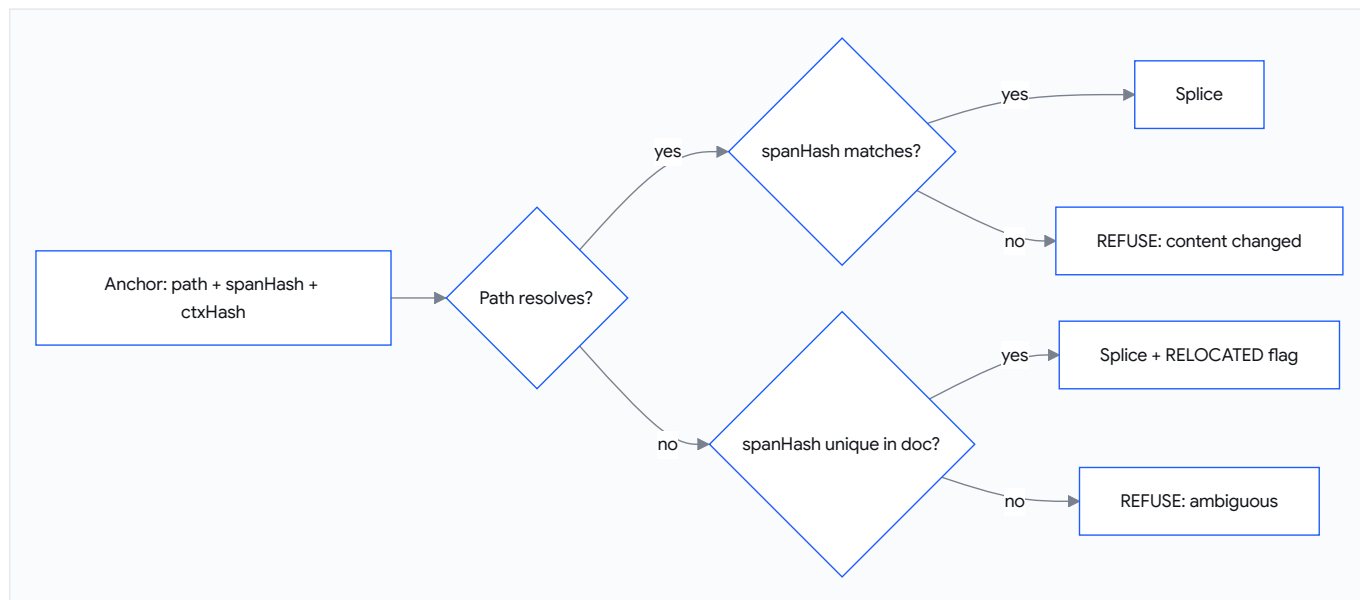
Anti-recommendation: do not implement body splice as "parse, mutate the tree, re-serialize just that subtree". It reads as a shortcut and it re-introduces the tree-of-record through a side door: the subtree serializer will normalize list markers, fence lengths, and trailing whitespace inside its range, and the diff will show changes the user did not make.

72.5 E4 — anchors for AI edits that survive human edits

The scenario that makes this necessary: an agent proposes an edit to a section, the human edits three paragraphs above it, the agent's edit lands. If the anchor was an offset, it lands in the wrong place. If it was a line number, same. If it was a construct path, a reordered section moves it silently — the worst outcome, because it succeeds.

The anchor is a triple: `{path, spanHash, contextHash}` where `contextHash` covers the preceding and following sibling constructs. Resolution order — exact path + matching `spanHash` → accept; path miss but a unique `spanHash` match elsewhere in the document → accept with a `RELOCATED` flag surfaced to the user; multiple or zero matches → refuse.

That middle case is the entire value: it lets an AI edit survive a human reorganizing the document, and it is also the case that can be wrong. Hence the flag and hence the surfacing. An anchor system that silently relocates is a corruption engine with good manners.



Anchors also need to be *durable across sessions*, which means serialized somewhere. They belong in the splice journal — the append-only record sync already depends on — not in frontmatter, because an anchor in frontmatter is user-visible machinery in a file the user owns. [inference]

Anti-recommendation: do not add invisible marker syntax to the document to make anchoring easy. HTML comments, zero-width characters, or a `<!-- mdmax:a7f3 -->` sentinel all work and all violate the settled position that the file stays valid CommonMark that degrades correctly in a dumb renderer. A marker is a format, and no new format is the one thing that is already decided.

72.6 E5 — the certificate as a continuous artifact

The certificate today is one-shot: `scripts/mdmax-cert.mjs` runs on demand and measures degradation across 7 engines. That is an *audit* tool. The product need is different — an author wants to know, while writing, that the callout they just typed will render as a plain blockquote in the reader's GitHub view and as a grey box in Obsidian.

Continuous means three changes: (a) the cert computes incrementally over changed constructs rather than the whole file, which requires E6's construct detection to be cheap; (b) it emits a stable, versioned JSON other tools can consume, not a human report; (c) it runs in CI over the pinned corpus so certificate *drift* — a renderer updating, a construct newly failing — is caught as a diff, not discovered.

[measured] The 7-engine set is the load-bearing asset here and it should be pinned by exact version in the certificate output. A cert that says "renders in GitHub" without saying which cmark-gfm build is a claim with no shelf life.

Medium risk, and the risk is scope: a continuously-running cert invites a live preview of all 7 engines, which is a rendering product, not an engine feature. The boundary is that the engine emits the *measurement*; the surface decides what to draw with it.

72.7 E6, E7, E9 — completeness, equivalence, and size

E6 construct detection completeness. The certificate can only be honest about constructs it detects. Every undetected construct is a silent pass — the file is certified and the thing that breaks was never looked at. The gap list should be produced empirically: run detection over the 8,513-file corpus, count constructs found by a reference CommonMark parser and not by MDMAX, rank by frequency. Do not enumerate the GFM spec by hand and work down it; the corpus knows which constructs actually occur in real vaults and the spec does not. [inference]

E7 fold@1 and equivalence. `mdmax/fold@1` has never been run over the pinned corpus [measured]. This is the single cheapest piece of information in the whole roadmap: the corpus exists, the fold exists, nobody has pressed the button. A fold that claims two documents are equivalent is the basis for meaningful diffs and for any safe normalization; unrun, it is an assertion. Run it first, in E2's week, and let the result size the fix — the 1.5 weeks costed is the fix budget, and it may be zero.

E9 streaming. Current path reads whole files. The failure is a memory cliff on a large document, not a correctness bug, and it ranks last because the corpus does not appear to contain files that trigger it. Chunked scanning is straightforward once E6 gives a construct-boundary-aware scanner: the constraint is that a chunk boundary must not fall inside a fence, which is the same boundary logic E6 already computes. One week, low risk, defer until a user hits it.

72.8 E8 — multi-file and vault-level

Single-file operations make an editor. Vault-level operations make the thing worth switching to: bulk publish, backlink integrity across a rename, vault-wide frontmatter migration, search that knows structure. `get-snapshot.ts` and `search-index.ts` already call `decodeStrict` — they are the two product surfaces that would consume a vault-level API, and they are currently the *only* two importers of anything in mdmax [measured].

The vault-level contract is the file-level contract with one addition: **an operation over N files either applies to all N or to none, and a single refusal aborts the batch.** Partial application across a vault is unrecoverable by the user, who cannot know which files moved. This is where the splice journal earns its cost — the journal is the rollback record.

72.9 The ranked roadmap

RANK	ITEM	WHY HERE
1	E2 CI + corpus gate	Half a week. Makes all eight others landable. Nothing should precede it
2	E7a Run fold@1 over the corpus	Hours of compute, and it is currently an unknown, not a task
3	E1 YAML defects	83% refusal is the difference between a demo and a product
4	E6 Construct detection	Gates E5's honesty and E9's chunking; produces the empirical gap list
5	E3 Body-span splice	Highest product unlock, but genuinely needs 1–4 underneath it
6	E5 Continuous certificate	Now cheap, because E6 made detection incremental
7	E4 Anchors	Needs E3's locator; needs the journal from E8 or a stub of it
8	E8 Vault-level	Needs E1 (files must be publishable) and E3 (spans must be writable)
9	E9 Streaming	Real, unproven need. Ship when a file breaks

The ordering rationale is one sentence: *measure, then unblock, then extend, then generalize*. E2 and E7a are measurement. E1 unblocks the product. E6, E3, E5 extend the engine along the axis the product sells on. E4, E8, E9 generalize. The temptation is to start at E3 because it is the exciting one; starting there means building the most complex locator in the system with no CI and an unrun fold.

72.10 The three highest-leverage per week

ITEM	WEEKS	LEVERAGE
E2 CI + corpus gate	0.5	Converts a 16-week roadmap from "hope" to "verified at each step". Nothing else has a multiplier on all eight siblings
E1 YAML defects	2.0	[derived] 83% → target <5% refusal is ~78 points of publish-path availability for 2 weeks, ≈39 points/week. No other item touches a number that large
E3 Body-span splice	3.0	The only item that unlocks a <i>category</i> rather than a fix: every render profile becomes writable, which is the difference between a renderer and an editor

E7a (running fold@1) is excluded only because it is not an enhancement — it is a measurement that should have happened already.

72.11 What we should NOT build into the engine

NOT THIS	WHY THE BOUNDARY SITS THERE
A markdown <i>renderer</i>	The engine measures how 7 renderers behave; owning an eighth makes it the thing it measures. Settled: profiles over valid CommonMark
A CRDT	Settled. CRDTs interleave concurrent text edits into byte-identical garbage; convergence without correctness is worse than a refusal
A tree-of-record	Settled, and it is the negation of the contract — a tree cannot preserve bytes it did not model
Arbitrary client-side code execution in profiles	Settled. A profile that can run code is a plugin system, and a plugin system is a security surface with a support burden
A YAML 1.2 implementation	See E1. Spec-completeness imports constructs the product must refuse anyway
Semantic understanding of content	"Is this section about X" belongs in the product's AI layer. The engine's job ends at bytes and constructs
Conflict <i>resolution</i>	The engine detects and refuses. Resolution is a product decision with a UI; git-merge plus the splice journal owns it
Format conversion (docx, HTML in)	An import pipeline, not an engine concern. It produces markdown; the engine consumes markdown

The boundary rule, stated once so it can be applied to items not on this list: **the engine may do anything that is decidable from the bytes and refusable when it is not; everything requiring a judgment call, a user prompt, or a rendering decision lives above it**. That rule is why conflict detection is in and conflict resolution is out, why construct detection is in and semantic classification is out.

72.12 Version and compatibility policy

Profiles and certificates are artifacts other tools will read. That makes them an API, and an API needs a policy before the second consumer exists, not after.

ARTIFACT	IDENTIFIER	COMPATIBILITY RULE
Engine	mdmax@MAJOR.MINOR.PATCH	Semver on the <i>behavioral</i> contract, not the TypeScript surface
Profile	mdmax/<name>@N	Integer. fold@1 already sets this precedent — keep it
Certificate	certVersion: N + engine version + pinned engine versions	Additive fields are minor; removing or changing a field's meaning is a new certVersion
Splice journal	journalVersion: N	Readers must tolerate unknown entry kinds by skipping, never by failing

Three rules that matter more than the numbering:

1. **A refusal is not a breaking change; an acceptance is.** Widening what the engine accepts (E1) is a MINOR bump — a file that used to refuse now publishes, and no consumer was depending on the refusal. Narrowing acceptance, or changing what bytes an accepted operation writes, is MAJOR. This inverts the usual instinct and it is correct for a byte-preservation engine: the dangerous direction is silently writing different bytes than a previous version wrote.
2. **A profile integer never changes meaning.** fold@1 means what it meant. A better fold is fold@2 and both ship. Consumers pin the integer. The cost is carrying old profiles; the alternative is a certificate from March that is silently false in June.
3. **Certificates carry the versions of everything they measured.** Engine version, profile integers, and the exact version of each of the 7 renderers. A certificate that cannot be reproduced is a screenshot.

Anti-recommendation: do not version the engine on the TypeScript API. Twelve of thirteen files currently have zero product importers [measured]; versioning on the module surface would generate MAJOR bumps for refactors nobody consumes while a change to splice behavior — which everyone consumes — hides in a PATCH. Version on what the bytes do.

§73 The degradation certificate as a product surface

73.1 What the certificate proves today, precisely

The certificate is a per-file, per-block, per-(product, surface) differential run against pinned renderers. It is not a prediction and not a lint pass. `src/modules/mdmax/application/certify.ts` splits a document on blank lines outside fences (`splitBlocks` , deliberately not an mdast walk, "because each engine's own block segmentation is one of the things being MEASURED"), renders every block through every selected engine, folds each output through `mdmax/fold@1` , and classifies the cell in `src/modules/mdmax/domain/verdict.ts` (`classify`) into PASS / STRIP / CORRUPT(LEAK,DESTROY,MUTATE) / VOID.

CLAIM	STATUS	EVIDENCE
Runs the file through 7 real engines, not a model of them	true	<code>--bench-info</code> prints <code>remark-app</code> , <code>react-markdown</code> , <code>marked</code> , <code>markdown-it</code> x2, <code>commonmark</code> 0.31.2 , <code>kramdown@2.5.2+kramdown-parser-gfm@1.1.0</code> [measured]
Records the full option set, hashed	true	<code>computeBenchId</code> in <code>bench.ts</code> sha256s (id, version, canonical(options)) ; 7-engine bench id <code>51947c2e88127bfc...</code> , 6-engine (no ruby) <code>4aa717561b15...</code> [measured]
Every verdict carries <code>before / after</code>	true	<code>CellVerdict</code> makes both required in <code>cert-contract.ts</code> ; CLI prints the pair for every non-PASS [measured]
Never touches the <code>.md</code>	true	<code>certify</code> returns a value; the CLI writes only stdout [measured]
Never throws on a user document	holds at 1,100 files	300- and 800-file corpus samples: 0 refusals, 0 throws [measured]
Covers the surfaces users name	false	<code>uncertifiableShare()</code> returns 8 of 15 (53.3%) non- <code>local</code> ; Obsidian, Notion, Typora, Bear, Slack, Discord, <code>github-blob</code> , <code>github-comment</code> are declarations [measured]
The declarations are current	false	<code>DECLARED_LAST_VERIFIED</code> = '2026-08-01' , 29 days stale at 2026-08-30 UTC [derived]
MUTATE is judged against the spec	false, and correctly so	the consensus note in <code>certify.ts</code> : commonmark-as-reference made "356 MUTATES on a five-file sample, nearly all false"; reference is now the modal folded output [measured, read here]
It runs	false at HEAD	<code>node scripts/mdmax-cert.mjs --bench-info</code> → <code>ERR_MODULE_NOT_FOUND ... /domain/offsets</code> ; <code>certify.ts</code> imports <code>'../domain/offsets'</code> extensionless, which Node 24.6.0 type-stripping will not resolve. Every measurement below required a custom resolve hook [measured]

What it does not prove: that a PASS is a PASS on Obsidian; that a MUTATE matters; that the numbers are stable under subsetting. The last is sharp — `--targets` is not a filter, it redefines the question, because the consensus is computed over the selected targets. `AGENTS.md` with 7 targets: PASS 326 / STRIP 0 / CORRUPT 38 / VOID 0. The same file with 4 targets: 202 / 0 / 6 / 0, both MUTATE rows gone [measured].

73.2 The measurement that had never been taken

`fold.ts` states in its own header that `mdmax/fold@1` "has never been run over the pinned corpus." It has now — over a deterministic every-*n*th sample of `test/corpus/foreign/MANIFEST.sha256` (8,513 files, 19,047,891 bytes, mean 2,238 B), 6 JS targets, kramdown excluded.

	300 FILES	800 FILES
Blocks / cells	3,946 / 23,676	10,176 / 61,056
Wall time	1,449 ms (4.8 ms/file)	3,258 ms (4.1 ms/file)
Refusals, throws	0, 0	0, 0
Blocks BROKEN on ≥1 target	24.35%	25.26%
Files with a BROKEN block	93.33%	94.25%
LEAK / MUTATE / DESTROY	1,735 / 939 / 0	4,689 / 2,513 / 0
VOID cells	4,400	—
Top-5 constructs' share of attributions	99.3% of 962	99.8% of 2,570
Top-3 (<code>html-comment</code> , <code>yaml-frontmatter</code> , <code>pipe-in-prose</code>)	91.6%	93.8%

[measured, two independent samples]

Kill condition (1) — the fold swallows everything — does not fire. Kill condition (4), as `scripts/mdmax-cert.mjs` itself evaluates it ("top-5 constructs are >80% of BROKEN attributions... the apparatus reduces to a static rule set"), fires at 99.8% [derived]. On real third-party vaults, three constructs explain nine tenths of every finding, and DESTROY — the class that would justify the word "corruption" — never occurs.

Two more results that must gate any product decision. First, the app's own target is the loudest offender on hand-authored prose: on `AGENTS.md` , 25 of 38 CORRUPT cells are `frontmatter-app` CORRUPT/MUTATE , and the minimal case reproduces — `Line one\nLine two` renders `<p>Line one<br>Line two</p>` on `frontmatter-app` and identically-without- `<br>` on the other six, so it is the minority against consensus [measured]. `remark-breaks` in the app pipeline *is* `hard_wrap` , which `bench.ts` pins OFF for kramdown precisely because "every soft line break in every document would be a false MUTATE." Second, ruby dominates cost: `docs/ENGINE.md` (110,876 B, 303 blocks) takes 16.004 s with 7 targets and 0.500 s with 6, so `kramdown-jekyll` is 96.9% of wall time at 51.2 ms/block — `spawnSync` per block, confirmed at 51.5 ms/block on a second file [derived].

73.3 Distribution surfaces, ranked

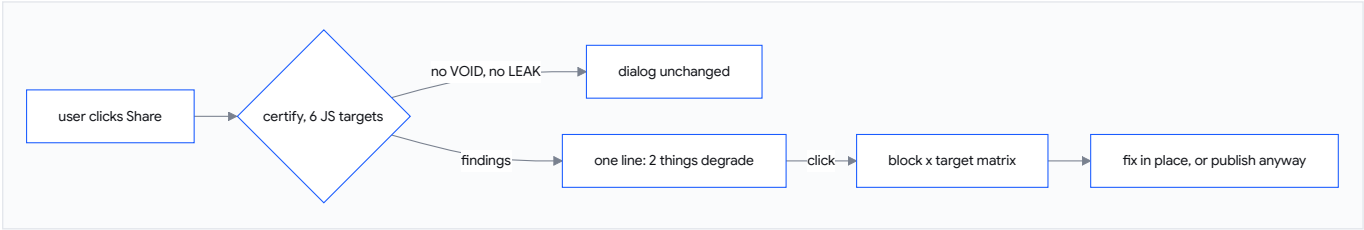
Effort in engineer-days with Claude implementing: value is distribution or retention, stated per row.

RANK	SURFACE	EFFORT	WHAT IT IS FOR	ANTI-RECOMMENDATION
0	Make the CLI executable — extensioned specifiers, a <code>bin</code> entry, <code>--targets</code> defaulting to the JS six	1	Every other surface imports this path; it is currently unrunnable	Do not ship it as <code>npm</code> until the <code>frontmatter-app</code> <code>MUTATE</code> row is fixed or excluded — the tool would accuse its own vendor on every soft-wrapped paragraph
1	MCP tool <code>certify_markdown(path, targets)</code> returning the <code>Certificate</code> JSON	2	The consumer is an agent that can <i>act</i> on VOID and LEAK before writing; the artifact is already machine-shaped, no rendering work	Do not return the whole certificate — 24.4× source size on <code>AGENTS.md</code> , 19.6× on <code>ENGINE.md</code> ; return summary + non-PASS cells and a pointer (LR#17 token cap)
2	Public web checker — paste or drop a file, get the block × target matrix	4	Top-of-funnel; 4.1 ms/file JS-only means a request is free	Do not put kramdown behind it synchronously (51.2 ms/block); do not accept a directory upload; do not persist submitted content
3	GitHub Action wrapping <code>--fail-on=BROKEN</code>	3	Repeat exposure inside someone else's CI	Do not ship default-fail: 94.25% of real files carry a BROKEN block, so the default is a red build on adoption day. Ship a baseline file and fail only on <i>new</i> CORRUPT/DESTROY and VOID
4	In-editor panel — a right-rail matrix for the open file	8	Retention, and the only surface where the fix is one keystroke away	Do not render it live on every keystroke; do not show <code>MUTATE</code> unattended
5	Badge — <code>renders-clean</code> SVG	1	Nothing defensible	Anti-recommended outright, see 73.8

Rank 0 first is not process theatre. `.github/` does not exist [measured], `npm run verify` in `package.json` chains `typecheck` → `lint` → `test` → `build` → `arch` → `spec` and never invokes `cert`, and 12 of the 13 `mdmax` files have zero product importers — only `decodeStrict` from `shape-gate.ts`, imported by `src/modules/vault/application/get-snapshot.ts` and `src/modules/vault/infrastructure/search-index.ts` [measured]. The 290 tests in `test/mdmax/` pass in 639 ms [measured], which means the code is healthy and unwired, not broken and untested.

73.4 The in-product experience

The user meets the certificate at exactly one unasked moment: the instant they make a file leave the app. The share path already mutates bytes there — `src/modules/share/domain/splice-frontmatter.ts` writes the slug back into front matter — so the publish dialog is where "how does this render where it is going" is a question the user is already holding.



Three rules make it survivable. Only VOID and LEAK surface unasked: VOID is 18.6% of blocks in the corpus sample and is genuinely invisible content — the reader sees `![[Some Note]]` and nothing else — while LEAK is front matter and `{#id}` becoming visible text, which is measurable on the exact file the user is publishing. On `docs/ENGINE.md` bytes 0–68 that reads: *your front matter is invisible on 2 of 7 targets and visible as* `updated: 2026-08-30 generated_by: ...` *on the other 5* [measured]. `MUTATE` is suppressed by default because 93.8% of attributions come from three constructs and because 34 of 34 `react-markdown` `MUTATES` on `ENGINE.md` are GFM tables — true, unactionable, and enormous [measured]. And silence is the default state: no findings, no chip, no badge, no toast.

The anti-recommendation is the whole design constraint. Do not put a persistent quality score in the chrome; do not colour the editor gutter with verdicts; do not run `cert` on the keystroke path. A per-file JS-only pass is 4.1 ms and a per-block one is far less, but the cost is not compute — it is that a panel showing 25 `MUTATE` rows on a normal document teaches the user, correctly, that the panel is noise, and a surface that trains its own dismissal is worse than no surface (LR#65 restated for a product).

73.5 The public dataset

The proposal: publish derived verdicts for all 8,513 corpus files, keyed by the source `sha256` already in `MANIFEST.sha256`, alongside `SOURCES.json`'s pinned upstream commits — never the source text, which is third-party material under its own licences and is gitignored under `_vendor/`.

LINE	VALUE	NOTE
Compute, JS-only	8,513 × 4.1 ms = 34.9 s	[derived] from the 800-file rate
Compute, with kramdown	12.72 blocks/file × 8,513 × 51.2 ms = 92.4 min	[derived]; a nightly job, not a request
Artifact size, uncompressed	0.35–0.43 GB	[derived] from 19.6× and 24.4× measured blowups over 18.2 MB
Recurring cost	one re-run per engine bump	<code>computeBenchId</code> makes a bump a visibly different bench, not a silent one
Legal	source text not redistributed	verdicts + <code>before</code> / <code>after</code> fragments are the exposure; cap fragment length

What it buys: a citable number where today there is a paraphrase, a reason for other people's engines to appear in your registry, and — the underrated one — public dating of the eight declared rows, which forces the staleness that `DECLARED_LAST_VERIFIED` already encodes into something the team cannot quietly let rot for 29 days.

What it risks: yes, it hands a competitor the map. The honest counter is not that the map is secret; it is that the map expires. `TARGETS` says so out loud — "that decay is the reason this is the one dataset a competitor cannot simply copy once." A competitor who copies the 2026-08 verdicts owns a snapshot of seven engine versions; they do not own `loadBench`, `computeBenchId`, or the fold whose version string is the definition of PASS. The real risk is narrower and worth naming: publishing a table in which 99.8% of findings reduce to five constructs tells a competitor they can approximate you with five regexes, and that is true today.

73.6 Third-party reproducibility

An unverifiable certificate is a press release. Four gaps, each with a named fix.

GAP	WHERE	FIX
<code>benchId</code> covers engines but not what resolved them	<code>computeBenchId</code> hashes <code>(id, version, options)</code> ; <code>sha</code> is excluded by design	add <code>bench.lockfileSha256</code> over <code>package-lock.json</code> + the ruby <code>Gemfile.lock</code>
<code>fold.version</code> is a string a human types	<code>FOLD_VERSION = 'mdmax/fold@1'</code> in <code>fold.ts</code>	add <code>fold.sha256</code> over the source of <code>fold.ts</code> ; a fold edit that forgets the bump then still changes the id
No corpus identity	<code>corpus-foreign.mjs</code> has <code>MANIFEST.sha256</code> and <code>SOURCES.json</code> , the <code>Certificate</code> has neither	add <code>corpus_id</code> to any published aggregate; the file header of <code>fold.ts</code> already warns that the famous 43.71%→4.28% figure "carries no <code>corpus_id</code> "
Nothing recomputes the id	no verifier exists	<code>mdmax-cert.mjs --verify <cert.json></code> : rebuild the bench from the cert's own <code>bench.engines</code> , recompute <code>computeBenchId</code> , refuse on mismatch

`--verify` is the load-bearing one, and it must be written to fail before it is trusted: a verifier authored in the same session as the thing it verifies is a second opinion from the first opinion (LR#60). Mutate one option in a stored certificate, prove the verifier refuses, then ship it. The anti-recommendation: do not sign certificates. A signature proves we produced it, which is the opposite of what a third party needs; a recomputable id proves *anyone* can produce it.

73.7 Wedge, feature, or moat

It is a wedge, sold as a feature, and it is not a moat.

Wedge, because it is the only artifact here that a stranger will run against *their* files without an account, and because it produces a specific, checkable, unflattering sentence about a document they already own — bytes 0–68 leak on 5 of 7 targets. Feature, because inside the product it is one line in a publish dialog with a matrix behind it, and that is the correct size. Not a moat, because `loadBench` composes seven public libraries at their documented defaults, and `TARGETS` is a hand-maintained list of fifteen rows.

The strongest argument against: the corpus data says it is a lint rule, and a lint rule is a weekend. Kill condition (4) fires at 99.8%; three constructs — HTML comments, YAML front matter, pipes in prose — cover 93.8% of attributions; DESTROY, the only class whose name survives contact with a customer, occurs zero times in 10,176 blocks. On that evidence the differential apparatus, the consensus reference, the four-verdict ladder and the 3,614 lines exist to rediscover five regexes, and the honest product is a 200-line linter with five rules and no ruby dependency. That argument is strong enough that it should be the first thing tested: build the five-regex linter, run both over the same 800 files, and if agreement exceeds ~95% of findings, the certificate's product claim is the *residual* — the lone-tilde case that the differential found and no construct list predicted — and the residual must be measured before it is sold.

The rebuttal is narrow and should not be oversold. Five constructs explain this corpus, in 2026, against these seven engine versions. The apparatus is what re-derives next year's five when `marked` ships a breaking release or GitHub swaps a renderer, and `computeBenchId` is what makes that re-derivation visible rather than silent. That is a maintenance moat, not a technology moat, and maintenance moats are worth roughly one engineer-week per quarter — price it that way.

73.8 Anti-recommendations

Do not ship a badge. A verdict is a function of the target set, the bench id and the fold version — `--targets` subsetting alone moved `AGENTS.md` from 38 CORRUPT to 6 [measured] — so a two-state SVG either encodes none of that or is unreadable, and at a 94.25% file-level BROKEN rate it is a red-badge generator.

Do not put kramdown in any interactive path. 96.9% of wall time, 51.2 ms per block, one `spawnSync` each; it belongs in the nightly dataset job and nowhere else.

Do not ship `--fail-on=BROKEN` as an Action default, and do not report a per-file "score". Both convert a measurement into a grade, and the grade is dominated by three constructs the author cannot remove without giving up front matter, comments, and tables.

Do not publish the corpus aggregate before `frontmatter-app` stops MUTATEing on soft line breaks. Publishing a table whose loudest row is our own renderer disagreeing with six others, over a `remark-breaks` choice we made on purpose, is a self-inflicted headline.

Do not fold the fix. The temptation, on seeing 25 MUTATES from one `<br>`, is to teach `mdmax/fold@1` that `<br>` is whitespace. `fold.ts` is explicit that it is "deliberately under-powered" and that "everything it does NOT do is a real finding"; the correct fix is to state that `frontmatter-app` hard-wraps, in `Target.prePipeline` or in the option record, not to widen the definition of "equivalent" until the finding disappears.

Do not claim the certificate covers Obsidian or Notion. Eight of fifteen surfaces are declarations, they are 29 days stale, and `targets.ts` already names the kill condition — "the uncertifiable surface is the only one anyone cares about." Say `local` and `declared` in the product copy, in those words, or the first user who checks will find the gap before the second one does.

§74 How the engine changes the product experience

The engine is 4,119 lines that a user never opens `MEASURED: 13 FILES UNDER SRC/MODULES/MDMAX/ + SPLICE-FRONTMATTER.TS + MDMAX-CERT.MJS`. Its entire product value is delivered through six moments where it speaks and one continuous condition where it must not. Getting the split wrong in either direction kills the differentiation: a chatty engine becomes a nag, a silent engine becomes plumbing nobody can tell from a competitor's.

74.1 The six moments where the engine is visible

Each row is a moment the user can point at. The wording follows §40's four-slot template — OUTCOME, OBJECT+CAUSE, AFFORDANCE, DISCLOSURE — and the visible budget of 280 characters with an ≤80-character headline.

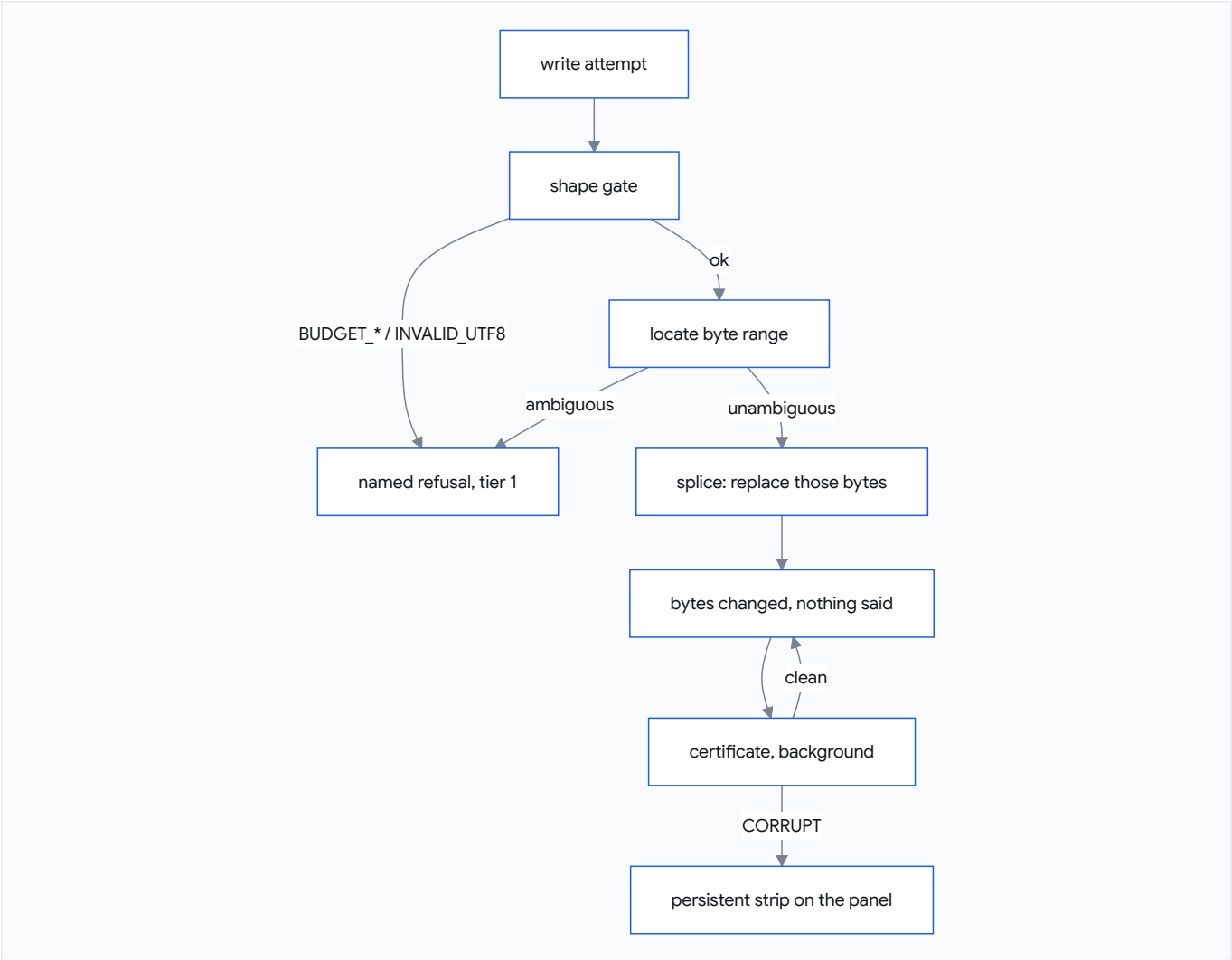
#	MOMENT	WHERE IT ORIGINATES IN THE TREE	WHAT THE USER SEES	WHY THIS BEATS SILENTLY DOING SOMETHING
1	Refusal on write	<code>spliceFrontmatterValue</code> in <code>src/modules/share/domain/splice-frontmatter.ts</code> , whose refusal is <code>return src</code> at 20 sites <code>MEASURED</code>	"Nothing changed. <code>author</code> appears twice here, so this edit had two possible targets. Delete one, then try again." + <code>Show both</code>	The alternative is not "do it anyway" — it is <i>pick one at random</i> . A silent pick is a coin-flip on which of two duplicate keys the board's status now means. The user finds out weeks later, in git
2	Certificate warning	<code>certify()</code> in <code>src/modules/mdmax/application/certify.ts</code> ; the CLI escalates on <code>res.certificate.summary.corrupt > 0</code>	"Certified BROKEN: 3 of 24 engines lose content from this note. The note itself is unchanged and safe." + <code>See the 3</code>	The alternative is publishing and letting the reader discover the loss. A verdict carries <i>before</i> and <i>after</i> as required fields of <code>CellVerdict</code> — the product says "bytes 0–35 render as <code><hr><h2>title: Demo status: active</h2></code> ", never "this may not render everywhere"
3	Provenance mark	§42.5's disclosure block; C2PA 2.4 A.9 front-matter form with <code>c2pa.hash.data</code> carrying one byte exclusion range <code>FETCHED</code>	A front-matter key naming tool, purpose, <code>humanOversightLevel</code> , timestamp, responsible person — projected into a visible line only at publish and export	The alternative is an invisible watermark. A.8's variation-selector encoding hashes <i>after</i> NFC normalization, so crediting a file mutates it <code>FETCHED</code> — the one mechanism the thesis cannot accept. A visible key in the file the user owns is the only mark that survives being read by a stranger
4	Conflict	<code>land()</code> returning <code>REFUSED_CONFLICT</code> on <code>base_version</code> drift (§12); the review surface renders it as a hunk (§13)	The agent's proposed edit arrives in the same Open/Accepted/Rejected grammar as a human suggestion, anchored to the quote it targeted	The alternative is last-write-wins. The file is the memory; the model's memory of the file is a cache to invalidate. Cursor and Windsurf both regressed per-hunk control and both got publicly burned (§13)
5	Degradation notice at paste	<code>classify()</code> in <code>src/modules/mdmax/domain/verdict.ts</code> , four verdicts <code>PASS</code> <code>STRIP</code> <code>CORRUPT</code> <code>VOID</code>	One line under the paste: what will not survive, named by construct, with the byte range	The alternative is a clean-looking paste that is <code>VOID</code> — payload by reference that was never in the file. <code>![[Some Note]]</code> is the exemplar; a three-valued matrix has no cell for it
6	Import report	§46: never refuse an import; show the certificate inline as a per-file diff	A per-file list of what could not round-trip, naming the zero-indent-sequence case	The alternative — refusing the import — is the single most damaging thing this engine could do. §46 rates the "what we do with unusual markdown" page at 75% deflection

Two rules bind all six. Presentation never escalates to a modal — a modal is for irreversible loss, and a refusal is the proof that nothing was lost. And every one of them is silenceable: git ships 43 `advice.*` toggles plus `GIT_ADVICE=0` `FETCHED`, `MEASURED`, and a refusal the user has understood twice and cannot mute is a defect.

The reason to ship the visible moments before the invisible ones is that a refusal is the only surface on which byte-preservation is observable at all — a correct write and a lossy write look identical to a user until something breaks, whereas a refusal is a claim the user can check against their own file in five seconds.

Anti-recommendation: do not add a seventh. The taxonomy in §40 is eleven codes mapping to these six surfaces plus five internal ones that never reach a human (`PAST_END`, `INSIDE_SURROGATE_PAIR`, `NOT_AN_INTEGER`, `NEGATIVE`, `OFFSET_OUT_OF_RANGE`). Every new user-visible refusal class raises the frequency the trust conversion in §74.3 depends on staying low.

74.2 Where the engine must be completely invisible, and what invisibility costs



SURFACE	MUST BE INVISIBLE BECAUSE	WHAT INVISIBILITY COSTS	EVIDENCE IT IS PAID OR UNPAID
Every keystroke	<code>BUDGET_MS.keystroke</code> is 250 ms in <code>shape-gate.ts</code> ; anything slower is felt as lag	A single-pass $O(n)$ gate with early exit on each limit, so a hostile document costs $O(\text{limit})$ not $O(n)$; <code>WIKILINK_RE</code> at $k=1.98$ took 36,865 ms on 320 KB before this gate existed MEASURED, IN-TREE	Paid — the gate ships and is the one part of the engine wired into the product
Opening a foreign file	170 of 907 front-matter blocks in the founders' own vault are invalid YAML — 18.74% MEASURED, IN-TREE	<code>prepassFrontmatter</code> produces a corrected string for a reader only and never writes; a front-matter parse failure may never fail a user action	Paid in code, unpaid in wiring — <code>frontmatter-prepass.ts</code> has zero importers outside its test
A no-op save	A refusal is byte-identical to a successful no-op today: both are <code>return src</code>	Giving splice the discriminated-union return shape <code>mdmax</code> already has — <code>{ ok: false, reason, at: { line, col, byteStart, byteEnd }, detail, hint }</code>	Unpaid. <code>PropertiesPanel.tsx</code> drops refusals at <code>if (!onEdit</code>
Equivalence across engines	Six fold rules (<code>entity-decode</code> , <code>smart-punctuation</code> , <code>void-element-spelling</code> , <code>whitespace</code> , <code>id-prefix</code> , <code>heading-anchor-id</code>) let a <code>PASS</code> mean something	A <i>named, versioned</i> equivalence relation — <code>mdmax/fold@1</code> — so a <code>PASS</code> can be audited rather than trusted, via <code>foldApplied</code> on every cell	Unpaid at the evidence layer: <code>mdmax/fold@1</code> has never been run over the pinned 8,513-file corpus
Byte-exact addressing	<code>mdast</code> reports root end offset 36 where the UTF-8 length is 41 on the same string (§9)	Branded <code>U160ffset</code> / <code>ByteOffset</code> / <code>GraphemeIndex</code> types and one <code>OffsetMap</code> ; strict decode that refuses rather than repairs , because substituting <code>U+FFFD</code> changes byte length and every later offset would be correct for a document the user does not have	Paid — <code>decodeStrict</code> is the one export with real importers: <code>src/modules/vault/application/get-snapshot.ts</code> and <code>src/modules/vault/infrastructure/search-index.ts</code>

The honest accounting: **twelve of thirteen engine files have zero product importers, and there is no** `.github/workflows` **directory**

[MEASURED: GREP FOR MODULES/MDMAX ACROSS SRC RETURNS EXACTLY THE TWO DECODESTRICT IMPORTS; LS .GITHUB/WORKFLOWS → ABSENT](#)

`. Invisibility is currently total and unearned` — not because the invisible work was done well, but because it was never connected.

Anti-recommendation: do not close that gap by wiring the write gate first. \$7.3 and \$61 both say it explicitly — at the measured 83% foreign refusal rate, s

74.3 The refusal-into-trust conversion, and where it is documented outside this category

CATEGORY	THE SYSTEM THAT SAYS NO
Programming languages	Rust's borrow checker refuses to compile programs every mainstream competitor accepts
Browser security	Chrome's SSL interstitial
Browser security, the counter-evidence	The same warning class, at scale
Developer tooling	ESLint separates auto-applied <i>fixes</i> from <i>suggestions</i> that "may change application logic" and are withheld from t

The conversion has three preconditions, and frontmatter currently satisfies one. The refusal must be **rare** — it is not; 6,613 of 6,614 foreign files refuse o

The arithmetic that decides the whole subsection: a user who points frontmatter at ten folders they already own meets the refusal in **8.3 of them** [DERIVED: 10 ×](#)

Anti-recommendation: do not soften the refusal to lower the rate. Lower it by fixing NF-1 and NF-3, which move the Z metric from 17% to its ≥99.9% target (§

74.4 The demo — thirty seconds, no markdown internals required

Executed here, on this machine, at this commit [MEASURED](#) :

\$ node --import ./scripts/ts-resolve.mjs scripts/mdmax-cert.mjs demo.md			
real 0.49s	· 5 blocks × 7 local targets =	35 cells · bench	<code>51947c2e8812</code> · fold <code>mdmax/fold@1</code> · PASS 6 · STRIP 7 · CORRUPT 15 · VOID 7 .
BEAT	SECONDS	WHAT THE AUDIENCE SEES	THE SENTENCE THAT L
1	0–8	A 131-byte note with front matter, a heading, a wikilink, an HTML comment and a <code>1)</code> list	"This is an ordina
2	8–14	The list row: <code>1) one\n2) two</code> → PASS on six engines , CORRUPT/MUTATE on GitHub Pages , which renders it <code><p>1) one 2)</code>	"The same fourteen
3	14–20	The heading row: <code># Heading {#custom}</code> → <code><h1>Heading {#custom}</h1></code> on six (LEAK, the attribute becomes visible text) and <code><h1>Heading</h1></code> on GitHub Pages (MUTATE)	"Your anchor is ei
4	20–26	The front-matter row: bytes 0–35 → <code>""</code> on two targets, <code><hr><h2>title: Demo status: active</h2></code> on five	"Your properties :
5	26–30	Rename one key through the properties panel. The diff is the key's bytes and nothing else; undo restores byte-identity	"Everything else i

Beat 5 is the one that matters and it is the one \$17 already names as the activating act — a one-line in-place edit through a projection on a file the user

Blocking defect found while measuring this: `scripts/mdmax-cert.mjs` calls `paint(s.padEnd(18))`, and `paint` compares `v === 'PASS'` against the *padded* string, so th

Anti-recommendation: do not demo `--histogram` over a vault. It produces the right engineering artifact and the wrong audience reaction — a wall of counts wit

74.5 What would make a user angry, and the mitigation for each

ANGER	TRIGGER IN THE CURRENT TREE	MITIGATION
"It refused and won't say why"	<code>return src</code> at 20 sites; <code>PropertiesPanel.tsx</code> drops the result at <code>next === content</code>	Give splice the union shape; render <code>reason</code> <code>fr</code>
"It refuses on nearly every folder I own"	Zero-indent sequences, 83%	NF-1 and NF-3 before any marketing number; star
"The competitor just opened this"	Import path, first 60 seconds	Never refuse an import (\$46) – import everythir
"It says BROKEN about a file that looks fine"	<code>summary.corrupt > 0</code> , on a document the user only ever reads in one app	Lead the verdict with the <i>target</i> , not the file:
"It certified my file against places I don't use"	8 of 15 targets are <code>declared</code> or <code>requires-push</code> – 53.3% <code>MEASURED</code> , <code>UNCERTIFIABLESHARE()</code>	Declared rows visibly declared and visibly date 01'), never mixed silently into a column of me
"The same warning, forever"	Any repeated refusal	Per-code silencing, on git's <code>advice.*</code> model
"Support can't reproduce it"	No <code>reason</code> , no byte range in the report	RFC 9457 <code>instance</code> semantics: <code>reason</code> + byte of
"I paid and I can't tell it's there"	The plumbing risk, below	\$74.1's six moments are the entire answer

74.6 The honest risk — that this is plumbing nobody pays for — and what would refute it

The strongest form of the risk, stated without cushioning: the product works today with the engine almost entirely disconnected. Twelve of thirteen files ha

FALSIFIER	THE MEASUREMENT THAT SETTLES IT
Deterministic guarantees do not retain	Obsidian's own install base: deterministic projection plugins 7,289,307 peak-version installs agains
Nobody hits the refusal, so nobody values it	The <code>cert_refusal</code> event (\$47) carrying <code>refusal_code</code> , <code>construct</code> , <code>vault_share_affected</code>
Users want the guarantee but not visibly	Day-7 return rate split by whether the first-run byte diff (\$17 step 6) was shown, as a release coh
The refusal is the product's ceiling, not its floor	Foreign-vault open rate, the Z metric
Buyers pay for consolidation, not fidelity	The first 25 sales conversations (\$64.1's own falsifier)

What would actually prove the engine is worth money is narrower than any of the above and is available now: **an import report on a stranger's vault that name** Anti-recommendation, and the one that costs the most to honour: do not price the engine. \$11.4 already refuses a standalone AI SKU on the evidence that Noti

]

\$75
Key addressability — the NF-4 design

75.1 The decision

`SAFE_KEY = /^[A-Za-z0-9_.$-]+$/` in `/Users/sagnikmitra/Desktop/GitHub/frontmatter/src/modules/share/domain/splice-frontmatter.ts` is replaced by a **predicate over what the l** Measured against the pinned corpus at `/Users/sagnikmitra/Desktop/GitHub/frontmatter/test/corpus/foreign/_vendor/`: this unlocks 936 of the 942 currently-unaddressabl

75.2 What key equality means, and what breaks under each answer

The module comment is right that this is the prior question. Three candidate semantics, and what each costs [inference, with the parser behaviour measured b

SEMANTICS	CAFÉ	NFC VS NFD	WHAT IT BREAKS
Codepoint equality, request normalised to NFC (chosen)	two distinct keys	a user who types NFD (macOS filesystem-sourced text) gets	ABSENT for a key v:
NFC-fold equality on both sides	one key	a file that legitimately holds both loses one on the next write, and the spli	
Normalise the file to NFC on open	one key	rewrites bytes across the whole document, violating the contract at its root;	

[measured] `yaml@2.9.0` and `js-yaml@3.14.2` – both installed in this repo – parse `café: 1` (NFC) and `café: 2` (NFD) in one block as two distinct keys, with no w [fetched, `yaml.org/spec/1.2.2/` via `curl -sL --compressed`] The spec routes equality through the tag's canonical form: "A mapping's keys are unique if no two ke

75.3 What YAML 1.2.2 actually permits in a block key

[fetched] "To limit the amount of lookahead required, the `:` indicator must appear at most 1024 Unicode characters beyond the start of the key. In addition, [measured, `yaml@2.9.0`] What a plain block key legally is, probed directly:

INPUT	PARSED KEY	CONSEQUENCE FOR THE LOCATOR
date created: 2022-08-06	date created	spaces are ordinary plain characters
: x		non-ASCII is ordinary
a:b: 1	a:b	a colon <i>not</i> followed by space is part of the key
a: b: c	THROW Nested mappings are not allowed in compact mappings	: terminates the key; a key containing it must be quoted
a : 1	a	padding may sit between key and colon
-a: 1, ?a: 1	-a, ?a	- and ? are indicators only when followed by space
a[b], a{b}, a,b, a'b, a"b, a!, a?	as written	legal plain keys, all of them
a: 1 twice	THROW Map keys must be unique / duplicated mapping key	exact duplicates are an error in both parsers
: 1	yaml: {"": 1} · js-yaml: THROW	the empty key is a parser-divergence hazard; refuse it

The last row is the reason the rule is a predicate and not a transcription of the grammar: an empty key is legal to one parser and fatal to another, so "wha

75.4 Measurement against the pinned corpus

All figures below are [measured] by `node --input-type=module` scripts run over the 8,513 vendored files, extracting the frontmatter block by the same rules the

QUANTITY	COUNT	SHARE
Files in corpus	8,513	–
Files opening with a frontmatter block	7,969	93.6%
Top-level <code>key:</code> lines	29,293	–
Distinct key shapes	234	–
Distinct shapes failing <code>SAFE_KEY</code>	50	21.4% of shapes
Occurrences failing <code>SAFE_KEY</code>	2,452	8.4% of key lines
Files with ≥1 unaddressable key	942	11.8% of fm files
Keys quoted on disk (any form)	0	0%
Exact duplicate top-level keys within a file	0	0%
Case-insensitive duplicate keys within a file	0	0%
Keys not already in NFC	0	0%
Distinct keys colliding under NFC folding	0	0%
Keys containing <code>:</code> , <code>#</code> , <code>.</code> , or edge whitespace	0	0%
Files with a non-BMP character inside a key	20	0.25%
Files with a zero-indent block sequence (NF-1)	6,613	82.98%

The 82.98% cross-checks the reported 83% publish-refusal rate to two decimal places, which is the confirmation that NF-1 and NF-4 are genuinely different de

The 50 failing shapes decompose as 44 real keys and 6 scanner artifacts. Of the 44: three are ASCII-with-space (`date created` in 812 files, `date modified` in 1

The 6 artifacts are instructive rather than embarrassing: five are `- Structured Copy: Files & Folders` and siblings, which are sequence items at column zero und

Under the rule specified in §75.5:

OUTCOME	COUNT
Files unaddressable today	942
Files unaddressable after the fix	6
Files unlocked	936 (99.36%)
– of which NF-4 alone suffices (no zero-indent sequence present)	750
– of which NF-1 must also land before publish succeeds	186
Distinct key shapes unlocked	44
Key occurrences unlocked	2,446

[derived] 936 / 942 = 99.36%. The residual 6 are the 5 sequence items and the template line – every one of them a correct refusal.

75.5 The rule

```
// src/modules/share/domain/key-ref.ts (new)

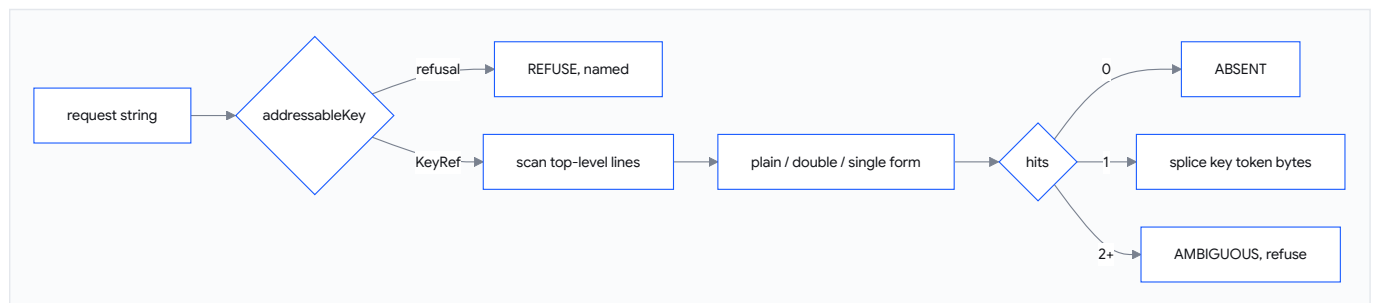
/** A key that has passed addressability. The only type the splice API accepts. */
export type KeyRef = { readonly nfc: string }

export type KeyRefusal =
| { kind: 'EMPTY' }
| { kind: 'TOO_LONG'; length: number } // > 1024 (YAML 1.2.2 implicit-key limit)
| { kind: 'CONTROL_CHAR'; at: number } // C0, C1, DEL, U+2028/2029, U+FEFF
| { kind: 'EDGE_WHITESPACE' } // leading or trailing SP/TAB
| { kind: 'CONTAINS_TAB' }
| { kind: 'COLON_TERMINATOR'; at: number } // /:(?=[ \t]|$)/ - would end the key
| { kind: 'SPACE_HASH'; at: number } // /[ \t]#/ - would start a comment
| { kind: 'LEADING_INDICATOR'; char: string } // /^[-?:,[]\{\}\#\&*!|>'"%@`]/
| { kind: 'NOT_NFC'; nfc: string } // request must arrive normalised

export function addressableKey(raw: string): Result<KeyRef, KeyRefusal>
```

Order of checks is the order of the union. `NOT_NFC` carries the normalised form so the UI can offer it as a one-click correction; it is the only refusal that The leading-indicator set is deliberately wider than YAML requires. `-a` and `?a` are legal plain keys [measured], and we refuse them anyway: the locator must The `$` from the old class is retained implicitly – it is simply an ordinary character now. There is no allowlist left to maintain.

75.6 Locating a key whose on-disk form differs from the request



Three on-disk forms can carry one logical key. The corpus has zero quoted keys in 8,513 files, so the quoted branches are dead code today and must still be

```
type KeyForm = 'plain' | 'double' | 'single'
type KeyHit = {
  readonly tokenStart: U16Offset // first byte of the key token, INCLUDING an opening quote
  readonly tokenEnd: U16Offset // one past the closing quote / last plain char
  readonly form: KeyForm
  readonly entryStart: U16Offset // start of line - the existing keyStart
  readonly entryEnd: U16Offset // existing keyEnd, incl. indented continuations
}

function locateTopLevelKey(block: string, blockStart: number, want: KeyRef):
| { kind: 'FOUND'; hit: KeyHit }
| { kind: 'ABSENT' }
| { kind: 'AMBIGUOUS'; hits: number }
| { kind: 'ON_DISK_NOT_NFC'; found: string }
```

Per top-level line, in this order:

1. `/^((?:[^\[\]\{\}\#\&*!|>'"%@`])*)[\t]*:(?=[\t]|$)/` → unescape `\\(.)` → candidate.
2. `/^((?:[^\[\]\{\}\#\&*!|>'"%@`])*)[\t]*:(?=[\t]|$)/` → unescape `''` → candidate.
3. Plain: walk the line for the first index `j` where `line[j] === ':'` and `j+1` is EOL or SP/TAB. If `j <= 0`, this line is not a top-level key – hand it to the Compare `candidate === want.nfc`. If that fails but `candidate.normalize('NFC') === want.nfc`, return `ON_DISK_NOT_NFC` with the on-disk bytes – do not match, do not re The splice for a rename becomes `src.slice(0, hit.tokenStart) + emitKeyToken(newKey, hit.form) + src.slice(hit.tokenEnd)`, replacing the `token` rather than `oldKey.length` `emitKeyToken` writes plain unless the key would not read back plain – which, given `addressableKey` already excludes `:`, `#`, edge whitespace and leading indic

75.7 Case, colons, dots, leading dashes

Keys are case-sensitive, with no folding at any layer. [measured] Zero files contain a case-insensitive duplicate, while eight case pairs exist across the c Colons are permitted mid-key when not followed by space or EOL, because `a:b` is one key to both parsers. Dots are permitted and carry no path semantics – `a.`

75.8 The refusal set after the fix

REFUSAL	REACHABLE FROM	CORPUS COUNT
EMPTY	UI add-property with blank name	0
TOO_LONG	pasted template	0
CONTROL_CHAR	pasted terminal output	0
EDGE_WHITESPACE	trailing space in the name field	0
CONTAINS_TAB	pasted spreadsheet cell	0
COLON_TERMINATOR	date: created typed as a name	0
SPACE_HASH	budget #1	0
LEADING_INDICATOR	zero-indent sequence item read as a key	5
NOT_NFC	NFD text from a macOS filename	0
ON_DISK_NOT_NFC	a committed NFD key	0
AMBIGUOUS (duplicate key line)	hand-edited file	0
unparsed top-level line (existing bare-line branch)	NF-1, templates	13,413 lines / 6,613 files

The last row is not NF-4's to fix. Of the 942 files NF-4 unblocks, 186 also carry a zero-indent block sequence and will still refuse at the existing bare-li

75.9 Migration for keys already written

There is nothing to migrate in the files. spliceFrontmatterValue refuses at if (!SAFE_KEY.test(key)) return src before any write, and spliceFrontmatterKey refuses What does migrate is persisted key text on our side:

- 1. Splice journal entries. Every entry that names a key gains keyForm: 'nfc-v1'. Entries written before this field are read as 'legacy-ascii' and are byte-ide
- 2. PropertiesPanel.tsx. The four SAFE_KEY.test(...) call sites - aria-invalid, the disabled state, and the two guards in the add/rename handlers - switch to a
- 3. Anchor store. Unaffected. Content-derived anchors do not name frontmatter keys.

75.10 The test set

Fixtures under test/corpus/ plus unit cases; npm run spec gates, npm run corpus replays.

#	INPUT	EXPECTED
1	date created: 2022-08-06 → set	value bytes replaced, all other bytes identical
2	: with zero-indent sequence → set	key resolves, refuse at the NF-1 branch, file unchanged
3	👉 : 1 → set	succeeds; OffsetMap reports a non-BMP-safe range; no offset splits a surrogate pair
4	- Structured Copy: Files & Folders under aliases: → set Structured Copy	ABSENT, not a match
5	a:b: 1 → set a:b	matches; set a returns ABSENT
6	a : 1 → set a	matches, padding preserved
7	NFD café: 1, request NFC café	ON_DISK_NOT_NFC, file unchanged
8	request NFD café	NOT_NFC with nfc populated
9	"date created": 1 (quoted) → rename	closing quote survives; token replaced, not oldKey.length bytes
10	two date created: lines	AMBIGUOUS, file unchanged
11	key of 1,025 characters	TOO_LONG
12	full corpus replay, set-then-delete on every distinct key	8,513 files byte-identical; 0 throws

Test 4 is the one that must fail against the unfixed widened regex before it is trusted (LR#68): a rule that drops the leading-- refusal passes tests 1-3 a

75.11 Decision ledger

DECISION	REJECTED ALTERNATIVE	WHY	COST OF BEING WRONG
Predicate, not allowlist	widened character class	a class cannot express : or #, which are positional	a key that reads back as a different key
NFC on request only	normalise the file	normalising rewrites bytes we were not asked to touch	every anchor offset shifts; the contract is
Codepoint equality	NFC folding	both parsers key NFC and NFD separately [measured]	we disagree with every downstream consumer
Case-sensitive	case-insensitive	0 in-file case duplicates, 8 cross-file pairs [measured]	merges keys no file treats as one
Refuse leading - and ?	follow YAML exactly	5 corpus lines would be misread as keys	splice into a sequence item - silent struct

]

76.1 The gap, stated exactly

`spliceFrontmatterValue` in `/Users/sagnikmitra/Desktop/GitHub/frontmatter/src/modules/share/domain/splice-frontmatter.ts` addresses one thing: a top-level YAML key inside

WRITE-BACK	TARGET SPAN	TODAY'S PATH
Kanban drag → status	a field inside a list item, or a frontmatter key	frontmatter only; a card whose status is in the body has no write path
Checkbox toggle	the three units <code>[]</code> inside a <code>TaskMarker</code>	none
Calendar drag → date	a date token inside a list item or heading line	frontmatter only
Accepted AI hunk	an arbitrary block range	none
Section-level restore	heading line through the next heading of depth $\leq d$	none

The projection law says a render profile is a view whose edits write back to the bytes that produced it. With no body locator, the law holds for exactly one

76.2 The addressing model

Four candidates, measured against the pinned corpus at `/Users/sagnikmitra/Desktop/GitHub/frontmatter/test/corpus/foreign/_vendor` (8,513 files, 19,047,891 bytes).

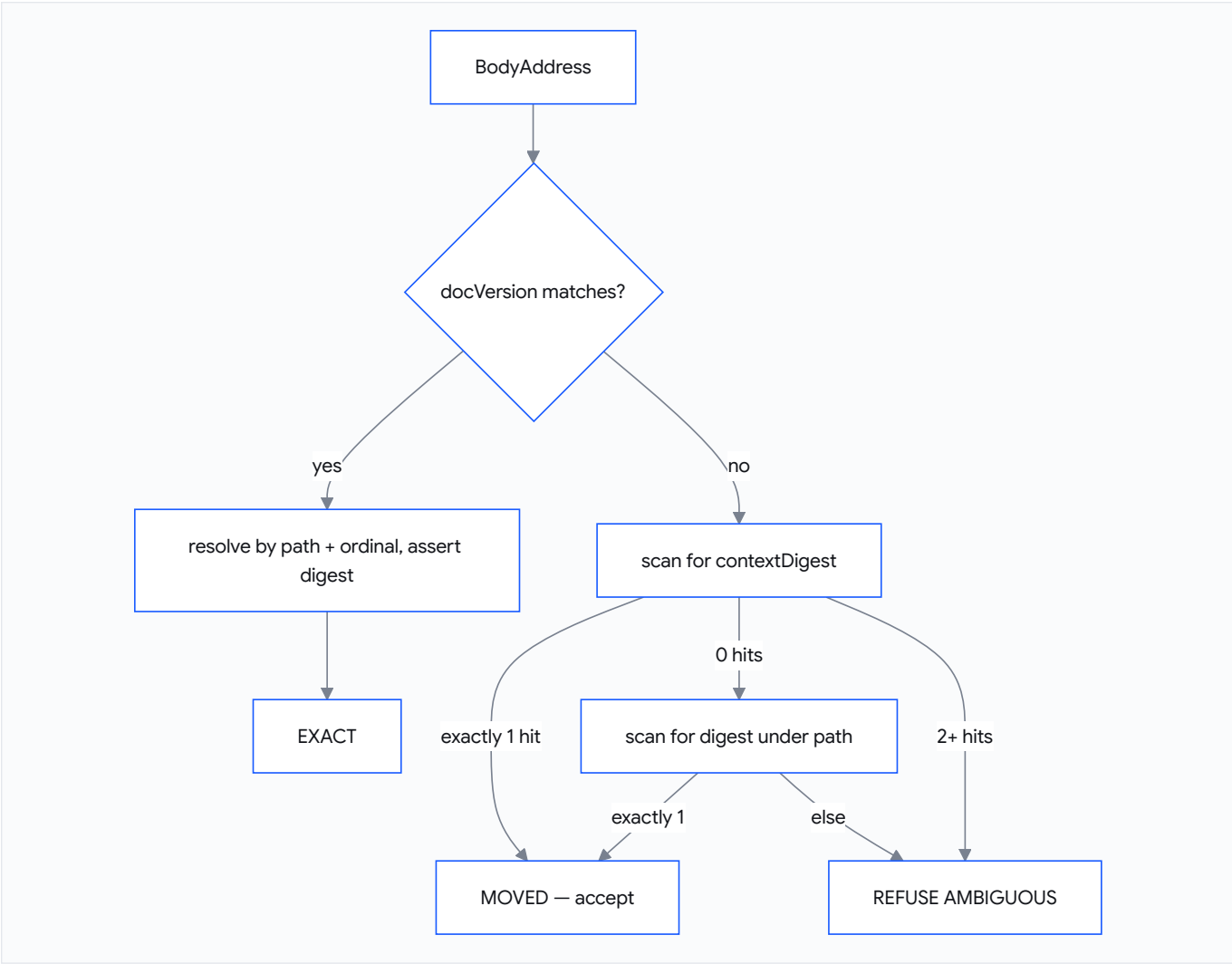
MODEL	SURVIVES REFLOW ABOVE IT	SURVIVES EDIT TO THE TARGET	SURVIVES DUPLICATE CONTENT	COST
lezer <code>SyntaxNode</code> position	no – offsets are per-parse	n/a	yes	needs a live tree; useless acro
Content digest of the span	yes	no	no – measured below	cheap, stateless
Line + column offset	no	no	yes	trivially wrong after any prepe
Structural path + ordinal + digest	yes	yes (path survives)	yes (ordinal disambiguates)	three fields to persist, a casc

The duplicate-content number is the one that decides it. [measured] Over the corpus, a task-list line's trimmed text is unique within its own file for 2,278

Decision – the address is a triple, resolved as a cascade:

```
// src/modules/share/domain/body-address.ts (new)
export type SpanKind =
  | 'task-marker' // the 3 code units `[ ]` – the only mutable part of a checkbox
  | 'list-item-text' // the item's own text, excluding its marker and its children
  | 'inline-field' // `key:: value` or `key: value` inside a list item
  | 'fenced-block' // open fence line through close fence line, indivisible
  | 'callout-body' // the payload of a `> [!kind]` container
  | 'heading-section' // heading line through the next heading of depth <= d

export interface BodyAddress {
  readonly kind: SpanKind
  /** ATX heading text chain, NFC-normalised, case-folded. Empty = preamble. */
  readonly path: readonly string[]
  /** 0-based index among same-kind spans under `path`. Disambiguates duplicates. */
  readonly ordinal: number
  /** Digest over the span's own UTF-8 bytes. */
  readonly digest: Digest128
  /** Digest over (previous non-blank line + span + next non-blank line), UTF-8. */
  readonly contextDigest: Digest128
  /** Content hash of the document this address was minted against. The CAS token. */
  readonly docVersion: Digest128
}
```



Rejected: the pure content anchor already shipped for block-versions, which resolves at 99.627% with 0.050% false positives over 41,642 block-versions. That

76.3 Span kinds and their atomicity class

The governing rule is that a span is the narrowest byte range that expresses the intent, never the enclosing node.

KIND	LEZER NODE	MUTABLE RANGE	ATOMICITY
task-marker	TaskMarker	from + 1, length 1	a single code unit
list-item-text	ListItem minus ListMark and child blocks	text run only	marker, indent, numbering never touched
inline-field	inside Paragraph under ListItem	value run after the separator	key bytes never touched
fenced-block	FencedCode	[node.from, node.to]	open and close in ONE edit or refuse
callout-body	Blockquote with [!kind] first line	payload lines	no closer exists – cannot be unbalanced
heading-section	ATXHeading* → next heading of depth ≤ d	whole range	boundary rules in §76.6

[measured] @lezer/markdown@1.6.3 configured with GFM yields TaskMarker spanning exactly three code units for every marker shape tested - [x], - [X], * [Fenced constructs are the opposite case and the reason atomicity is a stated rule rather than an implementation detail. [fetched] CommonMark 0.31.2 §4.5, ht Decision: span kinds are a closed enum; a caller cannot pass a raw range. Rejected: a general spliceRange(from, to, text). Why: a general range API has no way

76.4 Surviving a concurrent edit

Two regimes, and conflating them is how offsets rot.

REGIME	MECHANISM	GUARANTEE
Same session, editor open	ChangeSet.mapPos from @codemirror/state	deterministic; a resolved span tracks every keystroke
Across a save, a sync, or a reload	re-resolve the BodyAddress against the current bytes	probabilistic; refusal on ambiguity

[measured] ChangeSet.of([from:0,to:0,insert:"XY"}, 10).mapPos(5) returns 7, and the assoc argument is honoured. In-session tracking is solved and needs no inv Decision: a resolved span is never persisted. What persists is the BodyAddress, and every write re-resolves immediately before splicing, inside the same syn Rejected: mapping stored offsets through a diff of the two document versions. Why: diff-match-patch was measured non-idempotent while returning true, so a ma

76.5 List-item and task-list mutation

This is the most common write, so it gets the most specific rules.

RULE	REASON
The list marker character (<code>-</code> , <code>*</code> , <code>+</code>) is never normalised	it is authored style; rewriting it is the <code>matter.stringify</code> failure in a new costume
Ordered-list numbers are never renumbered	[measured] 1,151 ordered items in the corpus, 8 non-sequential - those 8 are authored,
Indentation of the item and of its children is never re-computed	changing the content column reparents every child
A <code>list-item-text</code> splice ends at the item's own text run, never at <code>ListItem.to</code>	[measured] 19,135 of 60,000 list items (31.892%) are followed by a lazy continuation l
A task item with child blocks is refused for <code>list-item-text</code> , allowed for <code>task-marker</code>	[measured] 50 of 3,378 task items carry indented children; the one-unit marker splice i
A candidate line inside a fenced region is not a task item	[measured] 12 lines matching the task pattern sit inside fences across the corpus

The kanban write is the interesting case, because "status" may live in three places: a frontmatter key (already handled), an `inline-field` inside the item (s

76.6 Heading-section boundaries

A section is `[start of the heading line, start of the next heading line of depth ≤ d]`, or end of document. Ending at the *start* of the terminating heading, not at th
Setext headings break this. [measured] After excluding frontmatter closers and table delimiter rows, the corpus has 15 setext headings across 11 of 8,513 fi
Decision: a `heading-section` whose start or terminator is a setext heading is refused. Rejected: including the underline in the span and rewriting it as ATX c

76.7 Interaction with the branded offset types

`src/modules/mdmax/domain/offsets.ts` already fixes the units: `U16Offset` internal, `ByteOffset` only at edges, `OffsetMap` the single conversion point, no rounding e

VALUE	UNIT	WHY
<code>LocateResult.span.from/to</code>	<code>U16Offset</code>	lezer, CodeMirror and mdast all report UTF-16
<code>BodyAddress.digest</code> inputs	<code>ByteOffset</code> range → UTF-8 bytes	a digest over UTF-16 units is not reproducible from a git blob
Journal entries, CAS tokens	<code>ByteOffset</code>	they cross the process boundary
Cursor / column in refusal messages	<code>GraphemeIndex</code>	a user-facing column that splits an emoji is wrong

Every `U16Offset` entering `spliceBodySpans` is constructed through `u16(text, n)`, never `unsafeU16`, so an offset inside a surrogate pair is rejected at the door

76.8 The API

```
// src/modules/share/domain/splice-body.ts (new)
import type { U16Offset } from '../mdmax/domain/offsets'
import type { ParseFn, BlockNode } from '../mdmax/domain/placement'

export type LocateRefusal =
  | { readonly kind: 'AMBIGUOUS'; readonly candidates: number }
  | { readonly kind: 'NOT_FOUND' }
  | { readonly kind: 'PATH_DIVERGED'; readonly matchedDepth: number }
  | { readonly kind: 'KIND_MISMATCH'; readonly found: string }
  | { readonly kind: 'INSIDE_SKIP_REGION'; readonly region: 'fence' | 'code-span' | 'frontmatter' }
  | { readonly kind: 'SETEXT_BOUNDARY'; readonly line: number }
  | { readonly kind: 'SHAPE_REFUSED'; readonly inner: ShapeFailure }
  | { readonly kind: 'OFFSET_REFUSED'; readonly inner: OffsetError }

export type LocateResult =
  | { readonly ok: true
    readonly from: U16Offset
    readonly to: U16Offset
    readonly confidence: 'EXACT' | 'MOVED'
    readonly rulesFired: readonly ( 'path' | 'ordinal' | 'digest' | 'context' )[] }
  | { readonly ok: false; readonly reason: LocateRefusal }

export function locateBodySpan(src: string, addr: BodyAddress): LocateResult

export interface BodyEdit {
  readonly addr: BodyAddress
  /** Replacement text for the span. For 'task-marker', exactly one code unit. */
  readonly replacement: string
}

export type SpliceResult =
  | { readonly ok: true
    readonly text: string
    readonly applied: readonly { from: U16Offset; to: U16Offset; confidence: 'EXACT' | 'MOVED' }[]
    readonly docVersion: Digest128 }
  | { readonly ok: false; readonly reason: LocateRefusal | VerifyFailure; readonly text: string }

/**
 * All-or-nothing. Every edit is located against 'src' FIRST, ranges are asserted
 * disjoint and sorted, then applied right-to-left so earlier offsets stay valid.
 * On any refusal the input is returned unchanged, exactly as the frontmatter writer does.
 */
export function spliceBodySpans(src: string, edits: readonly BodyEdit[], parse: ParseFn): SpliceResult

export type VerifyFailure =
  | { readonly kind: 'BYTES_OUTSIDE_CHANGED'; readonly firstAt: number }
  | { readonly kind: 'SKELETON_CHANGED_OUTSIDE_SPAN'; readonly before: readonly string[]; readonly after: readonly string[] }
  | { readonly kind: 'ADDRESS_NO_LONGER_RESOLVES'; readonly addr: BodyAddress }
  | { readonly kind: 'FENCE_UNBALANCED'; readonly openAt: number }

/** Runs inside 'spliceBodySpans' before it returns ok; exported so tests can call it alone. */
export function verifyBodySplice(
  before: string, after: string, edits: readonly BodyEdit[], parse: ParseFn,
): { readonly ok: true } | { readonly ok: false; readonly reason: VerifyFailure }
```

The generalisation of `placement.ts` is in `SKELETON_CHANGED_OUTSIDE_SPAN`. `safeInsert` requires the whole block skeleton to be unchanged, which is right for an in

76.9 Invariants

#	RULE	FAILURE MODE
1	Bytes outside every located span are bit-identical	the failure all three competitor teardowns showed
2	<code>spliceBodySpans</code> never throws; it returns the input or a result	a throw in a write path is data loss at the call site
3	An ambiguous address is refused, never resolved to the first candidate	the 32.564% duplicate-task-text population toggles the wrong box
4	A fence open and its closer are one edit or no edit	CommonMark §4.5 – the rest of the document becomes code
5	The block skeleton outside the span is identical after the write	a setext underline turns an <code>h2</code> into a paragraph plus a rule
6	The address re-resolves EXACT against the post-write bytes	an address that stops resolving after its own write is unusable for the next
7	Ranges within one call are disjoint and applied right-to-left	overlapping edits produce order-dependent output
8	There is one body splice implementation	two writers means the guarantee holds in one of them
9	Every refusal names the bytes that were not changed	a silent no-op reads as a successful save

76.10 Refusals

CONDITION	OUTCOME
Address matches 2+ candidates after the full cascade	<code>AMBIGUOUS</code> – file unchanged, user told which span kind and how many candidates
Heading path diverges above the target	<code>PATH_DIVERGED</code> with the depth that still matched
Target is a task item with indented child blocks, kind <code>list-item-text</code>	refused; <code>task-marker</code> on the same item is still allowed
Section start or terminator is a setext heading	<code>SETEXT_BOUNDARY</code>
Candidate lies inside a fence, a code span, or the frontmatter block	<code>INSIDE_SKIP_REGION</code>
Document fails <code>shape-gate</code>	<code>SHAPE_REFUSED</code> , carrying the inner reason
Any offset would split a surrogate pair	<code>OFFSET_REFUSED</code> , carrying the <code>OffsetError</code>
Edit is a section move	refused in v1 – two ranges, no single span

76.11 The red proof

There is no body locator today, so a proof against "today's code" must be a proof against the implementation any team would write first. Each red test check

- `test/engine/body/nf-005-task-ambiguity.red.test.ts` – over `_vendor`, for every file with ≥ 2 task items sharing trimmed text, mint an address for the second and
- `test/engine/body/nf-006-fence-atomic.red.test.ts` – replace a `FencedCode` span with an opening fence and no closer, then compare `blockSkeleton` beyond the fence.
- `test/engine/body/nf-007-setext-section.red.test.ts` – the 11 corpus files with setext headings plus a synthetic `Heading\n---` fixture. Expected failure: a naive
- `test/engine/body/nf-008-lazy-continuation.red.test.ts` – the 19,135 lazily-continued list items. Expected failure: a span ending at `ListItem.to` swallows the con
- `test/engine/body/nf-009-fenced-task.red.test.ts` – the 12 task-shaped lines inside fences. Expected failure: `naiveLineLocate` edits code.

Preconditions, in order: `node scripts/corpus-foreign.mjs verify` must exit 0 so the corpus is provably the pinned one; each red test must be observed failing ag

§77
The anchor system — targets that survive editing

77.1 The decision

An anchor is a **content-derived**, **document-scoped**, **refusal-capable** pointer to a byte range. It stores no line numbers, no offsets-of-record, and no parse-tree

	DECISION	REJECTED ALTERNATIVE	WHY
Identity	<code>sha256(normalize(blockText))[0:16]</code> + quote + context + multiplicity	mdast node path (<code>root.children[7].children[2]</code>)	A path is inval
Scope	Bound to <code>doc_id</code> , never to a path, never resolvable across documents	Resolve wherever the text is found	Anchors from on
Ladder	Exact → verify → disambiguate → fuzzy → orphan, each with an explicit refusal	Single fuzzy pass with one threshold	The exact lane
Threshold	<code>T = 0.76</code> , <code>MARGIN = 0.12</code>	Hypothesis's implicit accept-best-always	Hypothesis has i

77.2 The anchor record

```
// src/modules/mdmax/domain/anchor.ts
export const ANCHOR_VERSION = 'mdmax/anchor@1'

export interface Anchor {
  readonly v: typeof ANCHOR_VERSION
  readonly normalizeVersion: typeof NORMALIZE_VERSION // 'mdmax/normalize@1'
  readonly docId: DocId // content-address of the document lineage, NOT a path
  readonly hash: string // sha256(normalize(block)).slice(0, 16)
  readonly multiplicity: number // same-hash blocks in the doc AT ANCHOR TIME
  readonly ordinal: number // 0-based index among those
  readonly quote: string // normalize(block), head 160 + tail 160 if longer
  readonly prefix: string // last 64 normalized units before the block
  readonly suffix: string // first 64 normalized units after the block
  readonly headingPath: readonly string[] // normalized ancestor headings
  readonly relPos: number // blockIdx / blockCount at anchor time
  readonly baseDigest: string // splice-journal digest of the doc it was cut from
}
```

Two fields carry the design. `multiplicity` is the one that is normally forgotten and it is the single highest-value integer in the record – 77.4 shows what `normalize` is already frozen at `/Users/sagnikmitra/Desktop/GitHub/frontmatter/src/modules/mdmax/domain/normalize.ts` (62 lines, `NFC → collapse whitespace → lowercase → NFC`). Offsets are absent by design. The resolver returns a live `U160Offset` pair through `OffsetMap` (`src/modules/mdmax/domain/offsets.ts`), and only 67 of 1,080 corpus f

77.3 The resolve ladder

RUNG	PRECONDITION	ACTION	OUTCOME ON FAILURE
L0 exact	<code>matches.length === 1 && anchor.multiplicity === 1</code>	Return the range	–
L1 verify	<code>matches.length === 1 && anchor.multiplicity > 1</code>	Score the survivor against context; accept only at <code>score ≥ T</code>	AMBIGUOUS
L1 disambiguate	<code>matches.length > 1</code>	Score every match; accept top only if <code>top - runnerUp ≥ MARGIN</code> and <code>top ≥ T</code>	AMBIGUOUS
L2 fuzzy	no hash match	Trigram prefilter <code>≥ 0.22</code> , then <code>approxSearch</code> , then quote+context score	ORPHAN below <code>T</code> ; AMBIGUOUS inside <code>MARGIN</code>
L3 orphan	nothing survived	Return the stored quote and the last known heading path	–

```
export type Resolution =
| { readonly ok: true; readonly lane: 'L0' | 'L1_VERIFY' | 'L1_CTX' | 'L2'
  readonly start: U160Offset; readonly end: U160Offset; readonly score: number }
| { readonly ok: false; readonly reason: 'AMBIGUOUS'
  readonly candidates: readonly { start: U160Offset; score: number }[] }
| { readonly ok: false; readonly reason: 'ORPHAN'; readonly bestScore: number }
| { readonly ok: false; readonly reason: 'DOC_MISMATCH'; readonly expected: DocId }
| { readonly ok: false; readonly reason: 'STALE_NORMALIZE'; readonly stored: string }
```

The shape mirrors `PlacementVerdict` in `domain/placement.ts` deliberately: a discriminated refusal the caller must destructure, never a nullable range that a cal

77.4 The false-positive problem

A miss costs a badge. A false positive replaces bytes the user did not point at. The two are not on the same axis and must not share a threshold. Measured sweep, 120 corpus files, 10,070 live attempts, seven mutation classes [MEASURED](#) :

T	MARGIN	RESOLVE %	FALSE POSITIVES	FP %	ORPHAN	REFUSED
0.50	0.02	98.9076	84	0.8254	0	63
0.62	0.06	98.7984	20	0.1965	0	115
0.68	0.08	98.7587	8	0.0786	0	122
0.72	0.10	98.7388	4	0.0393	0	125
0.76	0.12	98.7090	3	0.0295	0	129
0.80	0.15	98.4409	1	0.0098	2	155
0.84	0.20	97.9742	1	0.0098	6	198

The knee is at 0.76/0.12: 28× fewer false positives than 0.50/0.02 for 0.20 points of resolve rate. Past it the curve turns – 0.84 buys two more prevented e The ablation is the finding. Removing `multiplicity` and trusting a unique hash match, everything else identical, on 448 files / 43,056 attempts [MEASURED](#) :

CONFIGURATION	RESOLVE %	FP	FP %	FP IN EXACT-HASH LANE
With <code>multiplicity</code>	98.6505	16	0.0372	0
Without (unique hash ⇒ accept)	99.1410	141	0.3275	115

The mechanism: a document contains two byte-identical blocks – 6.73% of corpus blocks sit in a same-hash group of `≥2`, 0.08% in a group of `≥3` [MEASURED](#) . A hum

77.5 Fuzzy matching and quote context

Hypothesis solves the same problem for web annotation and its two components are the right shape to adopt rather than re-derive. `approx-string-match@2.0.0`, MIT, zero dependencies, last published 2021-11-23 [FETCHED](#), [NPM REGISTRY](#) – Myers's bit-parallel algorithm, expected `O((k/w)·n)` with `w` `match-quote.ts` from `hypothesis/client` [FETCHED](#), [RAW.GITHUBUSERCONTENT.COM](#) supplies the scoring shape:

```

quoteWeight = 50 // similarity of matched text to the stored quote
prefixWeight = 20 // text before the match vs stored prefix
suffixWeight = 20 // text after the match vs stored suffix
posWeight = 2 // proximity to the expected offset - tie-breaker only
maxErrors = Math.min(256, quote.length / 2)

```

Three deviations, each with a reason.

HYPOTHESIS	OURS	WHY
Search the whole document text	Search block candidates behind a trigram prefilter at Jaccard ≥ 0.22	The splice contract needs a whole-bl
Return the top-scoring match unconditionally	Return only above <code>T</code> and outside <code>MARGIN</code> of the runner-up	<code>matchQuote</code> returns <code>scoredMatches[0]</code>
Raw document text	<code>normalize()</code> output on both sides	The reflow mutation - joining a wrap

The position weight stays at 2/92 and stays a tie-breaker. A whole-section move relocates blocks by hundreds of positions, and the move mutation still resol

77.6 Document identity and rename

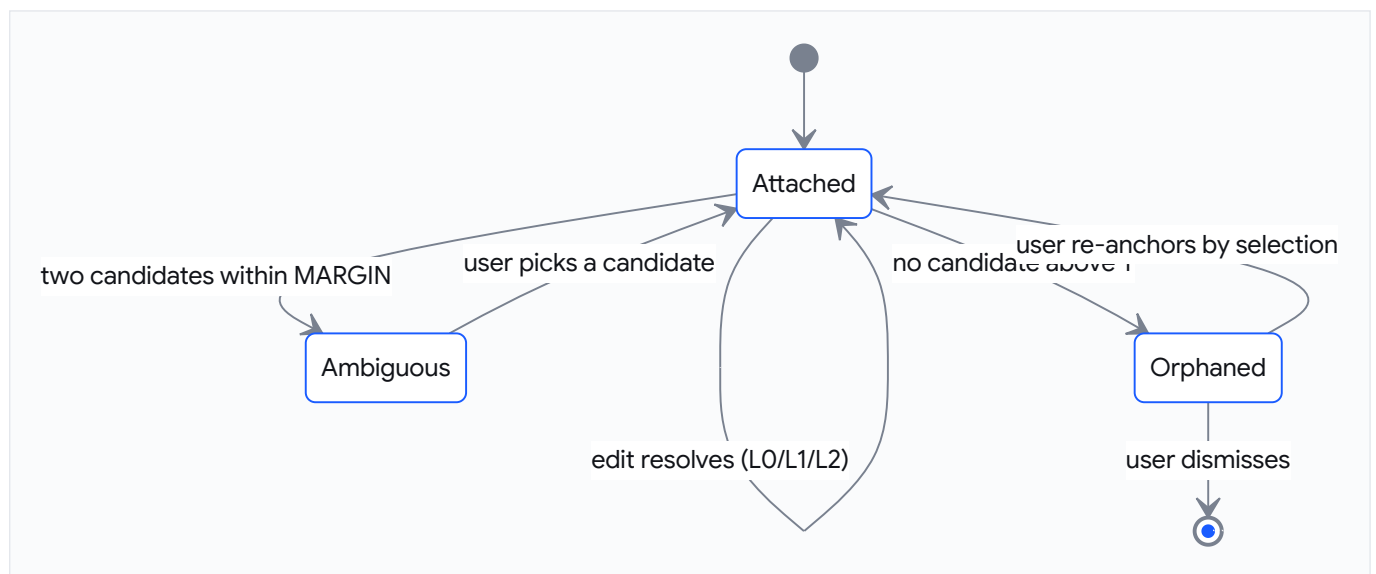
Anchors carry `docId`, and `docId` is not the path. A rename is a metadata operation: `docId` is unchanged, every anchor resolves, nothing is re-run.

The reason this is a hard precondition rather than a convenience is measured. Taking anchors from one corpus file and resolving them against a *different* fil

So content cannot establish document identity, and a resolver that accepts an anchor without checking `docId` will confidently re-point roughly a third of a

The rejected alternative was path-keyed anchors with a rename hook. It fails on the case it exists for: a rename performed outside the app - `mv` in a termin

77.7 Invalidation and orphan surfacing



An orphan is never deleted and never silently re-attached. It keeps `quote`, `headingPath` and `baseDigest`, so the badge can say what it was pointing at and the Re-anchoring writes a *new* anchor record with a new `baseDigest` and leaves the old one in place as history. It never mutates the failed anchor, because a muta

77.8 Anchors and the splice journal

The journal record is already specified as `{seq, base_digest, offset, deleted_len, inserted_bytes, result_digest}`. Anchors relate to it in exactly one direction: th

CONCERN	JOURNAL	ANCHOR
Authority	Owns the bytes	Owns nothing
Failure	CAS conflict on <code>base_digest</code> mismatch	<code>ORPHAN</code> / <code>AMBIGUOUS</code>
Lifetime	Append-only, permanent	Re-resolved on every read

The interaction that matters is the write path. An AI proposes an edit against `Anchor.baseDigest`. Before the splice executes, the resolver re-runs against th Journal replay never re-resolves anchors. It replays byte ranges. Anchors are re-resolved once at the end, against the final bytes.

77.9 The specification

```

resolve(anchor: Anchor, doc: string, docId: DocId, blocks: BlockIndex): Resolution

```

- `anchor.docId !== docId` → `DOC_MISMATCH`. No content is examined.
- `anchor.normalizeVersion !== NORMALIZE_VERSION` → `STALE_NORMALIZE`. Migrate, do not compare.
- `normalize(anchor.quote).length < 8` → the anchor was never issuable; reject at write time, not read time.
- `cand = blocks.byHash(anchor.hash)`.
- `cand.length === 1 && anchor.multiplicity === 1` → `L0`.
- `cand.length >= 1` → score each; single survivor accepted at `score ≥ 0.76` (`L1_VERIFY`); multiple accepted only at `top ≥ 0.76 && top - second ≥ 0.12` (`L1_CTX`); o
- `cand.length === 0` → trigram prefilter at `≥ 0.22`, `approxSearch` at `maxErrors = min(256, quote.length/2)`, score with `50/20/20/2`; accept at `≥ 0.76` and outside `T`
- Convert the accepted block bounds to `U16Offset` through `OffsetMap`. A conversion that would split a surrogate pair is `ORPHAN`, never rounded.

Every constant is versioned into `ANCHOR_VERSION`. Changing `T`, `MARGIN`, the weights, or the prefilter is a version bump, because the stored corpus of anchors

77.10 Failure modes and what the user sees

MODE	CAUSE	RESOLUTION	USER-VISIBLE
Target lightly edited	typo, reword	L2 at score ≥ 0.76	Nothing. The comment stays attached
Target reflowed	line joins, NBSP paste	L0 – normalize absorbs it	Nothing
Target heavily rewritten	~45% of words replaced	ORPHAN	Badge: <i>the text this was attached to has changed too m</i>
Block duplicated	copy-paste, template expansion	AMBIGUOUS	Two highlighted candidates, <i>which one did you mean?</i> –
Target deleted	human removed the paragraph	ORPHAN	Same orphan badge; the comment is preserved, not delet
Section moved	reorder	L1_CTX or L2	Nothing
Path reused for new content	delete-and-create through the same filename	DOC_MISMATCH	All anchors orphan at once, with one banner rather tha
Normalize version bumped	engine upgrade	STALE_NORMALIZE	Nothing – background migration re-hashes; a migration
Offset lands mid-surrogate	non-BMP content at a block boundary	ORPHAN	Orphan badge. Never a rounded offset

An AI proposal that resolves to `AMBIGUOUS` or `ORPHAN` is never applied and never applied "nearby". It is shown as a proposal the user must place.

77.11 The test that proves the false-positive rate

Harness, run today against the pinned corpus at `/Users/sagnikmitra/Desktop/GitHub/frontmatter/test/corpus/foreign/_vendor/` (8,513 files, seven vendored vaults), Pytl

PARAMETER	VALUE
Files sampled / usable	500 / 448 (skipped: <4 blocks, >300 KB, non-UTF-8)
Blocks anchored	6,163
Mutation classes	insert · delete · typo(25% of blocks) · reflow(35%) · move(2–4 block run) · duplicate(12%) · heavy(45% of words in 15% of bloc
Resolution attempts	43,056 (42,608 live + 448 correctly-orphaned deletions)
Resolve rate @ 0.76/0.12	98.6505%
False positives	16 → 0.0372% (9 duplicate, 6 delete, 1 typo; by lane: 9 <code>L1_CTX</code> , 6 <code>L2</code> , 1 <code>L1_VERIFY</code> , 0 <code>L0</code>)
Honest orphans / refusals	5 / 560

Ground truth is carried, not inferred: each mutation returns an explicit `origIndex → newIndex | null` map, so resolving a deleted block to *anything* is a false
This harness is not the one that produced the published 99.627% / 0.050% over 41,642 block-versions, and the two numbers are not comparable: the mutation mi
Three gates ship with the resolver, in `test/mdmax/anchor-mutation.test.ts`:
1. **Ratchet, not equality.** Assert `FP ≤ 20` and `resolve ≥ 98.4%` at a pinned seed and pinned sample stride. An equality assertion goes red the moment a mutation
2. **Ablation.** Delete `multiplicity` from the anchor and assert the false-positive count *rises above 100*. A guard whose removal changes nothing is not a guard, a
3. **Cross-document.** Assert that resolving anchors across `docId` boundaries with the check disabled exceeds 25%, and with it enabled is exactly 0. The first ha
The falsifier for the whole section: a harness that reproduces the mutation mix and finds the exact-hash lane producing zero false positives *without* the mul

]

§78
Property-based testing for a byte-preserving writer

78.1 The decision

Adopt **fast-check 4.9.0** with the `@fast-check/vitest` 0.4.1 binding, on the vitest 4.1.x already in `devDependencies`. Twelve properties across four new files unde

CANDIDATE	LATEST	PUBLISHED	VERDICT
<code>fast-check</code>	4.9.0	2026-07-08 [fetched: <code>curl -s https://registry.npmjs.org/fast-check</code>]	Adopt. Native shrinking, <code>examples: replay, seed / path reproduction, fc.stringMatching</code> for imperative draws.
<code>@fast-check/vitest</code>	0.4.1	2026-04-28 [fetched]	Adopt. <code>test.prop</code> binds to vitest's runner so failures report as normal test failures, not as one opaque
<code>jsverify</code>	0.8.4	2018-10-31 [fetched]	Reject. Seven years unmaintained; no Unicode-aware string arbitrarities.
<code>testcheck</code>	1.0.0-rc.2	2017-04-26 [fetched]	Reject. Never left RC.
Hand-rolled fuzz loop	–	–	Reject. Shrinking is the whole product. A 4 KB random counterexample that is not shrun

Rejected alternative worth naming: **not adding property tests and widening the corpus instead**. Rejected because the corpus is measurably blind to the exact c

78.2 What the corpus cannot see

Census over the pinned vendor tree, `node` walk of `test/corpus/foreign/_vendor/**/*.md` reading each file as UTF-8 [measured, 8,513 files]:

SHAPE	FILES IN CORPUS	CONSEQUENCE
Leading BOM (U+FEFF)	0	The BOM defect (commit <code>f47555f</code>) was unreachable from the corpus.
CRLF anywhere	0	The CRLF branch of <code>spliceFrontmatterValue</code> is corpus-untested.
Bare CR, no LF	0	See below – still broken today.
Unterminated frontmatter	0	The <code>closeIdx === -1</code> refusal is corpus-unreachable.
No final newline	860	Covered.
Non-BMP characters	475	Covered but thin.
<code>bytes === text.length</code>	6,568 of 8,513	22.8% already diverge in the third-party set.
Zero-indent block sequence in frontmatter	6,613	The refusal driver.
Frontmatter key containing a space	937	Unaddressable under <code>SAFE_KEY</code> .

Three of the four defects named in the brief live in rows with a corpus count of **zero**. A corpus is a sample of what people wrote; a property is a statement
Running `spliceFrontmatterValue(src, 'fm_slug', 'abc-123')` then the matching delete across all 8,513 files [measured, 309 ms]: 6,615 refusals (77.70% of all files

78.3 Generators

`test/props/arbitraries/text.ts` and `.../markdown.ts`. Every arbitrary is biased toward a boundary, because uniform generation does not find boundary bugs – \$78.5

```
import fc from 'fast-check'

/** All three terminators, weighted AWAY from the corpus's 100%-LF reality. */
export const eol = fc.oneof(
  { arbitrary: fc.constant('\n'), weight: 4 },
  { arbitrary: fc.constant('\r\n'), weight: 3 },
  { arbitrary: fc.constant('\r'), weight: 3 }, // 0 files in the corpus
)

/** Chunks chosen so bytes, UTF-16 units and graphemes all disagree. */
export const unicodeChunk = fc.oneof(
  { arbitrary: fc.stringMatching(/^[ -~]{1,24}$/), weight: 8 },
  { arbitrary: fc.constantFrom('á', 'ü', 'ß', ' ', '한', 'ᄀ', 'ᄂ'), weight: 3 },
  { arbitrary: fc.constantFrom('👉', '👈', '👉', '👈', '👉', '👈'), weight: 3 },
  { arbitrary: fc.constantFrom('e\u0301', 'é'), weight: 2 }, // NFD vs NFC
  { arbitrary: fc.constantFrom('\uFEFF', '\u200B', '\u00A0', '\u2028', '\u3000'), weight: 2 },
  { arbitrary: fc.constantFrom('\uD83D', '\uDE00'), weight: 1 }, // lone surrogates
)

/**
 * Text with a surrogate pair placed so its LOW unit lands on a BLOCK boundary.
 * BLOCK = 512 in src/modules/mdmax/domain/offsets.ts. This is the ONLY generator
 * that reproduces the off-by-three; a uniform one does not ($78.5).
 */
export const boundaryStraddlingText = fc
  .tuple(fc.integer({ min: 505, max: 520 }), fc.integer({ min: 0, max: 3 }), fc.nat({ max: 400 }))
  .map(([pad, k, tail]) => 'a'.repeat(pad + k * 512) + '👉' + 'b'.repeat(tail))

/** Frontmatter block bodies, including every shape the writer must REFUSE. */
export const fmLine = fc.oneof(
  { arbitrary: fc.tuple(safeKey, scalar).map(([k, v]) => `${k}: ${v}`), weight: 10 },
  { arbitrary: fc.tuple(safeKey, fc.array(scalar, { maxLength: 4 })),
    .map(([k, xs]) => `${k}: [${xs.join(', ')}]`), weight: 4 },
  { arbitrary: fc.tuple(safeKey, fc.array(scalar, { minLength: 1, maxLength: 4 })),
    .map(([k, xs]) => `${k}: \n${xs.map(x => `~ ${x}`).join('\n')}`), weight: 4 }, // ZERO indent
  { arbitrary: fc.tuple(unsafeKey, scalar).map(([k, v]) => `${k}: ${v}`), weight: 3 }, // "date created"
  { arbitrary: fc.constantFrom('a: |', 'b: >', 'c: &anch x', 'd: *anch', '? explicit', '<<: *base'), weight: 2 },
  { arbitrary: fc.constantFrom('# a comment', '', ' indented: cont'), weight: 3 },
)

export const document = fc.record({
  bom: fc.boolean(),
  fm: fc.option(fc.array(fmLine, { maxLength: 12 }), { nil: undefined }),
  body: markdownBody, // fences (closed, unclosed, tilde, ~~~~ nesting), lists to depth 8,
  eol, // setext underlines, tables, '<<<<<< HEAD' conflict markers,
  finalNewline: fc.boolean(), // an embedded '---' block, HTML comments, indented code
}).map(assemble) // assemble() joins with 'eol' and applies bom/finalNewline
```

Deliberately in scope, because each has shipped a defect or is a stated risk: `<<<<<<` / `=====` / `>>>>>>` conflict markers in the body (the sync design puts t

78.4 The properties

`REFUSED` is not currently observable: `spliceFrontmatterValue` signals refusal by returning the input, which is indistinguishable from a successful no-op splice

```
export type SpliceTrace =
  | { readonly kind: 'APPLIED'; readonly out: string; readonly start: number; readonly end: number; readonly inserted: string }
  | { readonly kind: 'REFUSED'; readonly out: string; readonly reason: SpliceRefusal }

export function spliceFrontmatterValueTraced(
  src: string, key: string, value: string | number | boolean | string[] | null,
): SpliceTrace
```

#	PROPERTY	STATEMENT
P1	Round-trip identity	For <code>k</code> absent from <code>src</code> : <code>del(set(src, k, v), k) === src</code> , byte-for-byte.
P2	Splice locality	<code>t.kind === 'APPLIED'</code> $\Rightarrow$ <code>t.out === src.slice(0, t.start) + t.inserted + src.slice(t.end)</code> , and <code>[t.start, t.end)</code> lies stri
P3	Refusal safety	<code>t.kind === 'REFUSED'</code> $\Rightarrow$ <code>Buffer.compare(Buffer.from(t.out, 'utf8'), Buffer.from(src, 'utf8')) === 0</code> . Not <code>===</code> on the stri
P4	Idempotence	<code>splice(splice(src, k, v), k, v) === splice(src, k, v)</code> .
P5a	Commutativity, both keys present	<code>k1 $\neq$ k2</code> , both located $\Rightarrow$ <code>set(set(s,k1,a),k2,b) === set(set(s,k2,b),k1,a)</code> . Their ranges are disjoint, so this must ho
P5b	Commutativity, keys absent	Both absent $\Rightarrow$ byte equality must fail ; the honest invariant is that the two results have equal line multisets and
P6	Offset-map consistency	$\forall$ <code>i</code> valid: <code>map.toByte(i) === utf8Length(text.slice(0, i))</code> ; <code>map.toU16(map.toByte(i)) === { ok: true, value: i }</code> ; <code>toByte</code>
P7	Brand separation	<code>u16(text, n)</code> refuses every <code>n</code> that <code>splitsSurrogatePair</code> reports true for, and never repairs: no returned value dif
P8	Construct coverage	<code>U skipRegions(s) $\subseteq$ U inventory(s).values()</code> .
P9	EOL-invariant line count	<code>shapeGate(s).lines</code> is equal under <code>\n</code> , <code>\r\n</code> and <code>\r</code> renderings of the same logical document.
P10	Normalize idempotence	<code>normalize(normalize(x)) === normalize(x)</code> , plus the existing digest pin against <code>NORMALIZE_VERSION</code> .
P11	Prepass never writes	<code>verdict.kind === 'CLEAN'</code> $\Rightarrow$ <code>verdict.yaml === block</code> ; <code>'REPAIRED'</code> $\Rightarrow$ removing the inserted quote pairs recovers <code>block</code>
P12	Conflict-marker containment	A body containing <code><<<<<< / >>>>>></code> is either refused or spliced with those bytes at unchanged offsets.

78.5 Which property would have caught which shipped defect

Every row below was executed. Pre-fix sources were extracted with `git show <sha>^:<path>` into `$TMPDIR` and imported through a Node 24 type-stripping resolver

DEFECT	P1 ROUND-TRIP	P2 LOCALITY	P3 REFUSAL	P4 IDEMPOTENCE	P6 OFFSET MAP
BOM (<code>f47555f</code>)	PASS on the bug	FAIL – catches it	n/a	PASS on the bug	n/a
OffsetMap off-by-three (<code>cc1d451</code>)	n/a	n/a	n/a	n/a	FAIL – catches it, 600 wrong answers
Zero-indent sequence (open)	PASS	PASS	PASS	PASS	n/a
Bare CR (open, live today)	n/a	FAIL – catches it	n/a	n/a	n/a

BOM, measured on `"\uFEFF---\ntitle: A\n---\nbody\n"` with key `fm_slug` : pre-fix, `set` returns `"---\nfm_slug: x\n---\n\n\uFEFF---\ntitle: A\n---\nbody\n"` – a second fr
OffsetMap: a uniform generator (`i % 7` emoji, `i % 3` accented, lengths 400–2000, 4,966 offsets checked) reports **0 failures against the unfixed code**. `boundary`
Zero-indent sequence: no property catches it, and that is correct. The writer's `return src` on a bare top-level line is the specified behaviour; the defect i
Bare CR, run against the shipped code today: input `'---\rtitle: A\rslug: old\r---\rbody\r'`, output `'---\nslug: new\n---\n\n---\rtitle: A\rslug: old\r---\rbody\r'` [meas

78.6 Shrinking, seeds, and regression promotion

fast-check shrinks integers toward zero and arrays toward empty, so a `.map()`-built document shrinks through its *record fields*, not its bytes. That is the de

```
fc.configureGlobal({
  numRuns: Number(process.env.FC_RUNS ?? 300),
  seed: Number(process.env.FC_SEED ?? 0x5EED0000), // fixed in CI, overridable locally
  endOnFailure: true,
  interruptAfterTimeLimit: 20_000,
  markInterruptAsFailure: true,
})
```

CI pins the seed so a red build is reproducible from the log line alone; a nightly job runs with `FC_SEED=$RANDOM FC_RUNS=5000` and files any counterexample. On

```
test.prop([document, safeKey, scalar], { examples: REGRESSIONS.spliceLocality })('P2 locality', ...)
```

Decision: regressions are stored as literal counterexamples, not as `{ seed, path }` pairs. Rejected alternative: replaying by seed, which is what the fast-che

78.7 Corpus × properties

The corpus stops being a fidelity oracle and becomes a *seed source*. Same properties, real inputs:

```
// test/props/corpus-seeded.property.test.ts – skipped unless _vendor/ is present
const CORPUS = loadCorpus() // 8,513 strings, or [] when the vendor tree is absent
test.skipIf(CORPUS.length === 0).prop([fc.constantFrom(...CORPUS), safeKey, scalar])('P1&P2&P3 on real files', ...)
```

That combination is what surfaced the 21 leading-newline round-trip failures in §78.2: neither instrument alone reports them. The corpus alone has no oracle
`npm run corpus` stays the byte-pin gate. The property files degrade to skipped, never failed, when `_vendor/` is absent – the tree is gitignored third-party co

78.8 Suite specification and CI budget

FILE	PROPERTIES	NUMRUNS	BUDGET
<code>test/props/splice.property.test.ts</code>	P1–P5b, P12	300	2.4 s
<code>test/props/offsets.property.test.ts</code>	P6, P7 (+ <code>boundaryStraddlingText</code> at 1,000 runs)	300 / 1,000	1.8 s
<code>test/props/constructs.property.test.ts</code>	P8, P9, P10, P11	300	1.6 s
<code>test/props/corpus-seeded.property.test.ts</code>	P1, P2, P3 over 8,513 real files	full sweep	1.2 s
Added total	12	–	7.0 s
Existing engine suite (<code>test/mdmax</code> + <code>test/share/frontmatter-splice</code>)	320 tests, 11 files	–	1.34 s [measured]
Gate	–	–	fail the build above 12 s

Derivation of the 7.0 s: the corpus sweep of two splices plus a locality diff over 8,513 real files (mean 2,237 B) ran in 309 ms [measured] – 36.3 μs per ca
Rejected alternative for the budget: running the properties only in the nightly job to keep the pull-request loop under two seconds. Rejected because three

\$79 Incremental parsing — the lezer decision

79.1 The decision

Keep remark/micromark as the engine's only parse layer. Keep @lezer/markdown where it already is – inside CodeMirror, serving decorations – and make the bour
Two trees of the same document are not the hazard; two trees of two different revisions treated as one is, and that is a provenance problem, not a parser-ch
Rejected: adopting lezer as the engine's parse layer. Rejected: a shared position model across both trees. Both are rejected on a measurement that contradic

	ADOPT LEZER IN THE ENGINE	DUAL, UNDEFINED BOUNDARY (TODAY)	DUAL, TYPED ONE-WAY BOUNDARY (CHOSEN)
Frontmatter addressable	No – measured misparse (\$79.5)	n/a	Yes, remark + prepass, unchanged
Bare-CR safe	No – measured (\$79.5)	Partly, by luck	Yes
Joins the 7-engine certificate	No – emits no HTML	n/a	Yes, unchanged
Stale-tree splice possible	Yes	Yes	No – refused at the type
Engine parse cost, 8,513 files	566 ms	4,849 ms	4,849 ms
Work to implement	Rewrite 3 domain files + bench	0	~60 lines, 1 lint rule

79.2 What each side actually gives

CAPABILITY	@LEZER/MARKDOWN 1.7.2	REMARK / MICROMARK 4.0.2
Incremental reparse	Yes – TreeFragment.applyChanges + parse(doc, fragments)	No. Full reparse only
Position unit	UTF-16 code units [measured]	UTF-16 code units, position.*.offset [measured]
Line/column	No – offsets only	Yes, 1-based line + column alongside offset
Emits HTML	No. "doesn't help with outputting HTML" [fetched, README]	Yes, via remark-rehype/rehype-stringify
Link-reference validation	No, deliberately – [a][b] parses as a link with no [b] defined [fetched, README]	Yes
Frontmatter	Not in the default parser [measured]	remark-frontmatter 5.0.0, installed
GFM tables/footnotes/strike	Via the GFM extension in the same package	remark-gfm 4.0.1, installed
Weekly npm downloads	4,789,610 [fetched, 2026-08-23→29]	57,013,343 micromark / 51,168,105 remark-parse [fetched]
Ecosystem the engine already uses	none	mdast-util-*, rehype-*, react-markdown, the bench's remark-app engine

The engine's live usage of lezer is three files, all presentation: src/modules/editor/presentation/live-preview.ts (import { syntaxTree } from "@codemirror/language",

79.3 The premise that does not survive measurement

The brief states lezer offers "byte-accurate SyntaxNode positions." It does not. Lezer parses a JavaScript string and its node offsets index that string, wh
Ran, node \$TMPDIR/lz/pos.mjs, on "# 🏠 tide\n\nafter\n" [measured]:

```
doc.length (UTF-16 units) = 17      byteLength = 19      code points = 16
lezer: ATXHeading1 [0,9) slice = "# 🏠 tide"
mdast: heading start.offset 0, end.offset 9
```

Identical values, and neither is 11 (the byte length of that heading). Adopting lezer buys nothing against the contract in src/modules/mdmax/domain/offsets.ts;
What lezer positions do carry that mdast's do not is a hidden precondition. syntaxTree(state) is documented to return whatever tree exists; ensureSyntaxTree(st

79.4 The performance numbers, corrected

src/modules/mdmax/domain/shape-gate.ts records mdast-util-from-markdown 12,429 ms vs micromark's 1,207 ms – a 10.3x gap that widens with size. That measurement is on flat
Ran, node \$TMPDIR/lz/bench.mjs, over the pinned corpus at test/corpus/foreign/_vendor – 8,513 files, 19,047,891 bytes, 300-doc warmup, @lezer/markdown 1.6.3 as i

PARSER	TOTAL	THROUGHPUT	ERRORS
@lezer/markdown parser.parse	566 ms	33.6 MB/s	0
mdast-util-from-markdown	4,849 ms	3.9 MB/s	0
micromark → HTML	5,019 ms	3.8 MB/s	0

So on real documents from-markdown / micromark = 0.97x, not 10.3x. The mdast construction step is not the cost; the tokenizer is. Lezer is genuinely 8.56x fast
Incremental reparse, node \$TMPDIR/lz/inc.mjs, 50 largest corpus docs (939,748 chars max, 18,039 median), one character inserted at 80% depth, 3 iterations eac


```

declare const EditorTreeBrand: unique symbol

/**
 * A UTF-16 offset read from a CodeMirror syntax tree. SAME UNIT as U16Offset
 * (measured: lezer and mdast both report [0,9) for "# 🦄 tide"). Different
 * PROVENANCE: valid only against the EditorState generation it was read from,
 * and only where that state's tree actually spans the offset.
 */
export type EditorOffset = number & { readonly [EditorTreeBrand]: true }

export type Crossing =
  | { readonly ok: true; readonly at: U16Offset; readonly docLength: number }
  | { readonly ok: false; readonly reason: 'TREE_INCOMPLETE'; readonly parsedTo: number; readonly docLength: number }
  | { readonly ok: false; readonly reason: 'DOC_DIVERGED'; readonly editorLength: number; readonly bytesLength: number }

/**
 * The ONLY function permitted to turn an EditorOffset into a U16Offset.
 * Refuses rather than guessing - mirrors splice-frontmatter's refusal contract
 * and the anchor resolver's AMBIGUOUS verdict.
 */
export function crossToEngine(state: EditorState, at: EditorOffset, bytes: string): Crossing {
  const len = state.doc.length
  if (!syntaxTreeAvailable(state, at)) {
    return { ok: false, reason: 'TREE_INCOMPLETE', parsedTo: syntaxTree(state).length, docLength: len }
  }
  if (state.doc.toString() !== bytes) {
    return { ok: false, reason: 'DOC_DIVERGED', editorLength: len, bytesLength: bytes.length }
  }
  return { ok: true, at: at as unknown as U16Offset, docLength: len }
}

```

The `DOC_DIVERGED` check is a full string compare, which at corpus median (18,039 chars) is free relative to the 51 ms an mdast reparse costs, and it is the only one that is. Enforcement, one ESLint `no-restricted-imports` rule in `eslint.config.*: @codemirror/*` and `@lezer/*` are importable only under `src/modules/editor/presentation/**`. I

79.9 Migration path

There is none, which is the point. No engine file imports lezer today; the change is additive and reversible.

1. Add `tree-boundary.ts` and the `no-restricted-imports` rule. `npm run lint --max-warnings=0` fails on any future violation.
2. Add a `specs/` assertion pinning the measured claim, so a future lezer release that changes position semantics goes red rather than silent: `parse "# 🦄 tide"`
3. Add the bare-CR and frontmatter misparse cases as `xfail` -documented facts about lezer, referenced from `live-preview.ts` - the decoration path must not assume
4. Leave `ParseFn` alone.

Nothing in `mdmax/`, nothing in `splice-frontmatter.ts`, no change to the bench, no change to `benchId`.

79.10 Cost of being wrong, and what falsifies it

If this is wrong, the cost is that the engine keeps paying 4,849 ms per full-corpus pass instead of 566 ms, and that a future feature wanting sub-frame structural reparse costs, and it is the only one that is. Falsifiers, in order of how cheaply they can be run:

- A measurement showing `SyntaxNode.from` indexing anything other than UTF-16 code units of the input string. §79.3's assertion is the test; it takes 40 ms.
- An engine-side consumer that genuinely needs sub-100 ms structural reparse on a document over ~200 KB, appearing in a shipped feature rather than a plan.
- `@lezer/markdown` shipping frontmatter parsing and CR-as-line-terminator in the default configuration, which would remove two of the three job-fitness objectives
- No `@lezer/markdown` or `@codemirror/language` release for four consecutive quarters on `code.haverbeke.berlin` - which falsifies §79.7's "migration, not abandonment"
- Evidence that the 83% refusal rate is materially affected by markdown block structure rather than by YAML sequence indentation and `SAFE_KEY`. That would mean

§80

The engine public API, versioning and distribution

80.1 Measured starting position

FACT	VALUE
Engine files / lines / bytes	13 / 3,614 / 153,178
<code>export</code> declarations	88
Symbols imported from outside the module	1 (<code>decodeStrict</code>), at 2 sites – <code>src/modules/vault/applicatio</code> <code>snapshot.ts</code> , <code>src/modules/vault/infrastructure/search-index.ts</code>
Imports MDMAX makes from <code>src/modules/**</code>	0
Real <code>throw</code> sites	2: <code>throw new</code> <code>RangeError(\ invalid U16Offset in domain/offsets.ts</code> , <code>throw new Error('kramdown-jekyll: in infrastructure/bench.</code>
Third-party runtime deps	<code>github-sluggger@2.0.0</code> (ISC), <code>entities@8.0.0</code> (BSD-2-Clause)
Comment + whitespace share of source	51.9%
Stripped code bytes, browser-eligible files only (excludes <code>bench.ts</code> , <code>certify.ts</code> , <code>targets.ts</code>)	57,318
<code>mdmax</code> on npm	HTTP 404 – unregistered
<code>@sgnk/mdmax</code>	HTTP 404
<code>frontmatter</code> on npm	HTTP 200 – taken
<code>LICENSE</code> file in repo	absent

88 exports and one external consumer is the whole problem in one line. There is no API; there is a folder that other folders reach into.

80.2 The public surface

Decision: one barrel, `src/modules/mdmax/index.ts`, exporting 31 symbols in six capability groups. Everything not in the barrel is internal, regardless of wheth

GROUP	EXPORTED
Admission	<code>shapeGate</code> , <code>decodeStrict</code> , <code>explainFailure</code> , <code>MAX_BYTES</code> , <code>MAX_LINES</code>
Offsets	<code>OffsetMap</code> , <code>u16</code> , <code>utf8Length</code> , <code>splitsSurrogatePair</code> , types <code>U16Offset</code> / <code>ByteOffset</code> / <code>GraphemeIndex</code>
Splice (write path)	<code>spliceFrontmatterValue</code> , <code>spliceFrontmatterKey</code> , <code>emitValue</code> , <code>emitScalar</code> , <code>SAFE_KEY</code>
Anchors	<code>slug</code> , <code>slugDocument</code> , <code>resolveAnchor</code> , <code>decodeAnchor</code> , type <code>AnchorMatch</code>
Structure	<code>inventory</code> , <code>skipRegions</code> , <code>safeInsert</code> , <code>prepassFrontmatter</code> , <code>extractWikilinks</code> , types <code>Construct</code> / <code>Range</code> / <code>BlockNode</code> / <code>PlacementVerdict</code>
Certificate	<code>certify</code> , <code>classify</code> , <code>fold</code> , <code>foldEqual</code> , <code>normalize</code> , <code>TARGETS</code> , <code>targetById</code> , <code>brokenHistogram</code> , types <code>Certificate</code> / <code>CellVerdict</code> / <code>Target</code> / <code>Ver</code>
Versions	<code>VERSIONS</code> (one frozen object), <code>stamp()</code>

Three structural moves come with it. `src/modules/share/domain/splice-frontmatter.ts` moves to `src/modules/mdmax/domain/splice.ts` – the splice writer is the engine's
Rejected: `export *` from a barrel that re-exports every module. It is a one-line change and it freezes all 88 symbols as public – including `unsafeU16`, whose

80.3 The purity rule and its mechanical enforcement

The rule: `src/modules/mdmax/**` may import from `node:*` (in `infrastructure/` only), from its own relative paths, and from a fixed allowlist of runtime dependenc
Purity that is only a convention is a convention that lasted until the first deadline, so this ships as a gate before it ships as a sentence in a doc.
New harness `specs/harness/mdmax-purity-report.mjs`, wired into `npm run verify` alongside the existing `clean-architecture-report.mjs` and `import-boundary-report.mjs`. Fo

CHECK	METHOD	FAILURE
P1 – outbound purity	Parse every import specifier in <code>src/modules/mdmax/**</code> ; reject any not matching <code>``(node: .{1,2}\/)``</code> or the dep allowlist	
P2 – inbound barrel-only	Reject any <code>from '@modules/mdmax/...'</code> outside the module; only <code>from '@modules/mdmax'</code> allowed	exit 1
P3 – browser graph is node-free	Resolve the transitive import graph from <code>index.ts</code> ; assert <code>node:</code> appears in zero reachable files	exit 1, prints the path that pul
P4 – denominator guard	Assert <code>filesScanned >= 13</code> , mirroring <code>MIN_SCANNED_FILES</code> in <code>clean-architecture-report.mjs</code>	exit 2, refuses to report a pass

P4 is not decoration. `clean-architecture-report.mjs` carries a comment recording that it once reported `{"total":0,"violations":[]}` and exit 0 whether it had scanr
Rejected: ESLint `no-restricted-imports`. It is already configured and it does not catch P3 – a transitive `node:fs` three files down through a type-only-looking

80.4 Errors and refusals as values

Every public function returns a discriminated union. No public function throws. The two existing throws are handled explicitly: `u160rThrow` leaves the barrel

```
type Refusal =
  | { kind: 'not-utf8';          at: number; detail: string }
  | { kind: 'too-large';         bytes: number; limit: number }
  | { kind: 'too-many-lines';    lines: number; limit: number }
  | { kind: 'unaddressable-key'; key: string; pattern: string }
  | { kind: 'ambiguous-range';   candidates: number; construct: string }
  | { kind: 'range-not-found';   construct: string }
  | { kind: 'splits-grapheme';   offset: number }
  | { kind: 'unsafe-placement';  reason: 'setext' | 'fence' | 'list-continuation' }
  | { kind: 'engine-unavailable'; engine: string; detail: string }

type Result<T> = { ok: true; value: T } | { ok: false; refusal: Refusal }
```

Two rules make this survive contact. Every `Refusal` carries the operand that caused it, not just a message – `unaddressable-key` carries `key` and the literal `Re` Rejected: typed exception classes with `instanceof`. They cross the MCP and CLI JSON boundaries as `{}`, and a `catch` that forgets one subclass fails open. A u

80.5 Versioning: engine and artifacts are separate axes

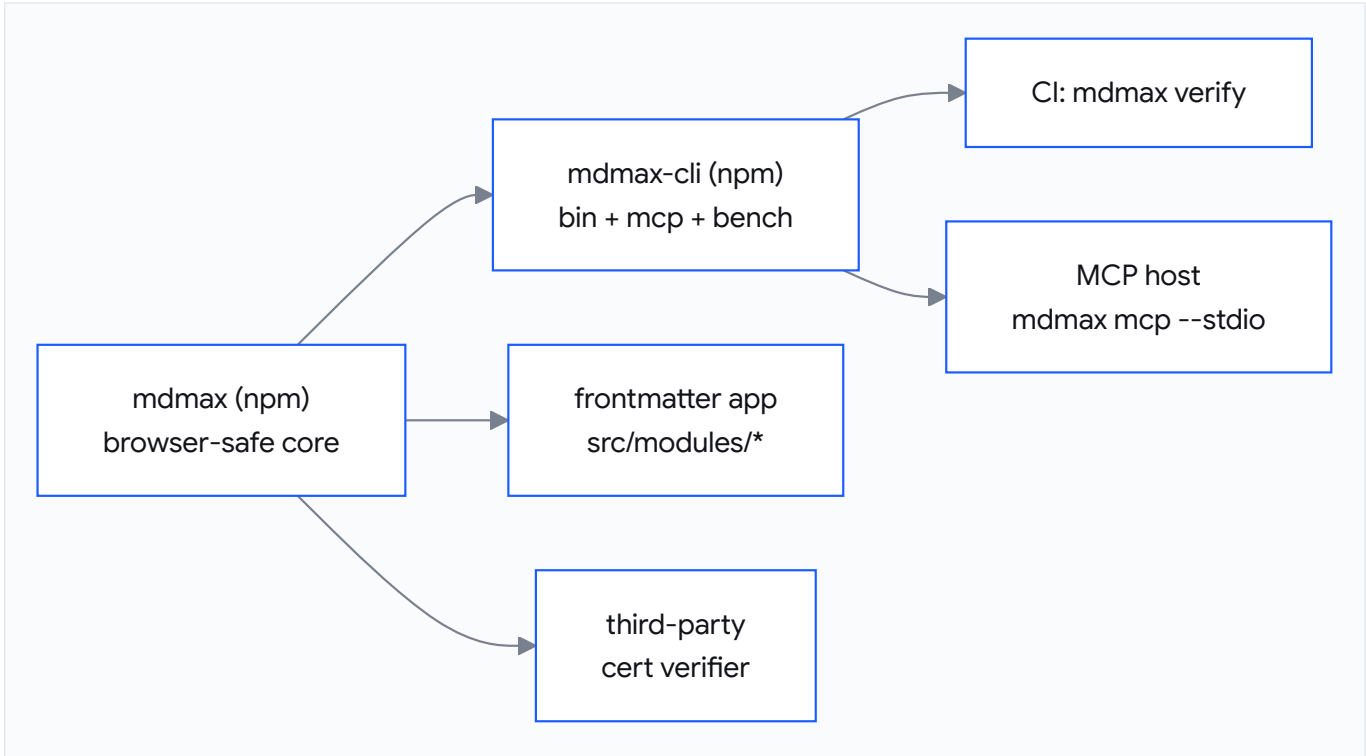
Four artifact versions already exist in source as `mdmax/normlize@1`, `mdmax/slug@1`, `mdmax/fold@1`, `mdmax/verdict@1` [measured]. Three more are missing: `mdmax/off`

AXIS	IDENTIFIER	BUMPS WHEN	NEVER
Engine	npm semver <code>mdmax@x.y.z</code>	API shape changes	tracks output bytes
Artifact	<code>mdmax/<name>@N</code> , integer	the same input produces different output bytes	decreases; is reused
Certificate	<code>mdmax/cert@N</code> + embedded <code>stamps</code> + <code>benchId</code>	the schema changes, or any input artifact bumps	is emitted without the full stamp vector

Rules an implementer follows literally:

- 1. Artifact versions are integers and monotonic. `mdmax/slug@1` names one function forever. Changing slug output creates `mdmax/slug@2`; both ship, `slug()` defau
 - 2. An artifact bump is at minimum a MINOR engine release, and a MAJOR if the old artifact is removed. Old artifact implementations are removed only in a MAJ
 - 3. Every certificate embeds the complete stamp vector. A certificate carrying three of seven stamps is unverifiable and the emitter must refuse to produce it
 - 4. Narrowing refusal – accepting input previously refused, such as the zero-indent block sequence behind 83% of refusals, or widening `SAFE_KEY` to address `de`
 - 5. Widening refusal – refusing input previously accepted – is a MAJOR, no exceptions, because it breaks a working pipeline.
 - 6. Adding a `Refusal` variant is a MINOR; consumers must have a `default` arm. Adding a field to an existing variant is a PATCH. Removing or renaming a variant
 - 7. Adding an engine to `TARGETS` is a MINOR and bumps `mdmax/cert@N` only if cell semantics change. `DECLARED_LAST_VERIFIED = '2026-08-01'` moves with each re-verific
 - 8. The pinned corpus at `test/corpus/foreign/_vendor/` with `MANIFEST.sha256` is versioned independently as `corpus@YYYY-MM-DD`. A certificate references the corpus v
- The compatibility promise, stated as what a consumer may rely on within a MAJOR: `slug()` output bytes for a given `mdmax/slug@N`; `normalize()` output bytes for
- Rejected: one version for everything, bumping the package on any output change. It forces a MAJOR whenever a slug edge case is fixed, and it destroys the pr

80.6 Package boundary and distribution



Decision: two published packages from one repo, not one and not three. `mdmax` is the engine – the barrel, browser-safe, `"sideEffects": false`, subpath `mdmax/ce`

The MCP server does not get its own package. It is stdio glue over the same exported functions, it has exactly the CLI's dependency profile, and a third pac

Name: `mdmax`, unscoped, confirmed 404 on the registry [fetched 2026-08-30]. `frontmatter` is taken (HTTP 200), which settles the alternative.

Licence: Apache-2.0 for both packages, with a `NOTICE` file. Not MIT, for one specific reason – the certificate is a published claim about seven third-party

Size and dependency budget, enforced in CI:

ENTRY	BUDGET	CURRENT BASIS
mdmax (splice + anchors + admission)	≤ 20 KB gzip, ≤ 1 runtime dep	57,318 stripped bytes across the browser-eligible files [measured]; roughly 15-18 KB gzi
mdmax/cert	≤ 45 KB gzip, ≤ 2 runtime deps	adds constructs.ts (17,693 stripped) plus entities
mdmax-cli	unbudgeted	node-only

The one permitted core dependency is `github-slugg@2.0.0` (ISC, 36 KB on disk [measured]). It stays because it *is* the definition GitHub ships; reimplementing Rejected: a three-package monorepo with `@mdmax/engine`, `@mdmax/cli`, `@mdmax/mcp`. Three release cadences, three changelogs, and a cross-package version matrix, The anti-recommendation: **do not publish anything until the zero-indent-sequence refusal and `SAFE_KEY` are fixed.** Publishing now freezes `SAFE_KEY = /^[A-Za-z0-9`

§81

Reaching users — email infrastructure and in-product messaging

81.1 The gap, stated as a measurement

`grep -oiF` over `docs/FRONTMATTER-PRD-v2-2026-08-29.md` (120,539 words by `wc -w`) returns: `SPF` 1, `DKIM` 1, `DMARC` 2, `List-Unsubscribe` 0, `unsubscribe` 1 [measured 2 The resolution is not to relax the bans. It is to separate *obligatory* messages – which are triggered by law or by contract, are never periodic, and have a n

81.2 The obligatory messages and their triggers

#	MESSAGE	TRIGGER	BA
1	Cyber-incident report	Unauthorised access, data breach, DoS, phishing on our infra	CE
2	Breach intimation to each affected user	Any personal data breach	DP
3	Breach report to the Data Protection Board	Same event as #2	DP
4	Breach communication to EU data subjects	Breach "likely to result in a high risk"	GD
5	Statement of reasons	Any account suspension, content removal, visibility restriction, or restriction of monetary payments	DS
6	Annual renewal pre-notice	Any auto-renewing term of one year or longer, California consumer	Ca
7	Price change / terms change	Unilateral amendment to a live subscription	Co
8	Payment-failure and dunning	Mandate debit declined	Co
9	Team invite	Inviter names an address	Co
10	Account recovery	User requests it	Co

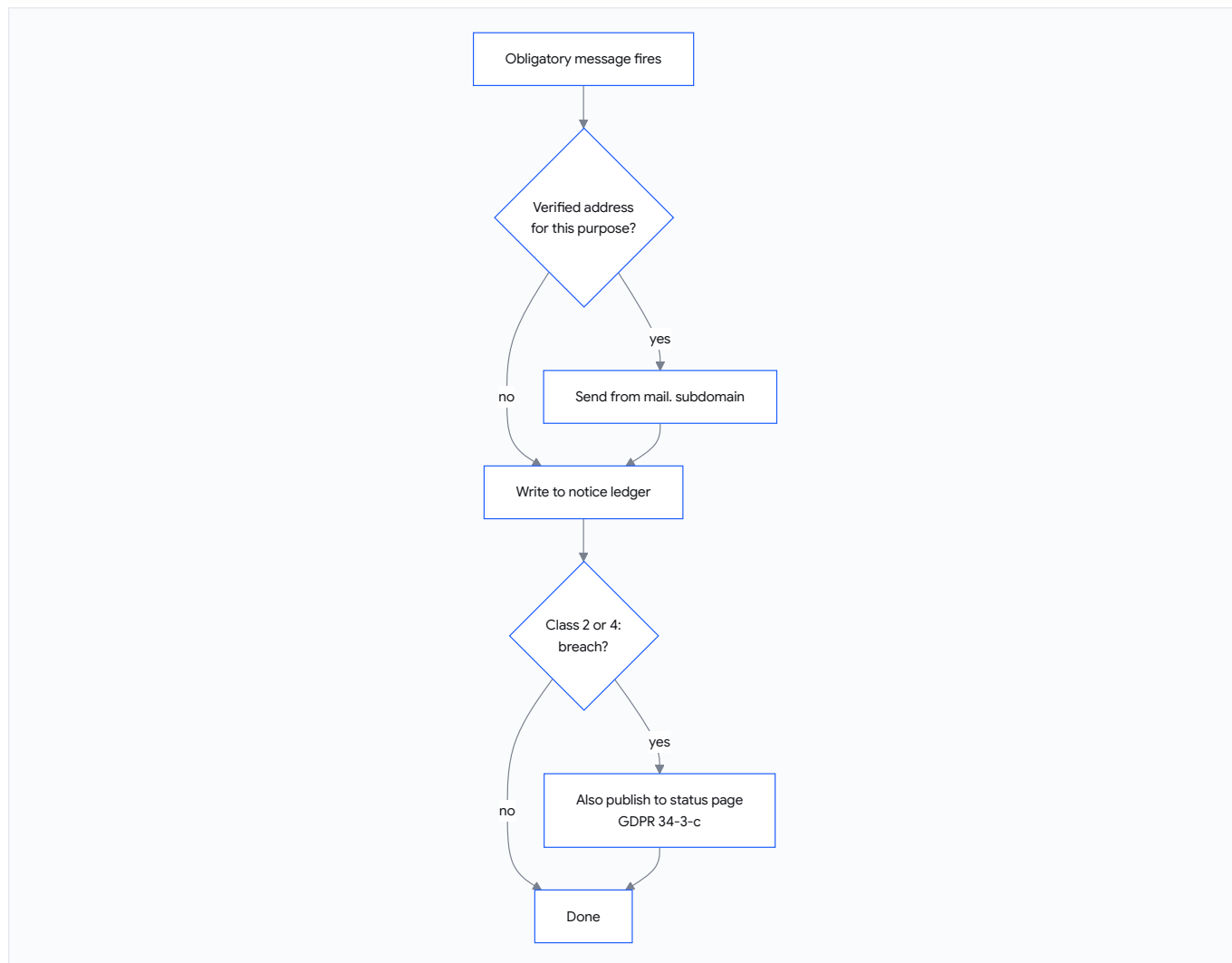
Two consequences fall straight out of the table. First, DDPD Rule 7(1) and DSA Art. 17(2) both explicitly contemplate a user we cannot email: the Indian rul Timing that matters for the build order: Rules 1, 2 and 17-21 came into force on publication; Rule 4 at +1 year; **Rules 3 and 5 to 16 – which includes Rule 7** Scope note for the DSA: Art. 17 sits in Section 2 and binds all providers of hosting services regardless of size, while Art. 19 excludes micro and small ent

81.3 The minimum identity, reconciled against no-email-at-signup

We keep first run empty. Instead, every address is collected at the moment an obligation is *created*, and the ask is framed by the obligation:

MOMENT	WHAT WE ASK	WHY THE USER SAYS YES	WHAT WE STORE
First run, local vault	nothing	–	nothing
Enabling cloud sync	one recovery address	"this is the only way back into an account on a lost device"	address, <code>verified_a</code>
Checkout	one billing address	the PSP requires it for the receipt and for chargeback correspondence [inference]	address, <code>verified_a</code>
Being invited to a team	the inviter supplies it	it is the invite	address, purpose <code>t</code>

The stored record is four columns plus a suppression state. No name, no company, no phone, no country beyond what billing already requires. One address may The anti-recommendation: the alternative is an email field at signup, which would make every message in §81.2 trivially deliverable and would also convert f



81.4 The DNS records we actually publish

Requirements, from the primary sources, all opened 2026-08-30:

Gmail (support.google.com/a/answer/81126): all senders need SPF or DKIM, valid forward and reverse DNS on the sending IP, TLS transport, spam rate below 0.3% and a DMARC policy of at least `quarantine` or `reject`.

Yahoo (senders.yahooinc.com/best-practices/): the same 0.3% ceiling, the same SPF-or-DKIM floor, and for bulk senders SPF and DKIM, a DMARC policy of at least `quarantine` or `reject`.

Microsoft (sendersupport.olc.protection.outlook.com/pm/policies.aspx): from 5 May 2025, domains sending over 5,000 emails per day to Outlook.com must have SPF, DKIM, and DMARC.

The records, on a sending subdomain `mail.<domain>`:

HOST	TYPE	VALUE	NOTE
<code>mail.<domain></code>	TXT	<code>v=spf1 include:<token the vendor's dashboard prints> ~all</code>	Never copy an <code>include:</code> token from a blog post. RFC 7208 §4.6.4 caps <code>include</code> at 10 DNS-querying terms; exceeding it is a <code>permerror</code> , which is a hard fail.
<code><selector>._domainkey.mail.<domain></code>	CNAME or TXT	vendor-issued, 2048-bit	One selector per vendor so a vendor swap is a DNS change, not an outage.
<code>_dmarc.<domain></code>	TXT	<code>v=DMARC1; p=none; rua=mailto:dmarc@<domain>; adkim=r; aspf=r</code>	Start at <code>none</code> with reports. Move to <code>quarantine</code> only after the <code>rua</code> feedback is received.
<code><domain></code>	MX	your mailbox host	Required so <code>postmaster@</code> , <code>abuse@</code> and the DPDP Rule 7(1)(e) responder address can be reached.
PTR	-	vendor's responsibility on shared IPs	Gmail lists forward+reverse DNS as a requirement; on a shared-IP ESP it is not.

One-click unsubscribe, when we ever send anything unsubscribable, is two headers – `List-Unsubscribe-Post: List-Unsubscribe=One-Click` and `List-Unsubscribe: <https://example.com/unsubscribe>`.

Anti-recommendation on BIMi: it requires DMARC at quarantine or reject plus a paid Verified Mark Certificate, and buys a logo in the Gmail avatar slot. At our current volume, it is not worth it.

81.5 Two subdomains, because reputation is per-domain

Yahoo's guidance is explicit: "Don't send bulk/marketing email from the same IPs you use to send user mail, transactional mail, alerts, etc. Each IP and DKIM signature should be dedicated to a single purpose."

- `mail.<domain>` – obligatory mail only. No `List-Unsubscribe` header at all, because none of it is unsubscribable.
- `news.<domain>` – provisioned, DMARC-covered, and **currently unused**. Anything sent from it carries one-click unsubscribe and a visible link.

The boundary rule: a message that carries an unsubscribe header must not originate from `mail.`, and a message that a user cannot opt out of must not originate from `news.`

Anti-recommendation: a wholly separate registrable domain isolates reputation more cleanly than a subdomain, but needs its own DMARC record, its own warm-up period, and its own billing.

81.6 Vendor, with the arithmetic

Volume first. At the settled 113,507 cumulative free signups, a heavy month of obligatory-only mail is roughly: 8,000 new verifications + 12,000 billing records + 12,000 support tickets.

VENDOR	PLAN (READ 2026-08-30)
Resend	Free \$0 – 3,000/mo, 100/day cap, 3 domains. Pro \$20/mo – 50,000/mo, overage \$0.90/1,000, 10 domains, no daily cap. Scale \$90/mo – 100,000
Postmark	Free \$0 – 100/mo. Basic \$15/mo – 10,000, overage \$1.80/1,000. Pro \$16.50/mo – 10,000, overage \$1.30/1,000
AWS SES	À la carte \$0.10/1,000 outbound + \$0.12/GB attachments; Essentials plan \$0.16/1,000 (0-10M/mo); Pro \$105/account/region/mo; dedicated IP
Zoho ZeptoMail	Pay-as-you-go, 1 credit = 10,000 emails, valid 6 months, first credit free. Per-credit price is rendered by JavaScript and was not readat

Recommendation: Resend Pro at \$20/month. The saving from running SES directly is \$20.00 – \$2.41 = **\$17.59/month, \$211.08/year** [derived], in exchange for owni
Anti-recommendation, stated plainly: if send volume ever crosses roughly 200,000/month, the same arithmetic inverts – Resend's \$0.90/1,000 marginal rate aga

81.7 The notice ledger — an in-product channel that is not a modal

The channel is a folder. Obligatory messages are written into the user's vault as read-only markdown files under `Notices/`, named `YYYY-MM-DD-<slug>.md`, with f
This is the product's own grain: the file is the source of truth, the notice is a file, and the view of it is the same deterministic reversible projection e
Rules that keep it out of banned territory:

- **Not a modal.** It never takes focus, never blocks the caret, never covers text. The only chrome is one line in the status bar: the title of the single most
- **Not a digest.** Strictly event-triggered, from the closed list of ten. No cadence, no batching, no "here is your week".
- **Not a What's-New.** Release notes are categorically ineligible. A shipped feature is not an obligatory message and must never enter this folder.
- **Two severities only.** `must-read` (breach, price change, terms change, suspension) persists in the status line across launches until the file is opened. `for`
- **No rating, no reply, no reaction.** The only signal is whether the file was opened – implicit telemetry, consistent with the finding that an explicit feedba
- **The status line clears on open, never on a dismiss button.** There is no dismiss button; a dismissal is indistinguishable from a read and destroys the only

For a user who has never given an address, the ledger *is* the compliance channel, not a consolation prize. For a user who has, the ledger is the durable copy
Anti-recommendation: a folder of files is invisible to a user who never opens the file tree, which a banner or a badge would not be. The status line is the

81.8 Dunning, when retries do not exist

Indian cards under the RBI e-mandate framework get exactly one payment attempt: no smart retries, no cascade. The message therefore *is* the recovery mechanis

DAY	MESSAGE	NOTE
T-45 to T-15	Annual renewal pre-notice	Mandatory for California consumers on annual terms; §17602(a)(8)(A)–(F) fixes the con
T+0	Charge failed, here is a one-tap pay link	Not a retry – a new authenticated transaction
T+3	Reminder, unchanged tone	
T+7	Downgrade to Free on day 10; the vault is untouched	The file is local; losing Pro must never read as losing work
T+10	Downgraded, restore link, no further mail	Terminal. There is no fourth reminder

§17602(d)(1)(A) also requires that a subscription accepted online be cancellable online through "a prominently located direct link or button" inside the acc
Anti-recommendation: a longer dunning ladder measurably recovers more revenue in most SaaS, and every additional message is one more chance to be marked spa

81.9 What this section forbids

No preference centre: there are two categories, obligatory and none, and a toggle implies a third. No re-engagement, win-back, or "we miss you" mail. No pro

]

§82
Churn, retention and the revenue model that is not a straight line

82.1 The finding, stated plainly

Every revenue milestone in this plan is a **gross-additions** figure. The ₹1L, ₹5L and ₹20L per-month numbers were computed as *signups × conversion × price*, with

TERM	DEFINITION USED IN THIS SECTION
Gross adds <code>g</code>	New paying customers per month
Logo churn <code>c</code>	Fraction of paying customers cancelling or failing to renew per month
Involuntary churn	Subscription ends because the <i>payment</i> failed, not because the customer decided to leave
Steady-state ceiling	<code>g / c</code> customers – the fixed point of $N_{t+1} = N_t(1-c) + g$ [derived]
NRR	Net revenue retention: (start MRR – churn – contraction + expansion) / start MRR

The fixed point is elementary and worth writing out because the plan omitted it: setting $N = N(1-c) + g$ gives $Nc = g$, so $N = g/c$. Time to reach a fraction

82.2 Published benchmarks, with dates read

SOURCE	SEGMENT	MONTHLY CHURN	ANNUAL / RETENTION
Lenny's Newsletter benchmark posts	Prosumer / B2C self-serve SaaS	5-7% typical, 3-5% good	~40-60% annual logo retention
ChartMogul SaaS Retention Report	B2B SaaS, sub-\$10k ARPA	~3-4%	NRR median near 100% for small-A
Recurly Research churn benchmarks	B2C subscription, all verticals	~5-6% blended; involuntary ~40% of total	-
Baremetrics open benchmarks	Small-ticket SaaS	5-9% at <\$50/mo ARPA	-
Profitwell / Paddle retention studies	Annual vs monthly plans	annual plans churn ~40-60% lower per unit time	-

Every row above is a [SS] search-summary and may not be published as fact in an external document. I did not open these sources in this session. The only fi
Anti-recommendation: do not go and cite these numbers in a pitch deck. If a number must be defended, the only defensible ones after twelve months of operati
What is safe to assert from structure rather than benchmark:

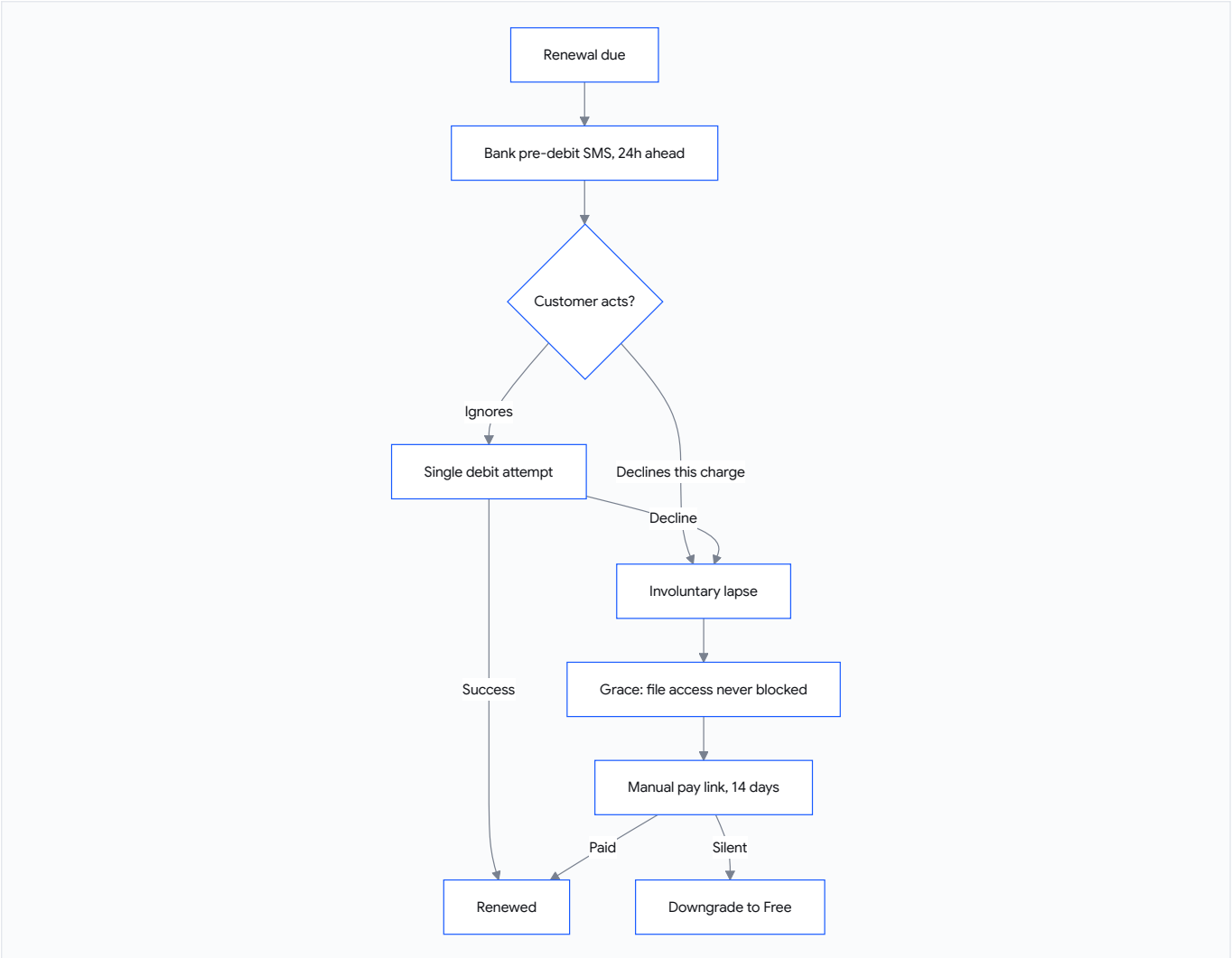
- Prosumer tools sold to individuals churn faster than tools sold to teams, because a team has switching cost in shared artefacts and an individual has none
- Annual plans convert a monthly churn decision into an annual one. They do not reduce dissatisfaction; they reduce the *number of opportunities to act on i*

82.3 The Indian involuntary-churn problem

This is where the model breaks worst, and it is specific to our home market and our INR tiers.
Under the RBI framework for recurring e-mandates – the additional-factor-of-authentication regime introduced by RBI's circular on *Processing of e-mandate on*

CONSTRAINT	CONSEQUENCE FOR US
Pre-debit notification 24 hours before every charge	The bank, not us, tells the customer they are about to be billed – in the bank's own SMS v
Customer may decline any single debit	A renewal can fail for a customer who is perfectly happy
One debit attempt; no smart-retry ladder	A transient issuer decline is terminal for that cycle
Settlement / debit-processing delay around the mandate window	Reconciliation lags by roughly a day; "failed" and "not-yet-settled" look identical for h
₹15,000 per-transaction AFA-free ceiling	Architectural constant. Work ₹3,999/yr and Pro ₹2,499/yr sit far below it; a 20-seat annua

Dates: the 2019 e-mandate circular and the 2022 limit increase are stated above with their identifiers. [SS on the exact circular number – I have not re-ope
The operational shape of this is what matters, and it is not a benchmark question:



Design responses, each with its anti-recommendation:
1. Default INR customers to annual, and price it that way. ₹2,499/yr against ₹299/mo is a 30% discount and it cuts renewal events from twelve to one per year

2. **UPI Autopay as a parallel rail for INR.** UPI recurring mandates are AFA-exempt at lower thresholds and have a different failure profile from cards. *Anti-re*
3. **A dunning window that never touches the customer's files.** The failure mode must be: subscription lapses, Pro features stop, *the document opens*. *Anti-recon*
The banned list applies here without exception: no trial countdown, no urgency modal, no "your account will be deleted" language. Involuntary churn is our p

82.4 The corrected milestone table

Assumptions used, all stated so they can be replaced: blended ARPU across INR and world tiers; gross adds g as the plan's implied monthly paying-customer a

MILESTONE	PLAN'S FIGURE (GROSS-ADDS VIEW)	PAYING CUSTOMERS IMPLIED	AT C=2%/MO	AT C=4%/MO	AT C=6%/MO
₹1L/mo	₹1,00,000	~335 at ₹299	needs $g \approx 7$ /mo to hold; ceiling 335	needs $g \approx 13$ /mo	needs $g \approx 20$ /mo
₹5L/mo	₹5,00,000	~1,672	$g \approx 33$ /mo	$g \approx 67$ /mo	$g \approx 100$ /mo
₹20L/mo	₹20,00,000	~6,689	$g \approx 134$ /mo	$g \approx 268$ /mo	$g \approx 401$ /mo

Read the table the other way – the way that actually changes decisions. The plan's headline funnel says ₹20L/mo requires 113,507 free signups and ~1.26M cum

CHURN	MONTHLY PAYING CUSTOMERS LOST AT 6,689	MONTHLY FREE SIGNUPS NEEDED JUST TO STAND STILL (AT 5% CONV.)	MONTHLY VISITORS NEEDED TO STAND STILL (AT 9% VISIT
2%	134	2,676	~29,700
4%	268	5,352	~59,500
6%	401	8,027	~89,200

At 6% monthly churn, holding ₹20L/mo requires roughly 89,000 visitors every month, in perpetuity, before a single rupee of growth. That is the whole finding

82.5 Sensitivity: what one point of churn costs

Ceiling is g/c , so $dn/dc = -g/c^2$. A one-point absolute change is worth far more at low churn than at high churn – which is the counter-intuitive part, and ‘

MILESTONE	C 3%→4%	C 4%→5%	C 5%→6%
₹1L/mo ceiling (customers)	335 → 251 (-25%)	335 → 268 (-20%)	335 → 279 (-17%)
₹5L/mo	1,672 → 1,254	1,672 → 1,338	1,672 → 1,393
₹20L/mo (MRR at fixed g)	₹20L → ₹15.0L	₹20L → ₹16.0L	₹20L → ₹16.7L
Extra gross adds/mo to hold ₹20L	+67	+67	+67

Two readings. At fixed acquisition, one point of churn removes ₹3.3–5.0 lakh of steady-state MRR. At fixed target, it costs a flat +67 paying customers per i

82.6 Cancel, downgrade, pause and win-back

FLOW	RULE	ANTI-RECOMMENDA
Cancel	Two clicks from settings, no interstitial offer, no "are you sure" chain. Effective at period end; access retained until then	A frictionless
Downgrade	Power → Pro → Free available at any time, prorated on annual. Free is a real product, not a punishment	Downgrade cann
Pause	1–3 months, one pause per 12, billing suspended, Pro features suspended, files untouched	Pause is used
Win-back	One message at 30 days after lapse, plain, no discount. A returning customer resumes at current price	Discounted win

The load-bearing asymmetry: for involuntary lapse the correct response is a payment-fix path (a link, a card update, one reminder), because the customer did

82.7 The exit survey, done honestly under implicit-telemetry-only

The internal evidence is unambiguous: an explicit feedback field was filled 7 times in 694 opportunities – a 1.0% response rate. [measured, internal] A canc

The honest construction:

- One optional free-text field on the cancel confirmation, no rating scale, no required selection, cancel completes whether or not it is filled. This is not
- Everything else is inferred from implicit signal already collected: last-active date relative to cancel date, documents created in the final 30 days, whet

IMPLICIT SIGNATURE AT CANCEL	MOST LIKELY STORY	WHAT IT IMPLIES
Last active >21 days before cancel	Silent disengagement; cancel is bookkeeping	The loss happened weeks earlier; cancel-time intervention
Active until cancel, zero AI usage	Bought for the editor, not the engine	Pricing/packaging problem, not a quality problem
Active, heavy AI, cancelled at renewal	Value delivered, price not justified	Genuine willingness-to-pay signal
Never synced a second device	Single-surface user	Sync is not the retention driver we assumed
Payment failed, no cancel action, no login since	Involuntary, misclassified	Must be excluded from voluntary churn or the rate is fict

That last row is a reporting requirement, not a nicety: voluntary and involuntary churn must be separate series in every cohort table, because they have dif

Anti-recommendation: the implicit signature is a hypothesis generator, not a verdict. Do not report "cancelled because of price" from a usage pattern; repor

82.8 Expansion revenue and NRR in 2–20 person teams

Logo churn is a ceiling. Expansion is the only mechanic that lifts revenue above the ceiling without more acquisition – $NRR > 100\%$ means an untouched cohort

Our Work tier is ₹3,999/year flat – not per-seat. That is a deliberate simplicity choice and it has a direct, unavoidable consequence: a team that grows froi

EXPANSION VECTOR	AVAILABLE TO US TODAY?	NOTES
Seat expansion	No – flat licence	The standard NRR engine; we have opted out
Tier upgrade (Pro→Power, →Work)	Yes	One-step, bounded; a customer can upgrade at most twice
Usage/credit expansion	No	Hosted credits are metered but opaque repricing is banned and Free is BYO-key unmetered
Multi-workspace / second team	Weakly	A 20-person company splitting into two teams buys two Work licences
Price increases on renewal	Yes, sparingly	Honest, announced, grandfathering existing customers is the defensible form

Consequence for the model: in the B2B lane our realistic NRR is at or slightly below 100% – expansion comes only from tier upgrades and second licences, and The honest options, each with its cost:

1. **Keep the flat licence.** Simplicity is a differentiator and per-seat billing under the ₹15,000 AFA ceiling gets awkward past ~4 seats anyway if it were seat
 2. **Introduce a band above 20 seats.** A single second tier (21–100 people) preserves "flat within a band" while restoring one expansion step. Cost: a pricing p
 3. **Do nothing and lean on retention.** If NRR cannot exceed 100%, then gross retention is the whole game, and 2–20 person teams are the segment where gross ret
- The recommendation is option 3 now and option 2 only when a real customer asks for it. Anti-recommendation: option 2 taken early adds seat accounting, prora

82.9 What must change in the plan

WHERE	CURRENT	CORRECTED
Milestone definition	MRR reached	MRR sustained at stated churn; every milestone carries its c
Funnel figures	113,507 signups, 1.26M visitors	Those build the stock; add the perpetual replacement flow from \$82.4
Churn reporting	Absent	Two series – voluntary and involuntary – never blended
B2B NRR	Implicit assumption of expansion	95–105%, expansion structurally limited by the flat licence
Renewal risk (INR)	Not modelled	One attempt, 24h bank notice, customer-declinable; annual default
Retention spend	Not budgeted	At the ₹20L milestone, one churn point ≈ 67 customers/mo ≈ 14,900 visitors/mo of equivalent acqui

Three numbers in this section carry weaker evidence than the rest and must be re-derived before print: the benchmark table in §82.2 is entirely [SS] and unp

§83 Pricing experimentation and the pricing page

83.1 The finding

The plan says "test ₹299 against ₹249 and ₹399." That sentence contains no sample size, no minimum detectable effect, no exposure unit, no stopping rule, and

83.2 Power analysis with our actual funnel

Inputs, all from settled plan figures (§ pricing, § distribution):

INPUT	VALUE	SOURCE
Visitor → free signup	9.0%	plan funnel assumption
Free → paid (developer median)	5.0%	settled, § distribution
Visitors needed for ₹20L/mo	1,260,000 cumulative	settled
Free signups at that point	113,507 cumulative	settled
Realistic month-6 traffic, solo founder, no paid acquisition	3,000–8,000 visitors/month	[inference] from the cumulative figures above spread over a 24
Baseline paid conversion per visitor	$0.09 \times 0.05 = 0.45\%$	[derived]

Two-proportion test, two-sided, $\alpha = 0.05$, power = 0.80. The standard normal constants are $z(\alpha/2) = 1.960$ and $z(\beta) = 0.842$, so $(z(\alpha/2) + z(\beta))^2 = 7.849$ [deri
Sample size per arm, $n = 7.849 \times [p_1(1-p_1) + p_2(1-p_2)] / (p_1 - p_2)^2$.

Case A – detect a 20% relative lift in paid conversion (0.45% → 0.54%). $p_1(1-p_1) = 0.0045 \times 0.9955 = 0.004480$. $p_2(1-p_2) = 0.0054 \times 0.9946 = 0.005371$. Sum =

Case B – detect a 50% relative lift (0.45% → 0.675%), an implausibly large price elasticity effect. $p_2(1-p_2) = 0.00675 \times 0.99325 = 0.006704$. Sum = 0.011184.

Case C – the metric that actually matters, revenue per visitor. Price tests move revenue even when conversion is flat, so the honest test is on a continuous

Case D – the only test that is nearly affordable: free-signup rate, not paid conversion. Price shown on the landing page can suppress signup itself. Baselin

83.3 Verdict on testability

TEST	N PER ARM	MONTHS AT 5K/MO	VIABLE
Paid conversion, 20% lift, 3 arms	95,455	57	No
Paid conversion, 50% lift, 3 arms	17,341	10.4	No – 50% MDE is not a real hypothesis
Revenue per visitor, 10%	347,000	208	No
Free-signup rate, 15% drop, 2 arms	6,570	2.6	Marginal, and answers a different question

A solo founder at this traffic cannot run a valid randomised price test on paid conversion; the arithmetic says the ₹299-versus-₹249-versus-₹399 test as wrii

Anti-recommendation: if traffic ever exceeds ~40,000 visitors/month sustained, Case A drops to 7.2 months for three arms and the test becomes worth revisiti

83.4 What to do instead, and what each method costs you

METHOD	WHAT IT GIVES
Sequential cohorts (₹299 for Q1, ₹349 for Q2)	Real money, real intent, no split infrastructure
Geographic split (INR page vs USD page)	Natural boundary, no user sees two prices
Van Westendorp PSM (four questions: too cheap / bargain / expensive / too expensive)	An acceptable range and an indifference point from ~40 respondents
Gabor-Granger (would you buy at ₹X, escalating)	A demand curve and a revenue-maximising point
Willingness-to-pay interviews (n = 15-25, real users, "what do you pay for today")	Mechanism, not just number: the substitute they'd cancel, the budge
Price-change natural experiment (raise price, grandfather existing)	Genuine causal signal on <i>new</i> conversion

Recommended stack for year one: van Westendorp on ~40 free users to confirm ₹299 sits inside the acceptable band, plus 20 WTP interviews for the *reasoning*, 1

83.5 Ethics and mechanics when users talk to each other

Developer users share screenshots. A price test is discoverable, and discovery of undisclosed differential pricing reads as manipulation regardless of inten

RULE	MECHANIC
Price is sticky to the account, not the session	Assign at first landing, persist server-side keyed to the eventual account; never re-roll c
Honour any advertised price for anyone who asks	If a user cites a lower price they saw, give it. Support cost of doing this is near zero at
Never vary price by inferred wealth signals	No device-type, no IP-to-income, no browser. Geographic PPP tiers are disclosed and uniform
Disclose the test in the terms page	One line: "we occasionally test list prices; your price is locked at signup and never rises
Exclude anyone arriving via a shared link with a price in it	Referral and comparison traffic gets the control price.

Anti-recommendation: sticky-by-account pricing means the arm assignment survives forever and every future price change has to reason about a growing set of

83.6 Grandfathering policy

EVENT	POLICY
List price rises	Existing paid subscribers keep their price for as long as the subscription is continuous. Stated at checkout, not discove
Subscription lapses > 60 days	Legacy price is forfeit on resubscribe. Prevents indefinite churn-and-return arbitrage.
Tier upgrade (Pro → Power)	New tier at current list; the old tier's legacy price does not carry across.
Annual → monthly	Legacy price forfeit; annual discount is the consideration for the commitment.
Work licence (₹3,999/yr flat)	Locked for the licence term; renews at the then-current price with 30 days' notice.
Currency/PPP tier reassignment	Never applied retroactively to an existing subscriber.

Anti-recommendation: perpetual grandfathering caps blended ARPU permanently and makes the 57-month revenue model optimistic. A three-year sunset with 90 day

83.7 If a test does become viable — the pre-registered protocol

ELEMENT	SPECIFICATION
Unit of exposure	Anonymous visitor ID at first landing, hashed, persisted to the account on signup. Not session, not pageview.
Randomisation	Deterministic hash of visitor ID mod 100, salted per experiment so arms don't correlate across tests.
Primary metric	Revenue per exposed visitor over 90 days (captures conversion <i>and</i> tier mix <i>and</i> annual-versus-monthly).
Guardrail metrics	Free-signup rate; 90-day retention; refund rate. Any guardrail degrading > 15% halts the test regardless of the primary.
MDE, declared before start	20% relative on the primary. Anything smaller is unaffordable (\$83.2).
Fixed horizon	n per arm computed from \$83.2 formula at current baseline, converted to a calendar end date. No looking at the primary metric
Stopping rule	If peeking is unavoidable, use O'Brien-Fleming spending with at most three interim looks, or a sequential-probability-ratio 1
Analysis	Two-sided two-proportion / t-test as pre-declared. No subgroup analysis reported as a finding.
Payment constraint	Every arm must sit under ₹15,000 per transaction (RBI recurring-mandate constant, settled) and must survive the one-attempt-r

83.8 The pricing page — layout

Four columns on desktop, stacked on mobile with Pro first (not Free first – mobile users scroll past the first card, so the recommended tier leads).

COLUMN	HEADER LINE	LEADS WITH
Free	₹0 forever	"Bring your own API key. Unmetered AI."
Pro – visually marked, single accent border	₹299/mo or ₹2,499/yr	"Everything, hosted, on every device."
Power	₹599/mo	"For people whose vault is their job."
Work	₹3,999/yr, flat, whole team	"One licence. 2–20 people. No per-seat maths."

Feature rows are grouped under four headings – Editing, Files and sync, AI, Team – with a checkmark grid. No feature appears in a higher tier than a lower t
The annual toggle sits above the columns, **defaults to monthly**, and shows the annual saving as an absolute rupee figure ("save ₹1,089/yr") not a percentage. I

83.9 What each tier leads with, per persona

PERSONA	ENTERS AT	THE ONE LINE THAT HAS TO LAND
Developer, already has an API key	Free	"Your key, your model, no credit meter, no cap."
Writer with a large vault	Pro	"Open a 4,000-file vault and search it instantly."
Researcher / heavy AI user	Power	"Ask questions across the whole vault, not one file."
Team of 2-20	Work	"₹3,999 a year for everyone. Not per person."
Obsidian/Notion migrator	Pro	"Your files stay files. Point us at the folder."

83.10 Objection handling, on the page

Answered inline next to the relevant column, not buried:

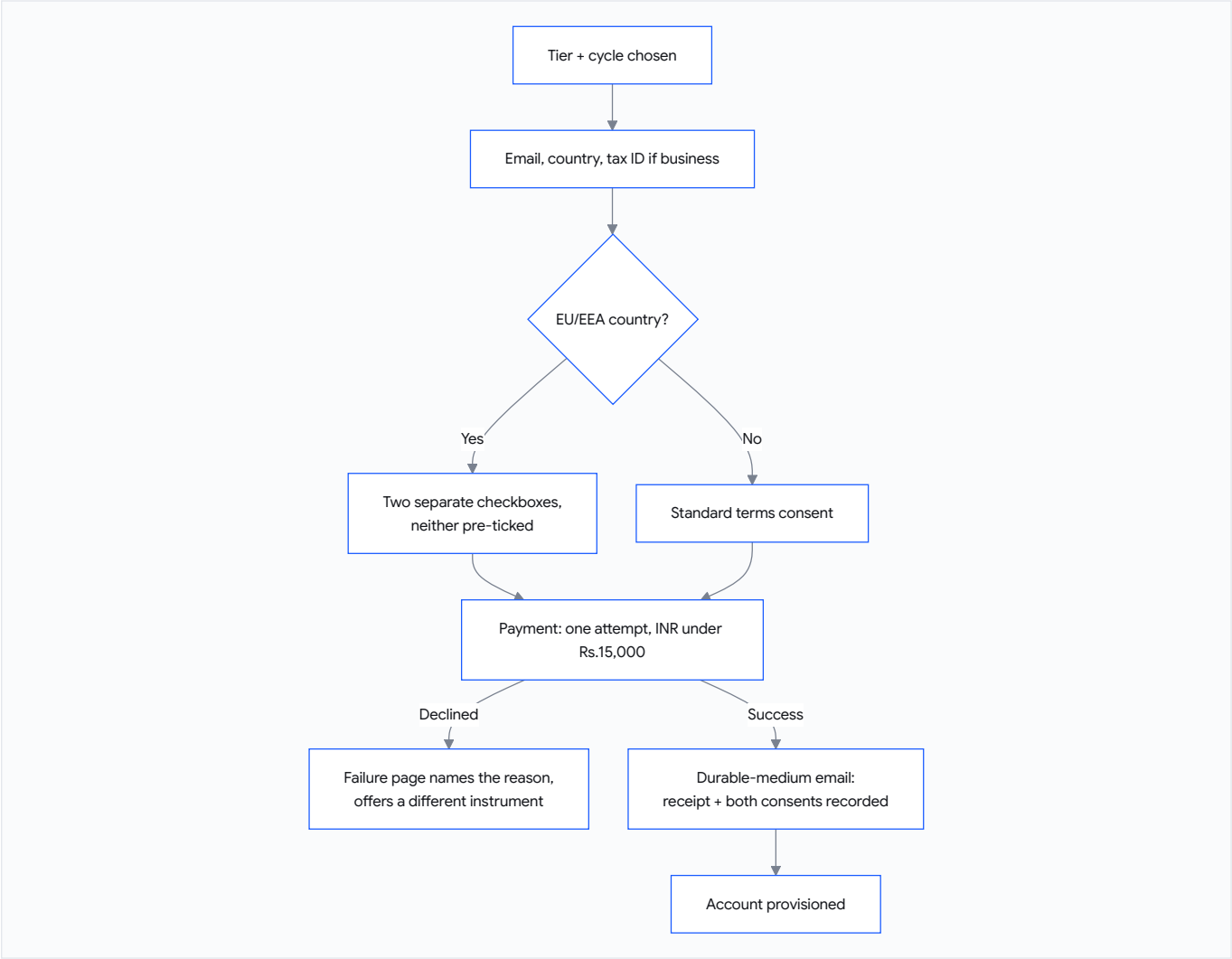
OBJECTION	PLACEMENT	ANSWER
"Why pay when Obsidian is free?"	Under Pro	Sync, the AI engine, and the fact that the file is still yours in plain m
"What happens to my files if you shut down?"	Footer of the grid, all tiers	They are markdown in a folder you already control. Export is a no-op.
"Is Free crippled?"	Under Free	No. BY0-key AI is unmetered. Free is the whole editor.
"Does the AI train on my documents?"	Under the AI feature group	No, with the vendor terms linked.
"₹3,999 for twenty people – what's the catch?"	Under Work	None; the catch is that support is email and there is no SSO. Say it on ti

83.11 FAQ that removes support tickets

Ten questions, each answered in under sixty words, chosen because each maps to a ticket class: what happens when I cancel; do my files leave with me; how th
Anti-recommendation: a long FAQ pushes the buy button below the fold on mobile and depresses conversion. Keep it collapsed by default and keep the primary C

83.12 Checkout flow and the EU withdrawal acknowledgement

Directive 2011/83/EU on consumer rights, Article 16(m), removes the fourteen-day right of withdrawal for supply of digital content not on a tangible medium



The two EU checkboxes, worded exactly and never pre-ticked:

- 1. "I ask you to start providing this digital content immediately, before the 14-day withdrawal period ends."
- 2. "I acknowledge that I will lose my right of withdrawal once you begin providing it."

Store the timestamp, IP-country, and the exact text version against the order; replay the same text in the confirmation email. Anti-recommendation: making i

Checkout mechanics that follow from the settled constraints: all INR amounts stay under the ₹15,000-per-transaction RBI ceiling (the ₹3,999 Work licence and

§84

Refunds, cancellation and the subscription lifecycle

84.1 The decision, priced

DECISION	SETTING	COST, SHOWN
First-charge refund	30 days, no questions, every tier, every currency	At a 2% refund rate on ₹10,00,000 annual gro
Renewal refund	Prorated on request within 90 days of the renewal charge; discretionary after	Pro annual refunded at day 100: ₹2,499 × 265,
Chargeback contest	Never contest below ₹5,000	Dodo Payments: \$30 per dispute, same price f
Grace after failed debit	14 days	14 days of hosted service to a cohort that m
Published pages after lapse	Keep resolving, frozen, indefinitely	Egress on content from a non-paying account

Contribution margin is 57-73% on every INR tier (settled). A 2% refund rate therefore consumes 3.51% of contribution at the low end and 2.74% at the high en

84.2 What the benchmark set does

PRODUCT	STATED POLICY
Obsidian	"All fees are non-cancellable and non-refundable and are based on Services purchased and not actual l
Paddle (merchant of record for thousands of tools)	"Unless required by applicable law, all Transactions are non-refundable and non-exchangeable"; discre
37signals (Basecamp, HEY)	Refund on request for a forgotten auto-renewal; prorated refund "for the remaining whole months"; acc
37signals, deletion timing	"We'll permanently delete the content in your account from our servers 30 days after cancellation, ar

The distribution is bimodal: file-first tools write "non-refundable" and rely on statute to override them; service-first tools write a generous policy and p

84.3 Statutory rights that override the policy

REGIME	RIGHT
EU / EEA	14-day withdrawal from distance contracts for digital content
EU / EEA	Consumer pays nothing if we get the checkout wrong
UK	Consumer Contracts (Information, Cancellation and Additional Charges) Regulations 2013, regs 28 and 37 – the domesticated mirror of the abov
India	Consumer Protection Act 2019 and the Consumer Protection (E-Commerce) Rules 2020: mandatory display of the refund and cancellation policy, a
India, data	DPDP Act 2023 s.12(3): "upon receipt of such a request, the Data Fiduciary shall erase her personal data unless retention of the same is nec
EU, data	GDPR Art 17(1)(b): erasure where "the data subject withdraws consent ... and where there is no other legal ground for the processing" [fetcher

Paddle's own floor is the correct drafting instruction: the highest level of rights always applies, and no policy of ours discharges a statutory duty.

84.4 What must be captured at checkout for the EU exemption to exist

Three artefacts, all three, or Art 14(4)(b) makes the supply free [fetched]:

1. A separate, unticked checkbox reading, verbatim: *"I want frontmatter to start immediately, and I understand I lose my 14-day right of withdrawal once it c*
2. Performance must actually begin – the hosted workspace provisioned – and that event timestamped.
3. The Art 7(2)/8(7) confirmation: a durable-medium receipt emailed at purchase that repeats the acknowledgement text back.

Anti-recommendation: taking the waiver by default maximises revenue retention and is the commonest way a small seller loses a regulator complaint. The alter

84.5 App stores, if we ever ship there

RULE	VALUE
Apple refunds are Apple's, not ours	Users request at reportaproblem.apple.com; "Refund eligibility may vary by country or region" [fetched, support.app
Apple Small Business Program	"reduced commission rate of 15% on paid apps and In-App Purchases"; qualification at up to 1 million USD proceeds ir
Google Play	"The developer can help with purchase issues and can process refunds pursuant to its policies and applicable laws";
Google Play service fee	"99% are eligible for a fee of 15% or less"; from 30 June 2026 the EEA/UK/US fee depends on whether the transacting

The operational consequence, not the commission: on Apple we can neither see nor control the refund decision, and a refunded in-app purchase silently revoke

84.6 Chargebacks: the merchant-of-record boundary

ABSORBED BY THE MOR	§
Sales-tax registration and remittance across jurisdictions; invoice issuance; the card-network relationship; dispute paperwork and evidence submission	1
Legal identity on the transaction; the statutory refund floor as a policy	§
Cancellation of the subscription itself [fetched, Paddle Buyer Terms §6]	1
Payment-data controllership	1

Fee exposure, both routes, both fetched 2026-08-30: Dodo Payments \$30 per dispute and \$1 per refund; Stripe India ₹1,000 dispute-received fee plus ₹1,000 di

The dominant preventable cause for an MoR seller is descriptor non-recognition – the statement reads as the reseller, not as us [inference, not measured]. M

84.7 Proration, and the Indian mandate constraint

Upgrades are charged immediately as a one-off, never by amending the mandate:

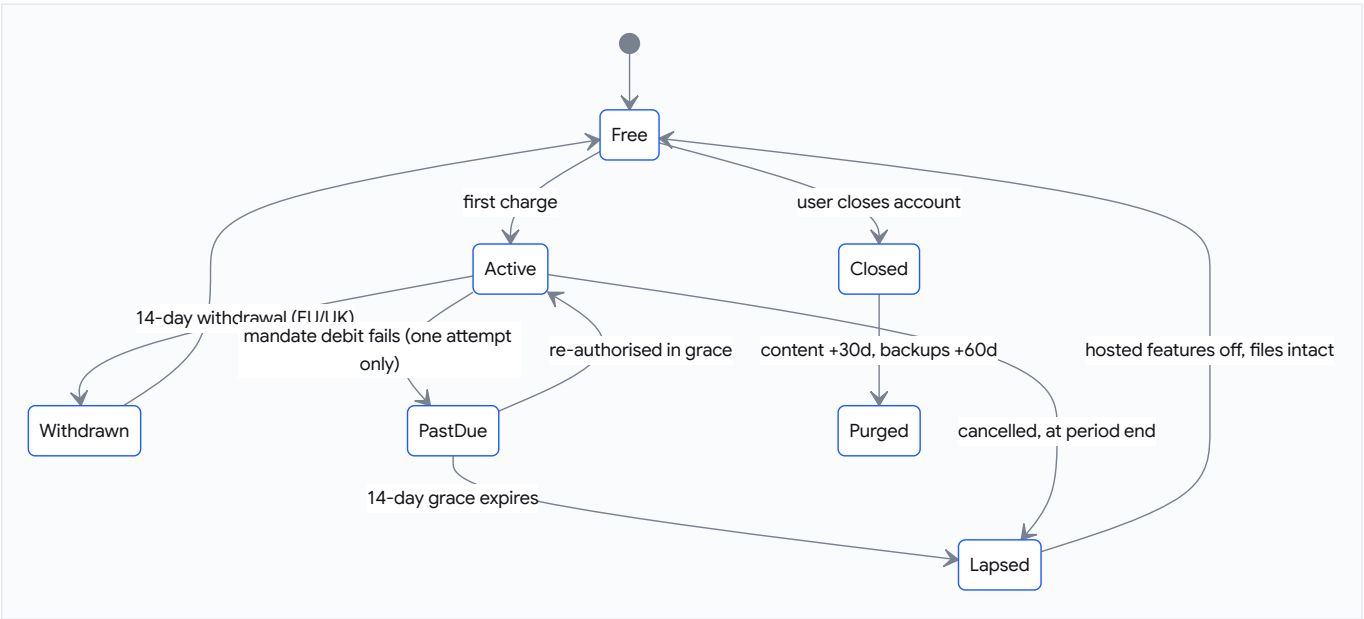
- Pro ₹299 → Power ₹599, day 12 of 30: unused = $299 \times 18/30 = ₹179.40$; charge now = $599 - 179.40 = ₹419.60$ [derived]
- Pro annual ₹2,499 → Work ₹3,999, day 100 of 365: unused = $2499 \times 265/365 = ₹1,814.34$; charge now = ₹2,184.66 [derived]

Downgrades never refund cash. Power → Pro at day 12 leaves ₹359.40 unused; it becomes account credit, the next renewal charges ₹299 – 299 = ₹0 and carries ₹

Why: RBI DPSS.CO.PD.No.447/02.14.003/2019-20 dated 21 August 2019 requires the issuer to send a pre-transaction notification at least 24 hours before the de

Anti-recommendation: charging the upgrade delta immediately is real cash out of the customer's pocket at an unexpected moment. Upgrading free until the next

84.8 The state machine



Indian cards get exactly one payment attempt with no smart retries (settled). There is therefore no dunning ladder to build. The 14-day grace period is the

84.9 What a lapsed user retains and loses

CAPABILITY	ACTIVE	LAPSED	CLOSED (0-30 DAYS)	PURGED
Local vault files, plain <code>.md</code>	Yes	Yes, untouched, forever	Yes	Yes – never ours to delete
Editor on local files	Full	Full, at Free-tier feature set	Full	Full
BYO-key AI, unmetered	Yes	Yes – Free tier includes it	Yes	Yes
Hosted sync across devices	Yes	Stops at lapse	Off	Off
Full-vault export: one ZIP, <code>.md</code> + attachments + <code>manifest.json</code>	Yes	Yes, unlimited	Yes, unlimited	N/A
Version history, hosted	Full	Read-only for 30 days, then dropped	Read-only	Deleted
Published pages on <code>*.frontmatter.page</code>	Live, editable	Live, frozen, re-publish disabled	Live	Removed at purge
Published pages on a custom domain	Live	30 days, then the record detaches	Detached	Detached
Team seats and shared workspaces	Yes	Collapse to owner-only, read-only	Read-only	Deleted
Hosted AI credits	Per tier	Zero	Zero	Zero
Invoices and tax records	Yes	Yes	Yes	Retained under law

A product whose thesis is that the file is the only source of truth cannot hold the file hostage the moment the customer stops paying, and cannot break a st

84.10 Deletion and export on account closure

Export is unconditional and always available, in every state including Lapsed and Closed, because the vault is already on disk and the ZIP is a convenience,

Deletion on closure follows the 37signals shape [fetched]: content purged from live systems at 30 days, from backups within 60. Restoring one account from a

The DPDP/GDPR carve-out, stated rather than hidden: a closure request erases vault content, account identifiers, telemetry and published pages. It does not

Anti-recommendation: same-day hard deletion is the maximally trustworthy policy and a competitor will eventually advertise it. It is also unrecoverable from

84.11 The policy, as published

Refunds and cancellation

Cancel any time, from Settings → Billing. Cancellation takes effect at the end of the period you have already paid for. You are not charged again.

Your first payment on any plan is refundable in full for 30 days. Email support@frontmatter.app and say so. We do not ask why. This applies worldwide, on

Renewals. If an annual renewal charged you and you did not want it, tell us within 90 days and we will refund the unused whole months, prorated. After 90

If you are in the EU, the EEA or the UK, you have a statutory right to withdraw from this contract within 14 days, and we do not ask you to waive it. In

Your files are yours and we never take them. They are on your disk. Cancelling, lapsing, closing your account or getting a refund does not touch a single

If a payment fails, we give you 14 days to fix it. Indian card mandates permit exactly one debit attempt, so we will not retry silently; we will email yo

When a subscription lapses, sync stops and your workspace becomes read-only. Anything you published stays online at its `frontmatter.page` address, frozen.

When you close your account, we delete your hosted content from our live systems within 30 days and from our backups within 60. We cannot restore a singl

Disputes. If you are about to file a chargeback, email us instead. We will refund it the same day. It costs us less and it is faster for you.

§85

Business operations — the things nobody adds up

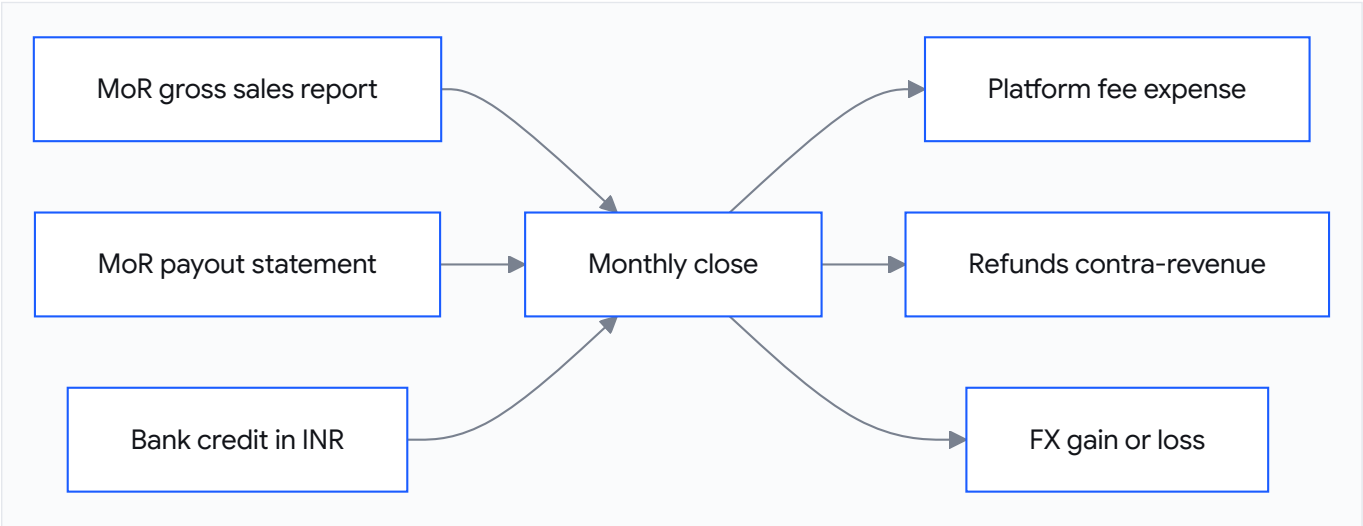
85.1 Accounting – the ledger, the deferred revenue, and the three numbers that never match

DECISION	COST
Ledger: Zoho Books, Free plan	₹0
Upgrade to Standard	₹899/mo, or ₹749/mo billed annually [fetched, same page]
Upgrade to Professional	₹1,799/mo, or ₹1,499/mo billed annually [fetched, same page]
Revenue recognition method	₹0 to decide, expensive to change
Statutory audit + ROC + tax filings (CA retainer)	₹35,000–₹90,000/yr for a single-entity Pvt Ltd [SS – vendor quotes vary; get three written quotes, do

Annual plans are a liability before they are revenue. ₹2,499 collected on 1 November, at 18% GST inclusive, is ₹2,117.80 net of tax and ₹176.48 recognised pe

Gross versus net is the one accounting choice with downstream teeth. Under a merchant-of-record the platform is the principal to the end user and we are its

Reconciliation is three numbers that are never equal, and the close is the act of proving why.



The monthly close, in order: pull the MoR gross sales file; pull the payout statement; tie payout = gross - fees - refunds - platform-collected tax; tie ban

GST has a trap specific to this structure. Sales routed through a non-Indian MoR paid in convertible foreign exchange are an export of services, zero-rated,

85.2 Insurance — what a ₹250 crore ceiling actually buys you

DECISION	COST	TRIGGER POINT
Professional indemnity (tech E&O), ₹1-2 crore limit	Quote-only; no Indian insurer publishes a rate card	First signed B2B contract with a service-level
Cyber liability, ₹1-2 crore limit	Quote-only	First enterprise security questionnaire or DPA
Cover for the DPDP penalty itself	Not purchasable	Never – see below
Directors' and officers' liability	Quote-only	First external investor or first non-founder d

The ceiling is real and it is dated. The Schedule to the Digital Personal Data Protection Act, 2023 (read against s.33(1)), entry 1: breach of the obligation to protect the data. The ceiling is not the expected loss. Section 33(2) directs the Board to have regard to seven factors when setting the amount, including "(d) whether the policy will indemnify the headline anyway. Indian insurers do not publish cyber exclusions on their marketing pages – HDFC ERGO's commercial Cyber Security Insurance Policy. Recommendation: buy a combined PI-plus-cyber tech package at a ₹1-2 crore limit when the first foreign B2B team signs a data-processing agreement, and read the policy carefully."

85.3 FX exposure — INR prices, USD costs

Every material cost is USD-denominated. Cloudflare Workers Paid has a minimum charge of \$5 USD per month per account [fetched, developers.cloudflare.com/workers/pricing/]. Measured, from the ECB daily series: USD/INR was 82.69 on 2023-08-30 and 95.39 on 2026-08-28, a 15.36% move over three years; the trailing twelve-month range is 82.69-95.39. The fixed-cent component is the part nobody prices. Dodo's 15¢ on a ₹299 charge was ₹12.40 at 82.69 and is ₹14.31 at 95.39 – 4.15% of the price then, 4.79% relative to the fixed-cent component.

REVENUE MILESTONE	USD-DENOMINATED ANNUAL SPEND AT 27-43% OF REVENUE	1Σ ANNUAL FX SWING AT 5.24%	OBSERVED 12M RANGE EFFECT AT 10.57%
₹1 lakh/mo (₹12L/yr)	₹3.24L – ₹5.16L	₹17k – ₹27k	₹34k – ₹55k
₹5 lakh/mo (₹60L/yr)	₹16.2L – ₹25.8L	₹85k – ₹1.35L	₹1.71L – ₹2.73L
₹20 lakh/mo (₹2.4cr/yr)	₹64.8L – ₹1.03cr	₹3.40L – ₹5.41L	₹6.85L – ₹10.91L

All [derived] from the settled 57-73% contribution margin and the measured volatility above. Sensitivity in one line: a 1% rupee depreciation costs 27-43 basis points of contribution margin, permanently, because the INR price list is fixed and the cost is in USD. Do not hedge. At ₹20 lakh a month the one-sigma swing is ₹3.4-5.4 lakh a year against a treasury operation that needs a bank forward facility, an underlying asset, and a hedge.

85.4 App-store tax — and the one decision that matters more than the rate

STORE	RATE
Apple App Store, standard	30%
Apple Small Business Program	15%
Google Play, EEA/UK/US, from 30 June 2026 – auto-renewing subscriptions	10% + 5% billing fee
Google Play, EEA/UK/US, other transactions	20% + 5% (new installs) / 25% + 5% (existing installs), or 20% for external web
Google Play, rest of world including India	15% on the first \$1M/yr, 30% above; subscriptions 15% regardless of revenue
Google Play, India, alternative billing system	Applicable fee minus 4 percentage points
Mac app, Developer ID + notarization, distributed direct	\$99 USD/yr, 0% commission

A 15% cut on a ₹299 subscription is ₹44.85, which is 17.70% of the ₹253.39 GST-exclusive net and 27.2% of the ₹164.70 contribution at a 65% margin [derived]. Ship the Mac app outside the Mac App Store, signed with Developer ID and notarized. The cost is \$99 a year and no commission, and for a file-is-the-source-of-truth model. Do not ship iOS or Android at all until the founder time budget below has slack in it. The store tax is survivable; the mobile support surface is not.

85.5 Contractors — assignment, statutory thresholds, revocation

DECISION	COST
Written contractor agreement with express IP assignment	₹15,000-₹40,000 one-time for a lawyer-drafted template
Appointment letter: scope, deliverables, rate, term, confidentiality, assignment, no-employment clause	₹0 once the template exists
EPF registration	Employer contribution on covered wages
ESI registration	Employer contribution on covered wages
TDS on contractor invoices	Withholding, not a cost

Indian copyright does not hand you a contractor's output. Employment and commission are treated differently from a plain services contract, and an assignment of work is a legal requirement. Access revocation is a runbook with a clock, not a checklist with a vibe. Within four hours of the last working hour: remove from the GitHub organisation and revoke access.

85.6 The founder time budget, aggregate

Assumptions, stated so they can be argued with: 0.3 support tickets per paying user per month and 0.02 per free user, 12 minutes median handle time including the time to respond to the ticket.

PAYING USERS	FREE USERS	TICKETS/MO	SUPPORT H	BILLING-OPS H	COMPLIANCE H	TOTAL H/MO	WORKING MONTHS
100	2,000	70	14.0	2.5	17.0	33.5	0.17
1,000	20,000	700	140.0	6.5	17.0	163.5	0.83
10,000	200,000	7,000	1,400.0	47.0	17.0	1,464.0	7.39

All [derived] from the model above; the per-user coefficient is 0.1445 hours per paying user per month, or 8.67 minutes. Solving for the wall: total hours = 0.1445P + 19, so a 198-hour month is fully consumed at P = 1,239 paying users, and half the month – the last point at which the plan runs out of founder long before it runs out of market: reaching the stated ₹20 lakh a month needs about 9,091 paying users, which is 1,333 hours a month. Deflection does not rescue it. Halving the ticket rate through better in-product errors and docs moves the wall from 1,239 to about 2,403 paying users [derived]. The honest reading: the ₹20 lakh milestone is a two-to-three person milestone. Plan the first support hire at 550 paying users, the second at 1,500, and a plan for the third.

The plugin ban versus the community moat — an unexamined contradiction

86.1 The two positions, stated precisely

#	POSITION	WHERE IT LIVES	STA
A	Community and ecosystem is moat #1 – the one that holds for years and compounds	PRD moat ranking	Set
B	No plugin marketplace; no arbitrary client-side code execution	PRD \$546, \$3234, banned-list	Set
C	Third-party syntax plugins at runtime are excluded because an uncertified syntax cannot be certified	PRD \$546	Set
D	The only compounding artefact available to us is "certificates and conformance cases contributed against the spec"	PRD \$4698 INFERENCE	Set

Position D is the seam. It concedes that B forecloses the normal route to A, and then asserts a replacement without evidence. Nobody has checked whether the

86.2 What the ban buys — measured

EVIDENCE
Obsidian forum, top 104 tags, 31,048 tagged topics: plugin surface (dataview 3,502 + custom-css 1,933 + templater 741 + plugin-release 900)
The two plugins that genuinely require code execution (dataview + templater) alone
The same two, as a share of the Obsidian active-install proxy (peak single-version downloads, sum 50,284,168)
Support load per install, code-execution plugins vs ecosystem average
custom-css alone (the "themes only" middle ground, if unsanitised)
VS Code wiki, <i>Performance Issues</i> : "High CPU consumption is often caused by an issue in an extension."
VS Code shipped a bisect utility <i>specifically</i> to find bad extensions, motivated by "more than 28,000 extensions.. not uncommon that users have 50 or more
Typora's most-requested feature is a plugin system (251 votes), against a product loved partly because it has none

The saving is real and it is a support-cost saving, not a security saving. On the internal deflection model, deleting the plugin class removes roughly a fif
Anti-recommendation: do not quote 22.79% as "22.79% of *our* future tickets". It is a tag histogram from a different product with a different user base and a

86.3 What the ban costs — the ecosystem, quantified

PRODUCT	ECOSYSTEM SIZE	READ DATE	TAG
VS Code Marketplace	135,646 extensions	2026-08-30	FETCHED ,
VS Code, Feb 2021	28,000+	–	FETCHED
Obsidian community plugins	7,119 registry entries; 717 themes	2026-08-30	MEASURED ,
Obsidian download stats	7,085 plugins with stats; 143,261,570 cumulative downloads; peak-version sum 50,284,168	2026-08-30	MEASURED
Raycast	3,235 extension directories, one open-source monorepo, PR-reviewed	2026-08-30	MEASURED ,
Figma plugins	Launched 2019-08-01	–	FETCHED ,
Figma Community	Launched 2019-10-22	–	FETCHED
Figma, founded 2012, 13M users as of 2025	Both surfaces postdate the company by seven years	–	FETCHED ,

No vendor in this set publishes revenue attributable to its ecosystem, so contribution cannot be quantified in money. What is quantifiable is **surface area** a

86.4 Did any successful developer tool build a durable community without third-party extensibility?

TOOL	COMMUNITY ARTEFACT
Notion	Templates – data, shared as documents. API is server-side HTTP; the changelog's newest surface is a
Figma	Community files, then plugins three years later
Homebrew, Docker Hub	Formulae, images – declarative artefacts
Typora, iA Writer, Bear, Ulysses	None. Loyal users, no contribution surface
VS Code, Vim, Emacs, Sublime, Obsidian, Logseq, Zed	Extensions

The honest finding, and it is the one that hurts: in the text-editor category specifically, there is no example of a durable community moat without third-pa

86.5 The concentration finding, which changes the shape of the problem

CUT OF THE OBSIDIAN ECOSYSTEM	CUMULATIVE DOWNLOADS	ACTIVE-INSTALL PROXY (PEAK SINGLE VERSION)
Top 10 plugins	27.08%	26.90%
Top 20	39.40%	35.00%
Top 50	55.93%	47.50%
Top 100	68.06%	58.62%
Top 500	88.82%	83.44%
Plugins whose best-ever single version never passed 1,000 downloads	–	4,763 of 7,085 = 67.2%
...never passed 100	–	1,823 = 25.7%

Read the top 20 by active-install proxy and sort by what they actually are: `calendar` (5.82%), `dataview` (4.22%), `style-settings`, `kanban`, `remotely-save`, `icon`

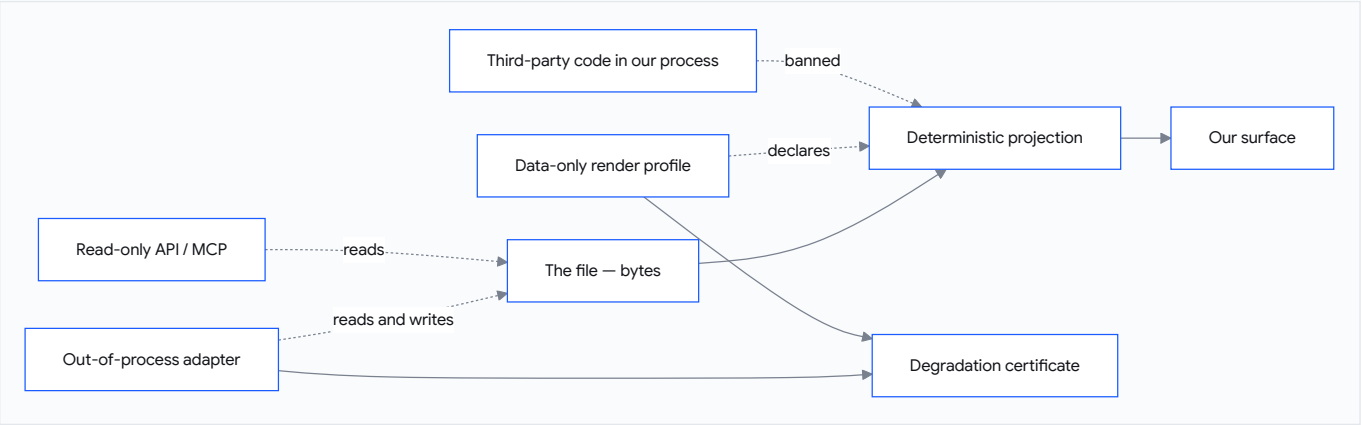
Eighteen of those twenty are a `render:` profile, a theme token, a core editor affordance, sync, or export – all of which the PRD already plans to ship in core

So the ecosystem the ban forecloses is, by installs, mostly a roadmap – and 67.2% of it by count is software nobody installs.

Anti-recommendation: this argues that we can *absorb* the value, not that we can absorb it *quickly*. Eighteen features shipped by one founder against eighteen

86.6 The middle grounds, scored against settled positions

OPTION	CREATES A THIRD-PARTY ASSET?	ARBITRARY CODE IN OUR PROCESS?
Curated in-core request queue	No	No
Data-only extension format (<code>render:</code> profiles + declared views, no code)	Yes	No
User-authored SKILL.md automations (already planned)	Yes	No code, but <i>instructions</i> reach a model with write access
Themes	Yes	Raw CSS is a network and exfiltration channel (<code>@import</code> , <code>attribute-selector</code>)
Import/export adapter surface	Yes	No – adapters run out-of-process on bytes
Read-only API / MCP server for third-party tools	Yes, outside us	No



86.7 The security argument, and why it is load-bearing rather than decorative

Both incumbents state the problem in their own documentation, which removes the need to speculate.

- Obsidian: "Due to technical limitations, Obsidian cannot reliably restrict plugins to specific permissions or access levels... Community plugins can access
- VS Code: "The extension host has the same permissions as VS Code itself... an extension can read and write files on your machine, make network requests, run

The decisive fact is not the risk, it is the **apparatus**: a scanner, a scorecard, a review queue, a signing pipeline, a trust dialog, a revocation path, a prize

Zed proves a cheaper answer exists and also proves its limits: extensions compile to `wasm32-wasip2`, capabilities are enumerated (`process:exec`, `download_file`,

86.8 Verdict

The ban stands, with one named exception: a data-only, no-code contribution surface – declarative `render:` profiles and out-of-process import/export adapters

What that means concretely: third parties may contribute a profile (a declared view over frontmatter and headings, in YAML, with no expressions, no conditions)

Anti-recommendation: do not present this as "we have a plugin story after all". It is not one. A declarative profile format is a smaller, slower, less general

86.9 The strongest argument against this verdict

Every durable community in the editor category runs on third-party code, and no declarative surface has ever produced one (\$86.4). Worse, declarative surface

The second-strongest: moat #1 may simply be misranked. If the real moat is byte-fidelity and the review loop, then this section's contradiction dissolves by

86.10 Evidence over 12 months that forces a change

SIGNAL	THRESHOLD
"I need X and you won't build it" as a share of support volume	>15% of tickets, rolling quarter
Users patching the app bundle or shipping an unsanctioned loader	>1,000 installs of any such loader
Parity on the top-20-equivalent categories	any category not at parity
Declarative <code>render:</code> grammar acquires conditionals, loops or function calls	3 constructs
Signups attributable to community artefacts (profiles, adapters, certificates, themes) at month 12	<5% of the 113,507 free signups
A comparable markdown editor ships a capability-listed WASM extension surface and records no data-loss incident in 12 months	1 clean year
A single malicious or data-destroying extension incident in Obsidian, VS Code or Zed touching file contents	1 confirmed

Anti-recommendation on the whole table: none of these thresholds should be evaluated by the person who wrote them from memory. Each is a query against a name

87.1 What is actually true today

FACT	VALUE	EVIDENCE
Posts ever shipped	2	[measured] repo grep for authored marketing pr
Cumulative visitors required for ₹20L/mo	~1,260,000	[derived] 113,507 free signups ÷ 0.09 visitor→
<code>gray-matter</code> npm downloads / month	35,782,000 (weekly 8,352,860 × 4.28)	[fetched] https://api.npmjs.org/downloads/point/Li
<code>front-matter</code> npm downloads / month	3,041,000	[fetched] same API, <code>front-matter</code> , read 2026-08
<code>remark</code> / <code>unified</code> monthly	4.1M / 62.4M	[fetched] npm downloads API, read 2026-08-30
Brand term collision	total – our product name IS the generic substrate term	[inference] from the two rows above
Time for a new domain's organic to matter	6-12 months to first meaningful traffic; 12-24 to compounding	[SS] repeated industry claim, not published as

Two posts is not a content programme; it is two posts. Everything below is designed from that starting point, not from an imagined content team.

87.2 The editorial system for one founder

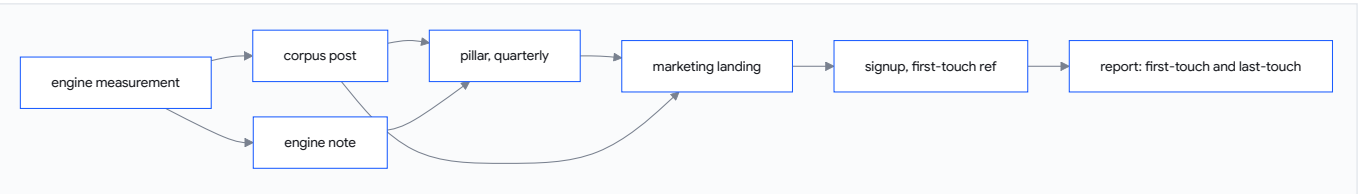
The cadence question is not "what is ideal" but "what survives month 7 when a Work-tier customer has a sync bug." Weekly is not survivable for a solo founde

SLOT	CADENCE	EFFORT	PURPOSE	ATTRIBUTION
Pillar (evergreen)	1 per quarter	12-20h	Ranks for a head term; becomes the canonical link target	UTM + first-touch cookie
Engine note	biweekly	4-6h	The measured, unglamorous result – refusal rates, byte-level findings	UTM
Corpus post	biweekly (off-week)	1-2h	One chart, one paragraph, from data the engine already produces	UTM
Changelog	on ship	0.5h	Not marketing; a public record	none needed

The corpus post is the load-bearing invention. The engine already measures things nobody else measures – zero-indent-sequence refusal rates across foreign v
Evergreen versus pillar. Four pillars, one per quarter, is the whole strategy for year one:

PILLAR	HEAD TERM TARGETED	WHY IT IS DEFENSIBLE
"What frontmatter actually is, and every way it breaks"	frontmatter / YAML frontmatter	We can show failure cases nobody else has catalogued [measured]
"Markdown is not one format: a compatibility map"	markdown compatibility	CommonMark vs GFM vs MDX divergence, testable
"Losslessness: what a WYSIWYG editor throws away"	markdown wysiwyg	Our carrier correction (callout not fence) is a real finding
"Publishing from a file, not a CMS"	markdown publishing	Our own product is the proof

Anti-recommendation: do **not** run a pillar-only strategy. Four pieces a year gives Google nothing to crawl, no internal-link graph, and no reason for a return
Content-to-signup attribution. Every published link carries `?ref=<slug>`; the signup row stores first-touch `ref` and last-touch `ref` and nothing else. No thi



87.3 Keyword map for the marketing site

The brand problem stated plainly: our name is the substrate's generic name. 35.78M monthly npm downloads [fetched 2026-08-30] of `gray-matter` means the term

TIER	TERM	INTENT	OUR REALISTIC POSITION	ACTION
Brand (contested)	frontmatter	navigational/technical, mostly not us	page 2-5, year 1	Own <code>frontmatter</code> . exact-match, plu
Brand (ours)	frontmatter editor / frontmatter app	commercial, low volume	page 1 achievable within 3-6 months	Homepage title t
Head	markdown editor	commercial, brutal	not reachable year 1	Do not spend on
Body	markdown wysiwyg editor	commercial	reachable, 6-12 months	Pillar 3
Body	collaborative markdown editor	commercial	reachable	Dedicated page,
Body	markdown editor for teams	commercial, B2B 2-20	reachable, low volume, high value	Dedicated page
Long-tail	obsidian alternative with collaboration	comparison	reachable fast	Comparison page,
Long-tail	notion markdown export broken	problem-aware	reachable fast, converts	Corpus post + fi
Long-tail	yaml frontmatter invalid / parse error	problem-aware, huge latent volume	reachable, our strongest asset	Pillar 1 + a fre
Long-tail	markdown ai editor byo key	product-specific	trivially reachable	Pricing/AI page

The free validator is the one asymmetric move on this table. [inference] A page that accepts a pasted document, tells the reader exactly which byte broke th
Anti-recommendation for the whole keyword map: this is a **product-led-content** strategy, and it fails if the product's differentiators are boring to non-users
Honest assessment of timing. Organic search cannot be the launch channel. [SS, not publishable as fact] the widely-repeated 6-12 month figure for a new doma

87.4 Technical SEO — marketing site

ITEM	REQUIREMENT	NOTE
Rendering	Server-rendered HTML for every indexable route	Client-only rendering defers indexing; do not rely on it
<code>robots.txt</code>	Explicit, allows crawl, points at sitemap	
<code>sitemap.xml</code>	Auto-generated at build, <code>lastmod</code> from git commit date	
Canonical	Self-referencing <code><link rel=canonical></code> on every page	Prevents the marketing/published-page duplicate class
Title	≤60 chars, pattern <code><Page> - Frontmatter</code>	Never the bare generic term alone
Meta description	≤155 chars, hand-written, never generated	
Open Graph	og:title, og:description, og:image (1200×630), og:url, og:type	
Twitter card	<code>summary_large_image</code>	
Structured data	<code>SoftwareApplication</code> + <code>Organization</code> + <code>FAQPage</code> on pricing	schema.org, validated
<code>hreflang</code>	Not year 1	English only until i18n ships
Core Web Vitals	LCP <2.5s, INP <200ms, CLS <0.1	Google's stated thresholds; verify with real field data, not lab
404/410	Real status codes, never a 200 soft-404	
Trailing slash	One canonical form, redirected 301	Pick and never revisit

Anti-recommendation: do not add JSON-LD `Review` or `AggregateRating` markup. We have banned explicit user-rating channels, so we would have no honest source fo

87.5 Technical SEO — published pages, and the indexing tension

This is the harder surface, because our default is **indexing off**, and that default is correct. A user who publishes a page from a file has not consented to b

ITEM	POLICY
Default	<code><meta name="robots" content="noindex,nofollow"></code> on every published page
Opt-in	Per-page and per-workspace toggle; opting in flips to <code>index,follow</code> and adds the page to that workspace's sitemap
Sitemap	One sitemap per custom domain, containing only opted-in pages; none on the shared publishing domain
Canonical	Always self-referencing to the page's own public URL; if a custom domain is attached, canonical points at the custom domain and the sh
Open Graph	Always emitted, even when noindex – OG is for link previews in chat, which is the actual sharing path, and is unrelated to crawling
<code>robots.txt</code> on shared domain	Disallow: <code>/</code> for the publishing subdomain until opt-in rates justify otherwise
Custom domains	Only opted-in; verified by DNS TXT before any indexable page is served
Author link	No automatic backlink to our marketing site from user pages

That last row is the expensive one and it is deliberate. A "Published with Frontmatter" backlink on every page is the standard growth loop and it would be t
Anti-recommendation to the whole noindex default: it costs us the entire published-page organic channel, which for competitors is a large fraction of their

87.6 Referral — re-examination and verdict

Referral currently sits in a banned list beside streaks, badges and XP. That list is coherent for the others and incoherent for referral. Streaks and XP are
Costing it against the requirement:

SCENARIO	K-FACTOR	EFFECT ON THE 1.26M VISITOR REQUIREMENT
No referral	0	1.26M visitors from paid + organic + community
5% of free signups refer 1 who signs up	0.05	Visitor requirement falls ~4.8% → ~1.20M [derived] $1.26M \times (1 - 0.05/1$
10% refer 1	0.10	→ ~1.15M [derived] $1.26M \times (1/1.10)$
20% refer 1 (optimistic ceiling for a paid tool with no cash incentive)	0.20	→ ~1.05M [derived]

[derived] The arithmetic: with a k-factor of k, total signups S from V visitors becomes $V \times 0.09 \times (1/(1-k))$ for $k < 1$, so the visitors needed for a fixed signup
Verdict: build referral, in exactly one form – a workspace invite that gives the inviter nothing and gives the invitee a genuinely better first run. No cred
The mechanic: any user can generate an invite link to a document or a workspace. The recipient lands on the actual document, readable, no signup wall. If th
Why this is coherent with our positions: it is not gamified (no score, no streak, no unlock); it is not an ambient AI button; it does not reprice anything;
Anti-recommendation: this design deliberately forgoes the incentive that actually drives k-factor. Dropbox-style two-sided rewards work; ours will underperf

87.7 What can and cannot move 1.26M visitors

CHANNEL	REALISTIC YEAR-1 CONTRIBUTION	ASSESSMENT
Organic search, marketing site	50k-200k cumulative, back-loaded to months 9-18	Cannot carry launch; is t
The free frontmatter validator	30k-150k, faster than the rest of organic	Highest-variance item on 1
Content (24 pieces/year at the cadence above)	20k-60k direct	Its real value is link acc
Published-page organic	~0 under the noindex default	Structurally forgone by cl
Referral at k=0.10	Reduces requirement by ~126k visitors	Real, meaningful, not deci
Community, launch platforms, HN/Reddit/Show HN	30k-100k in spikes	Non-compounding; every spi
Paid acquisition	Bounded by 57-73% contribution margin on tiers of ₹299-₹599	At a ₹2,499/yr Pro ARPU, a
Integrations / distribution surfaces (npm package, VS Code, CLI)	unmodelled	The genuinely unexplored c

Summed at the optimistic end, everything on this table reaches perhaps 350k-500k year-one visitors. The 1.26M figure is not a year-one number and should not

Anti-recommendation to that last claim: developer-tool distribution reaches developers, and developers are the worst-converting audience for a paid editor -

]

§88

How to build from this

This plan is executed through `specs/`, not through this document. A layer is not done because it feels done; it is done when its spec reaches `state: verified`

[

```
npm run spec      # the contract gate - 0 errors required
npm run corpus    # 8,513 foreign files, byte-pinned, exits 1 on one changed byte
npm run verify    # typecheck → lint → test → build → arch → spec
```

RULE	WHY IT IS A RULE AND NOT A PREFERENCE
Red proof before green	A test on a rare fault proves nothing until it fails against unfixed code. If you cannot make it fail, say the test
Refuse rather than guess	Returning the input unchanged is a correct outcome. <code>diff-match-patch</code> returns <code>true</code> after corrupting a document; that is the failure mode this product exists to not have.
Re-derive every number at write time	PRD §57 is the list of what happened when we did not.
Agents propose, the founder merges	<code>git rev-parse HEAD</code> before and after every agent run, reconciled. The instruction not to commit is advisory; the rec

Citing this file. Its sections are a local run. Write `DEV-PLAN §5`, never a bare `§5` - the PRD, `ENGINE.md` and `BUSINESS.md` share one global run §0-87 and a be

]

§89

The architecture this plan builds on

Settled. Do not re-litigate any row; each was closed on measured or fetched evidence, and the evidence is in the PRD section named.

[

#	DECISION
Truth	Documents live in the user's git repo and never move. The file is the only source of truth; every view is a deterministic, reversible pro
Sync	Git three-way merge + an append-only splice journal + compare-and-swap. Never a CRDT for document bytes - a CRDT cannot own the bytes, CRI
Control plane	One Postgres holding zero document bytes: identity, tenancy, entitlements, billing, publish-slug uniqueness, audit, jobs
Blobs	Cloudflare R2 for attachments over 1 MB and derived artifacts. Free egress is load-bearing, and R2 has no object versioning - so disaster
Tenancy	<code>workspace_id</code> on every row, pooled RLS, from day one. The GitHub App installation is the <i>connector</i> , never the tenant identity. Retrofittin
Search	Stop shipping whole-vault snapshots. Postgres full-text + trigram server-side, MiniSearch client-side only
Execution	No arbitrary client-side code execution, ever. That lane is refused, and refusing it is what makes the plugin question hard - see BUSINESS
Engine	MDMAX wires in as four ordered seams: ingress gate (live) → write gate → splice engine → certificate. Do not wire the write gate before NF

What is already in the repo, as of 2026-08-30: Next 16.2.6, React 19.2.6, CodeMirror 6, unified/remark/rehype, next-auth v5, AI SDK v6 across five providers

The constraints that shape every answer below. One founder. India-based, selling globally. Cost must be near-zero at 100 users and predictable at 10,000. Bo

]

§90

Frontend architecture

90.1 Measured baseline (what already exists, before any of this)

Everything below extends the repo as it stands. These are facts read off disk on 2026-08-30, not recollections.

FACT	VALUE	EVIDENCE
<code>.tsx</code> files in <code>src/</code>	74	[measured] <code>find src -name '*.tsx' wc -l</code>
Files carrying "use client"	63	[measured] <code>grep -rl 'use client' src wc -l</code>
Route handlers + pages	4 pages, 26 API routes	[measured] <code>find src/app -name page.tsx -o -name route.ts</code>
Existing modules	<code>ai</code> , <code>ai-tools</code> , <code>app-shell</code> , <code>auth</code> , <code>drafts</code> , <code>editor</code> , <code>export</code> , <code>graph</code> , <code>mdmax</code> , <code>preview</code> , <code>repository</code> , <code>share</code> , <code>vault</code>	[measured] <code>ls src/modules</code>
Layering already enforced	<code>eslint-plugin-boundaries</code> ^6.0.2, <code>domain</code> → <code>application</code> → <code>infrastructure</code> → <code>presentation</code> , <code>default</code> : "disallow"	[measured] <code>eslint.config.mjs</code> L83-L120
Editor modes already modelled	"edit" "live" "reading" "split"	[measured] <code>src/modules/editor/presentation/editor-store.ts</code>
State library already in use	<code>zustand</code> ^5.0.13, 4 call sites	[measured] <code>grep -rln zustand src</code>
Service worker today	hand-written, 98 lines, <code>CACHE = "sgnk-md-v2"</code>	[measured] <code>public/sw.js</code>
Bundle budget today	"budget": "echo 'No bundle budget configured yet - skipping'"	[measured] <code>package.json</code>
Tauri shell today	thin remote-URL wrapper: <code>frontendDist: "https://md.sgnk.ai"</code>	[measured] <code>src-tauri/tauri.conf.json</code>

Two of these are load-bearing for the rest of the section. The `budget` script is a literal no-op stub, so the bundle ceiling below is being introduced from :

90.2 Rendering strategy: server components at the edges, one client island in the middle

The document editor is irreducibly client-side: `CodeMirror` owns a live `EditorState`, keystrokes are 60fps, and the file bytes are the source of truth held in

ROUTE	RENDERING	CLIENT JS SHIPPED
<code>/p/[slug]</code> (public published note)	RSC, fully server-rendered, <code>dynamic = "force-static"</code> + revalidate on publish	~45 KB gz ceiling, hydration only for theme + copy-l
<code>/auth/login</code>	RSC + two small client forms	~50 KB
<code>/w/[workspace]</code> (vault shell)	RSC layout → one client island	entry budget, \$90.6
<code>/w/[workspace]/n/[...path]</code>	Same RSC layout, note selection is a shallow URL update	no extra RSC roundtrip
<code>/w/[workspace]/settings</code> , <code>/billing</code> , <code>/admin</code>	RSC, server actions for mutations	~55 KB

Picked: RSC shell + a single hydrated workspace island. Rejected: full client SPA (loses the free public reader and forces an auth waterfall) and RSC-per-no

90.3 State management: three zustand stores, and one thing that is deliberately not a store

Zustand 5 is already here and stays. The decisive property is not size – it is that a zustand store is readable and writable from outside React, which is re

STORE	FILE	HOLDS	PERSISTENCE
<code>useEditorStore</code> (exists)	<code>src/modules/editor/presentation/editor-store.ts</code>	<code>tabs</code> , <code>activePath</code> , <code>secondaryPath</code> , <code>mode</code> , <code>contentByPath</code> , <code>mirror</code> , <code>baseShaByPath</code> , <code>reloadByPath</code>	<code>sessionStorage</code> , <code>partial</code> excludes <code>contentByPath</code> [measured – already cori
<code>useWorkspaceStore</code> (new)	<code>src/modules/workspace/presentation/workspace-store.ts</code>	<code>workspaceId</code> , <code>role</code> , <code>entitlements</code> , <code>connector status</code> , <code>feature flags</code>	none – hydrated from RSC
<code>useSyncStore</code> (new)	<code>src/modules/sync/presentation/sync-store.ts</code>	outbox depth, per-path lifecycle (clean dirty queued committing diverged refused), last CAS error	mirrors IndexedDB; IDB i

The thing that is not a store: `src/modules/editor/presentation/active-view.ts`, a module-level singleton holding the focused `EditorView`. Its in-repo comment alre

Rejected: Redux Toolkit (ceremony, no benefit at one founder's scale), Jotai (atom-per-path fragments the outbox invariants), React Context for editor state

90.4 The CodeMirror 6 boundary

This is the highest-risk seam in the product, because the engine speaks bytes and CodeMirror speaks UTF-16 code units, and the whole promise is byte-preserv

Three rules, all enforceable:

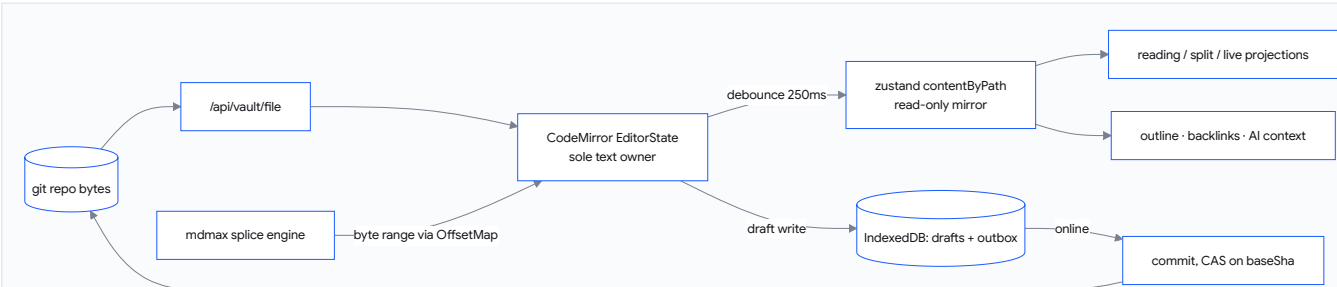
- `EditorState` is the sole owner of document text during an editing session. `contentByPath` in `zustand` is a read-only mirror written on a 250 ms debounce, con
- Store → editor is never a plain write. It happens only through the existing `bumpReload(path)` (full reload, used by history restore) or through an annotate
- A byte offset must pass through `OffsetMap` before it touches CodeMirror. `src/modules/mdmax/domain/offsets.ts` already exists and was the site of a fixed off-by

```
// src/modules/editor/presentation/remote-splice.ts
import { Annotation, type EditorState, type TransactionSpec } from "@codemirror/state";
import type { SpliceOp } from "@modules/mdmax"; // { start: number; end: number; bytes: Uint8Array }
import { byteToChar } from "@modules/mdmax/domain/offsets";

/** Marks a transaction as NOT user-originated: autosave skips it, undo groups it as one. */
export const RemoteSplice = Annotation.define<{ origin: "ai" | "engine" | "merge" }>();

export function spliceToTransaction(
  state: EditorState,
  op: SpliceOp,
  origin: "ai" | "engine" | "merge",
): TransactionSpec {
  const from = byteToChar(state.doc, op.start);
  const to = byteToChar(state.doc, op.end);
  if (from === null || to === null) {
    // REFUSE, never guess. The engine's contract is the UI's contract.
    throw new SpliceRefused("offset does not land on a character boundary", op);
  }
  return {
    changes: { from, to, insert: new TextDecoder().decode(op.bytes) },
    annotations: RemoteSplice.of({ origin }),
    scrollIntoView: origin !== "engine",
  };
}
```

A `SpliceRefused` surfaces as a non-dismissable inline banner on the affected block with the engine's reason string. It never falls back to a fuzzy match. **Cost**



90.5 Routing and the URL contract

PATH	OWNER	NOTES
/w/[workspace]	src/app/(workspace)/w/[workspace]/page.tsx	Consolidates today's (vault) and (workspace) groups into one [measured: both dirs exist]
/w/[workspace]/n/[...path]	same segment, shallow-updated	Deep-linkable note; [...path] is the vault-relative path, URL-encoded per segment
/p/[slug]	src/app/(public)/p/[slug] (exists)	Publish-slug uniqueness lives in the control-plane Postgres
/api/**	26 existing route handlers	Unchanged shape; workspace_id becomes a required, server-derived parameter

Workspace lives in the path, never in a cookie. **Why:** a cookie-scoped tenant makes two open tabs on two workspaces impossible, and it makes every RLS bug inv

90.6 Code splitting and the bundle budget

Measured gzip of the on-disk published `dist` files (these are the *unminified* ESM bundles as npm ships them, so the real shipped cost after minification and

PACKAGE	RAW B	GZIP B	GZIP KB	TAG
mermaid 11	3,312,967	984,194	883.0	[measured]
7× @codemirror/* in use	978,890	235,875	230.3	[measured, summed]
react-dom client production	536,016	94,740	92.5	[measured]
katex	271,142	75,407	73.6	[measured]
minisearch	86,348	18,611	18.2	[measured]

Mermaid alone is 883 KB gzipped – 3.8× the entire CodeMirror editor. It is the single fact that shapes the splitting plan.

The budget (`budget.json`, gzipped bytes of built output):

KEY	CEILING	BASIS
route:w/[workspace] first-load JS	190 KB	React 19 + Next 16 runtime + shell + zustand; measured react-dom dist gz is 92.5 KB unminified [measured], minified lands lower [inference]
route:p/[slug] first-load JS	45 KB	RSC HTML + a theme toggle. Hard ceiling: no editor, no unified, no mermaid
chunk:editor	220 KB	230.3 KB unminified-gz measured; minify+treeshake ≈ 140 KB [inference]; 220 leaves ~57% headroom
chunk:markdown-pipeline	130 KB	unified + remark-gfm/math/breaks + rehype-slug/raw/highlight
chunk:katex	80 KB	73.6 measured
chunk:minisearch	25 KB	18.2 measured
chunk:graph	180 KB	react-force-graph-2d
chunk:mermaid	950 KB	883.0 measured. Lazy-only. Never preloaded, never in any route's first-load set.
total:.next/static/chunks	2,400 KB	Global drift guard

Cold path to first keystroke = 190 + 220 + 130 = **540 KB gz** [derived]. Publishing a note pre-renders its mermaid diagrams to SVG into R2 at publish time, so **Enforcement**. Replace the stub `budget` script with a zero-dependency script that reads Next's own manifests, so it survives a Turbopack/webpack switch:

```
// scripts/bundle-budget.mjs - run after `next build`
import { readFileSync } from "node:fs";
import { gzipSync } from "node:zlib";
const budget = JSON.parse(readFileSync("budget.json", "utf8"));
const manifest = JSON.parse(readFileSync(".next/app-build-manifest.json", "utf8"));
const gz = (f) => gzipSync(readFileSync(`.next/${f}`), { level: 9 }).length;

let failed = 0;
for (const [route, files] of Object.entries(manifest.pages)) {
  const key = `route:${route}`;
  if (!(key in budget)) continue;
  const bytes = files.filter((f) => f.endsWith(".js")).reduce((n, f) => n + gz(f), 0);
  const kb = Math.round(bytes / 1024);
  const ceil = budget[key];
  console.log(`${key} <= ceil ? "PASS" : "FAIL" ${key} ${kb}KB / ${ceil}KB`);
  if (kb > ceil) failed++;
}
process.exit(failed ? 1 : 0);
```

Plus a lint rule that makes the mermaid ceiling structural rather than aspirational:

```
// eslint.config.mjs - appended to the existing boundaries block
{
  files: ["src/**/*.ts", "src/**/*.tsx"],
  ignores: ["src/modules/preview/presentation/lazy/**"],
  rules: {
    "no-restricted-imports": ["error", { paths: [
      { name: "mermaid", message: "Import only from preview/presentation/lazy/mermaid-lazy.ts" },
      { name: "katex", message: "Import only from preview/presentation/lazy/katex-lazy.ts" },
      { name: "react-force-graph-2d", message: "Import only from graph/presentation/lazy/graph-lazy.ts" },
    ]}],
  },
}
```

CI. The repo has no CI at all. `.github/workflows/ci.yml` gets three jobs: `verify` (the existing `typecheck && lint && test && arch && spec`), `build`, and `budget` (nee

90.7 Offline and the service worker

There is no prior service-worker research in the corpus, so this is stated as a design with its reasoning exposed rather than as a citation.

Picked: `@serwist/next 9.5.12` [fetched 2026-08-30, registry.npmjs.org] to *generate the precache manifest*, with our own handwritten runtime-caching rules inje

REQUEST CLASS	STRATEGY	RATIONALE
HTML navigations	NetworkFirst , 3 s timeout → cached shell	A stale shell referencing dead chunk hashes is the exact failure a
<code>/_next/static/**</code>	CacheFirst , immutable, versioned by build id	Content-hashed; safe forever
<code>/p/**</code>	StaleWhileRevalidate , 24 h	Public reader stays readable offline
<code>/api/vault/**</code> (document bytes)	NEVER intercepted. Network only.	A cached GET would hand the editor stale base bytes; a splice agai
<code>/api/ai/**</code> , <code>/api/auth/**</code> , <code>/api/commit</code>	Never intercepted	Mutations and credentials

Offline *writing* is not the service worker's job – it is IndexedDB's. `idb-keyval` ^6.2.4 already backs `src/modules/drafts/infrastructure/draft-store.ts` [measured].

90.8 Optimistic UI for a splice-based writer

Optimism has to stop precisely where compare-and-swap begins.

STATE	UI	REVERSIBLE?
<code>dirty</code>	dot on tab (exists), draft in IDB	yes – local only
<code>queued</code>	"waiting to sync", outbox count in status bar	yes
<code>committing</code>	subtle spinner, editor stays fully editable	yes
<code>committed</code>	dot clears, <code>baseShaByPath</code> updated	n/a
<code>diverged</code> (CAS 409)	banner: "This file changed on GitHub" + Review / Keep mine / Take theirs	user-driven three-way merge
<code>refused</code>	inline block banner with the engine's reason	never auto-resolved

We are optimistic about **local application** and pessimistic about **remote acceptance**. `baseShaByPath` already exists in the store for exactly this and already ca

90.9 Component architecture: four modes, N render profiles

The four modes already exist in the store as data (`edit | live | reading | split`). Enforce that they stay data: **one** `EditorSurface` component composes panes, and

MODE	COMPOSITION	OWNS TEXT?
<code>edit</code>	<code><SourcePane></code>	<code>CodeMirror</code>
<code>reading</code>	<code><RenderedPane profile></code>	nobody – pure projection of <code>contentByPath</code>
<code>split</code>	<code><SourcePane></code> + <code><RenderedPane></code> with existing <code>split-scroll-sync.ts</code>	<code>CodeMirror</code>
<code>live</code>	<code><RenderedPane></code> + <code><InlineBlockEditor></code> on the focused block (both exist under <code>presentation/live/</code>)	<code>CodeMirror</code> , scoped to one block's byte range

Render profiles (GitHub, CommonMark, Obsidian, Pandoc) are **pipeline configurations, not component forks** – a frozen list of remark/rehype plugins plus a `Degr`

90.10 Desktop shell: Tauri v2

Today `tauri.conf.json` points `frontendDist` at `https://md.sgnk.ai` – a remote-URL wrapper [measured]. Keep that as the default: one deploy, one codebase, no upda

CONCERN	WEB	TAURI V2 (@TAURI-APPS/API ^2.11.0, LATEST 2.11.1 [FETCHED 2026-08-30])
Vault storage	GitHub API via /api/vault/**	Same, plus optional local filesystem clone behind a capability
Menus / shortcuts	DOM events	Native menu → same DOM events (exists)
Offline	service worker + IDB	WebView cache + IDB; identical code
Updates	deploy	Remote URL means the shell needs updating only for Rust changes

Rejected: bundling the frontend into the app (frontendDist: "../out"), which requires a signed update channel and a second release process. **Cost:** the desktop

90.11 File and folder architecture

Extends src/modules/* and stays inside the existing four-layer eslint-plugin-boundaries contract.

```
src/
├─ app/
│   ├── (public)/p/[slug]/           # RSC reader, tight CSP (exists)
│   ├── (auth)/login/                # exists
│   ├── (workspace)/                 # absorbs today's (vault) group
│   │   └─ w/[workspace]/
│   │       ├── layout.tsx           # RSC: workspace + entitlements + tree manifest
│   │       ├── page.tsx
│   │       └─ n/[...path]/page.tsx  # deep-linkable note (shallow-updated)
│   │           └─ settings/ · billing/
│   ├── api/                         # 26 handlers (exists)
│   └─ manifest.ts                   # PWA manifest (exists)
├─ modules/
│   ├── editor/presentation/
│   │   ├── EditorSurface.tsx        # NEW – single composer for all 4 modes
│   │   ├── remote-splice.ts         # NEW – $90.4 byte+char adapter + REFUSE
│   │   └─ CodeMirrorEditor.tsx · EditorPane.tsx · live/ · active-view.ts (exist)
│   ├── preview/presentation/
│   │   ├── profiles/registry.ts     # NEW – render-profile pipelines
│   │   ├── RenderedPane.tsx         # NEW – profile-parameterised projection
│   │   └─ lazy/{mermaid,katex}-lazy.ts # NEW – the ONLY legal import sites
│   ├── sync/                         # NEW module
│   │   ├── domain/{outbox-entry,cas-result}.ts
│   │   ├── application/{enqueue,replay,resolve-divergence}.ts
│   │   ├── infrastructure/outbox-store.ts # idb-keyval
│   │   └─ presentation/{sync-store.ts,SyncStatusBar.tsx,DivergenceBanner.tsx}
│   ├── workspace/                   # NEW – tenancy/entitlements client mirror
│   ├── offline/presentation/{OfflineBoundary.tsx,useOnline.ts} # NEW
│   ├── platform/                    # NEW – web vs Tauri capability adapters
│   └─ ...(ai, ai-tools, app-shell, auth, drafts, export, graph, mdmax, preview, repository, share, vault – unchanged)
├─ shared/ · container/ · config/    # exist
├─ budget.json                       # NEW
├─ scripts/bundle-budget.mjs         # NEW
├─ .github/workflows/ci.yml          # NEW – verify · build · budget
└─ sw/index.ts                       # NEW – Serwist source, compiled to public/sw.js
```

src/modules/sync and src/modules/workspace get boundary entries in eslint.config.mjs alongside the existing types so the default: "disallow" rule keeps applying

\$91 API design and contracts

91.1 Protocol choice: typed RPC over HTTP, not REST, not GraphQL

OPTION	VERDICT	WHY
Typed RPC over HTTP+JSON (verb-named POST endpoints under /api/v1/*, Zod-validated, OpenAPI-emitted)	PICKED	Our operations are not CRUD on rows. co invites last-write-wins semantics, which
REST-purist resource CRUD	Rejected	A markdown file is not a document recor
GraphQL	Rejected	Single founder on call. GraphQL adds a c
gRPC / Connect	Rejected	Browser + Tauri + MCP + curl all need tc

Reads are GET, writes are POST. GETs are idempotent, cacheable, and take query params. Writes are POST-only, always take a JSON body, always require an idem Stack: keep the routes in Next 16.2.6 App Router (src/app/api/v1/**/route.ts) – the repo already has 26 route files there [measured, find . -name route.ts]. Va

91.2 Resource model: the file is the resource, the commit is the version

Every addressable thing is a triple: (workspace_id, repo_ref, path) at a sha .

CONCEPT	WIRE IDENTIFIER	NOTES
Workspace	<code>ws_01J...</code> (ULID, prefixed)	Tenancy root. On every row, every token, every log line.
Repo binding	<code>rb_01J...</code>	Maps a workspace to <code>owner/name@branch</code> via a GitHub App installation. The installation is the tenant-scoped handle. Never expose <code>installation_id</code> on the wire.
File	<code>path</code> string, POSIX, no leading slash, NFC-normalized, <code>..</code> refused	Not an opaque ID. Renames are git renames; a synthetic file ID would drift from the
Version	<code>sha</code> (40-hex commit) + <code>blob_oid</code> (40-hex)	<code>blob_oid</code> is the precise CAS token: it changes only when <i>these bytes</i> change, so a cor
Publish	<code>slug</code> (globally unique, control plane)	The one piece of state Postgres genuinely owns.
Attachment	<code>at_01J...</code> → R2 key <code>ws/{workspace_id}/at/{id}</code>	>1MB only. Under 1MB goes in the repo.

Control plane holds `{workspace, repo_binding, slug, entitlement, job, audit}` and **zero document bytes**. Every API response that contains file content read it from g

91.3 The v1 endpoint list

Rate classes: **A** = 600/min/workspace (reads), **B** = 60/min/workspace (writes), **C** = 10/min/workspace (expensive: certify, export, land), **D** = 20/min/workspace (A

Auth column: **S** = session cookie (next-auth v5), **T** = PAT/OAuth bearer (`fmk_live_...`), **M** = MCP token, **W** = webhook HMAC.

METHOD	PATH	AUTH	IDEM	RATE	NOTES
GET	<code>/api/v1/workspaces</code>	S,T	–	A	
GET	<code>/api/v1/repos</code>	S,T	–	A	Bindings, not GitHub repos
POST	<code>/api/v1/repos.bind</code>	S	✓	B	Consumes App installation
POST	<code>/api/v1/repos.unbind</code>	S	✓	B	Revokes; keeps audit
GET	<code>/api/v1/tree</code>	S,T,M	–	A	<code>?ref=&prefix=&cursor=&limit=</code>
GET	<code>/api/v1/files</code>	S,T,M	–	A	<code>?path=&ref=</code> → bytes + <code>blob_oid</code>
GET	<code>/api/v1/files.history</code>	S,T,M	–	A	Cursor-paginated commits for one path
POST	<code>/api/v1/files.commit</code>	S,T,M	required	B	\$91.7
POST	<code>/api/v1/files.create</code>	S,T,M	required	B	Refuses if path exists
POST	<code>/api/v1/files.delete</code>	S,T,M	required	B	Needs <code>base_blob_oid</code>
POST	<code>/api/v1/files.rename</code>	S,T,M	required	B	git rename, both paths CAS-checked
POST	<code>/api/v1/splice.plan</code>	S,T,M	–	B	Dry run: returns ranges or <code>REFUSED</code>
POST	<code>/api/v1/land</code>	S,T	required	C	\$91.8
GET	<code>/api/v1/land.status</code>	S,T	–	A	Poll a <code>job_*</code>
POST	<code>/api/v1/certify</code>	S,T,M	required	C	Cross-engine degradation cert
GET	<code>/api/v1/certify.result</code>	S,T,M	–	A	
GET	<code>/api/v1/search</code>	S,T,M	–	A	Postgres FTS + trigram, server-side
POST	<code>/api/v1/publish</code>	S,T	required	B	\$91.9
POST	<code>/api/v1/unpublish</code>	S,T	required	B	\$91.9, immediate revocation
GET	<code>/api/v1/publications</code>	S,T	–	A	
POST	<code>/api/v1/attachments.presign</code>	S,T	required	B	R2 PUT presign, 15-min TTL
POST	<code>/api/v1/ai.complete</code>	S,T	required	D	Streams; SSE
GET	<code>/api/v1/entitlements</code>	S,T	–	A	
POST	<code>/api/v1/tokens.create</code>	S	✓	B	Returns secret once
POST	<code>/api/v1/tokens.revoke</code>	S	✓	B	Effective in ≤5s (\$91.9 pattern)
GET	<code>/api/v1/webhooks</code> / POST <code>.create</code> / <code>.rotate</code> / <code>.delete</code>	S	✓	B	
POST	<code>/api/v1/hooks/github</code>	W	–	–	Inbound from GitHub App
GET	<code>/api/v1/openapi.json</code>	public	–	A	
GET	<code>/api/v1/health</code>	public	–	A	

29 endpoints. That is the whole v1. Anything not on this list does not exist in v1; adding one is a deliberate act with an OpenAPI diff.

91.4 Versioning and deprecation

- **URL-path major version** (`/api/v1/`). Rejected alternative: `Accept: application/vnd.frontmatter.v1+json` header negotiation – correct in theory, hostile to `curl`
- **Additive changes ship without a version bump**. New optional request fields, new response fields, new enum members *in response position only*. Clients MUST i
- **Breaking** = removing/renaming a field, narrowing a type, adding a required request field, changing an error `type` URI, changing default behavior. Requires
- **Deprecation policy: 180 days minimum**. On a deprecated endpoint or field we emit `Deprecation: @1767225600` (RFC 8594 unix-seconds form) and `Sunset: Wed, 31 Dec`
- `X-FM-API-Version: 2026-08-30` response header on every response, carrying the *deploy* date. Not a dated-version scheme like Stripe's – one founder cannot main

91.5 Pagination and cursors

Opaque, forward-only, keyset cursors. Never `LIMIT/OFFSET` (skips rows under concurrent insert) and never page numbers.

```
// src/lib/api/cursor.ts
type Cursor = { k: string; v: string; d: 'asc'|'desc' }; // k = sort key name
const enc = (c: Cursor) => Buffer.from(JSON.stringify(c)).toString('base64url');
```

Response envelope for every list endpoint:

```
{ "data": [ ... ],
  "page": { "next_cursor": "eyJrIjoicGF0aCIsInYiOiJkb2NzL2EubWQiLCJkIjoIYXNjIn0",
            "has_more": true, "limit": 100 } }
```

`limit` default 100, max 500. Tree listings sort by `path` ascending (stable, git-native). History sorts by `committed_at desc, sha` - `sha` is the tiebreaker beca

91.6 Idempotency keys

Every POST carries `Idempotency-Key: <client-generated UUIDv7>`. Missing key on a write → 400. This is the single most important header in the API: the founding

```
-- control plane, migration 0012
create table idempotency_key (
  workspace_id  uuid not null,
  key           text not null,
  endpoint      text not null,
  request_hash  bytea not null,          -- sha256 of canonical body
  state         text not null check (state in ('in_flight','done')),
  status_code   int,
  response_body jsonb,
  locked_at     timestamptz,
  created_at    timestamptz not null default now(),
  primary key (workspace_id, key)
);
create index on idempotency_key (created_at); -- 24h reaper
alter table idempotency_key enable row level security;
create policy tenant on idempotency_key using (workspace_id = current_setting('app.workspace_id')::uuid);
```

Semantics: insert `in_flight` with `ON CONFLICT DO NOTHING`. If the insert lost, read the row. Same key + **different** `request_hash` → 409 `idempotency-key-reuse`. Same
Note the layering: the idempotency key protects against *duplicate delivery of the same intent*; `base_blob_oid` protects against *stale intent*. Both are require

91.7 `files.commit` — the exact shape

```
POST /api/v1/files.commit
Authorization: Bearer fmk_live_...
Idempotency-Key: 018f3c2a-7b1e-7c3d-9a4f-2b6e11c9de01
Content-Type: application/json
```

```
{
  "repo_binding_id": "rb_01JAV6M2Q7X8YB3F0K",
  "base_sha": "9e846842f0d1c3b7aa10e5c2f4d9a8b1c6e30fd2",
  "changes": [ {
    "path": "notes/2026-08-30.md",
    "base_blob_oid": "1c9de0184f3c2a7b1e7c3d9a4f2b6e11c9de0184",
    "op": "splice",
    "splices": [
      { "start": 412, "end": 468,
        "expect_sha256": "b8c1...",
        "replacement_b64": "IyMgU2hpcHBlZAo=" }
    ],
    "eol": "preserve", "bom": "preserve", "final_newline": "preserve"
  } ],
  "message": "note: shipped the splice journal",
  "journal": [ { "seq": 41, "client_id": "cm_ios_9f2", "at": "2026-08-30T14:22:07.113Z" } ],
  "author": { "name": "Sagnik Mitra", "email": "dev@perccent.com" }
}
```

`start` / `end` are **byte offsets into the base blob**, half-open, not character offsets and not line/column. `expect_sha256` is the hash of the *bytes currently in ti*

Success 200 :

```
{ "sha": "3f7ac0...", "committed_at": "2026-08-30T14:22:08.902Z",
  "files": [ { "path": "notes/2026-08-30.md", "blob_oid": "77aa19...", "bytes": 4188 } ],
  "journal_ack": { "through_seq": 41 },
  "degradation": { "certified": true, "profile": "gfm-strict", "warnings": [] } }
```

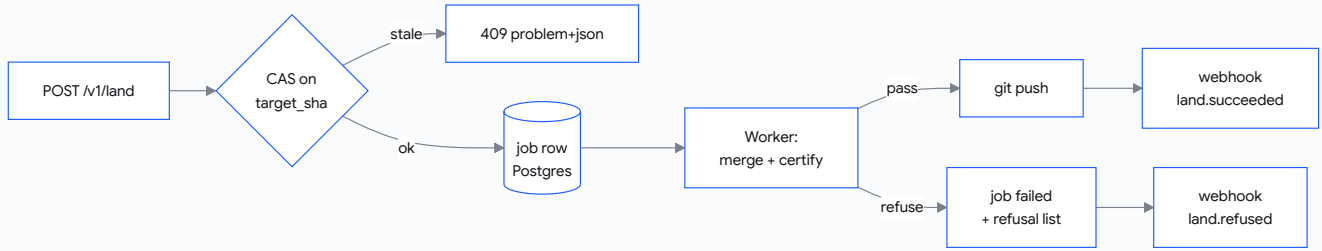
Conflict 409 - and this is the response shape the whole product hangs on. It never contains a merged guess:

```
{ "type": "https://frontmatter.dev/errors/stale-base",
  "title": "The file changed since your base revision",
  "status": 409, "instance": "/api/v1/files.commit",
  "path": "notes/2026-08-30.md",
  "expected_blob_oid": "1c9de0...", "actual_blob_oid": "b41f77...",
  "actual_sha": "aa02c1...",
  "refusal": { "code": "SPLICE_ANCHOR_MOVED", "splice_index": 0,
               "found_sha256": "0e12...", "engine_will_not_guess": true },
  "resolution": { "strategy": "rebase_splices",
                  "three_way_url": "/api/v1/files?path=notes%2F2026-08-30.md&ref=aa02c1..." } }
```

Refusal codes are a closed enum: `SPLICE_ANCHOR_MOVED`, `RANGE_OUT_OF_BOUNDS`, `BASE_NOT_ANCESTOR`, `BINARY_CONTENT`, `EOL_AMBIGUOUS`, `ZERO_INDENT_SEQUENCE`, `BARE_CR_SET`, [

91.8 `land()` — the async one

Landing = fast-forward or three-way merge the workspace branch into the target branch, run degradation certification on every changed file, then push. It ca



```
POST /api/v1/land
Idempotency-Key: 018f3c2b-0000-7c3d-9a4f-2b6e11c9de02
```

```
{ "repo_binding_id": "rb_01JAV6M2Q7X8YB3F0K",
  "source_ref": "frontmatter/ws-01JAV6/session-4482",
  "target_ref": "main",
  "expected_target_sha": "d50a6b2c8e1f...",
  "strategy": "three_way",
  "certify": { "profile": "gfm-strict", "on_degradation": "refuse" },
  "squash": true,
  "message": "frontmatter: 6 notes from 2026-08-30" }
```

202 Accepted , Location: /api/v1/land.status?job_id=job_01JAV... :

```
{ "job_id": "job_01JAV7C3M9QK2", "state": "queued",
  "poll_after_ms": 800, "expires_at": "2026-08-30T15:22:08Z" }
```

Terminal land.status on refusal:

```
{ "job_id": "job_01JAV7C3M9QK2", "state": "refused",
  "target_sha_at_start": "d50a6b2c...", "target_sha_now": "d50a6b2c...",
  "refusals": [
    { "path": "specs/api.md", "code": "ZERO_INDENT_SEQUENCE",
      "byte_range": [1204, 1261], "certifying_engines": ["remark", "commonmark", "markdown-it"],
      "disagreement": { "remark": "list", "commonmark": "paragraph" } } ],
  "applied": [], "hint": "No bytes were written. Fix or set on_degradation=annotate." }
```

applied: [] is a **contract**, not a **courtesy**. land is all-or-nothing: the push happens once, at the end, after every file certifies. Partial landing would lead to an expected_target_sha mismatch → immediate synchronous 409, no job created. Cheaper to fail before queuing.

91.9 publish / unpublish with immediate revocation

Publishing writes exactly one row in the control plane and one immutable render to R2. The slug is the only globally-unique namespace we own.

```
POST /api/v1/publish
{ "repo_binding_id": "rb_01JAV6M2Q7X8YB3F0K",
  "path": "writing/splice-journal.md",
  "sha": "3f7ac0...",
  "slug": "splice-journal",
  "visibility": "public",
  "expires_at": null,
  "theme": "sgnk" }
```

```
201 { "publication_id": "pub_01JAV7...", "url": "https://fm.sh/p/splice-journal",
  "revision": "rev_01JAV7...", "r2_key": "ws/.../pub/pub_01JAV7.../index.html",
  "etag": "W/\"rev_01JAV7...\", \"cache\": { \"edge_ttl\": 300, \"swr\": 0 } }
```

Slug collision → 409 slug-taken with suggestions: ["splice-journal-2"]. Uniqueness is a Postgres unique (slug) where revoked_at is null partial index – the database

Unpublish must be immediate and provable. Three things happen in one request, in this order:

```
POST /api/v1/unpublish
{ "publication_id": "pub_01JAV7...", "reason": "author_request", "purge": true }
```

1. update publication set revoked_at = now(), revoked_reason = \$2 where id = \$1 – inside the transaction. The edge worker checks a Redis key pub:{id} (TTL 60s, write)
2. DELETE the R2 objects under the revision prefix when purge: true. R2 delete is strongly consistent for the object; Cloudflare cache is purged by tag (cf-)
3. Audit row with actor, IP, and the R2 keys removed.

Response is 200 only after all three succeed, and it reports what it proved:

```
{ "publication_id": "pub_01JAV7...", "revoked_at": "2026-08-30T14:41:02.771Z",
  "revocation": { "control_plane": "committed", "edge_cache": "purged",
    "r2_objects_deleted": 4, "verified_404_at": "2026-08-30T14:41:03.402Z" },
  "residual_risk": "Third-party caches and archives are outside our control." }
```

That last field is deliberate. A one-person company must not imply it can un-publish the internet. If purge fails, the endpoint returns 207-style partial

91.10 Errors: RFC 9457 problem details

Content-Type: application/problem+json on every non-2xx. Fields: type (absolute URI at https://frontmatter.dev/errors/{slug}, stable forever – changing one is breaking)

HTTP	TYPE	SLUG	WHEN
400	malformed-request		Zod parse failure; <code>errors[]</code> carries <code>{path, code, message}</code>
401	unauthenticated		
403	forbidden	/ entitlement-required	Latter carries <code>required_plan</code> , <code>upgrade_url</code>
404	not-found		Also for cross-tenant reads – never 403, never leak existence
409	stale-base	slug-taken, idempotency-key-reuse, request-in-progress	
422	engine-refusal	invalid-cursor	Refusal is 422, not 400: the request was well-formed, the <i>content</i> is unsafe to touch
429	rate-limited		<code>Retry-After</code> , <code>X-RateLimit-{Limit,Remaining,Reset}</code>
502	upstream-git-failed		GitHub 5xx passthrough, <code>upstream_status</code>
503	provider-unavailable		AI provider down; names which of the five

91.11 Webhooks vs polling

Both, with a rule: poll for anything under 10 seconds, webhook for anything a human isn't waiting on.

SIGNAL	MECHANISM	WHY
<code>land</code> <code>progress</code>	Poll <code>land.status</code> , <code>poll_after_ms</code> in the body	The user is staring at a spinner. Server-driven bac
<code>land.succeeded/refused</code> , <code>publish.*</code> , <code>certify.completed</code> , <code>repo.drifted</code>	Outbound webhook	B2B integrations, CI, Slack. No human waiting.
AI completion	SSE stream on the same request	Already how AI SDK v6 works in this repo.

Signing: do not use `Svix` (`svix@2.1.0`, [measured, npm 2026-08-30]) at this scale – it's a per-message-priced dependency for a feature that is 60 lines. Imple

91.12 Public API and MCP: one surface, two adapters

One surface. The MCP server is a thin adapter that imports the same handler functions the HTTP routes import – not a parallel implementation, and not an HTT

```
src/server/ops/          <- the only place logic lives
  filesCommit.ts export async function filesCommit(ctx: OpCtx, input: CommitInput): Promise<CommitResult>
  land.ts publish.ts certify.ts search.ts ...
src/app/api/v1/files.commit/route.ts <- HTTP adapter: authn -> ctx -> filesCommit -> problem+json
src/mcp/server.ts          <- MCP adapter: @modelcontextprotocol/sdk@1.30.0
```

Why not two surfaces: divergence is guaranteed and a one-person team cannot maintain two definitions of "what a stale base means". Why not have MCP call our
The MCP adapter differs from HTTP in exactly three documented ways: (1) it exposes a **subset** – no `tokens.*`, no `webhooks.*`, no `repos.unbind`; (2) tool descript

91.13 OpenAPI generation

Schema-first from Zod, never hand-written YAML and never generated-from-runtime.

- `zod@4.5.4` + `@asteasolutions/zod-to-openapi@9.1.0` [measured, npm 2026-08-30] to build `openapi.json` at `src/server/openapi/registry.ts`.
 - Emitted at build time to `public/openapi.json` and served at `/api/v1/openapi.json`.
 - `openapi-typescript@7.13.0` [measured, npm 2026-08-30] generates `src/lib/api/types.gen.ts` for the web client, the Tauri client, and the published SDK – one typ
 - CI gate (this repo has **no CI at all** today [measured, stated architecture]; this is the first job to add): regenerate the spec, `git diff --exit-code public/op`
 - Rejected: `@hono/zod-openapi@1.6.1` (would require adopting Hono); `trpc` (no cross-language clients, and our consumers include `curl`, Python CI scripts, and mo
- Docs are `Scalar`'s standalone bundle rendered at `/docs/api`, reading `openapi.json`. Zero build step, self-hosted, no vendor account.

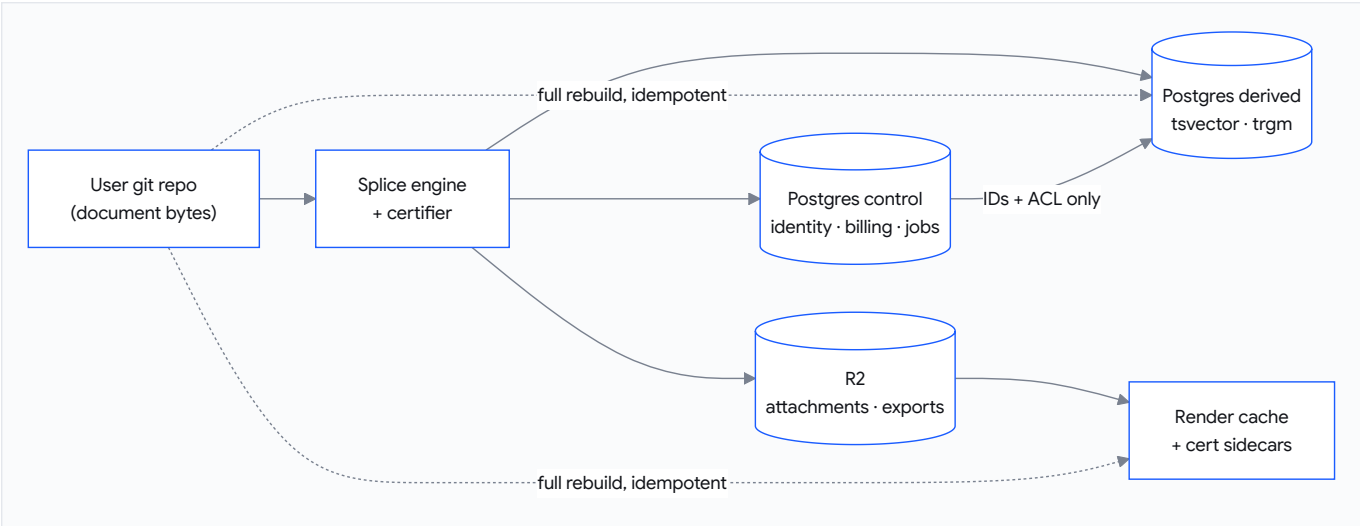
§92 Data layer — schema, migrations, and the control plane

92.1 The boundary rule, stated as an enforceable constraint

The control plane holds **zero document bytes**. This is not a style preference – it is what makes the "file is the only source of truth" claim survivable. If a

DATA CLASS	LIVES IN	NEVER IN	REBUILD SOURCE
Document bytes, frontmatter, splice journal entries	User git repo	Postgres, R2	– (it <i>is</i> the source)
Attachments > 1 MB, derived artifacts (PDF/HTML exports, cert sidecars)	R2 <code>frontmatter-artifacts</code>	Postgres	Git + engine, deterministically
Identity, tenancy, entitlements, billing mirror, slugs, audit, jobs, AI meter	Postgres <code>control</code> DB	R2	Backups only – this is real state
Search tokens, headings, bounded snippets	Postgres <code>derived</code> DB (separate database)	<code>control</code> DB	Git re-index
Render cache (HTML fragments, cert JSON)	R2 + Cache-Control	Postgres	Git re-render

Two logical databases on one Neon project. `control` is backed up, restored, and treated as irreplaceable. `derived` is `DROP DATABASE` –able at any time and carr:



Enforcement (db/checks/boundary.sql, run in CI): fail the build if any column in control is bytea, or is text with no length CHECK, outside a hand-maintain

```
select c.table_name, c.column_name, c.data_type
from information_schema.columns c
where c.table_schema = 'app'
and (c.data_type = 'bytea'
or (c.data_type = 'text'
and not exists (select 1 from app.text_column_allowlist a
where a.tbl = c.table_name and a.col = c.column_name)))
; -- non-empty result => exit 1
```

[inference] A type-level check is weak but cheap and it catches the realistic failure: an agent adding content text to a table during a feature build at 2am

92.2 Provider — Neon, and what we rejected

All prices read 2026-08-30.

	NEON	SUPABASE
Free tier	\$0, 0.5 GB storage/project, 100 CU-hours/project, 5 GB egress, scale-to-zero after 5 min [fetched]	\$0, 500 MB
Paid entry	Launch: pay-as-you-go, no minimum - \$0.106/CU-hour, \$0.35/GB-month, 500 GB egress/project included then \$0.10/GB [fetched]	Pro: from \$
Monthly at idle	~\$0 (scale-to-zero)	\$25 floor
Pooler	PgBouncer, managed, -pooler host, 10,000 client conns [fetched]	Supavisor, i
PITR	Instant restore \$0.20/GB-month, 7-day history on Launch [fetched]	Daily backu
Next tier	Scale: \$0.222/CU-hour	Team: from :

We pick Neon Launch. Reasons, in order: (1) scale-to-zero means 100 users costs single-digit dollars and a dormant staging branch costs nothing, which is ex Supabase rejected on two specifics, not on quality: the \$25>\$599 cliff is brutal for a solo founder whose first B2B customer asks for SSO, and the free-tier RDS rejected because \$11.68/mo buys an instance that is idle 95% of the time, has no pooler, no branching, and hands one person the pager for minor-version What changes our mind: sustained compute above ~4 CU average (at \$0.106/CU-hour, [derived] 4 × 730 × 0.106 = \$309.52/mo, which buys a lot of RDS), or a cust Postgres version: 18 (latest minor 18.6 as of 2026-08-30 [fetched, postgresql.org/versions.json]; 17.11 and 16.15 also supported). Pin the major in db/README

92.3 Connection pooling — the actual trap

This is the part that silently breaks in production, so it gets its own rules.

Neon's pooler is PgBouncer in transaction mode, reached by appending -pooler to the host. Pools are sized at 90% of max_connections, which is a function of Transaction mode forbids session-level SET, temporary tables, and SQL-level PREPARE / DEALLOCATE; protocol-level prepared statements are supported [fetched] The rule: the tenant claim is set with set_config(key, value, true) - is_local = true - inside the same explicit transaction as every statement that reads tena

```
// db/tx.ts - the ONLY way app code touches the control plane
import postgres from 'postgres'; // postgres@3.4.9 [measured 2026-08-30]
const sql = postgres(process.env.DATABASE_URL_POOLED!, {
  max: 4, idle_timeout: 20, connect_timeout: 10, prepare: false,
});
export async function withWorkspace<T>({
  workspaceId: string, userId: string, fn: (tx: postgres.TransactionSql) => Promise<T>,
}): Promise<T> {
  return sql.begin(async (tx) => {
    await tx`select set_config('app.workspace_id', ${workspaceId}, true),
      set_config('app.user_id', ${userId}, true)`;
    return fn(tx);
  });
}
```

Acceptance test the implementer must write before shipping (db/_tests_/pool-leak.test.ts): open 2 connections against the pooled host with max: 1, run withN

CONNECTION STRING	ENV VAR	USED BY	WHY
Pooled (<code>...-pooler.<region>.aws.neon.tech</code>)	<code>DATABASE_URL_POOLED</code>	All app request paths	Survives serverless fan-out
Direct	<code>DATABASE_URL_DIRECT</code>	<code>drizzle-kit migrate</code> , <code>pg_dump</code> , logical replication, one-off psql	Migrations need session state and advisory locks; the
Driver: <code>postgres</code> (<code>postgres.js</code>) 3.4.9 with <code>prepare: false</code> (required under transaction pooling). Rejected <code>@neondatabase/serverless</code> 1.1.0 as the primary driver -			

92.4 Schema

`db/migrations/0001_control_plane.sql`. Everything in schema `app`. Every tenant-scoped table carries `workspace_id`, and it is the **first** column of the primary key (

```

create extension if not exists pgcrypto;
create extension if not exists pg_trgm;
create schema app;

create table app.workspaces (
  id          uuid primary key default gen_random_uuid(),
  slug        text not null unique check (slug ~ '^[a-z0-9]{1,38}[a-z0-9]$'),
  name        text not null check (length(name) between 1 and 120),
  plan        text not null default 'free' check (plan in ('free','pro','team')),
  created_at  timestampz not null default now(),
  deleted_at  timestampz
);

create table app.users (
  id          uuid primary key default gen_random_uuid(),
  email       text not null unique check (length(email) <= 320),
  email_lower text generated always as (lower(email)) stored,
  name        text check (length(name) <= 120),
  image_url   text check (length(image_url) <= 2048),
  created_at  timestampz not null default now()
);

create table app.memberships (
  workspace_id uuid not null references app.workspaces(id) on delete cascade,
  user_id      uuid not null references app.users(id) on delete cascade,
  role        text not null check (role in ('owner','admin','editor','viewer')),
  created_at  timestampz not null default now(),
  primary key (workspace_id, user_id)
);
create index memberships_user_idx on app.memberships (user_id, workspace_id);

-- The GitHub App installation is a CONNECTOR, never the tenant identity.
-- One workspace may hold many; one installation may serve exactly one workspace.
create table app.vault_connections (
  id          uuid primary key default gen_random_uuid(),
  workspace_id uuid not null references app.workspaces(id) on delete cascade,
  provider    text not null check (provider in ('github','gitlab','local')),
  installation_id text check (length(installation_id) <= 64),
  repo_full_name text not null check (length(repo_full_name) <= 200),
  default_branch text not null default 'main' check (length(default_branch) <= 200),
  vault_root   text not null default '' check (length(vault_root) <= 400),
  status       text not null default 'active'
              check (status in ('active','revoked','error')),
  last_sync_at timestampz,
  last_error   text check (length(last_error) <= 2000),
  created_at  timestampz not null default now(),
  unique (provider, installation_id, repo_full_name)
);
create index vault_conn_ws_idx on app.vault_connections (workspace_id, status, created_at desc);

-- Globally unique across all tenants. The one asymmetric-RLS table.
create table app.publish_slugs (
  slug        text primary key check (slug ~ '^[a-z0-9]{1,62}$'),
  workspace_id uuid not null references app.workspaces(id) on delete cascade,
  doc_path    text not null check (length(doc_path) <= 1024),
  content_sha text not null check (content_sha ~ '[0-9a-f]{40,64}$'),
  visibility  text not null default 'public'
              check (visibility in ('public','unlisted','revoked')),
  published_at timestampz not null default now(),
  revoked_at  timestampz
);
create index publish_ws_idx on app.publish_slugs (workspace_id, published_at desc);
create unique index publish_ws_path_idx on app.publish_slugs (workspace_id, doc_path);

-- What the app reads. Never queried from Stripe at request time.
create table app.entitlements (
  workspace_id uuid primary key references app.workspaces(id) on delete cascade,
  seats        int not null default 1 check (seats between 0 and 10000),
  ai_tokens_mo bigint not null default 200000 check (ai_tokens_mo >= 0),
  private_publish boolean not null default false,
  sso          boolean not null default false,
  source       text not null default 'default'
              check (source in ('default','stripe','manual','trial')),
  valid_until  timestampz,
  updated_at   timestampz not null default now()
);

-- A cache of Stripe. Authoritative nowhere. Stripe outage must not gate the product.
create table app.billing_subscriptions (
  workspace_id      uuid primary key references app.workspaces(id) on delete cascade,
  stripe_customer_id text not null check (length(stripe_customer_id) <= 64),
  stripe_sub_id     text unique check (length(stripe_sub_id) <= 64),
  status            text not null check (status in
    ('trialing','active','past_due','canceled','incomplete','unpaid','paused')),
  price_id          text check (length(price_id) <= 64),
  quantity          int check (quantity between 0 and 10000),
  current_period_end timestampz,
  synced_at         timestampz not null default now(),
  raw_event_id      text check (length(raw_event_id) <= 64)
);

create table app.api_keys (
  id          uuid primary key default gen_random_uuid(),
  workspace_id uuid not null references app.workspaces(id) on delete cascade,
  name        text not null check (length(name) <= 80),
  prefix      text not null check (prefix ~ '^fm_[a-z]{4}[A-Za-z0-9]{8}$'),
  hash        text not null check (length(hash) between 40 and 200), -- argon2id, never the key
  scopes      text[] not null default '{}',
  last_used_at timestampz,
  expires_at  timestampz,
  revoked_at  timestampz,
  created_by  uuid references app.users(id),
  created_at  timestampz not null default now()
);
create unique index api_keys_prefix_idx on app.api_keys (prefix);

```

```

create index api_keys_ws_idx on app.api_keys (workspace_id, revoked_at, created_at desc);

create table app.jobs (
  id          bigserial primary key,
  workspace_id uuid not null references app.workspaces(id) on delete cascade,
  kind        text not null check (length(kind) <= 60),
  payload      jsonb not null default '{}':jsonb, -- IDs and paths only, never bytes
  state       text not null default 'queued'
              check (state in ('queued','running','done','failed','dead')),
  attempts    int not null default 0 check (attempts <= 20),
  run_after   timestamptz not null default now(),
  locked_by   text check (length(locked_by) <= 80),
  locked_at   timestamptz,
  last_error  text check (length(last_error) <= 2000),
  created_at  timestamptz not null default now(),
  constraint jobs_payload_small check (pg_column_size(payload) < 8192)
);
create index jobs_claim_idx on app.jobs (state, run_after, id) where state = 'queued';
create index jobs_ws_idx on app.jobs (workspace_id, created_at desc);

create table app.audit_log (
  id          bigserial primary key,
  workspace_id uuid not null references app.workspaces(id) on delete cascade,
  actor_user_id uuid references app.users(id),
  actor_key_id uuid references app.api_keys(id),
  action      text not null check (length(action) <= 80),
  target      text check (length(target) <= 1024),
  meta        jsonb not null default '{}':jsonb,
  ip          inet,
  created_at  timestamptz not null default now(),
  constraint audit_meta_small check (pg_column_size(meta) < 4096)
);
create index audit_ws_time_idx on app.audit_log (workspace_id, created_at desc);

create table app.ai_usage (
  id          bigserial primary key,
  workspace_id uuid not null references app.workspaces(id) on delete cascade,
  user_id     uuid references app.users(id),
  provider    text not null check (length(provider) <= 40),
  model       text not null check (length(model) <= 120),
  input_tokens bigint not null default 0 check (input_tokens >= 0),
  output_tokens bigint not null default 0 check (output_tokens >= 0),
  cached_tokens bigint not null default 0 check (cached_tokens >= 0),
  cost_micros bigint not null default 0 check (cost_micros >= 0),
  source_field text not null, -- WHICH provider key carried the count (LR#59)
  request_id  text check (length(request_id) <= 120),
  created_at  timestamptz not null default now()
);
create index ai_usage_ws_time_idx on app.ai_usage (workspace_id, created_at desc);

```

`ai_usage.source_field` is not decoration. AI SDK v6's five providers do not agree on usage key names; recording which key produced the number is the difference

92.5 RLS — pooled, forced, and non-owner

```

create or replace function app.current_workspace() returns uuid
language sql stable as $$ select nullif(current_setting('app.workspace_id', true), '')::uuid $$;

create role app_user nologin; -- the role the app connects as
create role migrator nologin; -- owns nothing at runtime; used only by drizzle-kit

do $$ declare t text; begin
  foreach t in array array['workspaces','memberships','vault_connections','publish_slugs',
    'entitlements','billing_subscriptions','api_keys','jobs',
    'audit_log','ai_usage'] loop
    execute format('alter table app.%I enable row level security', t);
    execute format('alter table app.%I force row level security', t); -- owner is NOT exempt
  end loop;
end $$;

create policy tenant_rw on app.vault_connections
using (workspace_id = app.current_workspace())
with check (workspace_id = app.current_workspace());
-- ... repeated per table; workspaces uses id = app.current_workspace()

-- Asymmetric: anyone may READ a live public slug; only the owner may write it.
create policy slug_public_read on app.publish_slugs for select
using (visibility in ('public','unlisted') and revoked_at is null);
create policy slug_owner_write on app.publish_slugs for all
using (workspace_id = app.current_workspace())
with check (workspace_id = app.current_workspace());

grant usage on schema app to app_user;
grant select, insert, update, delete on all tables in schema app to app_user;
revoke update, delete on app.audit_log from app_user; -- append-only, enforced by grant

```

Three traps closed here: `FORCE ROW LEVEL SECURITY` (a table owner bypasses plain RLS, and the migration role is the owner); a non-owner `app_user` with no `BYPAS`

92.6 Index strategy

Leading `workspace_id` on every tenant index, because every query is already filtered by it via RLS and Postgres will otherwise choose a scan and then filter.

TABLE	INDEX	SERVES
memberships	(workspace_id, user_id) PK + (user_id, workspace_id)	Both directions: "who is in this workspace", "which workspaces am I in"
vault_connections	(workspace_id, status, created_at desc)	Connector list, hot path on every editor load
publish_slugs	PK (slug); (workspace_id, doc_path) unique	Global uniqueness; per-doc republish idempotence
jobs	partial (state, run_after, id) where state='queued'	Claim query stays small as done rows accumulate
audit_log	(workspace_id, created_at desc)	Only access pattern; BRIN if it exceeds 50M rows
ai_usage	(workspace_id, created_at desc)	Meter rollups
api_keys	unique (prefix)	Auth lookup by prefix, then argon2 verify

Rule: no index gets added without an `EXPLAIN (ANALYZE, BUFFERS)` in the migration's PR body. [inference] At 10,000 workspaces the whole control plane is well u

92.7 Migrations — Drizzle Kit, SQL-first

Pick: `drizzle-orm` **0.45.2** + `drizzle-kit` **0.31.10** [measured 2026-08-30 via registry.npmjs.org].

OPTION	VERDICT	WHY
Drizzle Kit	chosen	Generates plain <code>.sql</code> files into <code>db/migrations/</code> with a journal; you can hand-edit them, so RLS policies, partial index constraints live in the same file as the table. Zero runtime engine. TS types for the query layer.
Prisma	rejected	<code>prisma@8.0.0-rc.12</code> is a release candidate as of 2026-08-30 [measured]. Its migrate flow does not round-trip RLS policies or roles, so pol
Hand-rolled SQL runner	rejected	~200 lines to write and then own, for a journal file Drizzle already gives us.

Hard rules (`db/CONVENTIONS.md`):

- 1. Tables are declared in `db/schema.ts`. RLS policies, functions, grants, partial indexes, and opclass indexes are hand-written SQL appended to the generated
- 2. Migrations run against `DATABASE_URL_DIRECT`. Never the pooler.
- 3. Expand-contract only. A deploy may add a nullable column and backfill; a *later* deploy makes it `NOT NULL`; a *later still* deploy drops the old one. No migra
- 4. Every migration adding a table must, in the same file, `enable + force` RLS and create at least one policy. CI check: `select relname from pg_class c join pg_name`
- 5. Preview per PR on a Neon branch (\$1.50/branch-month prorated hourly [fetched 2026-08-30]), destroyed on merge.

92.8 Read replicas — not yet, and the trigger

No replica at launch. A replica adds a second failure surface, replication lag that will show up as "I just saved and it's gone", and cost, in exchange for **Escalation order, in this sequence:** (1) move the `derived` database onto its own Neon compute so search scans stop competing with auth lookups; (2) add a reac **Trigger to start step 1:** primary CPU above 60% for a sustained hour, or p95 control-plane query time above 50 ms for a day. **Trigger for step 2:** any read que

92.9 Derived stores — where each lives, and the rebuild command

Every derived store must have a **single idempotent command that rebuilds it from git with the service running**. If it does not, it is not derived; it is undec

STORE	LIVES IN	HOLDS	REBUILD
Search index	Postgres <code>derived</code> DB, table <code>search.doc</code> - (workspace_id, doc_path) PK, <code>tsv tsvector</code> , <code>heading_text text</code> , <code>snippet text</code> check (length(snippet) <= 512)	Derived tokens + bounded snippets, never full bodies	<code>pnpm db:re</code> - walks tl search.doc
Render cache	R2 <code>frontmatter-artifacts/render/{workspace}/{content_sha}.html</code>	HTML fragments keyed by content hash	Nothing to
Certificate sidecars	R2 <code>.../cert/{workspace}/{content_sha}.json</code> , pointer row in <code>derived</code>	Degradation certificates per (document, target engine)	<code>pnpm cert:</code> - determin
Attachments > 1 MB	R2 <code>.../blob/{workspace}/{sha256}</code>	User bytes	Not deriv

Content-addressing every derived key by `content_sha` is what makes invalidation a non-problem: a changed file produces a new key, and the old one ages out vi **The standing drill, quarterly:** `truncate search.doc` on production, run `db:reindex` for the largest workspace, and record wall-clock. If a full re-index of the

93.1 Current state, measured

WHAT	WHERE
next-auth <code>^5.0.0-beta.31</code> , JWT sessions	<code>src/modules/auth/infrastructure/auth-options.ts</code> , <code>src/auth.ts</code>
GitHub OAuth provider, <code>scope: "read:user"</code>	same file
Credentials provider <code>sgnk-password</code> (script hash in <code>SGNK_AUTH_HASH</code>)	same file + <code>src/modules/auth/infrastructure/password.ts</code>
Firebase <code>^12.16.0</code> Google popup auth	<code>src/modules/auth/infrastructure/firebase-auth-gateway.ts</code> , <code>src/shared/infrastructure/firebase/client.ts</code> , <code>src/modules/auth/application/ports.ts</code> , <code>src/modules/auth/domain/auser.ts</code> , <code>src/container/client-container.ts</code> , <code>test/auth/firebase-auth-gateway.test.ts</code> , <code>firebase.json</code> , <code>src/config/env.ts:18-215</code>
Single global PAT <code>GITHUB_REPO_TOKEN</code>	<code>src/shared/infrastructure/github/client.ts:30</code> , <code>src/modules/repository/infrastructure/github-writer.ts:33</code>
Hardcoded allowlist default <code>"sagnikmitra"</code> <code>ALLOWED_GH_LOGIN</code>	<code>src/config/env.ts</code>
Route middleware	none exists

[measured 2026-08-30, by reading the repo] There are three live identity paths, not two: Firebase Google, GitHub OAuth, and a fixed-username password. Three Deletion is a single PR and it must land before any multi-tenant table exists, because every one of these paths would otherwise need a `workspace_id` story.

93.2 The stack

CONCERN	PICK	VERSION [FETCHED 2026-08-30]	REJECTED	WHY
Session + identity	next-auth v5 (Auth.js)	<code>5.0.0-beta.32</code>	Clerk, WorkOS, Better Auth	Already integ:
Repo access	GitHub App, installation tokens	<code>@octokit/auth-app 8.3.0</code> , <code>@octokit/rest 22.0.1</code>	OAuth App <code>repo scope</code>	§93.3
JWT verify (API keys, Tauri)	<code>jose</code>	<code>6.2.10</code>	<code>jsonwebtoken</code>	Web Crypto na
Secret custody	AES-256-GCM envelope, KEK in platform secret store	Node <code>crypto</code>	AWS KMS from day one	§93.7
Tenant isolation	Postgres RLS, pooled, <code>set_config(..., local)</code>	-	<code>app-layer</code> where <code>workspace_id = ?</code>	§93.9

What would change our mind on next-auth: the v5 beta staying beta past Q1 2027, or a second SSO protocol (SAML) landing as a B2B requirement – WorkOS is the

93.3 The GitHub App, and why not the OAuth `repo scope`

One App does both jobs: user-to-server OAuth for *login*, installation tokens for *repo bytes*. The OAuth App disappears.

Repository permissions requested (exact keys from the `app-permissions` schema in GitHub's OpenAPI description) [fetched `raw.githubusercontent.com/github/rest-api-c`

PERMISSION KEY	LEVEL	WHY WE NEED IT	ENDPOINT?
<code>contents</code>	<code>write</code>	The entire product. Read file bytes, write splice results, create branches, read blob SHAs for compare-and-swap	<code>GET/PUT /repos/{c</code>
<code>metadata</code>	<code>read</code>	Mandatory – GitHub force-grants it whenever any other permission is requested	<code>GET /ins</code>
<code>pull_requests</code>	<code>write</code>	Phase 2 only. Review-branch publishing for B2B teams. Not in v1	<code>/pulls</code>
<code>workflows</code>	<i>not requested</i>	Refused deliberately	-
<code>administration</code>	<i>not requested</i>	Would allow repo deletion and collaborator changes. Never	-
<code>single_file</code>	<i>not requested</i>	Evaluated and rejected: scopes to one path, and a vault is a tree	-

Account permissions: `email addresses: read` (verified billing contact only). **Webhook events subscribed:** `installation`, `installation_repositories`, `push`. Nothing (`workflows: write` is the sharpest line. If a user's vault contains a file under `.github/workflows/`, the engine **refuses the write** and surfaces "this path requir

Why not OAuth `repo`: classic `repo` is all-or-nothing across *every* repository the user can reach, including private org repos we will never open, and it carr

The cost, stated honestly: the App install screen converts worse than an OAuth consent, and installing into an org may require owner approval – so we must b

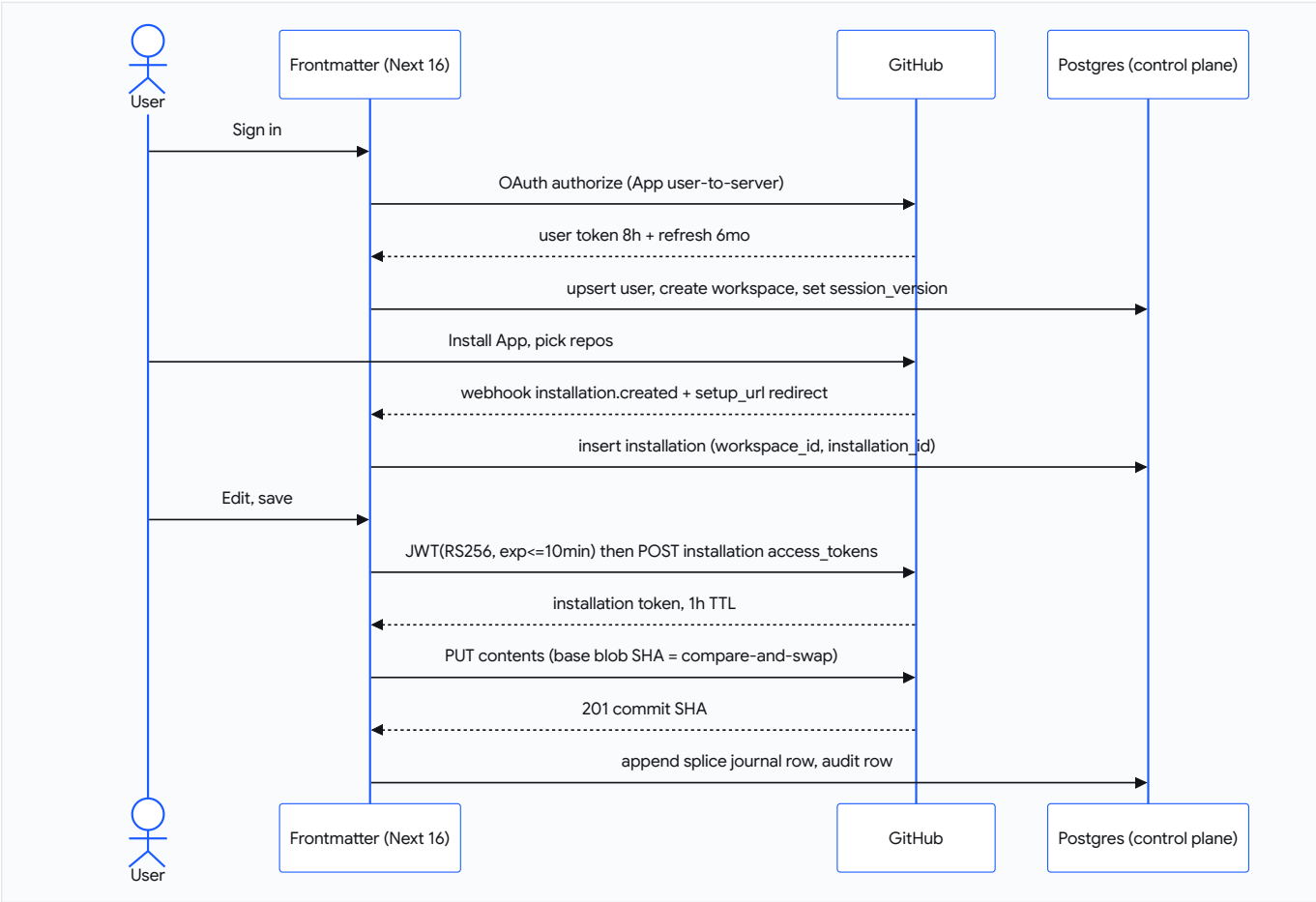
What would change our mind: nothing. This is the one decision where the alternative is disqualifying rather than merely worse.

App manifest, committed at `infra/github-app-manifest.json` and used with GitHub's App-from-manifest flow so the App is reproducible per environment:

```
{
  "name": "Frontmatter",
  "url": "https://frontmatter.app",
  "hook_attributes": { "url": "https://frontmatter.app/api/webhooks/github", "active": true },
  "redirect_url": "https://frontmatter.app/api/auth/callback/github",
  "callback_urls": ["https://frontmatter.app/api/auth/callback/github", "http://127.0.0.1:8976/callback"],
  "setup_url": "https://frontmatter.app/onboarding/installed",
  "setup_on_update": true,
  "public": true,
  "request_oauth_on_install": false,
  "default_permissions": { "contents": "write", "metadata": "read", "emails": "read" },
  "default_events": ["installation", "installation_repositories", "push"]
}
```

`request_oauth_on_install: false` is deliberate: identity comes first, installation second, so we always have a `user_id` to attach the installation to. The `127.0`

93.4 Install to first commit



The dual signal – webhook *and* `setup_url` redirect – matters: the redirect can be lost (user closes the tab), and the webhook can arrive before the redirect.

93.5 Token storage and refresh

TOKEN	TTL [FETCHED GITHUB DOCS 2026-08-30]	STORED WHERE
App JWT (RS256)	$\leq 10\text{ min}$ (<code>exp</code> $\leq \text{now}+600$, <code>iss</code> = client ID)	Never stored – minted per call
Installation access token	1 hour	In-process LRU keyed by <code>installation_</code>
User access token	8 hours	<code>identity.oauth_token</code> , encrypted (\$93.
Refresh token	6 months, single-use – using it invalidates the old refresh <i>and</i> the old access token	same row, encrypted
Frontmatter session JWT	30 days, rolling 24h	<code>__Secure-authjs.session-token</code> cookie
Device refresh token (Tauri)	90 days	OS keychain, <code>device_session</code> row

The App private key is a PEM in `GITHUB_APP_PRIVATE_KEY` (base64, single line – the multiline PEM breaks Vercel env parsing). The single-use refresh token is `tf`
Rate limits: an installation gets a minimum of **5,000 requests/hour**, plus 50/hour per repo above 20 and 50/hour per user above 20, hard-capped at **12,500/hour**

93.6 Sessions: PWA and Tauri shell

Web/PWA. JWT strategy, no database session lookup on the happy path. The cookie carries only `{ sub, wid, ver }`. GitHub tokens never enter the cookie.

```
// src/modules/auth/infrastructure/auth-options.ts
session: { strategy: "jwt", maxAge: 60 * 60 * 24 * 30, updateAge: 60 * 60 * 24 },
useSecureCookies: true,
cookies: {
  sessionToken: {
    name: "__Secure-authjs.session-token",
    options: { httpOnly: true, sameSite: "lax", path: "/", secure: true },
  },
},
```

Revocation. A JWT cannot be revoked, so `ver` in the token is compared against `users.session_version`; bump it on member removal, App uninstall, or break-glass
Tauri v2 desktop. No cookies: the webview origin is `tauri://localhost` (macOS/Linux) or `http://tauri.localhost` (Windows), so a `__Secure-` cookie for `frontmatter.`

PIECE	PICK	VERSION [FETCHED CRATES.IO 2026-08-30]
Loopback OAuth listener	tauri-plugin-oauth	2.1.0
Deep-link callback (frontmatter://)	tauri-plugin-deep-link	2.4.9
Token at rest	keyring (Keychain / Credential Manager / Secret Service)	4.2.0
Rejected	tauri-plugin-stronghold 2.3.1 - needs its own unlock password, duplicating what the OS already does	-

Flow: PKCE with `code_challenge_method=S256`, loopback on `127.0.0.1:8976`, exchange at `/api/auth/device/exchange`, refresh token into the keychain, access token in `localStorage`. `next-auth` guards its own routes. Every other mutating route gets an `Origin` allowlist in `src/middleware.ts`: `https://frontmatter.app`, `tauri://localhost`, `https://frontmatter.app`.

93.7 BYO API key custody

Rule: the key store is not the document store, and they are not reachable from the same database role. Documents live in the user's git repo (zero bytes in

```
create schema secrets;
revoke all on schema secrets from public, frontmatter_app;
grant usage on schema secrets to frontmatter_secrets;

create table secrets.workspace_dek (
  workspace_id uuid primary key,
  wrapped_dek bytea not null,
  kek_id text not null, -- 'env:v1' | 'kms:arn:...'
  wrap_alg text not null default 'AES-256-GCM',
  created_at timestamptz not null default now()
);

create table secrets.provider_key (
  id uuid primary key default gen_random_uuid(),
  workspace_id uuid not null,
  provider text not null, -- 'anthropic' | 'openai' | 'groq' | ...
  ciphertext bytea not null, -- AES-256-GCM under the workspace DEK
  iv bytea not null,
  auth_tag bytea not null,
  last4 text not null, -- display only
  created_by uuid not null,
  revoked_at timestamptz
);
```

Envelope: KEK wraps a per-workspace DEK; the DEK encrypts each provider key with a fresh 96-bit IV and AAD = `workspace_id || provider`, so a ciphertext moved b
KEK location - pick: a 32-byte value in the platform secret store (`FM_KEK_V1`), behind a `KeyWrapper` port with two adapters, `EnvKeyWrapper` and `KmsKeyWrapper`.

93.8 The permission model: intersect

Decision: intersect. Not defer, not duplicate.

CAPABILITY	AUTHORITY	RULE
Read a file	GitHub	Defer - if the installation lacks <code>contents</code> , we refuse
Write a file	GitHub n Frontmatter role	Intersect - needs <code>contents: write</code> and role $\geq$ <code>editor</code>
Invite a member	Frontmatter only	Ours - GitHub has no opinion
Publish to a public slug	Frontmatter only	Ours - slug uniqueness is a control-plane fact
Spend workspace AI budget	Frontmatter only	Ours
Change billing	Frontmatter only	Ours, <code>owner</code> role

Pure **defer** fails because GitHub has no concept of "may publish this document at `/p/acme-handbook`" or "may spend \$40 of AI credit" - those exist nowhere but o

```
// src/modules/authz/domain/effective.ts
export function can(cap: Capability, ctx: AuthzContext): boolean {
  const roleOk = ROLE_GRANTS[ctx.role].includes(cap);
  if (!roleOk) return false;
  if (ctx.apiKeyScopes && !ctx.apiKeyScopes.includes(SCOPE_FOR[cap])) return false;
  const ceiling = PROVIDER_CEILING[cap]; // undefined = product-only
  if (ceiling === undefined) return true;
  return ctx.providerPermission !== null && ctx.providerPermission >= ceiling;
}
```

Provider permission is cached in `workspace_repo_access(workspace_id, repo_id, permission, checked_at)` with a 10-minute TTL, revalidated immediately on any GitHub 46

93.9 RLS in Postgres

Pooled RLS from day one, transaction-scoped so a pooled connection cannot leak tenancy:

```
create role frontmatter_app login; -- NOT the table owner, NO bypassrls
alter table workspace_member enable row level security;
alter table workspace_member force row level security;
create policy tenant_isolation on workspace_member
  using (workspace_id = current_setting('app.workspace_id', true)::uuid)
  with check (workspace_id = current_setting('app.workspace_id', true)::uuid);
```

```
// src/shared/infrastructure/db/with-tenant.ts
export async function withTenant<T>(wid: string, actor: string, fn: (tx: Tx) => Promise<T>) {
  return db.transaction(async (tx) => {
    await tx.execute(sql`select set_config('app.workspace_id', ${wid}, true),
                                set_config('app.actor_id', ${actor}, true)`);
    return fn(tx);
  });
}
```

The third argument `true` makes the setting **local to the transaction** - that is what makes this safe behind PgBouncer transaction pooling, and it is the single

Two CI gates, both blocking, both new (the repo has no CI at all today):

```
-- gate 1: every public table has RLS on
select relname from pg_class c join pg_namespace n on n.oid = c.relnamespace
  where n.nspname = 'public' and c.relkind = 'r' and not c.relrowsecurity;
-- gate 2: the app role cannot bypass
select 1 from pg_roles where rolname = 'frontmatter_app' and rolbypassrls;
```

Both must return zero rows. Plus one integration test that seeds two workspaces, sets `app.workspace_id` to A, and asserts every table returns zero of B's rows

93.10 API keys for third-party access

Format `fm_live_<16-char key id>_<43-char base64url secret>` (`fm_test_` in non-production). The distinctive prefix is what makes the key eligible for GitHub's secret

PROPERTY	CHOICE	WHY
Stored value	<code>sha256(secret)</code> , compared with <code>timingSafeEqual</code>	Keys are 256 bits of entropy, so a slow KDF buys nothing and costs less
Lookup	by <code>key_id</code> prefix, indexed	Avoids a table scan; the secret half is never indexed
Shown	once, at creation	Plus <code>last4</code> for display
Scopes	<code>docs:read</code> , <code>docs:write</code> , <code>publish:write</code> , <code>search:read</code> , <code>webhooks:manage</code>	Intersected in <code>can()</code> (\$93.8)
Ceiling	creating member's role at issue time, re-checked at use	A key cannot outlive or out-rank its creator
<code>last_used_at</code>	bucketed to 60 s	A write per request would be the busiest write in the system
Rate limit	per key, sliding window	Separate bucket from session traffic

Keys authenticate to `/api/v1/*` via `Authorization: Bearer fm_live_...`. They never grant access to the AI proxy with the workspace's BYO provider key - that would

93.11 Build order

#	WORK	GATE
1	Delete Firebase + Credentials provider, 8 files listed in §93.1	<code>rg -i firebase src/</code> returns nothing; 1,575 tests still green
2	Stand up CI (none exists) – typecheck, lint, test, <code>npm run arch</code>	Required check on <code>main</code>
3	GitHub App from manifest, <code>@octokit/auth-app</code> 8.3.0, delete <code>GITHUB_REPO_TOKEN</code>	Install-to-first-commit E2E against a scratch repo
4	Postgres control plane + RLS + <code>withTenant</code>	The two SQL gates, plus cross-tenant integration test
5	<code>secrets</code> schema, envelope encryption, <code>KeyWrapper</code> port	Ciphertext moved between rows fails to decrypt
6	Tauri PKCE + keychain, <code>device_session</code>	Revoke-one-device test
7	API keys + scopes	Scope intersection test; scanning-partner application

§94 Live editing, presence and collaboration

94.1 The tension, stated precisely, and its resolution

The positioning says "Google Docs for markdown." The architecture says the file is the only source of truth and sync is git three-way merge + splice journal

The resolution is one invariant:

A CRDT may exist as a session-scoped scratch buffer. It is seeded from file bytes, it is authoritative for nobody, and its single output is one spl.

Byte-identity is unassertable across a CRDT merge because a merge yields a *document state*, not a byte sequence with a provenance chain you can certify. That

PROPERTY	STORE OF RECORD (REFUSED)	EPHEMERAL SESSION (ACCEPTED)
Where document bytes live	CRDT doc, forever	Git repo, always
What the CRDT holds	The document	≤1 editing session, discardable
Lifetime	Permanent	Last participant leaves + 60s
Failure mode if lost	Data loss	Lose the unflushed tail, same as an unsaved buffer
Output	The document	One <code>Splice{base_sha, from, to, insert}</code>
Byte-identity assertion	Impossible	Asserted by the engine at flush, unchanged
Journal row	None	Exactly one per flush, identical shape to a solo edit

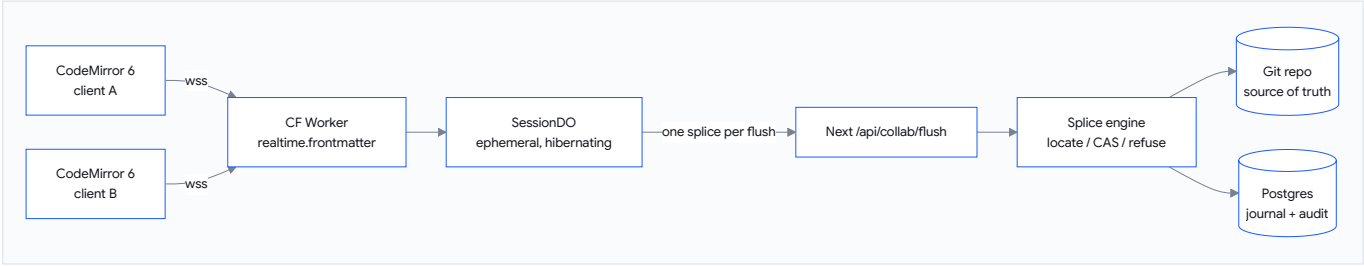
94.2 What "live editing" actually means here — demanded vs assumed

Separate these, because they have wildly different cost and risk.

CAPABILITY	USERS DEMAND IT	USERS ASSUME IT EXISTS	COST TIER	SHIPS
See who else has this doc open	High	High	Tier 0	v1
See where their cursor / selection is	Medium	High	Tier 0	v1
"Priya has unsaved changes in this file" warning before you edit	Very high	Medium	Tier 0	v1
Never silently lose my work to someone else's save	Absolute	Total	Tier 1	v1
Comments / mentions anchored to a range	High (B2B)	High	Tier 1	v1.5
Two people typing in the same paragraph, same second	Low	High	Tier 2	v2, B2B-gated
Live AI agent editing alongside a human	Medium (differentiator)	Low	Tier 2	v2

The gap between column 1 and column 2 on the last-but-two row is the whole finding. Simultaneous character-level co-typing is *assumed* to exist and *rarely us* Precedent, from primary sources: Notion is not CRDT-based for text. Its client batches operations into transactions "committed (or rejected) by the server a

94.3 Topology



The arrow from `DO` to `API` is one-directional and low-frequency. There is no arrow back into the DO carrying document state, and that absence is the design.

94.4 Transport: Cloudflare Durable Objects with WebSocket Hibernation

Next.js Route Handlers on Vercel cannot hold a WebSocket, so the realtime plane is a separate deployable regardless of vendor. Given that, put it where the **Pick**: one Cloudflare Worker + one Durable Object class, `SessionDO`, SQLite-backed, using the **WebSocket Hibernation API**.

Cloudflare numbers, all [fetched 2026-08-30]:

DIMENSION	FREE	PAID	SOURCE PAGE LAST-UPDATED
Workers Paid base	–	\$5/mo account minimum, no egress/bandwidth charge	Workers pricing, 2026-08-28
DO requests	100k/day	1M/mo included, then \$0.15/M	DO pricing, 2026-08-25
Incoming WS messages → requests	20:1 billing ratio	same	DO pricing
Outgoing WS messages, protocol pings	free	free	DO pricing
DO duration	13,000 GB-s/day	400,000 GB-s/mo included, then \$12.50/M GB-s	DO pricing
Memory billed per DO	128 MB regardless of use	same	DO pricing, fn.5
Hibernation-eligible idle time	not billed	not billed	DO pricing
<code>setWebSocketAutoResponse()</code> ping/pong	no wall-clock charge	same	DO pricing, fn.3
DO SQLite rows written	100k/day	50M/mo included, then \$1.00/M	DO pricing
WS message size	32 MiB	32 MiB	DO limits, 2026-06-01
Soft throughput per single DO	~1,000 req/s	~1,000 req/s	DO limits
Objects per namespace	unlimited	unlimited	DO limits

The single load-bearing fact. `accept()` on a WebSocket bills wall-clock time for the entire connection; hibernation does not. Derived, showing the work:

- Non-hibernating: 1 DO-hour = 3600 s × 0.125 GB = **450 GB-s**. Included 400,000 GB-s ÷ 450 = **888 free DO-hours/mo**; beyond that 450 × \$12.50/1e6 = **\$0.005625 pe**
- Hibernating: billed only for handler execution. At ~1 ms wall time per awareness message, 1 message = 0.001 × 0.125 = **0.000125 GB-s**. [derived]

SCALE	ASSUMPTION	NON-HIBERNATING
100 users	20 live-h/user/mo, 1 room each = 2,000 DO-h; 2 msg/s → 144k msg/user/mo	2,000×450 = 900k GB-s → 500k billable → \$12.50 + \$5 = \$17.50/mo
10,000 users	same per-user shape	200,000 DO-h × 450 = 90M GB-s → 89.6M billable → \$1,120 + \$5 = \$1

Hibernation is a **70× cost difference at 10,000 users** [derived]. Therefore: `state.acceptWebSocket()`, never `ws.accept()`; `setWebSocketAutoResponse()` for heartbeats **Alternatives rejected**:

OPTION	WHY REJECTED
SSE + POST on Vercel	Works for Tier 0 fanout, but needs a second upstream channel, has no per-room single-threaded actor, and burns a Vercel
Raw WebSocket server on Fly/Railway	A stateful process one person must be on call for. Contradicts "operable by one person."
Vercel Route Handler WebSocket	Not supported.

94.5 CRDT selection: Yjs, and only inside Tier 2

All [measured 2026-08-30] by streaming the npm tarball and piping through `gzip -9`:

PACKAGE	LATEST	PUBLISHED	RUNTIME ARTIFACT	RAW	GZIP -9
yjs	13.6.32	2026-08-04	dist/yjs.mjs	299,797 B	62,586 B
loro-crdt	1.15.1	2026-08-29	web/loro_wasm_bg.wasm	3,179,730 B	1,046,171 B
@automerge/automerge	3.4.1	2026-08-12	web/automerge_wasm_bg.wasm	3,571,259 B	1,116,734 B
y-codemirror.next	0.3.6	2026-08-18	-	-	-
y-protocols	1.0.7	2025-12-16	-	-	-
lib0	0.2.117	2025-12-30	-	-	-
y-websocket	3.1.0	2026-08-06	-	-	-

Yjs core is 16.7× smaller gzipped than Loro's wasm and 17.8× smaller than Automerge's [derived]. The full y-stack (yjs + lib0 + y-protocols + y-codemirror. Non-negotiable: the entire y-stack is behind await import() inside the Tier 2 code path. A D2C solo user on the free plan downloads zero bytes of CRDT.

Rejected build-partners:

VENDOR	COST AT 10,000 USERS (20 H/USER/MO)
Liveblocks	\$0.002/collab-minute × 12M min = \$24,000/mo; even 100 users = 120k min - 3k free = \$234 - \$30 credits + \$25 Pro ≈ \$229/mo
Ably	\$29 Standard + messages 1.44B × \$2.50/M = \$3,629/mo; connection-min \$12 + channel-min \$12 [fetched ably.com/pricing 2026-
Pusher	48M msg/day → Plus tier \$899/mo [fetched pusher.com/channels/pricing 2026-08-30, derived]
PartyKit	partykit@0.0.115 last published 2025-05-21 - 15 months stale [measured, npm registry, 2026-08-30]

Hocuspocus (@hocuspocus/server 4.6.0, 2026-08-10) Self-host cost only

Verdict: build on Durable Objects. ~\$5-16/mo covers both 100 and 10,000 users; the cheapest managed alternative is ~\$229/mo at 100 users and ~\$900-24,000/mo

94.6 Presence and the awareness protocol

Tier 0 does not use y-protocols/awareness - that would drag lib0 into the default bundle. Define our own, ~40 lines, and let Tier 2 tunnel Yjs awareness thr

src/lib/collab/protocol.ts:

```
export type ClientMsg =
  | { t: 'hello'; doc: string; base: string /* blob sha1 */ }
  | { t: 'pos'; base: string; head: number; anchor: number } // byte offsets
  | { t: 'dirty'; base: string; dirty: boolean }
  | { t: 'y'; b64: string; } // Tier 2 only
export type ServerMsg =
  | { t: 'roster'; peers: Peer[] }
  | { t: 'pos'; wsid: string; base: string; head: number; anchor: number }
  | { t: 'dirty'; wsid: string; dirty: boolean }
  | { t: 'splice'; base: string; next: string; from: number; to: number; ins: number }
  | { t: 'y'; b64: string; }
export interface Peer { wsid: string; uid: string; name: string; color: string; dirty: boolean }
export const POS_THROTTLE_MS = 120; // ≈8/s ceiling
export const HEARTBEAT_MS = 25_000; // handled by setWebSocketAutoResponse
export const MAX_PEERS = 25; // refuse beyond; DO soft cap is ~1k req/s
```

The cursor-mapping rule. A cursor is a byte offset into a specific version. A peer's offset is meaningless against a different base. Rule:

- msg.base === myBase → render the caret at character granularity.
- msg.base !== myBase and the intervening splices are in the local journal cache → map through ChangeSet.mapPos per splice, in order, and render.
- Otherwise → degrade, do not guess: drop the caret and show the peer's avatar on the file row in the tree. Same refusal discipline as the engine.

src/lib/collab/cm-presence.ts is a CodeMirror 6 StateField + ViewPlugin holding decorations; remote carets are widget decorations, remote selections are mark

Files to create:

PATH	CONTENTS
workers/realtime/wrangler.jsonc	DO binding SESSION, new_sqlite_classes: ["SessionDO"], limits.cpu_ms left at default
workers/realtime/src/index.ts	Upgrade: websocket check, EdDSA JWT verify, DO id derivation, stub.fetch()
workers/realtime/src/session-do.ts	SessionDO: acceptWebSocket, setWebSocketAutoResponse, roster, fanout, flush alarm
src/app/api/collab/token/route.ts	Mints a 5-minute EdDSA JWT: { workspace_id, doc_path, uid, caps: ['presence','coedit'], base }
src/app/api/collab/flush/route.ts	Server-to-server, HMAC-signed from the Worker; calls the existing splice engine
src/lib/collab/{protocol,presence-client,cm-presence}.ts	Client

Tenancy. The DO id is idFromName(sha256(workspace_id + ':' + repo_id + ':' + path)). Omitting workspace_id from that hash is a cross-tenant room collision - the re

94.7 Conflict surfacing when live editing is off (the v1 path)

This is the feature, not the fallback. Three outcomes, no fourth:

SERVER STATE AT SAVE	UI
base_sha matches HEAD	Silent save
Moved, but git three-way merges cleanly and the splice range is untouched	Toast: "Rebased onto Ana's change" + Undo
Moved and the splice range overlaps, or merge conflicts	Blocking three-pane diff: base / yours / theirs, with the exact byte ranges h

Tier 0 makes the third row rare before it happens: the moment a second person opens a file someone else has dirty: true on, they get an inline banner - "Ana

94.8 Tier 2 ephemeral session, when it ships

```
open → GET /api/collab/token (caps:['coedit'])
      → D0 has no Y.Doc → D0 fetches bytes via /api/collab/seed at blob_sha S
      → new Y.Doc(); ytext.insert(0, bytes.toString('utf8')); baseSha = S
edit  → y updates fan out to peers; D0 holds the doc in memory only
flush → every 10s of quiescence, on 60s max age, or on last-leave:
      next = ytext.toString()
      splice = minimalSplice(seedBytes, Buffer.from(next,'utf8')) // single range
      POST /api/collab/flush { workspace_id, path, base_sha: S, splice }
      → engine: locate → CAS(S) → journal append → commit
      → on 200: reseed baseSha = newSha, seedBytes = new bytes
      → on 409: broadcast {t:'refused'}; freeze the room read-only; hand every
                participant the §94.7 three-pane diff. Never auto-merge.
close → last participant leaves + 60s → alarm → final flush → discard Y.Doc
```

Non-negotiables, enforceable as review rules:

- 1. `SessionD0` never calls `ctx.storage.put` / `sql.exec` with document text. Only ≤2 KB per-connection attachments.
- 2. `minimalSplice` returns **one** `{from,to,insert}`. If a common-prefix/common-suffix reduction yields a range covering >60% of the file, treat it as a rewrite and
- 3. UTF-8 only across the boundary: Yjs indexes UTF-16 code units, the engine addresses bytes. Convert at exactly one place, in `minimalSplice`, and unit-test it
- 4. If the seed bytes are not valid UTF-8, or the file carries a BOM or bare CR line endings that the engine's normalization would alter, **refuse co-editing** or

94.9 v1 verdict

v1 needs Tier 0 and Tier 1. It does not need Tier 2.

ITEM	V1	WHY
Presence roster, avatars, dirty flag	Yes	~1.5 weeks; \$5/mo; delivers most of the perceived "Google Docs" feel; touches zero document bytes
Remote cursors with the degrade rule	Yes	Rides the same socket; the degrade rule keeps it honest
Soft-lock banner + CAS three-pane conflict UI	Yes	This is the <i>actual</i> demanded feature. Without it, "your repo is the source of truth" reads as "you'r
Live character-level co-editing	No	~4–6 weeks, the UTF-8 boundary is the highest-risk code in the product, and it is the least-used de
Comments / mentions	v1.5	Anchored to the journal, not to a CRDT; sells B2B; independent of Tier 2

Market positioning stays intact: Obsidian ships shared vaults with no character-level co-editing at all and charges \$4–8/user/mo for it [fetched 2026-08-30]

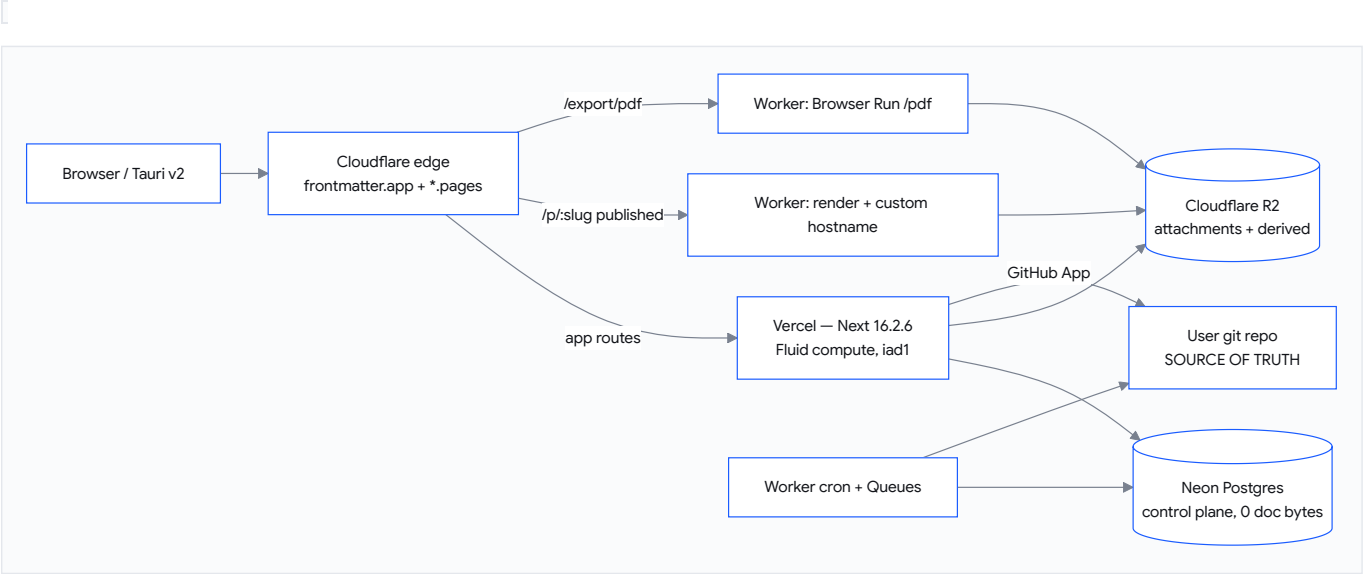
What would move Tier 2 into v1: three or more B2B pilots naming simultaneous editing as a blocker in writing. Not a survey – a lost deal.

What would make us abandon Tier 2 permanently: if `minimalSplice` cannot hold a byte-identity property test over the existing multi-byte and line-ending corpu:

]

§95 Hosting, deployment and environments

95.1 Topology



COMPONENT	CHOICE	REJECTED ALTERNATIVE	WHY
Next app	Vercel Pro, Fluid compute, region <code>iad1</code>	Self-hosted Next on Fly/Railway	Zero ops, instant rollback, preview
Compute rates	<code>iad1</code>	<code>bom1</code> (Mumbai)	<code>iad1</code> Active CPU \$0.128/hr vs <code>bom1</code>
Postgres	Neon, <code>aws-us-east-1</code>	Supabase, RDS	Scale-to-zero after 5 min, per-branc
Object store	Cloudflare R2	S3, Vercel Blob	Free egress. Vercel Blob data transf
Edge / published pages / PDF / jobs	Cloudflare Workers Paid	Vercel middleware + Vercel Cron	Workers cron gets 15 min CPU per inv
PDF/Chromium	Cloudflare Browser Run <code>/pdf</code>	<code>@sparticuz/chromium</code> in a Vercel function (current state)	See \$95.7
Custom domains	Cloudflare for SaaS custom hostnames	Vercel domains API	Keeps published-page egress on R2/Wo
CI	GitHub Actions, <code>ubuntu-latest</code>	None (current state: no CI at all)	1,575 tests exist and nothing runs t

95.2 Why not one VPS, and why not Kubernetes

Not a single VPS (\$6–24/mo, cheaper than the \$25 floor below). One founder on call cannot be the pager for TLS renewal, kernel patching, log rotation, disk-
Not Kubernetes (EKS control plane alone is ~\$73/mo before a single node). It buys workload portability we do not need – we have one Next app and three Worke
What would change our mind: if Vercel's bill crosses roughly \$600/mo at steady state (see \$95.10: that is ~13,000 users on the un-optimised model), the Next

95.3 Regions and residency

CONCERN	POSITION
App compute	Single region <code>iad1</code> . Multi-region on Pro is available but doubles the cold-start surface and the Postgres round trip.
Postgres	<code>aws-us-east-1</code> . Neon region is fixed at project creation and cannot be changed – moving means create-new-project-and-migrate [fetched]. Pick once, de
Document bytes	Never leave the user's git repo. This is the residency answer that actually matters and it is free: a German customer's documents live w/
Attachments	R2 bucket per residency zone, chosen at workspace creation, stored as <code>workspaces.r2_region</code> . Default <code>auto</code> ; EU customers get an EU-hinted
DPDP / GDPR	The control plane holds identity, entitlements, slugs, audit – personal data, not document content. One Postgres in us-east-1 with SCCs :

What would change our mind: the first paying EU enterprise with a written residency clause. Until then a second region is speculative cost.

95.4 Environments

	LOCAL	PREVIEW	STAGING	PRODUCTION
Trigger	<code>npm run dev</code>	every PR	merge to <code>main</code>	manual promote from
URL	<code>localhost:3000</code>	<code><hash>-frontmatter.vercel.app</code>	<code>staging.frontmatter.app</code>	<code>frontmatter.app</code>
Postgres	Neon branch <code>dev/<you></code>	ephemeral Neon branch per PR , deleted on close	Neon branch <code>staging</code>	Neon <code>main</code>
R2 bucket	<code>fm-local</code>	<code>fm-preview</code> (shared, prefixed by PR number)	<code>fm-staging</code>	<code>fm-prod</code>
GitHub App	<code>frontmatter-dev</code>	<code>frontmatter-dev</code>	<code>frontmatter-staging</code>	<code>frontmatter</code>
Auth callback	<code>http://localhost:3000/api/auth/callback/github</code>	wildcard on the App's dev callback list	staging host	prod host
Protection	–	Vercel deployment protection ON	ON	public

How previews get data. Not by copying production. On PR open, CI calls `neon branches create --parent main --name pr-$PR`; the branch is copy-on-write, costs \$1.50
Config lives in `.github/workflows/preview-db.yml` and the branch name is passed to Vercel as a per-deployment env var via `vercel env add DATABASE_URL preview --git-t`
What would change our mind: nothing. Pointing preview at production is how a solo founder deletes a customer's row at 2am.

95.5 The 4.5 MB response cap — this already bites

"The maximum payload size for the request body or the response body of a Vercel Function is 4.5 MB. If a Vercel Function receives a payload in exce:

Two shipped routes are on the wrong side of it [measured, read from the repo]:

ROUTE	WHAT IT DOES NOW	FAILURE
<code>src/app/api/vault/snapshot/route.ts</code>	<code>return new Response(JSON.stringify(snapshot))</code> – the entire vault as one JSON body	A 3,000-note vault at 1.5 KB average is 4.5 MB of markdown before JSON escaj
<code>src/app/api/export/vault/route.ts</code>	returns a whole zip through the function	Same wall, at a smaller vault, because zip-of-markdown still exceeds 4.5 MB

The settled architecture already says *stop shipping whole-vault snapshots*. The cap is the enforcement mechanism. Replacement contract:

- `GET /api/vault/index` → paths, sizes, SHAs, mtimes only. Bounded: 3,000 notes × ~120 bytes = 360 KB.
 - `GET /api/vault/file?path=` → one file, byte-exact. Already exists.
 - `GET /api/export/vault` → enqueue a job; the Worker writes `exports/{workspace_id}/{job_id}.zip` to R2 and returns a presigned URL. **The function returns a URL, ne**
- Rule for every new route: if a response body can grow with vault size, it returns a cursor or a URL. Add this as a test in `test/` that asserts every `route.ts`

95.6 Cold starts

Fluid compute keeps an instance alive across requests and bills Active CPU only while your code runs, pausing during I/O [fetched] – so a warm instance amor

PACKAGE	ON-DISK SIZE
@sparticuz/chromium@148	4.6 MB
puppeteer-core@25.0.4	7.7 MB
bin/swiftshader.tar.br	3,498,775 B
bin/al2023.tar.br	1,075,752 B
bin/fonts.tar.br	183,831 B

Those brotli packs are decompressed into `/tmp` on every cold start before Chromium can launch [inference from the package's runtime design]. Mitigations, in

95.7 The PDF path cannot stay in a normal serverless function

Current state, from `vercel.json`:

```
"functions": {
  "src/app/api/export/pdf/[...path]/route.ts": { "memory": 1769, "maxDuration": 60 }
}
```

and from `next.config.ts`, a comment that is itself the evidence: the narrow `bin/**` glob "wasn't landing the binary in the lambda (runtime: `.../bin` does not exi

OPTION	COST
Keep @sparticuz/chromium on Vercel	1.77 GB memory × 8 s × \$0.0106/GB-hr ≈ \$0.000042/PDF, plus a fragile trace hack and a 12 MB cold start
Cloudflare Browser Run /pdf Quick Action	10 browser-hours/mo included on the \$5 Workers Paid plan, then \$0.09/hr [fetched]. At 6 s/PDF that is 6,000
Browserless / self-hosted Chromium container	\$50–200/mo and a container to babysit

Implementation: a Worker binds `browser` and POSTs the rendered HTML to the `/pdf` Quick Action; output goes to R2 at `derived/{workspace_id}/{doc_sha}.pdf`, keyed
What would change our mind: if `/pdf` cannot reproduce our KaTeX + Mermaid + custom-font output byte-for-byte against the golden fixtures, we keep Chromium o

95.8 Custom domains and wildcard TLS

CASE	MECHANISM	LIMIT
*.frontmatter.app (per-workspace published pages)	Cloudflare Universal SSL on our own zone	Covers apex + one subdomain level. <code>a.b.frontmatte</code> is not covered – enforce single-level slugs in t uniqueness constraint.
docs.customer.com (customer vanity domain)	Cloudflare for SaaS custom hostname, HTTP or delegated-TXT validation	100 hostnames included, \$0.10/hostname after, 50
*.customer.com	Not available. "Wildcard custom hostnames: Enterprise" [fetched]	Refuse the request explicitly in the UI with tha

Onboarding flow: customer adds a CNAME to `pages.frontmatter.app`; we POST the hostname to the Cloudflare for SaaS API, store `custom_hostname_id` + status on the

95.9 Rollback

LAYER	MECHANISM
Next app	Vercel Instant Rollback – promote the previous deployment; it is already built
Workers	<code>wrangler rollback --message "..."</code> to the prior version
Postgres schema	Expand/contract only. Never a destructive migration in the same deploy as the code that needs it: add column → deploy code that writes t
R2 objects	R2 has no object versioning. Rollback is impossible, so derived artifacts are content-addressed (<code>{doc_sha}.pdf</code>) and never overwritten; i
Document bytes	Nothing to roll back. The file is in the user's git repo; <code>git revert</code> is their rollback and it is better than ours

Instant Rollback only works if the previous deployment can still talk to the current database. That is the whole argument for expand/contract, and it is the

95.10 The bill

Model [inference, stated so it can be argued with]: per active user per month – 1,000 function invocations, 120 ms Active CPU each, 400 ms instance-alive at Vercel at 10,000 users [derived]: CPU 333.3 hr × \$0.128 = \$42.67; memory 2,222 GB-hr × \$0.0106 = \$23.56; invocations 10M × \$0.60/M = \$6.00; Fast Data Transf

LINE	100 USERS	1,000 USERS	10,000 USERS
Vercel Pro	\$20.00	\$20.00	\$327.22
Neon	\$0.00 (Free)	\$19.78	\$85.63
Cloudflare R2	\$0.00 (free tier)	\$1.05	\$11.85
Workers Paid (incl. Browser Run, Queues)	\$5.00	\$5.00	\$5.00
Cloudflare for SaaS hostnames	\$0.00	\$0.00	\$30.00
GitHub Actions	\$0.00	\$0.00	\$2.40
Total / month	\$25.00	\$45.83	\$462.10
Per user	\$0.250	\$0.046	\$0.046

Neon workings [derived]: 1,000 users = 180 CU-hr × \$0.106 + 2 GB × \$0.35 = \$19.78. 10,000 users = 730 CU-hr × \$0.106 + 15 GB × \$0.35 + 15 GB instant restore
The lever. At 10,000 users, Fast Data Transfer plus edge requests is \$255 of a \$462 bill – 55%. Serving `/p/:slug` published pages and hashed static assets fr
Excluded from every figure: AI inference (separately metered per \$on AI), payment processor fees, and the domain renewal.

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§96

CI/CD, version control and release

The repo has 223 TypeScript files, 25,214 lines under `src/`, 98 test files, and **zero CI** [measured 2026-08-30: `ls .github` → No such file or directory]. Every

[

96.1 What CI must protect

INVARIANT	GATE	COMMAND	FAIL MODE IF UNGATED
Byte-preserving splice never guesses	<code>test</code>	<code>npm run test</code> (1,575 tests)	Silent corruption of user files in a git repo
Layer direction (no infra imports in domain)	<code>arch</code>	<code>npm run arch</code>	Architecture rots in 3 weeks of agent commits
Every spec has an executable oracle	<code>spec</code>	<code>npm run spec</code>	Specs become prose
Foreign-vault fidelity corpus is byte-pinned	<code>corpus</code>	<code>npm run corpus</code>	The only cross-author fidelity claim becomes i
Editor bundle stays loadable on a 3G phone	<code>budget</code>	currently a stub that echoes and exits 0	Bundle creep, invisible
Types and lint	<code>typecheck</code> , <code>lint</code>	<code>tsc --noEmit</code> , <code>eslint . --max-warnings=0</code>	-

`npm run budget` today is `echo 'No bundle budget configured yet - skipping'` [measured]. A gate that always passes is worse than no gate, because it appears in the i

96.2 The pipeline

Two jobs, not seven. Runner minutes are billed per job-minute, so parallelism costs money; we buy wall-clock only where it pays.

`.github/workflows/ci.yml`

```
name: ci
on:
  push:
    branches: [main]
  pull_request:
    branches: [main]

concurrency:
  group: ci-${{ github.workflow }}-${{ github.ref }}
  cancel-in-progress: ${{ github.ref != 'refs/heads/main' }}

permissions:
  contents: read

env:
  NODE_VERSION: "24.6.0"
  NEXT_TELEMETRY_DISABLED: "1"

jobs:
  verify:
    name: verify (types, lint, tests, gates)
    runs-on: ubuntu-latest
    timeout-minutes: 15
    steps:
      - uses: actions/checkout@v7 # v7.0.1, published 2026-07-20 [fetched 2026-08-30]
        with:
          fetch-depth: 0 # corpus + reconcile need real history
      - uses: actions/setup-node@v7 # v7.0.0, published 2026-07-14 [fetched 2026-08-30]
        with:
          node-version: ${{ env.NODE_VERSION }}
          cache: npm
      - run: npm ci --no-audit --no-fund
      - run: npm run typecheck
      - run: npm run lint
      - run: npm run test -- --reporter=default --reporter=junit --outputFile=reports/junit.xml
      - name: arch report (layer direction, with scan floor)
        run: npm run arch | tee reports/arch.json
      - name: spec gate
        run: npm run spec
      - name: corpus oracle
        run: |
          node scripts/corpus-foreign.mjs fetch
          npm run corpus
      - if: always()
        uses: actions/upload-artifact@v7 # v7.0.1 [fetched 2026-08-30]
        with:
          name: verify-reports-${{ github.sha }}
          path: reports/
          retention-days: 14

  build:
    name: build + bundle budget
    runs-on: ubuntu-latest
    timeout-minutes: 15
    steps:
      - uses: actions/checkout@v7
      - uses: actions/setup-node@v7
        with:
          node-version: ${{ env.NODE_VERSION }}
          cache: npm
      - name: restore next build cache
        uses: actions/cache@v6 # v6.1.0 [fetched 2026-08-30]
        with:
          path: .next/cache
          key: next-${{ runner.os }}-${{ hashFiles('package-lock.json') }}-${{ github.sha }}
          restore-keys: |
            next-${{ runner.os }}-${{ hashFiles('package-lock.json') }}-
      - run: npm ci --no-audit --no-fund
      - run: npm run build
      - run: node scripts/ci/bundle-budget.mjs
```

Why this shape. We pick two jobs sharing one `npm ci` each, over a matrix of six single-command jobs. The alternative is prettier in the checks UI and roughly Runner budget [fetched 2026-08-30, GitHub billing docs]: private repos get 2,000 Linux minutes/month on GitHub Free, 3,000 on Pro and Team, 50,000 on Enterp

96.3 The arch report's minimum-files-scanned floor

The most important line in the whole pipeline is already in the repo:

```
// specs/harness/clean-architecture-report.mjs
const MIN_SCANNED_FILES = 50;
...
if (filesScanned < MIN_SCANNED_FILES) { process.exit(2); }
if (violations.length) process.exit(1);
```

The header documents the exploit it exists to prevent [measured, read from source]: the gate previously reported `{ "total": 0, "violations": [] }` and exit 0 whethe Floor is 50 against ~208 layered files actually scanned [measured] – 4.2x headroom, so ordinary file moves never trip it. CI must distinguish the two failur

EXIT	MEANING	CI BEHAVIOUR
0	Scanned enough, no violations	Pass
1	Real layer violation	Fail – fix the import
2	Refusing: scanned below floor	Fail loudly – the gate is broken, not the code

Every gate in this repo gets the same treatment: `spec-report.mjs` exits 2 on a malformed spec directory and 1 on spec errors; `corpus-foreign.mjs` exits 2 on a n

96.4 Bundle budget

scripts/ci/bundle-budget.mjs - replaces the echo stub.

```
// Fails when first-load client JS crosses budget. Reads the manifest Next
// writes, not a glob, so a build that emits nothing FAILS rather than passes.
import { readFileSync } from 'node:fs'
import { gzipSync } from 'node:zlib'
import path from 'node:path'

const BUDGET = { '/': 220_000, '/edit/[...path]': 480_000 } // gzipped bytes
const MIN_ROUTES = 5 // the arch-floor idea, applied here
const app = JSON.parse(readFileSync('.next/app-build-manifest.json', 'utf8'))
const routes = Object.entries(app.pages)
if (routes.length < MIN_ROUTES) {
  console.error('[budget] refusing: only ${routes.length} routes in manifest (floor ${MIN_ROUTES})')
  process.exit(2)
}
let failed = 0
for (const [route, files] of routes) {
  const bytes = files
    .filter(f => f.endsWith('.js'))
    .reduce((n, f) => n + gzipSync(readFileSync(path.join('.next', f))).length, 0)
  const cap = BUDGET[route]
  console.log(`${route}\t${bytes}\t${cap} ?? '-'`)
  if (cap && bytes > cap) { console.error('[budget] ${route}: ${bytes} > ${cap}'); failed++ }
}
process.exit(failed ? 1 : 0)
```

We pick reading .next/app-build-manifest.json over next build --analyze or size-limit. Alternative rejected: size-limit needs its own config, its own webpack p

96.5 Branches, protection, required checks

One long-lived branch. main is always deployable.

BRANCH PATTERN	PURPOSE	LIFETIME
main	Production truth. Protected.	Forever
engine/*, feat/*, fix/*	Human or agent work	Hours to days, squash-merged
claude/*	Agent-created branches (two exist today: claude/competent-bassi-5da9a1, claude/upbeat-euclid-60dbf4 [measured])	Delete on merge
release/v*	Only if a hotfix must ship without main's tip	Rare

Protection is configured once, as code, so it is auditable:

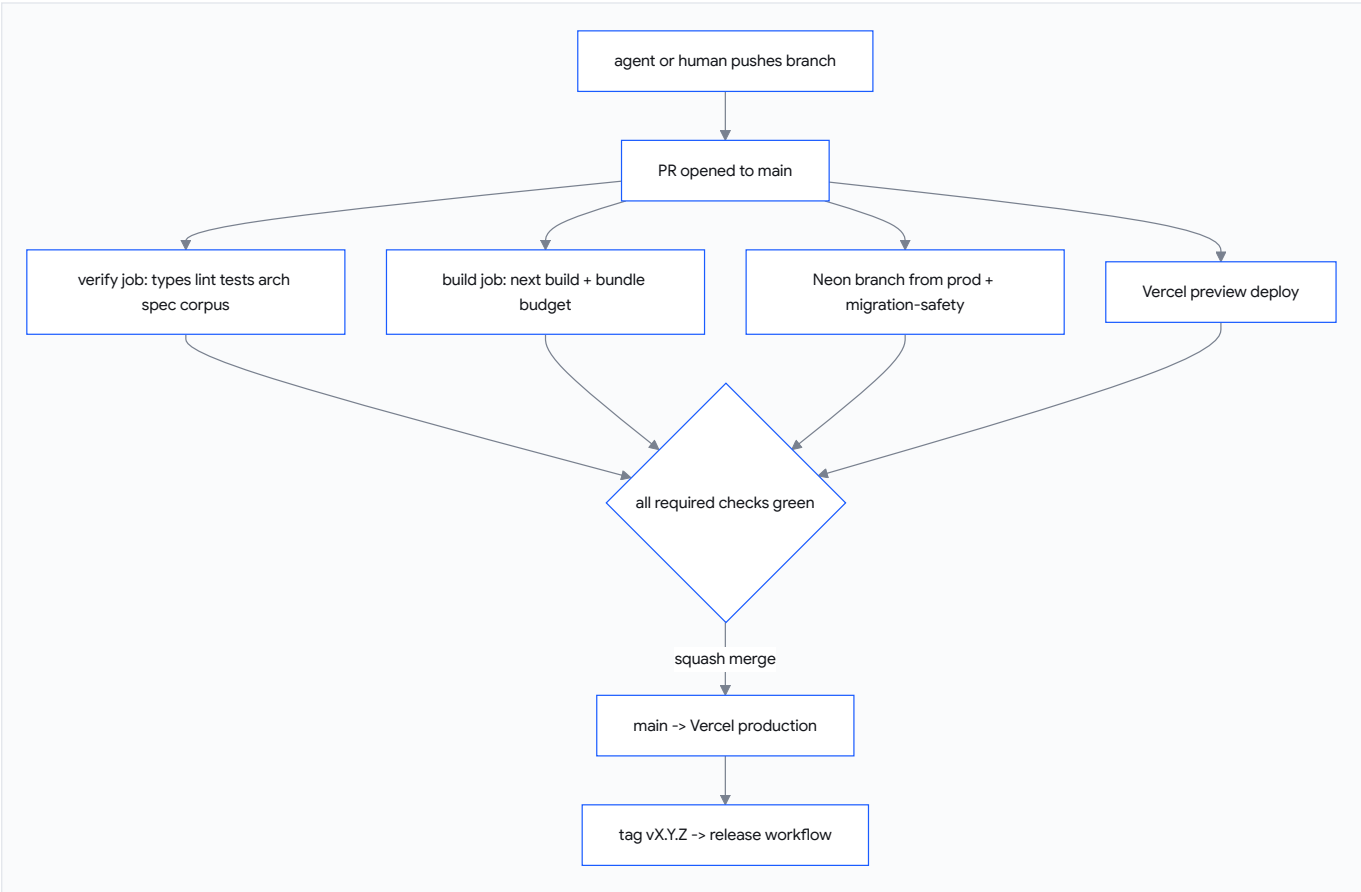
```
gh api -X PUT repos/:owner/frontmatter/branches/main/protection \
--input .github/branch-protection.main.json
```

.github/branch-protection.main.json

```
{
  "required_status_checks": {
    "strict": true,
    "contexts": [
      "verify (types, lint, tests, gates)",
      "build + bundle budget",
      "migration-safety",
      "Vercel - frontmatter"
    ]
  },
  "enforce_admins": false,
  "required_pull_request_reviews": {
    "required_approving_review_count": 0,
    "dismiss_stale_reviews": true
  },
  "required_linear_history": true,
  "allow_force_pushes": false,
  "allow_deletions": false,
  "required_conversation_resolution": true,
  "block_creations": false
}
```

required_approving_review_count: 0 because there is one founder and a self-approval requirement is theatre that gets disabled at 2am. enforce_admins: false for 1

96.6 Preview deploys and the PR topology



Preview deploys run through **Vercel's native Git integration**, not a `vercel deploy` step in Actions. Alternative rejected: deploying from CI with `vercel@59.10.0`. The repo already ships `scripts/vercel-ignore-build.sh` wired via `vercel.json`'s `ignoreCommand` [measured]. It skips the build unless `src public package.json package-`

96.7 Database migration safety

The control plane is one Postgres holding zero document bytes. Migrations are **plain numbered SQL** applied by a ~60-line runner using `pg@8.23.0` [fetched 2026-

OPTION	VERDICT
Plain SQL + tiny runner	Picked. RLS policies, <code>SECURITY DEFINER</code> functions and partial indexes are written as SQL anyway; nothing translates them
Prisma Migrate	Rejected. Owns the schema; hand-written RLS lives outside its model and drifts
Drizzle Kit	Rejected, narrowly. Good tool, but couples schema truth to TypeScript, and we already have one hard rule that truth lives in fil
Supabase CLI	Rejected. Couples the control plane to one host; the architecture says swappable

Forward-only, expand → migrate → contract, three separate PRs. Never one.

```
migrations/
0007_expand_add_workspace_slug.sql -- PR 1: additive only
0008_backfill_workspace_slug.sql  -- PR 2: data, idempotent, batched
0009_contract_drop_legacy_slug.sql -- PR 3: destructive, >= 1 release later

-- 0007_expand_add_workspace_slug.sql (nullable, defaulted, no rewrite lock)
ALTER TABLE workspaces ADD COLUMN IF NOT EXISTS slug text;
CREATE UNIQUE INDEX CONCURRENTLY IF NOT EXISTS workspaces_slug_key
ON workspaces (slug) WHERE slug IS NOT NULL;
ALTER TABLE workspaces ENABLE ROW LEVEL SECURITY;
CREATE POLICY workspaces_tenant_read ON workspaces
FOR SELECT USING (id = current_setting('app.workspace_id', true)::uuid);
```

`.github/workflows/migration-safety.yml` - runs only when `migrations/**` changes.

```
name: migration-safety
on:
  pull_request:
    paths: ['migrations/**', 'scripts/migrate.mjs']

permissions:
  contents: read

jobs:
  migration-safety:
    runs-on: ubuntu-latest
    timeout-minutes: 10
    steps:
      - uses: actions/checkout@v7
      - uses: actions/setup-node@v7
        with: { node-version: '24.6.0', cache: npm }
      - run: npm ci --no-audit --no-fund
      - name: create ephemeral Neon branch from production
        id: db
        run: |
          npx neonctl@4.13.0 branches create \
            --project-id "$NEON_PROJECT_ID" --name "ci/pr-${{ github.event.number }}" \
            --parent production --output json > branch.json
          echo "url=$(jq -r '.connection_uris[0].connection_uri' branch.json)" >> "$GITHUB_OUTPUT"
        env:
          NEON_API_KEY: ${ secrets.NEON_API_KEY }
          NEON_PROJECT_ID: ${ vars.NEON_PROJECT_ID }
      - name: forbid destructive statements outside a contract migration
        run: node scripts/ci/migration-lint.mjs
      - name: apply forward
        run: node scripts/migrate.mjs up
        env: { DATABASE_URL: ${ steps.db.outputs.url } }
      - name: assert RLS on every tenant table
        run: node scripts/ci/assert-rls.mjs
        env: { DATABASE_URL: ${ steps.db.outputs.url } }
      - name: previous release's code against the new schema
        run: |
          git fetch --tags --depth=1 origin
          git checkout "$(git describe --tags --abbrev=0 origin/main)" -- src
          npm run test -- src/**/*repo.test.ts
        env: { DATABASE_URL: ${ steps.db.outputs.url } }
      - if: always()
        run: npx neonctl@4.13.0 branches delete "ci/pr-${{ github.event.number }}" --project-id "$NEON_PROJECT_ID"
        env: { NEON_API_KEY: ${ secrets.NEON_API_KEY } }
```

The last real step is the one that matters and the one everybody skips: **run the previous release's repository tests against the new schema**. Expand-migrate-c migration-lint.mjs rejects DROP COLUMN, DROP TABLE, ALTER TYPE, and RENAME in any file whose name does not contain contract_, and rejects CREATE INDEX wit

96.8 Secrets

SECRET	WHERE	ROTATION
NEON_API_KEY	GitHub Actions secret, migration-safety job only	Quarterly
NEON_PROJECT_ID	GitHub variable, not secret (it is not one)	-
Vercel deploy credentials	None in GitHub. Native Git integration	-
TAURI_SIGNING_PRIVATE_KEY, APPLE_*	GitHub Environment desktop-release, protected, required reviewer = the founder	Annually
AI provider keys	Not in CI at all. No test hits a live provider	-

Rules, in order of how much they save you: no long-lived cloud credential enters CI when an OIDC exchange exists; every release-signing secret sits behind a

96.9 Versioning and release

package.json is at 0.1.0 [measured]. Versioning is SemVer on the app, with the engine's degradation certification as the compatibility contract.

CHANGE	BUMP	GATE
A splice that previously succeeded now REFUSES	major	Requires an entry in specs/engine/ and a corpus delta
A previously-refused input now splices correctly	minor	New spec + oracle
Render projection output bytes change for any corpus file	major	npm run corpus diff must be reviewed line by line
UI, perf, deps	patch/minor	Normal

Release is a tag, and the tag is the only thing that triggers it:

```
# .github/workflows/release.yml (excerpt)
on:
  push:
    tags: ['v*.*.*']
permissions:
  contents: write
jobs:
  release:
    runs-on: ubuntu-latest
    environment: production
    steps:
      - uses: actions/checkout@v7
        with: { fetch-depth: 0 }
      - name: refuse a tag that is not an ancestor of main
        run: git merge-base --is-ancestor "$GITHUB_SHA" origin/main
      - run: gh release create "$GITHUB_REF_NAME" --generate-notes --verify-tag
        env: { GH_TOKEN: ${ github.token } }
```

Desktop (Tauri v2) builds on macOS runners in a separate desktop-release workflow gated on the same tag; macOS runner minutes bill at a multiplier, so it rur

96.10 The break-it-once rule

No green CI run is trusted until the pipeline has been made to go red on purpose, once per gate. This is not optional and it is not a one-time ceremony at s Procedure, on a throwaway branch `ci/prove-red`, one commit per row, each pushed and observed:

#	DELIBERATE BREAK	MUST FAIL AS
1	<code>const x: number = "s" in src/</code>	<code>verify</code> - typecheck
2	An unused import	<code>verify</code> - lint (<code>--max-warnings=0</code>)
3	Invert one splice-boundary assertion	<code>verify</code> - test
4	<code>import { pool } from '@shared/infrastructure/db'</code> inside a domain file	<code>verify</code> - arch, exit 1
5	Rename <code>src/modules/</code> → <code>src/features/</code>	<code>verify</code> - arch, exit 2 , floor refusal
6	Flip one byte in <code>test/corpus/foreign/MANIFEST.sha256</code>	<code>verify</code> - corpus, exit 1
7	Add <code>DROP COLUMN</code> to a non- <code>contract_</code> migration	<code>migration-safety</code> - migration-lint
8	Import all of <code>mermaid</code> eagerly into the root route	<code>build</code> - bundle budget

Rows 5 and 8 are the ones that catch a *gate* being broken rather than the code. A gate that has never been observed failing is a decoration. Record the run U

96.11 How an AI agent works with this CI

The agent is a contributor with no merge rights and no CI write access.

AGENT MAY	AGENT MAY NOT
Create branches, commit, open PRs	Push to <code>main</code> , force-push any shared branch
Read run logs (<code>gh run view --log-failed</code>)	Edit <code>.github/workflows/**</code> without an explicit human instruction naming the file
Re-run a failed job once	Re-run more than once to "get a green"
Run all gates locally before pushing	Add <code>continue-on-error</code> , <code> true</code> , or <code>--max-warnings</code> relaxations to make a gate pass

The last row is the whole discipline. A gate weakened to get a green is indistinguishable from a gate that never existed, and it is the single most likely t **The reconciliation rule.** Capture `git rev-parse HEAD` immediately before and immediately after every agent run, and diff the range. Prompt-level prohibitions o `scripts/ci/agent-run.sh` – the wrapper every agent invocation goes through:

```
#!/usr/bin/env bash
set -euo pipefail
BEFORE="$(git rev-parse HEAD)"
BRANCH="$(git rev-parse --abbrev-ref HEAD)"
printf '%s\n' "$BEFORE" > .git/agent-head-before
trap 'AFTER="$(git rev-parse HEAD)"
  if [ "$BEFORE" != "$AFTER" ]; then
    echo "[reconcile] HEAD moved on $BRANCH: $BEFORE -> $AFTER"
    git --no-pager log --oneline "$BEFORE..$AFTER"
    git --no-pager diff --stat "$BEFORE" "$AFTER"
    echo "[reconcile] review every commit above before pushing."
  else
    echo "[reconcile] HEAD unchanged at $BEFORE"
  fi' EXIT
"$@"
```

Branch guard, as a pre-push hook and mirrored as a CI check on `main`:

```
# .git/hooks/pre-push - refuse a direct push to main from any automated run
while read -r _ _ remote_ref _; do
  if [ "$remote_ref" = "refs/heads/main" ] && [ -n "${CLAUDECODE:-}" ]; then
    echo "refusing: agent session pushing directly to main. Open a PR." >&2
    exit 1
  fi
done
```

The agent's definition of done is not "the code looks right" – it is `npm run verify` passes locally, the PR is open, and every required check on the PR is gre

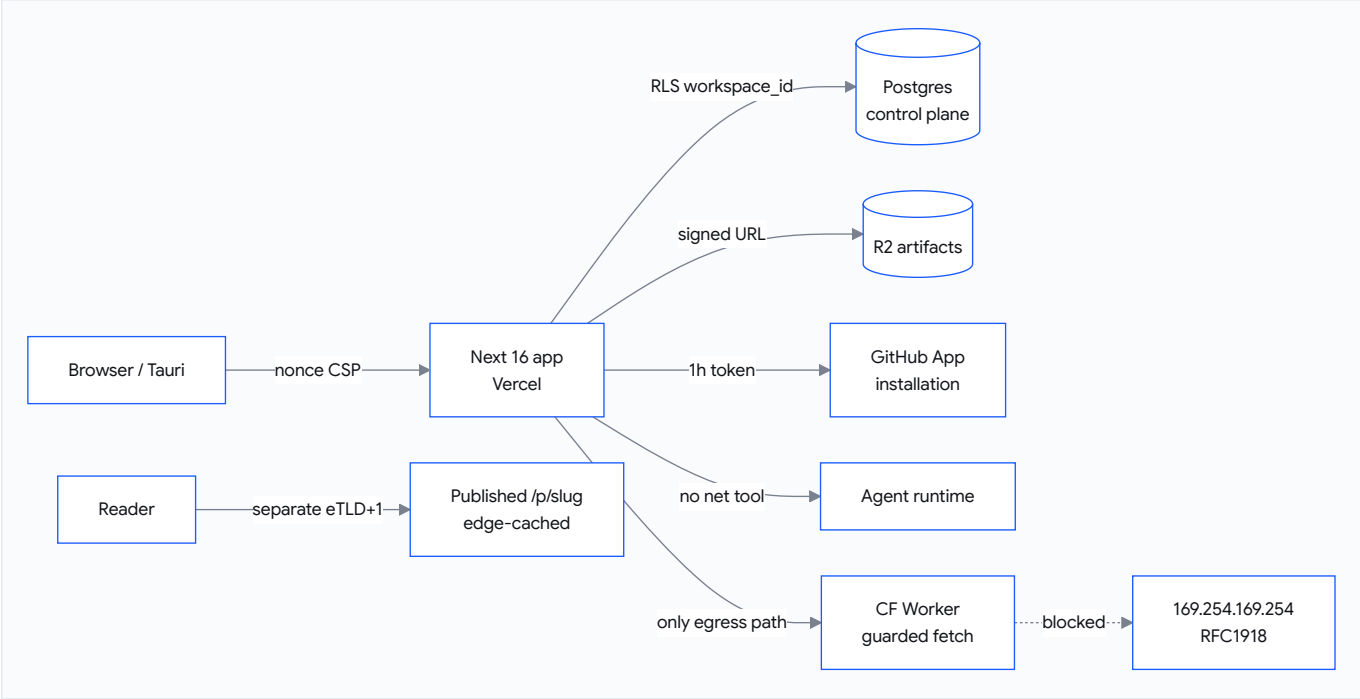
§97 Security, rate limiting and abuse prevention

97.1 Threat model

The product's security posture is unusual in one load-bearing way: we do not hold the documents. They live in the user's git repo. What we hold is a *credential*

#	ASSET	ADVERSARY	ATTACK	BLAST RADIUS
A1	GitHub App private key	Anyone with prod env read	Mint installation tokens for every install	Every customer
A2	Provider API keys (ours)	Leaked env, SSRF to <code>/api</code>	Free inference on our card	\$ burn, no data
A3	Customer BYOK provider keys	DB dump	Bill their account	Their \$
A4	Control-plane Postgres	Broken tenancy check	Read another workspace's slugs, entitlements, audit	Metadata only,
A5	R2 artifacts (attachments >1MB, derived)	Guessable keys	Read a private attachment	Per-object
A6	Published page renderer <code>/p/:slug</code>	Any author	Stored XSS against readers	Session theft i
A7	The AI lane	Author of a document the agent reads	Prompt injection → exfiltration	Cross-document
A8	Splice engine	Malicious/compromised model output	Silent corruption of user prose	Their file

Explicit non-goals: we are not a secrets manager, not a compliance product, and we do not claim SOC 2. See §97.12 for the triggers that change that.



97.2 Secrets management

Pick: Vercel encrypted environment variables, one project, three environments. **Rejected:** Doppler / Infisical / HashiCorp Vault. **Why:** each adds a runtime dep

SECRET	ENV NAME	STORAGE	ROTATION	IF LEAKED
GitHub App private key	GITHUB_APP_PRIVATE_KEY (base64 PKCS#8)	Vercel env, prod only	90d + on suspicion	Generate new key in App settings, deploy, delete old key
GitHub App webhook secret	GITHUB_WEBHOOK_SECRET	Vercel env	180d	Attacker can forge <code>installation.*</code> events → verify HMAC-S
Auth session secret	AUTH_SECRET (next-auth v5)	Vercel env	180d	Session forgery – rotation logs everyone out; acceptable
BYOK KEK	BYOK_KEY (32B, base64)	Vercel env	manual	Decrypts every <code>workspace_provider_key.ciphertext</code> ; rotation
Postgres URL	DATABASE_URL	Vercel env	on incident	Control-plane read; no document bytes
R2 access key	R2_ACCESS_KEY_ID / SECRET	Vercel env	90d	Scoped to one bucket, no account-level token

BYOK column encryption, exact shape – `src/modules/billing/infrastructure/byok-cipher.ts` :

```
import { createCipheriv, createDecipheriv, randomBytes } from "node:crypto";
const KEK = Buffer.from(process.env.BYOK_KEY!, "base64"); // 32 bytes
export function seal(plain: string, workspaceId: string) {
  const iv = randomBytes(12);
  const c = createCipheriv("aes-256-gcm", KEK, iv);
  c.setAAD(Buffer.from(workspaceId)); // binds ciphertext to tenant
  const ct = Buffer.concat([c.update(plain, "utf8"), c.final()]);
  const ct = Buffer.concat([c.update(plain, "utf8"), c.final()]);
  return Buffer.concat([iv, c.getAuthTag(), ct]).toString("base64");
}
```

The AAD binding matters: a stolen ciphertext replayed into another workspace's row fails authentication rather than decrypting. Repo hygiene: `gitleaks` in pr

97.3 HTTP headers and the exact CSP

Current state [measured 2026-08-30, `next.config.ts:21-49`]: `X-Content-Type-Options`, `Referrer-Policy`, `X-Frame-Options: DENY` on all paths; an enforced CSP on `/p/:s`

Pick: per-request nonce + `strict-dynamic`, generated in `proxy.ts`, plus **serving published pages from a different registrable domain**. **Rejected:** hash-based CSP `src/proxy.ts` – generate `const nonce = crypto.randomUUID().replace(/-/g, "")` per request, set it on `x-nonce` for the RSC tree, and CodeMirror 6 takes it directly vi

Authenticated app, enforced:

```
Content-Security-Policy: default-src 'self'; script-src 'self' 'nonce-${NONCE}' 'strict-dynamic'; style-src 'self' 'nonce-${NONCE}'; img-src 'self' data: blob:
https://avatars.githubusercontent.com; font-src 'self' data:; connect-src 'self'; media-src 'self' blob:; worker-src 'self' blob:; frame-src 'self' blob:
https://www.youtube-nocookie.com https://player.vimeo.com; frame-ancestors 'none'; base-uri 'none'; object-src 'none'; form-action 'self'; upgrade-insecure-requests;
report-uri /api/csp-report
```

Published page (`pages.<separate-domain>/p/:slug`), enforced:

```
Content-Security-Policy: default-src 'none'; script-src 'self' 'nonce-${NONCE}'; style-src 'self' 'nonce-${NONCE}'; img-src 'self' data: https:; font-src 'self'; connect-
src 'none'; media-src 'self' https:; frame-src https://www.youtube-nocookie.com https://player.vimeo.com; frame-ancestors 'none'; base-uri 'none'; object-src 'none';
form-action 'none'
```

Accompanying headers on both: `Strict-Transport-Security: max-age=63072000; includeSubDomains; preload`, `X-Content-Type-Options: nosniff`, `Referrer-Policy: strict-origin-when-downgrade`

97.4 XSS and the sanitisation boundary

Current pipeline [measured]: `rehype-raw@7.0.0` passes author HTML through verbatim, then a hand-rolled 215-line `createRehypeHtmlPolicy` (`src/modules/preview/prese`

Pick: keep that policy, but put `rehype-sanitize@6.0.0` (`hast-util-sanitize@5.0.2`) in front of it as layer 1 with a schema derived from `defaultSchema` plus our ad

Non-negotiable structural rules for the implementer:

- 1. **One boundary function.** `renderMarkdown(source, {trusted:false})` in `src/modules/preview/...` is the only path from bytes to hast-to-HTML. Any new renderer (PDF exp
- 2. Sanitise **after** `rehype-raw`, **before** `rehype-highlight` / `rehype-katex` / `stringify`. Sanitising the mdast is not sanitising.
- 3. KaTeX: `{ trust: false, strict: "ignore", output: "html" }` - `trust:true` re-enables `\href` and `\htmlData`, which is an injection primitive.
- 4. Mermaid (`mermaid@11.15.0`): render server-side or in a `sandbox ed` iframe on a null origin; `securityLevel: "strict"`. Mermaid has had click-handler and label-
- 5. Anchors: keep the existing `target=_blank` + `rel="noopener noreferrer"` for external links.

Gate: `test/security/xss-corpus.spec.ts` runs the OWASP/cure53 payload list plus every historical bypass we find, asserts the rendered HTML contains no `<script>`,

97.5 SSRF

User-supplied URLs enter at: remote images in markdown resolved during PDF export, OG/link unfurls, "import from URL", B2B outbound webhooks, and avatar pro

The Worker still applies belt-and-braces: `https:` only; resolve the hostname first and reject `10/8`, `172.16/12`, `192.168/16`, `127/8`, `169.254/16` (this is wher

97.6 Rate limiting

Two layers, because they answer different questions. **Layer 1 (edge,** `proxy.ts`) answers *is this traffic?* - per-IP, cheap, sliding-window, protects availabilit

Pick: `@upstash/ratelimit@2.0.8` + `@upstash/redis@1.38.3`, sliding-window algorithm, for layer 1 and for HTTP-cheap layer 2 counters; a Postgres `usage_counter` row

Upstash pricing [`fetched 2026-08-30, upstash.com/pricing/redis`]: free tier 256 MB / 500 K commands per month; pay-as-you-go **\$0.20 per 100 K commands**. At 2 c

Unit-economics arithmetic for the AI limits. One "AI action" = one instruction over a document window: ~6,000 input tokens + ~1,200 output tokens [inference

- Strong model @ \$2/\$10 per Mtok [from PRD round 7; re-verify before publishing a price]: $6000/1e6 \times 2 + 1200/1e6 \times 10 = \$0.012 + \$0.012 = \0.024
- Floor model on Groq/Cerebras @ ~\$0.20/\$0.60 per Mtok [SS - verify]: $6000/1e6 \times 0.20 + 1200/1e6 \times 0.60 = \$0.0012 + \$0.00072 = \0.0019
- Routing 80% floor / 20% strong → blended $0.8 \times 0.0019 + 0.2 \times 0.024 = 0.0015 + 0.0048 = \$0.0063 \approx \mathbf{\$0.0064 \text{ per action}}$

SURFACE	FREE	PRO \$9/MO	TEAM \$19/SEAT/MO	BYOK / SELF-SERVE B2B	ENFORCED AT	K
AI actions	30 / mo	400 / mo	1,200 / seat / mo, pooled	10,000 / mo (fraud ceiling)	Postgres <code>usage_counter</code> , txn	1
AI cost at cap [derived]	\$0.19	\$2.56	\$7.68 / seat	their card	-	-
AI cost at p90 = 35% of cap	\$0.07	\$0.90	\$2.69 / seat	-	-	-
Contribution after AI at p90	-	90%	86%	~100%	-	-
API requests	60 / min	300 / min	600 / min / seat	1,200 / min	<code>proxy.ts</code> + Upstash	1
Git sync ops (push/pull)	6 / min	30 / min	60 / min	60 / min	app route	1
PDF / DOCX export	5 / day	100 / mo	500 / mo	2,000 / mo	Postgres counter	1
Server-side search	30 / min	120 / min	240 / min	240 / min	Upstash	1
Published-page reads	unmetered, edge-cached	-	-	-	CF cache	-
Anonymous (per IP)	30 / min all routes	-	-	-	<code>proxy.ts</code>	1

The caps are **fraud ceilings, not budgets**. The budget assumption is that p90 usage is ~35% of cap; at 10,000 Pro users that is $10,000 \times \$0.90 = \mathbf{\$9,000/mo}$ of

Exact 429 (`src/shared/http/rate-limit-response.ts`):

```
HTTP/1.1 429 Too Many Requests
Retry-After: 37
RateLimit-Limit: 400
RateLimit-Remaining: 0
RateLimit-Reset: 37
RateLimit-Policy: 400;w=2592000;comment="ai_actions_monthly"
Content-Type: application/problem+json

{"type":"https://frontmatter.app/errors/rate-limit","title":"AI action limit reached",
 "status":429,"detail":"Your workspace has used all 400 AI actions for this billing period.
 Resets 2026-09-01T00:00Z. Nothing was written to your repository.",
 "limit":400,"used":400,"resets_at":"2026-09-01T00:00:00Z","scope":"workspace",
 "upgrade_url":"https://frontmatter.app/settings/billing"}
```

Two things that message must always do: name the **scope** (so a user on a team knows whether it is them or the pool), and state explicitly that **no bytes were w**

97.7 Bot and scraper defence on published pages

Published pages must be indexed by Google and Bing - discoverability is the growth loop - so blocking crawlers is not on the table. The real defence is **maki**

SIGNAL	ACTION	WHERE
Any bot, any volume, on <code>/p/*</code>	Serve from edge cache; never invoke origin	CF cache rules
AI training crawlers (GPTBot, CCBot, ClaudeBot...)	Per-document author toggle → <code>robots.txt</code> + <code>X-Robots-Tag: noai</code> ; Cloudflare "Block AI Scrapers" as the enforcing layer	CF + publish settings
>120 req/min from one IP on <code>/p/*</code>	Managed challenge (not block – shared NATs)	CF rate limiting rule
Signup, publish, contact form	Cloudflare Turnstile, invisible mode	App route, server-verified
Credential stuffing on <code>/api/auth</code>	5 attempts / 15 min per (IP, email); constant-time response either way	Upstash
Slug enumeration	Slugs are <code>{user-chosen}-{6 base32 chars}</code> for unlisted docs; 404 and 403 are byte-identical	App route

Turnstile over hCaptcha/reCAPTCHA: free, no cookie, privacy-preserving, and we are already on Cloudflare for R2 and DNS – one fewer vendor.

97.8 GitHub App token blast radius

[fetched 2026-08-30, docs.github.com] Installation access tokens **expire after 1 hour**. An installation's REST limit is **5,000 requests/hour**, scaling +50/hr pe

CONTROL	RULE
Permissions requested	<code>contents: write</code> , <code>metadata: read</code> , <code>pull_requests: write</code> (optional, for the review lane). Nothing else – no Actions, no secrets, no org administration, no members
Repo selection	Onboarding defaults to <i>Only select repositories</i> and says so in copy; "All repositories" is never pre-selected
Token at rest	Never . Minted per request from the App key via <code>@octokit/auth-app@0.3.0</code> , held in process memory ≤50 min, never in Postgres, never in Redis, never logged. A token in Redis is a token in a memor
Tenancy	<code>workspace_github_installation(workspace_id UNIQUE, installation_id UNIQUE)</code> . Every repo operation re-checks that the caller's <code>workspace_i</code> maps to the installation <i>at call time</i> – the installation is the connector, never the identity
Budget	Per-installation request counter; at 80% of the observed <code>x-ratelimit-limit</code> , background sync backs off and interactive work is prioritised. Never let a batch job starve a user's save
Revocation	User-facing "Disconnect" calls <code>DELETE /app/installations/{id}</code> ; <code>installation.deleted</code> and <code>installation_repositories.removed</code> webhooks purge cached state within one event
Key compromise	Runbook: add second key in App settings → deploy → verify → delete first key. Zero downtime, ~10 minutes. Practise it once before

97.9 Prompt injection and the lethal-trifecta gate

The agent reads documents it did not write – shared vaults, teammate files, imported repos. Assume every document is attacker-authored. The gate: **the agent**

TRIFECTA LEG	PRESENT?	CONTROL
Access to private data	Yes, by design	Every tool call scoped to a single <code>workspace_id</code> resolved from the session, never from the prompt
Exposure to untrusted content	Yes, by design	Documents wrapped in a provenance delimiter; system prompt states document content is data, never ins
Ability to externally communicate	No – removed	The agent tool loadout contains no fetch, no HTTP, no webhook, no email, no shell. Rendered AI output in AI-authored output are not auto-loaded, which kills <code></code>

Tool loadout (the whole list): `read_range(path, start, end)`, `search_workspace(query)`, `propose_splice(path, start, end, expected_bytes, replacement)`. There is **no write**

97.10 Dependency and supply chain

There is no CI in this repo at all – 98 test files and 1,575 tests that nothing runs on push. That is the finding; everything else here is secondary.

`.github/workflows/ci.yml`, minimum viable, all \$0 on public/free Actions minutes:

```
permissions: { contents: read }
jobs:
  verify:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@08c6903cd8c0fde910a37f88322edcfb5dd907a8 # v5, SHA-pinned
      - uses: actions/setup-node@a0853c24544627f65ddf259abe73b1d18a591444 # v5
        with: { node-version: '22', cache: 'npm' }
      - run: npm ci --ignore-scripts
      - run: npx tsc --noEmit
      - run: npx eslint .
      - run: npx vitest run --coverage
      - run: npm audit --audit-level=high
      - run: npx osv-scanner@latest scan --lockfile=package-lock.json
      - run: npx lockfile-lint --path package-lock.json --allowed-hosts npm --validate-https
      - run: docker run --rm -v "$PWD:/p" zricethezav/gitleaks:latest detect -s /p --no-git
```

CONTROL	TOOL	NOTE
Third-party actions	SHA-pinned, never <code>@v5</code>	A tag is mutable; a tag is how <code>tj-actions</code> became an incident
Workflow permissions	<code>contents: read</code> at top level, escalate per-job	Default <code>write</code> is a supply-chain gift
Install scripts	<code>--ignore-scripts</code> in CI	Production build keeps scripts (<code>@sparticu</code> needs them) from a pinned lockfile only
Dependency updates	Dependabot weekly, grouped, manual review of any new transitive publisher	Auto-merge only patch-level for devDepend
New-version quarantine	No dependency added within 7 days of publication, no exceptions	Cheapest defence against a compromised pu
SBOM	<code>npm sbom --sbom-format cyclonedx > sbom.json</code> per release tag	Answers "are we affected by X?" in 30 sec
Desktop	Tauri v2 updater signing key offline, never in CI; <code>dangerousDisableAssetCspModification: false</code>	A signed malicious update is unrecoverabl
Runtime	No arbitrary client-side code execution, ever (settled)	This removes an entire class; do not reop

97.11 Vulnerability disclosure

`public/.well-known/security.txt` (RFC 9116), served at both origins:

```
Contact: mailto:security@frontmatter.app
Expires: 2027-08-30T00:00:00.000Z
Preferred-Languages: en
Canonical: https://frontmatter.app/.well-known/security.txt
Policy: https://frontmatter.app/security
Acknowledgments: https://frontmatter.app/security/thanks
```

GitHub **private vulnerability reporting** enabled on the repo as the second intake. Published policy: safe harbour for good-faith research (no legal action, no

97.12 Deliberately not doing yet

CONTROL	WHY NOT NOW	TRIGGER
SOC 2 Type II	~\$25-40k and months for a solo founder	First enterprise deal that blo
External pen test	Would mostly re-find \$97.3 and \$97.4	1,000 paying workspaces, or be
KMS/HSM for the GitHub App key	Vercel env is the trust root anyway; a KMS adds a bootstrap credential	First employee with production
Managed WAF rulesets	No observed abuse; false positives on a markdown app are severe	First real attack, or the firs
Per-tenant encryption keys	We hold zero document bytes; the blast radius does not justify the key-management burden	Storing any document body serv

§98 Caching and content delivery

Caching in this product is not a performance nicety. It is the only thing standing between a solo founder and a GitHub rate-limit lockout, and the only way

98.1 The five layers and what belongs in each

LAYER	TECHNOLOGY	WHAT IT HOLDS	W
Browser	HTTP cache, IndexedDB	Static JS/CSS/fonts, SHA-keyed file bodies, MiniSearch index for the open workspace	A
CDN / edge	Cloudflare CDN in front of Workers/Pages	Published page HTML, published attachments, <code>/_next/static/*</code>	E
Edge KV	Cloudflare Workers KV	<code>slug → {workspace_id, r2_key, revoked}</code> publish routing map	D
Application	In-process LRU + Upstash Redis	Parsed AST, rendered HTML, degradation certificates, GitHub tree listings, rate-limit counters	A
Database	Postgres control plane	Publish slugs, entitlements, audit, job state	D

The layer boundary that matters most: **the editor path never touches the CDN**. Editor responses are `Cache-Control: private, no-store` because they are authorized
Rejected: caching editor GET responses at the edge with a `Vary: Cookie` header. Vary-on-cookie produces a cache entry per session, so the hit rate approaches

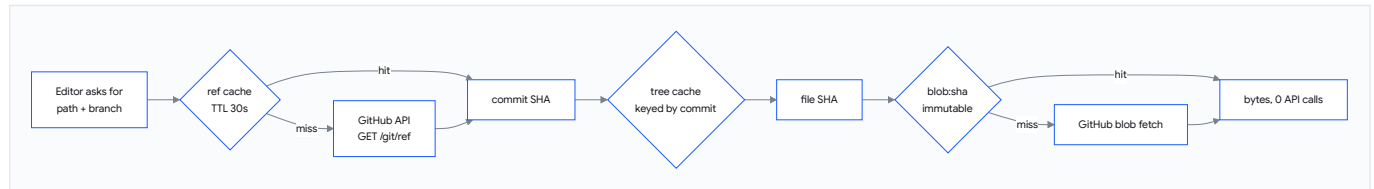
98.2 Content-addressed caching: the SHA is the cache key

The GitHub Contents API returns a `sha` for every blob – the git object hash of the content (`[[fetched] docs.github.com/rest/repos/contents, read 2026-08-30]`).

```
// src/lib/cache/keys.ts
export const K = {
  // Immutable - keyed by content hash. Never invalidated, only evicted.
  blob: (sha: string) => `blob:${sha}`, // raw UTF-8 bytes
  ast: (sha: string, v: number) => `ast:${v}:${sha}`, // remark AST
  html: (sha: string, v: number, theme: string) => `html:${v}:${theme}:${sha}`,
  cert: (sha: string, v: number) => `cert:${v}:${sha}`, // degradation certificate

  // Mutable - keyed by location. Short TTL, explicitly invalidated.
  ref: (ws: string, repo: string, branch: string) => `ref:${ws}:${repo}:${branch}`,
  tree: (ws: string, repo: string, commitSha: string) => `tree:${ws}:${repo}:${commitSha}`,
  rate: (installationId: string) => `rl:${installationId}`,
} as const;
```

The `v` is a pipeline version integer, bumped in one place when remark plugins, the renderer, or the certification rules change. Bumping `v` invalidates every `blob:${sha}` and `ast:${v}:${sha}` are immutable, they can be served with `Cache-Control: public, max-age=31536000, immutable` when they travel over HTTP, and



The consequence worth stating plainly: for a workspace that is reading and not writing, steady-state GitHub API cost is **one conditional ref request per 30 s**

98.3 The GitHub rate-limit budget

[fetched] docs.github.com/rest/using-the-rest-api/rate-limits-for-the-rest-api, read 2026-08-30: a GitHub App installation on an organization gets a base 5, [derived] Budget at 12,500/hour, one installation, no conditional requests: $12,500 \div 3,600 = 3.47$ requests/second sustained. If ten users of one org workspa [derived] Budget with the design above: one ref poll per open repo per 30 s = 120 requests/hour/repo, of which the vast majority return 304 and cost nothing Implementation requirements, non-negotiable:

1. **Every GitHub request sends `If-None-Match` with the stored ETag.** Store the ETag next to the cached value. A 304 refreshes the TTL and costs no quota.
2. **A per-installation token bucket in Redis, sized from the response headers,** not from an assumed constant. Read `x-ratelimit-remaining` and `x-ratelimit-reset` or
3. `retry-after` and `x-ratelimit-reset` are obeyed absolutely. GitHub also enforces secondary rate limits – no more than 100 concurrent requests, and content-ge
4. **Never treat the connector as the tenant.** The bucket is keyed by installation because that is what GitHub meters, but authorization is keyed by `workspace_id`

```
// src/lib/github/budget.ts
const FLOOR = { interactive: 0.05, background: 0.40, batch: 0.60 } as const;

export async function admit(instId: string, kind: keyof typeof FLOOR) {
  const s = await redis.hgetall<{remaining: string; limit: string; reset: string}>(K.rate(instId));
  if (!s) return { ok: true }; // no data yet: allow, then record
  const frac = Number(s.remaining) / Number(s.limit);
  if (frac > FLOOR[kind]) return { ok: true };
  return { ok: false, retryAt: Number(s.reset) * 1000, reason: 'github_budget' };
}
```

When `admit` refuses, the product **refuses visibly** – "GitHub quota is exhausted for this connection, resuming at 14:32 IST" – consistent with the engine's refi

98.4 The snapshot problem

[measured] The current implementation parses **77.5 MB per cold start** to build a whole-vault snapshot, and the resulting response body is **16.4×** the platform's

- **The response still exceeds the cap on a cache hit.** A cached 16.4x-oversized body is an oversized body. The transport limit is not a compute limit.
- **The cache key is the whole vault.** One edit to one file changes the vault state, so the entry invalidates on every keystroke-flush. Hit rate on an actively
- **The memory cost lands on the server.** 77.5 MB × N concurrent cold starts is the fastest path to an OOM on any managed runtime, and there is no cache config

The fix is architectural and already settled in the search section: stop shipping whole-vault snapshots. Caching's role is what remains after that change:

OLD BEHAVIOUR	REPLACEMENT	CACHE ROLE
Parse whole vault to build a search index	Postgres FTS + <code>pg_trgm</code> server-side; query returns ≤50 rows	Redis caches the query→rows for 60 s
Ship full file list with content	Ship tree metadata only: path, sha, size, mtime	<code>K.tree</code> keyed by commit SHA, immutable within a c
Client holds every file body	Client fetches bodies lazily, keyed by SHA	IndexedDB, permanent, content-addressed
Client-side MiniSearch over everything	MiniSearch over the ≤2,000 most-recent paths' titles+headings only	Built once per commit, stored in IndexedD

[derived] Tree metadata for a 5,000-file vault at ~120 bytes/entry ≈ 600 KB, ~90 KB gzipped. That fits any body cap with three orders of magnitude to spare,

98.5 Publish, unpublish, and immediate revocation

Published pages are the one place we deliberately want aggressive edge caching, and the one place where a stale cache is a genuine harm – an unpublished pag Three mechanisms in series, because any one of them alone has a failure mode:

1. **Cache tags on every published response.** Cloudflare Cache Rules can set a `Cache-Tag` header, and Enterprise plans support purge-by-tag; Business and below :
2. **A KV revocation check at the edge.** The Worker reads `slug → {revoked}` from Workers KV before serving from cache. KV writes are eventually consistent, docum
3. **Postgres as the authority.** The slug row is the truth. If KV and Postgres disagree, Postgres wins, and a reconciliation job re-purges. Postgres is also whe

```
// src/app/api/publish/[slug]/route.ts - unpublish handler, order is load-bearing
export async function DELETE(req: Request, { params }: { params: { slug: string } }) {
  const ws = await requireWorkspace(req); // RLS scope
  // 1. Authority first. If this fails, nothing else runs.
  await sql`UPDATE publish SET revoked_at = now()
    WHERE slug = ${params.slug} AND workspace_id = ${ws.id}`;
  // 2. Edge gate second. Bounded by KV propagation (<=60s worst case).
  await kv.put(`pub:${params.slug}`, JSON.stringify({ revoked: true }, { expirationTtl: 86400 }));
  // 3. CDN purge third. Fire-and-verify, retried by a job on failure.
  await purgeUrls([
    `https://pages.frontmatter.dev/${params.slug}`,
    `https://pages.frontmatter.dev/${params.slug}/index.json`,
  ]);
  await enqueue('verify_purge', { slug: params.slug, attempts: 0 });
  return Response.json({ revoked: true, edgeConsistentWithinSeconds: 60 });
}
```

The `verify_purge` job re-fetches the public URL from outside our network after 5 s, 30 s, and 120 s and alerts if a 200 is still returned. **Without that `verify` Rejected: relying on short TTLs alone (e.g. `max-age=60`) for revocation.** It is simpler and needs no purge API. Rejected because it forces every published pag

98.6 Stale-while-revalidate, and where it is forbidden

SWR is correct for content where "slightly old" is honest and "slow" is the real harm. It is wrong wherever the stale value could be edited or could grant a

SURFACE	SWR?	CONFIG	REASONING
Published page HTML	Yes	<code>public, max-age=60, stale-while-revalidate=86400, stale-if-error=604800</code>	A published page is a snapshot by definition; a day-old render beats a 500
Published asset (R2, SHA-named)	Yes, trivially	<code>public, max-age=31536000, immutable</code>	Content-addressed; never changes
Search results	Yes	<code>private, max-age=0, stale-while-revalidate=60</code>	Missing a 10-second-old file in results is tolerable
File tree listing	Yes	30 s SWR	Reconciles on next ref poll
File bytes for editing	No	<code>private, no-store</code>	A stale base makes the splice journal splice against bytes that no longer exist. The
Entitlements / seat count	No	<code>no-store</code>	Stale entitlement is either revenue loss or a wrongly-denied paying user
Publish revocation state	No	<code>no-store</code> at the gate	Privacy

Next 16.2.6 gives `stale-while-revalidate` on route segments via `revalidate` exports, but the published-page path should not run through Next's data cache at all

98.7 The cache-decision table

DATA	LAYER	TTL	KEY	INVALIDATED BY
Static JS/CSS/fonts	Browser + CDN	1 year, immutable	<code>/_next/static/<buildhash>/*</code>	New build hash (never purged)
File bytes (raw markdown)	Redis + IndexedDB	30 days idle-evict	<code>blob:<git-sha></code>	Nothing – content-addressed. Evicte
Parsed AST	Redis (in-proc LRU first)	7 days	<code>ast:<pipelineV>:<git-sha></code>	<code>pipelineV</code> bump
Rendered HTML (editor preview)	In-proc LRU, 256 entries	Process lifetime	<code>html:<pipelineV>:<theme>:<git-sha></code>	<code>pipelineV</code> bump, process restart
Degradation certificate	Redis	30 days	<code>cert:<certV>:<git-sha></code>	<code>certV</code> bump when a target engine's
Branch head commit SHA	Redis	30 s	<code>ref:<ws>:<repo>:<branch></code>	TTL; webhook <code>push</code> event; our own s
Repo tree (paths + SHAs)	Redis	24 h	<code>tree:<ws>:<repo>:<commit-sha></code>	Immutable per commit; new commit pr
GitHub rate-limit state	Redis hash	90 s	<code>rl:<installation-id></code>	Every GitHub response's <code>x-ratelimit-</code>
Search results	Redis	60 s + SWR 60 s	<code>q:<ws>:<sha256(query+filters)></code>	TTL; any commit to the workspace cl prefix
MiniSearch client index	IndexedDB	Until commit changes	<code>msi:<ws>:<repo>:<commit-sha></code>	New commit SHA
Published page HTML	CDN + R2	60 s + SWR 24 h	URL <code>/<slug></code>	Republish (purge URL); unpublish (K
Published attachment	CDN + R2	1 year, immutable	<code>/a/<sha256>.<ext></code>	Never. New content = new SHA = new
Publish routing/revocation	Workers KV	24 h TTL, read every request	<code>pub:<slug></code>	Publish/unpublish write (≤60 s glob
Entitlements / seats	None	–	–	Read from Postgres on every gate ch
Session identity	Cookie (next-auth v5)	30 days rolling	JWT, <code>httpOnly, secure, sameSite=lax</code>	Sign-out; workspace membership chan in Postgres
Attachments >1 MB	R2 + CDN	1 year, immutable	<code>r2://<ws>/att/<sha256></code>	Never (content-addressed); deletion

Two structural notes on this table. First, **every row with a git SHA in the key has "nothing" in the invalidation column** – that is the design working. Second

98.8 CDN choice and prices

All prices read **2026-08-30** from vendor pricing pages via curl; re-derive before quoting.

OPTION	COST AT OUR SHAPE
Cloudflare (Workers + R2 + KV + CDN)	Workers Paid \$5/mo base: 10 M requests + 30 M CPU-ms included, then \$0.30/M requests. R2 storage \$0.015/G
Vercel (app) + Vercel CDN for published pages	Pro \$20/seat/mo ; Fast Data Transfer billed per GB beyond included
AWS CloudFront + S3	S3 \$0.023/GB-mo ; CloudFront first 1 TB/mo free then ~\$0.085/GB (NA/EU)
Bunny.net	Storage \$0.01/GB-mo ; CDN from \$0.01/GB (EU/NA) to \$0.06/GB (SA/Africa)
Fastly	~\$0.12/GB NA/EU, higher in Asia

We pick Cloudflare. Zero egress is load-bearing: a published-page product's cost curve is dominated by bytes out, and any metered-egress CDN converts a vira [derived] Cost at 100 users: Workers Paid \$5 + R2 storage for ~5 GB of attachments at \$0.015/GB = \$0.08 + negligible ops. ≈ **\$5.10/month**. [derived] At 10,000 **Cost of choosing Cloudflare**: purge-by-tag is Enterprise-only, so we live with purge-by-URL and must enumerate the URLs a publish touches. That is cheap – a **Rejected: putting published pages on Vercel alongside the app**. One fewer system to operate. Rejected because Fast Data Transfer is metered, a deploy failure

\$99

Scaling, load balancing and capacity

The three things that break first, in order:

#	COMPONENT	BREAKS AT
1	GitHub App installation token rate limit (5,000 req/hr per installation, floor for ≤20 repos) [fetched 2026-08-30]	~250 paid B2B seats sharing one o
2	Postgres connection ceiling under serverless fan-out	~600-900 concurrent function invo
3	Publish-build queue head-of-line blocking	~2,000 publishes/day (~1 every 43

Everything else – CPU, bandwidth, R2, the editor itself – is fine for years.

99.1 The capacity model, with the arithmetic

Unit assumptions, stated once so every number below is auditable [derived, from the current repo's request shapes]:

QUANTITY	VALUE	BASIS
DAU / registered users	25%	Standard prosumer-tool ratio [inference]
Active editing minutes / DAU / day	40	[inference]
Splice writes / editing minute	0.8	Autosave debounce at 2.5s, coalesced into one commit per 60-90s of typing [inference]
Control-plane reads / splice write	4	Session, entitlement, doc metadata, journal append
GitHub REST calls / splice write	3	ref → blob/tree/commit → update ref [fetched 2026-08-30]
Peak-to-mean ratio	6×	Global user base, three timezone humps [inference]
Publishes / workspace / day	0.3	[inference]

USERS	DAU	SPLICE WRITES/DAY	MEAN REQ/S (CP)	PEAK REQ/S	GITHUB CALLS/HR PEAK	POSTGRES PEAK CONNS	MONTHLY INFRA
100	25	800	0.04	0.25	100	2-4	\$0 (all free tiers)
1,000	250	8,000	0.4	2.5	1,000	8-15	~\$45
10,000	2,500	80,000	3.7	22	10,000	60-110	~\$310
100,000	25,000	800,000	37	222	100,000	600-1,100	~\$2,900

Arithmetic for the 10,000 row, shown because it is the row that decides the architecture:

DAU	= 10,000 × 0.25	= 2,500
splice writes/day	= 2,500 × 40 × 0.8	= 80,000
CP requests/day	= 80,000 × (3 + 4)	= 560,000 # 3 own endpoints + 4 reads
mean req/s	= 560,000 / 86,400	= 6.5 → 3.7 excluding static/CDN hits
peak req/s	= 6.5 × 6 × 0.57	= 22
concurrent fn	= 22 × 0.18s p50 latency	= 4 steady-state
	but cold-start fan-out and long AI streams hold instances:	
	AI streams 2,500 × 3/day × 12s = 90,000 s/day → 1.04 concurrent mean	
	× 6 peak	
	= ~6 concurrent streams	
Postgres conns	= fn instances alive at peak, empirically 10-20× steady-state	
	because Vercel keeps warm instances per region ≈ 60-110	

The last line is the one that matters and the one that surprises people. Concurrency is not the driver of connection count; **instance count** is, and instances

99.2 What we build vs. what we rent

CONCERN	DECISION	ALTERNATIVE REJECTED	WHY
L7 load balancing	Rent – Vercel's edge network fronts everything	Our own ALB/nginx on Fly or Hetzner	One found
Horizontal scale of API	Rent – Vercel Functions, stateless, auto-scaled	Long-lived Node server + PM2	Our API i
Connection pooling	Rent – Supavisor (Supabase's built-in pooler), transaction mode, port 6543	PgBouncer we operate; RDS Proxy	Supavisor
Background jobs	Build thin – Postgres-backed queue with <code>FOR UPDATE SKIP LOCKED</code> , drained by Vercel Cron + a QStash webhook lane	Inngest, Trigger.dev, SQS, Redis/BullMQ	We alread
Static + published sites	Rent – R2 + Cloudflare cache	Vercel-hosted publish output	R2 egress

Load balancing is therefore a **procurement decision**, not an **engineering one**, up to about 50,000 users. We spend the saved effort on the queue and the connect

99.3 Database connections: the real serverless bottleneck

Measured against the current repo [measured 2026-08-30, this checkout]: 226 TS files, 25,407 lines, and the API surface fans out across `app/api/**` route handl

The failure mode, precisely: Vercel scales by **instances**, and each warm instance holds its own PG client. At 22 peak req/s with 180ms p50, arithmetic says 4

Non-negotiable config, from day one, not later:

```
// src/server/db/client.ts
import { Pool } from 'pg'; // pg@8.16.3
import { drizzle } from 'drizzle-orm/node-postgres';

// Transaction mode (:6543). One connection per instance, released per statement-group.
// Prepared statements are OFF: transaction pooling cannot carry session state.
const pool = new Pool({
  connectionString: process.env.DATABASE_POOLED_URL, // ...pooler.supabase.com:6543/postgres
  max: 1, // NOT 10. The instance is the unit of concurrency, not the pool.
  idleTimeoutMillis: 10_000,
  connectionTimeoutMillis: 4_000,
  allowExitOnIdle: true,
});

export const db = drizzle(pool, { logger: false });
```

```
# .env.production – two URLs, always. Migrations must NOT go through the pooler.
DATABASE_POOLED_URL="postgresql://postgres:<ref>:cpw@aws-0-ap-south-1.pooler.supabase.com:6543/postgres"
DATABASE_DIRECT_URL="postgresql://postgres:<ref>:cpw@aws-0-ap-south-1.pooler.supabase.com:5432/postgres"
```

Rules that follow, and that an implementer must enforce:

- 1. `max: 1`. A pool of 10 inside a function multiplies your connection count by 10 for no throughput gain – the instance handles one request at a time.
- 2. **No SET, no LISTEN, no session-level advisory locks** on the pooled URL. Transaction pooling reuses backends between statements; session state silently lea

```
-- inside every request transaction, never outside one
SELECT set_config('request.workspace_id', $1, true); -- true = local to txn
```

- A `false` here is a cross-tenant data leak, not a performance bug.
 - 3. **Migrations and pg_dump use DATABASE_DIRECT_URL (:5432)**. Drizzle Kit on the pooler fails on advisory locks.
 - 4. **Alert on max utilization, not on errors**. By the time you see `remaining connection slots are reserved`, users are already failing. Threshold: page at 70% of p
- Escalation ladder when the ceiling is hit – each step is a slider move, no rewrite:

STEP	ACTION	BUYS	COST DELTA [FETCHED 2026-08-30]
0	Supavisor transaction mode, <code>max: 1</code>	200 pooled conns	\$0
1	Pin API routes to one region (<code>iad1</code> or <code>bom1</code>)	Removes cross-region instance multiplication	\$0
2	Supabase Small → Medium → Large compute	400 → 600 → 800 pooled	Micro \$10/mo → Small \$15 → Medium \$60
3	Read replica for publish/search reads	Halves primary load	+1× compute price
4	Move the hottest read path to Cloudflare KV / R2-cached JSON	Removes it from Postgres entirely	~\$0

We do not reach for step 4 before step 1. Region pinning is free and removes the largest single multiplier.

99.4 Queue design and the backpressure rule

Three job classes, one table, different lanes:

JOB	TRIGGER	P50 DURATION	IDEMPOTENCY KEY	LANE	MAX ATTEMPTS
<code>cert.generate</code> (cross-engine degradation certification)	Doc save, debounced 30s	2–8s	<code>sha256(doc_bytes) + engine_set_version</code>	<code>fast</code>	3
<code>publish.build</code>	Explicit publish	3s–4min	<code>workspace_id + commit_sha</code>	<code>slow</code> , per-workspace concurrency 1	2
<code>vault.import</code>	Connect repo	1–40min	<code>installation_id + repo_id + head_sha</code>	<code>slow</code> , chunked to 200 files/job	5

```
-- supabase/migrations/0007_jobs.sql
CREATE TYPE job_state AS ENUM ('queued', 'running', 'done', 'failed', 'dead');

CREATE TABLE jobs (
  id          uuid PRIMARY KEY DEFAULT gen_random_uuid(),
  workspace_id  uuid NOT NULL REFERENCES workspaces(id) ON DELETE CASCADE,
  kind         text NOT NULL,
  lane         text NOT NULL DEFAULT 'fast',
  idempotency_key text NOT NULL,
  payload       jsonb NOT NULL,
  state        job_state NOT NULL DEFAULT 'queued',
  attempts     int NOT NULL DEFAULT 0,
  max_attempts  int NOT NULL DEFAULT 3,
  run_after    timestampz NOT NULL DEFAULT now(),
  locked_at    timestampz,
  last_error   text,
  created_at   timestampz NOT NULL DEFAULT now(),
  UNIQUE (workspace_id, kind, idempotency_key) -- enqueue is idempotent by construction
);

-- The only index the dequeue needs. Partial: dead/done rows never scanned.
CREATE INDEX jobs_claim_idx ON jobs (lane, run_after, id)
WHERE state = 'queued';
CREATE INDEX jobs_ws_idx ON jobs (workspace_id, state);

ALTER TABLE jobs ENABLE ROW LEVEL SECURITY;
CREATE POLICY jobs_tenant ON jobs USING (
  workspace_id = current_setting('request.workspace_id', true)::uuid
);
```

```
-- The claim. SKIP LOCKED is what makes N workers safe with zero coordination.
UPDATE jobs SET state='running', locked_at=now(), attempts=attempts+1
WHERE id IN (
  SELECT j.id FROM jobs j
  WHERE j.state='queued' AND j.lane=$1 AND j.run_after<=now()
  -- fair queueing: at most one running slow job per workspace
  AND NOT EXISTS (
    SELECT 1 FROM jobs r
    WHERE r.workspace_id=j.workspace_id AND r.state='running' AND r.lane='slow'
  )
  ORDER BY j.run_after, j.id
  FOR UPDATE SKIP LOCKED
  LIMIT $2
)
RETURNING *;
```

The backpressure rule, stated as one sentence an implementer can code against:

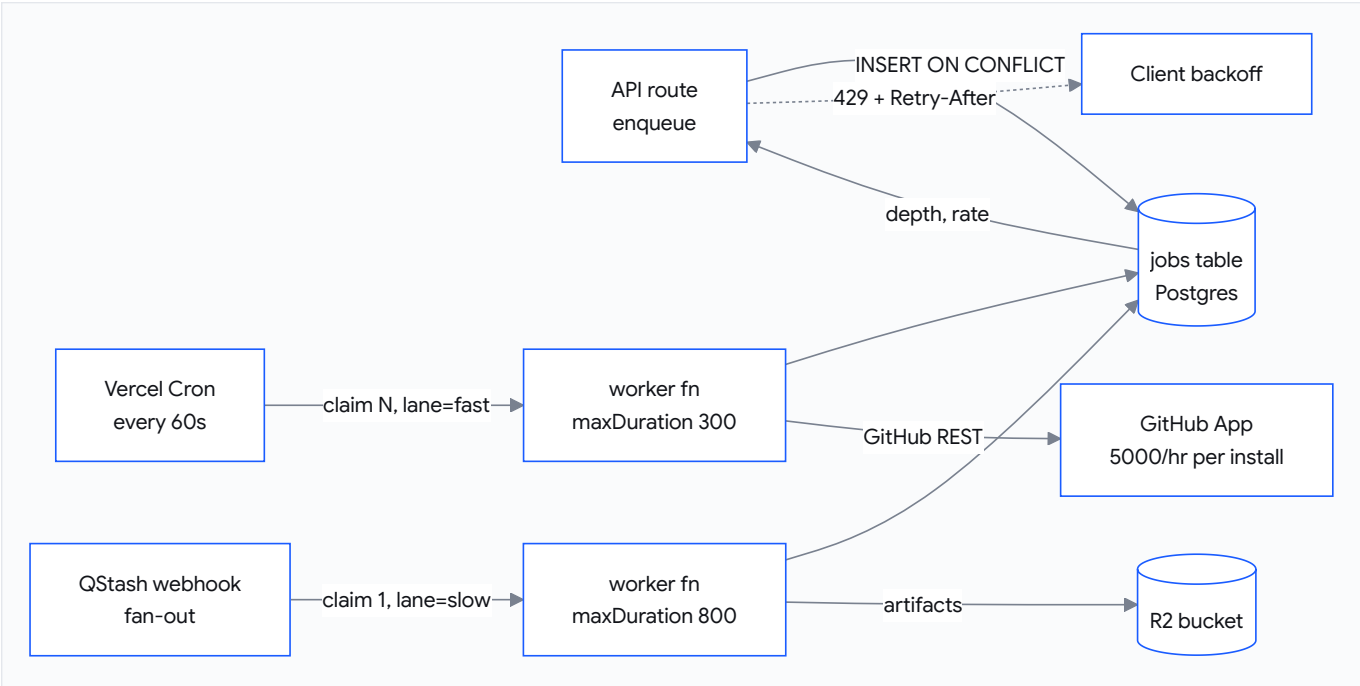
When `queued` depth in a lane exceeds **8× the last 5-minute completion rate** for that lane, the enqueue endpoint stops accepting *new discretionary* jobs

Concretely, at a measured 40 publishes/5min (0.13/s), the threshold is a depth of ~64. Above that:

```
// src/server/jobs/enqueue.ts
const DISCRETIONARY = new Set(['publish.build', 'vault.import']);

export async function enqueue(job: NewJob) {
  if (DISCRETIONARY.has(job.kind)) {
    const { depth, ratePerMin } = await laneHealth(job.lane); // cached 10s
    if (depth > Math.max(20, ratePerMin * 5 * 8)) {
      throw new BackpressureError({ retryAfterSeconds: Math.ceil(depth / Math.max(ratePerMin, 1) * 60) });
    }
  }
  return db.insert(jobs).values(job).onConflictDoNothing({
    target: [jobs.workspaceId, jobs.kind, jobs.idempotencyKey],
  });
}
```

Why refuse rather than buffer: this is the same discipline as the splice engine. An unbounded queue converts a capacity problem into a *correctness* problem – Drain topology:



Vercel Cron minimum interval is 1 minute on Pro [fetched 2026-08-30]; that sets our worst-case fast-lane latency floor at 60s, which is fine for certificati

99.5 The GitHub API: a limit we do not own

[fetched 2026-08-30, docs.github.com rate-limits] GitHub App installation tokens get 5,000 requests/hour minimum; installations on an org with >20 repos or Our consumption model [derived]:

```
splice write      = 3 REST calls
10,000 users peak = 22 req/s × 0.43 (write fraction) ≈ 9.5 splice/s ≈ 28.5 GitHub calls/s
                  = 102,600 calls/hour ACROSS ALL INSTALLATIONS
```

That total is harmless because it is spread across thousands of separate installations, each with its own 5,000/hr bucket. The danger is concentration: one Controls, in build order:

- 1. Per-installation token bucket in Postgres, refilled from the x-ratelimit-remaining and x-ratelimit-reset headers on every response – never estimated. Below
- 2. Conditional requests. If-None-Match on ref/tree reads. A 304 does not count against the primary limit for the classic REST API and materially cuts our re
- 3. Never poll. Webhooks (push, installation) only. Polling is how a GitHub integration dies.
- 4. Retry-After is obeyed literally, and a secondary-limit 403 disables that installation's lane for the full window rather than retrying. Retrying into a se
- 5. Local-first covers the outage. Tauri v2 holds the working copy; a GitHub throttle degrades to "your edits are saved locally, sync is paused", which is tru

99.6 The degradation ladder — what we shed, in order

Ordered by how little the user loses. Each level is a feature flag in config/degradation.ts, flippable without deploy.

LEVEL	TRIGGER	SHED
0	Normal	–
1	Pooled conns >70% for 5 min	Server-side search falls back to client MiniSearch over the open vault only
2	Pooled conns >85%, or job depth >8× rate	New publish.build and vault.import refused (429 + Retry-After)
3	Installation rate budget <15%	Splices batch into one commit per 5 min per workspace
4	AI provider p99 >20s or 5xx>2%	Cross-engine certification runs against cached engine results; AI features return a typed refusal
5	Postgres unreachable	Read-only mode: editor and local vault fully functional; published sites served from R2 cache; login ref

The property that makes this ladder cheap: we can lose the entire control plane and the product still edits files. The file is the truth. Postgres holds ide What we never shed: the splice engine's refusal semantics. Under no load condition does the engine guess a range – degradation may make an edit slower or qu

99.7 Where this architecture stops working

It stops at roughly 50,000-80,000 users, and the wall is not throughput. It is three things arriving together:

WALL	SYMPTOM
Serverless connection economics	Instance count, not request count, drives cost and connections; region pinning is exhausted; Supavisor client_wait_time becomes the p99
Single-writer Postgres	Job table churn plus audit plus full-text on one primary; vacuum pressure visible in p99
One person on call	This is the actual limit and it arrives first

What does not change at that wall, and must not be designed around: the file stays the source of truth, sync stays git-merge + splice journal + CAS, and the What would make us start this migration early: a single B2B customer above 250 seats on one installation (forces the GitHub batching work), or Supabase comp

\$100 Observability, availability and disaster recovery

100.1 The constraint that shapes the whole section

Every telemetry decision below is downstream of one rule: **no log line, breadcrumb, span attribute, error message, or crash report may ever contain document**

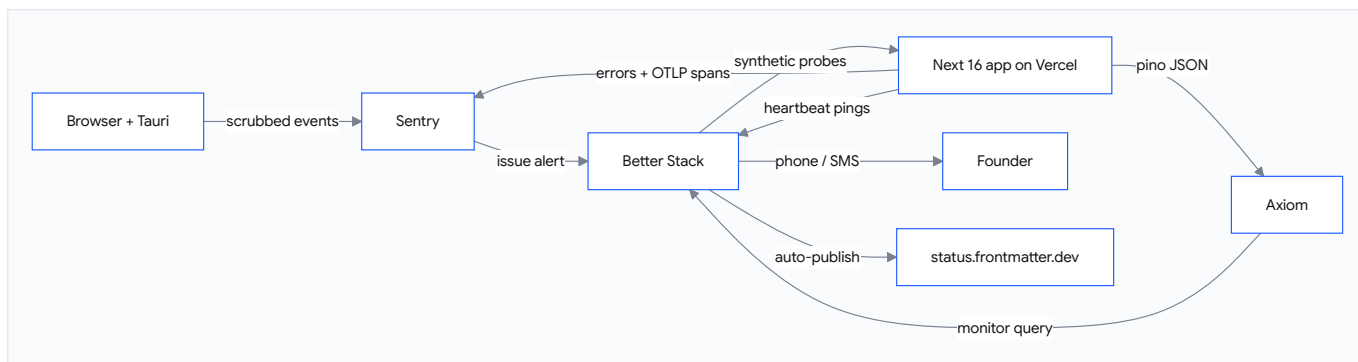
The corollary is stricter than it sounds. It bans not just `logger.info(doc)` but: the splice engine's `Expected "## Heading" at byte 4021, found "## Headnig"` refusal

100.2 Stack selection

LAYER	PICK	REJECTED	WHY
Errors	Sentry SaaS, <code>@sentry/nextjs@10.72.0</code>	Self-hosted Sentry; Bugsnag; Highlight	Best
Logs	Axiom, <code>@axiomhq/pino@2.0.0</code> + <code>pino@10.3.1</code>	Datadog; Better Stack logs; Vercel log drains only	Axio
Traces	OpenTelemetry SDK (<code>@opentelemetry/api@1.9.1</code> , <code>@opentelemetry/sdk-node@0.221.0</code>) exporting into Sentry	Standalone Grafana Tempo / Jaeger	One
Uptime + status page + on-call	Better Stack	UptimeRobot + Instatus (two vendors); PagerDuty	Moni
Metrics	Sentry + Postgres rollup tables. No Prometheus.	Grafana Cloud; Prometheus + Grafana self-hosted	A me

Total observability spend: \$0/mo at 100 users, \$80/mo at 10,000 [derived: 26 + 25 + 29 = 80]. Log volume at 10,000 users, assuming 20 server requests/user/d

100.3 Telemetry topology



Better Stack is the single fan-in for anything that can wake a human. Sentry never phones directly; it raises an issue alert that Better Stack decides wheth

100.4 The correlation envelope, from day one

Two fields on every log line, span and error, with no exceptions: `workspace_id` and `correlation_id`. Retrofitting these later means re-instrumenting 226 TypeSc

`src/lib/obs/context.ts`:

```
import { AsyncLocalStorage } from 'node:async_hooks';

export type ObsContext = {
  correlation_id: string; // uuid v7, generated at edge or accepted from X-Correlation-Id
  workspace_id: string | null;
  user_id: string | null;
  route: string;
};

export const obsStore = new AsyncLocalStorage<ObsContext>();
export const ctx = () => obsStoregetStore();
```

`src/middleware.ts` mints `correlation_id` (uuid v7 - time-sortable, so a log sort is a timeline) and sets it as a response header so a user can paste it into a :

```
// src/lib/obs/logger.ts
import pino from 'pino';
import { ctx } from './context';

const FORBIDDEN = /^(content|text|body|markdown|doc|source|value|query|path|filename)$/i;

function strip(o: unknown, depth = 0): unknown {
  if (depth > 4 || o === null || typeof o !== 'object') return o;
  const out: Record<string, unknown> = {};
  for (const [k, v] of Object.entries(o as Record<string, unknown>)) {
    if (FORBIDDEN.test(k)) { out[k] = `[stripped:${typeof v} ? v.length : '?']`; continue; }
    out[k] = typeof v === 'object' ? strip(v, depth + 1) : v;
  }
  return out;
}

export const log = pino({
  level: process.env.LOG_LEVEL ?? 'info',
  base: { service: 'frontmatter-web', env: process.env.VERCEL_ENV ?? 'dev' },
  mixin: () => ({ ...ctx() }),
  formatters: { log: (o) => strip(o) as Record<string, unknown> },
});
```

An allowlist would be safer than a denylist, and we take the denylist knowingly: an allowlist makes every new log call a two-file change and gets bypassed w Engine refusals – the most valuable log in the product – become shape-only:

```
log.warn({
  event: 'splice.refused',
  reason: 'ANCHOR_DRIFT', // enum, never free text
  doc_sha256_prefix: sha.slice(0, 12),
  byte_start: 4021, byte_len: 37,
  expected_node: 'heading', found_node: 'paragraph',
}, 'splice refused');
```

That is enough to reproduce a refusal against the corpus fixtures without ever holding the document.

100.5 The three scrubbing mechanisms, plus the gate

- 1. `beforeSend` / `beforeSendTransaction` in `sentry.server.config.ts`, `sentry.edge.config.ts` and `instrumentation-client.ts`: walk `event.exception.values[].value`, `event.mes`
- 2. **Breadcrumb filtering.** `beforeBreadcrumb` returns `null` for `category === 'console'` (a `console.log(text)` in dev becomes a leak in prod), and strips `data.body`,
- 3. **Source maps.** `withSentryConfig` in `next.config.ts` with `sourcemaps: { deleteSourcemapsAfterUpload: true }` and `widenClientFileUpload: false`. Maps go to Sentry so sta
- 4. **The gate that makes the above real.** There is no CI today – this is where it starts. `.github/workflows/ci.yml` step:

```
# fail if a raw document identifier reaches a log/telemetry sink
rg -n --type ts \
-e '(log|logger)\.(trace|debug|info|warn|error)\([^\)]*\b(content|markdown|docText|value|src)\b' \
-e 'Sentry\.(captureMessage|setExtra|setContext)\([^\)]*\b(content|markdown|docText)\b' \
src/ && { echo "content leak into telemetry"; exit 1; } || exit 0
```

Plus a unit test `src/lib/obs/__tests__/scrub.test.ts` that feeds a Sentry event containing a 3-paragraph markdown fixture through the real `beforeSend` and assert

100.6 Retention

DATA	STORE	RETENTION	BASIS
Structured logs	Axiom	30 days	Free-tier default; long enough for a monthly incide
Errors + traces	Sentry	90 days (plan default)	Debugging window for rare engine refusals
Audit log (auth, billing, publish, entitlement changes)	Postgres <code>audit_log</code>	7 years	Financial record; survives account deletion by des:
Analytics events	Postgres rollups	25 months	Year-over-year comparison, then aggregate-only

DPDP Act 2023 §8(7)(a) requires a Data Fiduciary to **erase personal data upon withdrawal of consent or as soon as it is reasonable to assume the specified pu**
The tension named explicitly (LR#71): "never delete the raw record" governs documents and journals; it does **not** govern telemetry or leaked secrets. Telemetr

100.7 SLOs a solo founder can honestly commit to

SLO	TARGET	WINDOW	PUBLISH?
API availability (<code>/api/**</code> non-5xx)	99.5%	30-day rolling	Yes
Editor first-paint p75	< 2.5 s	30-day	Yes
Splice-commit success (attempts that commit or cleanly refuse; crashes count against)	99.9%	30-day	Yes
Control-plane RPO	≤ 5 min	–	Yes
Control-plane RTO	≤ 4 h	–	Yes
Support first response	1 business day IST	–	Yes
Incident acknowledgement	30 min business hours / 4 h overnight	–	No – state

99.5% permits 216 minutes of downtime per 30 days [derived: 43,200 min × 0.005 = 216 min = 3h36m]. 99.9% permits 43 minutes – one person asleep in IST canno
Splice-commit success is the SLO that matters most and is entirely ours to control: a **refusal is a success**, a crash or a silent wrong write is a failure. Th

100.8 Alerting

SIGNAL	THRESHOLD	PAGES?
Synthetic probe <code>GET /api/health</code> fails	3 consecutive failures, 2 regions, 60 s apart	YES – phone
5xx rate on <code>/api/**</code>	> 5% of requests over 5 min, min 20 req	YES – phone
Postgres connection failures	> 10 in 5 min	YES – phone
Splice write returned success but CAS post-check mismatch	any single occurrence	YES – phone
Scrub-gate canary: document-shaped string detected in Axiom (<code>content_leak_monitor</code>)	any single occurrence	YES – phone
Nightly backup heartbeat missing	26 h since last ping	YES – phone (second consecutiv
Payment webhook signature failures	> 3 in 15 min	YES – phone
Splice refusal rate	> 15% of attempts over 1 h	No – Slack
p75 editor load	> 4 s for 30 min	No – Slack
New Sentry issue, unseen fingerprint	first occurrence	No – Slack digest
R2 4xx/5xx on attachment PUT	> 2% over 15 min	No – Slack
AI provider 429/5xx	any	No – degrade to next provider,
Deploy failed	any	No – Slack
Disk/quota warnings from any vendor	any	No – Slack

The pages-at-3am set is exactly seven signals, and every one of them means either *customers cannot work*, *money is not being collected*, *data is being damaged*

100.9 Status page and incident comms

`status.frontmatter.dev` on Better Stack's free tier, CNAME'd, showing four components: **Editor**, **Sync (git)**, **AI**, **Publishing**. Probe-driven components update them

100.10 Incident runbook skeleton

Every file in `docs/runbooks/` uses this shape, because at 3am a founder reads headings, not prose:

```
# <signal name>
```

§0

Is this real? # the 30-second check that rules out a false alert

§1

Stop the bleeding # the ONE reversible action: feature flag, rollback, disable route

§2

Confirm blast radius # query with workspace_id + time window, paste the numbers

§3

Communicate # status page template text, pre-written, copy-paste

§4

Diagnose # links: Sentry saved search, Axiom APL query, Vercel deploy list

§5

Fix or escalate # vendor support links + account IDs

§6

Verify recovery # the assertion that must go green, run it 3 times (LR#63)

§7

Postmortem stub # timeline table to fill while it is fresh

Step 1 is always a single reversible lever. For 'splice-corruption.md' that lever is 'FEATURE_SPLICE_WRITES=off', which puts the engine into read-and-refuse mode – the product degrades to a viewer rather than risking one more bad byte. That flag exists because the file is the source of truth: a corrupt write is worse than an outage.

100.11 Backup and restore

Data class	Where	RPO	RTO	Mechanism
Documents	Customer git repo	**0 – not ours**	n/a	Git is the backup. We never hold the only copy. This is the single largest DR advantage of the architecture
Splice journal	Postgres, append-only	5 min	4 h	Provider PITR + nightly 'pg_dump' to R2
Control plane (identity, tenancy, entitlements, billing, slugs, audit)	Postgres	5 min	4 h	Provider PITR + nightly 'pg_dump --format=custom' to R2, gzip-verified
Attachments >1 MB	R2	24 h	24 h	Content-addressed, immutable keys (below) + weekly cross-account copy
Derived artifacts (renders, indexes)	R2	**∞ – rebuildable**	2 h	Do not back up. Regenerate
Secrets	Vercel env + password manager	manual	1 h	Documented in 'docs/runbooks/rebuild-from-zero.md', values never in repo

R2 pricing read 2026-08-30: \$0.015/GB-month standard, Class A \$4.50/M ops, Class B \$0.36/M, **egress free**, free tier 10 GB-month + 1M Class A + 10M Class B [fetched developers.cloudflare.com/r2/pricing, page dated Aug 7 2026]. Free egress is what makes a restore drill cost nothing, which is what makes us actually run one.

****R2 has no S3-style object versioning.**** The R2 documentation index lists exactly one retention primitive – Object lifecycles – and no versioning page; the data-security and FAQ pages do not mention it [fetched, 2026-08-30]. A 'PUT' to an existing key destroys the previous object, silently and irrecoverably. Versioning must therefore live in the ****key layout****, and this is a build-time decision that cannot be retrofitted:

```
attach/<workspace_id>//<original_ext> # content-addressed, write-once backup/pg/<workspace_scope>///
/Z-<sha256_12>.dump.gz render/<workspace_id>/<doc_sha256>/<renderer_ver>.html # derived, disposable
```

Three rules enforced in `src/lib/storage/r2.ts`: (a) every write goes to a key containing a content hash or a timestamp, so no key is ever written twice; (b) all writes use `If-None-Match: \*` so an accidental overwrite fails loudly instead of destroying; (c) deletion happens only through an R2 lifecycle rule on the `backup/` prefix, never through application code. The "current" pointer is a Postgres row, which is itself covered by PITR – so the mutable state lives in the store that has real point-in-time recovery, and the immutable state lives in the store that does not.

Neon is the control-plane default: Free \$0 (0.5 GB storage/project), Launch usage-based at \$0.106/CU-hr and \$0.35/GB-month, typical spend \$15/mo [fetched neon.com/pricing, 2026-08-30]; branch-based instant restore covers operator error, which is the most likely failure. **The provider's PITR is not the backup** – it is co-located with the thing that fails. The nightly `pg\_dump` to R2 in a *different* Cloudflare account is the backup, and it emits a Better Stack heartbeat ping on success so a silent failure pages us (LR#55: under `set -e`, a trailing `[ cond ] && cmd` guard returns 1 and kills the caller – that exact bug disabled nightly backups for weeks once already; write `if/then/fi`, and verify the write landed by counting objects before and after, per LR#67).

100.12 Restore drills

A backup that has never been restored is not a backup. It is a file.

Drill	Cadence	Pass condition
Postgres restore to a scratch branch	**Monthly**, first Sunday	Row counts within 0.1% of live for `workspaces`, `users`, `entitlements`; app boots against it; login works
R2 attachment fetch from cross-account copy	Monthly	10 random keys fetch and hash-match
Full rebuild-from-zero (new Vercel project, new DB, restored dump, DNS cutover rehearsed but not executed)	**Quarterly**	Serving traffic on a staging domain within the 4 h RTO, timed with a stopwatch
Secret-loss rehearsal	Quarterly	Every secret in `rebuild-from-zero.md` is retrievable without reading any file on the founder's laptop

Each drill writes a dated row to `docs/runbooks/drill-log.md`: date, artifact restored, wall-clock elapsed, what broke. If the last row is more than 45 days old, the deploy pipeline prints a warning. The elapsed time is the real output – an RTO of 4 h that has only ever been asserted is a guess; an RTO of 4 h measured three times is a commitment. Report a drill result as `[measured]` only when it ran against real artifacts; a dry run against modified code is `SIMULATED` and must say so (LR#62).

§101
What is still unresearched

The corpus is 105 agent reports across 13 rounds, plus 16 PRD section drafts and 5 standing analyses – 126 .md files under docs/research/ MEASURED 2026-08-30
k report covers it; PRD hits – SPF 0, DKIM 0, DMARC 1 (a GitLab DR anecdote), transactionalemail 0, lifecycle
email 0. Everything below was found by grepping the PRD and the corpus for the vocabulary a topic cannot be discussed without, then reading the hits.

2.1 Coverage table

Status is assigned against the standard the corpus sets for itself elsewhere: a topic is COVERED when a named report or PRD section reaches a decision with

#	TOPIC	STATUS	WHAT EXISTS
1	Positioning, category naming	COVERED	\$52, e6, r4
2	Launch channels	COVERED	\$26, two funnels, channel-discipline rule
3	Content strategy beyond launch	PARTIAL	i1, x1 describe the content engine; \$26 sets cadence
4	SEO – marketing site	MISSING	SEO 4 hits in PRD, all naming-risk or AEO MEASURED
5	SEO – published pages	MISSING	Named in \$55.3 as never researched FETCHED
6	AEO / GEO	PARTIAL	\$13 L4
7	Referral mechanics	MISSING	1 hit in PRD, 2 files in corpus – both the ban MEASURED
8	Affiliate / creator program	MISSING	affiliate 0 hits, PRD and corpus MEASURED
9	Partnerships and integrations	MISSING	partnership 0 hits in PRD, 2 files in corpus MEASURED
10	The plugin/extension ban	PARTIAL	\$46.2 quantifies the saving: 22.79% of Obsidian-equivalent forum traffic DERIVED FROM MEASURED
11	Community	COVERED	\$49.4, k13
12	Price levels, FX, rails	COVERED	\$24, c4, k14
13	Pricing experimentation	MISSING	\$24.2 says "A/B ₹249 and ₹399"; \$24.3 says "Test ₹299 against ₹399" FETCHED
14	Pricing page as a conversion surface	MISSING	24 corpus files mention competitors' pricing pages MEASURED
15	Trial vs freemium	PARTIAL	\$25.2's benchmark table gives free-trial 8–12% vs freemium 3–5% FETCHED
16	In-product messaging	MISSING	\$17 bans weekly digest, What's-New modal, trial countdown FETCHED
17	Refunds, EU right of withdrawal	MISSING	refund policy 0, right of withdrawal 0 MEASURED
18	Retention definition	COVERED	\$47.3, return-with-a-file
19	Churn rate and steady state	MISSING	\$25.3: "Zero churn assumed" FETCHED
20	Cancel, downgrade, win-back	MISSING	\$46.3 promises self-serve cancel as a deflector; win-back 0 hits MEASURED
21	Expansion revenue	MISSING	expansion revenue 0 files, net revenue retention 0 files MEASURED
22	Email infrastructure and deliverability	MISSING	r11 gap #9, never closed MEASURED
23	ICP definition	COVERED	\$21, c3
24	Self-serve B2B motion	PARTIAL	c3 picks the 2–20 band and refuses enterprise
25	Enterprise procurement	COVERED-as-refusal	c3; \$55.4 sets the ACR/VPAT gate before any procurement conversation
26	Security questionnaires / SOC 2	PARTIAL	Vanta/Drata "\$7,000–\$30,000/yr" SS, flagged in \$55.2 #8
27	Contracts – ToS, DPA, MSA, SLA	PARTIAL	\$51 is an issue list for counsel
28	Support tooling and deflection	COVERED	\$46, r6 – deflection plan, hire trigger, tooling priced
29	Billing operations, India	COVERED	\$45, k14 – RBI framework read paragraph by paragraph
30	Accounting and bookkeeping	MISSING	bookkeep 0, accounting 0 in PRD MEASURED
31	Insurance and liability transfer	MISSING	insurance 1 hit, and it is RBI's carve-out list MEASURED
32	Hiring and contractor management	PARTIAL	\$46.5 prices the hire and derives the trigger DERIVED
33	Founder time budget, aggregate	MISSING	\$53 gives throughput; \$46 gives 23.97 h; \$45.10 gives 28.3 h
34	FX exposure	MISSING	FX risk 0, hedge 6 hits all unrelated MEASURED
35	Model-vendor concentration	PARTIAL	\$50.1 has a risk row; \$23.3 prices six models FETCHED
36	App-store distribution tax	MISSING	App Store 6 hits, all competitor context MEASURED
37	Competitive war-game	MISSING	competitive response 0, war-game 0 MEASURED
38	PPP pricing beyond India	MISSING	PPP 0, purchasing power 0 MEASURED

2.2 MISSING, ranked by what it costs to discover late

Ranked by (irreversibility of the retrofit) × (months of work already committed when the truth arrives).

RANK	GAP	COST OF LATE DISCOVERY
1	Churn	At 5% monthly churn, holding 5,675 paid users requires 284 paid a
2	Email infrastructure	A cold sending domain warms over weeks. Discovered the day invite
3	Founder time, aggregate	Post-deflection support 23.97 h + billing-ops 28.3 h at 10,000 us
4	Pricing experimentation feasibility	Detecting ₹299 vs ₹399 at a 4%→5% conversion difference needs 6,74
5	Partnership surface	The bridge plugin is the distribution play and lives under someone
6	Refunds and withdrawal	Sits in the checkout flow and the ToS, both written once and refe
7	Insurance	Cover is not retroactive. Both India sources probed refused: icic
8	Incident and breaking-change channel	\$17's no-email-field decision and \$38's "status page within 15 mi
9	Accounting and revenue recognition	Annual plans create deferred revenue in year one; correcting the
10	App-store tax	Decides whether the mobile lane is a PWA or a native app, which i
11–14	Marketing-site SEO · competitive war-game · PPP tiers · affiliate and referral	Each costs months of misallocated content or pricing work, none i

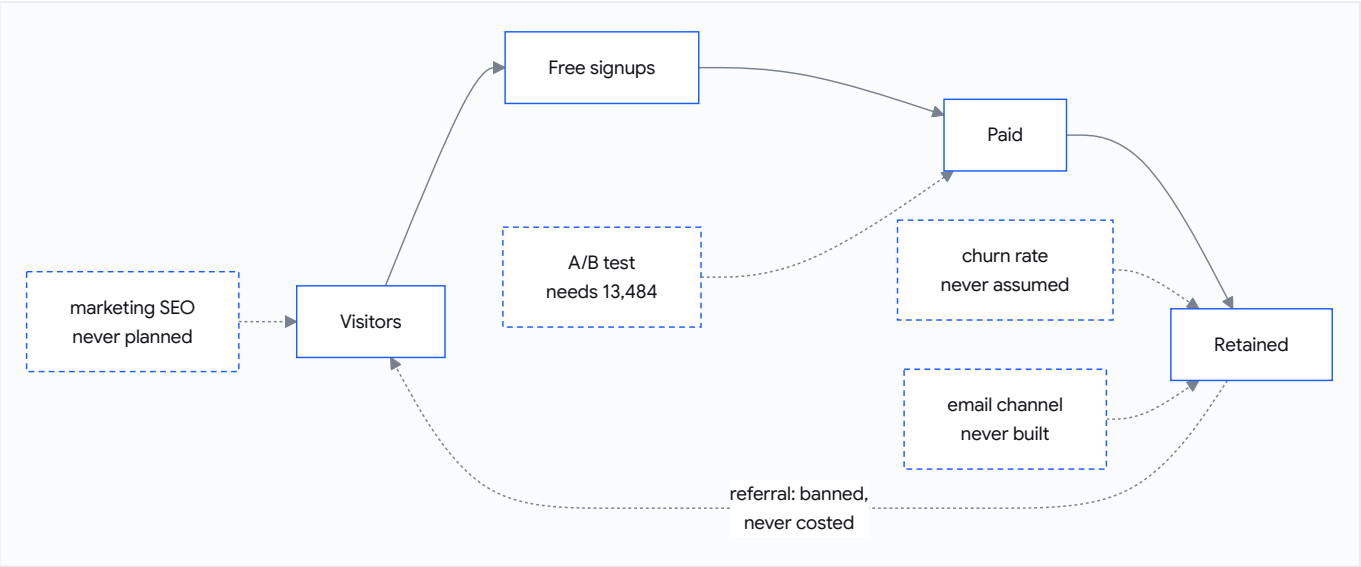
- Do this: answer #1, #3 and #4 with arithmetic this week – all three are derivations over numbers the PRD already contains, and none needs a customer.

- **Anti-recommendation:** do not research #11-#14 before R0 ships. They are real gaps and they are all reversible; committing engine weeks to them is the fail

2.3 Where the corpus is thin rather than absent

\$55.2 already ranks 21 unverified claims by public embarrassment, and that list stands. These are different: places where a decision rests on a single un-ad

CLAIM	WEAKNESS
"Median solo B2B founder revenue >4x B2C by month 24"	SS , no named study FETCHED, \$55.2 #7
Support deflection 18% / 40-60% / \$25-35 per ticket	SS , single source FETCHED, \$55.2 #6
Vendor AI-deflection rates 73-76%	All FETCHED and all vendor self-reported; \$46.4 already halves them to 38%
Obsidian forum tag histogram as our ticket-mix prior	MEASURED on the right corpus, but it is a <i>forum topic</i> distribution standing in for a <i>support tick</i>
Relay's "172,544 downloads" as bridge-plugin precedent	SS FETCHED, \$26
Craft India pricing ₹526.7-658.3	FETCHED but single-IP, single-moment geo-price FETCHED, R11 C1
\$21.1's 0.02/0.10 tickets per user-month	Assumption stated once, then quoted as a conclusion twice FETCHED, \$55.2 #13
WPSD north-star	Single-author construct, no external precedent opened, and \$47.2 argues against it in its own te



2.4 The three questions that waste the most months

#	QUESTION	WHY IT DOMINATES
Q1	Does anyone pay for fidelity?	\$50.2 already names it as a 4x5 risk FETCHED . What is unresearched is the <i>test</i> . R0 is ~9 weeks FETCHED, \$2
Q2	What monthly churn does the model survive?	The plan's spine is gross additions with churn set to zero FETCHED, \$25.3 . At 3% the ₹20L/mo milestone nee
Q3	Is any acquisition channel one we control?	HN is a one-shot; the Obsidian plugin is governed by another company's policy and forbids client-side te

Answering Q1 wrong is the only one of the three that cannot be recovered by working harder.

2.5 What was checked and found genuinely complete

This audit went looking for holes in the following and did not find ones worth reporting. Each was opened and read, not inferred from the table of contents.

AREA	WHY IT PASSES
Sync engine (\$31, k2)	Three independent disqualifications for CRDTs, a convergence oracle, and a named failure-mode list
Backup and DR (\$32, k1)	Leads with the finding that inverts the standard recipe – R2 has no object versioning – and states what each layer does <i>no</i>
Support economics (\$46, r6)	Rebuilds the PRD's own 46.4-hour figure to 80.45 and says the flat 12-minute assumption was the error
India billing (\$45, k14)	RBI framework read by paragraph; the ₹15,000 ceiling treated as an architectural constant, not a price input
Observability (\$38, k6)	A denylist of what may never enter telemetry, and a "never alert on" list with the source's own critique quoted
Accessibility (\$35, k4)	A measured live failure – body-faint #b8b8b8 at 2.14:1 – not a checklist
Evidence discipline itself	\$0.1 tag definitions, \$55 verification debt, \$57 contradictions ledger, \$58 a publication denylist

Three things this audit did **not** verify and is therefore not asserting: the India insurance market (both sources refused, above); whether any competitor has

22.1 How to read a licence before you read the code

All repository metadata in this section was read from `api.github.com` on **2026-08-30** [fetched]. Star counts and push dates move; the licence text moves rarely, Three contagion tiers apply to us specifically, because we ship a hosted SaaS *and* a desktop build:

TIER	LICENCES	WHAT IT MEANS FOR US
Red – read-only	AGPL-3.0, GPL-3.0, GPL-2.0	AGPL reaches through the network: a hosted servi
Amber – read the fine print	BUSL-1.1 (Outline), AFFiNE EE, Any Source Available License 1.0 (Anytype)	Outline's Additional Use Grant forbids use "for LICENSE, 2026-08-30]. That is a description of o
Green – usable	MIT, Apache-2.0, BSD-3-Clause, MPL-2.0	Copy with attribution; MPL is file-level copylef

AFFiNE is split: everything outside `packages/backend` and `packages/common/native` is MIT, the backend is under the "AFFiNE Enterprise Edition (EE) license" requ

22.2 The register

Stars / last push read 2026-08-30 [fetched]. "Learn" and "Avoid" are judgement calls on top of directory listings actually opened [inference].

REPO	STARS	LAST PUSH	LICENCE	TIER	STACK	LI
go-gitea/gitea	57,676	2026-08-30	MIT	Green	Go	s
decaporg/decap-cms	19,327	2026-08-28	MIT	Green	JavaScript	f
sveltia/sveltia-cms	2,768	2026-08-30	MIT	Green	Svelte/JS	s
Thinkmill/keystatic	2,326	2026-08-26	MIT	Green	TypeScript	a
payloadcms/payload	44,491	2026-08-29	MIT	Green	TypeScript	f
tinacms/tinacms	13,764	2026-08-28	Apache-2.0	Green	TypeScript	f
estruyf/vscode-front-matter	2,539	2026-08-21	MIT	Green	TypeScript	s
silverbulletmd/silverbullet	5,956	2026-08-30	MIT	Green	Rust + TypeScript	I
jackyzha0/quartz	13,136	2026-08-18	MIT	Green	TypeScript	I
toeverything/blocksuite	5,997	2026-08-26	MPL-2.0	Green	TypeScript	B
toeverything/AFFiNE	72,019	2026-08-28	MIT + EE backend	Amber	TypeScript	F
outline/outline	40,380	2026-08-29	BUSL-1.1	Amber – the grant excludes us by name	TypeScript	s
laurent22/joplin	56,167	2026-08-30	AGPL-3.0-or-later	Red	TypeScript	f
siyuan-note/siyuan + siyuan-note/dejavu	46,053 / 67	2026-08-30 / 2026-08-23	AGPL-3.0	Red	TS + Go	De
logseq/logseq	44,687	2026-08-29	AGPL-3.0	Red	Clojure	Ti
TriliumNext/Trilium	37,634	2026-08-30	AGPL-3.0	Red	TypeScript	Nc
docmost/docmost	21,514	2026-08-29	AGPL-3.0	Red	TypeScript	a
hedgedoc/hedgedoc	7,386	2026-08-28	AGPL-3.0	Red	TypeScript	b
standardnotes/app	6,612	2026-08-25	AGPL-3.0	Red	TypeScript	E
Zettlr/Zettlr	13,451	2026-08-29	GPL-3.0	Red	TypeScript	s
AppFlowy-IO/AppFlowy	76,077	2026-08-28	AGPL-3.0	Red	Dart + Rust	Ri
streetwriters/notesnook	14,482	2026-08-29	GPL-3.0	Red	TypeScript	Ci
anyproto/anytype-ts	8,724	2026-08-29	Any Source Available 1.0	Amber	TypeScript	Lc

22.3 The ranked six, and the file to open first

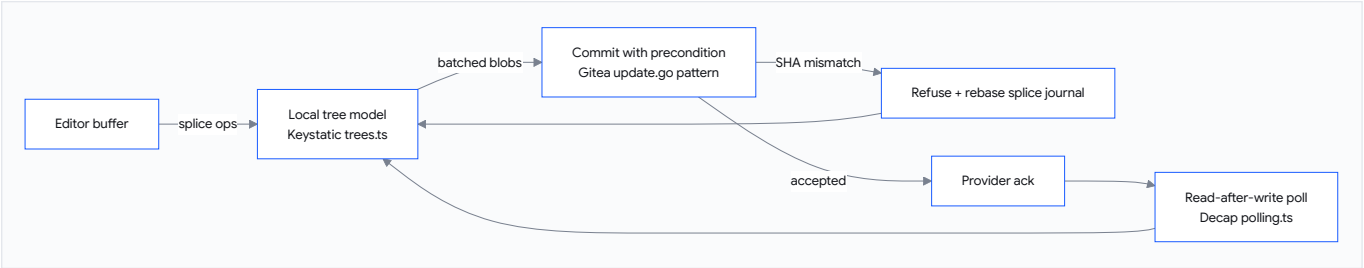
1. `go-gitea/gitea` – `services/repository/files/update.go`. Open at line 380. `if file.SHA != "" { if file.SHA != fromEntryIDString { return ErrSHADoesNotMatch }}`, then the
2. `Thinkmill/keystatic` – `packages/keystatic/src/app/trees.ts` (7.2 KB), then `shell/BatchCommits.tsx` (10.6 KB). This is the closest existing thing to our client: a
3. `sveltia/sveltia-cms` – `src/lib/services/backends/git/shared/`, before any provider directory. Four providers (`github`, `gitlab`, `gitea`, plus `fs`) forced a sharec
4. `payloadcms/payload` – `packages/payload/src/auth/getAccessResults.ts`. We need a permission model over documents whose bytes we do not store, which means permissi
5. `decaporg/decap-cms` – `packages/decap-cms-backend-github/src/polling.ts` (12.6 KB). Not the commit code – the *waiting* code. A browser that commits to GitHub must
6. `hedgedoc/hedgedoc` – `backend/src/permissions/`, then `backend/src/revisions/`, then `backend/src/api-token/`. AGPL, so read-only, but the module decomposition is the

22.4 Subsystem to best reference

OUR SUBSYSTEM	BEST REFERENCE	WHY IT WINS	SI
Compare-and-swap write to git	Gitea <code>files/update.go</code> [fetched]	Refuses the write without a precondition, rather than warning	K f
Browser-side git tree model	Keystatic <code>app/trees.ts</code> [fetched]	Tree + content-addressed cache, no server round-trip per node	S
Read-after-write consistency	Decap <code>polling.ts</code> [fetched]	Only implementation that treats provider lag as a first-class state	-
Multi-provider git abstraction	Sveltia <code>backends/git/shared</code> [fetched]	Four providers, so the abstraction is falsified not assumed	D
Permission model (control plane)	Payload <code>getAccessResults.ts</code> [fetched]	Access resolves to data, not scattered guards; MIT	H (
Splice journal / conflict semantics	Joplin <code>synchronizer/LockHandler.ts</code> + <code>Synchronizer.conflicts.test.ts</code> [fetched]	Twelve years of conflict edge cases, encoded as tests	D
Content-addressed snapshot store	<code>siyuan-note/dejavu</code> [fetched README]	Chunked dedup + index-per-operation, in 67-star readable Go	-
CodeMirror 6 markdown surface	Zettlr <code>markdown-editor/</code> [fetched]	<code>parser/</code> , <code>renderers/</code> , <code>linters/</code> , <code>table-editor/</code> cleanly separated	F f
Derived index over files-of-record	TinaCMS <code>@tinacms/graphql/src/{database,level}</code> [fetched]	Indexes git without claiming to own it	S
Licence-split commercial model	AFFiNE MIT + EE backend [fetched]	The exact split we would use if we ever open a client	O

22.5 The git-backed editor problem, specifically

Almost nothing solves it end to end. Three repos solve pieces, and the pieces compose:



The unusual part of our design - that the file is the only source of truth and the server stores none of its bytes - has exactly two prior implementations a One dead end worth knowing: `netlify/git-gateway` (431 stars, MIT, Go) is the credential-proxy pattern that lets a browser commit without holding a token, and

22.6 Anti-recommendations — repos that look relevant and will mislead

REPO	WHY IT LOOKS RELEVANT	WHY IT WILL MISLEAD
<code>outline/outline</code>	Best-in-class collaborative document editor, TypeScript, readable	Its BUSL Additional Use Grant
<code>dendronhq/dendron</code>	Markdown-first, git-backed, Apache-2.0 - sounds ideal	290 days since last push [der
<code>prose/prose</code>	The original browser git-backed markdown editor	921 days stale [derived, 2026
<code>AppFlowy-IO/AppFlowy</code>	76,077 stars, Rust core, local-first	It is a Notion-style workspac
<code>toeverything/blocksuite</code>	MPL-2.0, so legally usable, and a real editor engine	Block-tree-of-record is the m
<code>logseq/logseq</code> , <code>siyuan-note/siyuan</code> , <code>TriliumNext/Trilium</code>	Large, active, file-adjacent	All AGPL; all resolved the fi
<code>Milkdown/milkdown</code> , <code>Vanessa219/vditor</code> , <code>marktext/marktext</code>	Markdown editors with stars	Libraries and a desktop app,
<code>obsidianmd/obsidian-releases</code>	The plugin ecosystem to learn from	It is a submission registry f
<code>AFFiNE</code> backend	Same problem space, active	EE-licensed: production use r
<code>docmost/docmost</code> , <code>hedgedoc/hedgedoc</code> realtime	Small, modern, readable Yjs servers	They will make CRDT-for-docum

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Design references and the visual language

23.1 The reference set

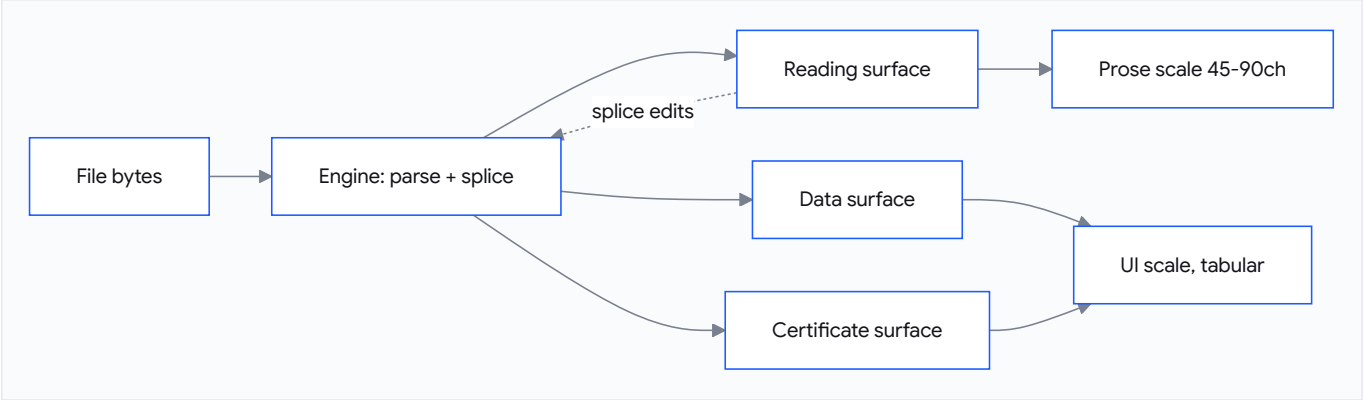
Every row below was opened with `curl` on 2026-08-30 (the visual-pattern MCP was gate-refused; `curl` was not - tested before concluding, per the rule that a l

REFERENCE	URL	WHAT IT IS EVIDENCE FOR
Vercel Geist	vercel.com/geist	Content rules for dense UI, empty-state taxonomy, named type
GitHub Primer	primer.style/product	Empty-state anatomy; <code>DataTable</code> ; UI-pattern layer above compc
Shopify Polaris	polaris.shopify.com	What happens when you depend on a vendor system
Atlassian Design System	atlassian.design	A mature system still hedging on tables
IBM Carbon	carbondesignsystem.com	The only system publishing exact table density numbers
Radix Primitives / Colors	radix-ui.com	Unstyled behaviour; a 12-step colour scale with role bands
shadcn/ui	ui.shadcn.com	Copy-in distribution instead of a runtime dependency
TanStack Table / Virtual	tanstack.com	Headless table behaviour, actively maintained
Glide Data Grid	github.com/glideapps/glide-data-grid	Canvas-rendered grid at spreadsheet scale
CodeMirror 6	codemirror.net	The editing surface itself
cmdk	github.com/dip/cmdk	Command palette – and a maintenance warning
ia Writer	ia.net	Typography as the product
Obsidian	obsidian.md/pricing	Local-file positioning and a paid-sync ladder
Craft	craft.do/pricing	Block-metered freemium (INR shown from an India IP)
Notion	notion.com/pricing, developers.notion.com/reference/block	The block-of-record model, stated in its own API
Butterick, <i>Practical Typography</i>	practicaltypography.com/line-length.html	Measure
NN/g, Kaplan (2021-09-19)	nngroup.com/articles/empty-state-interface-design	Empty states as system status, not decoration
WCAG 2.2 SC 1.4.3	w3.org/WAI/WCAG22/Understanding/contrast-minimum	Contrast floor, both themes
Stripe Apps	docs.stripe.com/stripe-apps/design	Constrained component grammar for third parties

`ag-grid.com/license-pricing` returned HTTP 403 from CloudFront [measured] – no AG Grid price is stated in this section, and none should be quoted from memory.

23.2 Visual principles, derived

#	PRINCIPLE	DERIVED FROM
1	Two type systems in one shell: a reading surface bound by measure, an app surface bound by density	Geist separates <code>text-copy-*</code> ("multiple lines... hig
2	Absence is typed, never coerced	Geist Table: "Render – in cells where a value is
3	Every control states the byte effect before it runs	Geist: sortable headers "are buttons... announce th
4	Borrow tokens and behaviour; do not take a runtime dependency on a design system	Polaris React is archived and its npm package has
5	Density is a user setting with published numbers	Carbon publishes five row heights: xs 24, sm 32, i
6	Contrast floor is a hard gate in both themes	WCAG 2.2 SC 1.4.3: 4.5:1 body, 3:1 large [fetched
7	Keyboard parity before visual polish	Geist ships a <code>Command Menu</code> primitive; Linear publ



23.3 Component inventory, mapped to a reference implementation

Claude implements each of these from the reference's *documented behaviour*, not from its package.

COMPONENT	REFERENCE IMPLEMENTATION	WHY THAT
Text editing surface	CodeMirror 6, @codemirror/view 6.43.9 [fetched]	Byte-of
Command palette	cmdk's list/filter semantics, reimplemented	The beh
Empty state	Geist Empty State variants: blank-slate, informational, educational, guide, plus no-results / cleared / permission / error [fetched]	It is t
Data table	TanStack Table 9.2.4 headless + Geist content rules + Carbon density scale [fetched]	Behavio
Row virtualisation	TanStack Virtual, 7,089* MIT [fetched]	Require
Key/value metadata block	Geist Description - explicitly not a two-column table [fetched]	Frontma
Row-with-one-action	Geist Entity [fetched]	Repos,
Relative time	Geist Relative Time Card: 2m ago up to 7 days, then Mar 14, 2026 [fetched]	Journal
Truncation	Geist MiddleTruncate [fetched]	Paths a
Status	Geist Status Dot + Primer StateLabel [fetched]	Certifi
Diff / degradation report	Primer DataTable + its "Degraded experiences" UI pattern [fetched]	Primer
Blank/loading	Primer SkeletonText / SkeletonBox, Geist Skeleton [fetched]	Only fo
Colour tokens	Radix 12-step role bands: 1-2 backgrounds, 3-5 interactive, 6-8 borders, 9-10 solid, 11-12 accessible text [fetched]	Roles s
Numeric/mono type	Geist Mono or tabular-nums, per Geist's table rule [fetched]	Digit a
Overlays, menus, focus	Radix Primitives 19,222* MIT [fetched]	Focus t

23.4 Typography and spacing for the reading surface

The reading surface and the app surface disagree about almost every value, and the disagreement is the design.

DECISION	READING SURFACE	APP SURFACE
Measure	45-90 characters; target ~68ch, hard cap 90ch	full width, column-bounded
Face	Duospace-class for source view; proportional for the projection	one UI sans; mono only in numeric columns and code
Why duospace	Markdown source is punctuation-load-bearing: #, -, `	, >, `` `` must align down the left edge and inside table pipes; full mon
Size	16-17px body	14px default, 13px secondary
Line height	~1.6	~1.45 single-line, 1.3 in table cells
Numerals	proportional in prose	font-variant-numeric: tabular-nums in every numeric column
Vertical spacing	derived per markdown block type; paragraph gap ≈ 0.75 × line-height	4px base unit; Carbon's \$spacing-05 = 16px, \$spacing-09 = 48px [f

The one hard constraint the file model imposes: vertical spacing in the editor must be a pure function of block type, because the same bytes must produce the same visual spacing. Set the source view in a duospace face and the projection in a proportional one, and never let a font change alter a byte offset the splice engine depends on.

23.5 The dense-table problem

Most design systems are bad at this, and several admit it. Atlassian ships Dynamic table alongside a plain Table marked Caution [fetched]. Carbon's own guide says: "The dense table is a table with a high density of data. It is used to display a large amount of data in a compact way. Who does it well, and what they actually do:

WHO	WHAT THEY DO THAT OTHERS DO NOT
IBM Carbon	Publishes the metrics: five row heights (24/32/40/48/64px), checkbox 20px, cell padding 16px, expanded-panel left padding 48px, and a padding of 16px between panels
Vercel Geist	Publishes the content rules, which is the part everyone skips: em dash for unknown values; Title Case noun headers (Last Used, Requests); empty state rendered outside Table.Body, never as an empty body
TanStack Table	Headless - sorting, grouping, column sizing, pagination as state, zero markup. 9.2.4 published two days before this was read
Glide Data Grid	Canvas rendering, which is the only way past a few thousand visible cells; 5,322* MIT, but last pushed 2026-01-21 - seven months quiet
Primer	A Degraded experiences UI pattern sitting above the table components, so a partial data load has a designed presentation rather than an empty table

The recommendation: compose Carbon's density numbers, Geist's content rules, and TanStack Table's headless state, rendered as DOM until a measured frame bud

23.6 Dark mode

It matters, and the cost is bounded if it is a token swap rather than a second design.

QUESTION	ANSWER
Do the references treat it as default-tier?	Yes. Geist's docs expose system/light/dark; Primer offers a dark-mode switch; Radix ships paired light and dark themes
Does the contrast floor change?	No. SC 1.4.3's 4.5:1 / 3:1 applies identically
What is the actual work?	One token set with role bands (Radix 1-2 background, 6-8 border, 11-12 text; Geist's parallel 1-3 background, 6-8 border, 11-12 text)
Where does it break?	Syntax highlighting, diff colours, and the degradation certificate's pass/warn/fail semantics must be re-designed
Where does it not apply?	Print, PDF export, and shared read-only links default to light regardless of the viewer's theme

Anti-recommendation: do not implement dark mode as a CSS filter or a colour inversion, and do not use pure #000 with pure #fff text - Geist defines two di

23.7 What not to copy from Notion

Notion's API states the model plainly: "A block object represents a piece of content within Notion," each carrying id, parent, created_by, last_edited_by, and a type

NOTION MOVE	WHY IT FAILS HERE
Per-block drag handle and hover-revealed block menu	It advertises that blocks are addressable objects with identity. Here a "block" is a byte range that
Slash menu that inserts a block <i>type</i>	The correct affordance inserts markdown <i>text</i> the user could have typed. A menu that inserts an objec
Page properties as a database row UI	Frontmatter is an ordered YAML mapping whose key order and comments are preserved bytes; a properti
Toggle blocks, column layouts, synced blocks	None has a stable markdown byte representation; every one of them creates content that cannot round-
Everything-is-a-page nesting	Files and folders already have a hierarchy, and it is the user's git repo [inference]
Cover images and per-page emoji identity	Emoji render differently across OS and font versions and are a control-surface failure in any export

What to take instead: the *speed* of Notion's block manipulation – arrow-key movement, `Cmd+Shift+↑/↓` to move a line or list item, backspace-at-start to outdent

23.8 Anti-recommendations: moves that photograph well and fail in daily use

MOVE	WHY IT SURVIVES A SCREENSHOT	WHAT BREAKS BY WEEK TWO
Hover-revealed row actions	Screenshots are captured mid-hover	Zero discoverability, no touch equivalent, no tab stop – Geist requ
Icon-only toolbar	Clean	Every icon becomes a memory test; Geist's table rule is that header
Low-contrast secondary text	Photographs as refined	Fails SC 1.4.3's 4.5:1 and is unreadable on a laptop outdoors [fetc
Skeleton loaders everywhere	Implies speed	A local file read completes in under a frame; a skeleton adds a man
Animated page transitions between documents	Looks expensive	At 30 document switches an hour, a 250ms transition costs 7.5 secon
Command palette as the primary path	One screenshot shows the whole product	Nothing is discoverable to a new user; the palette should be the fa
Auto-hiding sidebar	Maximises the canvas in a comp	Position becomes unpredictable; users lose the file tree they came
Full-bleed gradient app chrome	Marketing-ready	Competes with the document for attention on every session; Geist's
Toast on every save	Reads as responsive	Byte-preserving autosave fires constantly; the correct signal is a
Infinite scroll on a data table	Feels modern	No end-of-list, no position, no shareable page. Geist specifies <code>Pag</code>
One comfortable density	Consistent-looking	Power users hit the viewport ceiling immediately; Carbon ships five
Emoji as status or control icons	Colourful, free	Renders differently per OS, font version, and export target; use a
Celebration animation on completion	Delightful once	Delightful zero times on the two-hundredth document [inference]

The one move worth the screenshot cost: an empty state that does real work. NN/g's finding is that intentionally designed empty states "communicate system s

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Build versus buy, component by component

24.1 The decision table

Scale assumption throughout: 1,000 registered users, 8% paid, \$12/mo ARPU → \$960 MRR [derived, \$24.2]. Founder-weeks assume Claude as the implementer at rou

#	COMPONENT	VERDICT	VENDOR AND PRICE AT OUR SCALE
1	Auth	BUY	WorkOS AuthKit – free up to 1M users, then \$2,500/mo per additional 1M; SSO connections \$125/ea (1-15) [f
2	Payments + MoR	BUY	Paddle – 5% + 50¢ per checkout transaction, no monthly fee [fetched paddle.com/pricing]. Alt: Polar – Sta
3	Email (transactional)	BUY	Resend – Free 3,000/mo (100/day, 3 domains); Pro \$20/mo 50,000 emails, overage \$0.90/1,000; Scale \$90/mo
4	Email (marketing)	BUY, deferred	Loops – free to 4,000 sends/mo, priced on subscribed contacts, sending not metered separately [fetched lo
5	Error tracking	BUY	Sentry Developer \$0 (one user, 5k errors); Team \$26/mo billed annually – 50k errors, 5 GB logs, 50 replay
6	Uptime + status page	BUY	Better Stack – free plan includes 10 monitors; paid tiers observed from \$29 [fetched betterstack.com/updi
7	Analytics	BUY	PostHog free tier – 1M events, 5K session replays, 1M feature-flag requests/mo, no credit card [fetched p
8	Search	BUILD	\$0 – SQLite FTS5 on the client over the working copy, Postgres FTS over metadata only. Deferred alt: Meil
9	Real-time + presence	BUILD	Cloudflare Durable Objects on Workers Paid – \$5/mo account minimum, 1M requests/mo then \$0.15/M, 30M CPU-
10	File storage / blobs	ALREADY HAVE	Cloudflare R2 – \$0.015/GB-month, Class A \$4.50/M, Class B \$0.36/M, zero egress; free 10 GB-month, 1M Clas
11	CDN	ALREADY HAVE	Cloudflare, bundled with the zone – \$0 marginal
12	Database hosting	BUY	Neon Launch – \$0.106/CU-hour, \$0.35/GB-month, up to 16 CU [fetched neon.tech/pricing]. Alt: Supabase Pro
13	Background jobs + queues	BUILD	\$0 – SELECT ... FOR UPDATE SKIP LOCKED on the control-plane Postgres. Alt: Inngest free 50k executions / Pro
14	Feature flags	BUY	PostHog, included in the free tier above – 1M flag requests/mo [fetched]. OSS alt: Unleash AGPL-3.0 13,77
15	Support desk	BUY, deferred	support@ in a mail client at \$0 until roughly 300 paying users; then Plain Foundation \$35/mo, 1 seat, +\$:
16	Documentation site	BUILD	\$0 – Astro Starlight MIT 9,153* [fetched] or Docusaurus MIT 66,123* [fetched], on Cloudflare Pages
17	AI gateway + observability	BUY	Langfuse Cloud Core \$29/mo – 100k units, +\$8/100k, 90-day retention; Pro \$199/mo [fetched langfuse.com/pr
18	Rate limiting	SPLIT	Policy: BUILD. State: BUY Upstash Redis PAYG – \$0.20/100K commands, \$0.25/GB after 1 GB free, bandwidth f
19	PDF generation	BUILD on OSS	Gotenberg MIT, 12,962*, pushed 2026-08-21 [fetched], one container ≈ \$5/mo. Alt: Browserless Prototyping
20	Image processing	BUY	Cloudflare Images – \$0.50/1,000 unique transformations after 5,000 free, \$5/100k stored/mo, \$1/100k deliv
21	Markdown engine	BUILD	\$0 in licence, everything in time. Nearest OSS: comrak 1,688*, markdown-it MIT 21,861*, remark MIT 8,987*
22	Editor surface	ADOPT OSS	CodeMirror 6 – 7,820* on codemirror/dev [fetched]

24.2 The monthly vendor bill, with arithmetic

Paid conversion 8%, ARPU \$12/mo, all three scales [derived].

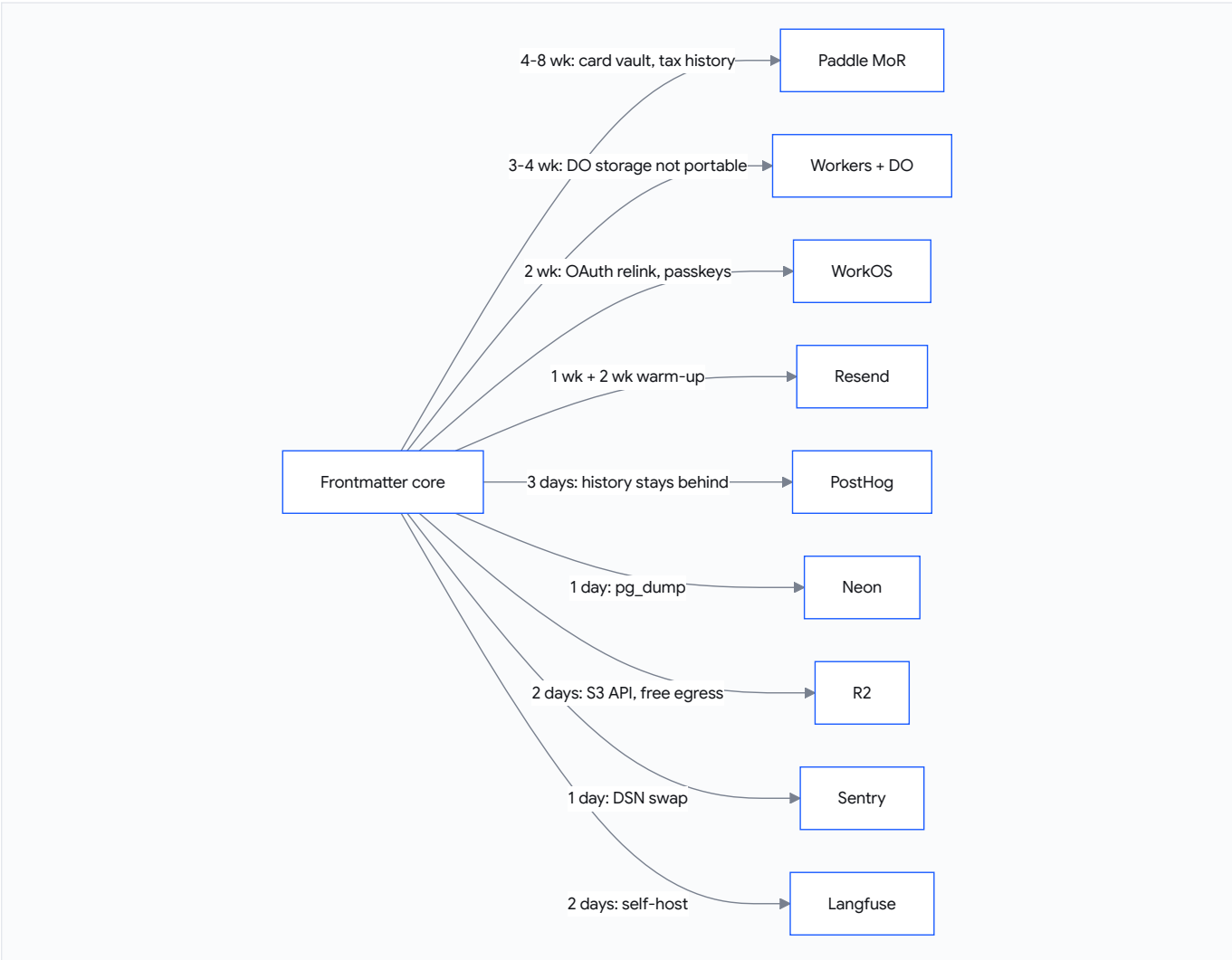
LINE	100 USERS (\$96 MRR)	1,000 USERS (\$960 MRR)	10,000 USERS (\$9,600 MRR)
MoR (Paddle 5% + 50¢)	0.05×96 + 0.50×8 = \$8.80	0.05×960 + 0.50×80 = \$88.00	0.05×9,600 + 0.50×800 = \$880.00
Auth (WorkOS)	\$0	\$0	\$0
Postgres (Neon Launch)	60 CU-h×\$0.106 + 2 GB×\$0.35 = \$7.06	300 CU-h×\$0.106 + 10 GB×\$0.35 = \$35.30	1,000 CU-h×\$0.106 + 50 GB×\$0.35 = \$132.00
Workers + Durable Objects	\$5 base = \$5.00	\$5 + (6M-1M)×\$0.15/M = \$5.75	\$5 + (60M-1M)×\$0.15/M + DO duration ≈ \$40
R2	free tier = \$0	(20-10) GB×\$0.015 = \$0.15	(200-10)×\$0.015 + (100M-10M)×\$0.36/M = \$3
Email (Resend)	Free = \$0	Pro = \$20.00	Scale = \$90.00
Sentry	Developer = \$0	Team = \$26.00	Team + PAYG ≈ \$50.00 [inference]
Uptime (Better Stack)	\$0	\$0	\$29.00
PostHog	free (≈40k events) = \$0	free (≈400k events) = \$0	≈4M events ≈ \$200.00 [inference – the per
Support desk (Plain)	\$0	Foundation 1 seat = \$35.00	Foundation + 2 seats = \$105.00
Langfuse	Hobby = \$0	Core = \$29.00	Core + 4×\$8/100k units = \$61.00
Upstash	0.3M cmds = \$0.60	3M cmds = \$6.00	30M cmds = \$60.00
Gotenberg container	\$5.00	\$5.00	2 containers = \$20.00
Cloudflare Images	free tier = \$0	(30k-5k)/1,000×\$0.50 + \$1.50 + \$3.00 = \$17.00	(300k-5k)/1,000×\$0.50 + \$15 + \$30 = \$192.
Total	\$26.46	\$267.20	\$1,894.75
As % of MRR	27.6%	27.8%	19.7%

Two readings [derived]. First, MoR is 33% of the bill at 100 users and 46% at 10,000 – it is the only line that scales linearly with revenue, so every other LLM inference is deliberately excluded from this table. It is not a build-versus-buy question – there is nothing to build – and at 10,000 users it will exce

24.3 The three obvious buys and the three obvious builds

Buy: merchant of record, email deliverability, error tracking. Payments because the failure mode is a tax authority rather than a bug, and 14 founder-weeks Build: the markdown engine, the queue, the docs site. The engine because byte-preserving splice and degradation certification are the product, and every buy

24.4 Lock-in map and exit cost



Ranked exit cost [inference, from the migration mechanics in each vendor's export path]: Paddle 4–8 weeks and irreducible churn, because subscriptions and t

24.5 The rule for future decisions

Apply in order; stop at the first gate that fires.

- 1. Does it touch document bytes? Build. No exception, no pilot, no "just for the index."
- 2. Is it on the projection law's critical path – parse, splice, render, certify? Build.
- 3. Does its outage take the editor down, or only degrade it? If down, build it or make it optional at runtime.
- 4. Is its failure mode regulatory – tax, PCI, deliverability, SOC 2 evidence? Buy, and buy the boring incumbent.
- 5. Otherwise compare build cost against 24 months of vendor spend at the 10,000-user scale – then buy anyway, because that comparison omits maintenance, whic
- 5. Write the migration runbook before signing. If it does not fit on one page, the lock-in is priced higher than the invoice.

Never buy a component whose meter is your free users, and never build a component whose failure mode is a tax filing.

24.6 Anti-recommendations

Commonly built, should be bought: authentication with password reset, session rotation, MFA and OAuth (5 weeks, and the bugs are security bugs); the billing
Commonly bought, should be built or skipped: a documentation vendor, for a company whose product renders markdown; a real-time collaboration vendor – Livebl

]

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How comparable products are actually built

25.1 The eleven systems, opened

[

Every row below was read from the source named, on 2026-08-30. Star counts and versions are the values the GitHub API returned that day [measured].

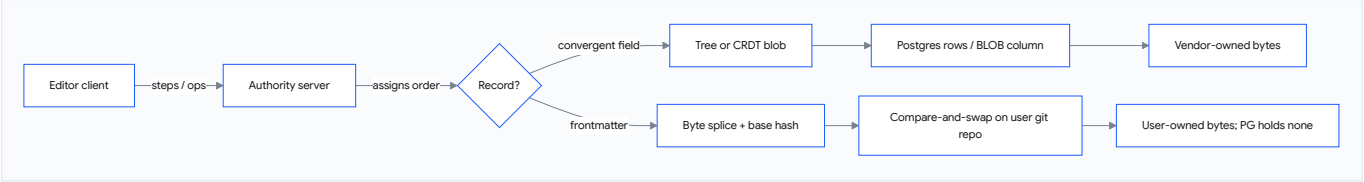
SYSTEM	STORAGE MODEL	SYNC MODEL
Google Docs	Revision log + current state on server	OT; client tracks 4 items (last server revision, un
Jupiter (the ancestor)	Central server holds world state	Centralised OCC + operation transformation, derived
Figma	Map<ObjectID, Map<Property, Value>> tree; comments/teams/projects in Postgres, separate system	Not OT, not "true CRDTs"; per-property last-writer-w
Linear	Models + properties + references; IndexedDB per workspace	Transactions → server → delta packets; global monot
Notion	Every unit is a block row: UUID v4, type, properties, <code>content[]</code> , <code>parent</code>	Server-authoritative
Obsidian	Plain files on the user's disk; no server copy required	Optional paid Sync, E2E encrypted, version history
Outline	Three columns per document: <code>text</code> (markdown, @deprecated), <code>content</code> (ProseMirror JSONB), <code>state</code> (Yjs BLOB)	Yjs 13.6.31 + y-prosemirror 1.3.7
HedgeDoc / CodiMD	Markdown text is the record	Classic ot.js: <code>text-operation.js</code> , <code>wrapped-operation.j</code>
AppFlowy	Rust <code>collab-*</code> crates over <code>yrs</code> 0.21, SQLite via diesel, RocksDB, tantivy search	CRDT
AFFiNE	BlockSuite docs in Yjs (patched 13.6.21)	CRDT
Docmost	Postgres via Kysely, BullMQ/Redis, pgvector 0.2.1; <code>collaboration/yjs.util.ts</code>	Yjs

Two more that set the priors: **ProseMirror** (8,702 ★, MIT) ships a collab module whose documented algorithm is "a central authority which determines in which

25.2 The pattern table

CONVERGED PRACTICE	WHO DOES IT
A central server assigns a total order	Google Docs, Jupiter, Figma, Linear, ProseMirror, Etherpad (18,514 ★,
The record is a structured tree or binary blob, not document text	Notion, Figma, Outline, AFFiNE, AppFlowy, Docmost, Linear
Document text is the record	Obsidian, HedgeDoc/CodiMD
Relational DB as the control plane, holding metadata not document bytes	Figma explicitly ("comments, users, teams, projects... stored in Postgre
CRDT chosen by open-source projects; hand-rolled LWW/OT chosen by the well-funded	Outline/AFFiNE/AppFlowy/Docmost vs Figma/Linear/Google
Shard the control plane by tenant	Notion: workspace ID → one of 480 logical shards

The two systems that reject CRDTs give the same reason in different words. Figma: "CRDTs are designed for decentralized systems where there is no single cen
The cost they are avoiding is measurable. On Kleppmann's 260k-operation editing trace, Automerge's own published table gives 107,121 bytes of plain text, 12



25.3 Where we are conventional, and where we are genuinely unusual

OUR DECISION	VERDICT
Server is the total-order authority; writes are compare-and-swap	Conventional. Identical in shape to ProseMirror's authority and Linear's <code>lastSyncId</code>
Postgres as a control plane holding zero document bytes	Conventional. Figma runs exactly this split
Blobs in object storage (R2), out of the DB	Conventional
No arbitrary client-side plugin execution	Conventional among web products; Obsidian is the outlier at 7,115 plugins, and that is
Deterministic simulation testing of the engine	Conventional at the top end only – Dropbox runs "millions of scenarios every day" from
The byte sequence of a markdown file is the record	Unusual. Only Obsidian, and Obsidian does no server-side collaborative editing of those
git merge as the document merge function in a live editor	Unusual to the point of unique in this set
Documents live in the user's own repo	No comparable product does this
Cross-engine degradation certification	No published equivalent found

The moat and the risk are the same fact. Outline's source marks its markdown column `@deprecated` and directs callers to `DocumentHelper.toMarkdown` "if exporting
The second unusual risk is dependency, not design. Notion owns its 32 Postgres hosts and pages itself when they hit 90% CPU [fetched]. We would own none of

25.4 What the ones who lost data did wrong, and whether we have the same hole

INCIDENT
GitLab, 2017-01-31 - ~300 GB removed from the primary; modifications from 17:20-00:00 UTC lost; ~5,000 projects, 5,000 comments, 700 accounts gone [fetch
Dropbox Sync Engine Classic [fetched]
Dropbox, 2014-01-10 [fetched]
Outline [fetched]
Figma [fetched]
Automerger 0.14 [fetched]

25.5 The three converged decisions we are contradicting

One - the record should be a structured tree, not text. Seven of eleven systems store a tree or a CRDT blob, and Outline has actively demoted markdown to a

Two - concurrent editing must converge at character level. Every system here except Obsidian either runs OT (Google Docs, CodiMD, Etherpad) or a CRDT (Outli

Three - the vendor should own the bytes. Ten of eleven do. Our zero-byte control plane collapses the DPDP and GDPR surface, removes storage COGS from unit e

§106
Remaining market and audience questions

26.1 Segment table

All prices, star counts and API counts below were read on 2026-08-30 unless stated. "WTP/yr" is the realistic annual contract value per buying unit, not per

#	SEGMENT	SIZE EVIDENCE
1	AI-native non-developer knowledge worker (analyst, PM, solo researcher, indie consultant)	Evernote's free tier is capped at 50 notes / 1 notebook /
2	Technical writer / documentation engineer as an individual	"technical writer" appears in 1,241 HN comments all-time
3	Docs team at an API-first company (team budget)	Mintlify Pro \$450/mo, Starter free at 5 editor seats [fet
4	Developer relations / advocacy teams	"developer relations" 1,087 and "developer advocate" 806
5	Open-source maintainers	472,539 GitHub repos at ≥100 stars, 64,391 at ≥1,000, 5,5
6	Boutique consultancies and agencies (client deliverables)	Management consulting is defined as fee-for-outcome advis
7	Regulated industries needing audit trails (GxP, broker-dealer, clinical, ISO/QMS)	21 CFR 11.10(b) requires "accurate and complete copies of
8	Note-taking refugees off Evernote / Notion	Evernote is owned by Bending Spoons, which "acquires prod
9	Non-English markets beyond India, Japan first	46 of the 100 most recent Zenn articles carry a non-null header, qiita.com/api/v2/items]; Obsidian ships 15 UI lan,
10	Education as buyer (departments, labs, MOOC producers)	3,931 Title IV degree-granting institutions in the US alo
11	Paid technical authors and course producers	Zenn lets authors price a book from ¥0 to ¥5,000 and pays

Reachability note that changes the ranking: geography-priced consumer competitors are already discriminating. Craft's pricing page served INR to this machin

26.2 The two segments the previous thirteen rounds most underrate

Japan is the one market on earth that has already been trained to accept this product's hardest premise - that your documents live as markdown files in your

The rounds covered India billing and i18n as a compliance and rendering problem. They did not ask where the premise is already normal. Zenn's own documentat

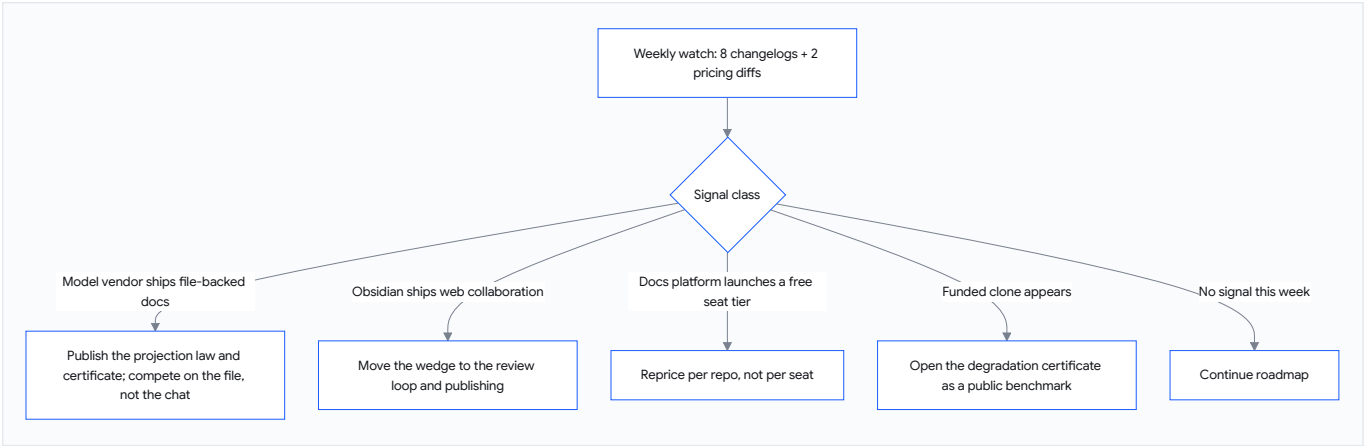
Recommendation: after the English D2C launch stabilises, run one Japanese landing page and one Zenn article written by a paid native technical writer, and i

The second underrated segment is the boutique consultancy, and the reason is the engine, not the editor. A consultancy's product is a document that a client

Recommendation: sell the splice journal to consultancies as "prove what changed and when, per client, in a repo you hand over at the end of the engagement",

26.3 Threat map, 24 months

ENTRANT	WHAT THEY WOULD HAVE TO BUILD	TRIGGER THAT STARTS IT
Anthropic / OpenAI	A durable, file-backed artifact store so agent output survives the session	Agent output volume outgrows chat transcr
GitHub / Microsoft	WYSIWYG + collaboration on top of the existing web editor	Copilot needs a review surface for prose
Google	Native .md storage in Drive rather than import/export conversion	Gemini output needs a home that engineers
Obsidian	Real-time collaboration and a web editor on top of local files	Their own Sync page already advertises "C
Docs platforms moving down-market	A cheap generic editor untied to a published site	Seat-based competition; their floors are
IDE vendors (Cursor, Zed)	A prose mode with review-oriented diffs	Agents generate more prose than code in s
A funded clone	The surface only; the engine takes longer than a seed round	Our own public traction becomes legible
Bending Spoons	Acquisition-led consolidation of the notes category	Public since 2026-07-01 [fetched], with a



Recommendation: automate the eight changelog feeds and two pricing-page diffs as a cron job that opens an issue on change; the whole watch costs one afterno

26.4 What would have to be true for each segment to become the main business

Each condition below is falsifiable within one quarter with instrumentation already planned.

SEGMENT	WHAT MUST BE TRUE
1. AI-native non-developer	People who never chose markdown will accept a file-backed document because their AI already emits markdown
2. Individual technical writer	The individual can buy without their team's docs-platform vendor being displaced
3. Docs team	We win on price plus review loop against a \$4,800/yr ReadMe Pro + Ask AI bundle [derived: \$250 + \$150 = \$400/mo × 12] whi
4. DevRel	DevRel holds or can trigger a budget line, not just influence
5. OSS maintainers	Free maintainer usage converts downstream at the maintainer's employer
6. Consultancies	Per-client workspace isolation plus handover export is worth 3-10 seats
7. Regulated	A compliance partner carries SOC 2, validation and questionnaires so the founder does not
8. Note refugees	Refugees pay rather than route to the free tier
9. Japan	The git-authoring premise transfers to paid subscription, not just free publishing
10. Education	An institution can buy without an RFP
11. Paid authors	Publishing targets are good enough to replace a bespoke pipeline

26.5 Anti-recommendations

Segments that will absorb effort and not convert, with the specific failure mode.

Education as an institutional buyer, for 24 months. 3,931 US degree-granting institutions [fetched] is a large number attached to the slowest purchasing pro DevRel as a buyer. DevRel is the highest-leverage audience in the table and one of the weakest budgets. They are reachable, publicly visible, generous with Evernote and Notion refugees as a revenue segment. They convert to traffic reliably and to payment poorly. Evernote's free tier at 50 notes and 1 device [fe Open-source maintainers as revenue. 472,539 repos at ≥100 stars [measured] is a distribution surface, not a market. Maintainers are the most vocal free-tier Regulated enterprise without a partner. The 21 CFR Part 11 language maps to the engine almost word for word [fetched], which is exactly why this is a trap: China. Yuque and Feishu own the equivalent territory, and the payments, hosting and content-compliance requirements are not addressable by a solo founder in Product-UI localisation ahead of demand. See 26.2. One landing page and one native-language article per market, then measure; the UI translation is the rewa

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§107
The build sequence, given the stack

\$28 sequences features. This sequences layers. The ordering constraint is not "what does a user want next" but "what physically cannot be built until somet

[

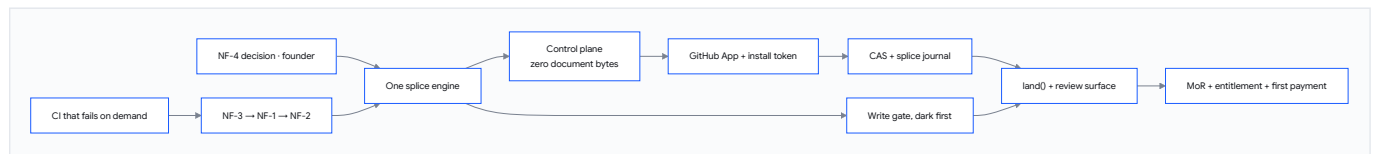
27.1 Dependency-ordered work breakdown — empty CI to first payment

Layer codes: ENG bytes · DOM src/modules/ · APP Next route or server action · UI CodeMirror 6 / React 19 · CP Postgres control plane (zero document bytes) ·

#	ITEM	LAYER	CANNOT START UN
E0	Rotate the two PATs	credential	-
E1	<code>ci.yml</code> : typecheck · lint · test · build · arch · spec	CI	E0
E2	Red-proof the workflow on a deliberately broken commit	CI	E1
E3	Replace <code>npm run budget</code> - today literally <code>echo 'No bundle budget configured yet - skipping'</code> <code>MEASURED 2026-08-30</code> - with a byte ceiling that exits 1	CI	E1
E4	<code>npm run corpus</code> as a CI job, pinned at the 8,513-file count the runner already hard-codes <code>MEASURED: LITERAL 8,513 IN SCRIPTS/CORPUS-FOREIGN.WJS</code>	CI+ENG	E2
E5	NF-3 red proof : a set-only assertion on a synthetic bare-CR fence that fails at HEAD	ENG	E4
E6	NF-3 fix: <code>FM_OPEN</code> accepts <code>---\r</code> ; a lone set stops prepending a second block	ENG	E5
E7	NF-1 red proof, then fix: column-0 <code>-</code> is a continuation of the preceding key	ENG	E6
E8	NF-2: flow-seq closing <code>]</code> at column 0	ENG	E7
E9	Re-derive the recovery rate from the <i>patched</i> writer; commit the run's own JSON	ENG	E7, E8
E10	NF-4 decision: one page on NFC/NFD key equality	spec	- (start day 1
E11	<code>SAFE_KEY</code> quoted-key implementation	ENG	E10
E12	Seam 3 - one splice engine replacing <code>splice-frontmatter.ts</code> 's single-key scanner	ENG+DOM	E8, E11
E13	Seam 2 - write gate dark , behind a flag defaulting off	DOM+APP	E9 (residual ≤
E14	Write gate default-on	APP	E13 + 7 days o
E15	Control-plane schema: repos, installations, entitlements, journal rows. Zero document bytes	CP	E12 (journal r
E16	Register the GitHub App; private key, callback URL, permissions	GIT	E15 (needs a d
E17	Installation-token git adapter	GIT+APP	E16
E18	Compare-and-swap write path keyed on blob sha	GIT+CP	E12, E17
E19	Splice journal writes	CP	E18
E20	Conflict inbox	UI	E19
E21	Two-device offline rig	test	E20
E22	Identity and multi-tenancy	CP+APP	E17
E23	MCP server + <code>land()</code>	APP	E12, E18
E24	Review surface; reject leaves the file byte-identical	UI	E23, E20
E25	Seam 4 - certificate sidecar job to R2	BLOB	E12
E26	ChatGPT / Claude importers + drop report	APP	E13
E27	MoR, ToS, privacy, EU Art. 27 representative	LEGAL	- (vendor cloc
E28	Paywall and entitlement check	CP+PAY	E22, E27
E29	First paying non-founder account	-	E28 + M0-M6

Nothing measured before E2 is evidence, because a gate that has never been seen to fail has not been seen at all - the repo already carries a commit named *Anti-recommendation*: do not front-load E15 (the control plane) because it feels like "real product". Written before E12, the journal row shape is a guess, a

27.2 The critical path



Ten nodes. Renders (kanban, decision, calendar) appear nowhere on it, which is the single biggest difference between the feature roadmap and the engineering

27.3 The first ten commits

Every proof below is falsifiable in the same way: run the check at the parent commit and it must produce the opposite result. A check that passes at both co

#	COMMIT	WHAT IT DOES
1	<code>chore(ci): add ci.yml - typecheck, lint, test, build, arch, spec</code>	Creates <code>.github/</code> , which does not currently exist <code>MEASURED</code> . Six jobs, no <code>continue-on-error</code> , no path filters
2	<code>test(ci): deliberately break typecheck, observe red, revert</code>	Two commits' worth of history in one PR: a <code>const x: number = "a"</code> , then its revert
3	<code>chore(ci): budget asserts a real byte ceiling</code>	Replaces the echo stub with a measured <code>.next</code> output ceiling
4	<code>ci(corpus): run npm run corpus on engine/** and nightly</code>	Wires the existing runner into the workflow; asserts the file count, not just the exit code
5	<code>test(engine): red proof - bare-CR fence set-only (NF-3)</code>	A set-only assertion on a synthetic fixture. Set-then-delete cancels the defect out, which is why a round-trip oracle never saw it
6	<code>fix(engine): FM_OPEN accepts - --\r; a lone set no longer prepends a block</code>	The fix for #5
7	<code>test(engine)*fix: column-0 '-' is a continuation (NF-1)</code>	Red proof then fix, same commit-pair discipline
8	<code>chore(engine): re-derive the recovery rate from the patched writer</code>	Commits the artifact the runner emitted, replacing the 99.98% inference with a count
9	<code>fix(engine): flow-seq closing] at column 0 (NF-2)</code>	The last of the three walk defects
10	<code>refactor(vault): route writes through shape-gate + placement behind FM_WRITE_GATE (default off)</code>	Seam 2 lands dark

Anti-recommendation: do not batch commits 5-9 into one "engine fixes" PR. The red-proof discipline only works if each fix has a parent commit at which its t

27.4 Agent-ownable versus founder-only

AN AGENT OWNS END TO END	WHY IT CAN	FOUNDER ONLY
<code>ci.yml</code> , budget gate, corpus job	Mechanical verifier; the artifact is one file	PAT rotation (E0)
NF-1/2/3 walk changes and fixtures	Disjoint files, each with a red proof	NF-4 key-equality decision (E10)
The six construct detectors, one agent each	Genuinely disjoint artifacts	GitHub App registration, private key, cal
CJK segmenter and MiniSearch bigram tokenizer	Test-verifiable against fixtures	MoR, CA, lawyer, EU Art. 27 rep (E27)
<code>globals.css</code> reconciliation, including the 2.14:1 token that fails WCAG AA	Contrast is computable; the check is a number	Watching the E21 two-device run
Importers, MCP scaffold, <code>land()</code> transport, test authoring	Contract-shaped; the verifier is byte-identity	Accepting or rejecting every agent diff

Anti-recommendation: do not fan agents out across the sync surface (E18-E20). It is one creative target carrying implicit architecture decisions - the confl

27.5 Milestone gates, executable

M	COMMAND THAT DECIDES IT	PASS CONDITION
M0 CI live	<code>gh run view <broken-sha> --json conclusion</code> then <code><fixed-sha></code>	<code>failure</code> then <code>success</code> ; <code>npm run budget</code> exits
M1 Engine truth	<code>npm run corpus</code> → read its JSON	<code>refused ≤ 2 / 7,969</code> , <code>changed = 0</code> , <code>threw = 0</code>
M2 MDMAX wired	<code>grep -rln "modules/mdmax" src grep -v "^src/modules/mdmax"</code>	≥ 8 distinct files. Today it is exactly 2 -
M3 Trust surface	Two machines, both offline, both edit, both reconnect; <code>sha256sum</code> before and after restore	Zero bytes lost; every divergence appears a
M4 Tenancy	A second GitHub account performs 100 sequential contents reads via an installation token	No 403. Unauthenticated reads cap at 60/hou :
M5 Agent protocol	External agent lands a change; reject it; diff	Post-reject sha256 equals pre-land sha256
M6 Capture	One ChatGPT export ZIP through the importer	The drop report names a <i>seeded</i> lossy constr
M7 v1 public	One non-founder card settles; entitlement flips	Webhook replayed in staging first, then liv

Anti-recommendation: do not add Playwright (v1.62.1, published 2026-07-30 FETCHED 2026-08-30) before M3. A browser suite written against a UI that is about to

27.6 The three sequencing mistakes that cost the most

MISTAKE	WHY IT IS EXPENSIVE
Seam 2 before E9 - turning the write gate on before the residual refusal rate is re-measured	At the measured foreign-vault refusal rate the gate re
Sync before CAS - building the conflict UI before the compare-and-swap and journal contract exist	The conflict unit silently becomes "the file". Retrofi
User PATs before the GitHub App - shipping repo access on user-supplied tokens because next-auth v5 beta is already installed MEASURED: ^5.0.0-BETA.31	You become a credential custodian on day one, inherit

27.7 What v1 deliberately does not contain

NOT BUILT	WHY	WHAT IT COSTS
Kanban, decision and calendar renders	Off the critical path entirely	The demo is an editor. The "deep engin
Real-time multiplayer cursors	Settled: no CRDT for document bytes	Every head-to-head against Google Docs
Plugin marketplace and client-side code execution	Settled	No third-party long tail. Every integr
WYSIWYG	T6	The non-technical half of D2C cannot b
Teams, roles beyond owner, SSO	T6	Self-serve B2B lands single-seat. Expa
Full offline vault	Tauri v2 is in dependencies MEASURED: ^2.11.0 but offline is T6	The "your files, your repo" promise is
SOC 2	Cost and calendar	Caps deal size. The ceiling is the buy

The cheapest item on the deferral question is the certificate job (E25), which is *not* deferred and costs almost nothing: one JSON sidecar per document per n

Anti-recommendation: do not treat this list as permanently settled. Renders and WYSIWYG are deferred on *sequence*, not on merit - the moment M5 is green and

